Claudius Gros

Complex and Adaptive Dynamical Systems

A Comprehensive Introduction

Fifth Edition

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May our own thoughts surprise us.

Preface

Complexity, Emergence and Self Organization Complex systems theory deals with dynamical systems showing a large variety of emergent phenomena. Examples relevant to all realms of science abound.

– Powerlaw distributions in evolving networks.
– Deterministic chaos in one-dimensional maps.
– Strange attractors in adaptive dynamical systems.
– Spontaneous pattern formation in reaction-diffusion systems.
– Self-organized criticality in cellular automata.
– Organisms living at the edge of chaos.
– The rise of life in terms of hypercycles.
– Spontaneous synchronization of interacting oscillators.
– Neural networks underpinning modern machine learning algorithms.

These are a few examples treated in this textbook. Complex systems theory is a fascinating subject becoming increasingly important in our socio-scientific culture, with its rapidly growing degrees of complexity. It is also a deeply interdisciplinary subject, providing the mathematical basis for applications in fields ranging from the social sciences and psychology on one side to the neurosciences and artificial intelligence on the other side.

Readership and Preconditions This primer is intended for students and scientists from the natural and social sciences, engineering, informatics, experimental psychology, or neuroscience. Technically, the reader should have a basic knowledge of mathematics as it is used in the natural sciences or engineering. This textbook is suitable both for studies in conjunction with teaching courses as well as for the individual reader.

Course Material and the Modular Approach When used for teaching, this primer is suitable for a course running over one or two terms, depending on the pace and on the number of chapters covered. In general, there should be no problem to follow the mathematics involved. Considerable care has been taken to perform the respective derivations on a step-by-step basis.

Individual chapters may be skipped, whenever time considerations demand it. The book follows a modular approach and the individual chapters are, as far as possible, independent of each other. Notwithstanding, cross references between the different chapters are included throughout the text, since interrelations between distinct topics may be helpful for a thorough understanding.

Style This interdisciplinary primer sets a high value on conveying concepts and notions within their respective mathematical settings. Believing that a concise style helps the reader to go through the material, this textbook mostly abstains from long text passages with general background explanations or philosophical considerations. Widespread use has been made of paragraph headings with the intention to facilitate scientific reading.

A Primer to Scientific Common-Sense Knowledge To a certain extent, one can regard this textbook as a primer to a wide range of scientific common-sense knowledge regarding complex systems. A basic expertise about life's organizational principles, to give an example such as the notion of "life at the edge of chaos", is important in today's world to an educated scientist. Other areas of scientific common-sense knowledge discussed in this primer include network theory, which has applications ranging from social networks to gene expression networks, the fundamentals of evolution, cognitive systems theory, as well as the basic principles of dynamical systems, self organization and information theory.

Exercises and Suggestions for Further Readings Towards the end of each chapter, a selection of exercises is presented. The respective solutions can be found in a dedicated chapter. In addition, the section "Further Reading" at the end of each chapter contains references to standard introductory textbooks and review articles. Also listed are selected articles for further in-depth studies dealing with specific issues treated within the respective chapter.

Content of the Fifth Edition Since its first printing in 2008, this textbook has seen several extensions. The present fifth edition has been completely revised. As a main addition, the complex systems theory of modern machine learning architectures has been included in a new chapter. In addition, a range of new sections have been added to existing chapters, each dealing with an exciting topic. Such as "Rate Induced Tipping", "Partially Predictable Chaos", the "Tragedy of the Commons", and "Piecewise Linear Dynamical Systems", to mention a few. The field of complex systems is continuously evolving, and so is this book.

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Frankfurt/Main, Germany Claudius Gros
January 2024

Contents

1 Network Theory

Dynamical and adaptive networks are the backbone of many complex systems. Examples range from ecological prey–predator networks to the gene- and protein expression networks on which all living creatures are based. The basis of our own identity is provided by the highly sophisticated neural network we carry in our heads. On a social level we interact through social and technical networks like the Internet. Networks are ubiquitous throughout the domain of all living creatures.

A good understanding of network theory is of basic importance for complex systems theory. In this chapter we will discuss the most important notions of graph theory, like clustering and degree distributions, together with basic network realizations. Central concepts like percolation, the robustness of networks with regard to failure and attacks, and the "rich-get-richer" phenomenon in evolving social networks will be treated.

1.1 Properties of Real-World Networks

1.1.1 The Small World Effect

Eight or more billion humans live on earth and it might seem that the world is a big place. But, as an Italian proverb states,

> Tutto il mondo è paese. – The world is a village.

The network of who knows whom—the network of acquaintances—is indeed quite densely webbed. Modern scientific investigations mirror this century-old proverb.

Social Networks Stanley Milgram performed a by now famous experiment in the 1960s. He distributed a number of letters addressed to a stockbroker in Boston to a random selection of people in Nebraska. The task was to send these letters to the

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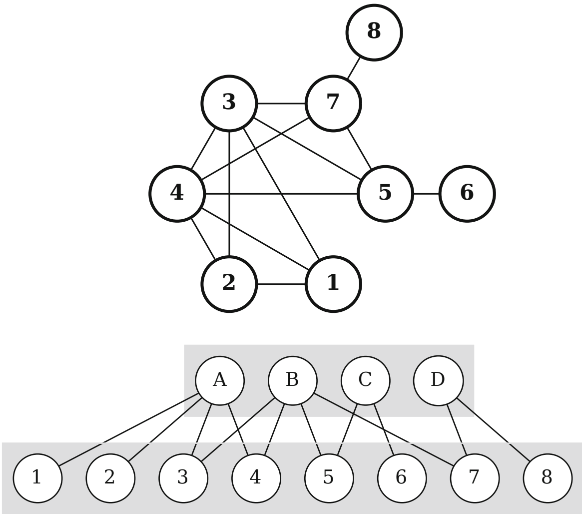

Fig. 1.1 **Bottom:** A network with two kind of nodes, a bipartite networks. The letters A–D may stand for distinct companies, the numerals 1–8 for managers serving on the respective boards of directors. **Top:** Managers serving on the same boards of directors are acquainted, forming a densely webbed social network

addressee (the stockbroker) via mail to an acquaintance of the respective sender. In other words, the letters were to be sent via a social network.

The initial recipients of the letters clearly did not know the Boston stockbroker on a first-name basis. Their best strategy was to send their letter to someone whom they felt was closer to the stockbroker, socially or geographically: perhaps someone they knew in the financial industry, or a friend in Massachusetts.

Six Degrees of Separation About 20% of Milgram's letters did eventually reach their destination. Milgram found that it had only taken an average of six steps for a letter to get from Nebraska to Boston. This result, dubbed "six degrees of separation", suggests that it should be possible to connect any two persons on earth via social networks in a similar number of steps.

> SMALL-WORLD EFFECT The "small-world effect" denotes the result that the average distance linking two nodes belonging to the same network can be orders of magnitude smaller than the number of nodes making up the network.

The small-world effect occurs in all kinds of networks. Milgram examined the networks of friends, which may originate from participating in common activities. This is the case also for other social networks, as illustrated in Fig. 1.1. An example are managers serving together on the board of directors of the same company, which results in a dense network of acquaintances between the managers.

Networks Are Everywhere Social networks are but just one important example of a communication network. Most human communication takes place directly among individuals. The spreading of news, rumors, jokes and of diseases takes place

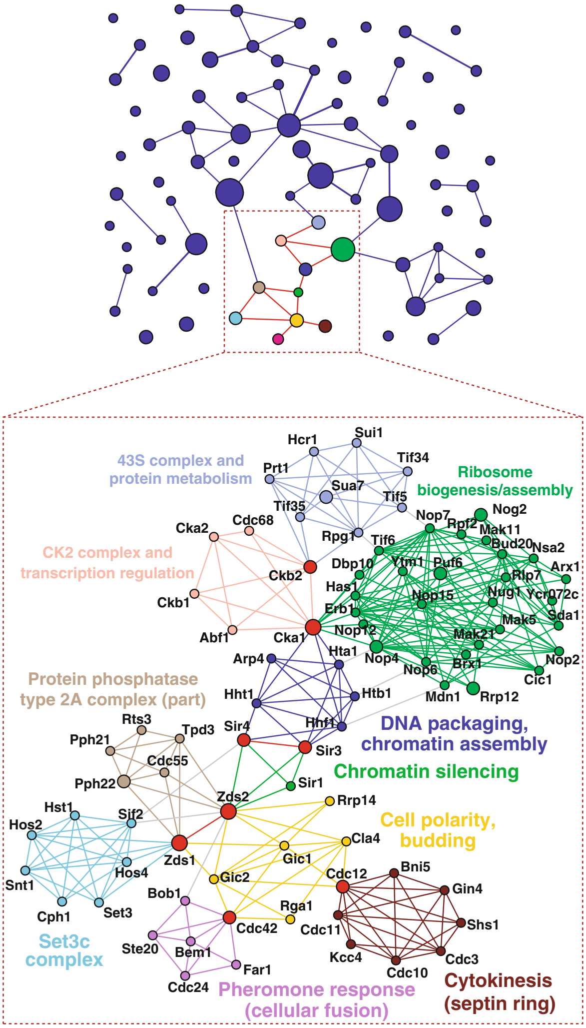

Fig. 1.2 A protein interaction network, showing a complex interplay between highly connected hubs and communities of subgraphs with increased densities of edges. Reprinted from Palla et al. (2005) with permissions, © 2005 Nature Publishing Group

by contacts between individuals. And we are all aware that rumors and epidemic infections can spread fast in densely webbed social networks.

Communication networks are ubiquitous. Well known examples are the Internet and the world-wide web. Inside a cell the constituent proteins form an interacting network, as illustrated in Fig. 1.2. The same holds for artificial neural networks and for the scaffolding of our brain, neural networks. It is therefore important to understand the statistical properties of core network classes.

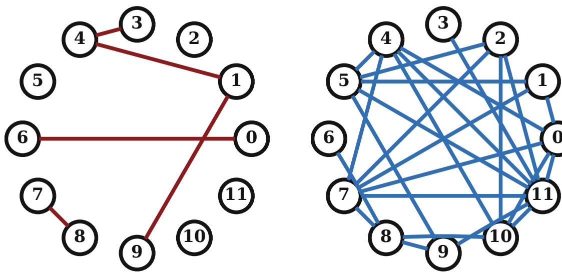

Fig. 1.3 Random graphs with $N = 12$ nodes and connection probabilities $p = 0.0758$ and $p = 0.3788$ (left/right). Three mutually connected vertices, such (0,1,7), contribute to the clustering coefficient

1.1.2 Basic Graph-Theoretical Concepts

We start with some basic concepts allowing to characterize graphs and real-world networks. We will use the terms graph and network as interchangeable, vertex, site and node as synonyms, and either edge or link.

Degree of a Vertex A graph is made out of N vertices connected by edges, with the degree k of a vertex being given by the number of edges linking to it. Nodes having a degree k substantially above the average are denoted "hubs", they are the VIPs of network theory.

Coordination Number The simplest type of network is the random graph. It is characterized by only two numbers: By the number of vertices N and by the average degree z, also called the coordination number.

> COORDINATION NUMBER The coordination number z is the average number of links per vertex, viz the average degree.

A graph with an average degree z has $Nz/2$ connections. Alternatively we can define with p the probability to find a given edge.

> CONNECTION PROBABILITY The probability that a given edge occurs is called the connection probability p.

Erdös–Rényi Graphs We can construct a specific type of a random graph with N nodes by drawing $Nz/2$ lines, the edges, between randomly chosen pairs of nodes, compare Fig. 1.3. This type of random graph is called an "Erdös–Rényi" random graph after two mathematicians who studied this type of graph extensively. In Sect. 1.3 we will introduce and study other types of random graphs.

For Erdös–Rényi random graphs we have

$$p = \frac{Nz}{2}\frac{2}{N(N-1)} = \frac{z}{N-1} \tag{1.1}$$

for the relation between the coordination number z and the connection probability p. In (1.1) we divided the total number of active links, $Nz/2$, by the total number of possible links, $N(N-1)/2$.

The Thermodynamic Limit Mathematical graph theory is often concerned with the thermodynamic limit.

> THE THERMODYNAMIC LIMIT The limit where the number of elements making up a system diverges to infinity is called the "thermodynamic limit" in physics. A quantity is extensive if it is proportional to the number of constituting elements, and intensive if it scales to a constant in the thermodynamic limit.

We note that $p = z/(N-1) \to 0$ in the thermodynamic limit $N \to \infty$ for Erdös–Rényi random graphs when the coordination number is intensive, viz when $z \sim O(N^0)$.

For real-world networks, the thermodynamic limit makes sense for very large numbers of nodes. This is the case for the world-wide web, the network of hyperlinks, which is so large that its size can only be estimated. In order of magnitude, the number of documents available online did grow from $N \simeq 0.8 \times 10^9$ in 1999 to $N \simeq 3.4 \times 10^{15}$ in 2023.

Network Diameter and the Small-World Effect A basic parameter characterizing networks is the diameter D, which specifies the maximal separation between all possible pairs of vertices. For a random network with N vertices and coordination number z we have,

$$z^D \approx N, \qquad D \propto \log N / \log z \,, \tag{1.2}$$

in order of magnitude, since nodes have z neighbors, z^2 next-nearest neighbors, and so on. The logarithmic increase in the number of degrees of separation with the size of the network is characteristic of small-world networks. A logarithm increases very slowly with its argument, which implies that network diameters remains small even for networks containing large numbers of nodes.

For large networks, the diameter is closely related to the average distance.

> AVERAGE DISTANCE The average distance ℓ is the average of the minimal path length between all pairs of nodes of a network.

Stochastic consideration will use in general the average distance ℓ.

Clustering in Networks Real-world networks, as the protein network reproduced in Fig. 1.2, contain substantial local recurrent connections. Distinct topological elements and loops are hence frequent.

> CLUSTERING COEFFICIENT The clustering coefficient C is the average fraction of pairs of neighbors of a node that are also neighbors of each other.

The clustering coefficient is a normalized measure of loops of length three. In a fully connected network, in which everyone knows everyone else, $C = 1$.

In a random graph a typical site has $z(z-1)/2$ pairs of neighbors. The probability of an edge to be present between a given pair of neighbors is $p = z/(N-1)$, see (1.1). The clustering coefficient, which is just the probability of a pair of neighbors to be interconnected is therefore

$$C_{\text{rand}} = \frac{z}{N-1} \approx \frac{z}{N} . \tag{1.3}$$

The clustering coefficient scales to zero in the thermodynamic limit, viz for large networks. In Table 1.1 the respective clustering coefficients for some real-world networks and for the corresponding random networks are listed for comparison.

Cliques The clustering coefficient measures the normalized number of triples of fully interconnected vertices. Cliques generalize the notion of clustering to larger subgraphs, see Fig. 1.4.

CLIQUES A clique is a set of vertices for which (a) every node is connected by an edge to every other member of the clique and (b) no node outside the clique is connected to all members of the clique.

The term "clique" comes from social networks. A clique is a group of friends where everybody knows everybody else. A clique corresponds, in terms of graph theory, to a maximal fully connected subgraph. For Erdös–Rényi graphs with N vertices and linking probability p, the number of cliques is

$$\binom{N}{K} p^{K(K-1)/2} \left(1 - p^K\right)^{N-K} ,$$

for cliques of size K. Here

- $\binom{N}{K}$ is the number of sets of K vertices,
- $p^{K(K-1)/2}$ is the probability that all K vertices within the clique are mutually interconnected and

Table 1.1 The number of nodes N, average degree of separation ℓ, and clustering coefficient C, for four real-world networks. The last column is the value $C_{\text{rand}} = z/(N-1)$ the clustering coefficient would take for a random graph with the same size and coordination number. From Watts and Strogatz (1998) and Ludueña et al. (2013)

Network	N	ℓ	C	C_{rand}
Movie actors	225,226	3.65	0.79	0.00027
Neural network	282	2.65	0.28	0.05
Power grid	4941	18.70	0.08	0.0005
Internet domains	6,242,195	–	0.20	0.000006

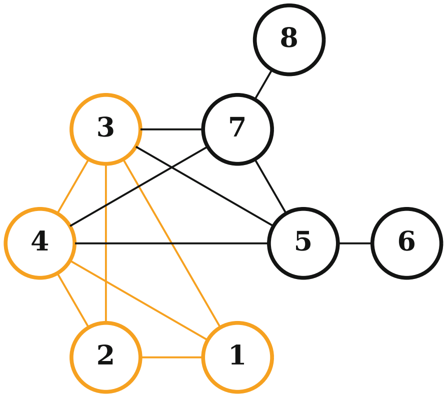

Fig. 1.4 The social graph of Fig. 1.1, with the four-site clique (1,2,3,4) highlighted. The network contains three additional cliques, (3,4,5,7), (5,6) and (7,8)

– $\left(1-p^K\right)^{N-K}$ the probability that every of the $N-K$ out-of-clique vertices is not connected to all K vertices of the clique.

The only cliques occurring in random graphs in the thermodynamic limit have the size two, since $p = z/N$.

Communities Another term used is "community". It is mathematically not as strictly defined as 'clique', denoting subgraphs with above-the-average densities of edges. Communities naturally emerge when decimating a bipartite graph stochastically. For the example shown in Fig. 1.1 one would assume that managers are not automatically acquainted when serving on the same board of directors, but with a certain probability, say 80%. Instead of cliques, the respective network of acquaintances between managers will be characterized by a dense community structure.

Clustering for Real-World Networks Most real-world networks have clustering coefficients that are substantially larger than the one a random network of the same size would have. This is immediately evident for the protein network shown in Fig. 1.2, which is characterized by a rich community structure and an elevated clustering coefficient.

In Table 1.1, we list C and the average distance ℓ for four different networks.

– A network of collaborations between movie actors.
– The neural network of the worm *C. Elegans*.
– A power grid of the United States.
– The networks of hyperlinks between domains (not webpages).

Also given in Table 1.1 are the clustering coefficients C_{rand} of random graphs of identical size and coordination number. Note that the real-world value is systematically higher than that of random graphs. The small values for the average distances ℓ given in Table 1.1 for three of the four networks relects their small-world nature.

Erdös–Rényi random graphs obviously do not match the properties of real-world networks well. In Sect. 1.3 we will discuss generalizations of random graphs that approximate the properties of real-world graphs better. Before, we will discuss several general properties of random graphs.

Correlation Effects The degree distribution p_k captures the statistical properties of nodes as if they where all independent of each other. In general, the property of a given node will however be dependent on the properties of other nodes, e.g. of its neighbors. When this happens one speaks of "correlation effects", with the clustering coefficient C being an example.

Another example for a correlation effect is what one calls "assortative mixing". A network is assortatively correlated whenever large-degree nodes, the hubs, tend to be mutually interconnected and assortatively anti-correlated when hubs are predominantly linked to low-degree vertices. Social networks tend to be assortatively correlated, in agreement with the everyday experience that the friends of influential persons, the hubs of social networks, tend to be VIPs themselves.

Tree Graphs Most real-world networks show strong local clustering and loops abound. An exception are metabolic networks which contain typically only very few loops since energy is consumed and dissipated when biochemical reactions go around in circles.

For many types of graphs commonly considered in graph theory, like Erdös–Rényi graphs, the clustering coefficient vanishes in the thermodynamic limit, and loops become irrelevant. One denotes a loopless graph a "tree graph".

Bipartite Networks Many real-world graphs have an underlying bipartite structure, see Fig. 1.1.

> BIPARTITE GRAPH A bipartite graph has two kinds of vertices with links only between vertices of unlike kinds.

Examples are networks of managers, where one kind of nodes is a company and the other kind of nodes the managers belonging to the board of directors. When eliminating one kind of vertices, in this case it is customary to eliminate the companies, one retains a social network; the network of directors, as illustrated in Fig. 1.1. This network has a high clustering coefficient, as all boards of directors are mapped onto cliques of the respective social network.

1.1.3 Network Degree Distribution

So far we considered mostly averaged quantities of random graphs, like the clustering coefficient or the average coordination number z. We will now develop tools allowing for a more sophisticated characterization of graphs.

Degree Distribution The basic description of general random and non-random graphs is given by the degree distribution p_k.

> DEGREE DISTRIBUTION If X_k is the number of vertices having the degree k, then $p_k = X_k/N$ is the degree distribution, where N is the total number of nodes.

The degree distribution is a probability distribution function and hence normalized, $\sum_k p_k = 1$.

Degree Distribution for Erdös–Rényi Graphs For an Erdös–Rényi network, the probability of a node to have k edges is given by the Binomial distribution,

$$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}, \tag{1.4}$$

where p is the link connection probability. For large $N \gg k$ we can approximate the degree distribution p_k by

$$p_k \simeq \mathrm{e}^{-pN} \frac{(pN)^k}{k!} = \mathrm{e}^{-z} \frac{z^k}{k!}, \tag{1.5}$$

where z is the average coordination number. We have used

$$\lim_{N\to\infty} \left(1 - \frac{x}{N}\right)^N = \mathrm{e}^{-x}, \qquad \binom{N-1}{k} = \frac{(N-1)!}{k!(N-1-k)!} \simeq \frac{(N-1)^k}{k!},$$

and $(N-1)^k p^k = z^k$, see (1.1). Equation (1.5) is a Poisson distribution with mean

$$\langle k \rangle = \sum_{k=0}^{\infty} k\, \mathrm{e}^{-z} \frac{z^k}{k!} = z\, \mathrm{e}^{-z} \sum_{k=1}^{\infty} \frac{z^{k-1}}{(k-1)!} = z\,.$$

The variance of the Poisson distribution is z, which implies that the relative width $\langle k \rangle / \sigma = 1/\sqrt{k}$ becomes progressively smaller when the coordination number z increases.

Ensemble Fluctuations In general, two specific realizations of random graphs differ. Their properties coincide on the average, but not on the level of individual links. With "ensemble" one denotes the set of possible realizations. The set of all possible realization of three-site networks is shown in Fig. 1.5.

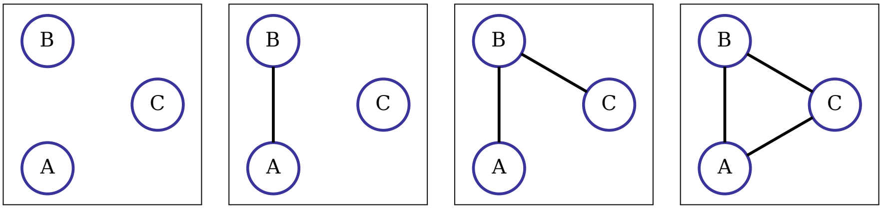

Fig. 1.5 Four networks with three nodes exist, respectively with 0/1/2/3 edges (from left to right). Given a link probability p, the probability to generate networks with 0/1/2/3 edges is $(1-p)^3$, $3p(1-p)^2$, $3p^2(1-p)$ and p^3. One speaks of ensemble fluctuations, compare (1.7)

In an ensemble of random graphs with fixed p and N the degree distribution X_k/N will be slightly different from one realization to the next. On the average it will be given by

$$\frac{1}{N}\langle X_k\rangle = p_k\,. \tag{1.6}$$

Here $\langle\ldots\rangle$ denotes the ensemble average. One can go one step further and calculate the probability $P(X_k = R)$ that in a realization of a random graph the number of vertices with degree k equals R. It is given in the large-N limit by

$$P(X_k = R) = \mathrm{e}^{-\lambda_k}\,\frac{(\lambda_k)^R}{R!}, \qquad \lambda_k = \langle X_k\rangle\,. \tag{1.7}$$

Note the similarity to (1.5) and that the mean $\lambda_k = \langle X_k\rangle$ is in general extensive while the mean z of the degree distribution (1.5) is intensive.

Power Law Distribution A special case are degree distributions decaying for large degrees as a power law,

$$p_k \sim \frac{1}{k^\alpha}, \qquad k\to\infty\,. \tag{1.8}$$

Typically, for real-world graphs, the scaling $\sim k^{-\alpha}$ holds only for large degrees k. The divergence at $k\to 0$ is avoided by the "Tsallis–Pareto distribution" $p_k = (\alpha-1)(1+k)^{-\alpha}$. For the normalization on $\boldsymbol{R}^+$ we consider $k\in[0,K]$, with $K\to\infty$,

$$\sum_{k=0}^{K} p_k \;\approx\; (\alpha-1)\int_0^K \frac{dk}{(1+k)^\alpha} = \frac{-1}{(1+k)^{\alpha-1}}\Big|_{k=0}^{K} = 1-\frac{1}{(1+K)^{\alpha-1}}\,.$$

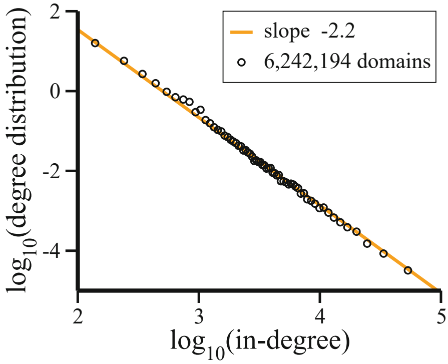

Fig. 1.6 Log-log plot of the distribution of the in-degrees of the hyperlink network between 6,242,194 domains. To a fair degree, the degree distribution is scale invariant, $p_k \sim 1/k^\alpha$. The value of the exponent, $\alpha \approx 2.2$, implies that the mean degree is finite, with a diverging variance. At the time of the crawl, $\langle k \rangle = 35.8$ was found. Data from Ludueña et al. (2013)

Powerlaw distributions are hence normalizable only if $\alpha > 1$. The average degree of the Tsallis–Pareto distribution is

$$\langle k \rangle = (\alpha - 1) \int_0^K dk \, \frac{k + 1 - 1}{(1 + k)^\alpha} = \frac{\alpha - 1}{2 - \alpha} (1 + k)^{2-\alpha} \Big|_0^K - 1 \,,$$

which diverges when $\alpha \leq 2$. Otherwise, when $\alpha > 2$, one has $\langle k \rangle = 1/(\alpha - 2)$. The variance is finite in analogy when $\alpha > 3$.

Scale-Free Graphs Power-law functional relations are "scale free", since any rescaling $k \to a\,k$ can be reabsorbed into the normalization constant. A famous example of a scale-invariant degree distribution is that of the in-degree of the hyperlink network between domains, as shown in Fig. 1.6.

Scale-free functional dependencies are considered to be 'critical', since they occur generally at the critical point of a phase transition. We will come back to this issue recurrently in the following chapters.

1.2 Spectral Properties

In order to understand a given real-world network it is desirable to develop characterization schemes relying just on a few key observables, e.g. like the coordination number z or the clustering coefficient C. As a further step in this direction we study now spectral properties.

Adjacency Matrix Any graph G with N nodes can be represented by a matrix encoding the topology of the network, the adjacency matrix.

ADJACENCY MATRIX The $N \times N$ adjacency matrix $\hat{A}$ has elements $A_{ij} = 1$ if nodes i and j are connected, with $A_{ij} = 0$ if they are not connected.

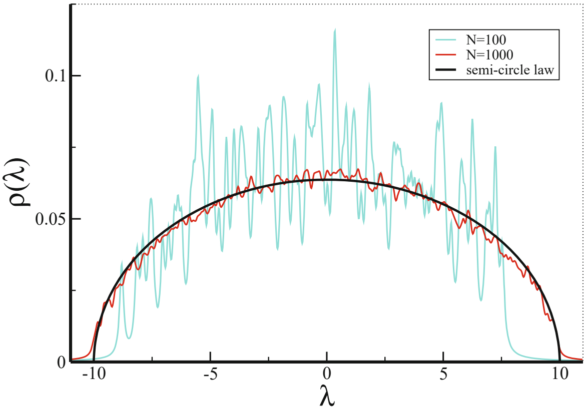

Fig. 1.7 The spectral densities (1.9) for Erdös–Rényi graphs with a coordination number $z = 25$ and $N = 100/1000$ nodes respectively. For comparison the expected result for the thermodynamic limit $N \to \infty$, the semi-circle law (1.15), is given. A broadening of $\epsilon = 0.1$ has been used, as defined by (1.11)

The adjacency matrix is symmetric and consequently has N real eigenvalues.

SPECTRUM OF A GRAPH The spectrum of a graph G is given by the set of eigenvalues λ_i of the adjacency matrix $\hat{A}$.

A graph with N nodes has N eigenvalues λ_i and it is useful to define the corresponding "spectral density"

$$\rho(\lambda) = \frac{1}{N}\sum_j \delta(\lambda - \lambda_j), \qquad \int d\lambda\, \rho(\lambda) = 1\,, \tag{1.9}$$

where $\delta(\lambda)$ is the Dirac delta function. In Fig. 1.7 the spectral density of an Erdös–Rényi graph is given.

Green's Function The spectral density $\rho(\lambda)$ can be evaluated once the Green's function[1] $G(\lambda)$,

$$G(\lambda) = \frac{1}{N}\,\mathrm{Tr}\left[\frac{1}{\lambda - \hat{A}}\right] = \frac{1}{N}\sum_j \frac{1}{\lambda - \lambda_j}\,, \tag{1.10}$$

is known. Here $\mathrm{Tr}[\ldots]$ denotes the trace over the matrix $(\lambda - \hat{A})^{-1} = (\lambda\mathbf{1} - \hat{A})^{-1}$, where $\mathbf{1}$ is the identity matrix. In complex plane, we can decompose the inverse,

$$\lim_{\varepsilon\to 0} \frac{1}{\lambda - \lambda_j + i\epsilon} = P\,\frac{1}{\lambda - \lambda_j} - i\pi\delta(\lambda - \lambda_j)\,. \tag{1.11}$$

[1] The reader without prior experience with Green's functions may skip the following derivation and pass directly to the result, namely to (1.15).

where P denotes the principal part, which takes into account that the positive and the negative contributions of the $1/\lambda$ divergence cancel. We find the relation

$$\rho(\lambda) = -\frac{1}{\pi}\lim_{\varepsilon\to 0} \mathrm{ImG}(\lambda + i\varepsilon)\,. \tag{1.12}$$

Semi-Circle Law The graph spectra can be evaluated for random matrices[2] for the case of small link densities $p = z/N$, where z is the average connectivity. We note that the Green's function (1.10) can be expanded into powers of $\hat{A}/\lambda$,

$$G(\lambda) = \frac{1}{N\lambda}\,\mathrm{Tr}\left[1 + \frac{\hat{A}}{\lambda} + \left(\frac{\hat{A}}{\lambda}\right)^2 + \dots\right]. \tag{1.13}$$

Traces over powers of the adjacency matrix $\hat{A}$ can be interpreted, for random graphs, as random walks which need to come back to the starting vertex.

Starting from a random site we can connect on the average to z neighboring sites and from there on to $z-1$ next-nearest neighboring sites, and so on. This consideration allows to recast the Taylor expansion (1.13) for the Green's function into a recursive continued fraction,

$$G(\lambda) = \frac{1}{\lambda - \frac{z}{\lambda - \frac{z-1}{\lambda - \frac{z-1}{\lambda - \dots}}}} \approx \frac{1}{\lambda - z\,G(\lambda)}\,, \tag{1.14}$$

where we used the approximation $z-1 \approx z$ in the last step. Equation (1.14) constitutes a self-consistency equation for $G = G(\lambda)$, with the solution

$$G^2 - \frac{\lambda}{z}G + \frac{1}{z} = 0, \qquad G = \frac{\lambda}{2z} - \sqrt{\frac{\lambda^2}{4z^2} - \frac{1}{z}}\,,$$

since $\lim_{\lambda\to\infty} G(\lambda) = 0$. The spectral density (1.12) then takes the form

$$\rho(\lambda) = \begin{cases} \sqrt{4z-\lambda^2}/(2\pi z) & \text{if} \quad \lambda^2 < 4z \\ 0 & \text{if} \quad \lambda^2 > 4z \end{cases} \tag{1.15}$$

of a half-ellipse also known as "Wigner law", or the "semi-circle law".

Loops and the Clustering Coefficient The total number of triangles, viz the overall number of loops of length three in a network is, on the average, $C(N/3)(z-1)z/2$, where C is the clustering coefficient. This relation holds for

[2] Random matrix theory will be further discussed in Sect. 10.2.1 of Chap. 10.

large networks. One can relate the clustering coefficient C to the adjacency matrix A via

$$C\frac{z(z-1)}{2}\frac{N}{3} = \text{number of triangles}$$
$$= \frac{1}{6}\sum_{i_1,i_2,i_3} A_{i_1 i_2}A_{i_2 i_3}A_{i_3 i_1} = \frac{1}{6}\mathrm{Tr}\left[A^3\right],$$

since three sites i_1, i_2 and i_3 are interconnected only when the respective entries of the adjacency matrix are unity. The sum of the right-hand side of above relation is a moment of the graph spectrum. The factors $1/3$ and $1/6$ account for overcountings. Above formula holds only when all node have identical degrees $k_i \equiv z$. Otherwise one needs to take an average, $z(z-1) \to \sum_i k_i(k_i-1)/N$.

Moments of the Spectral Density The graph spectrum is directly related to certain topological features of a graph via its moments. The lth moment of $\rho(\lambda)$ is given by

$$\int d\lambda\, \lambda^l \rho(\lambda) = \frac{1}{N}\sum_{j=1}^{N} \left(\lambda_j\right)^l$$
$$= \frac{1}{N}\mathrm{Tr}\left[A^l\right] = \frac{1}{N}\sum_{i_1,i_2,\ldots,i_l} A_{i_1 i_2}A_{i_2 i_3}\cdots A_{i_l i_1}, \tag{1.16}$$

as one can see from (1.9). The lth moment of $\rho(\lambda)$ is therefore equivalent to the number of closed paths of length l, the number of all paths of length l returning to the starting point.

1.2.1 Graph Laplacian

Differential operators, like the derivative df/dx of a function $f(x)$, may be generalized to network functions. We start by recalling the definitions of the first,

$$\frac{d}{dx}f(x) = \lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x},$$

and of the second derivative,

$$\frac{d^2}{dx^2}f(x) = \lim_{\Delta x\to 0}\frac{f(x+\Delta x)+f(x-\Delta x)-2f(x)}{\Delta x^2}.$$

Graph Laplacians Consider a function f_i on a graph with N sites, with $i = 1, \ldots, N$. One defines the graph Laplacian $\hat{\Lambda}$ via

$$\Lambda_{ij} = \left(\sum_l A_{il}\right)\delta_{ij} - A_{ij} = \begin{cases} k_i & i = j \\ -1 & i \text{ and } j \text{ connected} \\ 0 & \text{otherwise} \end{cases}, \tag{1.17}$$

where the $\Lambda_{ij} = (\hat{\Lambda})_{ij}$ are the elements of the Laplacian matrix, A_{ij} the adjacency matrix, and where k_i is the degree of vertex i.

Apart from a sign convention, $\hat{\Lambda}$ corresponds to a straightforward generalization of the usual Laplace operator $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$. To see this, just apply the Laplacian matrix Λ_{ij} to a graph-function $\mathbf{f} = (f_1, \ldots, f_N)$. The graph Laplacian is hence intrinsically related to diffusion processes on networks.[3] Alternatively one defines by

$$L_{ij} = \begin{cases} 1 & i = j \\ -1/\sqrt{k_i\, k_j} & i \text{ and } j \text{ connected} \\ 0 & \text{otherwise} \end{cases}, \tag{1.18}$$

the "normalized graph Laplacian", where $k_i = \sum_j A_{ij}$ is the degree of vertex i. The eigenvalues of the normalized graph Laplacian have a straightforward interpretation in terms of the underlying graph topology. One needs however to remove first all isolated sites from the graph, which do not generate entries to the adjacency matrix. The respective L_{ij} would be ill defined.

Eigenvalues of the Normalized Graph Laplacian Of interest are the eigenvalues λ_l of the normalized graph Laplacian (1.18).

– The normalized graph Laplacian is positive semidefinite,

$$0 = \lambda_0 \le \lambda_1 \le \cdots \le \lambda_{N-1} \le 2\,.$$

– The lowest eigenvalue λ_0 is always zero, corresponding to the eigenfunction

$$\mathbf{e}(\lambda_0) = \frac{1}{\sqrt{C}}\left(\sqrt{k_1}, \sqrt{k_2}, \ldots, \sqrt{k_N}\right), \tag{1.19}$$

where C is a normalization constant and where the k_i are the respective vertex degrees.

[3] See Sect. 3.3.1 of Chap. 3 for an indepth discussion of diffusion processes.

- The degeneracy of λ_0 is given by the number of disconnected subgraphs contained in the network. The eigenfunctions of λ_0 vanish on all subclusters beside one, where it has the functional form (1.19).
- The largest eigenvalue λ_{N-1} is $\lambda_{N-1} = 2$, if and only if the network is bipartite. Generally, a small value of $2 - \lambda_{N-1}$ indicates that the graph is nearly bipartite. The degeneracy of $\lambda = 2$ is given by the number of bipartite subgraphs.
- The inequality

$$\sum_l \lambda_l \;\le\; N$$

  holds generally. The equality holds for connected graphs, viz when λ_0 has degeneracy one.

Examples of Graph Laplacians The eigenvalues of the normalized graph Laplacian can be evaluated analytically for some reference graphs.

- For a complete N-site graph, viz when all sites are mutually interconnected, the eigenvalues are

$$\lambda_0 = 0, \qquad \lambda_l = N/(N-1), \qquad l = 1, \ldots, N-1\,.$$

- For a complete bipartite graph (all sites of one subgraph are connected to all other sites of the other subgraph) the eigenvalues are

$$\lambda_0 = 0, \qquad \lambda_{N-1} = 2, \qquad \lambda_l = 1, \qquad l = 1, \ldots, N-2\,.$$

  The eigenfunction for $\lambda_{N-1} = 2$ has the form

$$\mathbf{e}(\lambda_{N-1}) = \frac{1}{\sqrt{C}}\big(\underbrace{\sqrt{k_A}, \ldots, \sqrt{k_A}}_{\text{A sublattice}}, \underbrace{-\sqrt{k_B}, \ldots, -\sqrt{k_B}}_{\text{B sublattice}}\big)\,. \tag{1.20}$$

  Denoting with N_A and N_B the number of sites in two sublattices A and B, with $N_A + N_B = N$, the degrees k_A and k_B of vertices belonging to sublattice A and B respectively are $k_A = N_B$ and $k_B = N_A$ for a complete bipartite lattice.

A densely connected graph will therefore have a substantial number of eigenvalues close to unity. For real-world graphs one may therefore plot the spectral density of the normalized graph Laplacian in order to gain an insight into its overall topological properties. The information obtained from the spectral density of the adjacency matrix and from the normalized graph Laplacian are somewhat distinct.

1.3 Percolation in Generalized Random Graphs

The most random of all graphs are Erdös–Rényi graphs. One can relax the extent of randomness somewhat and construct random networks with an arbitrarily given degree distribution. This procedure is also denoted "configurational model". We will use the configurational model to examine the "percolation transition", which occurs as a function for link probability p. Graphs decompose into a set of disconnected subgraphs when containing only a few links, but not for high link densities. The respective transition is the percolation transition.

1.3.1 Graphs with Arbitrary Degree Distributions

In order to generate random graphs having non-Poisson degree distributions, we may choose a specific set of degrees.

> DEGREE SEQUENCE A degree sequence is a specified set $\{k_i\}$ of the degrees for the vertices $i = 1 \ldots N$.

The degree sequence is also the first information one obtains when examining real-world networks and hence the foundation for all further analysis.

Construction of Networks with Arbitrary Degree Distribution The degree sequence can be chosen in such a way that the fraction of vertices having degree k will tend to the desired degree distribution in the thermodynamic limit,

$$p_k, \qquad N \to \infty .$$

For the construction, two steps suffice.

– Assign k_i "stubs" (ends of edges emerging from a vertex) to all vertices, $i = 1, \ldots, N$.
– Iteratively choose pairs of stubs at random and join them together to make complete edges.

When all stubs have been used up, the resulting graph is a random member of the ensemble of graphs with the desired degree sequence. Figure 1.8 illustrates the construction procedure.

Average Degree and Clustering The mean number of neighbors is the coordination number

$$z = \langle k \rangle = \sum_k k p_k .$$

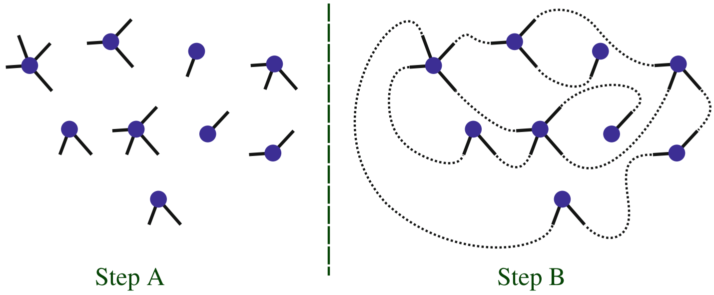

Fig. 1.8 Construction procedure of a random network with nine vertices and degrees $X_1 = 2$, $X_2 = 3$, $X_3 = 2$, $X_4 = 2$. In step A (left) the vertices with the desired number of stubs (degrees) are constructed. In step B (right) the stubs are connected randomly

The probability that one of the second neighbors of a given vertex is also a first neighbor scales as N^{-1} for random graphs, regardless of the degree distribution. Loops can be ignored in the limit $N \to \infty$.

Degree Distribution of Neighbors Consider a given vertex A and a vertex B that is a neighbor of A, i.e. A and B are linked by an edge.

We are now interested in the degree distribution for vertex B, viz in the degree distribution of a neighbor vertex of A, where A is an arbitrary node of a random network with degree distribution p_k. As a first step we consider the average degree of a neighbor node.

A high-degree vertex has more edges connected to it. There is then a higher chance that any given edge on the graph will be connected to it, with this chance being directly proportional to the degree of the vertex. The probability distribution of the degree of the vertex to which an edge leads is thus proportional to kp_k and not just to p_k.

Excess Degree Distribution When we are interested in determining the size of loops or the size of connected components in a random graph, we are normally interested not in the complete degree of the vertex reached by following an edge from A, but in the number of edges emerging from such a vertex that do not lead back to A, because the latter contains all information about the number of second neighbors of A.

The number of new edges emerging from B is just the degree of B minus one and its correctly normalized distribution is therefore

$$q_{k-1} = \frac{kp_k}{\sum_j jp_j}, \qquad q_k = \frac{(k+1)p_{k+1}}{\sum_j jp_j}, \tag{1.21}$$

since kp_k is the degree distribution of a neighbor, compare Fig. 1.9. The distribution q_k of the outgoing edges of a neighbor vertex is the “excess degree distribution”.

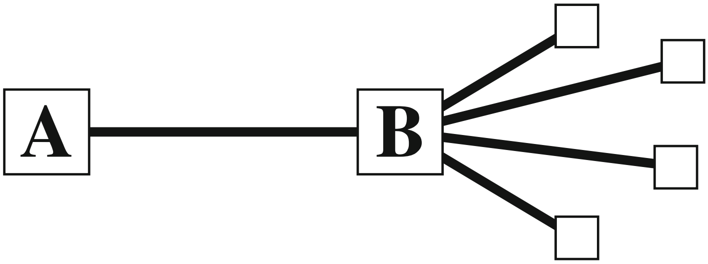

Fig. 1.9 The excess degree distribution q_{k-1} is the probability of finding k outgoing links of a neighboring site. Here the site B is a neighbor of site A and has a degree $k = 5$ and $k - 1 = 4$ outgoing edges, compare (1.21). The probability that B is a neighbor of site A is proportional to the degree k of B

The average number of outgoing edges of a neighbor vertex is then

$$\sum_{k=0}^{\infty} kq_k = \frac{\sum_{k=0}^{\infty} k(k+1)p_{k+1}}{\sum_j jp_j} = \frac{\sum_{k=1}^{\infty}(k-1)kp_k}{\sum_j jp_j}$$
$$= \frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}, \tag{1.22}$$

where $\langle k^2\rangle$ is the second moment of the degree distribution.

Number of Next-Nearest Neighbors We denote with

$$z_m, \qquad z_1 = \langle k\rangle \equiv z$$

the average number of m-nearest neighbors. Equation (1.22) gives the average number of vertices two steps away from the starting vertex A via a particular neighbor vertex. Multiplying this by the mean degree of A, namely $z_1 \equiv z$, we find that the mean number of second neighbors z_2 of a vertex is

$$z_2 = \langle k^2\rangle - \langle k\rangle\,. \tag{1.23}$$

Note that z_2 is not given by the variance of the degree distribution,[4] which would be $\langle(k - \langle k\rangle)^2\rangle = \langle k^2\rangle - \langle k\rangle^2$.

[4] Compare Sect. 5.1 of Chap. 5.

z_2 for the Erdös–Rényi Graph The degree distribution of an Erdös–Rényi graph is the Poisson distribution, $p_k = e^{-z} z^k/k!$, see (1.5). For the average number of second neighbors, Eq. (1.23), we obtain

$$z_2 = \sum_{k=0}^{\infty} k^2 e^{-z} \frac{z^k}{k!} - z = z e^{-z} \sum_{k=1}^{\infty} (k-1+1) \frac{z^{k-1}}{(k-1)!} - z$$
$$= z^2 = \langle k \rangle^2 .$$

The mean number of second neighbors of a vertex in an Erdös–Rényi random graph is just the square of the mean number of first neighbors. This is a special case however. For most degree distributions, Eq. (1.23) will be dominated by the term $\langle k^2 \rangle$, which implies that the number of second neighbors is roughly the mean square degree, rather than the square of the mean. For broad distributions these two quantities can be very different.

Number of Far Away Neighbors The average number of edges emerging from a second neighbor, and not lead ing back to where it came from, is also given by (1.22), and indeed this is true at any distance m away from vertex A. The average number of neighbors at a distance m is then

$$z_m = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} z_{m-1} = \frac{z_2}{z_1} z_{m-1} , \tag{1.24}$$

where $z_1 \equiv z = \langle k \rangle$ and z_2 are given by (1.23). Iterating this relation we find

$$z_m = \left[\frac{z_2}{z_1} \right]^{m-1} z_1 . \tag{1.25}$$

Giant Connected Cluster Depending on whether z_2 is greater than z_1 or not, Eq. (1.25) will either diverge or converge exponentially as m becomes large.

$$\lim_{m \to \infty} z_m = \begin{cases} \infty & \text{if } z_2 > z_1 \\ 0 & \text{if } z_2 < z_1 \end{cases} . \tag{1.26}$$

The percolation point is therefore $z_1 = z_2$. Below the transition, the total number of neighbors is

$$\sum_m z_m = z_1 \sum_{m=1}^{\infty} \left[\frac{z_2}{z_1} \right]^{m-1} = \frac{z_1}{1 - z_2/z_1} = \frac{z_1^2}{z_1 - z_2} .$$

The total number of sites connected to the starting node is finite below the percolation transition, which implies that the network decomposes into nonconnected components. Beyond the percolation transition one or more clusters with macroscopic numbers of nodes form.

> GIANT CONNECTED COMPONENT When the largest cluster of a graph encompasses a finite fraction of all vertices in the thermodynamic limit, it is said to form a giant connected component.

If the total number of neighbors is infinite, then there must be a giant connected component. When the total number of neighbors is finite, there can be no giant connected component.

Percolation Threshold When a system has two or more possibly macroscopically different states, one speaks of a phase transition.

> PERCOLATION TRANSITION A percolation transition occurs when the structure of an evolving graph transits from a state in which two (far away) sites are on the average connected/not connected.

This phase transition occurs when $z_2 = z_1$. Making use of (1.23), $z_2 = \langle k^2 \rangle - \langle k \rangle$, we find that this condition is equivalent to

$$\langle k^2 \rangle - 2\langle k \rangle = 0, \qquad \sum_{k=0}^{\infty} k(k-2)\, p_k = 0\,. \tag{1.27}$$

Because of the factor $k(k-2)$, vertices of degree zero and degree two do not contribute to the sum. The number of vertices with degree zero or two therefore affects neither the phase transition nor the existence of a giant component.

- Vertices of degree zero are not connected to any other node, they do not contribute to the network topology.
- Vertices of degree one constitute the sole negative contribution to the percolation condition (1.27). No further path is available when arriving at a site with degree one.
- Vertices of degree two act as intermediators between two other nodes. Removing vertices of degree two does not change the topological structure of a graph.

One can therefore remove (or add) vertices of degree two or zero without affecting the existence of the giant component.

Clique Percolation Edges correspond to cliques with $Z = 2$ sites, compare Fig. 1.4. The percolation transition can hence be interpreted as a percolation of cliques having size two and larger. It is then clear that the concept of percolation can be generalized to that of percolation of cliques with Z sites.

Average Vertex–Vertex Distance Below the percolation threshold the average vertex–vertex distance ℓ is finite and the graph decomposes into an infinite number of disconnected subclusters.

DISCONNECTED SUBCLUSTERS A disconnected subcluster or subgraph constitutes a subset of vertices for which (a) there is at least one path in between all pairs of nodes making up the subcluster and (b) there is no path between a member of the subcluster and any out-of-subcluster vertex.

Well above the percolation transition, ℓ is given approximately by the condition $z_\ell \simeq N$,

$$\log(N/z_1) = (\ell - 1)\log(z_2/z_1), \qquad \ell = \frac{\log(N/z_1)}{\log(z_2/z_1)} + 1\,, \tag{1.28}$$

when using $z_l = z_1(z_2/z_1)^{l-1}$, as given by (1.25). For Erdös–Rényi random graphs one has $z_1 = z$ and $z_2 = z^2$, which leads to

$$\ell = \frac{\log N - \log z}{\log z} + 1 = \frac{\log N}{\log z}\,,$$

as derived previously, see (1.2).

Clustering Coefficient of Generalized Random Graphs The clustering coefficient C denotes the probability that two neighbors i and j of a particular vertex A have stubs that do interconnect. The probability that two given stubs are connected is $1/(zN-1) \approx 1/zN$, since zN is the total number of stubs. We have

$$\begin{aligned} C &= \frac{\langle k_i\rangle_q \langle k_j\rangle_q}{Nz} = \frac{1}{Nz}\left[\sum_k k q_k\right]^2 \qquad (1.29)\\ &= \frac{1}{Nz}\left[\frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}\right]^2 = \frac{z}{N}\left[\frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle^2}\right]^2\,, \end{aligned}$$

where the notation $\langle\ldots\rangle_q$ indicates that the average is to be taken with respect to the excess degree distribution $q = q_k$, as given by (1.21).

As expected, the clustering coefficient vanishes in the thermodynamic limit $N \to \infty$. However, it may have a very big leading coefficient, especially for degree distributions with fat tails. The differences listed in Table 1.1, between the measured clustering coefficient C and the value $C_{\text{rand}} = z/N$ for Erdös–Rényi graphs, are partly due to the fat tails in the degree distributions p_k of the corresponding networks.

1.3.2 Probability Generating Function Formalism

Network theory is about the statistical properties of graphs. A powerful method from probability theory is the generating function formalism, which we will discuss now and apply later on.

Probability Generating Functions We define with

$$G_0(x) = \sum_{k=0}^{\infty} p_k x^k \tag{1.30}$$

the "generating function" $G_0(x)$ for the probability distribution p_k. The generating function $G_0(x)$ contains all information present in p_k. We can recover p_k from $G_0(x)$ by differentiation,

$$p_k = \frac{1}{k!} \frac{\mathrm{d}^k G_0}{\mathrm{d}x^k}\bigg|_{x=0} . \tag{1.31}$$

One says that G_0 "generates" the probability distribution p_k.

Generating Function for the Excess Degree Distribution For later use we define the generating function $G_1(x)$ for the distribution $q_k = (k+1)p_{k+1}/z$ measuring the number of non-returning edges leaving a neighboring vertex,

$$\begin{aligned} G_1(x) = \sum_{k=0}^{\infty} q_k x^k &= \frac{\sum_{k=0}^{\infty}(k+1)p_{k+1}x^k}{z} = \frac{\sum_{k=1}^{\infty} k p_k x^{k-1}}{z} \\ &= \frac{G_0'(x)}{z}, \qquad\qquad z = \langle k \rangle , \end{aligned} \tag{1.32}$$

where $G_0'(x)$ denotes the first derivative of $G_0(x)$ with respect to its argument.

Properties of Generating Functions Probability generating functions have a couple of important properties.

– NORMALIZATION
The normalization of p_k implies

$$G_0(1) = \sum_k p_k = 1 . \tag{1.33}$$

– MEAN
Straightforward differentiation,

$$G_0'(1) = \sum_k k\, p_k = \langle k \rangle\,, \tag{1.34}$$

yields the average degree $\langle k \rangle$.
– MOMENTS
The nth moment $\langle k^n \rangle$ of the distribution p_k is given by

$$\langle k^n \rangle = \sum_k k^n p_k \;=\; \left[\left(x \frac{\mathrm{d}}{\mathrm{d}x}\right)^n G_0(x)\right]_{x=1}. \tag{1.35}$$

Generating Function for Independent Random Variables Let us assume that we have two random variables, such as two dice. Throwing the two dice correspond to two independent random events. The joint probability to obtain $k = 1, \ldots, 6$ with the first dice and $l = 1, \ldots, 6$ with the second dice is $p_k\, p_l$. This probability function is generated by

$$\sum_{k,l} p_k p_l\, x^{k+l} = \left(\sum_k p_k\, x^k\right)\left(\sum_l p_l\, x^l\right),$$

i.e. by the product of the individual generating functions. This is the reason why generating functions are so useful in describing combinations of independent random events.

As an application consider n randomly chosen vertices. The sum $\sum_i k_i$ of the respective degrees has a cumulative degree distribution, which is generated by

$$\Big[\, G_0(x)\,\Big]^n .$$

Generating Function of the Poisson Distribution The generating function $G_0 = \sum_k p_k x^k$ of the Poisson distribution $p_k = \mathrm{e}^{-z}\, z^k/k!$ is

$$G_0(x) = \mathrm{e}^{-z} \sum_{k=0}^{\infty} \frac{z^k}{k!}\, x^k = \mathrm{e}^{z(x-1)}\,,$$

with z being the average degree. Compare (1.5). The generating function $G_1(x)$ for the excess degree distribution q_k is

$$G_1(x) = \frac{G_0'(x)}{z} = \mathrm{e}^{z(x-1)}\,, \tag{1.36}$$

compare (1.32). As expected, we find $G_1(x) = G_0(x)$ for the Poisson distribution.

Generating Function of the Exponential Distribution As a second example, consider a graph with an exponential degree distribution:

$$p_k = (1 - \mathrm{e}^{-1/\kappa})\, \mathrm{e}^{-k/\kappa}, \qquad \sum_{k=0}^{\infty} p_k = \frac{1 - \mathrm{e}^{-1/\kappa}}{1 - \mathrm{e}^{-1/\kappa}} = 1\,, \tag{1.37}$$

where κ is a constant.The generating function is

$$G_0(x) = (1 - \mathrm{e}^{-1/\kappa}) \sum_{k=0}^{\infty} \mathrm{e}^{-k/\kappa} x^k = \frac{1 - \mathrm{e}^{-1/\kappa}}{1 - x\mathrm{e}^{-1/\kappa}}\,,$$

together with

$$z = G_0'(1) = \frac{\mathrm{e}^{-1/\kappa}}{1 - \mathrm{e}^{-1/\kappa}}, \qquad G_1(x) = \frac{G_0'(x)}{z} = \left[\frac{1 - \mathrm{e}^{-1/\kappa}}{1 - x\mathrm{e}^{-1/\kappa}}\right]^2 .$$

Stochastic Sum of Independent Variables Let's assume that the random variables $k_1,\ k_2, \ldots$ are identically distributed, which implies that they have the same generating functions $G_0(x)$. The powers

$$G_0^2(x), \qquad G_0^3(x), \qquad G_0^4(x)\,,$$

are the generating functions for

$$k_1 + k_2, \qquad k_1 + k_2 + k_3, \qquad k_1 + k_2 + k_3 + k_4\,,$$

and so on. Next consider that the number of times n this stochastic process is executed is distributed as p_n. As an example consider throwing a dice several times, with a probability p_n to throw n times. The distribution of the results obtained is then generated by

$$\sum_n p_n G_0^n(x) = G_N\left(G_0(x)\right), \qquad G_N(z) = \sum_n p_n z^n\,. \tag{1.38}$$

We will make use of this relation further on.

1.3.3 Distribution of Component Sizes

Absence of Closed Loops For networks below the percolation transition we are interested in the distribution of the sizes of the individual subclusters. The calculations will depend on the absence of closed loops, or any other kind of

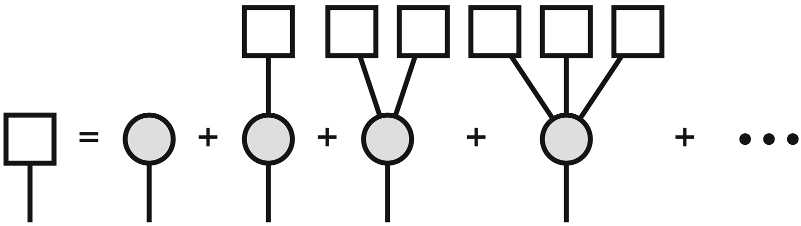

Fig. 1.10 Graphical representation of the self-consistency (1.39) for the generating function $H_1(x)$, represented by the open box. A single vertex is represented by a hashed circle. The subcluster connected to an incoming vertex can be either a single vertex or an arbitrary number of subclusters of the same type connected to the first vertex. In analogy to Newman et al. (2001)

clustering, for generalized random graphs in the thermodynamic limit. In physics jargon, large random graphs are "tree-like".

We recall that the number of closed loops of length three corresponds to the clustering coefficient C, viz to the probability that two of your friends are also friends of each other. For random networks $C = [\langle k^2\rangle - \langle k\rangle]^2/(z^3N)$, see (1.29), which tends to zero as $N \to \infty$.

Generating Function for the Size Distribution of Components We define by

$$H_1(x) = \sum_m h_m^{(1)} x^m$$

the generating function for the distribution of cluster sizes containing a given vertex j, with the condition that j is linked to a specific incoming edge, see Fig. 1.10. That is, $h_m^{(1)}$ is the probability that the such-defined cluster contains m nodes.

Self-Consistency Condition for $\boldsymbol{H_1(x)}$ We note the following.

- The first vertex j belongs to the subcluster with probability 1, its generating function is x.
- The probability that the vertex j has k outgoing stubs is q_k.
- At every stub outgoing from vertex j there is a subcluster.
- The total number of vertices consists of those generated by $[H_1(x)]^k$ plus the starting vertex.

The number of outgoing edges k from vertex j is described by the distribution function q_k, see (1.21). The total size of the k clusters is generated by $[H_1(x)]^k$, as a consequence of the multiplication property of generating functions discussed in

Sect. 1.3.2. The self-consistency equation for the total number of vertices reachable is then

$$H_1(x) = x \sum_{k=0}^{\infty} q_k \,[H_1(x)]^k = x\, G_1(H_1(x))\,, \tag{1.39}$$

where we made use of (1.32) and (1.38).

Embedding Cluster Distribution Function The quantity that we actually want to know is the distribution of the sizes of the clusters to which the entry vertex belongs.

– The number of edges emanating from a randomly chosen vertex is distributed according to the degree distribution p_k.
– Every edge leads to a cluster whose size is generated by $H_1(x)$.

The size of a complete component is thus generated by

$$H_0(x) = x \sum_{k=0}^{\infty} p_k \,[H_1(x)]^k = x\, G_0(H_1(x))\,, \tag{1.40}$$

where the prefactor x corresponds to the generating function of the starting vertex. The complete distribution of component sizes is given by solving (1.39) self-consistently for $H_1(x)$ and then substituting the result into (1.40).

Mean Component Size The calculation of $H_1(x)$ and $H_0(x)$ in closed form is not possible. We are, however, interested only in the first moment, viz the mean component size, see (1.34).

The component size distribution is generated by $H_0(x)$, Eq. (1.40). The the mean component size below the percolation transition is therefore

$$\begin{aligned} \langle s\rangle &= H_0'(1) = \Big[\, G_0(H_1(x)) + x\, G_0'(H_1(x))\, H_1'(x)\,\Big]_{x=1} \\ &= 1 + G_0'(1) H_1'(1)\,, \end{aligned} \tag{1.41}$$

where we made use of the normalization

$$G_0(1) = H_1(1) = H_0(1) = 1$$

of generating functions, see (1.33). The value of $H_1'(1)$ can be calculated from (1.39) by differentiating:

$$H_1'(x) = G_1(H_1(x)) + x\,G_1'(H_1(x))\,H_1'(x), \tag{1.42}$$
$$H_1'(1) = \frac{1}{1 - G_1'(1)}\,.$$

Substituting this into (1.41) we find

$$\langle s \rangle = 1 + \frac{G_0'(1)}{1 - G_1'(1)}\,. \tag{1.43}$$

Percolation Transition On a closer look, above expression for the mean component size reproduces the percolation condition $z_2 = z_1$ derived in Sect. 1.3. We make use of (1.23) and

$$G_0'(1) = \sum_k k\,p_k = \langle k \rangle = z_1, \tag{1.44}$$
$$G_1'(1) = \frac{\sum_k k(k-1)p_k}{\sum_k kp_k} = \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} = \frac{z_2}{z_1}\,,$$

where z_1 and z_2 are the mean numbers of nearest- and next-nearest neighbors. Substitution into (1.43) gives the average component size below the transition as

$$\langle s \rangle = 1 + \frac{z_1^2}{z_1 - z_2}\,. \tag{1.45}$$

This expression has a divergence at $z_1 = z_2$. The mean component size diverges at the percolation threshold, compare Sect. 1.3, beyond which the giant connected component forms.

Size of the Giant Component Above the percolation transition the network contains a giant connected component, which contains a finite fraction S of all sites N. Formally we can decompose the generating functional for the component sizes into a part generating the giant component, $H_0^*(x)$, and a part generating the remaining components,

$$H_0(x) \;\to\; H_0^*(x) + H_0(x), \qquad H_0^*(1) = S, \qquad H_0(1) = 1 - S\,,$$

and equivalently for $H_1(x)$. The self-consistency (1.39) and (1.40),

$$H_0(x) = xG_0(H_1(x)), \qquad H_1(x) = xG_1(H_1(x)),$$

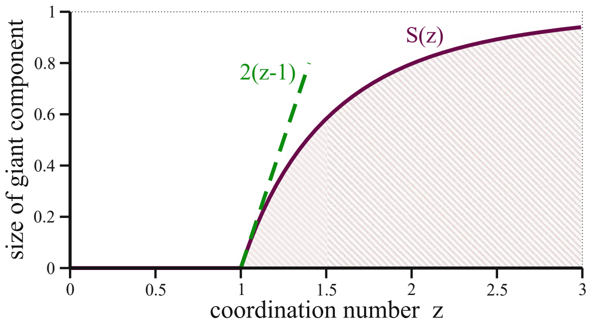

Fig. 1.11 For Erdös–Rényí graphs, the size $S(z)$ of the giant component vanishes linearly, at the percolation transition occurring at the critical coordination number $z_c = 1$. Compare (1.47) and (1.48)

then take the form

$$1 - S = H_0(1) = G_0(u), \qquad u = H_1(1) = G_1(u)\,. \tag{1.46}$$

For Erdös–Rényí graphs we have $G_1(x) = G_0(x) = \exp(z(x-1))$, see (1.36). This leads to $1 - S = u$ in (1.46), which takes with

$$1 - S = u = \mathrm{e}^{z(u-1)} = \mathrm{e}^{-zS}, \qquad S = 1 - \mathrm{e}^{-zS} \tag{1.47}$$

the form of a self-consistency condition for the mean size S of the giant component. Percolation takes place for $z \geq 1$.

Critical Behaviour The Taylor expansion for the exponential,

$$S = 1 - \mathrm{e}^{-zS} \approx zS - \frac{(zS)^2}{2}, \qquad S(z) \approx \frac{2(z-1)}{z^2} \tag{1.48}$$

shows that the $S(z)$ increases linearly in the percolating regime, as shown in Fig. 1.11. It follows from (1.47) that the giant component engulfs the entire net exponentially fast for large degrees z, namely that $\lim_{z\to\infty} S(z) \to 1$.

1.4 Robustness of Random Networks

Degree distributions of real-world networks often have “fat tails”, a vaguely defined notion that a given distribution decays only slowly when the degree becomes large. In general, fat tails increase the robustness of a network. That is, the network retains functionality even when a certain number of vertices or edges is removed. The Internet remains functional, to give an example, even when a substantial number of Internet routers fail.

Removal of Vertices We consider a graph model in which each vertex is either “active” or “inactive”. Inactive vertices are nodes that have either been removed, or are present but non-functional. We denote with

$$b = b_k$$

the probability that a vertex of degree k is active. The generating function

$$F_0(x) = \sum_{k=0}^{\infty} p_k\, b_k\, x^k, \qquad F_0(1) = \sum_k p_k\, b_k \ \le\ 1\,, \tag{1.49}$$

generates the probabilities that a vertex has degree k and is present. Strictly speaking, $F_0(x)$ is not a proper generating function because the normalization $F_0(1)$ is not unity, but the fraction of all vertices that are active. This shortcoming could be rectified by adding via

$$\Big(1 - \sum_k p_k\, b_k\Big) + \sum_{k=0}^{\infty} p_k\, b_k\, x^k\,,$$

a constant to $F(x)$, which corresponds to the number of inactive nodes, which are in turn equivalent to nodes with degree zero. The normalization can hence be disregarded. By analogy with (1.32) we define by

$$F_1(x) = \frac{\sum_k k\, p_k\, b_k\, x^{k-1}}{\sum_k k\, p_k} = \frac{F_0'(x)}{z} \tag{1.50}$$

the generating function for the degree distribution of neighbor sites. Also $F_1(x)$ is not proper normalized.

Distribution of Connected Clusters The distributions of the sizes of connected clusters reachable from a given vertex, $H_0(x)$, or from a given edge, $H_1(x)$, are generated respectively by the normalized functions

$$\begin{aligned} H_0(x) &= 1 - F_0(1) + x\,F_0(H_1(x)), \qquad H_0(1) = 1, \\ H_1(x) &= 1 - F_1(1) + x\,F_1(H_1(x)), \qquad H_1(1) = 1\,, \end{aligned} \tag{1.51}$$

which are the logical equivalents of (1.39) and (1.40).

Random Failure of Vertices First we consider the case of random failures of vertices. In this case, the probability

$$b_k \equiv b\ \le 1, \qquad F_0(x) = b\,G_0(x), \qquad F_1(x) = b\,G_1(x)$$

of a vertex being present is independent of the degree k and just equal to a constant, b, which means that

$$H_0(x) = 1 - b + bxG_0(H_1(x)), \qquad H_1(x) = 1 - b + bxG_1(H_1(x)), \tag{1.52}$$

where $G_0(x)$ and $G_1(x)$ are the standard generating functions for the degree of a vertex and of a neighboring vertex, see (1.30) and (1.32). This implies that the mean size of a cluster of connected and present vertices is

$$\begin{aligned}\langle s \rangle &= H_0'(1) = b + bG_0'(1)\, H_1'(1) \\ &= b + \frac{b^2 G_0'(1)}{1 - bG_1'(1)} = b\left[1 + \frac{bG_0'(1)}{1 - bG_1'(1)}\right],\end{aligned}$$

where we followed the derivation presented in (1.42) in order to derive the expression $H_1'(1) = b/(1 - bG_1'(1))$, which we used. With (1.44) for $G_0'(1) = z_1 = z$ and $G_1'(1) = z_2/z_1$, we obtain the generalization

$$\langle s \rangle = b + \frac{b^2 z_1^2}{z_1 - bz_2}$$

of (1.45). The model has a phase transition at the critical value of b,

$$b_c = \frac{z_1}{z_2} = \frac{1}{G_1'(1)}. \tag{1.53}$$

If the fraction b of the vertices present in the network is smaller than the critical fraction b_c, then there will be no giant component. This is the point at which the network ceases to be functional in terms of connectivity. When there is no giant component, connecting paths exist only within small isolated groups of vertices, but no long-range connectivity exists. For a communication network such as the Internet, this would be fatal.

For networks with fat tails, however, we expect that the number of next-nearest neighbors z_2 is large compared to the number of nearest neighbors z_1 and that b_c is consequently small. The network is robust as one would need to take out a substantial fraction of the nodes before it would fail.

Random Failure of Vertices in Scale-Free Graphs We consider a pure power-law degree distribution

$$p_k \sim \frac{1}{k^\alpha}, \qquad \int \frac{\mathrm{d}k}{k^\alpha} < \infty, \qquad \alpha > 1,$$

see (1.8) and Sect. 1.6. The first two moments are

$$z_1 = \langle k \rangle \sim \int \mathrm{d}k\,(k/k^\alpha), \qquad \langle k^2 \rangle \sim \int \mathrm{d}k\,(k^2/k^\alpha)\,.$$

Noting that the number of next-nearest neighbors $z_2 = \langle k^2 \rangle - \langle k \rangle$, we can identify three regimes:

– $1 < \alpha \le 2$, with $z_1 \to \infty$ and $z_2 \to \infty$.
  $b_c = z_1/z_2$ is arbitrary in the thermodynamic limit $N \to \infty$.
– $2 < \alpha \le 3$, with $z_1 < \infty$ and $z_2 \to \infty$.
  $b_c = z_1/z_2 \to 0$ in the thermodynamic limit. Any number of vertices can be randomly removed with the network remaining above the percolation limit. The network is extremely robust.
– $3 < \alpha$, with $z_1 < \infty$ and $z_2 < \infty$.
  $b_c = z_1/z_2$ can acquire any value and the network has normal robustness.

Biased Failure of Vertices What happens when somebody sabotages the most important sites of a network? This is equivalent to removing vertices in decreasing order of their degrees, starting with the highest degree vertices. The probability that a given node is active then takes the form

$$b_k = \theta(k_c - k)\,, \tag{1.54}$$

where $\theta(x)$ is the Heaviside step function

$$\theta(x) = \begin{cases} 0 \text{ for } x < 0 \\ 1 \text{ for } x \ge 0 \end{cases}. \tag{1.55}$$

This form of targeted attacks correspond to setting the upper limit of the sum in (1.49) to k_c. Differentiating (1.51) with respect to x yields

$$H_1'(1) = F_1(H_1(1)) + F_1'(H_1(1))\,H_1'(1), \qquad H_1'(1) = \frac{F_1(1)}{1 - F_1'(1)}\,,$$

as $H_1(1) = 1$. The phase transition occurs when $F_1'(1) = 1$,

$$\frac{\sum_{k=1}^{\infty} k(k-1)p_k b_k}{\sum_{k=1}^{\infty} k p_k} = \frac{\sum_{k=1}^{k_c} k(k-1)p_k}{\sum_{k=1}^{\infty} k p_k} = 1\,, \tag{1.56}$$

where we used the definition (1.50) for $F_1(x)$.

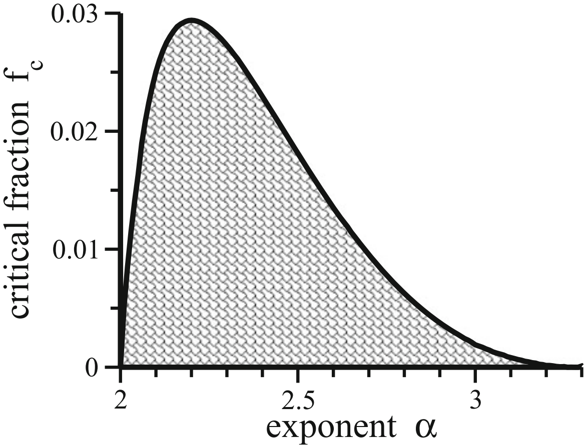

Fig. 1.12 For a scale-free network with a power-law degree distribution $p_k \sim 1/k^\alpha$ the critical fraction f_c of vertices, as defined by (1.59). Removing a fraction greater than f_c of high degree vertices drives the network below the percolation limit. For a smaller loss of highest degree vertices the giant connected component remains intact (shaded area)

Biased Failure of Vertices for Scale-Free Networks Scale-free networks have a power-law degree distribution, $p_k \propto k^{-\alpha}$. We can then rewrite (1.56) as

$$H_{k_c}^{(\alpha-2)} - H_{k_c}^{(\alpha-1)} = H_\infty^{(\alpha-1)}\,, \tag{1.57}$$

where $H_n^{(r)}$ is the nth harmonic number of order r,

$$H_n^{(r)} = \sum_{k=1}^{n} \frac{1}{k^r}\,. \tag{1.58}$$

The number of vertices present is $F_0(1)$, see (1.49), or $F_0(1)/\sum_k p_k$, since the degree distribution p_k is normalized. If we remove a certain fraction f_c of the vertices we reach the transition determined by (1.57),

$$f_c = 1 - \frac{F_0(1)}{\sum_k p_k} = 1 - \frac{H_{k_c}^{(\alpha)}}{H_\infty^{(\alpha)}}\,. \tag{1.59}$$

It is impossible to determine k_c from (1.57) and (1.59) to get f_c in closed form. However one can, solve (1.57) numerically for k_c and substitute it into (1.59). The results are shown in Fig. 1.12, as a function of the exponent α. The network is very susceptible with respect to a biased removal of highest-degree vertices.

– A removal of more than about 3% of the highest degree vertices always leads to a destruction of the giant connected component. Maximal robustness is achieved for $\alpha \approx 2.2$, which is actually close to the exponents measured in some real-world networks. Compare Fig. 1.6.

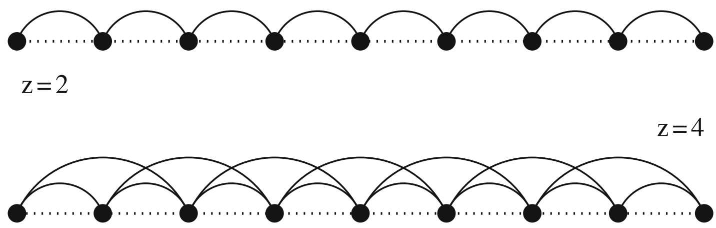

Fig. 1.13 Regular linear graphs with connectivities $z = 2$ (top) and $z = 4$ (bottom)

– Networks with $\alpha > \alpha_c = 3.4788$ never have a giant connected component. The critical exponent α_c is given by the percolation condition $H_\infty^{(\alpha-2)} = 2H_\infty^{(\alpha-1)}$, see (1.27).

1.5 Small-World Models

Random graphs and random graphs with arbitrary degree distribution show no clustering in the thermodynamic limit, in contrast to real-world networks. It is therefore important to find methods to generate graphs that have a finite clustering coefficient and, at the same time, the small-world property.

Clustering in Lattice Models Lattice models and random graphs are two extreme cases of network models. In Fig. 1.13 we illustrate a simple one-dimensional lattice with connectivity z. For periodic boundary conditions, viz when the chain wraps around itself in a ring, one can evaluate the clustering coefficient C analytically.

– ONE DIMENSION
  For number of clusters one finds

$$C = \frac{3(z-2)}{4(z-1)}, \tag{1.60}$$

  which tends to $3/4$ in the limit of large z.
– GENERAL DIMENSION
  Square or cubic lattices have dimension $d = 2, 3$, respectively. The clustering coefficient for general dimensions d is

$$C = \frac{3(z-2d)}{4(z-d)}, \tag{1.61}$$

  which generalizes (1.60). We note that the clustering coefficient tends to $3/4$ for $z \gg 2d$ for regular hypercubic lattices in all dimensions.

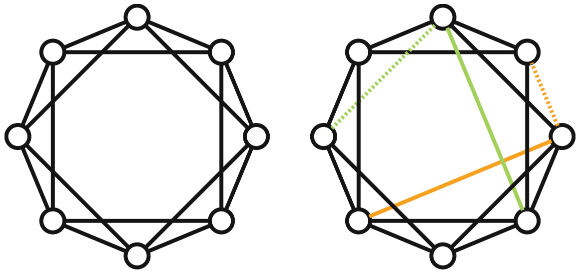

Fig. 1.14 Crossover from a regular lattice to a small-world network via rewiring. **Left:** Regular lattice with periodic boundary conditions. **Right:** Two rewired links, original and new links are color coded (dashed/solid)

Distances in Lattice Models Regular lattices do not show the small-world effect. A regular hypercubic lattice in d dimensions with linear size L has $N = L^d$ vertices. The average vertex–vertex distance increases as L, or equivalently as

$$\ell \approx N^{1/d} .$$

The Watts and Strogatz Model Watts and Strogatz proposed a small-world model that interpolates smoothly between a regular lattice and an Erdös–Rényi random graph. The construction starts with a one-dimensional periodic lattice, see Fig. 1.14. There are several possibilities

- One goes through all the links of the lattice and rewires the link with a given probability p_{new}.
- A single edge is selected and rewired, with the procedure repeated N_{new} times.
- As a variation one may add links, instead of rewiring. The advantage is that networks may not become disconnected.

For small p_{new} and/or N_{new}, this process produces a graph that is still mostly regular, possessing however a few connections stretching long distances across the lattice, as illustrated in Fig. 1.14. The average coordination number z of the lattice remains constant. The number of neighbors of any particular vertex can, however, be greater or smaller than z.

Social Network Interpretation The small-world models such as the ones illustrated in Fig. 1.14, have an intuitive justification for social networks. Most people are friends with their immediate neighbors. Neighbors on the same street, people that they work with, or their relatives. However, some people are also friends with a few far-away persons. Far away in a social sense, like people in other countries, people from other walks of life, acquaintances from previous eras of their lives, and so forth. These long-distance acquaintances are represented by the long-range links in small-world models.

In Fig. 1.15 the clustering coefficient is shown as a function of network diameter when rewiring is performed edge by edge. The key result is that a few steps are sufficient to achieve a drastic reduction of intra-network distances, as measured by

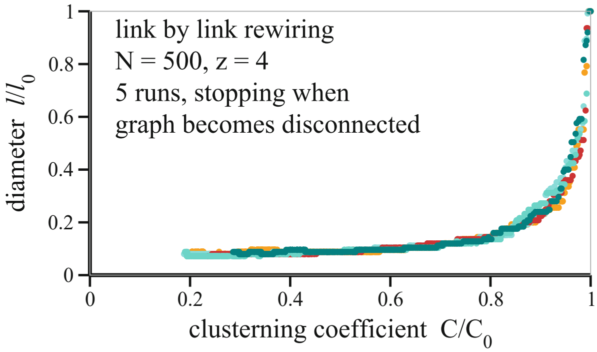

Fig. 1.15 The scaled clustering coefficient C/C_0 as as a function of the scaled diameter l/l_0, where $C_0 = 3/4$ and $l = N/4$ are the initial values, for a one-dimensional ring with $N = 500$ sites and $z = 4$, compare Fig. 1.14. Points correspond to the consecutive random rewiring of a single edge

the graph diameter. At the same time, the clustering coefficient remains high, as observed in many real-world networks.

1.6 Scale-Free Graphs

Evolving Networks Most real-world networks are open, i.e. they are formed by the continuous addition of new vertices to the system. The number of vertices, N, increases throughout the lifetime of the network, as it is the case for the world wide web, which grows exponentially by the continuous addition of new web pages. The small world networks discussed in Sect. 1.5 are, however, constructed for a fixed number of nodes N, growth is not considered.

Preferential Connectivity Random network models assume that the probability that two vertices are connected is random and uniform. In contrast, most real networks exhibit the "rich-get-richer" phenomenon.

> PREFERENTIAL CONNECTIVITY New vertices prefer to connect to existing nodes which already have high degrees.

A newly created web page, to give an example, will include links to well-known sites with a quite high probability. Popular web pages will therefore have both a high number of incoming links and a high growth rate for incoming links. The growth of vertices in terms of edges is therefore in general not uniform.

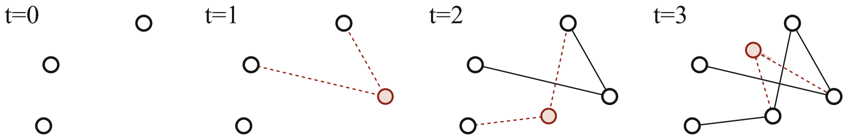

Fig. 1.16 Illustration of the preferential attachment model for an evolving network. At $t = 0$ the system consists of $N_0 = 3$ isolated vertices. At every time step a new vertex (shaded circle) is added, which is connected to $m = 2$ vertices, preferentially to the vertices with high connectivity, determined by the rule (1.62)

Barabási–Albert Model We start at time $t = 0$ with $N(0) = N_0$ unconnected vertices. The preferential attachment growth process can then be carried out in two steps.

– Growth
  At every time step a new vertex is added and connected with $m \le N_0$ links to existing nodes.
– Preferential Attachment
  The new links are determined by connecting the m stubs of the new vertex to nodes with degrees k_i with probability

$$\Pi(k_i) = \frac{k_i + C}{a(t)} \qquad a(t) = 2m(t-1) + C(N_0 + t - 1)\,, \tag{1.62}$$

  where the normalization $a(t)$ has been chosen such that the overall probability to connect is unity, $\sum_i \Pi(k_i) = 1$.

The attachment probability $\Pi(k_i)$ is linearly proportional to the number of links already present, modulo an offset C. Other functional dependencies for $\Pi(k_i)$ are of course possible, but not considered here.

After t time steps above two steps leads to a network with $N(t) = t + N_0$ vertices and mt edges, see Fig. 1.16. We will now show that preferential attachment leads to a scale-free degree distribution

$$p_k \sim k^{-\gamma} \qquad \gamma = 3 + \frac{C}{m}\,. \tag{1.63}$$

This scaling relation is valid for large degrees k_i in the limit $t \to \infty$.

Time-Dependent Connectivities The time dependence of the degree of a given vertex can be calculated analytically using a mean-field approach. Only nodes with large degrees matter, as the scaling relation (1.63) applies asymptotically in the limit $k \to \infty$, We may therefore assume k_i to be continuous,

$$\Delta k_i(t) \equiv k_i(t+1) - k_i(t) \approx \frac{\partial k_i}{\partial t} = A\,\Pi(k_i) = A\,\frac{k_i + C}{a(t)}\,, \tag{1.64}$$

where $\Pi(k_i)$ is the attachment probability. The overall number of new links is proportional to a normalization constant A, which is hence determined by the sum rule

$$\sum_i \Delta k_i(t) \equiv m = A \sum_i \Pi(k_i) = A\,,$$

since the overall probability to attach, $\sum_i \Pi(k_i)$, is unity. Making use of $a(t) = 2m(t-1) + C(N_0 + t - 1)$, we obtain

$$\frac{\partial k_i}{\partial t} = m\frac{k_i + C}{(2m+C)t + a_0} \approx \frac{m}{2m+C}\frac{k_i + C}{t}\,, \tag{1.65}$$

where did neglect $a_0 = C(N_0 - 1) - 2m$ for large times t. The solution of (1.65) is given by

$$\frac{\dot{k}_i}{k_i + C} = \frac{m}{2m+C}\,\frac{1}{t}, \qquad k_i(t) = \big(C + m\big)\left(\frac{t}{t_i}\right)^{m/(2m+C)} - C\,, \tag{1.66}$$

where t_i is the time at which the vertex i had been added, $k_i(t) = 0$ for $t < t_i$.

Adding Times We specialize to the case $C = 0$, the general result for $C > 0$ can be obtained subsequently from scaling considerations. A vertex added at time $t_i = N_i - N_0$ has initially $m = k_i(t_i)$ links, in agreement with (1.66), which reads

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{0.5}, \qquad t_i \;= t\,m^2/k_i^2 \tag{1.67}$$

for $C = 0$. Older nodes, i.e. those with smaller t_i, increase their connectivity faster than the younger vertices, viz those with bigger t_i, see Fig. 1.17. For social networks this mechanism is dubbed the rich-gets-richer phenomenon.

The number of nodes $N(t) = N_0 + t$ is identical to the number of adding times,

$$t_1, \ldots, t_{N_0} = 0, \qquad t_{N_0+j} \;= j, \qquad j = 1, 2, \ldots\,,$$

where we did define the initial N_0 nodes to have adding times zero.

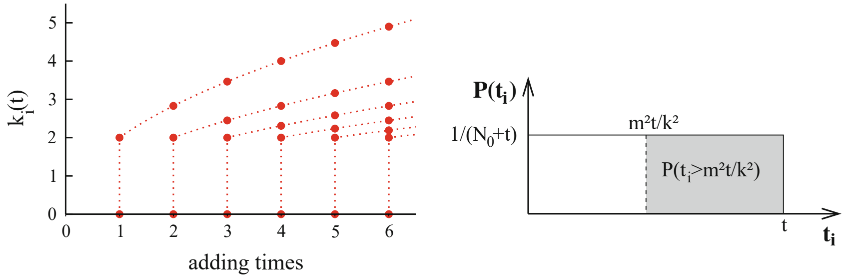

Fig. 1.17 **Left:** Time evolution of the connectivities for vertices with adding times $t = 1, 2, 3, \ldots$ and $m = 2$, following (1.67). **Right:** The integrated probability, $P(k_i(t) < k) = P(t_i > tm^2/k^2)$, see (1.68)

Integrated Probabilities Using (1.67), the probability that a vertex has a connectivity $k_i(t)$ smaller than a certain k, $P(k_i(t) < k)$ can be written as

$$P(k_i(t) < k) = P\left(t_i > \frac{m^2 t}{k^2}\right) . \tag{1.68}$$

Adding times are uniformly distributed, compare Fig. 1.17, which implies that the probability $P(t_i)$ to find an adding time t_i is

$$P(t_i) = \frac{1}{N_0 + t} , \tag{1.69}$$

viz just the inverse of the total number of adding times, which coincides in turn with the total number of nodes. $P(t_i > m^2 t/k^2)$ is therefore the cumulative number of adding times t_i larger than $m^2 t/k^2$, multiplied with the probability $P(t_i)$ to add a new node,

$$P\left(t_i > \frac{m^2 t}{k^2}\right) = \left(t - \frac{m^2 t}{k^2}\right) \frac{1}{N_0 + t} . \tag{1.70}$$

Scale-Free Degree Distribution The degree distribution p_k follows from (1.70) via a simple differentiation,

$$p_k = \frac{\partial P(k_i(t) < k)}{\partial k} = \frac{\partial P(t_i > m^2 t/k^2)}{\partial k} = \frac{2m^2 t}{N_0 + t} \frac{1}{k^3} , \tag{1.71}$$

in accordance with (1.63).

The degree distribution (1.71) has a well defined limit $t \to \infty$, approaching a stationary distribution. We note that the exponent $\gamma = 3$ is independent of the number m of added links per new site. This result indicates that growth and preferential attachment play an important role for the occurrence of a powerlaw scaling in the degree distribution. To verify that both ingredients are really necessary, we investigate variants of above model.

One can repeat above calculation for a finite offset $C > 0$. The exponent γ is identical to the inverse of the scaling power of k_i with respect to time t in (1.66), plus one, which leads to $\gamma = (2m + C)/m + 1 = 3 + C/m$.

Growth with Random Attachment We examine whether growth alone can result in a scale-free degree distribution, assuming random instead of preferential attachment. The growth equation for the connectivity k_i of a given node i, compare (1.65), then takes the form

$$\frac{\partial k_i}{\partial t} = \frac{m}{N_0 + t} \,. \tag{1.72}$$

The m new edges are linked randomly at time t to the $(N_0 + t - 1)$ nodes present at the previous time step. Solving (1.72) for k_i, with the initial condition $k_i(t_i) = m$, we obtain

$$k_i = m\left[\ln\left(\frac{N_0 + t}{N_0 + t_i}\right) + 1\right], \tag{1.73}$$

which is a logarithmic increase with time. The probability that vertex i has connectivity $k_i(t)$ smaller than k is

$$\begin{aligned} P(k_i(t) < k) &= P\left(t_i > (N_0 + t)\,\mathrm{e}^{1-k/m} - N_0\right) \\ &= \left[t - (N_0 + t)\,\mathrm{e}^{1-k/m} + N_0\right]\frac{1}{N_0 + t}\,, \end{aligned} \tag{1.74}$$

where we assumed that we add the vertices uniformly in time to the system. Using

$$p_k = \frac{\partial P(k_i(t) < k)}{\partial k}$$

for large times t, we find

$$p_k = \frac{1}{m}\,\mathrm{e}^{1-k/m} = \frac{\mathrm{e}}{m}\,\exp\left(-\frac{k}{m}\right). \tag{1.75}$$

Growing networks with random attachment are characterized hence by

$$k^* = m\,, \tag{1.76}$$

which is identical to half of the average connectivities of the vertices in the system, since $\langle k\rangle = 2m$. Random attachment does not lead to a scale-free degree distribution. Note that p_k in (1.75) is not properly normalized, nor in (1.71), since we used a large-k approximation during the respective derivations.

Internal Growth with Preferential Attachment The original preferential attachment model yields a degree distribution $p_k \sim k^{-\gamma}$ with $\gamma = 3$. Most social networks such as the Internet and the Wikipedia network, however, have exponents $2 < \gamma < 3$, with the exponent γ being relatively close to two. It is also observed that new edges are mostly added in between existing nodes, albeit with internal preferential attachment.

We can then generalize the preferential attachment model discussed above.

– At every time step a new vertex is added.
– At every time step m new edges are added.
– With probability $r \in [0, 1]$ any one of the m new edges is added between the new vertex and an existing vertex i, which is selected with a probability $\propto \Pi(k_i)$, see (1.62).
– With probability $1 - r$ any one of the m new edges is added in between two existing vertices i and j, which are selected with a probability $\propto \Pi(k_i)\,\Pi(k_j)$.

The model reduces to the original preferential attachment model in the limit $r \to 1$. The scaling exponent γ can be evaluated along the lines used above for the case $r = 1$. One finds

$$p_k \sim \frac{1}{k^\gamma}, \qquad \gamma = 1 + \frac{1}{1 - r/2}. \tag{1.77}$$

The exponent $\gamma = \gamma(r)$ interpolates smoothly between two and three, with $\gamma(1) = 3$ and $\gamma(0) = 2$. For most real-world graphs r is quite small, viz most links are added internally. Note, that the average connectivity $\langle k \rangle = 2m$ remains constant, since one new vertex is added for $2m$ new stubs.

Exercises

(1.1) <u>NETWORK OF CLIQUES</u>
Consider $i = 1, \ldots, 9$ managers sitting on the boards of six companies with (1,9), (1,2,3), (4,5,9), (2,4,6,7), (2,3,6) and (4,5,6,8) being the respective board compositions. Draw the graphs for the managers and companies, by eliminating from the bipartite manager/companies graph one type of node. The result can be interpreted as a network of cliques. Evaluate for both networks the average degree z, the clustering coefficient C and the graph diameter D.

(1.2) <u>ENSEMBLE FLUCTUATIONS</u>
Derive (1.7) for the distribution of ensemble fluctuations. In the case of difficulties Albert and Barabási (2002), can be consulted. Alternatively, check (1.7) numerically.

(1.3) <u>CLUSTERING COEFFICIENT AND DEGREE SEQUENCE</u>
Rewrite the expression (1.29) for the clustering coefficient C as a sum over the degree sequence $\{k_i\}$. Show that the resulting formula, valid only for statistically uncorrelated graphs, may violate the strict bound $C \leq 1$, by considering the degree sequence of a star-shaped network with a single central site connected to $N - 1$ otherwise isolated sites.

(1.4) <u>TSALLIS–PARETO DISTRIBUTION</u>
Evaluate mean and variance of the Tsallis–Pareto distribution,

$$p(x) = \frac{\alpha - 1}{(1 + x)^\alpha}\,, \tag{1.78}$$

on the interval $x \in [0, \infty]$.

(1.5) <u>KRONECKER GRAPHS</u>
A Kronecker graph K of two graphs G and H is defined as the outer product $K = G \otimes H$, in terms of the respective adjacency matrices, with a node X_{ij}^K of K being linked to a node X_{kl}^K of K if X_i^G is linked to X_k^G in G and X_j^H is linked to X_l^H in H.
Start by constructing the Kronecker graph of two very small networks of your choice. Express then generically the degree distribution $p^K(k)$ of the Kronecker graph in terms of $p^G(k)$ and $p^H(k)$, the respective degree distributions of G and H. How does the coordination number scale?

(1.6) <u>SELF-RETRACING PATH APPROXIMATION</u>
Look at Brinkman and Rice (1970) and prove (1.14). This derivation is only suitable for readers with a solid training in physics.

(1.7) <u>PROBABILITY GENERATING FUNCTIONS</u>
Prove that the variance σ^2 of a probability distribution p_k with a generating functional $G_0(x) = \sum_k p_k\, x^k$ and average $\langle k\rangle$ is given by $\sigma^2 = G_0''(1) + \langle k\rangle - \langle k\rangle^2$.
Consider furthermore a cumulative process generated by $G_C(x) = G_N(G_0(x))$, compare (1.38). Calculate the mean and the variance of the cumulative process and discuss the result.

(1.8) <u>CLUSTERING COEFFICIENT OF REGULAR LATTICES</u>
Prove (1.60) for the clustering coefficient of one-dimensional lattice graphs. Facultatively, generalize this formula to a d-dimensional lattice with links along the main axis.

(1.9) <u>ROBUSTNESS OF GRAPHS AGAINST FOCUSED ATTACKS</u>
Using the formalism of Sect. 1.4 calculate the robustness of Erdös–Rényi networks against focused attacks with $b_k = \theta(k_c - k)$ being the probability of nodes remaining active. For which values of k_c, as a function of the

coordination number z, does the network remain functional? Evaluate the corresponding critical fraction f_c of inactive nodes.

(1.10) EPIDEMIC SPREADING IN SCALE-FREE NETWORKS
Consult "R. Pastor-Satorras and A. Vespigiani, *Epidemic spreading in scale-free networks*, Physical Review Letters, Vol. 86, 3200 (2001)". The task is to solve a simple molecular-field approach to the SIS model for the spreading of diseases in scale-free networks by using the excess degree distribution discussed in Sect. 1.3.1, where S and I stand respectively for susceptible and infective individuals.

Further Reading

For further studies several books and review articles on general network theory are recommended, Estrada (2012), Kadushin (2012), and Albert and Barabási (2002).

The interested reader might delve into some of the original literature, including the original Watts and Strogatz (1998) small-world model, the mean-field solution of the preferential attachment model, Barabasi et al. (1999), a study regarding the community structure of real-world networks, Palla et al. (2005), or the mathematical basis of graph theory, Erdös and Rényi (1959). A good starting point is the account by Milgram (1967) of his by now famous experiment, which led to the law of "six degrees of separation", Guare (1990).

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Bifurcations and Chaos in Dynamical Systems 2

Complex systems theory deals with dynamical systems containing often large numbers of variables. It extends dynamical systems theory, which treats dynamical systems containing a few variables. A good understanding of dynamical systems theory is therefore a prerequisite when studying complex systems.

In this chapter we introduce core concepts, like attractors and Lyapunov exponents, bifurcations, and deterministic chaos from the realm of dynamical systems theory. An introduction to catastrophe theory will be provided together with the notion of rate-induced tipping and colliding attractors.

Most of the chapter will be devoted to ordinary differential equations and maps, the traditional focus of dynamical systems theory, venturing however towards the end into the intricacies of time delay dynamical systems.

2.1 Basic Concepts of Dynamical Systems Theory

Dynamical systems theory deals with the properties of coupled differential equations, determining the time evolution of a few, typically a handful of variables. We present a concise overview covering the most important concepts and phenomena.

Fixpoints and Limit Cycles In order to illustrate methodologies typical for dynamical systems theory, we start with an elementary non-linear rotator. For a two-dimensional systems $\mathbf{x} = (x, y)$, or

$$x(t) = r(t)\cos(\varphi(t)), \qquad y(t) = r(t)\sin(\varphi(t)) \tag{2.1}$$

in polar coordinates, we postulate the non-linear differential equations

$$\dot{r} = (\Gamma - r^2)\, r, \qquad \dot{\varphi} = \omega \tag{2.2}$$

C. Gros, *Complex and Adaptive Dynamical Systems*,
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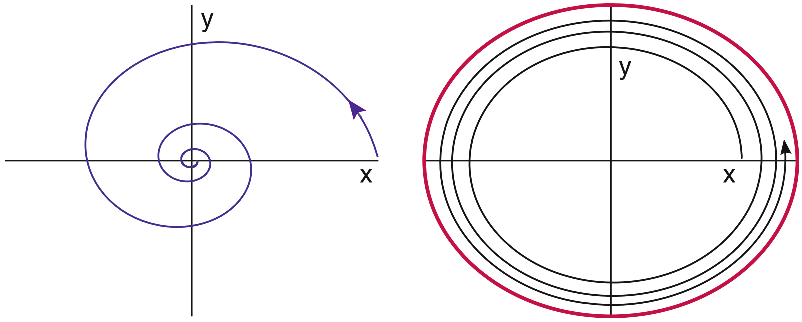

Fig. 2.1 The solution of the non-linear rotator, compare (2.1) and (2.2). For $\Gamma < 0$ the orbit is attracted by a stable fixpoint (left), for $\Gamma > 0$ by a stable limit cycle (right)

govern the dynamical behavior. The typical orbits $(x(t), y(t))$ are illustrated in Fig. 2.1. The limiting behavior of (2.2) is

$$\lim_{t\to\infty} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{cases} \begin{bmatrix} 0 \\ 0 \end{bmatrix} & \Gamma < 0 \\ \begin{bmatrix} r_c \cos(\omega t) \\ r_c \sin(\omega t) \end{bmatrix} & r_c^2 = \Gamma > 0 \end{cases} \tag{2.3}$$

In the first case, $\Gamma < 0$, trajectories are attracted by the stable fixpoint $\mathbf{x}_0^* = (0, 0)$. In the second case, $\Gamma > 0$, the dynamics approaches stable periodic orbit.

LIMIT CYCLE Trajectories retrace themselves with a period T for limit cycles, viz one has $\mathbf{x}(t + T) = \mathbf{x}(t)$.

Limit cycles can be attracting or repelling. In (2.3) one has an attracting limit cycle for $\Gamma > 0$.

Bifurcation Of particular interest are situations where the behavior of a dynamical system changes qualitatively.

BIFURCATION The long-term limiting behavior of a dynamical system described by a set of parameterized differential equations may change qualitatively as a function of an external parameter. The resulting change in terms of fixpoints, limit cycles and/or chaotic attractors constitutes a bifurcation.

The dynamical system (2.1) and (2.2) shows a bifurcation at $\Gamma = 0$, at which a fixpoint turns into a limit cycle. One denotes this specific type of bifurcation a "Hopf bifurcation", as discussed lateron in detail

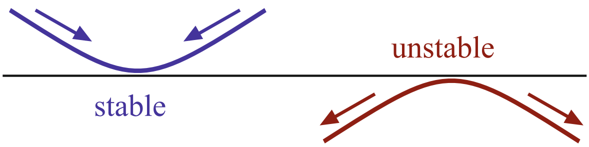

Fig. 2.2 A fixpoint is stable (unstable) when orbits are attracted (repelled)

Stability of Fixpoints The dynamics of orbits close to fixpoints or limiting orbits is determined by its stability.

> STABILITY CONDITION A fixpoint is stable (unstable) if nearby orbits are attracted (repelled) by the fixpoint, and metastable if the distance does not change.

An illustration is given in Fig. 2.2. The stability of fixpoints is closely related to their Lyapunov exponents, as discussed in Sect. 2.4.

One can examine the stability of a fixpoint $\mathbf{x}^*$ by linearizing the equation of motions for $\mathbf{x} \approx \mathbf{x}^*$. For the fixpoint $r^* = 0$ of (2.2) we find

$$\dot{r} = \left(\Gamma - r^2\right) r \approx \Gamma r, \qquad r \ll 1\,,$$

and $r(t)$ decreases (increases) for $\Gamma < 0$ ($\Gamma > 0$). For a d-dimensional system $\mathbf{x} = (x_1, \ldots, x_d)$ the stability of a fixpoint $\mathbf{x}^*$ is determined by calculating the d eigenvalues of the linearized equations of motion. The system is stable if all eigenvalues are negative and unstable if at least one eigenvalue is positive.

Phase Space A given dynamical systems lives in its phase space.

> PHASE SPACE The phase space is the space spanned by all allowed values of the variables entering a set of first-order differential equations defining the dynamical system.

For a two-dimensional system (x, y) the phase space is $\mathbb{R}^2$, but in the polar coordinates one has

$$\left\{ (r, \varphi) \,\middle|\, r \in [0, \infty],\ \varphi \in [0, 2\pi[\right\}.$$

The representation of a phase space changes upon coordinate transformations.

Attracting States and Manifolds Attracting states play a center role in dynamical systems theory.

> ATTRACTOR A bounded region in phase space to which orbits with certain initial conditions come arbitrarily close is called an attractor.

Attractors can be isolated points, fixpoints, limit cycles or more complex objects like attracting manifolds, viz subsets of phase space, or chaotic attractors.

BASIN OF ATTRACTION The set of initial conditions in phase space leading to orbits approaching a certain attractor arbitrarily closely is the basin of attraction.

Most of the times the extend of a basin of attraction can be evaluated only numerically.

First-Order Differential Equations Let us consider the third-order differential equation

$$\frac{d^3}{\mathrm{d}t^3}x(t) = f(x, \dot{x}, \ddot{x})\,. \tag{2.4}$$

Using

$$x_1(t) = x(t), \qquad x_2(t) = \dot{x}(t), \qquad x_3(t) = \ddot{x}(t)\,, \tag{2.5}$$

we can rewrite (2.4) as a first-order differential equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ f(x_1, x_2, x_3) \end{bmatrix}.$$

One can reduce any set of coupled differential equations to a set of first-order differential equations by introducing an appropriate number of additional variables. We therefore consider in the following only first-order, ordinary differential equations.

Autonomous Systems The generic form for a dynamical system is

$$\frac{\mathrm{d}\mathbf{x}(t)}{\mathrm{d}t} = \mathbf{f}(\mathbf{x}(t)), \qquad \mathbf{x}, \mathbf{f} \in \mathbb{R}^d, \qquad t \in [-\infty, +\infty]\,, \tag{2.6}$$

when time is continuous, or, equivalently, maps such as

$$\mathbf{x}(t+1) = \mathbf{g}(\mathbf{x}(t)), \qquad \mathbf{x}, \mathbf{g} \in \mathbb{R}^d, \qquad t = 0, 1, 2, \ldots\,, \tag{2.7}$$

when time is discrete. Together with the time evolution equation one has to set the initial condition $\mathbf{x}_0 = \mathbf{x}(t_0)$. An evolution equation of type (2.15) is denoted "autonomous", since it does not contain an explicit time dependence. A system of type $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$ is dubbed "non-autonomous".

A particular solution $\mathbf{x}(t)$ of a dynamical system in phase space is denoted "trajectory", or "orbit". Orbits are uniquely determined by the set of initial conditions, $\mathbf{x}(0) \equiv \mathbf{x}_0$, a consequence of dealing with first-order differential equations.

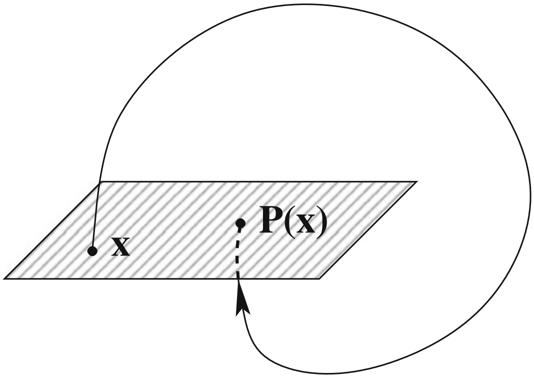

Fig. 2.3 The Poincaré map, $\mathbf{x} \to \mathbf{P}(\mathbf{x})$, which maps an intersection $\mathbf{x}$ of the trajectory with a hyperplane (shaded) to the consecutive intersection $\mathbf{P}(\mathbf{x})$

Poincaré Map It is in general difficult to illustrate graphically the motion of $\mathbf{x}(t)$ in d dimensions. Our retina, as well as our print media, are two-dimensional and it is therefore convenient at times to consider a plane Σ in $\mathbb{R}^d$ together with the points $\mathbf{x}^{(i)}$ marking the consecutive intersections of an orbit with Σ, as illustrated in Fig. 2.3. As an example consider the plane

$$\Sigma = \{ (x_1, x_2, 0, \ldots, 0) \,|\, x_1, x_2 \in \mathbb{R} \}$$

and the sequence of intersections (see Fig. 2.3)

$$\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)}, 0, \ldots, 0), \qquad (i = 1, 2, \ldots)$$

which define the "Poincaré map"

$$\mathbf{P} : \ \mathbf{x}^{(i)} \ \mapsto \ \mathbf{x}^{(i+1)} \,. \tag{2.8}$$

The Poincaré map is a discrete map, compare (2.7), which can be constructed for continuous-time dynamical systems like (2.15). The Poincaré map is very useful, since we can print and analyze it directly. In the simplest case, a periodic orbit would show up in the Poincaré map as the identity mapping.

Constants of Motion and Ergodicity Dynamical systems derived in physics from Hamilton- or Lagrange formalism are "mechanical systems". The vast majority of dynamical systems we will examine are non-mechanical. There are however several foundational concepts from the realm of mechanical systems that are generically relevant.

– CONSTANT OF MOTION
  A function $F(\mathbf{x})$ on phase space $\mathbf{x} = (x_1, \ldots, x_d)$ is a "constant of motion" if it is conserved under the time evolution of the dynamical system, i.e. when

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\mathbf{x}(t)) = \sum_{i=1}^{d} \left(\frac{\partial}{\partial x_i} F(\mathbf{x}) \right) \dot{x}_i(t) \equiv 0$$

  holds for all times t. In many mechanical systems the energy is a conserved quantity.
– ERGODICITY
  A dynamical system in which orbits come arbitrarily close to any allowed point in the phase space, irrespective of the initial condition, is called ergodic.

All conserving systems of classical mechanics are ergodic. The ergodicity of a mechanical system is closely related to "Liouville's theorem".[1]

Ergodicity holds only modulo conserved quantities, which is the case for the energy in many mechanical systems. Then, only points in the phase space having the same energy as the trajectory considered are approached arbitrarily close. It is clear that ergodicity and attractors are mutually exclusive: An ergodic system cannot have attractors and a dynamical system with one or more attractors cannot be ergodic.

Mechanical Systems and Integrability A dynamical system of type

$$\ddot{x}_i = f_i(\boldsymbol{x}, \dot{\boldsymbol{x}}), \qquad i = 1, \ldots, f$$

is a mechanical system, since equations of motion in classical mechanics are of this form, e.g. Newton's law. f is called the degree of freedom, which implies that mechanical systems can be written as a set of coupled first-order differential equations with $2f$ variables constituting the phase space,

$$(x_1 \ldots x_f, v_1 \ldots v_f), \qquad v_i = \dot{x}_i, \qquad i = 1, \ldots, f\,,$$

with $\mathbf{v} = (v_1, \ldots, v_f)$ being the generalized velocity. A mechanical system is "integrable" if there are $\alpha = 1, \ldots, f$ independent constants of motion $F_\alpha(\boldsymbol{x}, \dot{\boldsymbol{x}})$, such that

$$\frac{d}{\mathrm{d}t} F_\alpha(\boldsymbol{x}, \dot{\boldsymbol{x}}) = 0, \qquad \alpha = 1, \ldots, f\,.$$

The motion in the $2f$-dimensional phase space $(x_1 \ldots x_f, v_1 \ldots v_f)$ is then restricted to an f-dimensional subspace, which is an f-dimensional torus, see Fig. 2.4.

[1] Liouville's theorem will be discussed in Sect. 3.1.1 of Chap. 3.

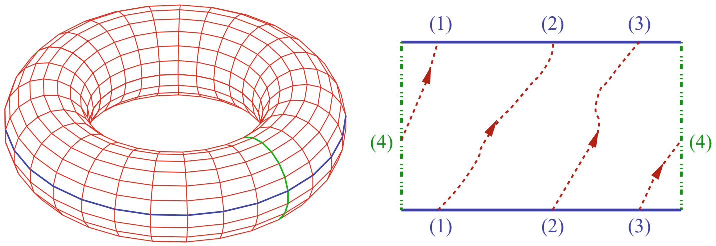

Fig. 2.4 **Left:** A KAM-torus can be cut along two lines (vertical/horizontal) and unfolded. **Right:** A closed orbit on the unfolded torus with $\omega_1/\omega_2 = 3/1$. The numbers indicate points that coincide after refolding (periodic boundary conditions)

An example of an integrable mechanical system is the Kepler problem, viz the motion of earth around the sun. Integrable systems are very rare, they constitute however important reference points for the understanding of more complex dynamical systems. A classical example of a non-integrable mechanical system is the three-body problem, viz the combined motion of earth, moon and sun around each other.

KAM Theorem Kolmogorov, Arnold and Moser (KAM) examined the question of what happens to an integrable system when it is perturbed. Let us consider a two-dimensional torus, as illustrated in Fig. 2.4. Orbit wraps around the torus respectively with frequencies ω_1 and ω_2. A key quantity is the ratio of revolution frequencies ω_1/ω_2, which can be rational or irrational.

We recalr that irrational numbers r may be approximated with arbitrary accuracy by a sequence of quotients

$$\frac{m_1}{s_1},\ \frac{m_2}{s_2},\ \frac{m_3}{s_3},\ \dots \qquad\qquad s_1 < s_2 < s_3 < \dots$$

with ever larger denominators s_i. A number r is "very irrational" when it is difficult to approximate r by such a series of rational numbers, viz when very large denominators s_i are required to achieve a certain given accuracy for $|r - m/s|$.

The KAM theorem states that orbits with rational ratios of revolution frequencies ω_1/ω_2 are the most unstable under a perturbation of an integrable system and that tori are most stable when this ratio is very irrational.

Gaps in the Saturn Rings A spectacular example for the instability of rational KAM-tori are the gaps in the rings of the planet Saturn.

The time a hypothetical particle in Cassini's gap (between the A- and the B-ring, $r = 118{,}000$ km) would need to orbit Saturn is half the orbiting period of the 'shepherd-moon' Mimas. The quotient of these two revolving frequencies is 2:1. Particle orbiting in Cassini's gap are therefore unstable against perturbation caused by Mimas and thrown out of their orbit.

2.2 Fixpoints, Bifurcations and Stability

We start our systematic discussions by considering the stability of fixpoint x^* of a one-dimensional dynamical system

$$\dot{x} = f(x), \qquad f(x^*) = 0\,. \tag{2.9}$$

A fixpoint is the most simplest invariant manifold.[2] Generically, invariant manifolds are sets of phase space that that are invariant under the dynamical flow.

Fixpoints are the only possible invariant manifolds for $d = 1$ dimension, in two dimensions fixpoints and limiting cycles are possible, with more complicated objects, such as strange attractors, becoming accessible in three and higher dimensions.

Lyapunov Exponents The stability of a fixpoint is determined by the direction of the flow close to it, which can be determined by linearizing the time evolution equation $\dot{x} = f(x)$ around the fixpoint x^*,

$$\frac{d}{dt}\big(x - x^*\big) = \dot{x} \approx f(x^*) \,+\, f'(x^*)(x - x^*) \,+\, \dots\,, \tag{2.10}$$

where $f'()$ denotes the first derivative. We rewrite (2.10) as

$$\frac{d}{dt}\Delta x = f'(x^*)\,\Delta x, \qquad \Delta x = x - x^*\,,$$

where we neglected terms of order $(x - x^*)^2$ and higher and where we made use of the fixpoint condition $f(x^*) = 0$. This equation has the solution

$$\Delta x(t) = \Delta x(0)\,\mathrm{e}^{tf'(x^*)} \;\;\rightarrow\;\; \begin{cases} \infty & f'(x^*) > 0 \\ 0 & f'(x^*) < 0 \end{cases}\,. \tag{2.11}$$

The perturbation Δx decreases/increases with time and the fixpoint x^* is hence stable/unstable for $f'(x^*) < 0$ and $f'(x^*) > 0$ respectively.

LYAPUNOV EXPONENT The time evolution close to a fixpoint x^* is generically exponential, $\sim \exp(\lambda t)$, and one denotes by $\lambda = f'(x^*)$ the Lyapunov exponent.

The Lyapunov exponent controls the direction of the flow close to a fixpoint. Orbits are exponentially repelled/attracted for $\lambda > 0$ and for $\lambda < 0$ respectively. For more than a single variable one has to find all eigenvalues of the linearized problem,

[2] Invariant manifolds are touched upon further in in Sect. 3.2.1 of Chap. 3.

as discussed further below, with the fixpoint being stable only when all eigenvalues are negative.

Fixpoints of Discrete Maps For a discrete map of type

$$x(t+1) = g(x(t)), \qquad x^* = g(x^*) \tag{2.12}$$

the stability of a fixpoint x^* can be determined by an equivalent linear analysis,

$$x(t+1) = g(x(t)) \approx g(x^*) \; + \; g'(x^*)(x(t) - x^*) \, .$$

Using the fixpoint condition $g(x^*) = x^*$ we write above expression as

$$\Delta x(t+1) = x(t+1) - x^* = g'(x^*)\, \Delta x(t) \, ,$$

which has the solution

$$\Delta x(t) = \Delta x(0) \left[g'(x^*)\right]^t, \qquad |\Delta x(t)| = |\Delta x(0)|\, \mathrm{e}^{\lambda t} \, . \tag{2.13}$$

The Lyapunov exponent,

$$\lambda = \log |g'(x^*)| = \begin{cases} < 0 & \text{for} \quad |g'(x^*)| < 1 \\ > 0 & \text{for} \quad |g'(x^*)| > 1 \end{cases} , \tag{2.14}$$

controls the stability of the fixpoint. Note the differences in the relation of the Lyapunov exponent λ to the derivatives $f'(x^*)$ and $g'(x^*)$ for differential equations and maps respectively, as given by (2.11) and (2.14).

2.2.1 Fixpoints Classification and Jacobian

We assume that the general d-dimensional dynamical system

$$\frac{\mathrm{d}\mathbf{x}(t)}{\mathrm{d}t} = \mathbf{f}(\mathbf{x}(t)), \qquad \mathbf{x}, \mathbf{f} \in \mathbb{R}^d, \qquad \mathbf{f}(\mathbf{x}^*) = 0 \tag{2.15}$$

has a fixpoint $\mathbf{x}^*$.

Jacobian and Lyapunov Exponents For a stability analysis of the fixpoint $\mathbf{x}^*$ one linearizes (2.15) around the fixpoint, using $x_i(t) \approx x_i^* + \delta x_i(t)$, with small $\delta x_i(t)$. One obtains

$$\frac{d\delta x_i}{dt} = \sum_j J_{ij}\, \delta x_j, \qquad J_{ij} = \left.\frac{\partial f_i(x)}{\partial x_j}\right|_{\mathbf{x}=\mathbf{x}^*} . \tag{2.16}$$

The matrix J_{ij} of all possible partial derivatives is the "Jacobian" of the dynamical system (2.15). The Jacobian allows to generalizes the definition of the Lyapunov exponent for one-dimensional systems, as given previously.

LYAPUNOV SPECTRUM The set of eigenvalues $\{\lambda_i\}$ of the Jacobian $i = 1, .., d$ is the spectrum characterizing the fixpoint $\mathbf{x}^*$.

Lyapunov exponents $\lambda_n = \lambda_n' + i\lambda_n''$ may have real λ_n' and imaginary λ_n'' components, which leads to the time evolution

$$\mathrm{e}^{\lambda_n t} = \mathrm{e}^{\lambda_n' t}\,\mathrm{e}^{i\lambda_n'' t}$$

of infinitesimal perturbations around the fixpoint. A Lyapunov exponent λ_n is attracting/neutral/repelling when λ_n' is negative/zero/positive respectively.

HYPERBOLIC FIXPOINT The flow is well defined in linear order when all $\lambda_i' \neq 0$. In this case the fixpoint is hyperbolic.

For a non-hyperbolic fixpoint at least one of the Lyapunov exponents is neutral. All Lyapunov exponents are neutral for a vanishing Jacobian.

Pairwise Conjugate Exponents With $\lambda = \lambda' + i\lambda''$ also its conjugate $\lambda^* = \lambda' - i\lambda''$ in an eigenvalue of the Jacobian, which is a real matrix. It follows that λ and λ^* differ when $\lambda'' \neq 0$. In this caser there are (at least) two eigenvalues having the same real part λ'.

Fixpoint Classification for $d = 2$ In Fig. 2.5 some example trajectories are shown for several fixpoints in $d = 2$ dimensions.

- NODE
  Both eigenvalues are real and have the same sign, which is negative/positive for a stable/unstable node.
- SADDLE
  Both eigenvalues are real and have opposite signs.
- FOCUS
  The eigenvalues are complex conjugate to each other. The trajectories spiral in/out for negative/positive real parts.

Fixpoints in higher dimensions are characterized by the number of respective attracting/neutral/repelling eigenvalues of the Jacobian, which may be in turn either real or complex.

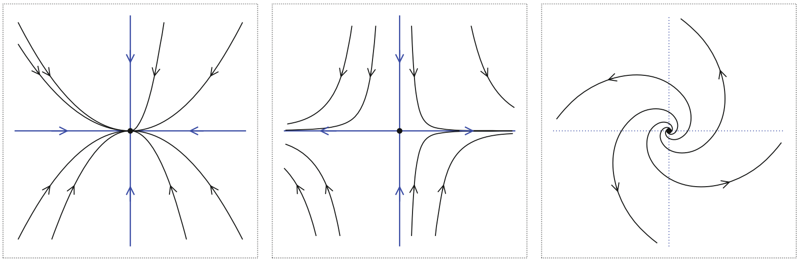

Fig. 2.5 Example trajectories for a stable node (left), with a ratio $\lambda_2/\lambda_1 = 2$, for a saddle (middle) with $\lambda_2/\lambda_1 = -3$, and for a unstable focus (right)

Stable and Unstable Manifolds For real eigenvalues $\lambda_n \neq 0$ of the Jacobian J, with eigenvectors $\mathbf{e}_n$ and a sign $s_n = \lambda_n/|\lambda_n|$, we define via

$$\lim_{t\to -s_n\infty} \mathbf{x}_n(t) = \mathbf{x}^* + \mathrm{e}^{t\lambda_n}\mathbf{e}_n, \qquad J\mathbf{e}_n = \lambda_n\mathbf{e}_n\,, \tag{2.17}$$

trajectories $\mathbf{x}_n(t)$ that leave/approach the fixpoint parallel to an eigendirection. These are stable manifolds (for $\lambda_n < 0$) and unstable manifolds (for $\lambda_n > 0$).

For a neutral Lyapunov exponent with $\lambda_n = 0$ one define a "center manifold" which we will discuss in the context of catastrophe theory in Sect. 2.3.2. The term manifold denotes in mathematics, loosely speaking, a smooth topological object.

Stable and unstable manifolds control the flow infinitesimal close to the fixpoint along the eigendirections of the Jacobian and may be continued to all positive and negative times t. Typical examples are illustrated in Figs. 2.5 and 2.6.

Heteroclinic Orbits One speaks of a "heteroclinc orbit" when the unstable manifold of one fixpoint connects to the stable manifold of another fixpoint, and viceversa. As an example, we consider a two dimensional dynamical system defined by

$$\begin{aligned} \dot{x} &= 1 - x^2 \\ \dot{y} &= yx + \epsilon\big(1 - x^2\big)^2 \end{aligned} \qquad J(x^*) = \begin{pmatrix} -2x^* & 0 \\ 0 & x^* \end{pmatrix}, \tag{2.18}$$

which has two saddles $\mathbf{x}^*_\pm = (x^*, 0)$, where $x^* = \pm 1$. The eigenvectors of the Jacobian $J(x^*)$ are aligned with the x and the y axis respectively, for all values of the control parameter ϵ.

The flow diagram, as illustrated in Fig. 2.6, is invariant when inverting both $x \leftrightarrow (-x)$ and $y \leftrightarrow (-y)$. The system contains additionally the $y = 0$ axis as a mirror line for a vanishing $\epsilon = 0$ and there is a heteroclinic orbit connecting an unstable manifold of $\mathbf{x}^*_-$ to one of the stable manifolds of $\mathbf{x}^*_+$. A finite ϵ removes the mirror

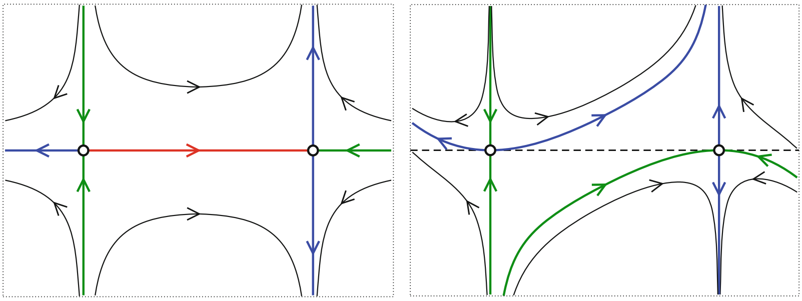

Fig. 2.6 Sample trajectories of the system (2.18), for $\epsilon = 0$ (left) and $\epsilon = 0.2$ (right). Shown are the stable manifolds (thick green lines), the unstable manifolds (thick blue lines) and the heteroclinic orbit (thick red line)

line $y = 0$, present at $\epsilon = 0$, destroying the heteroclinic orbit. Real world systems are often devoid of symmetries and heteroclinic orbits hence rare.

2.2.2 Bifurcations and Normal Forms

The nature of the solution to a dynamical system, as defined by a suitable first order differential equation (2.15), may change qualitatively as a function of a given control parameter. When this happens, a bifurcation occurs. Here we list the most important classes of what is called a "local bifurcation". The distinction to their counterpart, "global bifurcations", will be elucidated in Sect. 2.3.

A local bifurcations can be characterized by a simple archetypical equation, the "normal form", to which a more complex dynamical systems will generically reduce close to the transition point.

Saddle-Node Bifurcation We consider the dynamical system defined by

$$\frac{dx}{dt} = a - x^2, \tag{2.19}$$

for a real variable x and a real control parameter a. The fixpoints $\dot{x} = 0$

$$x_+^* = +\sqrt{a}, \qquad x_-^* = -\sqrt{a}, \qquad a > 0 \tag{2.20}$$

exist only for positive control parameters, $a > 0$; there are no fixpoints for negative $a < 0$. For the flow we find

$$\frac{dx}{dt} = \begin{cases} < 0 & \text{for } x > \sqrt{a} \\ > 0 & \text{for } x \in [-\sqrt{a}, \sqrt{a}] \\ < 0 & \text{for } x < -\sqrt{a} \end{cases} \tag{2.21}$$

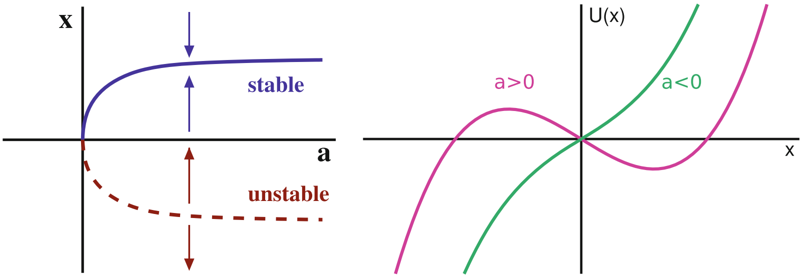

Fig. 2.7 The saddle-node bifurcation, as described by (2.19). There are two fixpoints for $a > 0$, an unstable branch $x_-^* = -\sqrt{a}$ and a stable branch $x_+^* = +\sqrt{a}$. **Left:** The phase diagram together with the flow (arrows). **Right:** The bifurcation potential $U(x) = -ax + x^3/3$, compare (2.26)

when $a > 0$. The upper branch x_+^* is hence stable and the lower branch x_-^* unstable, as illustrated in Fig. 2.7.

For a saddle-node bifurcation a stable and an unstable fixpoint collide and annihilate each other, one speaks also of a fold, tangential or blue sky bifurcation.

Transcritical Bifurcation We now consider the dynamical system

$$\frac{dx}{dt} = x(a - x)\,, \tag{2.22}$$

again for a real variable x and a real control parameter a. The two fixpoint solutions $\dot{x} = 0$,

$$x_0^* = 0, \qquad x_a^* = a, \qquad \forall a \tag{2.23}$$

exist for all values of the control parameter. The direction of the flow $\dot{x}$ is positive for x in between the two solutions and negative otherwise, see Fig. 2.8. The respective stabilities of the two fixpoint solutions exchange consequently at $a = 0$.

Pitchfork Bifurcation The "supercritical" pitchfork bifurcation is described by

$$\frac{dx}{dt} = ax - x^3, \qquad x_0^* = 0, \qquad x_+^* = +\sqrt{a}, \qquad x_-^* = -\sqrt{a}\,. \tag{2.24}$$

The trivial fixpoint $x_0^* = 0$ becomes unstable at criticality, $a = 0$, and two symmetric stable fixpoints appear, see Fig. 2.9.

Bifurcation Potentials In many cases one can write the dynamical system under consideration as

$$\frac{dx}{dt} = -\frac{d}{dx}U(x)\,, \tag{2.25}$$

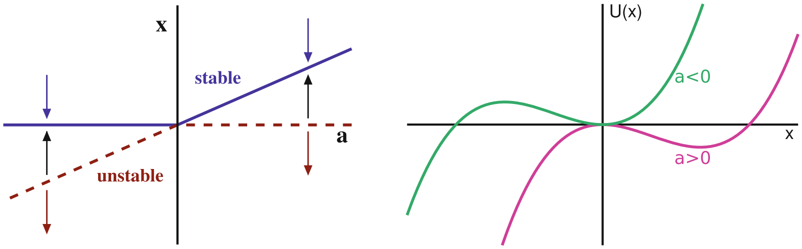

Fig. 2.8 The transcritical bifurcation, see (2.22). The two fixpoints $x_0^* = 0$ and $x_a^* = a$ exchange stability at $a = 0$. **Left:** Phase diagram and flow (arrows). **Right:** The bifurcation potential $U(x) = -ax^2/2 + x^3/3$, compare (2.26)

where $U(x)$ is a potential in analogy to Newton's equation of motion in classical mechanics. Local minima of the potential correspond to stable fixpoints, compare Fig. 2.2. The potentials for the saddle-node and the transcritical bifurcation are

$$U_{\text{saddle}}(x) = -ax + \frac{1}{3}x^3, \qquad U_{\text{trans}}(x) = -\frac{a}{2}x^2 + \frac{1}{3}x^3, \tag{2.26}$$

respectively, see the definitions (2.19) and (2.22). The bifurcation potentials, as shown in Figs. 2.7 and 2.8, bring immediately to evidence the stability of the respective fixpoints. The bifurcation potential of the pitchfork bifurcation,

$$U_{\text{pitch}}(x) = -\frac{a}{2}x^2 + \frac{1}{4}x^4, \tag{2.27}$$

is identical to the Landau-Ginzburg potential describing second-order phase transitions in statistical physics.[3] Of relevance is also the subcritical pitchfork transition, defined by $\dot{x} = ax + x^3$.

Bifurcation Symmetries The three bifurcation scenarios discussed above, saddle-node, transcritical and pitchfork, are characterized by their symmetries close to the critical point, which has been set to $x = 0$ and $a = 0$ in all three cases. The normal forms and their respective bifurcation potentials constitute the simplest formulations consistent with the defining symmetry properties.

[3] The Landau-Ginzburg theory of phase transitions will be treated at length in the context of self-organized criticality, in Sect. 6.1. of Chap. 6.

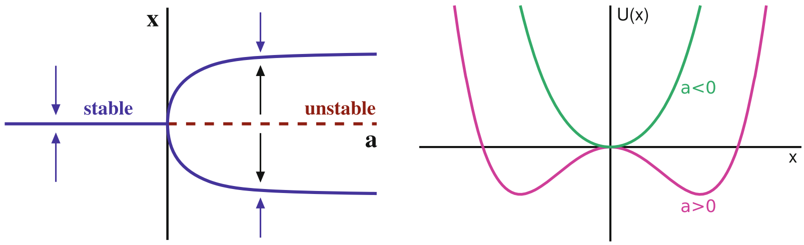

Fig. 2.9 The supercritical pitchfork bifurcation, as defined by (2.24). The t$x_0^* = 0$ becomes unstable for $a > 0$ and two new stable fixpoints, $x_+^* = +\sqrt{a}$ and $x_-^* = -\sqrt{a}$ appear. **Left:** Phase diagram and flow (arrows). **Right:** The bifurcation potential $U(x) = -ax^2/2 + x^4/4$, compare (2.26)

The bifurcation potentials of the saddle-node and the pitchfork transitions are respectively antisymmetric and symmetric under a sign change $x \leftrightarrow -x$ of the dynamical variable, compare (2.26) and (2.27).

$(+) \leftrightarrow (-)$	saddle-node	transcritical	pitchfork
x	anti	–	symm
a, x	–	anti	–

The bifurcation potential of the transcritical bifurcation is, on the other hand, antisymmetric under the combined symmetry operation $x \leftrightarrow -x$ and $a \leftrightarrow -a$, compare (2.26).

2.2.3 Hopf Bifurcations and Limit Cycles

Hopf Bifurcation A Hopf bifurcation occurs when a fixpoint changes its stability together with the appearance of an either stable or unstable limiting cycle, e.g. as for non-linear rotator illustrated in Fig. 2.1. The canonical equations of motion are[4]

$$\begin{aligned} \dot{x} &= -y + d(\Gamma - x^2 - y^2)\,x \\ \dot{y} &= x + d(\Gamma - x^2 - y^2)\,y \end{aligned} \tag{2.28}$$

[4] In complex plane, with $z = x + iy$, Eq. (2.28) takes the form of a Stuart–Landau oscillator, $\dot{z} = i\omega z + d(\Gamma - |z|^2)z$, with $\omega = 1$.

in Euclidean phase space $(x, y) = (r\cos\varphi, r\sin\varphi)$. The standard non-linear rotator (2.2) is recovered when setting $d = 1$. There are two steady-state solutions for $\Gamma > 0$,

$$(x_0^*, y_0^*) = (0, 0), \qquad (x_\Gamma^*, y_\Gamma^*) = \sqrt{\Gamma}\,(\cos(t), \sin(t))\,, \tag{2.29}$$

a fixpoint and a limit cycle. The limit cycle disappears for $\Gamma < 0$.

Super- vs. Sub-critical Hopf Bifurcation For $d > 0$ the bifurcation is "supercritical". The fixpoint $\mathbf{x}_0^* = (x_0^*, y_0^*)$ is stable/unstable for $\Gamma < 0$ and $\Gamma > 0$ and the limit cycle $\mathbf{x}_\Gamma^* = (x_\Gamma^*, y_\Gamma^*)$ is stable, as illustrated in Fig. 2.1.

The direction of flow is reversed for $d < 0$, with the limit cycle $\mathbf{x}_\Gamma^*$ becoming repelling. The fixpoint $\mathbf{x}_0^*$ is then unstable/stable for $\Gamma < 0$ and $\Gamma > 0$ and one speaks of a "subcritical" Hopf bifurcation.

Hopf Bifurcation Theorem One may be interested to find out whether a generic two dimensional system,

$$\dot{x} = f_\mu(x, y), \qquad \dot{y} = g_\mu(x, y)\,, \tag{2.30}$$

can be reduced to the normal form (2.28) for a Hopf bifurcation, where μ is the bifurcation parameter. Without loss of generality one can assume that the fixpoint $\mathbf{x}_0^*$ stays at the origin for all values of μ and that the transition takes place for $\mu = 0$.

To linear order the normal form (2.28) and (2.30) are equivalent if the Jacobian of (2.30) has a pair of complex conjugate eigenvalues, with the real value crossing $\mu = 0$ with a finite slope, which corresponds to a transition from a stable to an unstable focus.

Comparing (2.28) and (2.30) to quadratic order one notices that quadratic terms are absent in the normal form (2.28) but not in (2.30). One can however show, with the help of a suitable non-linear transformation, that it is possible to eliminate all quadratic terms from (2.30). One finds that the nature of the bifurcation is determined by a combination of partial derivatives up to cubic order,

$$\begin{aligned} a = {} & \big[f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy}\big]/16 \\ & + \big[f_{xy}(f_{xx} + f_{yy}) - g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} - f_{yy}g_{yy}\big]/(16\omega) \end{aligned}$$

where $\omega > 0$ is the imaginary part of the Lyapunov exponent at the critical point $\mu = 0$ and where the partial derivatives as f_{xy} are to be evaluated at $\mu = 0$ and $\mathbf{x} \to \mathbf{x}_0^*$. The Hopf bifurcations is supercritical and subcritical respectively for $a < 0$ and $a > 0$.

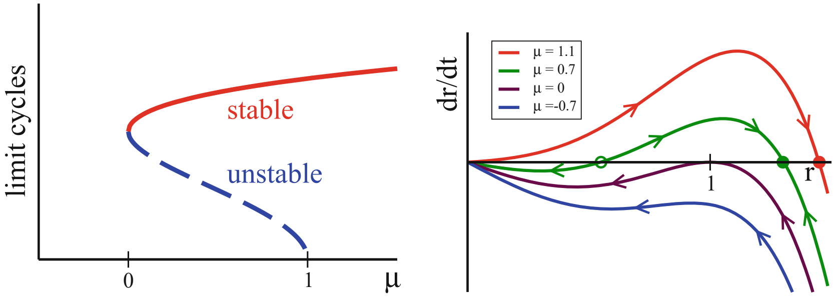

Fig. 2.10 **Left:** For a fold bifurcation of limit cycles, the locations $R_\pm = \sqrt{1 \pm \sqrt{\mu}}$ of the stable and unstable limit cycles, R_- and R_+, as defined by (2.31) and (2.32). At $\mu = 0$ a fold bifurcation of limit cycles occurs and a subcritical Hopf bifurcation at $\mu = 1$. **Right:** The respective flow diagram. The filled/open circles denote stable/unstable limit cycles. For $\mu \to 1$ the unstable limit cycle vanishes, a subcritical Hopf bifurcation. The stable and the unstable limit cycle collided for positive $\mu \to 0$ and annihilate each other, a fold bifurcation of limit cycles

Interplay Between Multiple Limit Cycles A dynamical system may dispose of a number of distinct limit cycles, which can merge or disappear as a function of a given parameter. We consider the simplest case, generalizing the non-linear rotator (2.2) to the next order in r^2,

$$\dot{r} = -(r^2 - \gamma_-)(r^2 - \gamma_+)\, r, \qquad \dot{\varphi} = \omega, \qquad \gamma_- \le \gamma_+ \,. \tag{2.31}$$

Real-world physical or biological systems have bounded trajectories, which implies that $\dot{r}$ must be negative for large radii $r \to \infty$. This requirement has been taken into account in (2.31), the "Bautin normal form".

For $\gamma_- < 0$ the first factor $(r^2 - \gamma_-)$ is smooth for $r^2 \ge 0$ and does not influence the dynamics qualitatively. In this case (2.31) reduces to a standard supercritical Hopf bifurcation, as a function of the bifurcation parameter γ_+.

Phenomenological Parametrization The roots $\gamma_\pm$ of r^2 in (2.31) typically result from some determining relation. As a possible simple assumption we consider the form

$$\gamma_\pm = 1 \pm \sqrt{\mu}\,, \tag{2.32}$$

where μ will be our bifurcation parameter. For $\mu \in [0, 1]$ we have two positive roots and consequently also two limit cycles, a stable and an unstable one. For $\mu \to 1$ the unstable limit cycle vanishes in a subcritical Hopf bifurcation, compare Fig. 2.10.

Fold Bifurcation of Limit Cycles For a saddle-node bifurcation of fixpoints, aka fold bifurcation, a stable and an unstable fixpoint merge and annihilate each other, as illustrated in Fig. 2.7. The equivalent phenomenon occurs for limit cycles, as shown in Fig. 2.10, happening in our model when μ becomes negative,

$$\gamma_\pm = 1 \pm i\sqrt{|\mu|}, \qquad \dot{r} = -\big[(r^2-1)^2 + |\mu|\big]r\,.$$

No limit cycle exists for $\mu \le 0$, only a stable fixpoint at $r_0^* = 0$ remains.

2.3 Global Bifurcations

The bifurcations discussed in Sect. 2.2 are termed "local" as they are based on Taylor expansions around a local fixpoint, which implies that the dynamical state changes smoothly at the bifurcation point. There are, on the other hand, bifurcations characterized by the properties of extended orbits. These kinds of bifurcations are hence of "global" character.

Taken–Bogdanov System We consider a mechanical system with a cubic potential $V(x)$ and velocity-dependent forces,

$$\begin{array}{rlrl} \ddot{x} &= (x-\mu)\dot{x} - V'(x) & \qquad \dot{x} &= y \\ V(x) &= x^3/3 - x^2/2 & \qquad \dot{y} &= (x-\mu)y + x(1-x) \end{array}\,. \tag{2.33}$$

The conservative contribution to the force is $-V'(x) = x(1-x)$, as illustrated in Fig. 2.11. The energy, defined by

$$E = \frac{\dot{x}^2}{2} + V(x), \qquad \frac{dE}{dt} = \big[\ddot{x} + V'(x)\big]\dot{x} = (x-\mu)\dot{x}^2\,, \tag{2.34}$$

is dissipated when $x < \mu$, viz when the term $(x-\mu)\dot{x}$ in (2.33) reduces the velocity. The energy increases however for $x > \mu$, which means that the term $(x-\mu)\dot{x}$ induces accelerated a movement. This interplay between energy dissipation and uptake is typical for adaptive systems.[5]

Fixpoints and Jacobian The Taken–Bogdanov system (2.33) has two fixpoints $(x^*, 0)$, with $x^* = 0, 1$. The eigenvalue of the Jacobian,

$$J = \begin{pmatrix} 0 & 1 \\ (1-2x^*) & (x^*-\mu) \end{pmatrix}, \qquad \begin{array}{rl} \lambda_\pm(0,0) = & -\mu/2 \pm \sqrt{\mu^2/4 + 1} \\ \lambda_\pm(1,0) = & (1-\mu)/2 \pm \sqrt{(1-\mu)^2/4 - 1} \end{array}$$

show that $(0,0)$ is always a saddle, a consequence of the quadratic maximum of mechanical potential $V(x)$ at $x = 0$.

[5] Per definition, adaption implies that a system may both dissipate energy and increase its own reservoir, as discussed further in Sect. 3.2 of Chap. 3.

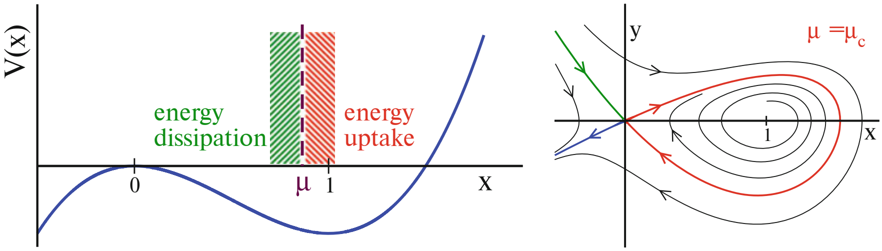

Fig. 2.11 **Left:** The potential $V(x)$ of the Taken–Bogdanov system (2.33). Energy is dissipated to the environment for $x < \mu$ and taken up for $x > \mu$. For $\mu < 1$ the local minimum $x = 1$ of the potential becomes an unstable focus. **Right:** The flow for the critical $\mu = \mu_c \approx 0.8645$. The stable and unstable manifolds form an homoclinic loop (red line)

The local minimum $(1, 0)$ of the potential is a stable/unstable focus for $\mu > 1$ and $\mu < 1$ respectively, with $\mu = 1$ being the locus of a supercritical Hopf bifurcation. We consider now $\mu \in [0, 1]$ and examine the further evolution of the resulting limit cycle.

Escaping the Potential Well We consider a particle starting with a vanishing velocity close to $x = 1$, the local minimum of the Potential well.

When the particle takes up enough energy from the environment, due to the velocity dependent force $(x - \mu)v$, it may be able to reach the local maximum at $x = 0$ and escape to $x \to -\infty$. This is the case when $\mu < \mu_c \approx 0.8645$. All orbits will escape when the region of energy uptake did expand to μ_c.

The particle remains trapped in the local potential well if dissipation dominates, which is the case for $\mu > \mu_c$. The particle is both trapped in the local well and repelled, at the same time, from the unstable minimum at $x = 1$, when $\mu_c < \mu < 1$. The orbit is forced in this case to perform an oscillatory motion around $x = 1$, which is equivalent to a stable limit cycle. This limit cycle increases in size for decreasing μ, exactly touching $(0, 0)$ for $\mu = \mu_c$, breaking apart byond, when $\mu < \mu_c$.

The bifurcation occurring at $\mu = \mu_c$ depends non-locally on the overall energy balance, as illustrated in Fig. 2.12. The "homoclinic transition" taking place in the Taken–Bogdanov system is therefore an example of a global bifurcation.

Homoclinic Bifurcation With a "homocline" one denotes a loop formed by joining a stable and an unstable manifold of the same fixpoint. Homoclines may generically only occur if either forced by symmetries or for special values of bifurcation parameters, with the later being the case for the Taken–Bogdanov system, which undergoes a homoclinic bifurcation.

An unstable and a stable manifold cross at $\mu = \mu_c$, compare Figs. 2.11 and 2.12, forming a homocline. The homocline is also the endpoint of the limit cycle present for $\mu > \mu_c$, which disappears for $\mu < \mu_c$. The limit cycle is hence destroyed at the

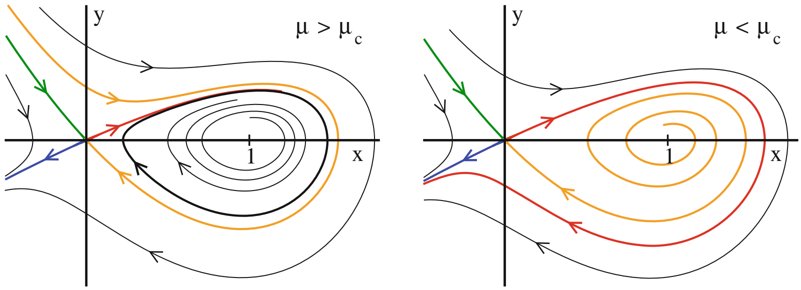

Fig. 2.12 The flow for the Taken–Bogdanov system (2.33). **Left:** The flow in the subcritical region, for $\mu = 0.9 > \mu_c \approx 0.8645$, together with the resulting limit cycle (thick black line). For $\mu \to \mu_c$ the unstable (red) and stable (orange) manifold join to form a homoclinic loop, which is identical to the locus of the limit cycle for $\mu \to \mu_c$. **Right:** The flow in the supercritical region, for $\mu = 0.8 < \mu_c$. The limit cycle did break after touching the saddle at (0, 0)

point of its maximal size for a homoclinic bifurcation, and not when vanishing, as for a supercritical Hopf bifurcation.

2.3.1 Infinite Period Bifurcation

For a further example of how a limit cycle may disappear discontinuously we consider two coupled oscillators characterized by their individual phases, respectively θ_1 and θ_2.[6] A typical evolution equation for the phase difference $\varphi = \theta_1 - \theta_2$ is

$$\dot{\varphi} = 1 - K\cos(\varphi)\,. \tag{2.35}$$

We can interpret (2.35) via

$$\dot{r} = \left(1 - r^2\right) r, \qquad x = r\cos(\varphi), \qquad y = r\sin(\varphi) \tag{2.36}$$

within the context of a two dimensional limit cycle, compare (2.2).

The phase difference φ continuously increases for $|K| < 1$, with the system settling into a limit cycle. For $|K| > 1$ two fixpoints for $\dot{\varphi} = 0$ appear, a saddle and a stable node, as illustrated in Fig. 2.13. At this point, which correspond to a saddle-node bifurcation on an invariant cycle, the limit cycle breaks up.

[6] This formulation parallels the Kuramoto model, which is treated detail in Sect. 9.1 of Chap. 9.

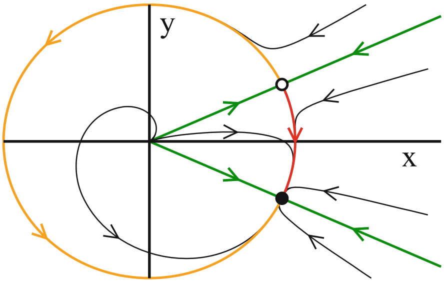

Fig. 2.13 The flow for a system showing an infinite period bifurcation, as defined by (2.35) and (2.36). Here $K = 1.1$. The stable fixpoint (filled circle) merges with the saddle (open circle) in the limit $K \to 1$, which leads to a saddle-node bifurcation on an invariant cycle

Infinite Period Bifurcation The limit cycle for $|K| < 1$ has a revolution period T of

$$T = \int_0^T dt = \int_0^{2\pi} \frac{dt}{d\varphi} d\varphi = \int_0^{2\pi} \frac{d\varphi}{\dot{\varphi}} = \int_0^{2\pi} \frac{d\varphi}{1 - K\cos(\varphi)} = \frac{2\pi}{\sqrt{1-K^2}} ,$$

which diverges in the limit $|K| \to 1$. The transition occurring at $|K| = 1$ is hence a "infinite period bifurcation", being characterized by a diverging time scale.

2.3.2 Catastrophe Theory

A catchy terminology for potentially discontinuous bifurcations in dynamical systems is "catastrophe theory", especially when placing emphasis on a geometric interpretation. Catastrophe theory is interested in bifurcations with "codimension" two or higher.

CODIMENSION The degrees of freedom characterizing a bifurcation diagram.

The codimension corresponds, colloquially speaking, to the number of parameters one may vary such that something interesting happens. The bifurcation normal forms discussed in Sect. 2.2.2 have a codimension of one.

Symmetry Broken Pitchfork Bifurcation We consider the one-dimensional system

$$\dot{x} = h + ax - x^3 . \tag{2.37}$$

For $h = 0$ the system reduces to the pitchfork normal form of (2.24), becoming invariant under the parity transformation $x \leftrightarrow -x$.

Parity is broken whenever $h \neq 0$. Equation (2.37) can hence be used to study the influence of symmetry breaking onto a bifurcation diagram. There are two free parameters, h and a, the codimension is two.

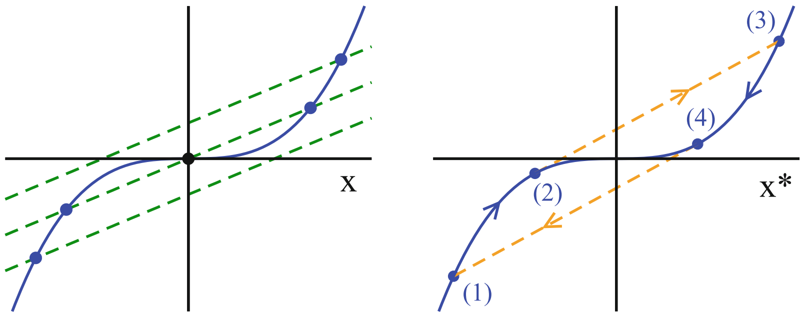

Fig. 2.14 **Left:** The self-consistency condition $x^3 = h + ax$ for the fixpoints x^* of the symmetry broken pitchfork system (2.37), for various fields h and a positive $a > 0$. The unstable fixpoint at $x = 0$ would becomes stable for $a < 0$, compare Fig. 2.9. **Right:** The hysteresis loop (1) $\rightarrow$ (2) $\rightarrow$ (3) $\rightarrow$ (4) $\rightarrow$ (1) occurring for $a > 0$ as function of the field h

Thermodynamic Phases The generalized pitchfork system (2.37) has a close relation to the theory of thermodynamic phase transitions, when assuming that

$$a = a(T) = a_0(T_c - T), \qquad a_0 > 0\,,$$

where T is the temperature. In the absence of an external field h, only the trivial fixpoint $x^* = 0$ exists for $T > T_c$, viz for temperatures above the critical temperature T_c. In the ordered state, for $T < T_c$, there are two possible phases, characterized by the positive and negative stable fixpoints x^*.

Hysteresis and Memory The behavior of the phase transition changes when an external field h is present. Switching the sign of the field is accompanied, in the ordered state for $T < T_c$, by a "hysteresis-loop"

$$(1) \rightarrow (2) \rightarrow (3) \rightarrow (4) \rightarrow \ldots,$$

as illustrated in Fig. 2.14.

- The field h changes from negative to positive values along (1) $\rightarrow$ (2), with the fixpoint x^* remaining negative.
- At (2) the negative stable fixpoint disappears and the system makes a rapid transition to (3), the catastrophe.
- Lowering eventually again the field h, the system moves to (4), jumping in the end back to (1).

The system retains its state, x^* being positive or negative, to a certain extend and one speaks of a memory in the context of catastrophe theory.

Center Manifold A d-dimensional dynamical system with a fixpoint $\mathbf{x}^*$ and a Jacobian J,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \qquad J_{ij} = \frac{\partial f_i}{\partial x_j}\Big|_{\mathbf{x}=\mathbf{x}^*}, \qquad J\mathbf{e}_n = \lambda_n \mathbf{e}_n ,$$

may have a number of neutral Lyapunov exponents with vanishing eigenvalues $\lambda_i = 0$ of the Jacobian.

> CENTER MANIFOLD The space spanned by the neutral eigenvectors $\mathbf{e_i}$ is the center manifold.

Catastrophe theory deals with fixpoints having a non-vanishing center manifold.

Center Manifold of the Pitchfork System The Lyapunov exponent

$$\lambda = a - 3\left(x^*\right)^2$$

of the pitchfork system (2.37) vanishes at the jump-points (2) and (4) at which the catastrophe occurs, compare Fig. 2.14. At the jump-points $h + ax$ is per definition tangent to x^3, having identical slopes,

$$\frac{d}{dx}x^3 = \frac{d}{dx}\big(h + ax\big), \qquad 3x^2 = a . \tag{2.38}$$

At these transition points the autonomous flow becomes hence infinitesimally slow, since $\lambda \to 0$, a phenomenon called "critical slowing down" in the context of the theory of thermodynamic phase transitions.

Center Manifold Normal Forms The program of the catastrophe theory consists of finding and classifying the normal forms for the center manifolds of stationary points $\mathbf{x}^*$, by expanding to the next, non-vanishing order in $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}^*$. The aim is hence to classify the types of dynamical behavior potentially leading to discontinuous transitions.

Catastrophic Fixpoints A generic fixpoint $\mathbf{x}^* = \mathbf{x}^*(\mathbf{c})$ may depend on control parameters $\mathbf{c} = (c_i, .., c_S)$ of the equations of motion, with S being the codimension. The flow is however smooth around a generic fixpoint and a finite center manifold arises only for certain sets $\{c_i^c\}$ of control parameters.

The controlling parameters of the pitchfork system (2.37) are h and a, in our example system, and a center manifold exists only when (2.38) is fulfilled, viz when

$$3\big[x^*(h^c, a^c)\big]^2 = a^c, \qquad x^*(h^c, a^c) = x_c^*$$

holds, which determines the set of "catastrophic fixpoints" x_c^*.

Changing the Controlling Parameters How does the location $\mathbf{x}^*$ of a fixpoint change upon variations $\delta\mathbf{c}$ around the set of parameters $\mathbf{c}^c$ determining the catastrophic fixpoint $\mathbf{x}_c^*$? With

$$\mathbf{x}_c^* = \mathbf{x}^*(\mathbf{c}^c), \qquad \mathbf{x}^* = \mathbf{x}^*(\mathbf{c}), \qquad \delta\mathbf{c} = \mathbf{c} - \mathbf{c}^c$$

we expand the fixpoint condition $\mathbf{f}(\mathbf{x}, \mathbf{c}) = 0$ and obtain

$$J\delta\mathbf{x}^* + P\delta\mathbf{c} = 0, \qquad J_{ij} = \frac{\partial f_i}{\partial x_j}, \qquad P_{ij} = \frac{\partial f_i}{\partial c_j}, \tag{2.39}$$

which we can invert if the Jacobian J is non-singular,

$$\delta\mathbf{x}^* = -J^{-1}P\delta\mathbf{c}, \qquad \text{if} \qquad |J| \neq 0\,. \tag{2.40}$$

The fixpoint may change however in a discontinuous fashion whenever the determinant $|J|$ of the Jacobian vanishes, viz in the presence of a center manifold. This is precisely what happens at a catastrophic fixpoint $\mathbf{x}_c^*(\mathbf{c}^c)$.

Catastrophic Manifold The set $\mathbf{x}_c^* = \mathbf{x}_c^*(\mathbf{c}^c)$ of catastrophic fixpoints is determined by two conditions, by the fixpoint condition $\mathbf{f} = 0$ and by $|J| = 0$. For the pitchfork system (2.37) we find,

$$a_c = 3\left(x_c^*\right)^2, \qquad h_c = \left(x_c^*\right)^3 - a_c x_c^* = -2\left(x_c^*\right)^3,$$

when using (2.38). Solving for $x_c^* = \sqrt{a_c/3}$ we can eliminate x_c^* and obtain

$$h_c = -2\,(a_c/3)^{3/2}, \qquad (h_c/2)^2 = (a_c/3)^3, \tag{2.41}$$

which determines the "catastrophic manifold" (a_c, h_c) for the pitchfork transition, as illustrated in Fig. 2.15.

Classification of Perturbations The control parameters (a_c, h_c) of the pitchfork transition may be changed in two qualitatively distinct ways, namely along the catastrophic manifold (2.41), or perpendicular to it.

It would hence be useful to dispose of a list of canonical perturbations characterizing all possible distinct routes to change catastrophic fixpoints $x_c^* = x^*(\mathbf{c}^c)$ qualitatively, upon small changes $\delta\mathbf{c} = \mathbf{c} - \mathbf{c}^c$ of the control parameters $\mathbf{c}$. It is the aim of catastrophe theory to develop such a canonical classification scheme for the perturbations of center manifolds.

Gradient Dynamical Systems At times the flow $\mathbf{f}(\mathbf{x})$ of a dynamical system may be represented as a gradient of a bifurcation potential $U(\mathbf{x})$,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) = -\nabla U(\mathbf{x})\,. \tag{2.42}$$

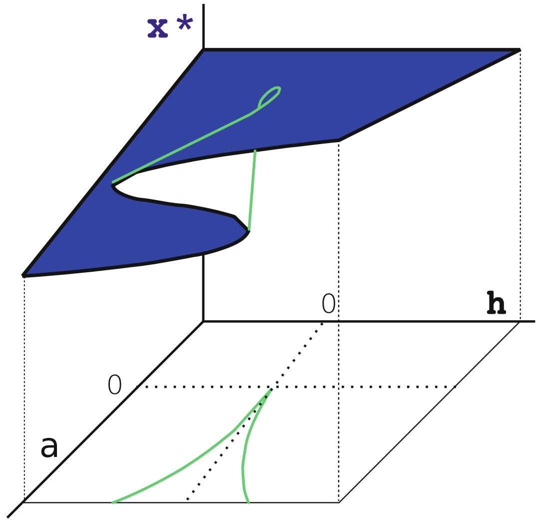

Fig. 2.15 The fixpoints x^* (upper folded plane) of the symmetry broken pitchfork system (2.37), as a function of the control parameters a and h. The catastrophic manifold (a_c, h_c), compare (2.41), has a cusp-like form (green lines)

This is generically the case for one-dimensional systems, as discussed in Sect. 2.2.2, but otherwise not. For the gradient representation

$$\begin{aligned} \dot{x} &= g(x, y) \qquad & g &= -U_x(x, y) \\ \dot{y} &= h(x, y) \qquad & h &= -U_y(x, y) \end{aligned}$$

of a two-dimensional system to be valid, to give an example, the cross-derivatives $g_y = -U_{xy} = -U_{yx} = h_x$ would need to coincide. This is however normally not the case.

For gradient dynamical systems one needs to discuss only the properties of the bifurcation potential $U(\mathbf{x})$, as scalar quantity, and they are hence somewhat easier to investigate than a generic dynamical system of the form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. Catastrophe theory is mostly limited to gradient systems.

2.3.3 Rate Induced Tipping

For the bifurcation scenarios studied hitherto, we took as a precondition that controlling parameters changed slowly, on time scales substantially exceeding that of the primary dynamics. This is the “adiabatic limit”.

Bifurcation vs. Tipping A rapid transfer between distinct stable manifolds may occur when systems reach a bifurcation point upon adiabatic parameter changes, as illustrated in Fig. 2.14. Systems undergoing self-accelerating instabilities are said “to tip”, which is a somewhat broader terminology. Instead of a description in terms

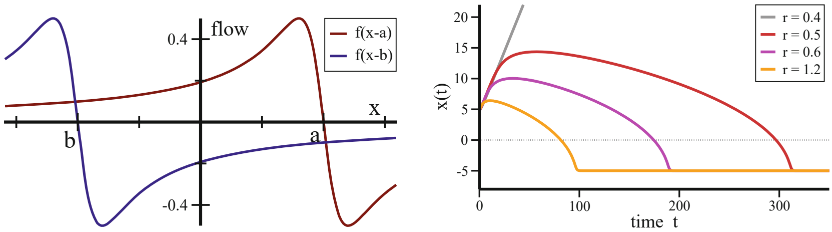

Fig. 2.16 Rate-induced tipping. **Left:** The fixpoints $x^* = a/b$ are stable for the individual flows (2.43), here for $a = 5$ and $b = -5$. The stationary points shift to $\pm\sqrt{5^2 - 1} \approx \pm 4.9$ when the two flows are superimposed, as given by (2.45). **Right:** The orbits $x(t)$ under the influence of a forcing $\dot{a} = r$. For small rates the system manages to adapt, as for $r = 0.4$, for larger rates not. For $r = 0.5/0.6/1.2$ the system tips towards the alternative steady state

of an abstract bifurcation, 'tipping' emphasizes the role of positive feedback effects. This view is particularly important in climatology and ecology.

Rate Induced Tipping Individual components of a dynamical system may evolve on vastly different time scales. In climatology, a widely discussed slow component is the Greenland iceshield, which takes millenia to grow or melt. In ecology, adaption to changing environmental condition via Darwinian evolution is likewise a slow process, extending over geological periods.[7] Systems with slow components may show a reduced resilence against shocks in terms of rapid parameter changes, as typical for climatic and ecological systems facing anthropogenic forcing. When the rate matters, at which control parameters change, "rated induced tipping" may occur.

Competing Fixpoints Close to an attracting state the flow is linear. At larger distances z, the numerical size of the flow $f = f(z)$ may however fade away, f.i. as described by

$$\dot{x} = f(x - a), \qquad f(z) = \frac{-z}{1 + z^2}\,. \tag{2.43}$$

In this example the flow decays inversely with the distance to the fixpoint $x^* = a$, as illustrated in Fig. 2.16, after being maximal in magnitude at $x = a \pm 1$. For two attracting states the dynamics takes the form

$$\dot{x} = f(x - a) + f(x - b)\,, \tag{2.44}$$

[7] See Chap. 8.

which makes the system bistable. The two flows compete with each other, which shifts the two stable fixpoints,

$$(x-a)\big[1+(x-b)^2\big] = -(x-b)\big[1+(x-a)^2\big]\,,$$

as determined by $\dot{x} = 0$, or

$$(x-a)(x-b) = -1, \qquad\qquad x = \pm\sqrt{a^2-1}\,, \tag{2.45}$$

where we specialized in the last step to $b = -a$. The two stable fixpoints are seperated by an unstable fixpoint at $x = 0$. Equation (2.44) describes a situation in which the influence of attracting states decays in phase space, which is typical for dynamical systems composed of interacting real-world identities. An example would be physical objects, whose sphere of influence is often limited to the immediate surrounding.

Adaption Catastrophy In (2.44) the dynamical variable $x = x(t)$ reflects the state of the system. In the following we take $x \approx a$ to correspond to a healthy system, say a pristine ecosystem, and states $x \approx b$ to a disrupted system. External impacts may change the control parameter $a = a(t)$, e.g. with a constant rate r,

$$\dot{a} = r\,.$$

The orbit cannot follow when a moves to the right too fast, which happens when $\dot{x} < \dot{a} = r$. Given that $f(x-a)$ is bounded, this necessarily happens with increasing r. At this point the system fails to adapt to the changing environment, with the orbit reverting to a disrupted state close to b. Rate induced tipping occurs in the form of an adaption catastrophy. Extended transients are observed close to the critical rate r, as shown in Fig. 2.16.

2.4 Logistic Map and Deterministic Chaos

The notion of "chaos" plays an important role in dynamical systems theory. A chaotic system is defined as a system that cannot be predicted within a given numerical accuracy. At first sight this seems to be a surprising concept, since differential equations of the type of (2.15), which do not contain any noise or randomness, are perfectly deterministic. Once the starting point is known, the resulting trajectory can be calculated for all times. Chaotic behavior can arise nevertheless, due to an exponential sensitivity to initial conditions.

> DETERMINISTIC CHAOS A deterministic dynamical system is chaotic when it shows exponential sensibility with respect to initial conditions.

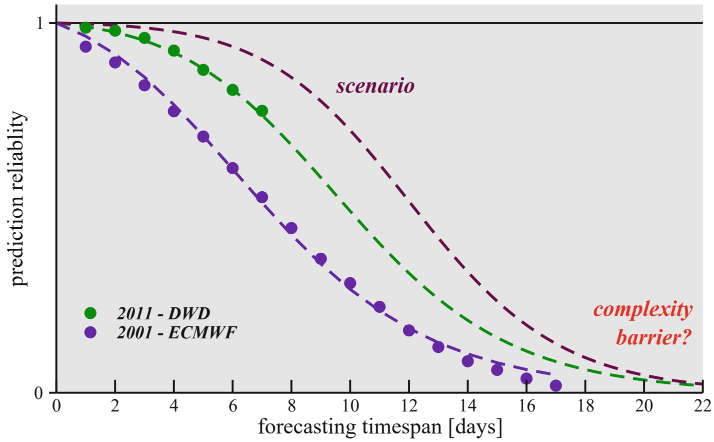

Fig. 2.17 The average accuracy of weather forecasting, normalized to [0, 1], decreases rapidly with increasing prediction timespan, due to the chaotic nature of weather dynamics. Increasing the resources devoted for improving the prediction accuracy results in decreasing returns close to the resulting complexity barrier. Reprinted from Gros (2012) under CC-BY-4.0 license, © 2012 Complex Systems Publications, Inc.

This means that a very small change in the initial setting can blow up even after a short time. When considering real-world applications, when models need to be determined from measurements containing inherent errors and limited accuracies, an exponential sensitivity can result in unpredictability. A well known example is the problem of long-term weather prediction, as shown in Fig. 2.17.

Logistic Map One of the most cherished models in the field of deterministic chaos is the logistic map of the interval [0, 1] onto itself,

$$x_{n+1} = g(x_n) \equiv r\, x_n\, (1 - x_n), \qquad x_n \in [0, 1], \qquad r \in [0, 4]\,, \tag{2.46}$$

where we made use of the notation $x_n = x(n)$, for discrete times $n = 0,\ 1,\ \ldots$. The functional dependence is illustrated in Fig. 2.18. Despite its apparent simplicity, the logistic map shows an infinite series of bifurcations that culminate in a transition to chaos.

Biological Interpretation We may consider $x_n \in [0, 1]$ as standing for the population density of a reproducing species in the year n. In this case the factor $r(1 - x_n) \in [0, 4]$ corresponds to the number of offspring per year and animal, which is limited for high population densities $x \to 1$, viz when resources become scarce. The classical example is that of a herd of reindeer on an island.

Knowing the population density x_n in a given year n we may predict via (2.46) the population density for all subsequent years exactly; the system is deterministic.

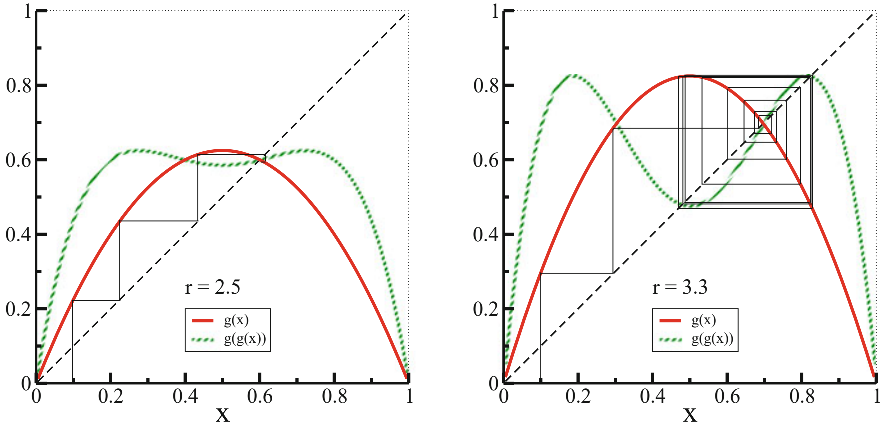

Fig. 2.18 The logistic map $g(x) = rx(1-x)$ (red), and the iterated logistic map $g(g(x))$ (green); for $r = 2.5$ (left) and $r = 3.3$ (right). Also shown are iterations of $g(x)$, starting from $x = 0.1$ (thin solid line). Note, that the fixpoint $g(x) = x$ is stable/unstable for $r = 2.5$ and $r = 3.3$, respectively. The orbit is attracted to a fixpoint of $g(g(x))$ for $r = 3.3$, corresponding to a cycle of period two for $g(x)$

Nevertheless the population shows irregular behavior for certain values of r, which is hence chaotic.

Fixpoints of the Logistic Map We start with the fixpoints of $g(x)$,

$$x = rx(1-x) \qquad \Longleftrightarrow \qquad x = 0 \quad \text{or} \quad 1 = r(1-x)\,.$$

The non-trivial fixpoint is

$$1/r = 1 - x, \qquad x^{(1)} = 1 - 1/r, \qquad r_1 < r, \qquad r_1 = 1\,. \tag{2.47}$$

It is present for $r_1 < r$, with $r_1 = 1$, due to the restriction $x^{(1)} \in [0, 1]$.

Stability Analysis For maps, fixpoints are stable when the derivative g' of the flow is smaller than one in magnitude, see (2.14). For $x^{(1)} = 1 - 1/r$ this translates to

$$g'(x) = r(1-2x), \qquad g'\left(x^{(1)}\right) = 2 - r\,. \tag{2.48}$$

Noting that $r \in [1, 4]$, we obtain

$$|2-r| < 1 \qquad \Longleftrightarrow \qquad r_1 < r < r_2 \qquad \text{with} \quad \begin{cases} r_1 = 1 \\ r_2 = 3 \end{cases} \tag{2.49}$$

for the region of stability of $x^{(1)}$.

Fixpoints of Period Two For $r > 3$ a fixpoint of period two appears. This fixpoint is a fixpoint of the iterated function

$$g(g(x)) = rg(x)(1 - g(x)) = r^2x(1 - x)(1 - rx(1 - x)) .$$

The fixpoint equation $x = g(g(x))$ leads to the cubic equation

$$\begin{aligned} 1 &= r^2(1 - rx + rx^2) - r^2x(1 - rx + rx^2), \\ 0 &= r^3x^3 - 2r^3x^2 + (r^3 + r^2)x + 1 - r^2 . \end{aligned} \tag{2.50}$$

In order to find the roots of (2.50) we use the fact that $x = x^{(1)} = 1 - 1/r$ is a stationary point of both $g(x)$ and $g(g(x))$, see Fig. 2.18. Dividing (2.50) by the root $(x - x^{(1)}) = (x - 1 + 1/r)$ one obtains

$$\begin{gathered} (r^3x^3 - 2r^3x^2 + (r^3 + r^2)x + 1 - r^2) : (x - 1 + 1/r) \ = \\ r^3x^2 - (r^3 + r^2)x + (r^2 + r) . \end{gathered}$$

The two fixpoints of $g(g(x))$ are therefore the roots of

$$x^2 - \left(1 + \frac{1}{r}\right)x + \left(\frac{1}{r} + \frac{1}{r^2}\right) = 0 ,$$

which leads to

$$x_{\pm}^{(2)} = \frac{1}{2}\left(1 + \frac{1}{r}\right) \pm \sqrt{\frac{1}{4}\left(1 + \frac{1}{r}\right)^2 - \left(\frac{1}{r} + \frac{1}{r^2}\right)} . \tag{2.51}$$

Period Doubling Bifurcation We have three fixpoints of period two for $r > 3$ (two stable ones and one unstable), and only a single fixpoint for $r < 3$. What happens at $r = 3$?

$$\begin{aligned} x_{\pm}^{(2)}(r = 3) &= \frac{1}{2}\frac{3 + 1}{3} \pm \sqrt{\frac{1}{4}\left(\frac{3 + 1}{3}\right)^2 - \left(\frac{3 + 1}{9}\right)} \\ &= \frac{2}{3} = 1 - \frac{1}{3} = x^{(1)}(r = 3) . \end{aligned}$$

At $r = 3$ the fixpoint splits into two stable and one unstable branch, see Fig. 2.19, akin to the pitchfork bifurcations discussed in Sect. 2.2. Given that a period doubling occurs at $r = 3$, the transition can be seen alternatively as the discrete analog of a supercritical Hopf bifurcation.

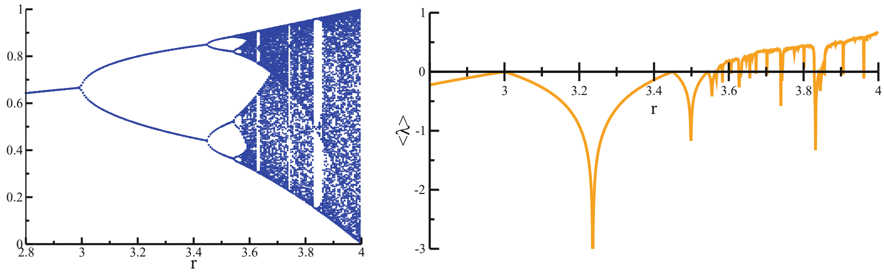

Fig. 2.19 **Left:** The values x_n for the iterated logistic map (2.46). For $r < r_\infty \approx 3.57$ the x_n go through cycles with finite but progressively longer periods. For $r > r_\infty$ the plot would be fully covered in most regions if all x_n would be shown. **Right:** The corresponding maximal Lyapunov exponents, as defined by (2.55). Positive Lyapunov exponents λ indicate chaotic behavior

Bifurcation Cascade One may carry out a stability analysis for $x_\pm^{(2)}$, just as for $x^{(1)}$. One finds a critical value $r_3 > r_2$ such that

$$x_\pm^{(2)}(r)\ \text{stable} \qquad \Longleftrightarrow \qquad r_2 < r < r_3\,. \tag{2.52}$$

Going further on one finds an r_4 such that there are four fixpoints of period four, that is of $g(g(g(g(x))))$, for $r_3 < r < r_4$. In general there are critical values r_n and r_{n+1} such that there are

$$2^{n-1}\ \text{fixpoints } x^{(n)} \text{ of period } 2^{n-1} \qquad \Longleftrightarrow \qquad r_n < r < r_{n+1}\,.$$

The logistic map therefore shows iterated bifurcations. This, however, is not yet chaotic behavior.

Chaos in the Logistic Map There exists a critical r_∞ at which the period-doubling series converges,

$$\lim_{n\to\infty} r_n \;\to\; r_\infty, \qquad\qquad r_\infty = 3.5699456\ldots$$

No stable fixpoint of even period exists in the region

$$r_\infty \;<\; r \;<\; 4\,.$$

In order to characterize the sensitivity of (2.46) with respect to the initial condition, we consider two slightly different starting points x_1 and x_1':

$$x_1 - x_1' = y_1, \qquad\qquad |y_1| \;\ll\; 1\,.$$

The key question is then whether the difference

$$y_m = x_m - x'_m, \qquad y_{m+1} \approx \left.\frac{dg(x)}{dx}\right|_{x=x_m} y_m \tag{2.53}$$

between the two respective orbits is still small after m iterations. For (2.53) we used $x'_m = x_m - y_m$, neglecting terms $\sim y_m^2$. We hence obtain

$$y_{m+1} = r(1 - 2x_m)\, y_m \equiv \epsilon\, y_m\,.$$

For $|\epsilon| < 1$ the map is stable, as two initially different populations close in with time passing. For $|\epsilon| > 1$ they diverge; the map is chaotic.

Lyapunov Exponents We remind ourselves of the definition

$$|\epsilon| = \mathrm{e}^{\lambda}, \qquad \lambda = \log\left|\frac{dg(x)}{dx}\right| \tag{2.54}$$

the Lyapunov exponent $\lambda = \lambda(r)$ of a map, as introduced in Sect. 2.2. For positive Lyapunov exponents the time development is exponentially sensitive to the initial conditions and shows chaotic features,

$$\lambda < 0 \;\Leftrightarrow\; \text{stability}, \qquad \lambda > 0 \;\Leftrightarrow\; \text{instability}\,.$$

This is indeed observed in nature, e.g. for populations of reindeer on isolated islands, as well as for the logistic map for $r_\infty < r < 4$, compare Fig. 2.19.

Maximal Lyapunov Exponent The Lyapunov exponent, as defined by (2.54), provides a description of the short time behavior. For a corresponding characterization of the long time dynamics one defines the "maximal Lyapunov exponent"

$$\lambda^{(\max)} = \lim_{n\gg 1} \frac{1}{n} \log\left|\frac{dg^{(n)}(x)}{dx}\right|, \qquad g^{(n)}(x) = g(g^{(n-1)}(x))\,. \tag{2.55}$$

Using (2.54) for the short time evolution we can decompose $\lambda^{(\max)}$ into an averaged sum of short time Lyapunov exponents. $\lambda^{(\max)}$ is the "global Lyapunov exponent".

One needs to select advisedly the number of iterations n in (2.55). On one side n should be large enough such that short-term fluctuations of the Lyapunov exponent are averaged out. The available phase space is however generically finite, for the logistic map we have $y \in [0, 1]$. Therefore, two initially close orbits cannot diverge ad infinitum. One needs hence to avoid phase-space restrictions, evaluating $\lambda^{(\max)}$ for large but finite numbers of iterations n.

Routes to Chaos The chaotic regime $r_\infty < r < 4$ of the logistic map connects to the regular regime $0 < r < r_\infty$ with increasing period doubling. One speaks of a "route to chaos via period-doubling". The study of chaotic systems is a wide field of research and a series of routes leading from regular to chaotic behavior have been found. Two important alternative routes to chaos are:

– INTERMITTENCY ROUTE TO CHAOS
  The trajectories are almost periodic and interdispersed with regimes of irregular behaviour. The occurrence of these irregular bursts increases until the system becomes irregular.
– RUELLE–TAKENS–NEWHOUSE ROUTE TO CHAOS
  A strange attractor appears in a dissipative system after two (Hopf) bifurcations. As a function of an external parameter a fixpoint evolves into a limit cycle (Hopf bifurcation), which then turns into a limiting torus, which subsequently turns into a strange attractor.

2.4.1 Colliding Attractors

Bifurcations are generically the result of colliding invariant manifolds, viz of stable and unstable fixpoints, limit cycles and chaotic sets. Examples are the saddle-node bifurcation (2.19), at which a stable and an unstable fixpoint merge, and the collision between a limit cycle and a saddle within the Taken–Bogdanov system (2.33).

Odd Logistic Map The collision between two chaotic attractors can be studied within the odd logistic map

$$x_{n+1} = g(x_n) = x_n\big(\lambda - x_n^2\big), \qquad \lambda > 0\,, \tag{2.56}$$

which is invariant under inversion $x \leftrightarrow (-x)$. The local maximum $g_m = g(x_m)$ is given by

$$g' = 0, \qquad x_m = \sqrt{\frac{\lambda}{3}}, \qquad g_m = 2\sqrt{\frac{\lambda^3}{3^3}}\,, \tag{2.57}$$

as illustrated in Fig. 2.20.

Merging Chaotic Attractors For $\lambda < 1$ only the trivial fixpoint exists. Afterwards, for $1 < \lambda < \lambda_s$, there are two attracting states that are related via inversion symmetry, as shown in Fig. 2.20. Both branches undergo a period-double transition to chaos, in equivalence to the one observed for the standard logistic map, compare

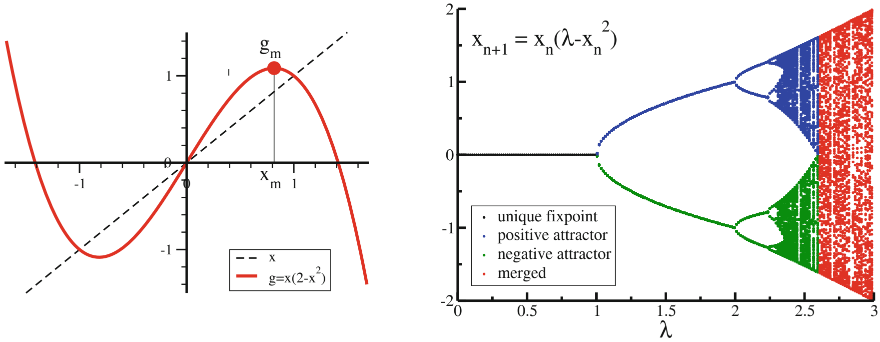

Fig. 2.20 **Left:** For $\lambda = 2$, the odd logistic map (2.56). The local maximum g_m is reached for $x_m = \sqrt{\lambda/3} \approx 0.82$. **Right:** The bifurcation diagram of the odd logistic map as a function of λ. One has a unique fixpoint $x_n = 0$ (black) for $\lambda < 1$ and two equivalent attractors (blue/green) for $1 < \lambda < \lambda_s$, which merge at $\lambda_s = \sqrt{27}/2 \approx 2.6$. Inversion symmetry is restored at λ_s (red). A period-doubling transition to chaos is observable in the symmetry broken phase

Fig. 2.19. At λ_s, determined by

$$g(g_m) = 0, \qquad \lambda_s = 4\,\frac{\lambda_s^3}{3^3}, \qquad \lambda_s = \frac{\sqrt{27}}{2}\,, \tag{2.58}$$

the two attractors merge. Only a single chaotic attractor remains for $\lambda > \lambda_s$.

Chaotic Attractors in Crisis In general, different types of attractors may collide. A Hopf bifurcation, f.i., occurs when limit cycles contract to a point, which is equivalent to the merger of limit cycles with fixpoints. One speaks of a "crisis" when chaotic attractors collide with unstable fixpoints or limit cycles. For the logistic map (2.46), a crisis occurs at $r = 4$, namely when the chaotic attractor hits the unstable fixpoint $x = 0$. Orbits diverge for $r > 4$.

2.5 Dynamical Systems with Time Delays

The dynamical systems we considered so far all had instantaneous dynamics, being of the type

$$\frac{d}{dt}y(t) = f(y(t)), \qquad t > 0 \tag{2.59}$$
$$y(t = 0) = y_0\,,$$

when denoting with y_0 the initial condition. This is the simplest case: one dimensional (a single dynamical variable only), autonomous ($f(y)$ is not an explicit function of time) and deterministic (no noise).

Time Delays In many real-world applications the couplings between different subsystems and dynamical variables is not instantaneous. Signals and physical interactions need a certain time to travel from one subsystem to the next. Time delays are therefore encountered commonly and become important when the delay time T is comparable to the intrinsic time scales of the dynamical system. We consider here the simplest case, a noise-free one-dimensional dynamical system with a single delay time,

$$\frac{d}{dt}y(t) = f(y(t), y(t-T)), \qquad t > 0 \tag{2.60}$$

$$y(t) = \phi(t), \qquad t \in [-T, 0]\,.$$

Due to the delayed coupling we need now to specify an entire initial function $\phi(t)$. Differential equations containing one or more time delays need to be considered carefully, with the time delay introducing additional dimensions to the problem. We discuss a several basic examples.

Linear Couplings We start with the linear delay equation

$$\frac{d}{dt}y(t) = -ay(t) - by(t-T), \qquad a, b > 0\,. \tag{2.61}$$

The only constant solution for $a + b \neq 0$ is the trivial state $y(t) \equiv 0$. The trivial solution is stable in the absence of time delays, $T = 0$, whenever $a + b > 0$. The question is, whether a finite T may change this.

We may expect the existence of a certain critical T_c, such that $y(t) \equiv 0$ remains stable for small time delays $0 \leq T < T_c$. In this case the initial function $\phi(t)$ will affect the orbit only transiently, in the long run the motion would be damped out, approaching the trivial state asymptotically for $t \to \infty$.

Delay Induced Hopf Bifurcation Trying our luck with the usual exponential ansatz, we find

$$\lambda = -a - b\mathrm{e}^{-\lambda T}, \qquad y(t) = y_0\mathrm{e}^{\lambda t}, \qquad \lambda = p + iq\,. \tag{2.62}$$

Separating into a real and an imaginary part we obtain

$$\begin{aligned} p + a &= -b\mathrm{e}^{-pT}\cos(qT), \\ q &= \quad b\mathrm{e}^{-pT}\sin(qT)\,. \end{aligned} \tag{2.63}$$

For $T = 0$ the solution is $p = -(a+b)$, $q = 0$, as expected, and the trivial solution $y(t) \equiv 0$ is stable. A numerical solution is shown in Fig. 2.21 for $a = 0.1$ and $b = 1$.

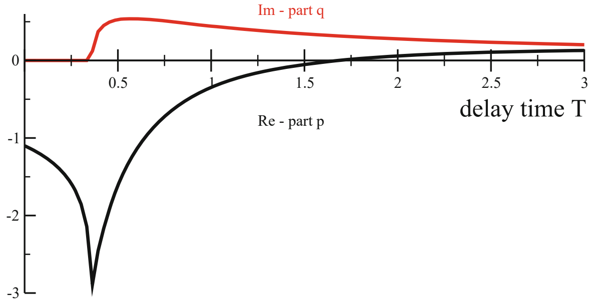

Fig. 2.21 For $a = 0.1$ and $b = 1$, the components p and q of the solution $e^{(p+iq)t}$ of the delay system given by (2.61). The state $y(t) \equiv 0$ becomes unstable whenever the real part p turns positive. The imaginary part q is given in units of π

The crossing point $p = 0$ is determined by

$$a = -b\cos(qT), \qquad q = b\sin(qT). \tag{2.64}$$

The first condition in (2.64) can be satisfied only for $a < b$. Taking the squares in (2.64) and eliminating qT one has

$$q = \sqrt{b^2 - a^2}, \qquad T \equiv T_c = \arccos(-a/b)/q.$$

One therefore finds a Hopf bifurcation at $T = T_c$, which implies that the trivial solution becomes unstable for $T > T_c$. For $a = 0$ the transition point is defined by $q = b$, together with $T_c = \pi/(2b)$. Note, that there is a Hopf bifurcation only for $a < b$, viz whenever the time delay dominates, and that q becomes non-zero well before the bifurcation point, compare Fig. 2.21. One has therefore a region of damped oscillatory behavior with $q \neq 0$ and $p < 0$.

Discontinuities For differential equations with time delays one may specify arbitrary initial functions $\phi(t)$, also initial functions with discontinuities, which will induce discontinuities in the derivatives of the respective trajectories. As an example we consider the case $a = 0$, $b = 1$ of (2.61), with a non-zero constant initial function,

$$\frac{d}{dt}y(t) = -y(t-T), \qquad \phi(t) \equiv 1. \tag{2.65}$$

The solution can be evaluated by stepwise integration,

$$y(t) - y(0) = \int_0^t dt'\dot{y}(t') = -\int_0^t dt' y(t'-T) = -\int_0^t dt' = -t, \;\; 0 < t < T.$$

The first derivative in consequently discontinuous at $t = 0$,

$$\lim_{t\to 0-}\frac{d}{dt}y(t) = 0, \qquad \lim_{t\to 0+}\frac{d}{dt}y(t) = -1.$$

In analogy one show that the second derivative has a discontinuity at $t = T$, the third derivative at $t = 2T$, and so on.

Injectivity Ordinary differential equations are injective in the sense that distinct initial conditions lead to distinct trajectories. This holds regardless of the presence of attracting manifolds, which determine solely the long-term behavior.

Delay systems are not necessarily injective with respect to the initial function. Consider logistic growth with a delayed growth rate,

$$\frac{d}{dt}y(t) = y(t-T)\big(y(t)-1\big), \qquad \phi(t=0) = 1\,. \tag{2.66}$$

For any $\phi(t)$ with $\phi(0) = 1$ the solution is $y(t) \equiv 1$ for all $t \in [0, \infty]$. Distinct initial functions lead to identical orbits.

Non-constant Time Delays Things may become rather weird when the time delays are not constant, as for

$$\begin{aligned} &\tfrac{d}{dt}y(t) = y\big(t - T(y)\big) + \tfrac{1}{2}, \qquad T(y) = 1 + |y(t)|, \\ &\phi(t) = \begin{cases} 1 & t < -1 \\ 0 & t \in [-1, 0] \end{cases} \;. \end{aligned} \tag{2.67}$$

It is easy to see, that the two functions

$$y(t) = \frac{t}{2}, \qquad y(t) = \frac{3t}{2}, \qquad t \in [0, 2]\,,$$

are both solutions of (2.67), with appropriate continuations for $t > 2$. Two different solutions of the same differential equation and identical initial conditions, this cannot happen for ordinary differential equations. It is evident, that special care must be taken when examining dynamical systems with time delays numerically.

2.5.1 Distributed Time Delays

Basic delay differential systems contain a single time delay T, like (2.61), which corresponds to an instantaneous memory process. In general, the memory $y_M(t)$ of past trajectories will be a convolution,

$$y_M(t) = \int_0^\infty K(\tau)y(t-\tau)d\tau, \qquad \int_0^\infty K(\tau)d\tau = 1\,, \tag{2.68}$$

where we defined with $K(\tau)$ the delay kernel. For a sharply peaked delay kernel, $K(\tau) = \delta(\tau - T)$, one recovers $y_M(t) = y(t - T)$.

Exponentially Distributed Delays Explicitly we consider with

$$K(\tau) = \frac{1}{T}\,\mathrm{e}^{-\tau/T} \tag{2.69}$$

exponentially distributed time delays. One has

$$\frac{d}{dt}y_M(t) = \int_0^\infty d\tau K(\tau)\frac{d}{dt}y(t-\tau) = -\int_0^\infty d\tau K(\tau)\frac{d}{d\tau}y(t-\tau), \tag{2.70}$$

which allows to integrate the last expression by parts when using (2.69). The resulting closed expression,

$$\dot{y}_M = \frac{-1}{T}\int_0^\infty d\tau \mathrm{e}^{-\tau/T}\frac{d}{d\tau}y(t-\tau) = \frac{y-y_M}{T}\,, \tag{2.71}$$

corresponds to a trailing average, with the memory variable y_M trying to approach y.

Kernel Series Framework Equation (2.71) implies that a delay differential equation with exponentially distributed delays can be mapped exactly to a system of ordinary differential equations by adding an additional variable, namely $y_M(t)$. For generic kernels $K(\tau)$ in (2.68) one can generalize this concept by adding a diverging number of memory variables. With this approach, denoted "kernel series framework", one can map any time delay system to a N-dimensional system of ordinary differential equations. For the case of a single time delay one speaks of the "linear chain trick". In general one has $N \to \infty$, which reflects the notion that delay systems are formally infinite dimensional.

Exercises

(2.1) <u>JACOBIANS WITH DEGENERATE EIGENVALUES</u>
Jacobians with degenerate eigenvalues may not dispose of a corresponding number of eigenvectors, being non-symmetric. Consider the linear system

$$\dot{x} = -x + y, \qquad \dot{y} = -ry, \qquad r > 0\,, \tag{2.72}$$

and determine the eigenvalues of the Jacobian for the fixpoint (0, 0) and its eigenvectors. What happens in the limit $r \to 1$?

(2.2) <u>CATASTROPHIC SUPERLINEAR GROWTH</u>
The growth of a resource variable $x(t) \geq 0$, as defined by

$$\dot{x} = x^\beta - \gamma x, \qquad \gamma = 0, 1\,, \tag{2.73}$$

is sub/super-linear for $\beta < 1$ and $\beta > 1$ respectively. Start by solving (2.73) analytically for $\gamma = 0$. Is there a singularity, viz a finite time $t_s < \infty$ for which $\lim_{t \to t_s} x(t)$ diverges catastrophically? Continue then by discussing qualitatively the behavior for $\gamma = 1$.
Next consider sublinear decay, viz $\dot{x} = -x^{\beta}$, with $\beta < 1$. Can the stable fixpoint $x^* = 0$ be reached in finite time?

(2.3) HETEROCLINIC ORBITS AND SYMMETRIES
Discuss, in analogy to (2.18), the fixpoints and the stable and unstable manifolds of

$$\dot{x} = 1 - x^2, \qquad \dot{y} = -y + \epsilon\left(1 - x^2\right)^2 . \tag{2.74}$$

Is there a heteroclinic orbit?

(2.4) THE SUBCRITICAL PITCHFORK TRANSITION
Discuss, in analogy to Sect. 2.2, the fixpoints, and their stability, of the subcritical pitchfork transition described by the canonical equation of motion $\dot{x} = ax + x^3$.

(2.5) ENERGY CONSERVATION IN LIMIT CYCLES
Evaluate, for the Taken–Bogdanov system (2.33) and $\mu \in [\mu_c, 1]$, the energy balance, as defined by (2.34), by integrating numerically dE/dt. How large is the overall energy dissipation and uptake when completing one cycle? Good numerical accuracy is important, you will probably need to use the Runge-Kutta method.

(2.6) MAXIMAL LYAPUNOV EXPONENT
Show, that the maximal Lyapunov exponent (2.55) can be written as the time-averaged local exponent, see (2.54).

(2.7) LYAPUNOVEXPONENTS ALONG PERIODIC ORBITS
Evaluate the longitudinal Lyapunov exponent λ_{φ} along the $r = 1$ trajectory of (2.35). Consider both the case when the coupling K is smaller or larger than the critical $K_c = 1$ of the infinite period bifurcation. For $K < K_c$ the orbit is periodic and the integral of the longitudinal Lyapunov exponent over one period T vanishes. Why is this generically the case for periodic orbits?

(2.8) GRADIENT DYNAMICAL SYSTEMS
Is it possible to add a term to the evolution equation $\dot{x} = \ldots$ of the dynamical system (2.74), such that it becomes a gradient dynamical system, as defined in (2.42)?

(2.9) LIMIT CYCLES IN GRADIENT SYSTEMS
Show that gradient dynamical systems, compare (2.42), have no limit cycles.

(2.10) DELAY DIFFERENTIAL EQUATIONS
The delay equation (2.61) allows for harmonically oscillating solutions for certain sets of parameters a and b. Which are the conditions? In addition, specialize for the case $a = 0$.

(2.11) PERIOD-3 CYCLES IN THE LOGIST MAP
For the logistic map, $g(x) = rx(1 - x)$, a window of stability for period-3 cycles starts at $r^* = 1 + 2\sqrt{2} \approx 3.8284$, viz inside the otherwise chaotic

regime. Compare Fig. 2.19. Find the values for

$$x_1 = g(x_3), \qquad x_2 = g(x_1), \qquad x_3 = g(x_3)\,, \tag{2.75}$$

together with Lyapunov exponent. Plot the iterated map for $r = r^*$ and values a bit smaller/larger. The occurring saddle-node bifurcation is also called 'tangent transition' Why?

(2.12) CAR-FOLLOWING MODEL
A car moving with velocity $\dot{x}(t)$ follows another car driving with velocity $v(t)$ via

$$\ddot{x}(t+T) = \alpha\big(v(t) - \dot{x}(t)\big), \qquad \alpha > 0\,, \tag{2.76}$$

with $T > 0$ being the reaction time of the driver. Prove the stability of the steady-state solution for a constant velocity $v(t) \equiv v_0$ of the preceding car.[8]

Further Reading

For further studies we refer to introductory texts to classical dynamical systems, Goldstein (2002), to bifurcation theory, Kielhöfer (2012), to chaos theory, Devaney (2018), Poston and Stewart (2014), and to the generalization of chaos to quantum mechanics, Gutzwiller (2013). Readers interested in the KAM theorem may consult Ott (2002).

An easy to follow introduction to time delays system can be found in Wernecke et al. (2019), for more details see Kharitonov (2012). The kernel series framework for delay dynamical systems is elaborated in Nevermann and Gros (2023), rate induced tipping in Ritchie et al. (2023). The interplay between complexity barriers and chaotic dynamics is discussed in Gros (2012).

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[8] More realistic car-following models are discussed in Sect. 4.4 of Chap. 4.

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Dissipation, Noise and Adaptive Systems

3

Most dynamical systems are not isolated, but interacting with an embedding environment that may add stochastic components to the evolution equations. The internal dynamics slows down when energy is dissipated to the outside world, approaching attracting states which may be regular, such as fixpoints or limit cycle, or irregular, such as chaotic attractors. Adaptive systems alternate between phases of energy dissipation and uptake, until a balance between these two opposing processes is achieved.

In this chapter an introduction to adaptive, dissipative and stochastic systems will be given together with important examples from the realm of noise controlled dynamics, like diffusion, random walks and stochastic escape and resonance. We will discuss to which extent chaos, a regular guest of adaptive systems, may remain predictable.

3.1 Chaos in Dissipative Systems

The time evolution of deterministic systems can be computed exactly, at least as matter of principle, once the initial conditions are known.[1] We now turn to "stochastic systems", i.e. dynamical systems that are influenced by noise and fluctuations. On a mean level, the impact of noise shows up as "dissipation".

3.1.1 Phase Space Contraction and Expansion

Friction and Dissipation Friction plays an important role in real-world systems, causing energy to be dissipated to the environment.

[1] For references, Chap. 2 is devoted to determinstic dynamics.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_3

Energy is conserved when the contributions from all constituent parts of the overall system are taken into account. Friction just stands for a transfer process of energy; when energy is transferred from a system we observe, like a car on a motorway with the engine turned off, to a system not under observation, such as the surrounding air. In this case the combined kinetic energy of the car and the thermal energy of the air body remains constant; the air heats up a little bit while the car slows down.

Mathematical Pendulum As an example we consider the damped mathematical pendulum,

$$\ddot{\phi} + \gamma\dot{\phi} + \omega_0^2 \sin\phi = 0\,, \tag{3.1}$$

which describes a pendulum with a rigid bar, capable of turning over completely, with ϕ corresponding to the angle between the bar and the vertical.

The mathematical pendulum reduces to the damped harmonic oscillator for small $\phi \approx \sin\phi$, which is damped/critical/overdamped for $\gamma < 2\omega_0$, $\gamma = 2\omega_0$ and $\gamma > 2\omega_0$. In the absence of damping, $\gamma = 0$, the energy

$$E = \frac{\dot{\phi}^2}{2} - \omega_0^2 \cos\phi \tag{3.2}$$

is conserved,

$$\frac{d}{dt}E = \dot{\phi}\,\ddot{\phi} + \omega_0^2\dot{\phi}\sin\phi = \dot{\phi}\left(\ddot{\phi} + \omega_0^2\sin\phi\right) = -\gamma\dot{\phi}^2\,,$$

when using (3.1).

Normal Coordinates Transforming the damped mathematical pendulum (3.1) to a set of coupled first-order differential equations via $x = \phi$ and $\dot{\phi} = y$ one gets

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -\gamma y - \omega_0^2 \sin x\,. \end{aligned} \tag{3.3}$$

The phase space for $\mathbf{x} = (x, y)\ \boldsymbol{R}^2$. For all $\gamma > 0$ the motion approaches for $t \to \infty$ one of the equivalent global fixpoints $(2\pi n, 0)$, where $n \in \boldsymbol{Z}$.

Phase Space Contraction By definition, phase space contracts close to an attractor. For a three-dimensional phase space, $\mathbf{x} = (x, y, z)$, the quantity

$$\begin{aligned} \Delta V(t) &= \Delta x(t)\Delta y(t)\Delta z(t) \\ &= (x(t) - x'(t))\,(y(t) - y'(t))\,(z(t) - z'(t)) \end{aligned}$$

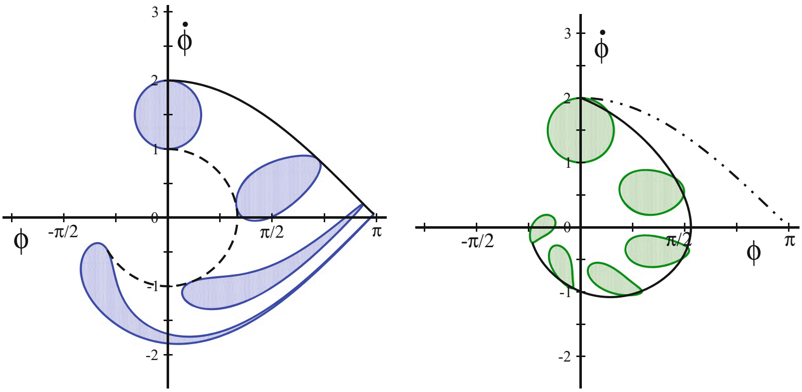


Fig. 3.1 Simulation of the mathematical pendulum $\ddot{\phi} = -\sin(\phi) - \gamma\dot{\phi}$, illustrating the evolution of the phase space volume for consecutive times (shaded regions), starting with $t = 0$ (top). **Left:** The dissipationless case $\gamma = 0$. The energy, see (3.2), is conserved as well as the phase space volume (Liouville's theorem). Shown are trajectories for $E = 1$ and $E = -0.5$ (solid/dashed line). **Right:** For $\gamma = 0.4$. Note the consecutive contraction of the phase space volume

corresponds to a small volume in phase space when $|\mathbf{x} - \mathbf{x}'|$ is small. Its time evolution is given by

$$\frac{d}{dt}\Delta V = \Delta\dot{x}\Delta y\Delta z + \Delta x\Delta\dot{y}\Delta z + \Delta x\Delta y\Delta\dot{z}\,,$$

or

$$\frac{\Delta\dot{V}}{\Delta x\Delta y\Delta z} = \frac{\Delta\dot{x}}{\Delta x} + \frac{\Delta\dot{y}}{\Delta y} + \frac{\Delta\dot{z}}{\Delta z} = \boldsymbol{\nabla}\cdot\dot{\mathbf{x}}\,, \tag{3.4}$$

with the right-hand side corresponding to the trace of the Jacobian.

In Fig. 3.1 the time evolution of a phase space volume is illustrated for the case of the mathematical pendulum. Volumes of the phase space remain connected under the effect of the time evolution, undergoing however at times substantial deformations.

DISSIPATIVE AND CONSERVING SYSTEMS A dynamical system is dissipative, if its phase space volume contracts continuously, viz when $\boldsymbol{\nabla}\cdot\dot{\mathbf{x}} < 0$ for all $\mathbf{x}(t)$. The system is said to be conserving if the phase space volume is a constant of motion, viz if $\boldsymbol{\nabla}\cdot\dot{\mathbf{x}} \equiv 0$.

Mechanical systems, i.e. systems described by Hamiltonian mechanics, are conserving in above sense. One denotes this result from classical mechanics as "Liouville's theorem".

Depending on the energy, mechanical systems may have bounded or non-bounded orbits. Planets run through bounded orbits around the sun, to give an example, with certain comets leaving the solar system for ever on unbounded trajectories. One can easily deduce from Liouville's theorem, i.e. from phase space conservation, that bounded orbits are "ergodic". They come arbitrarily close to all points in phase space having the identical conserved energy.

Global Dissipation Dissipation may occur everywhere in phase space, or only locally. The first holds for the damped mathematical pendulum (3.3),

$$\frac{\partial \dot{x}}{\partial x} = 0, \qquad \frac{\partial \dot{y}}{\partial y} = \frac{\partial[-\gamma y - \omega_0^2 \sin x]}{\partial y} = -\gamma\,,$$

which implies

$$\nabla \cdot \dot{\mathbf{x}} = -\gamma < 0\,.$$

The damped harmonic oscillator is consequently globally dissipative. The basin of attraction of the fixpoint $(0, 0)$ covers the full phase space (modulo 2π). Selected trajectories are illustrated in Fig. 3.1 together with phase space contraction.

Local Dissipation For the non-linear rotator defined by (2.2),

$$\dot{r} = (\Gamma - r^2)\,r, \qquad \dot{\varphi} = \omega\,, \tag{3.5}$$

we have

$$\frac{\partial \dot{r}}{\partial r} + \frac{\partial \dot{\varphi}}{\partial \varphi} = \Gamma - 3r^2 = \begin{cases} < 0 \text{ for } \Gamma < 0 \\ < 0 \text{ for } \Gamma > 0 \text{ and } r > r_c/\sqrt{3} \\ > 0 \text{ for } \Gamma > 0 \text{ and } 0 < r < r_c/\sqrt{3} \end{cases}\,, \tag{3.6}$$

where $r_c = \sqrt{\Gamma}$ is the radius of the limit cycle existing when $\Gamma > 0$. The system might either dissipate or take up energy, which is typical behavior of adaptive systems, as we will discuss further in Sect. 3.2.

Phase space contracts globally when $\Gamma < 0$, and locally, close to the limit cycle, when $\Gamma > 0$. In the latter case, when $\Gamma > 0$, phase space expands around the unstable fixpoint $(0, 0)$.

Coordinate Transformation The time development of a small phase space volume, as defined by (3.4), depends on the coordinate system chosen to represent the variables. As an example we reconsider the non-linear rotator (3.5) in terms of the Cartesian coordinates $x = r\cos\varphi$ and $y = r\sin\varphi$.

The respective infinitesimal phase space volumes are related via the Jacobian,

$$dx\,dy = r\,dr\,d\varphi\,,$$

and we find

$$\frac{\Delta\dot{V}}{\Delta V} = \frac{\dot{r}\Delta r\Delta\varphi + r\Delta\dot{r}\Delta\varphi + r\Delta r\Delta\dot{\varphi}}{r\Delta r\Delta\varphi} = \frac{\dot{r}}{r} + \frac{\partial\dot{r}}{\partial r} + \frac{\partial\dot{\varphi}}{\partial\varphi} = 2\Gamma - 4r^2\,.$$

Comparing with (3.6) we see that the amount and even the sign of phase space contraction can depend on the choice of the coordinate system. However, phase space will always contract close to an attractor, regardless of the coordinate system selected. This hold for $(0, 0)$ when $\Gamma < 0$ and for the limit cycle at $r = \sqrt{\Gamma}$ when $\Gamma > 0$.

Divergence of the Flow and Lyapunov Exponents For a dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ the local change in phase space volume is given by the divergence of the flow,

$$\frac{\Delta\dot{V}}{\Delta V} = \nabla\cdot\mathbf{f} = \sum_i \frac{\partial f_i}{\partial x_i} = \sum_i \lambda_i\,, \tag{3.7}$$

and hence by the trace of the Jacobian $J_{ij} = \partial f_i/\partial x_j$, as mentioned before. The trace of a matrix corresponds to the sum $\sum_i \lambda_i$ of its eigenvalues λ_i. Phase space hence contracts when the sum of the local Lyapunov exponents is negative.

3.1.2 Strange Attractors and Dissipative Chaos

Lorenz Model A rather natural question regards the possible existence of attractors with irregular behaviors, i.e. which are different from stable fixpoints, periodic or quasi-periodic motion. For this question we examine the Lorenz model

$$\begin{aligned}\dot{x} &= -\sigma(x - y),\\ \dot{y} &= -xz + rx - y,\\ \dot{z} &= xy - bz\,.\end{aligned} \tag{3.8}$$

The classical values are $\sigma = 10$ and $b = 8/3$, with r being the control variable.

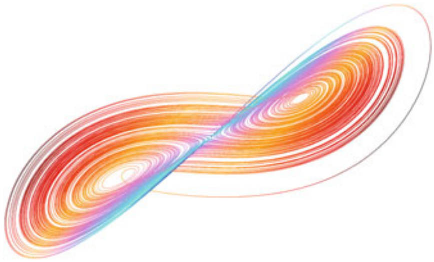


Fig. 3.2 A typical trajectory of the Lorenz system (3.8), for the classical set of parameters, $\sigma = 10$, $b = 8/3$ and $r = 28$. The chaotic orbit loops around the remnants of the two fixpoints (3.9), which are unstable for the selected set of parameters. Color coding with respect to z, projected to the $z = 0$ plane

Fixpoints of the Lorenz Model A trivial fixpoint is $(0, 0, 0)$. The non-trivial fixpoints are

$$\begin{aligned} 0 &= -\sigma(x-y), & x &= y, \\ 0 &= -xz + rx - y, & z &= r-1, \\ 0 &= \quad xy - bz, & x^2 = y^2 &= b\,(r-1)\,. \end{aligned}$$

It is easy to see by linear analysis that the fixpoint $(0, 0, 0)$ is stable for $r < 1$. For $r > 1$ it becomes unstable via a pitchfork bifurcation and two new fixpoints appear,

$$C_{\pm} = \left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1\right)\,. \tag{3.9}$$

These are stable for $r < r_c = 24.74$ (for $\sigma = 10$ and $b = 8/3$), at which point a subcritical Hopf bifurcation occurs. Generally one has $r_c = \sigma(\sigma+b+3)/(\sigma-b-1)$. For $r > r_c$ the behavior becomes more complicated and generally non-periodic.

Strange Attractors One can show, that the Lorenz model has a positive Lyapunov exponent for $r > r_c$. It is chaotic with sensitive dependence on the initial conditions. A typical orbit is illustrated in Fig. 3.2. The Lorenz model is at the same time globally dissipative, since

$$\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -(\sigma + 1 + b) < 0, \qquad \sigma, b > 0\,. \tag{3.10}$$

The consequence is that the attractor of the Lorenz system cannot be a smooth surface. Close to the attractor phase space contracts. At the same time two nearby orbits are repelled due to the positive Lyapunov exponents. One finds a self-similar structure for the Lorenz attractor with a fractal dimension 2.06 ± 0.01, as defined further below. Such a structure is called a "strange attractor".

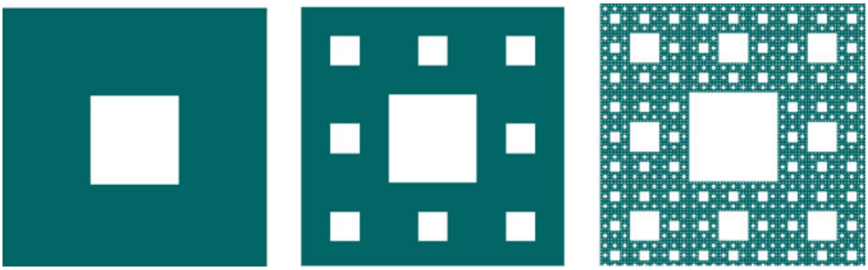


Fig. 3.3 The Sierpinski carpet and its iterative construction

Dissipative Chaos and Strange Attractors Strange attractors can only occur in dynamical system of dimension three and higher, in one dimension fixpoints are the only possible attracting states and one needs at least two dimensions for limit cycles.

The Lorenz model has an important historical relevance for the development of chaos theory It is considered a paradigmatic model, since chaos in dissipative and deterministic dynamical systems is closely related to the emergence of strange attractors. Chaos may arise in one dimensional maps, as we see in Sect. 2.4, but continuous-time dynamical systems need to be at least three dimensional in order to show chaotic behavior.

Hyperchaos An attractor is chaotic when at least a single Lyapunov exponent is positive. As a function of the position on the attracting manifold, local Lypunov exponets may vary substantially, which implies that the occurance of chaos is determined by the sign of the maximal Lypunov, compare the corresponding definition (2.55) for maps. Several Lypunov exponents may become positive for systems with larger numbers of dynamical variables, such as dynamical networks. One speaks of "hyperchaos" when this happens.

Fractals Strange attractors often show a high degree of self-similarity, being fractal. Fractals can be defined on an abstract level by recurrent geometric rules, prominent examples are the Cantor set, the Sierpinski triangle and the Sierpinski carpet illustrated in Fig. 3.3. Strange attractors are normally "multi fractal", i.e. fractals with non-uniform self similarity.

Hausdorff Dimension An important notion in the theory of fractals is the "Hausdorff dimension". We consider a geometric structure defined by a set of points in d dimensions and the number $N(l)$ of d-dimensional spheres of diameter l needed to cover this set. If $N(l)$ scales like

$$N(l) \propto l^{-D_H}, \qquad \text{for} \qquad l \to 0, \tag{3.11}$$

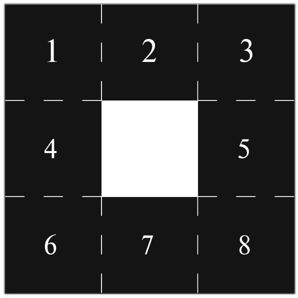


Fig. 3.4 The fundamental unit of the Sierpinski carpet, compare Fig. 3.3. It contains eight squares which can be covered by discs of appropriate diameter

then D_H is called the Hausdorff dimension of the set. Alternatively we can rewrite (3.11) as

$$\frac{N(l)}{N(l')} = \left(\frac{l}{l'}\right)^{-D_H}, \qquad D_H = -\frac{\log[N(l)/N(l')]}{\log[l/l']}, \tag{3.12}$$

which is useful for self-similar structures (fractals).

The d-dimensional spheres necessary to cover a given geometrical structure will generally overlap. The overlap does not affect the value of the fractal dimension as long as the degree of overlap does not change qualitatively with decreasing diameter l.

Hausdorff Dimension of the Sierpinski Carpet For the Sierpinski carpet we increase the number of points $N(l)$ by a factor of eight when we decrease the length scale l by a factor of three,

$$D_H \to -\frac{\log[8/1]}{\log[1/3]} = \frac{\log 8}{\log 3} \approx 1.8928 .$$

See Fig. 3.4 for an illustration.

3.1.3 Partially Predictable Chaos

The strange attractor of the Lorenz model has the form of a 'butterfly', but otherwise seemingly no further internal structure. Orbits exponentially diverge close to the attractor, remaining however bounded by its overall size. The only feature remaining predictable is that orbits will not diverge indefinitely. The sistuation is more complex for most real-world chaotic systems.

As an example of a real-world system with substantial chaotic components we may take weather forecasting. When a hurricane approaches a coastline, one may

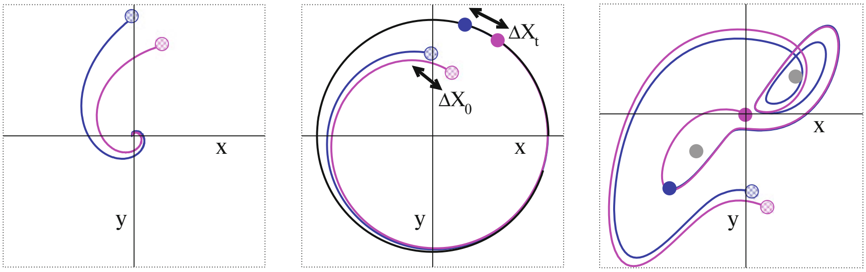


Fig. 3.5 The evolution of two initially close trajectories (blue/magenta lines). The respective initial states (shaded circles) are evolved for the identical period t (filled circles). Shown are orbits for the non-linear rotator (3.5), for $\Gamma < 0$ (left panel) and $\Gamma > 0$ (middle panel), which converge respectively to a stable focus and limit cycle (black ring). For the Lorenz model (right panel), trajectories diverge on the attractor, with one orbit circling one of the unstable fixpoints (grey circles) an additional time. Parameters as in Fig. 3.2

not be able to predict the location of landfall precisely, nor the strength of the hurricane at that point. Nobody will doubt however, that a full-blown hurricane is approaching. Certain features can remain predictable even in otherwise chaotic systems. This is the notion of "partially predictable chaos", which we will examine in the following.

For a precise definition of partially predictable chaos one needs an observational tool allowing to determine the presence of chaos directly from the properties of typical orbits, without resorting to the evaluation of Lyapunov exponents. This tool, a "0-1 test for chaos", is developed first.

Initially Close Orbits For a given dynamical system we examine the long-term fate of two initially close orbits, $\mathbf{x} = \mathbf{x}(t)$ and $\mathbf{x}' = \mathbf{x}'(t)$, with

$$\Delta X_0 = ||\mathbf{x} - \mathbf{x}'||_{t=0}$$

being small. We are interested in the average long-term inter-orbit distance,

$$\Delta X_\infty = \left\langle ||\mathbf{x} - \mathbf{x}'|| \right\rangle_{t>T}, \tag{3.13}$$

which is defined as the average distance between $\mathbf{x}$ and $\mathbf{x}'$ for times $t > T$, where T is larger than the time scales of the defining dynamical system.

0-1 Test for Chaos In Fig. 3.5 the evolution for two initial close orbits is illustrated for the case that both orbits approach the same attractor, which may be a fixpoint, limit cycle, or a chaotic attractor.

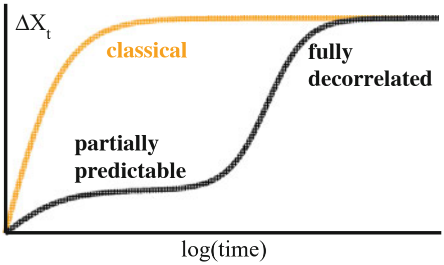


Fig. 3.6 Chaotic dynamics is partially predictable (black curve) when the distance ΔX_t of two initially close orbits remains at a temporary plateau after the starting divergence. The fully decorrelated final state is reached only after a substantial delay. Information is lost within a single process for classical chaos (orange curve)

– FIXPOINT
  One has $\Delta X_\infty \to 0$ independently of ΔX_0.
– LIMIT CYCLE
  When ΔX_0 is small, substantially smaller than the size of the attracting limit cycle, one observes $\Delta X_\infty \propto \Delta X_0$. The two orbits will follow each other indefinitely after having entered the limit cycles at slightly different points, with the average distance ΔX_∞ scaling linearly with ΔX_0
– CHAOTIC ATTRACTOR
  Orbits fully decorrelate on the attractor, with ΔX_∞ approaching the average two-point distance irrespectively of ΔX_0.

Above rules constitute a "0-1 test" for chaos. Operationally, one integrates the system in question for pairs of initially close orbits. The resulting average final distance ΔX_∞ is plotted relative to its initial value. The attracting state is a limit cycle if ΔX_∞ scales with ΔX_0, otherwise a fixpoint or a chaotic attractor.

Partially Predictable Chaos Decorrelation is characterized by a single time scale for classical chaotic attractors, such as the butterfly attractor illustrated in Fig. 3.2. The incurring loss of information is captured by the time development of the inter-orbit distance, ΔX_t, as illustrated in Fig. 3.6. Chaos remains however partially predictable when decorrelation is limited for prolonged periods. In this case, when two orbits retain a certain finite distance, one can evaluate their relative development with an accuracy given by the respective ΔX_t. One can still predict what is going to happen for extended time spans, albeit with a finite accuracy. The whole process is shown in Fig. 3.6.

As a function of the control parameter r, a variety of attracting states appear in the Lorenz model. For $r = 28$ the classical butterfly attractor is observed, see Fig. 3.2. For larger values of r, windows with period-doubling limit cycles appear. Interestingly, for certain values of r, like $r = 180.78$, a distinct type of chaotic attractors is observed, in the form of chaotic braids. Topologically, these braids correspond to chaotically broadened limit cycles. Decorrelation slows down by orders of magnitude once ΔX_t becomes comparable to the width of the braids. The subsequent diffusive decorrelation along the former limit cycle is very slow. This is a typical example of partially predictable chaos.

3.2 Adaptive Systems

In general, complex systems are neither fully conserving nor fully dissipative. Adaptive systems will have phases where they take up energy and periods where they give energy back to the environment. An example is the non-linear rotator defined in (3.5), see also (3.6).

One affiliates with the term "adaptive system" often to the notion of complexity and adaption. Strictly speaking, any dynamical system is adaptive if $\nabla \cdot \dot{\mathbf{x}}$ may take both positive and negative values. In practice, however, it is usual to reserve the term adaptive system to dynamical systems showing a certain complexity, such as emerging behavior.

Van der Pol Oscillator Circuits or mechanisms built for the purpose of controlling an engine or machine are intrinsically adaptive. An example is the Van der Pol oscillator,

$$\ddot{x} - \epsilon(1-x^2)\dot{x} + x = 0, \qquad \begin{array}{l} \dot{x} = y \\ \dot{y} = \epsilon(1-x^2)y - x \end{array} \tag{3.14}$$

where $\epsilon > 0$ and where we did use the phase space variables $\mathbf{x} = (x, y)$. The evolution of the phase space volume is

$$\nabla \cdot \dot{\mathbf{x}} = \epsilon\,(1-x^2)\,.$$

The oscillator takes up/dissipates energy for $x^2 < 1$ and $x^2 > 1$, respectively. A simple mechanical example for a system with similar properties is illustrated in Fig. 3.7

Secular Perturbation Theory We consider a perturbation expansion in ϵ. For $\epsilon = 0$, the solution of (3.14) describes harmonic oscillations,

$$x_0(t) = a\,\mathrm{e}^{i(\omega_0 t+\phi)} + c.c., \qquad \omega_0 = 1\,. \tag{3.15}$$

Amplitude a and phase ϕ are arbitrary, as usual for harmonic oscillators. The perturbation $\epsilon(1-x^2)\dot{x}$ may change both the amplitude and the unperturbed frequency $\omega_0 = 1$ by an amount $\propto \epsilon$. In order to account for this "secular perturbation" we make the ansatz

$$x(t) = \left[A(T)\mathrm{e}^{it} + A^*(T)\mathrm{e}^{-it}\right] + \epsilon x_1, \qquad A(T) = A(\epsilon t)\,, \tag{3.16}$$

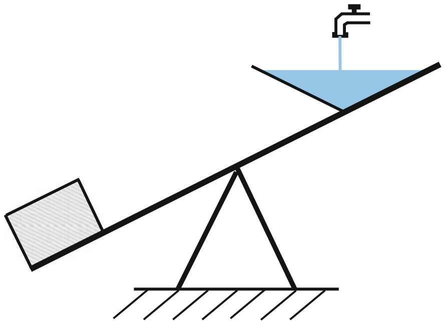


Fig. 3.7 The seesaw with a water container at one end; an example of an oscillator that takes up and disperses energy periodically

which differs from the usual expansion $x(t) \to x_0(t) + \epsilon x'(t)$ of the full solution $x(t)$ of a dynamical system with respect to a small parameter ϵ. The yet to be determined time-dependent complex prefactor A allows for an adaption of the original frequency. The ansatz $A = A(\epsilon t)$ ensures that the prefactor is constant when $\epsilon \to 0$.

Expansion The goal is to expand the Van der Pol oscillator together with (3.16) with respect to the perturbation[2] We start be evaluating several expressions involing $x(t)$ up to order $O(\epsilon^1)$, namely

$$
\begin{aligned}
x^2 &\approx A^2 \mathrm{e}^{2it} + 2|A|^2 + (A^*)^2 \mathrm{e}^{-2it} + 2\epsilon x_1 \left[A\mathrm{e}^{it} + A\mathrm{e}^{-it} \right] \\
\epsilon(1 - x^2) &\approx \epsilon(1 - 2|A|^2) - \epsilon \left[A^2 \mathrm{e}^{2it} + (A^*)^2 \mathrm{e}^{-2it} \right], \\
\dot{x} &\approx \left[(\epsilon A_T + iA)\, \mathrm{e}^{it} + c.c. \right] + \epsilon\, \dot{x}_1, \qquad A_T = \frac{\partial A(T)}{\partial T} \\
\epsilon(1 - x^2)\dot{x} &= \epsilon(1 - 2|A|^2) \left[iA\mathrm{e}^{it} - iA^* \mathrm{e}^{-it} \right] \\
&\quad - \epsilon \left[A^2 \mathrm{e}^{2it} + (A^*)^2 \mathrm{e}^{-2it} \right] \left[iA\mathrm{e}^{it} - iA^* \mathrm{e}^{-it} \right]
\end{aligned}
$$

and

$$
\begin{aligned}
\ddot{x} &= \left[\left(\epsilon^2 A_{TT} + 2i\epsilon A_T - A \right) \mathrm{e}^{it} + c.c. \right] + \epsilon\, \ddot{x}_1 \\
&\approx \left[(2i\epsilon A_T - A)\, \mathrm{e}^{it} + c.c. \right] + \epsilon \ddot{x}_1 .
\end{aligned}
$$

[2] The following derivations are informative, but somewhat advanced. In case, the reader may skip directly to the result, Eq. (3.18).

Substituting these expressions into (3.14), we obtain

$$\ddot{x}_1 + x_1 = \left(-2iA_T + iA - i|A|^2 A\right) e^{it} - iA^3 e^{3it} + c.c.\,, \tag{3.17}$$

to order $O(\epsilon^1)$.

Solvability Condition The time dependencies

$$\sim \mathrm{e}^{it} \qquad \text{and} \qquad \sim \mathrm{e}^{3it}$$

of the two terms on the right-hand side of (3.17) are proportional to the unperturbed frequency $\omega_0 = 1$ and to $3\omega_0$, respectively.

Equation (3.17) is identical to that of a driven harmonic oscillator.[3] The term $\sim \mathrm{e}^{it}$ is therefore exactly at resonance and would induce a diverging response $x_1 \to \infty$, in contradiction to the perturbative assumption made by ansatz (3.16). Its prefactor must therefore vanish,

$$A_T = \frac{\partial A}{\partial T} = \frac{1}{2}\left(1 - |A|^2\right) A, \qquad \frac{\partial A}{\partial t} = \frac{\epsilon}{2}\left(1 - |A|^2\right) A\,, \tag{3.18}$$

where we used $T = \epsilon t$. The solvability condition (3.18) can be written as

$$\dot{a}\, \mathrm{e}^{i\phi} + i\dot{\phi}\, a\, \mathrm{e}^{i\phi} = \frac{\epsilon}{2}\left(1 - a^2\right) a\, \mathrm{e}^{i\phi}\,,$$

in phase-magnitude representation $A(t) = a(t)\mathrm{e}^{i\phi(t)}$, or

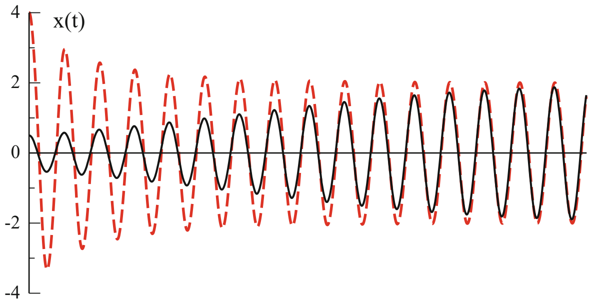
$$\dot{a} = \epsilon\left(1 - a^2\right) a/2, \qquad \dot{\phi} \sim O(\epsilon^2)\,. \tag{3.19}$$

The system takes up energy for $a < 1$ and the amplitude a increases till the saturation limit $a \to 1$, the conserving point. For $a > 1$ the system dissipates energy to the environment and the amplitude a decreases, approaching unity for $t \to \infty$, just as we discussed in connection with (3.5).

In the limit $\epsilon \to 0$ the long-term solution of the Van der Pol oscillator is $x(t) \approx 2a\cos(t)$, compare (3.16) and (3.19), which constitutes an amplitude-regulated oscillation. This behavior, compare Fig. 3.8, is relevant for technical control tasks.

[3] The harmonic oscillator is resonant only when the frequency of the perturbation matches the internal frequency. Non-harmonic oscillators may however unstable against rational frequency ratios, as discussed in Sect. 2.1 of Chap. 2 in the context of the KAM theorem, with regard to the gaps in the Saturn rings.

Fig. 3.8 Two solutions of the Van der Pol oscillator (3.14), for small ϵ and two different initial conditions. Note the self-generated amplitude stabilization



Liénard Variables For large ϵ it is convenient to define with

$$\epsilon \frac{\mathrm{d}}{\mathrm{d}t} Y(t) = -x(t) = \ddot{x}(t) - \epsilon \left(1 - x^2(t)\right) \dot{x}(t) \tag{3.20}$$

the Liénard variable $Y(t)$, where the second equality is just the definition of the Van der Pol oscillator, see (3.14). With the $X(t) \equiv x(t)$ we rewrite (3.20) as

$$\epsilon \dot{Y} = \ddot{X} - \epsilon \left(1 - X^2\right) \dot{X}, \qquad X(t) = x(t)\,,$$

which we integrate with respect to t,

$$\epsilon Y = \dot{X} - \epsilon \left(X - \frac{X^3}{3}\right)\,,$$

where we did set the integration constant to zero. Together with (3.20) we obtain

$$\begin{aligned} \dot{X} &= c\left(Y - f(X)\right), \qquad f(X) = X^3/3 - X, \\ \dot{Y} &= -X/c\,, \end{aligned} \tag{3.21}$$

with $c \equiv \epsilon$, as we are now interested in the case $c \gg 1$.

Relaxation Oscillations For a large driving c, we can discuss the solution of the Van der Pol oscillator (3.21) graphically, as illustrated in Fig. 3.9. Of relevance is the flow $(\dot{X}, \dot{Y})$ in phase space (X, Y). For $c \gg 1$ there is a separation of time scales,

$$(\dot{X}, \dot{Y}) \sim (c, 1/c), \qquad \dot{X} \gg \dot{Y}\,,$$

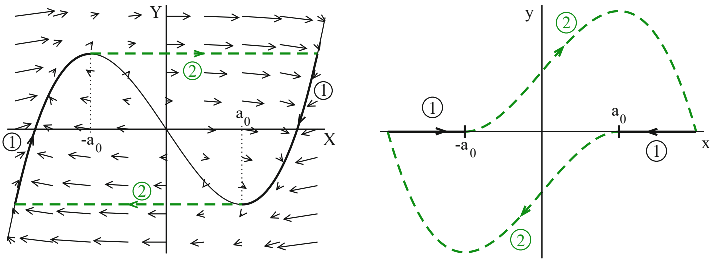


Fig. 3.9 Van der Pol oscillator for a large driving $c \equiv \epsilon$. **Left:** Relaxation oscillations with respect to the Liénard variables (3.21). Indicated is the flow $(\dot{X}, \dot{Y})$ (arrows), for $c = 3$, see (3.21). Also shown is the $\dot{X} = 0$ isocline $Y = -X + X^3/3$ (solid line) and the limit cycle, which includes a non-constant part (dashed line) and a section of the isocline. **Right:** The limit cycle in terms of the original variables $(x, y) = (x, \dot{x}) = (x, v)$. Note that $X(t) = x(t)$

which leads to the following dynamical behavior:

- Starting at a general $(X(t_0), Y(t_0))$ the orbit develops very fast $\sim c$ and nearly horizontally until it hits the "isocline"[4]

$$\dot{X} = 0, \qquad Y = f(X) = -X + X^3/3\,. \tag{3.22}$$

- Once the orbit is close to the $\dot{X} = 0$ isocline $Y = -X + X^3/3$ the motion slows down, proceeding with a velocity $\sim 1/c$ close-to (but not exactly on) the isocline (3.22).
- Once the slow motion reaches one of the two local extrema $X = \pm a_0 = \pm 1$ of the isocline, it cannot follow the isocline any more and makes a rapid transition towards the other branch of the $\dot{X} = 0$ isocline, with $Y \approx$ const. Note, that trajectories may cross the isocline vertically, which implies that $\dot{Y}|_{X=\pm 1} = \mp 1/c$ is small but finite right at the extrema.

The orbit therefore relaxes rapidly towards a limiting oscillatory trajectory, illustrated in Fig. 3.9, with the time needed to perform a whole oscillation depending on the relaxation constant c; therefore the term "relaxation oscillation". Relaxation oscillators represent an important class of cyclic attractors, allowing to model systems going through several distinct and well characterized phases during the course of one cycle.[5]

[4] The term isocline stands for "equal slope" in ancient Greek.

[5] We will discuss relaxation oscillators further in Sect. 9.4.2 of Chap. 9.

3.2.1 Conserving Adaptive Systems

Per definition, conserving dynamical systems conserves the volume of phase space enclosed by a set of trajectories, as illustrated in Fig. 3.1. In contrast, phase space expands and contracts along a given orbit when the system is adaptive, as we discussed for the case of the Van der Pol oscillator, as defined by (3.14), and for the Taken-Bogdanov system.[6] Adaptive systems cannot conserve phase space volume, but they may conserve other quantities, like a generalized energy functional.

Energy Conservation in Mechanical Systems Newton's equation

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{v}\\ \dot{\mathbf{v}} &= -\nabla V(\mathbf{x})\end{aligned} \qquad\qquad E(\mathbf{x},\mathbf{v}) = \frac{\mathbf{v}^2}{2} + V(\mathbf{x}) \tag{3.23}$$

for a mechanical systems with a potential $V(\mathbf{x})$ conserves energy, $E = E(\mathbf{x},\mathbf{v})$,

$$\frac{dE}{dt} = \frac{\partial E}{\partial \mathbf{x}}\dot{\mathbf{x}} + \frac{\partial E}{\partial \mathbf{v}}\dot{\mathbf{v}} = \big(\nabla V + \dot{\mathbf{v}}\big)\mathbf{v} = 0\,.$$

Energy is an instance of a "constant of motion", viz of a conserved quantity.

Lotka–Volterra Model for Rabbits and Foxes Evolution equations for one or more interacting species are termed "Lotka–Volterra" models. A basic example is that of a prey (rabbit) with population density x being hunted by a predator (fox) with population density y,

$$\begin{aligned}\dot{x} &= \quad Ax - Bxy\\ \dot{y} &= -Cy + Dxy\end{aligned}\,. \tag{3.24}$$

The population x of rabbits can grow by themselves but the foxes need to eat rabbits in order to multiply. All constants A, B, C and D, are positive.

Fixpoints The Lotka–Volterra equation has two fixpoints,

$$\mathbf{x}_0^* = (0,0), \qquad\qquad \mathbf{x}_1^* = (C/D, A/B)\,. \tag{3.25}$$

with the respective Jacobians,[7]

$$J_0 = \begin{pmatrix} A & 0\\ 0 & -C\end{pmatrix}, \qquad\qquad J_1 = \begin{pmatrix} 0 & -BC/D\\ AD/B & 0\end{pmatrix}.$$

[6] See Sect. 2.3 of Chap. 2). for an in-depth treament of the Taken-Bogdanov equations.

[7] We recall that the Jacobian is the matrix of all possible partial derivatives, see Sect. 2.2.1, of Chap. 2.

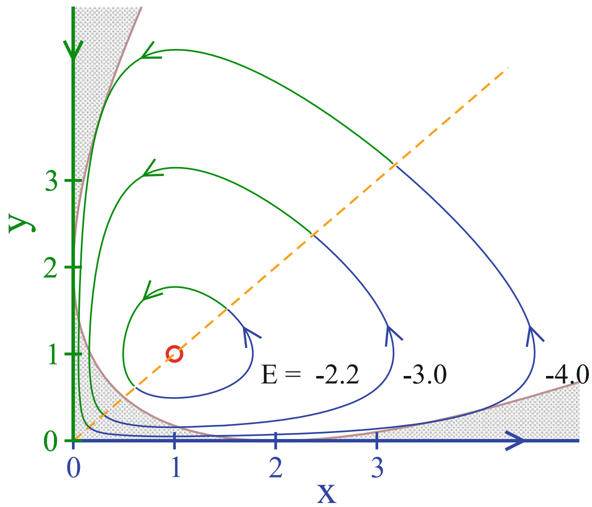


Fig. 3.10 In the space of population densities (x, y), the flow of the fox and rabbit Lotka–Volterra model, see (3.24) and (3.26). The flow expands (contracts) for $x > y$ $(x < y)$, as indicated (blue/green orbits). Trajectories coincide with the iso-energy lines of the conserved function E, as defined by (3.27), with (0, 0) being a saddle and (1, 1) a neutral focus (open red circle). Lyapunov exponents are real in the shaded region and complex otherwise

The trivial fixpoint $\mathbf{x}_0^*$ is hence a saddle and $\mathbf{x}_1^*$ a neutral focus with purely imaginary Lyapunov exponents $\lambda = \pm i\sqrt{CA}$. The trajectories circling the focus close onto themselves, as illustrated in Fig. 3.10 for $A = B = C = D = 1$.

Phase Space Evolution We now consider the evolution of phase space volume, as defined by (3.4),

$$\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{v}}{\partial v} = A - By - C + Dx\,. \tag{3.26}$$

The phase space expands/contracts for y smaller/larger than $(A + Dx - C)/B$, the tell sign of an adaptive system.

Constant of Motion The function

$$E(x, y) = A\log(y) + C\log(x) - By - Dx \tag{3.27}$$

on phase space (x, y) is a constant of motion for the Lotka–Volterra model (3.24), since

$$\begin{aligned}\frac{dE}{dt} &= A\dot{y}/y \,+\, C\dot{x}/x \,-\, B\dot{y} \,-\, D\dot{x} \\ &= A(-C + Dx) + C(A - By) - B(-C + Dx)y - D(A - By)x \\ &= 0\,.\end{aligned}$$

The prey-predator system (3.24) does hence dispose of a non-trivial constant of motion. Note that $E(x, y)$ has no biological significance which would be evident per se.

ISO-Energy Manifolds The flow of a d-dimensional dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ disposing of a conserved functional $E(\mathbf{x})$, we call it here a generalized energy, is always restricted to an iso-energy manifold defined by $E(\mathbf{x}) = \text{const}$,

$$\frac{dE}{dt} = \nabla E \cdot \dot{\mathbf{x}} = 0, \qquad\qquad \nabla E \perp \dot{\mathbf{x}}\,. \tag{3.28}$$

The flow $\dot{\mathbf{x}}$ is consequently perpendicular to the gradient ∇E the conserved generalized energy. Orbits are therefore confined to an iso-energy manifold, which is one-dimensional, given that phase space is two-dimensional. Trajectories coincided hence with the iso-energy manifold for the fox and rabbit Lotka–Volterra, as illustrated in Fig. 3.10.

Lyapunov Exponent We evaluate the Jacobian for a generic point (x, y) in phase space,

$$\begin{pmatrix} (1-y) & -x \\ y & (x-1) \end{pmatrix}, \qquad\qquad \lambda_\pm = \frac{x-y}{2} \pm \frac{1}{2}\sqrt{(x-y)^2 - 4(x+y-1)}\,,$$

where we did set $A = B = C = D = 1$. The Lyapunov exponents $\lambda_\pm$ are real close to the axes and complex further away, with the separatrix given by

$$(x-y)^2 = 4(x+y-1), \qquad\qquad y = x + 2 \pm \sqrt{8x}\,. \tag{3.29}$$

There is no discernible change in the flow dynamics across the separatrix (3.29), which we included in Fig. 3.10, viz when the Lyapunov exponents acquire finite imaginary components.

Invariant Manifolds Fixpoints and limit cycles are examples of invariant subsets of phase space.

> INVARIANT MANIFOLD A subset of phase space invariant under the flow for all times $t \in [-\infty, \infty]$ is denoted an invariant manifold.

All trajectories of the fox and rabbit Lotka–Volterra model, apart from the stable and the unstable manifolds of the saddle $(0, 0)$, are closed and constitute hence invariant manifolds.

Fixpoints and limit cycles are invariant manifolds with dimensions zero and one respectively, strange attractors, see Sect. 3.1.1, have generically a fractal dimension.

Closed Invariant Manifolds The evolution of phase space on an invariant manifold (viz inside the manifold) with dimension m is determined by m Lyapunov exponents whenever the manifold has a smooth topology (which is not the case for fractals).

Phase space cannot expand or contract forever for orbits on closed invariant manifolds M, which are finite. This is the case for all trajectories of the fox and rabbit Lotka–Volterra model and manifestly evident for the case $A = B = C = D = 0$ discussed above, for which the real part of the Lyapunov exponent $\lambda_\pm$ is anti-symmetric under the exchange $x \leftrightarrow y$.

Lotka–Volterra System with Resource Limitation The reproduction of the prey in the original Lotka–Volterra model (3.24) is not limited. In practice, the population density x of the rabbits will be bounded by the carrying capacity $x_{\max}$ of the supporting environment, such that $x < x_{\max}$. The modified model,

$$\dot{x} = Ax\left(1 - \frac{x}{x_{\max}}\right) - Bxy, \qquad \dot{y} = (Dx - C)y\,.$$

has the non-trival fixpoint

$$x^* = \frac{C}{D}, \qquad y^* = \frac{A}{B}\left(1 - \frac{C}{Dx_{\max}}\right), \tag{3.30}$$

which exists for $C < Dx_{\max}$. The population of rabbits is never large enough to support a finite population of foxes in the opposite case, when $C > Dx_{\max}$, which leads to $x \to x_{\max}$ and $y \to 0$. When existing, the steady state defined by (3.30) is stable.

The stability of (3.30) can be shown via a direct evaluation of the respective Jacobian. On a general level one can argue that the resource-limiting factor $1 - x/x_{\max}$ adds a contracting element. It is hence not surprising that the previously neutral fixpoint is stabilized.

3.3 Diffusion and Transport

Deterministic vs. Stochastic Time Evolution So far we discussed concepts of deterministic dynamical systems, governed by sets of coupled differential equations without noise or randomness. At the other extreme are diffusion processes for which the random process dominates the dynamics.

Dissemination of information through social networks is one of many examples where diffusion processes plays a paramount role. The simplest model of diffusion is Brownian motion, which describes the erratic movement of grains suspended in a liquid observed by the botanist Robert Brown as early as 1827. Brownian motion became the prototypical example of a stochastic process with the seminal works of Einstein and Langevin at the beginning of the twentieth century.

3.3.1 Random Walks, Diffusion and Lévy Flights

One-Dimensional Diffusion We start with the random walk of particles along a line, with each particle having an equal probability $1/2$ to move left/right at every time step. The probability

$$p_t(x), \qquad x = 0, \pm 1, \pm 2, \ldots\ldots, \qquad t = 0, 1, 2, \ldots$$

to find the particle at time t at position x obeys the master equation

$$p_{t+1}(x) = \frac{p_t(x-1) + p_t(x+1)}{2}\,. \tag{3.31}$$

Next we take the limit of continuous time and space by generalizing (3.31) to discrete steps Δx and Δt in space and time,

$$\frac{p_{t+\Delta t}(x) - p_t(x)}{\Delta t} = \frac{(\Delta x)^2}{2\Delta t}\,\frac{p_t(x+\Delta x) + p_t(x-\Delta x) - 2p_t(x)}{(\Delta x)^2}\,, \tag{3.32}$$

where we subtracted on both sides the current distribution $p_t(x)$. Taking the limit $\Delta x, \Delta t \to 0$ in such a way that $(\Delta x)^2/(2\Delta t)$ remains finite, we obtain the diffusion equation

$$\frac{\partial p(x,t)}{\partial t} = D\,\frac{\partial^2 p(x,t)}{\partial x^2}, \qquad D = \frac{(\Delta x)^2}{2\Delta t}\,, \tag{3.33}$$

with D being the diffusion constant. Note that the diffusion equation can be cast into the form of a continuity equation,

$$\dot{p} + \nabla \cdot \mathbf{j} = 0, \qquad \mathbf{j} = -D\,\nabla p\,, \tag{3.34}$$

with the diffusion current $\mathbf{j}$ encoding the diffusive transport of particle from high- to low concentrations.

Solution of the Diffusion Equation The solution $\Phi(x,t)$ of the diffusion equation (3.33) is given by

$$\Phi(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right), \qquad \int_{-\infty}^{\infty} \mathrm{d}x\, \Phi(x,t) = 1\,, \tag{3.35}$$

which holds[8] for a localized initial state $\Phi(x, t = 0) = \delta(x)$. For the derivation one enters the appropiate derivates,

$$\dot{\Phi} = \frac{-\Phi}{2t} + \frac{x^2\Phi}{4Dt^2}, \qquad \Phi' = \frac{-x\Phi}{2Dt}, \qquad \Phi'' = \frac{-\Phi}{2Dt} + \frac{x^2\Phi}{4D^2t^2}\,.$$

into the diffusion equation (3.33).

Diffusive Transport As a function of the coordinate x, the solution (3.35) of the diffusion equation corresponds to a Gaussian with variance $\sigma^2 = 2Dt$. One hence concludes that the variance of the displacement follows diffusive behavior, i.e.

$$\langle x^2(t)\rangle = 2D\,t, \qquad \bar{x} = \sqrt{\langle x^2(t)\rangle} = \sqrt{2D\,t}\,, \tag{3.36}$$

where we assumed that the mean $\langle x\rangle = 0$ vanishes. Diffusive transport is therefore characterized by transport sublinear in time, in contrast to ballistic transport following $x = vt$. Compare Fig. 3.11.

Green's Function for Diffusion For general initial distributions $p_0(x) = p(x,0)$ of walkers the diffusion equation (3.33) is solved by

$$p(x,t) = \int dy\, \Phi(x-y,t)\, p_0(y)\,, \tag{3.37}$$

[8] Note that $\int \mathrm{e}^{-x^2/a}\mathrm{d}x = \sqrt{a\pi}$, together with $\lim_{a\to 0} \exp(-x^2/a)/\sqrt{a\pi} = \delta(x)$.

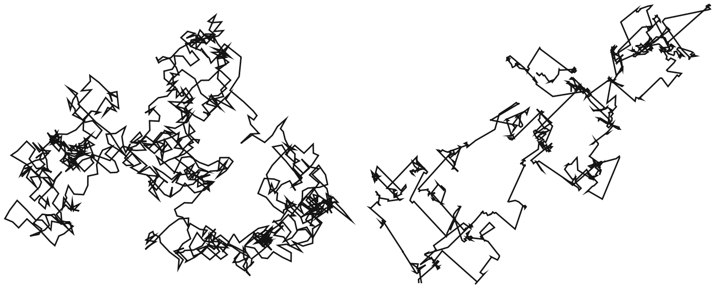


Fig. 3.11 Examples of random walkers with scale-free distributions $\sim 1/|\Delta x|^{1+\beta}$ for real-space jumps, see (3.39). **Left:** $\beta = 3$, which falls into the universality class of standard Brownian motion. **Right:** $\beta = 0.5$, a typical Levy flight. Note the occurrence of longer-ranged jumps in conjunction with local walking

since $\lim_{t\to 0} \Phi(x-y,t) = \delta(x-y)$. An integral kernel allowing to construct the solution of a differential equation for arbitrary initial conditions is a "Green's function".

First Passage Time When starting from the origin, which is the typical time t_y a random walker would need to reach a certain distance $y > 0$ for the first time? We define the survival probability

$$S_y(t) = \int_{-\infty}^{y} dx \left[\Phi(x,t) - \Phi(x-2y,t)\right], \qquad S_y(0) = 1, \qquad S_y(\infty) = 0\,,$$

which denotes the probability that the walker is below y at time t, without having ever crossed the 'cliff' $x = y$. Here we used the solution (3.35) of the diffusion equation, as describing walkers starting respectively at $x = 0$ and $x = 2y$. Importantly, the kernel $\Phi(x,t) - \Phi(x-2y,t)$ vanishes for $x = y$. The survival probability is monotonically decreasing with increasing time. The first passage time $t = t_F$ is not fixed, but distributed as

$$F_y(t) = -\frac{d}{dt} S_y(t) = -D \int_{-\infty}^{y} dx\, \frac{d^2}{d^2x}\left[\Phi(x,t) - \Phi(x-2y,t)\right],$$

where we used the diffusion equation $\dot{\Phi} = D\Delta\Phi$ in one dimension below the integral. Direct integration yields

$$
\begin{aligned}
F_y(t) &= \frac{-D}{\sqrt{4\pi Dt}} \int_{-\infty}^{y} dx\, \frac{d^2}{dx^2}\Big[\mathrm{e}^{-x^2/(4Dt)} - \mathrm{e}^{-(x-2y)^2/(4Dt)}\Big] \\
&= \frac{-D}{\sqrt{4\pi Dt}} \frac{d}{dx}\Big[\mathrm{e}^{-x^2/(4Dt)} - \mathrm{e}^{-(x-2y)^2/(4Dt)}\Big]_{x=y} \\
&= \frac{y}{\sqrt{4\pi Dt^3}} \exp\left(-\frac{y^2}{4Dt}\right), \qquad (3.38)
\end{aligned}
$$

which is a "Lévy distribution".[9]

For fixed y, one has that (3.38) scales as $\sim t^{-\alpha}$ for large t, with $\alpha = 3/2$. The first moment of the first passage time density $F_y(t)$ diverges consequently.[10] An average first passage time is not defined.

Lévy Flights One can generalize the concept of a random walker, which is at the basis of ordinary diffusion, and consider a random walk with distributions $\rho(\Delta t)$ and $\rho(\Delta x)$ for waiting times Δt_i and jumps Δx_i, at time step $i = 1, 2, \ldots$ of the walk, as illustrated in Fig. 3.12. One may assume scale-free distributions

$$
\rho(\Delta t) \sim \frac{1}{(\Delta t)^{1+\alpha}}, \qquad \rho(\Delta x) \sim \frac{1}{(\Delta x)^{1+\beta}}, \qquad \alpha, \beta > 0\,. \qquad (3.39)
$$

If $\alpha > 1$ (finite mean waiting time) and $\beta > 2$ (finite variance), nothing special happens. In this case the central limiting theorem for well behaved distribution functions is valid for the spatial component and one obtains standard Brownian diffusion. Relaxing the above conditions one finds four regimes: normal Brownian diffusion, "Lévy flights", fractional Brownian motion, also denoted "subdiffusion" and generalized Lévy flights termed "ambivalent processes". Their respective scaling laws are listed in Table 3.1, with two examples being shown in Fig. 3.11.

Lévy flights occur for a wide range of processes, such as for the flight patterns of wandering albatrosses. Human travel habits seem to be characterized by a generalized Lévy flight with $\alpha, \beta \approx 0.6$.

[9] The Lévy distribution $\sqrt{c/(2\pi)}\,\exp(-\frac{c}{2t})/t^{3/2}$ is normalized on the interval $t \in [0, \infty]$.

[10] The moments of powerlaw distributions are discussed in Sect. 1.1.3 of Chap. 1.

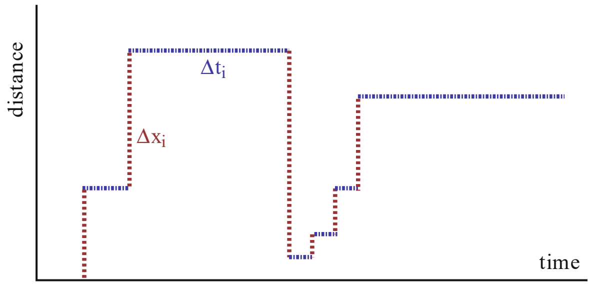


Fig. 3.12 A random walker with distributed waiting times Δt_i and jumps Δx_i may become a generalized Lévy flight, compare (3.39)

3.3.2 Markov Chains

For many common stochastic processes $x_1 \to x_2 \to x_3 \to \ldots$ the probability to visit a state $x_{t+1} = y$ depends solely on the current state $x_t = x$.

MARKOV PROPERTY A stochastic process is "markovian" if it has no memory.

A memory would be present, on the other hand, if the transition rule $x_t \to x_{t+1}$ would be functionally dependent on earlier $x_{t-1}, x_{t-2}, \ldots$ elements of the process.

Absorbing States The transition probabilities $p(x, y)$ to visit a state $x_{t+1} = y$, when being at $x_t = x$, are normalized,

$$1 = \sum_y p(x, y), \qquad p(x, y) \geq 0\,, \tag{3.40}$$

since one always arrives to some state $x_{t+1} = y$ when starting from a given $x_t = x$. A process may stay in place, which occurs with the probability $p(x, x)$. A state x^* is "absorbing" whenever

$$p(x^*, x^*) = 1, \qquad p(x^*, y) = 0, \qquad \forall y \neq x^*\,. \tag{3.41}$$

A stochastic process can be viewed as being terminated, or "extinct", when reaching an absorbing state. The extinction probability is then the probability to hit x^* when

Table 3.1 The four regimes of a generalized walker with distribution functions, Eq. (3.39), characterized by scalings $\sim (\Delta t)^{-1-\alpha}$ and $\sim (\Delta x)^{-1-\beta}$ for the waiting times Δt and jumps Δx, as depicted in Fig. 3.12

$\alpha > 1$	$\beta > 2$	$\bar{x} \sim \sqrt{t}$	Ordinary diffusion
$\alpha > 1$	$0 < \beta < 2$	$\bar{x} \sim t^{1/\beta}$	Lévy flights
$0 < \alpha < 1$	$\beta > 2$	$\bar{x} \sim t^{\alpha/2}$	Subdiffusion
$0 < \alpha < 1$	$0 < \beta < 2$	$\bar{x} \sim t^{\alpha/\beta}$	Ambivalent processes

starting from a given state x_0. A famous example is the Galton-Watson process, which describes the extinction probabilities of family names.[11]

Master Equation We consider density distributions $\rho_t(x)$ of walkers, with each walker having the same transition probabilities $p(x, y)$. For discrete times $t = 0, 1, \ldots$, the evolution of the density of walkers is given by the "master equation"

$$\begin{aligned} \rho_{t+1}(y) &= \rho_t(y) + \textstyle\sum_x \big[\, \rho_t(x)p(x, y) - \rho_t(y)p(y, x)\,\big] \\ &= \rho_t(y) + \textstyle\sum_x \rho_t(x)p(x, y) - \rho_t(y) \\ &= \textstyle\sum_x \rho_t(x)p(x, y)\,, \end{aligned} \tag{3.42}$$

where we took into account that the number of walkers is conserved, namely that $\sum_y p(x, y) = 1$. Random walks and any other stochastic time series, are described by their defining master equations.

Stationarity A Markov process becomes stationary when the distribution of walkers does not change anymore with time, viz when

$$\rho^*(y) = \rho_{t+1}(y) = \rho_t(y), \qquad \sum_x \rho^*(x)p(x, y) = \rho^*(y), \qquad \rho^* P = \rho^*\,,$$

where P is the transition matrix $p(x, y)$. The stationary distribution of walkers ρ^* is consequently a left eigenvector of P.

General Two-State Markov Process As an example we define with

$$P = \begin{pmatrix} \alpha & 1-\alpha \\ 1-\beta & \beta \end{pmatrix}, \qquad \alpha, \beta \in [0, 1] \tag{3.43}$$

the transition matrix P for the general two-state Markov process, compare Fig. 3.13. The eigenvalues λ of the left eigenvectors $\rho^* = (\rho_1, \rho_2)$ of P are determined by

$$\begin{array}{rl} \alpha\,\rho_1 + (1-\beta)\rho_2 &= \lambda\rho_1 \\ (1-\alpha)\rho_1 + \qquad \beta\,\rho_2 &= \lambda\rho_2 \end{array}, \tag{3.44}$$

which has the solutions

$$\lambda_1 = 1, \qquad \rho^*_{\lambda_1} = \frac{1}{N_1}\begin{pmatrix} 1-\beta \\ 1-\alpha \end{pmatrix}, \qquad N_1 = \sqrt{(1-\alpha)^2 + (1-\beta)^2}$$

[11] The Galton-Watson process will be treated in Sect. 6.5.2 of Chap. 6.

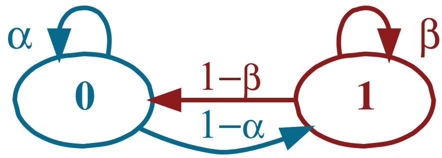


Fig. 3.13 The general two-state Markov chain, as defined by (3.43)

and

$$\lambda_2 = \alpha + \beta - 1, \qquad \rho^*_{\lambda_2} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The first eigenvalue dominates generically, $\lambda_1 = 1 > |\alpha + \beta - 1| = |\lambda_2|$, with the contribution to $\rho^*_{\lambda_2}$ dying out. Absorbing states are present whenever $\alpha/\beta = 1$, see Fig. 3.13.

Random Surfer Model A famous diffusion process is the "random surfer model" which tries to capture the behavior of Internet users. This model is at the basis of the original Page & Brin Google page-rank algorithm.

Consider a network of $i = 1, \dots, N$ Internet hosts connected by directed hyperlinks characterized by the adjacency matrix A_{ij}.[12] We denote with

$$\rho_i(t), \qquad \sum_{i=1}^{N} \rho_i(t) = 1$$

the probability of finding an Internet surfer visiting host i at time t. The surfers are assumed to perform a markovian walk on the Internet by clicking randomly any available out-going hyperlink, giving raise to the master equation

$$\rho_i(t+1) = \frac{c}{N} + (1-c)\sum_j \frac{A_{ij}}{\sum_l A_{lj}} \rho_j(t)\,. \tag{3.45}$$

Normalization is conserved,

$$\sum_i \rho_i(t+1) = c + (1-c)\sum_j \frac{\sum_i A_{ij}}{\sum_l A_{lj}} \rho_j(t) = c + (1-c)\sum_j \rho_j(t)\,.$$

Hence $\sum_i \rho_i(t+1) = 1$ whenever $\sum_j \rho_j(t) = 1$.

[12] More about the adjaceny matrix in Sect. 1.2 of Chap. 1.

Google Page Rank The parameter c in the random surfer model regulates the probability to randomly enter the Internet:

- For $c = 1$ the adjacency matrix and hence the hyperlinks are irrelevant. We can interpret therefore c as the uniform probability to enter the Internet.
- For $c = 0$ a surfer never enters or leaves the Internet, continuing to click around forever, c is hence also the probability to stop clicking hyperlinks.

The random surfer model (3.45) can be solved iteratively. Convergence is fast for not too small c. At every iteration authority is transferred from one host j to other hosts i through its outgoing hyperlinks A_{ij}. The steady-state density ρ_i of surfers can hence be considered as a measure of host authority and is equivalent to the orginal Google page rank, which was an important score for ranking search results at the dawn of the Internet.

Relation to Graph Laplacian The continuous time version of the random surfer model can be derived, for the case $c = 0$, from

$$\frac{\rho_i(t+\Delta t) - \rho_i(t)}{\Delta t} = \sum_j \frac{A_{ij}}{k_j}\rho_j(t) - \rho_i(t), \qquad k_j = \sum_l A_{lj}\,,$$

where k_j is the out-degree of host j and Δt the time step. Taking the limit $\Delta t \to 0$ yields

$$\frac{d}{dt}\rho = \tilde{\Lambda}\,\rho, \qquad \tilde{\Lambda}_{ij} = -\frac{\Lambda_{ij}}{k_j}, \qquad \Lambda_{ij} = k_j\delta_{ij} - A_{ij}\,, \tag{3.46}$$

where Λ_{ij} is the Graph Laplacian.[13] Equation (3.46) corresponds to a generalization of the diffusion equation (3.33) to networks.

3.4 Stochastic Systems

3.4.1 Langevin Equation

Diffusion as a Stochastic Process Langevin proposed to describe the diffusion of a particle by the stochastic differential equation,

$$m\,\dot{v} = -m\,\gamma\,v + \xi(t), \qquad \langle\xi(t)\rangle = 0, \qquad \langle\xi(t)\xi(t')\rangle = Q\delta(t-t')\,, \tag{3.47}$$

where $v(t)$ is the velocity of the particle and $m > 0$ its mass.

[13] The Graph Laplacian is treated in Sect. 1.2.1 of Chap. 1.

– The term $-m\gamma v$ on the right-hand-side of (3.47) corresponds to a damping term, the friction being proportional to $\gamma > 0$.
– $\xi(t)$ is a stochastic variable, viz noise. The brackets $\langle \ldots \rangle$ denote ensemble averages, i.e. averages over different noise realizations.
– As "white noise" (in contrast to "colored noise") one denotes noise with a flat power spectrum (as white light), viz $\langle \xi(t)\xi(t')\rangle \propto \delta(t-t')$.
– The constant Q is a measure for the strength of the noise.

All stochastic systems have a noise term on the right-hand side.

Solution of the Langevin Equation Considering a specific noise realization $\xi(t)$, one finds

$$v(t) = v_0\,\mathrm{e}^{-\gamma t} + \frac{\mathrm{e}^{-\gamma t}}{m}\int_0^t dt'\,\mathrm{e}^{\gamma t'}\,\xi(t') \tag{3.48}$$

for the formal solution of the Langevin equation (3.47), where $v_0 \equiv v(0)$.

Mean Velocity Taking the ensemble average $\langle v(t)\rangle$ of the velocity leads to

$$\langle v(t)\rangle = v_0\,\mathrm{e}^{-\gamma t} + \frac{\mathrm{e}^{-\gamma t}}{m}\int_0^t dt'\,\mathrm{e}^{\gamma t'}\,\underbrace{\langle\xi(t')\rangle}_{0} = v_0\,\mathrm{e}^{-\gamma t}\,. \tag{3.49}$$

The average velocity decays hence exponentially to zero.

Mean Square Velocity For the ensemble average $\langle v^2(t)\rangle$ of the velocity squared one finds

$$\begin{aligned}
\langle v^2(t)\rangle &= v_0^2\,\mathrm{e}^{-2\gamma t} + \frac{2\,v_0\,\mathrm{e}^{-2\gamma t}}{m}\int_0^t dt'\,\mathrm{e}^{\gamma t'}\,\underbrace{\langle\xi(t')\rangle}_{0} \\
&\quad + \frac{\mathrm{e}^{-2\gamma t}}{m^2}\int_0^t dt'\int_0^t dt''\,\mathrm{e}^{\gamma t'}\,\mathrm{e}^{\gamma t''}\,\underbrace{\langle\xi(t')\xi(t'')\rangle}_{Q\,\delta(t'-t'')} \\
&= v_0^2\,\mathrm{e}^{-2\gamma t} + \frac{Q\,\mathrm{e}^{-2\gamma t}}{m^2}\underbrace{\int_0^t dt'\,\mathrm{e}^{2\gamma t'}}_{(\mathrm{e}^{2\gamma t}-1)/(2\gamma)}\,,
\end{aligned}$$

and finally

$$\langle v^2(t)\rangle = v_0^2\,\mathrm{e}^{-2\gamma t} + \frac{Q}{2\,\gamma\,m^2}\left(1-\mathrm{e}^{-2\gamma t}\right)\,. \tag{3.50}$$

For long times the average squared velocity

$$\lim_{t\to\infty}\langle v^2(t)\rangle = \frac{Q}{2\,\gamma\,m^2} \tag{3.51}$$

becomes, as expected, independent of the initial velocity v_0. Equation (3.51) shows explicitly that the dynamics is driven exclusively by the stochastic process $\propto Q$ for long time scales.

Langevin Equation and Diffusion The Langevin equation is formulated in terms of the particle velocity. In order to make connection with the time evolution of a real-space random walker, see (3.36), we multiply the Langevin equation by x, take at the same time the ensemble average,

$$\langle x\,\dot{v}\rangle = -\gamma\,\langle x\,v\rangle + \frac{1}{m}\langle x\,\xi\rangle\,. \tag{3.52}$$

We note that

$$x\,v = x\,\dot{x} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{x^2}{2}, \qquad x\,\dot{v} = x\,\ddot{x} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{x^2}{2} - \dot{x}^2$$

and

$$\langle x\xi\rangle = \left\langle \xi(t)\int_0^t v(t')dt'\right\rangle = \int_0^t dt'\int_0^{t'} dt''\,\frac{\mathrm{e}^{-\gamma(t'-t'')}}{m}\,\underbrace{\left\langle \xi(t)\xi(t'')\right\rangle}_{Q\delta(t-t'')} = 0\,,$$

where we have used (3.48) in the limit of large times and that $t'' < t$. We then find

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{\langle x^2\rangle}{2} - \langle v^2\rangle \;=\; -\gamma\,\frac{\mathrm{d}}{\mathrm{d}t}\frac{\langle x^2\rangle}{2}$$

for (3.52), or

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\langle x^2\rangle \;+\; \gamma\,\frac{\mathrm{d}}{\mathrm{d}t}\langle x^2\rangle \;=\; 2\langle v^2\rangle \;=\; \frac{Q}{\gamma m^2}\,, \tag{3.53}$$

the latter with the help of the long-time result (3.51) for $\langle v^2\rangle$. The solution of (3.53) is

$$\langle x^2\rangle \;=\; \left[\gamma t - 1 + \mathrm{e}^{-\gamma t}\right]\frac{Q}{\gamma^3 m^2}\,. \tag{3.54}$$

For long times we find we

$$\lim_{t\to\infty} \langle x^2 \rangle \simeq \frac{Q}{\gamma^2 m^2}\, t \equiv 2Dt, \qquad D = \frac{Q}{2\gamma^2 m^2} \tag{3.55}$$

that the solutions of the Langevin equation show diffusive behavior, compare (3.36). This result, that $D \propto Q$, underpins the notion that diffusion is microscopically due to a stochastic process.

Massless Limit We add an external force $F(x)$ to the Langevin equation (3.47),

$$\dot{x} = v, \qquad m\,\dot{v} = -m\,\gamma\,v + F(x) + \xi(t)\,, \tag{3.56}$$

where we included the definition $v = \dot{x}$ of the velocity. Keeping $\Gamma = m\gamma$ constant while taking the limit $m \to 0$ leads to a well defined diffusion constant $D = Q/(2\Gamma^2)$, as given by (3.55). In this limit, the Langevin equation reduces to

$$\Gamma\dot{x} = F(x) + \xi(t)\,. \tag{3.57}$$

For stochastic processes in non-physical settings, like in finance, one usually starts with (3.57), which may be further adapted to the problem at hand.

3.4.2 Stochastic Calculus

Stochastic effects may depend functionally on the location x, e.g. via

$$\Gamma\dot{x} = F(x) + b(x)\xi(t)\,, \tag{3.58}$$

which is called the "non-linear Langevin equation". While looking like a fairly innocuous term, $b(x)\xi$ is actually not uniquely defined. The random kicks the system receives at time t depend via $b(x) = b(x(t))$ on a yet undefined position $x(t)$. Equation (3.58) needs hence to be supplemented with a rule on how to treat the last term. This is usually done by looking at an integral over a small time inveral $[t, t + \Delta t]$.

Ito Stochastic Calculus Assuming that the strength of the kicks received is determined by the position of the sytem immediately before the kick takes place is consistent with the substitution

$$\int_t^{t+\Delta t} dt'\, b(x(t'))\,\xi(t') \quad\to\quad b(x(t)) \int_t^{t+\Delta t} dt'\,\xi(t')\,, \tag{3.59}$$

which is known as the "Ito stochastic calculus".

Stratonovich Stochastic Calculus Alternatively, one may assume that the strength of the individual kicks depend on a suitable average of the position. A possibility to model this situation is to substitute $\int_t^{t+\Delta t} dt'\, b(x(t'))\, \xi(t')$ by

$$\frac{b(x(t)) + b(x(t+\Delta t))}{2} \int_t^{t+\Delta t} dt'\, \xi(t')\,, \tag{3.60}$$

which is known as the "Stratonovich stochastic calculus". The two formalisms, Stratonovich and Ito, lead in general to different results. In physics, Stratonovich is usually the correct choice, with Ito describing what happens in finance.

3.5 Noise-Controlled Dynamics

Stochastic Systems A set of first-order differential equations with a stochastic term is generally denoted a "stochastic system". The Langevin equation (3.47) is a prominent example. Depending on the circumstances, the stochastic component may determine the long-term dynamical behavior alltogether. Some examples.

– NEURAL NETWORKS
  Networks of interacting neurons are responsible for the cognitive information processing in the brain. They must remain functional also in the presence of noise and be stable as stochastic systems. In this case the introduction of a noise term to the evolution equation should not change the dynamics qualitatively. This postulate should be valid for the vast majorities of biological networks.
– DIFFUSION
  The Langevin equation reduces, in the absence of noise, to a damped motion without an external driving force, with $v = 0$ acting as a global attractor. The stochastic term is therefore essential in the long-time limit, leading to diffusive behavior, with $\langle x^2 \rangle \propto t$.
– STOCHASTIC ESCAPE AND STOCHASTIC RESONANCE
  A particle trapped in a local minimum may escape the minimum by a noise-induced diffusion process; a phenomenon denoted "stochastic escape". Stochastic escape in a driven bistable system leads to an even more subtle consequence of noise-induced dynamics, "stochastic resonance".

In the following we detail out both stochastic escape and stochastic resonance.

3.5.1 Fokker–Planck Equation

Drift Velocity We add an external potential $V(x)$ to the Langevin equation (3.47),

$$m\,\dot{v} = -m\,\gamma\,v + F(x) + \xi(t), \qquad F(x) = -V'(x) = -\frac{d}{dx}V(x)\,, \tag{3.61}$$

where v and m are the velocity and the mass of the particle, $\langle \xi(t) \rangle = 0$ and $\langle \xi(t)\xi(t') \rangle = Q\delta(t-t')$. In the absence of damping and noise, when $\gamma = 0 = Q$, Eq. (3.61) reduces to Newton's law.

We consider for a moment a constant force $F(x) = F$ and the absence of noise, $\xi(t) \equiv 0$. The system reaches an equilibrium for $t \to \infty$ when relaxation and force cancel each other:

$$m\,\dot{v}_D = -m\,\gamma\,v_D + F \equiv 0, \qquad v_D = \frac{F}{\gamma m}\,. \tag{3.62}$$

v_D is called the "drift velocity". A typical example is the motion of electrons in a metallic wire. As described by Ohm's law, an applied voltage leads to an electric field along the wire, which induces in turn an electrical current. As a result one has drifting electrons that are continuously accelerated by the electrical field, while bumping into lattice imperfections or colliding with the lattice vibrations, i.e. with phonons.

Continuity Equation Generalizing to an ensemble of particles diffusing in an external potential, we denote with $P(x,t)$ the density of particles at location x and time t. Particle number conservation defines the particle current density $J(x,t)$ via the continuity equation

$$\frac{\partial P(x,t)}{\partial t} + \frac{\partial J(x,t)}{\partial x} = 0. \tag{3.63}$$

There are two contributions, J_D and J_ξ, to the total particle current density, $J = J_D + J_\xi$, induced respectively by diffusion and by stochastic motion. We derive these two contributions in two steps.

Drift and Diffusion Currents In a first step we disregard noise in (3.61) and set $Q = 0$. In the stationary limit particles move in this case uniformly, with the drift velocity v_D. The respective current density is

$$J_D = v_D\,P(x,t)\,.$$

In a second step we derive the contribution J_ξ of the noise term $\sim \xi(t)$ to the particle current density by setting the force term to zero, $F = 0$. For this purpose we rewrite the diffusion equation (3.33) with

$$\frac{\partial P(x,t)}{\partial t} = D\,\frac{\partial^2 P(x,t)}{\partial x^2} \equiv -\frac{\partial J_\xi(x,t)}{\partial x} \qquad \frac{\partial P(x,t)}{\partial t} + \frac{\partial J_\xi(x,t)}{\partial x} = 0$$

as a continuity equation, which allows us to determine the functional form of J_ξ,

$$J_\xi = -D\frac{\partial P(x,t)}{\partial x}\,. \tag{3.64}$$

Fokker–Planck Equation We recall the relation $D = Q/(2\gamma^2 m^2)$ between the diffusion constant D and the amplitude of the noise term Q, see (3.55). Adding both current contributions, we obtain

$$\begin{aligned} J(x,t) &= v_D\, P(x,t) - D\,\frac{\partial P(x,t)}{\partial x} \\ &= \frac{F}{\gamma m}\, P(x,t) - \frac{Q}{2\gamma^2 m^2}\,\frac{\partial P(x,t)}{\partial x} \end{aligned} \tag{3.65}$$

for the total current density $J = J_D + J_\xi$. Substituting this expression for the total particle current density into the continuity equation, see (3.63), one obtains the "Fokker–Planck" or "Smoluchowski" equation

$$\frac{\partial P(x,t)}{\partial t} = -\frac{\partial v_D\, P(x,t)}{\partial x} + \frac{\partial^2 D\, P(x,t)}{\partial x^2}\,, \tag{3.66}$$

aka the master equation of the density distribution $P(x,t)$. The first and second term on the right-hand side correspond respectively to ballistic and diffusive transport. Without going into details, we note that above expression for the Fokker–Planck equation is consistent with the Ito stochastic calculus, as defined by (3.59).

3.5.2 Stochastic Escape

Harmonic Potential One can solve the Fokker–Planck equation (3.66) analytically for a harmonic confining potential,

$$V(x) = \frac{f}{2}\,x^2, \qquad\qquad F(x) = -f\,x\,.$$

We are interested in particular in the stationary density distribution,

$$\frac{dP(x,t)}{dt} = 0 \qquad \Longrightarrow \qquad \frac{dJ(x,t)}{dx} = 0\,,$$

where the second equation follows from the continuity condition (3.63). With (3.65) for the total current density we find

$$\frac{d}{dx}\left[\frac{f\,x}{\gamma\, m} + \frac{Q}{2\gamma^2 m^2}\,\frac{d}{dx}\right] P(x) = 0 = \frac{d}{dx}\left[\beta f\,x + \frac{d}{dx}\right] P(x)\,,$$

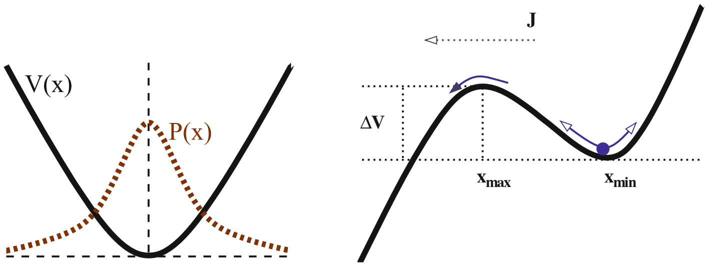


Fig. 3.14 **Left:** Stationary distribution $P(x)$ of diffusing particles in a harmonic potential $V(x)$. **Right:** Stochastic escape from a local minimum, with $\Delta V = V(x_{\max}) - V(x_{\min})$ being the potential barrier height and J the escape current

where we used $F = -fx$ in (3.65). We defined $\beta = 2\gamma m/Q$ together with the stationary limit $P(x) = \lim_{t\to\infty} P(x,t)$ for the distribution of random walkers. The system is confined, which implies that the steady-state current vanishes, which implies in turn that

$$0 = \left[\beta f x + \frac{d}{dx}\right] P(x) \tag{3.67}$$

holds. The corresponding solution is

$$P(x) = A\,\mathrm{e}^{-\beta f x^2/2} = A\,\mathrm{e}^{-\beta V(x)}, \qquad A = \frac{1}{\sqrt{2\pi\sigma^2}}, \tag{3.68}$$

where $\sigma^2 = 1/(\beta f) = Q/(2f\gamma m)$, with normalization condition $\int dx\, P(x) = 1$ being fulfilled. The density of diffusing particles in a harmonic trap is Gaussian-distributed, see Fig. 3.14.

Escape Current We now consider particles in a local minimum, as depicted in Fig. 3.14. A typical partially confining potential has a functional form like

$$V(x) \sim -x + x^3. \tag{3.69}$$

Without noise, the particle will oscillate around the local minimum eventually coming to a standstill under the influence of friction, with $x \to x_{\min}$. With noise, the particle will have a small but finite probability

$$\propto \mathrm{e}^{-\beta \Delta V}, \qquad \Delta V = V(x_{\max}) - V(x_{\min})$$

to reach the next saddle, where ΔV is the potential difference between the saddle and the local minimum, see Fig. 3.14.

The solution (3.68) for the stationary particle distribution in a confining potential $V(x)$ has a vanishing total current J. For non-confining potentials, like (3.69), the particle current $J(x,t)$ never vanishes. Stochastic escape occurs when starting with a density of diffusing particles close the local minimum, as illustrated in Fig. 3.14. The escape current will be nearly constant whenever the escape probability is small. In this case the escape current

$$J(x,t)\Big|_{x=x_{\max}} \propto \mathrm{e}^{-\beta\,[V(x_{\max})-V(x_{\min})]},$$

will be proportional to the probability a particle has to reach the saddle. Functionally, we did approximate $P(x)$ with one valid for a perfect harmonic potential, as given by (3.68).

Kramer's Escape When the escape current is finite, there is a finite probability per unit of time for the particle to escape the local minima, the "Kramer's escape rate", r_K,

$$r_K = \frac{\omega_{\max}\omega_{\min}}{2\pi\,\gamma}\,\exp\big[-\beta\,(V(x_{\max})-V(x_{\min}))\big], \tag{3.70}$$

where $\beta = 2\gamma m/Q$ and where the prefactors can be derived from a more detailed calculation, with $\omega_{\min}^2 = |V''(x_{\min})|/m$ and $\omega_{\max}^2 = |V''(x_{\max})|/m$.

Stochastic Escape in Evolution Stochastic escape occurs in many real-world systems. Noise allows the system to escape from a local minimum where it would otherwise remain stuck for eternity. As an example, we mention stochastic escape from a local fitness maximum (in evolution fitness is to be maximized) by random mutations that play the role of noise.[14]

3.5.3 Stochastic Resonance

Driven Double-Well Potential We consider diffusive dynamics in a driven double-well potential, see Fig. 3.15,

$$\dot{x} = -V'(x) + A_0\cos(\Omega t) + \xi(t), \qquad V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4\,. \tag{3.71}$$

Several remarks.

[14] Stochastic escape from local fitness maxima will be discussed in more detail in Sect. 8.4 of Chap. 8.

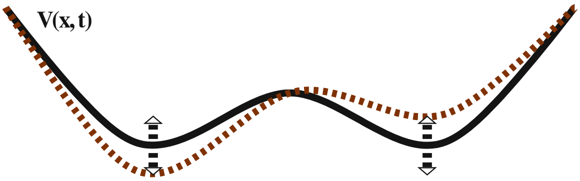


Fig. 3.15 The driven double-well potential, $V(x) - A_0 \cos(\Omega t)x$, compare (3.71). The driving force is small enough to retain the two local minima

- Equation (3.71) is an example for a Langevin equation in the massless limit (3.57), here with $\Gamma = m\gamma = 1$. Friction has hence been taken to diverge.
- For the potential $V(x)$ a normal form has been taken, which one can always achieve by rescaling variables appropriately.
- The potential has two stable minima x_0,

$$-V'(x) = 0 = x - x^3 = x(1 - x^2), \qquad x_0 = \pm 1 .$$

  The local maximum at $x_0 = 0$ is unstable.
- We assume that the periodic driving $\propto A_0$ is small enough, such that the effective potential $V(x) - A_0 \cos(\Omega t)x$ retains two minima at all times, compare Fig. 3.15.

Transient State Dynamics The system will stay close to one of the two minima, $x \approx \pm 1$, for most of the time when both A_0 and the noise strength are weak, as illustrated in Fig. 3.16. The dynamics is therefore characterized by rapid transitions between transiently stable states.

Switching Times An important question is then: How often does the system switch between the two preferred states $x \approx 1$ and $x \approx -1$? There are two time scales present.

- STOCHASTIC ESCAPE
  In the absence of external driving, $A_0 \equiv 0$, the transitions are noise driven and irregular, with the average switching time given by Kramer's lifetime $T_K = 1/r_K$, see Fig. 3.16. The system is translational invariant with respect to time and the ensemble averaged expectation value

$$\langle x(t) \rangle \; = \; 0$$

  therefore vanishes in the absence of an external force.
- EXTERNAL FORCING
  When $A_0 \neq 0$ the external force induces a reference time together with a non-zero response $\bar{x}$,

$$\langle x(t) \rangle = \bar{x} \, \cos(\Omega t - \bar{\phi}) \, , \tag{3.72}$$

  which follows the time evolution of the driving potential with a certain phase shift $\bar{\phi}$, see Fig. 3.17.

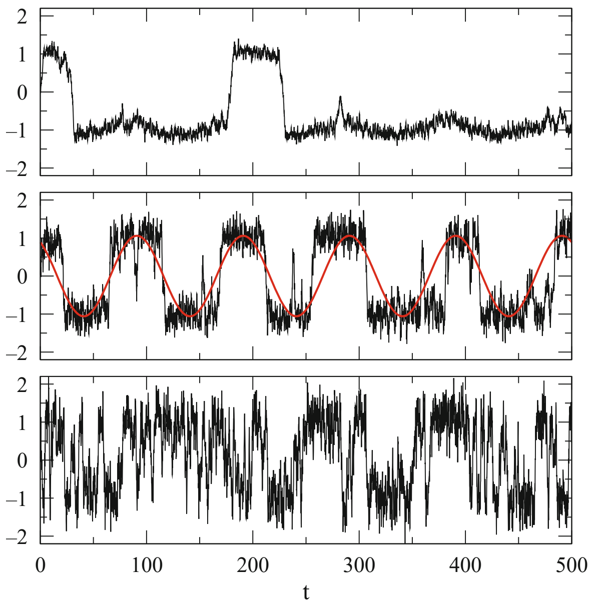


Fig. 3.16 Example trajectories $x(t)$ for the driven double-well potential. The strength and the period of the driving potential are $A_0 = 0.3$ and $2\pi/\Omega = 100$, respectively. The noise level Q is 0.05, 0.3 and 0.8 (top/middle/bottom), see (3.71)

The phenomenon of stochastic resonance regards the size of the response, as measured by $\bar{x}$

Resonance Condition When the time scale $2T_K = 2/r_K$ to switch back and forth due to the stochastic process equals the period $2\pi/\Omega$, we expect a large response $\bar{x}$, see Fig. 3.17. The time-scale matching condition

$$\frac{2\pi}{\Omega} \approx \frac{2}{r_K}$$

depends via (3.70) on the noise-level Q, which enters the Kramer's escape rate r_K. For otherwise constant parameters, the response $\bar{x}$ first increases with rising Q, decreasing however for elevated noise levels Q. This is the telltale characteristic of "stochastic resonance", as shown in Fig. 3.17.

Ice Ages The average temperature T_e of the earth differs by about $\Delta T_e \approx 10\,°\mathrm{C}$ in between a typical ice age and interglacial periods. Both states of the climate are locally stable.

- Ice Age
  A substantial ice covering increases the albedo of the earth, which leads in turn to a larger part of sunlight to be reflected back to space. Earth remains cool.

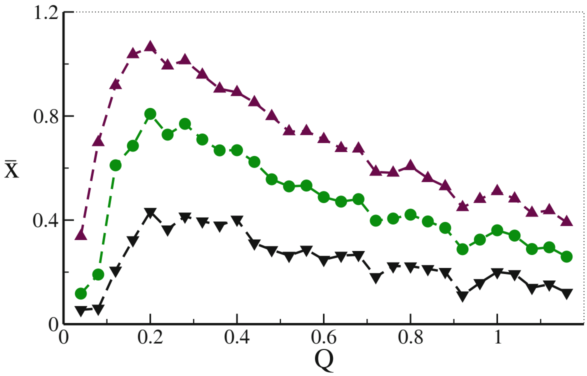


Fig. 3.17 The gain $\bar{x}$, see (3.72), as a function of noise level Q. The strength of the driving amplitude A_0 is 0.1, 0.2 and 0.3 (bottom/middle/top curves), see (3.71) and the period $2\pi/\Omega = 100$. The response $\bar{x}$ is very small for vanishing noise $Q = 0$, when the system performs only small-amplitude oscillations in one of the local minima

– INTERGLACIAL PERIOD
  The ice covering is reduced. A larger portion of the sunlight is absorbed by the oceans and land, with earth remaining warm.

A parameter of the orbit of planet earth, the eccentricity, varies slightly with a period $T = 2\pi/\Omega \approx 10^5$ years. The intensity of the incoming radiation from the sun therefore varies with the same period. Long-term climate changes can therefore be modeled by a driven two-state system, i.e. by (3.71). The driving force, viz the variation of the energy flux the earth receives from the sun, is however small. The increase in the amount of incident sunlight is too weak to pull the earth out of an ice age into an interglacial period or vice versa. Random climatic fluctuation, like variations in the strength of ocean circulations, are needed to finish the job. The alternation of ice ages with interglacial periods may therefore be modeled as a stochastic resonance phenomenon.

Beyond Stochastic Resonance Resonance phenomena generally occur when two frequencies, or two time scales, match as a function of some control parameter. For the case of stochastic resonance these two time scales correspond to the period of the external driving and to the average waiting time for Kramer's escape respectively, with the latter depending directly on the level of the noise. The phenomenon is denoted as stochastic resonance since one of the time scales involved is controlled by noise.

One generalization of stochastic resonance is "coherence resonance". In this case one has a dynamical system with two internal time scales, say t_1 and t_2. These two time scales can be affected to a different degree by an additional source of noise. An additional stochastic term may therefore change the ratio t_1/t_2, leading to internal resonance phenomena.

Exercises

(3.1) SHIFT MAP
Using the representation $x_n = (1 - \cos(\pi\theta_n))/2$ show that the logistic map (2.46) at $r = 4$ is equivalent to the shift map

$$\theta_{n+1} = (2\theta_n)\%1 = \begin{cases} 2\theta_n & \text{for} \quad 0 < \theta < 0.5 \\ 2\theta_n - 1 & \text{for} \quad 0.5 < \theta < 1 \end{cases} \tag{3.73}$$

where the %-sign denotes the modulus operation. This representation can be used to evaluate analytically the distribution $p(x)$ of finding x when iterating the logistic map at $r = 4$ ad infinitum.

(3.2) FIXPOINTS OF THE LORENZ MODEL
Perform the stability analysis of the fixpoint $(0, 0, 0)$ and of $C_{+,-} = (\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1)$ for the Lorenz model (3.8) with $r, b > 0$.

(3.3) HAUSDORFF DIMENSION OF THE CANTOR SET
Calculate the Hausdorff dimension of the Cantor set, which is generated by removing consecutively the middle-1/3 segment of a line having a given initial length. Start with the Hausdorff dimension of simple straight line.

(3.4) DRIVEN HARMONIC OSCILLATOR
Solve the driven, damped harmonic oscillator

$$\ddot{x} + \gamma\,\dot{x} + \omega_0^2\,x = \epsilon\cos(\omega t)$$

in the long-time limit. Discuss the behavior close to the resonance $\omega \to \omega_0$.

(3.5) MARKOV CHAIN OF UMBRELLAS
Lady Ann has four umbrellas which she uses whenever it rains, to go from work to home, or vice versa. She takes only an umbrella with her whenever it rains, leaving the umbrellas otherwise in the office and at home respectively. It rains with probability $p \in [0, 1]$. How often does Lady Ann get wet?

(3.6) BIASED RANDOM WALK
Generalize the derivation of the diffusion equation (3.33) for a random walker jumping with probabilities $(1 \pm \alpha)/2$ either to the right or to the left, with $\alpha \in [-1 : 1]$. How does α needs to scale such that a non-trivial contribution is retained in the limit $\Delta t \to 0$ and $\Delta x \to 0$? What kind of solutions does one find for a vanishing diffusion constant $D \to 0$?

(3.7) CONTINUOUS-TIME LOGISTIC EQUATION
Consider the continuous-time logistic equation

$$\dot{y}(t) = \alpha y(t)\Big[1 - y(t)\Big].$$

Find the general solution. Furthermore, compare to the logistic map (2.46) for discrete times $t = 0,\ \Delta t,\ 2\Delta t,\ \ldots$

(3.8) NOISE-INDUCED QUASI-PERIODIC ORBITS

Consider with

$$\dot{r}(t) = g(1-r), \qquad \dot{\varphi} = 1 - K\cos(\varphi) \tag{3.74}$$

a system showing a saddle-node bifurcation on a limit cycle, as illustrated in Fig. 2.13. Write a script and demonstrated that noise may allow the orbit to pass from the stable to the unstable fixpoint. Use the case of noisy measurements, which corresponds to substitute

$$r \;\to\; r(1+\xi), \qquad \varphi \;\to\; \varphi(1+\xi), \tag{3.75}$$

on the right-hand-side of (3.74). Here $\xi = \xi(t)$ is normal distributed with mean zero and standard deviation $\sigma \ll 1$.

Further Reading

For further studies we refer to introductory texts, on dynamical systems theory, Datseris and Parlitz (2022), containing Julia code snippets, on chaos theory and fractals, Layek (2015), and to textbooks on stochastic systems, Karatzas and Shreve (2012), Kulkarni (2002).

Overview articles potentially of interest for the reader are Benzi (2010), for stochastic resonance as a real-world phenomenon, Ginoux and Letellier (2012) for the history of Van der Pol and relaxation oscillators, and Baronchelli and Radicchi (2013) for the occurrence of Lévy flights. Partially predictable chaos is treated in Wernecke et al. (2017).

For insights into some of the original literature we recommend Einstein (1905) and Langevin (1908) on Brownian motion, or the first formulation and study of the Lorenz (1963) model.

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Self Organization

4

Self-organized pattern formation occurs when complex spatio-temporal structures result from seemingly simple dynamical evolution processes. The formation of animal markings can be understood in this context in terms of a Turing instability. Zebra stripes, and other biological patterns, emerge in reaction-diffusion systems, which will be discussed together with the notion of self-stabilizing wavefronts, as observed for the Fisher equation.

Further prominent examples of self-organizing processes treated in this chapter involve collective decision making and swarm intelligence, as occurring in social insects and flocking birds, information offloading in terms of stigmergy, opinion dynamics and the physics of traffic flows, including the ubiquitous phenomenon of self-organized traffic congestions.

4.1 Interplay Between Diffusion and Reaction

Processes characterized by the random motion of particles or agents are said to be diffusive. Consequently, they are described by the diffusion equation[1]

$$\frac{\partial}{\partial t}\rho(\mathbf{x},t) = D\Delta\rho(\mathbf{x},t), \qquad \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \tag{4.1}$$

with $D > 0$ denoting the diffusion constant, Δ the Laplace operator and $\rho(\mathbf{x},t)$ the density of walkers at any given point $(\mathbf{x},t)$ in space and time.

[1] Diffusion is treated in depth in Sect. 3.3 of Chap. 3.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_4

Reaction-Diffusion Systems The diffusion Eq. (4.1) describes a conserving process, which implies that the overall number of particles, $\int d\mathbf{x}\rho(\mathbf{x},t)$, remains constant. With

$$\frac{\partial}{\partial t}\rho(\mathbf{x},t) = R(\rho) + D\Delta\rho(\mathbf{x},t) \tag{4.2}$$

a "reaction-diffusion system" is formulated, where $R(\rho)$ constitutes the reaction of the system to the current state $\rho = \rho(\mathbf{x},t)$.

In biological settings ρ typically stands for the population density and $R(\rho)$ for reproduction processes, in the context of chemical reactions ρ is a measure for the relative concentration of a given substance whereas $R(\rho)$ functionally describes effective reaction rates. Reaction-diffusion systems are fundamental for describing the self-organized formation of spatial-temporal patterns.

Fisher Equation Considering a one-dimensional system and logistic growth for the reaction term, one obtains the "Fisher equation"

$$\dot{\rho} = r\rho(1-\rho) + D\rho'', \qquad \rho \in [0,1]\,, \tag{4.3}$$

which describes a reproductive and diffusive species in an environment with spatially local resource limitation.

Normal Form One-component reaction-diffusion equations can be cast into a dimensionless normal form, by rescaling the time and space coordinates appropriately via

$$t = \alpha\tilde{t}, \qquad x = \beta\tilde{x}, \qquad \frac{\partial}{\partial t} = \frac{1}{\alpha}\frac{\partial}{\partial \tilde{t}}, \qquad \frac{\partial^2}{\partial x^2} = \frac{1}{\beta^2}\frac{\partial^2}{\partial \tilde{x}^2}\,.$$

The rescaled Fisher equation takes the form

$$\frac{\partial\rho}{\partial\tilde{t}} = \alpha r\rho(1-\rho) + \frac{\alpha D}{\beta^2}\frac{\partial^2\rho}{\partial\tilde{x}^2}, \qquad \alpha r = 1, \qquad \frac{\alpha D}{\beta^2} = 1\,.$$

It is hence sufficient to consider the normal form

$$\dot{\rho} = \rho(1-\rho) + \rho'' \tag{4.4}$$

of the Fisher equation.

Saturation and Wavefront Propagation The reaction term $R(\rho) = \rho(1-\rho)$ of the Fisher equation is strictly positive for $\rho \in [0,1]$, with the consequence that

$$\lim_{t\to\infty}\rho(x,t) = 1, \qquad \forall x\,,$$

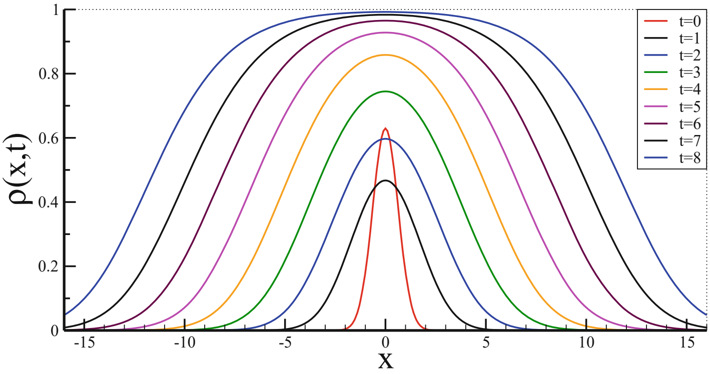

Fig. 4.1 Simulation of the Fisher reaction-diffusion equation (4.4). Plotted is $\rho(x,t)$ for $t = 0,\ldots,8$. For the initial distribution $\rho(x,0)$ a Gaussian has been taken, which does not correspond to the natural line-form, but already at $t = 1$ the system has relaxed. The wavefronts propagate asymptotically with velocities ± 2

viz that the system saturates. The question of interest is however how saturation is achieved when starting from a local population $\rho(x,0)$, a simulation is presented in Fig. 4.1. The system is seen to develop wavefronts with a characteristic shape and velocity. In a biological setting this corresponds to an expansion wave allowing an initial local population to invade ballistically the uninhabited regions of the surrounding ecosystem.

This is an interesting observation, since diffusion alone, viz in the absence of a reaction term, would lead to a sublinear expansion $\sim \sqrt{t}$. In contrast, ballistic propagation is linear in time.

4.1.1 Travelling Wavefronts in the Fisher Equation

In order to describe the propagation of wavefronts in a reaction-diffusion system, we examine the properties of plane-wave solutions of the form

$$\rho(x,t) = u(x-ct), \qquad \dot{\rho} = -cu', \qquad \rho'' = u''. \tag{4.5}$$

For the Fisher equation (4.4), this ansatz leads to the two-component ordinary differential equation

$$u'' + cu' + u(1-u) = 0, \qquad \left\{ \begin{array}{l} u' = v \\ v' = -cv - u(1-u) \end{array} \right\}, \tag{4.6}$$

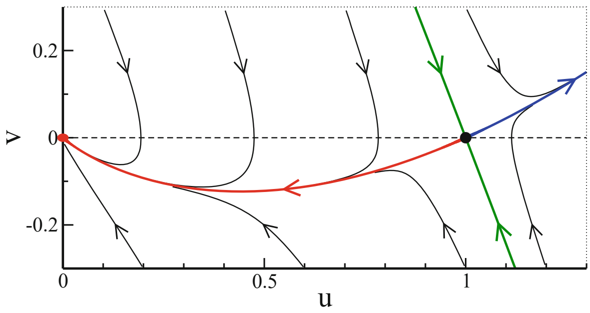

Fig. 4.2 Phase space trajectories of the travelling-wave solution $\rho(x, t) = u(x - ctz)$, with $v = u'$, as given by (4.6). The shape of the propagating wavefront is determined by the heteroclinic trajectory (red line) emerging from the saddle $(1, 0)$ and leading to the stable fixpoint $(0, 0)$

with fixpoints $\mathbf{u}^* = (u^*, v^*)$,

$$\mathbf{u}_0^* = (0, 0), \qquad \mathbf{u}_1^* = (1, 0) . \tag{4.7}$$

Minimal Propagation Velocity For the stability of the trivial fixpoint $\mathbf{u}_0^* = (0, 0)$ one expands (4.6) for small (u, v),

$$\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -c \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \qquad \lambda(\lambda + c) + 1 = 0 ,$$

where λ is an eigenvalue of the Jacobian[2] given by

$$\lambda_\pm = \frac{1}{2}\left(-c \pm \sqrt{c^2 - 4}\right), \qquad c \geq 2 . \tag{4.8}$$

A complex eigenvalue would lead to a spiral around $\mathbf{u}_0^* = (0, 0)$, which is not admissible since $u \in [0, 1]$ needs to be strictly positive. Hence $c = 2$ is the minimal occurring propagation velocity. The trivial fixpoint $(0, 0)$ is stable, since $\sqrt{c^2 - 4} < c$ for $c \geq 2$ and hence $\lambda_\pm < 0$. The situation is illustrated in Fig. 4.2.

Saddle Restpoint The eigenvalues of the $\mathbf{u}_1^* = (1, 0)$ fixpoint are given by

$$\frac{d}{dz}\begin{pmatrix} u - 1 \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -c \end{pmatrix} \begin{pmatrix} u - 1 \\ v \end{pmatrix}, \qquad \lambda_\pm = \frac{1}{2}\left(-c \pm \sqrt{c^2 + 4}\right) ,$$

when denoting $u = u(z)$. The fixpoint $(1, 0)$ is hence a saddle, with $\lambda_- < 0$ and $\lambda_+ > 0$.

[2] The entries of Jacobian matrix are the derivatives of the flow, see Sect. 2.2 of Chap. 2.

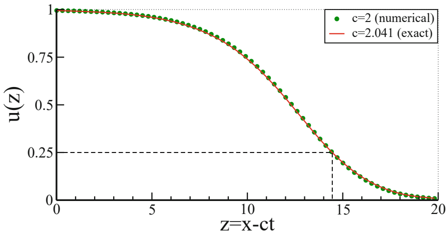

Fig. 4.3 Numerical result for the minimal velocity ($c = 2$) wavefront solution $u^*(z)$, compare (4.9), of the Fisher equation (4.4). For comparison, the wavefront for a slightly larger velocity $c = 5/\sqrt{6} \approx 2.041$ has been plotted, for which the shape can be obtained analytically, see (4.10). Indicated is $u = 1/4$ (dashed lines), which has been used to align the respective horizontal offsets

The unstable direction $u^*(z)$, emerging from the saddle and leading to the stable fixpoint $(0, 0)$, viz the heteroclinic trajectory, is the only trajectory in phase space fulfilling the conditions

$$\lim_{z\to-\infty} u^*(z) = 1, \qquad \lim_{z\to\infty} u^*(z) = 0, \qquad \lim_{z\to\pm\infty} v^*(z) = 0 \tag{4.9}$$

characterizing a propagating wavefront. The lineshape $u^*(z)$ of the wavefront can be evaluated numerically, see Fig. 4.3.

Exact Particular Solution of the Fisher Equation A special travelling-wave solution for the Fischer equation (4.4) is given by

$$\rho^*(x,t) = \sigma^2(x - ct), \qquad \sigma(z) = \frac{1}{1 + e^{\beta z}}, \tag{4.10}$$

where $\sigma(\beta z)$ is the “sigmoidal” or “Fermi function”. It’s derivatives are

$$\sigma' = \frac{-\beta e^{\beta z}}{\left(1 + e^{\beta z}\right)^2} = -\beta\sigma(1-\sigma), \qquad \sigma'' = \beta^2(1-2\sigma)\sigma(1-\sigma), \tag{4.11}$$

which leads to

$$\frac{\partial \rho^*}{\partial t} = 2c\beta\sigma^2(1-\sigma)$$

and

$$\frac{\partial^2 \rho^*}{\partial x^2} = \frac{\partial}{\partial x}(-2\beta\sigma^2)(1-\sigma) = \beta^2(4\sigma - 6\sigma^2)\sigma(1-\sigma).$$

The solution ρ^* fulfills the Fisher equation,

$$\frac{\partial\rho^*}{\partial t} - \frac{\partial^2\rho^*}{\partial x^2} = \sigma^2(1-\sigma)\underbrace{\left[2c\beta - \beta^2(4-6\sigma)\right]}_{\stackrel{!}{=}(1+\sigma)} \stackrel{!}{=} \sigma^2(1-\sigma^2) \equiv \rho^*(1-\rho^*)\,,$$

if

$$1 = 6\beta^2, \qquad 1 = 2c\beta - 4\beta^2 = 2c\beta - \frac{4}{6}, \qquad 2c\beta = \frac{10}{6} = \frac{5}{3}\,.$$

This last condition determines the two free parameters β and c as

$$\beta = \frac{1}{\sqrt{6}}, \qquad c = \frac{5}{6\beta} = \frac{5}{\sqrt{6}} \approx 2.041\,, \tag{4.12}$$

which shows that the propagation velocity c of $\rho^*(x,t)$ is very close to the lower bound $c = 2$ for the allowed velocities. The lineshape of the exact particular solution is nearly identical to the numerically-obtained minimal-velocity shape for the propagating wavefront, as shown in Fig. 4.3.

Exponential Falloff of Wavefronts The population density $\rho(x,t) = u(z)$ becomes very small for large $z = x - ct$, with the consequence that the quadratic term in (4.6) may be neglected. The resulting linear equation,

$$u'' + cu' + u \approx 0\,,$$

has the solution

$$u(z) = \mathrm{e}^{-az}, \qquad a^2 - ca + 1 = 0, \qquad c = a + \frac{1}{a}\,. \tag{4.13}$$

The forward tails of all propagating solutions are hence exponential, with the minimal velocity c corresponding to $a = 1$. Relation (4.13) holds also for the exact particular solution (4.10), for which $a = 2\beta$,

$$c = \frac{1}{2\beta} + 2\beta = \frac{1+4\beta^2}{2\beta} = \frac{6+4}{2\sqrt{6}} = \frac{5}{\sqrt{6}}\,,$$

in agreement with (4.12), when using $\beta = 1/\sqrt{6}$.

4.1.2 Sum Rule for the Shape of the Wavefront

The travelling-wave Eq. (4.6),

$$c(u')^2 = -\big(u(1-u)+u''\big)u', \qquad \lim_{z\to\infty} u(z) = 0, \qquad \lim_{z\to-\infty} u(z) = 1\,, \tag{4.14}$$

may be used to derive a sum rule for the shape $u(z)$ of the wavefront. For this purpose one integrates (4.14),

$$c\int_{-\infty}^{z} \big(u'(w)\big)^2 dw = A + \frac{u^3(z)}{3} - \frac{u^2(z)}{2} - \frac{\big(u'(z)\big)^2}{2}\,. \tag{4.15}$$

The integration constant A is determined by considering $z \to -\infty$ and taking into account that $\lim_{z\to\pm\infty} u'(z) = 0$,

$$A = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \qquad \int_{-\infty}^{\infty} \big(u'(w)\big)^2 dw = \frac{1}{6c}\,, \tag{4.16}$$

where the second equation is the sum rule for u'. The wavefront is steepest for the minimal velocity $c = 2$.

Sum Rule for Generic Reaction Diffusion Systems Sum rules equivalent to (4.16) can be derived for any integrable reaction term $R(\rho)$ in (4.2). One finds, by generalizing the derivation leading to (4.16),

$$\int_{-\infty}^{\infty} \big(u'(w)\big)^2 dw = \frac{1}{c}\int_0^1 R(\rho)d\rho\,. \tag{4.17}$$

In biological setting the reaction term corresponds to the reproduction rate, it is hence generically non-negative, $R(\rho) \ge 0$, which makes the right-hand side of the sum rule (4.17) positive. Interestingly, the diffusion constant D has no influence on the sum rule.

Sum Rule for the Exact Particular Solution We verify that the sum rule (4.16) for the Fisher equation is satisfied by the particular solution (4.10),

$$u_*(z) = \sigma^2(z), \qquad \sigma(z) = \frac{1}{1+\mathrm{e}^{\beta z}} \qquad \sigma' = -\beta\sigma(1-\sigma)\,.$$

With $u_*' = -2\beta\sigma^2(1-\sigma)$ we integrate

$$\int_{-\infty}^{\infty} \left(u_*'(w)\right)^2 dw = \int_{-\infty}^{\infty} (-4\beta)\sigma^3(1-\sigma)\,\underbrace{(-\beta)\sigma(1-\sigma)}_{\sigma'}\,dw$$
$$= 4\beta\left(\frac{1}{4}-\frac{1}{5}\right) = \frac{\beta}{5} = \frac{\beta^2}{5\beta} = \frac{1}{6c}\,,$$

where we used $\beta^2 = 1/6$ and $c = 5\beta$, see (4.12). Note that only the lower bound $z \to -\infty$ contributes to above integral.

4.1.3 Self-Stabilization of Travelling Wavefronts

The Fisher equation $\dot\rho = \rho(1-\rho) + \rho''$ supports travelling wavefront solutions $\rho(x,t) = u(x-ct)$ for any velocity $c \geq 2$. The observed speed of a wavefront may either depend on the starting configuration $\rho(x,0)$, or may self-organize to a universal value. We use perturbation theory in order to obtain an heuristic insight into this issue.

Solution of the Linearized Fischer Equation The solution of the linearized Fischer equation,

$$\dot\rho = r\rho + \rho'', \qquad r = 1\,, \tag{4.18}$$

can be constructed for arbitrary initial conditions $p_0(x) = \rho(x,0)$, using via

$$\rho_0(x,t) = \int dy\,\Phi(x-y,t)\,p_0(y), \qquad \Phi(z,t) = \frac{1}{\sqrt{4\pi t}}\mathrm{e}^{-z^2/(4t)} \tag{4.19}$$

the Green's function $\Phi(z,t)$ for diffusion systems,[3] which obeys $\lim_{t\to 0}\Phi(z,t) = \delta(z)$. Note that $\Phi(z,t)$ is the generic solution of the diffusion equation,viz of (4.18) with $r \to 0$. One can easily verify that

$$\rho(x,t) = \mathrm{e}^{rt}\rho_0(x,t), \qquad \rho(x,0) = p_0(x) \tag{4.20}$$

is the solution of the linearized Fisher equation (4.18), propagating the initial distribution $p_0(x)$ to finite times. Equation (4.20) describes exponentially growing diffusive behavior.

[3] Diffusion Green's functions are introduced in Sect. 3.3 of Chap. 3.

Velocity Stabilization and Self Organization Using (4.19) for $\rho_0(x,t)$ in (4.20) leads to terms of the type

$$
\begin{aligned}
e^t \Phi(z,t) \;\propto\; e^{-z^2/(4t)+t} &= e^{-(z^2-4t^2)/(4t)} \\
&= e^{-(z-2t)(x+2t)/(4t)}
\end{aligned}
\tag{4.21}
$$

in the kernel, viz inside the integral, with $z = x - y$. Equation (4.21) describes left- and right propagating fronts, travelling with propagation speeds $c = \pm 2$. The envelopes of the respective wavefronts are time dependent and do not show the simple exponential tail (4.13), as exponential falloffs are observed only for solutions $\rho(x,t) = u(x-ct)$ characterized by a single velocity c.

– *Ballistic Transport*
  The expression (4.21) shows that the interplay between diffusion and exponential growth, the reaction of the system, leads to ballistic transport.
– *Velocity Selection*
  The perturbative result (4.21) indicates, that the minimal velocity $|c| = 2$ is achieved for the wavefront when starting from an arbitrary localized initial state $p_0(x)$, since $\lim_{t\to 0} e^t \Phi(x,t) = \delta(x)$.
– *Self Organization*
  Propagating wavefronts with any $c \geq 2$ are stable solutions of the Fisher equation, but the system settles to $c = 2$ for localized initial conditions. The stabilization of a non-trivial dynamical property for a wide range of starting conditions is an example of a self-organizing process.

Above considerations concern the speed c of the travelling wavefront, but do not make any direct statement regarding the lineshape.

Stability Analysis of the Wavefront In order to examine the stability of the shape of the wavefront we consider

$$
\rho(x,t) = u(z) \;+\; \epsilon\psi(z)e^{-cz/2}\, e^{-\lambda t}, \qquad z = x - ct\,,
\tag{4.22}
$$

where the second term with $\epsilon \ll 1$ is a perturbation to the solution $u(z) = u(x-ct)$ of the travelling wave Eq. (4.6). The particular form of above ansatz is motivated by yet to be derived final expression.

With the help of the derivatives,

$$
\begin{aligned}
\dot\rho &= \; -cu' + \epsilon\big[(c^2/2-\lambda)\psi - c\psi'\big]e^{-cz/2}e^{-\lambda t}, \\
\rho' &= \; u' + \epsilon\big[-c\psi/2 + \psi'\big]e^{-cz/2}e^{-\lambda t}, \\
\rho'' &= \; u'' + \epsilon\big[c^2\psi/4 - c\psi' + \psi''\big]e^{-cz/2}e^{-\lambda t}\,,
\end{aligned}
$$

we obtain

$$
\begin{aligned}
\dot{\rho} - \rho'' &= -\big[cu' + u''\big] + \epsilon\big[(c^2/4 - \lambda)\psi - \psi''\big]\mathrm{e}^{-cz/2}\mathrm{e}^{-\lambda t} \\
\rho(1-\rho) &\approx u(1-u) + \epsilon\big[1 - 2u\big]\psi\mathrm{e}^{-cz/2}\mathrm{e}^{-\lambda t}\,.
\end{aligned}
$$

Wavefront Self-Stabilization With above results the Fisher equation $\dot{\rho} - \rho'' = \rho(1-\rho)$ reduces with

$$
\left[-\frac{d^2}{dz^2} + V(z)\right]\psi(z) = \lambda\psi(z), \qquad V(z) = 2u(z) + \frac{c^2}{4} - 1\,, \tag{4.23}
$$

to a differential equation for $\psi(z)$, which holds to order $O(\epsilon)$. This expression corresponds to a time-independent one-dimensional Schrödinger equation[4] for a particle with a mass $m = \hbar^2/2$ moving in a potential $V(z)$ that is strictly positive since $u(z) \in [0, 1]$,

$$
V(z) \;\geq\; 0, \qquad \text{for} \qquad c \geq 2\,.
$$

the eigenvalues λ are hence also positive. The perturbation term in (4.22) consequently decays with $t \to \infty$, which implies that that the wavefront self-stabilizes.

This result would be trivial if it were known a priori that the wavefront equation $u''+cu'+u(1-u) = 0$ has a unique solution $u(z)$. In this case, all states of the form of (4.22) would need to contract to $u(z)$. The stability condition (4.23) indicates that the travelling-wavefront solution to the Fisher equation is indeed unique.

Finally, we remark that the procedure followed here, to derive a differential equation for a perturbation containing a yet unknown wavefunction, here $\psi(z)$, is analogous to secular perturbation theory, which we got to know in the context of the Van der Pol oscillator.[5]

4.2 Interplay Between Activation and Inhibition

In chemical reaction systems one reagent may activate or inhibit the production of the other components, leading to non-trivial chemical reaction dynamics. Chemical reagents typically also diffuse spatially and the interplay between the diffusion process and the chemical reaction dynamics may give rise to the development to spatially structured patterns.

[4] Recall the expression $\mathrm{i}\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\psi + V\psi$ for the time-dependent one dimensional Schrödinger equation. Equation (4.23) is recovered for an exponential time dependency $\sim \exp(-i\lambda t/\hbar)$ of the wavefunction ψ.

[5] For the Van der Pol oscillator, secular perturbation theory is developed in Sect. 3.2 of Chap. 3.

4.2.1 Turing Instability

The reaction-diffusion system (4.2) contains additively two terms, the reaction and the diffusion term. Diffusion alone leads generically to an homogeneous steady state.

For the reaction terms considered in Sect. 4.1 the reference state $\rho = 0$ was unstable against perturbations. Will now consider reaction terms for which the reference homogeneous state is however stable. Naively one would expect a further stabilization of the reference state, but this is not necessarily the case.

TURING INSTABILITY The interaction of two processes, which separately would stabilize a given homogeneous reference state, can lead to an instability.

The Turing instability is thought to be the driving force behind the formation of spatio-temporal patterns observed in many physical and biological settings, such as the stripes of a zebra.

Turing Instability of Two Stable Foci As a first example we consider a linear two-dimensional dynamical system composed of two subsystem, which are per se stable foci[6]

$$\dot{\mathbf{x}} = A\mathbf{x}, \qquad A_1 = \begin{pmatrix} -\epsilon_1 & 1 \\ -a & -\epsilon_1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} -\epsilon_2 & -a \\ 1 & -\epsilon_2 \end{pmatrix}. \tag{4.24}$$

Here $0 < a < 1$ and $\epsilon_\alpha > 0$, for $\alpha = 1, 2$. The eigenvalues for the matrices A_α and $A = A_1 + A_2$ are

$$\lambda_\pm(A_\alpha) = -\epsilon_\alpha \pm i\sqrt{a}, \qquad \lambda_\pm(A) = -(\epsilon_1 + \epsilon_2) \pm (1 - a).$$

The origin $\mathbf{x} = 0$ of the system $\dot{\mathbf{x}} = (A_1 + A_2)\mathbf{x}$ hence is consequently a saddle if $1 - a > \epsilon_1 + \epsilon_2$, an instance of a Turing instability: Superposing two stable vector fields may generate unstable directions.

Eigenvalues of Two-Dimensional Matrices We remind ourselves that the eigenvalues $\lambda_\pm$ of a 2×2 matrix A can be written in terms of the trace $a + b$ and of the determinant $\Delta = ab - cd$,

$$A = \begin{pmatrix} a & d \\ c & b \end{pmatrix}, \qquad \lambda_\pm = \frac{a+b}{2} \pm \frac{1}{2}\sqrt{(a+b)^2 - 4\Delta}. \tag{4.25}$$

For negative determinants $\Delta < 0$ the system is a saddle, having both an attracting and a repelling eigenvalue.

[6] As a reminder, note that a node/focus has real/complex Lypunov exponents, as defined in Sect. 2.2 of Chap. 2.

Superposing a Stable Node and a Stable Focus We start by investigating what may happen when superimposing a stable focus and a stable node, with the focus later being defined as

$$A_1 = \begin{pmatrix} -\epsilon_a & 1 \\ -1 & \epsilon_b \end{pmatrix}, \qquad \Delta_1 = 1 - \epsilon_a \epsilon_b > 0\,, \tag{4.26}$$

with

$$(\epsilon_b - \epsilon_a)^2 < 4\Delta_1, \qquad \epsilon_a > \epsilon_b > 0\,.$$

The system $\dot{\mathbf{x}} = A_1 \mathbf{x}$ has a stable focus at $x = 0$ when the trace $\epsilon_b - \epsilon_a < 0$ is negative, compare (4.25), viz when it has two complex conjugate eigenvalues with negative real parts. Possible values are, e.g. $\epsilon_a = 1/2$ and $\epsilon_b = 1/4$.

Next we add a stable node

$$A_2 = \begin{pmatrix} -a & 0 \\ 0 & -b \end{pmatrix}, \qquad A = \begin{pmatrix} -\epsilon_a - a & 1 \\ -1 & \epsilon_b - b \end{pmatrix}. \tag{4.27}$$

Can we select $a, b > 0$ such that $A = A_1 + A_2$ becomes a saddle? In this case, the determinant Δ_{12} of A,

$$\Delta_{12} = 1 - (\epsilon_a + a)(\epsilon_b - b)\,,$$

should then become negative, compare (4.25). This is clearly possible for $b < \epsilon_b$ and a large enough a.

Turing Instability and Activator-Inhibitor Systems The interplay of one activator and one inhibitor often results in a stable focus, as described by (4.26). Diffusion processes correspond, on the other side, to stable nodes, such as A_2 in (4.27). The Turing instability possible when superimposing a stable node and a stable focus is hence of central relevance for chemical reaction-diffusion systems.

4.2.2 Pattern Formation

We now consider a generic reaction-diffusion system of type (4.2)

$$\begin{aligned} \dot{\rho} &= f(\rho, \sigma) + D_\rho \Delta \rho \\ \dot{\sigma} &= g(\rho, \sigma) + D_\sigma \Delta \sigma \end{aligned} \tag{4.28}$$

containing two interacting components, $\rho = \rho(\mathbf{x}, t)$ and $\sigma = \sigma(\mathbf{x}, t)$, characterized by respective diffusion constants $D_\rho,\ D_\sigma > 0$. We assume, that the reaction system $\mathbf{R} = (f, g)$ has a stable homogeneous solution with steady-state densities ρ_0 and σ_0

respectively, together with the stability matrix

$$A_1 = \begin{pmatrix} f_\rho & f_\sigma \\ g_\rho & g_\sigma \end{pmatrix}, \qquad g_\rho f_\sigma < 0, \qquad f_\rho + g_\sigma < 0 \tag{4.29}$$

which we assume to model an activator-inhibitor system characterized by $g_\rho f_\sigma < 0$. Here $f_\rho = \partial f/\partial\rho$, etc, denotes the relevant partial derivatives. We may always rescale the fields ρ and σ such that $|g_\rho| = 1 = |f_\sigma|$ and (4.29) is then identical with the previously used representation, namely (4.26).

Fourier Expansion of Perturbations The reaction term $\mathbf{R} = (f, g)$ is independent of of the spatial coordinate, $\mathbf{x}$. We may hence expand the perturbation of the fields around the equilibrium state in terms of a spatial Fourier analysis, with

$$\begin{aligned} \rho(\mathbf{x}, t) &= \rho_0 + \mathrm{e}^{-i\mathbf{k}\cdot\mathbf{x}}\,\delta\rho(t) \\ \sigma(\mathbf{x}, t) &= \sigma_0 + \mathrm{e}^{-i\mathbf{k}\cdot\mathbf{x}}\,\delta\sigma(t) \end{aligned} \tag{4.30}$$

describing the deviation from the equilibrium state (ρ_0, σ_0) by a single Fourier component. We obtain

$$\begin{pmatrix} \delta\dot\rho \\ \delta\dot\sigma \end{pmatrix} = A \begin{pmatrix} \delta\rho \\ \delta\sigma \end{pmatrix}, \qquad A_2 = \begin{pmatrix} -D_\rho k^2 & 0 \\ 0 & -D_\sigma k^2 \end{pmatrix}, \tag{4.31}$$

where the overall stability matrix $A = A_1 + A_2$ is given by the linear superposition of a stable focus A_1 and a stable node A_2.

Spatio-Temporal Turing instability For concreteness we assume

$$f_\rho < 0, \qquad g_\sigma > 0, \qquad 0 > g_\sigma f_\rho = -|g_\sigma f_\rho|,$$

in analogy with (4.26). The determinant Δ of A is then

$$\Delta = |f_\sigma g_\rho| - \big(|f_\rho| + D_\rho k^2\big)\big(|g_\sigma| - D_\sigma k^2\big)\,. \tag{4.32}$$

A Turing instability occurs when the determinant Δ becomes negative, which is possible if the respective diffusion constants D_ρ and D_σ are different in magnitude.

- When the determinant (4.32) is reduced, the homogeneous fixpoint solution changes first from a stable focus to a stable node as evident from expression (4.25). The instability at $\Delta = 0$ is hence a transition from a stable focus to a saddle, as illustrated in Fig. 4.4.
- The contribution A_2 of the diffusion induces the transition and can be hence regarded as a bifurcation parameter.

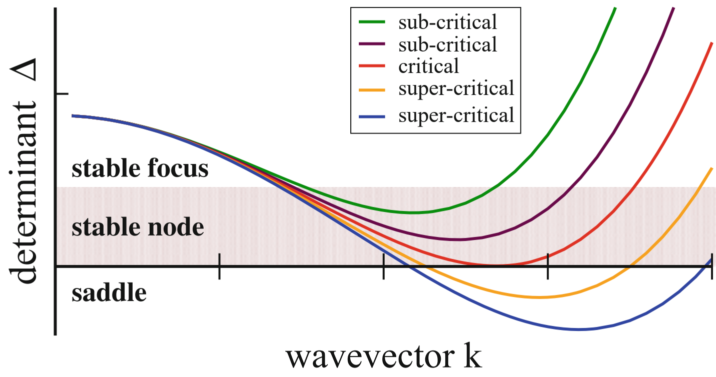

Fig. 4.4 The determinant Δ of the Turing bifurcation matrix, see (4.32), for a range $d = D_\sigma/D_\rho$ of ratios of the two diffusion constants. For large $d \to 1$ the determinant remains positive, becoming negative for a finite interval of wavevectors k when d becomes small enough. With decreasing size of the determinant the fixpoint changes from a stable focus (two complex conjugate eigenvalues with negative real components) to a stable node (two negative real eigenvalues) and to a saddle (a real positive and a real negative eigenvalue), compare (4.25)

Diffusion processes may act as bifurcation parameters also within other bifurcation scenarios, like a Hopf bifurcation, as we will discuss in more detail in Sect. 4.2.3.

Mode Competition For a real-world chemical reaction system the parameters are fixed. A range of Fourier modes with negative determinants (4.32), and corresponding positive Lyapunov exponents (4.25), will start to grow and compete with each others. The shape of the steady-state pattern, if any, will be determined in the end by the non-linearities of the reaction term.

4.2.3 Gray–Scott Reaction Diffusion System

Several known examples of reaction-diffusion models showing instabilities towards the formation of complex spatio-temporal patterns are characterized by contributions $\pm\rho\sigma^2$ to the reaction term,

$$\begin{aligned} \dot\rho &= -\rho\sigma^2 + F(1-\rho) + D_\rho \Delta\rho, \\ \dot\sigma &= \quad \rho\sigma^2 - (F+k)\sigma + D_\sigma \Delta\sigma\,, \end{aligned} \tag{4.33}$$

corresponding to a chemical reaction needing two σ-molecules and one ρ-molecule. The remaining contributions to the reaction term in (4.33) serve for overall mass balance, which can be implemented in various ways. The replenishment rate for the reactant $\rho < 1$ is determined in (4.33) by F, with σ particles being lost to the environment at a rate $K = F + k$. A slightly different choice for the mass conservation terms would lead to the "Brusselator model", the parametrization used in (4.33) corresponds to the "Gray–Scott system".

Saddle-Node Bifurcation The reaction term of the Gray–Scott system (4.33) supports three fixpoints $\mathbf{p}_i^* = (\rho_i^*, \sigma_i^*)$, $i = 0, 1, 2$. The first one is $\mathbf{p}_0^* = (1, 0)$, with the other two being determined by

$$\rho_i^* \sigma_i^* = K, \qquad F\rho_i^* + K\sigma_i^* = F, \qquad i = 1,\ 2\,, \tag{4.34}$$

where $K = F + k$. The resulting steady-state densities are

$$\rho_{1,2}^* = \frac{1 \pm a}{2}, \qquad \sigma_{1,2}^* = \frac{1 \mp a}{2K} F, \qquad a^2 = 1 - \frac{4K^2}{F}\,.$$

The trivial fixpoint $\mathbf{p}_0^*$ has a diagonal Jacobian with eigenvalues $-F$ and $-K$ respectively. It is always stable, even in the presence of diffusion.

Critical Loss of Particles The two non-trivial fixpoints $\rho_{1,2}^*$ merge when $a = 0$, which happens at

$$F = 4K^2 = 4(k + F)^2, \qquad k_c = \sqrt{F_c}/2 - F_c\,. \tag{4.35}$$

We recall that two fixpoints merge via a saddle-node bifurcation,[7] which implies that the two non-trivial fixpoints $\rho_{1,2}^*$ cease to exists when the loss rate $K = F + k$ of σ-particles becomes too large. Beyond, $\mathbf{p}_0^* = (1, 0)$ remains as a global attractor. The critical line (k_c, F_c) is shown in Fig. 4.5, where the phase diagram of the Gray–Scott system is presented.

Saddle Fixpoint $\mathbf{p}_1^*$ The Jacobians and determinants of the non-trivial restpoints $\mathbf{p}_{1,2}^*$ are

$$\begin{pmatrix} -(F + (\sigma_{1,2}^*)^2) & -2K \\ (\sigma_{1,2}^*)^2 & K \end{pmatrix}, \qquad \begin{aligned} \Delta_{1,2} &= K\left((\sigma_{1,2}^*)^2 - F\right) \\ &= KF\left((\sigma_{1,2}^*)^2/F - 1\right) \end{aligned}\,. \tag{4.36}$$

Inserting the steady-state densities (4.34) leads to

$$\frac{\Delta_{1,2}}{KF} = \left(\frac{1 \mp \sqrt{1 - 4\gamma^2}}{2\gamma}\right)^2 - 1, \qquad \gamma = \frac{K}{\sqrt{F}} \in [0, 1/2]\,. \tag{4.37}$$

The restpoint corresponding to the minus sign in above expression, $\mathbf{p}_1^*$, is always a saddle, given that Δ_1 is negative for all $\gamma^2 \in [0, 1/2]$, viz when $\mathbf{p}_1^*$ exists. Compare (4.25).

[7] Bifurcation theory is discussed in Sect. 2.2.2 of Chap. 2.

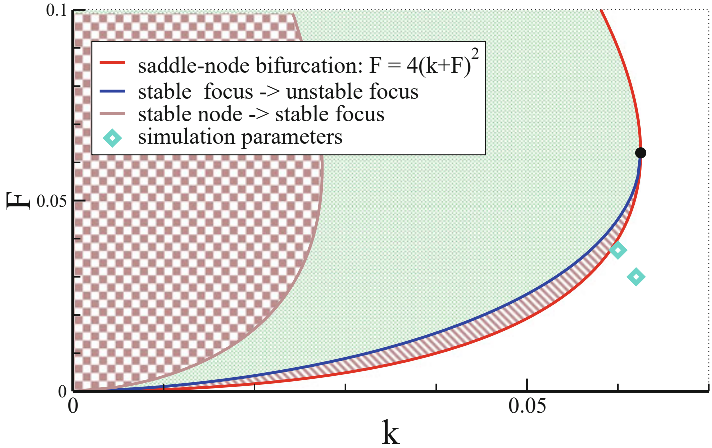

Fig. 4.5 Phase diagram for the reaction term of the Gray–Scott model (4.33). Inside the saddle-node bifurcation line (solid red line), see (4.35) there are three fixpoints $\mathbf{p}_i^*$ ($i = 0, 1, 2$), outside the saddle-node bifurcation line only the stable trivial restpoint $\mathbf{p}_0^*$ is present. $\mathbf{p}_1^*$ is a saddle and $\mathbf{p}_2^*$ a stable node (checkerboard brown area), a stable focus (light shaded green area) or an unstable focus (shaded brown area), the later two regions being separated by the focus transition line (solid blue line), as defined by (4.39). Also include is conjunction point $\mathbf{p}_c = (1, 1)/16$ (black filled circle) and the parameters used for the simulation presented in Fig. 4.7 (open diamonds)

Focus Transition for $\mathbf{p}_2^*$ Eigenvalues are complex when the discriminant $D = (\text{trace})^2 - 4\Delta$ is negative, compare (4.25). For the Gray–Scott system one has

$$D = \left[k - (\sigma_{1,2}^*)^2\right]^2 - 4K(\sigma_{1,2}^*)^2 + 4KF\,,$$

where we used $k = K - F$. The discriminant is positive for small and large densities $\sigma_{1,2}^*$, but negative in between. At the transition, $\mathbf{p}_2^*$ changes from a stable node to a stable focus. The explicit analytic expression is not of relevance for the following, the transition line is however shown in Fig. 4.5.

Stability of $\mathbf{p}_2^*$ The focus $\mathbf{p}_2^*$ changes stability when the trace of its Jacobian (4.36),

$$K_f - F_f - (\sigma_2^*)^2 = 0, \qquad \sigma_2^* = \sqrt{K_f - F_f} = \sqrt{k_f}\,, \tag{4.38}$$

changes sign. The respective focus line (k_f, F_f) has been included in Fig. 4.5. Inserting this relation, $\sigma_2^* = \sqrt{k_f}$, into the fixpoint conditions (4.34)

$$\frac{K}{\sigma_2^*} = \rho_2^* = 1 - \frac{K}{F}\sigma_2^*, \qquad K\,\frac{F + (\sigma_2^*)^2}{F} = \sigma_2^*\,,$$

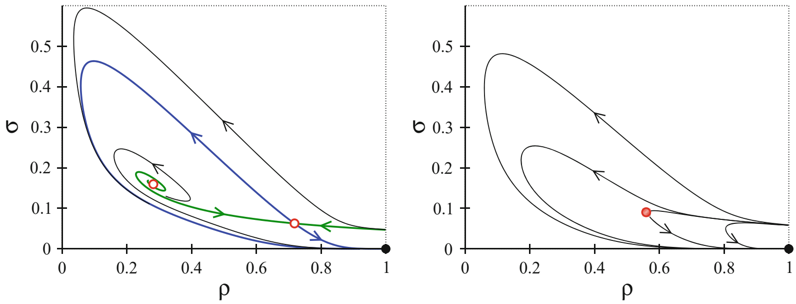

Fig. 4.6 Flow of the reaction term of the Gray–Scott model (4.33) in phase space (ρ, σ), for $F = 0.01$. The stable fixpoint $\mathbf{p}_0^* = (1.0)$ (black filled circle) is globally attracting. The focus transition (4.39) occurs at $k_f = 0.0325$ and the saddle-node transition (4.35) at $k_c = 0.04$ **Left:** $k_f < k = 0.035 < k_c$, with $\mathbf{p}_1^*$ and $\mathbf{p}_2^*$ (red circles) being a saddle an unstable focus respectively. **Right:** $k_c < k = 0.045$. The locus of the attractor relict (shaded circle) corresponds to the local minimum of $q = \mathbf{f}^2$, see (4.41), which evolves into the bifurcation point for $k \to k_c$

we obtain

$$(F_f + k_f)^2 = F_f \sqrt{k_f}\,, \tag{4.39}$$

where we also used $K = F + k$. One can verify that the Lyapunov exponents are complex along (4.39), becoming however real for lower values of k, as illustrated in Fig. 4.5. The endpoint of (4.39), determined by $\partial k_f / \partial F_f = 0$, is

$$\mathbf{p}_c = (1/16, 1/16) = (0.0625, 0.0625) \tag{4.40}$$

in the (k, F) plane, which coincides with the turning point of the saddle-node line (4.35).

Merging of an Unstable Focus and a Saddle In Fig. 4.6 we illustrate the flow of the reaction term of the Gray–Scott model for two sets of parameters (k, F). The unstable focus $\mathbf{p}_2^*$ and the saddle $\mathbf{p}_1^*$ annihilate each other for $k \to k_c$. One observes that the large-scale features of the flow are remarkable stable similar for $k < k_c$ and $k > k_c$, as all trajectories, apart from the stable manifolds of the saddle, flow to the global attractor $\mathbf{p}_0^* = (1, 0)$.

Attractor Relict Dynamics Close to the outside of the saddle-node line (k_c, F_c) the dynamics slows down when approaching a local minimum of

$$q(\mathbf{x}) = \mathbf{f}^2(\mathbf{x}), \qquad \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})\,, \tag{4.41}$$

with q being a measure for the velocity of the flow, which vanishes at a fixpoint.

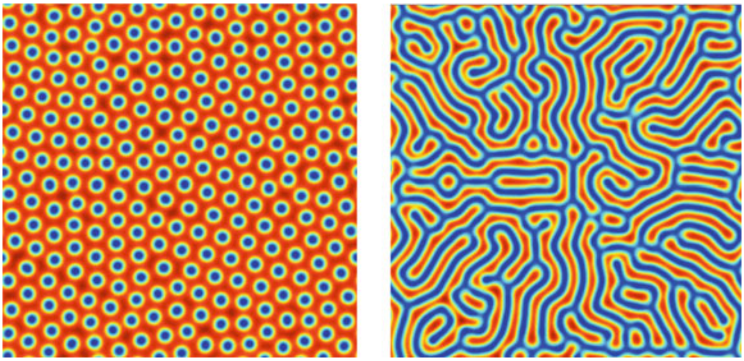

Fig. 4.7 Dynamical patterns for the Gray–Scott model (4.33). Shown is $\rho(x, y, t)$, the diffusion constants are $D_\rho/2 = D_\sigma = 10^{-5}$. The simulation parameters are $(k, F) = (0.062, 0.03)$ (left) and $(k, F) = (0.06, 0.037)$ (right), as indicated in the phase diagram, Fig. 4.5 (Illustrations courtesy P. Bastani)

ATTRACTOR RELICT A local, non-zero minimum of $q(\mathbf{x})$, close to a bifurcation point, and which turns into a fixpoint at the bifurcation, is denoted an "attractor relict" or a "slow point".

Beyond the transition, for $k > k_c$, the attractor relict still influences the flow strongly, as indicated in Fig. 4.6, determining a region in phase space where the direction of the flow turns sharply.

Dynamical Pattern Formation The Gray–Scott system shows a wide range of complex spatio-temporal structures close to the saddle-node line, as illustrated in Fig. 4.7, ranging from dots growing and dividing in a cell-like fashion to more regular stripe-like patterns.

The generation of non-trivial pattern occurs even, though not exclusively, when only the trivial fixpoint $(1, 0)$ exists, with

$$A_1 = \begin{pmatrix} -F & 0 \\ 0 & -(F+k) \end{pmatrix} \tag{4.42}$$

being the respective Jacobian. There is no way to add a diffusion term A_2, compare expression (4.31), such that $A_1 + A_2$ would have a positive eigenvalue. Pattern formation in the Gray–Scott system is hence not due to a Turing instability.

4.3 Collective Phenomena and Swarm Intelligence

When a system is composed out of many similar or identical constituent, parts,[8] their mutual interaction may give rise to interesting phenomena. Several distinct concepts have been developed in this context, each carrying its own specific connotation.

– *Collective Phenomena*
  Particles like electrons obey relatively simple microscopic equations of motions, like Schrödinger's equation, interacting pairwise. Their mutual interactions may lead to phase transitions and to emergent macroscopic collective properties, like superconductivity or magnetism, not explicitly present in the underlying microscopic description.
– *Emergence*
  At times heavily loaded with philosophical connotations, "weak emergence" is equivalent to collective behavior, e.g. such as occurring in physics. There is also "strong emergence", which denotes the emergence of higher-level properties which cannot be traced back causally to microscopic laws. Strong emergence is a popular playball in philosophy, it transcends however the realm of scientific investigations.
– *Self Organization*
  When generic generative principles give rise to complex behavior for a wide range of environmental and/or starting conditions, one speak of self organization in the context of complex systems theory. The resulting properties may be interesting, biologically relevant or emergent.
– *Swarm Intelligence*
  In biological settings, with the agents being individual animals, one speaks of swarm intelligence whenever the resulting collective behavior is of behavioral relevance. Given that evolutionary optimized behavior is performance oriented, one can use the principles underlying swarm intelligence as an algorithm to solve certain computational problems.

Collective phenomena arise, loosely speaking, when "the sum is more than the parts", just as mass psychology transcends the psychology of the constituent individuals.

4.3.1 Phase Transitions in Social Systems

Phase transitions occur in many physical systems,[9] being of central importance also in biology, climatology and sociology. A well known psychological phenomenon is,

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[8] A prototypical example are the boolean networks discussed in Chap. 7.

[9] The generic framework for phase transitions,, the Ginzburg-Landau theory, is developed in Sect. 6.1 of Chap. 6.

in this context, the transition from normal crowd behavior to collective hysteria. As a basic example we consider here the nervous rats problem.

Calm and Nervous Rats Consider N rats constrained to live in an area A, with an overall population density $\rho = N/A$. There are N_c calm and N_n nervous rats with $N_c + N_n = N$, together with the respective densities $\rho_c = N_c/A$ and $\rho_n = N_n/A$.

Comfort Zone Each rat has a zone $a = \pi r^2$ around it, with r being a characteristic radius. A calm rat will get nervous if at least one nervous rat comes too close, entering its comfort zone a, with a nervous rat calming down when having its comfort zone all for itself.

Master Equation The time evolution for the density of nervous rats is given by

$$\dot{\rho}_n = P_{c\to n}\,\rho_c - P_{n\to c}\,\rho_n\,, \tag{4.43}$$

with transition probabilities

$$P_{c\to n} = 1 - \left(1 - \frac{a}{A}\right)^{N_n}, \qquad P_{n\to c} = \left(1 - \frac{a}{A}\right)^{N-1}, \tag{4.44}$$

since $1 - a/A$ is the probability for a rat being out of a given comfort zone. A calm rat become nervous when there is at least another nervous rat in her comfort zone, calming down when being all alone.

Thermodynamic Limit For constant comfort areas a we make take with

$$\lim_{A\to\infty}\left(1 - \frac{a}{A}\right)^{\rho_n A} = \mathrm{e}^{-\rho_n a}, \qquad \lim_{A\to\infty}\left(1 - \frac{a}{A}\right)^{\rho A} = \mathrm{e}^{-\rho a}$$

the thermodynamic limit $A \to \infty$ for (4.44), where we made use of $(N-1) \approx N$, $N_n = \rho_n A$ and $N = \rho A$.

Stationary Solution For the stationary solution $\dot{\rho}_n = 0$ of the master equation (4.43) we obtain

$$\left(1 - \mathrm{e}^{-\rho_n a}\right)\rho_c = \mathrm{e}^{-\rho a}\rho_n, \qquad \rho_c = \rho - \rho_n\,,$$

which has a trivial solution $\rho_n = 0$ for all population densities ρ and comfort zones a. Multiplying with a yields

$$\sigma_n = \mathrm{e}^{\sigma}\left(1 - \mathrm{e}^{-\sigma_n}\right)(\sigma - \sigma_n), \qquad \sigma = \rho a, \qquad \sigma_n = \rho_n a\,, \tag{4.45}$$

where the dimensionless densities σ and σ_n correspond to the average numbers of rats within a comfort area a.

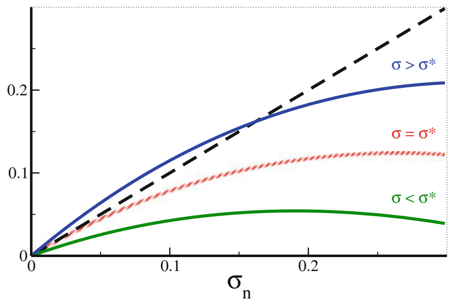


Fig. 4.8 Solution of the nervous rats problem (4.43). Shown is the left- and the right-hand side of the self-consistency condition (4.45), for various numbers of rats σ in the comfort zone a, as function of the average number of nervous rats σ_n per a. The transition occurs at σ^* (dashed line)

Critical Rat Density The graphical representation of (4.45) is given in Fig. 4.8. A non-trivial solution $\sigma_n > 0$ is possible only above a certain critical number σ^* of rats per comfort zone. At $\sigma = \sigma_*$ the right-hand side of (4.45) has unitary slope with respect to σ_n, for $\sigma_n \to 0$,

$$1 = e^{\sigma^*} \sigma^*, \qquad \sigma^* \approx 0.56713\,, \tag{4.46}$$

with σ^* corresponding to roughly half-a-rat per comfort zone. A finite fraction of nervous rats is present roughly whenever a rat has less than two comfort areas for itself, whereas all rats calm eventually down whenever the average population density ρ is smaller than $\rho^* = \sigma^*/a$. Note, that one can rewrite (4.46) approximatively as

$$\sigma^* = e^{-\sigma^*} \approx 1 - \sigma^*, \qquad \sigma^* \approx 1/2\,,$$

which yield a first-order estimate for σ^*.

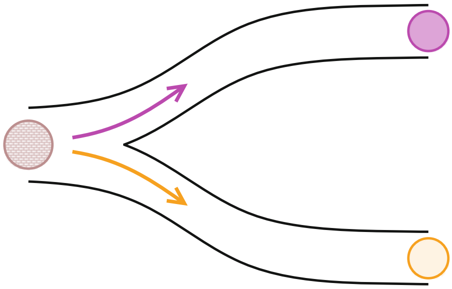
4.3.2 Collective Decision Making and Stigmergy

Herding animals and social insects are faced at times with the problem of taking decisions collectively. When selecting a new nest for swarming honey bees or a good foraging site for ants, no single animal will compare two prospective target sites. Individual opinions regarding the quality of prospect sites are instead pooled together into groups of alternative decisions with a competitive dynamical process reducing the number of competing opinions until a single target site remains.

Swarming Bees Bees communicate locally by dancing, both when they are in the nest and communicate prospective foraging sites as well when a new queen leaves the old nest together with typically about 10.000 workers, searching for a suitable location for building a new nest.

The swarm stays huddled together with a small number of typically 5% of bees scouting in the meantime for prospective new nest sites. Scouts coming back to

Fig. 4.9 Illustration of binary decision making for foraging ants starting from the nest (to the left). The colony needs to decide which of the two food sources (upper/lower paths) is more profitable



the waiting swarm will advertise new locations they found by dancing, with the duration of the dance being proportional to the estimated quality of the prospective nest location.

New scouts ready to fly will observe the dancing returning scouts. The probability that departing scouts will target advertised sites is consequently proportional to their quality. This mechanism leads to a competitive process with lower quality prospective nest sites receiving fewer consecutive visits by scouting bees.

Opinion pooling is inherent in this process as there are many scout bees flying out for site inspection. The whole consensus process is an extended affair, taking up to a few days.

Foraging Ants Social animals gain in survival efficiency by making collective decisions when it comes to key tasks. A celebrated example are ant colonies which profit from exploiting the richest food sources available. The corresponding decision problem is illustrated schematically in Fig. 4.9.

The quality of a food resource is given by its distance to the nest and its utility for the species. When the ant returns to the nest it lays down a path of pheromones, with the amount of pheromone released being proportional to the quality of the food site discovered.

Ants leaving the nest in the search for food will tend to follow the pheromone gradient and such arrive predominantly to high quality food sites. Returning they will reenforce the existing pheromone path with an intensity appropriate to their own assessment of the food site. With a large number of ant going forth and back this process will lead to an accurate weighting of foraging sites, with the best site eventually winning in the end over all other prospective food resources.

Information Encoding and Stigmergy Bees and ants need an observable scalar quantity in order to exchange information about the quality of prospective sites. This quantity is time for the case of bees and pheromone intensity for the ants.

STIGMERGY When information is off-loaded to the environment for the purpose of communication one speaks of “stigmergy”.

Producing pheromone traces ants manipulate the environment for the purpose of information exchange.

Ant and Pheromone Dynamics For the binary decision process of Fig. 4.9 we denote with $\rho_{1,2}$ the densities of travelling ants along the two routes, with $\varphi_{1,2}$ and $Q_{1,2}$ the respective pheromone densities and site qualities. The process

$$\begin{aligned} T\dot{\rho}_1 &= (\rho-\rho_1)\varphi_1 - \Psi \qquad & \dot{\varphi}_1 &= -\Gamma\varphi_1 + Q_1\rho_1 \\ T\dot{\rho}_2 &= (\rho-\rho_2)\varphi_2 - \Psi \qquad & \dot{\varphi}_2 &= -\Gamma\varphi_2 + Q_2\rho_2 \end{aligned} \tag{4.47}$$

models the dynamics of $\rho = \rho_1 + \rho_2$ ants foraging with T setting the real-world time needed for the foraging and Γ being the pheromone decay constant. The update rates $\dot{\rho}_i$ for the individual number of ants in (4.47) are relative to an average update rate Ψ/T.

Ant Number Conservation The updating rules $\dot{\rho}_{1,2}$ need to obey conservation of the overall number $\rho = \rho_1 + \rho_2$ of ants. This can be achieved by selecting the flux Ψ in (4.47) appropriately as

$$\begin{aligned} 2\Psi &= \rho(\varphi_1+\varphi_2) - \rho_1\varphi_1 - \rho_2\varphi_2 \qquad & 2T\dot{\rho}_1 &= \rho_2\varphi_1 - \rho_1\varphi_2 \\ &= \rho_2\varphi_1 + \rho_1\varphi_2 \qquad & 2T\dot{\rho}_2 &= \rho_1\varphi_2 - \rho_2\varphi_1 \end{aligned}, \tag{4.48}$$

which results from demanding $\dot{\rho}_1 + \dot{\rho}_2$ to vanish.

Pheromone Conservation Considering with $\Phi = Q_2\varphi_1 + Q_1\varphi_2$ a weighted pheromone concentration, we find

$$\dot{\Phi} = -\Gamma\Phi + Q_1Q_2(\rho_1+\rho_2), \qquad \Phi \;\to\; \rho\, Q_1Q_2/\Gamma\,,$$

when using (4.47). Any initial weighted pheromone concentration Φ will hence relax fast to $\rho Q_1 Q_2/\Gamma$. It is hence enough to consider the time evolution in the subspace spanned by $\varphi_2 = (\Phi - Q_2\varphi_1)/Q_1$,

$$\begin{aligned} 2T\dot{\rho}_1 &= (\rho-\rho_1)\varphi_1 - \rho_1(\Phi - Q_2\varphi_1)/Q_1 \\ \dot{\varphi}_1 &= -\Gamma\varphi_1 + Q_1\rho_1 \end{aligned}, \tag{4.49}$$

which leaves us with two dynamical variables, ρ_1 and φ_1.

Binary Decision Making here are two fixpoints of (4.49), given by

$$\rho_1 = \rho, \qquad \rho_2 = 0, \qquad \varphi_1 = Q_1\rho/\Gamma, \qquad \varphi_2 = 0\,,$$

and viceversa, with $1 \leftrightarrow 2$. The Jacobian of (4.49) for the fixpoint $\rho_1 = 0 = \varphi_1$ is

$$\begin{pmatrix} -\Phi/Q_1 & \rho \\ Q_1 & -\Gamma \end{pmatrix}, \qquad \Delta = \Phi\Gamma/Q_1 - Q_1\rho = (Q_2 - Q_1)\rho\,,$$

when setting $2T = 1$ for simplicity. The trace $-(\Gamma + \Phi/Q_1)$ of the Jacobian is negative and the fixpoint is hence stable/unstable, compare (4.25), when the determinant Δ is positive/negative, hence when $Q_2 > Q_1$ and $Q_1 > Q_2$ respectively.

The dynamics (4.49) hence leads to binary decision process with all ants proceeding in the end along the path with the higher quality factor Q_j.

Ant Colony Optimization Algorithm One can generalize (4.48) to a network of connections with the quality factors Q_i being proportional to the inverse travelling times. This setup, the "ant colony optimization algorithm", has a close connection to the travelling salesman problem. It can be used to find shortest paths on networks.

4.3.3 Collective Behavior and Swarms

In a large body of moving agents, like a flock of birds in the air, a school of fishes in the ocean, or cars on a motorway, agents will adapt individually their proper velocity according to the perceived positions and movements of other close-by agents. For modelling purposes one can consider the behavior of the individual agents, as we will do. Alternatively, for a large enough number of agents one may also use a hydrodynamic description together with a corresponding master equation for the density of agents.

Individual Decision Making For a swarm of birds, the individual birds take their decisions individually, an explicit mechanism for collective decision making is not present, f.i. regarding the overall direction the swarm should be heading to. The resulting group behavior may nevertheless have high biological relevance, such as predator avoidance.

Newton's Equations for Birds The motion of $i = 1, \ldots, N$ birds with positions $\mathbf{x}_i$ and velocities $\mathbf{v}_i$ can be modelled in analogy to the dynamics of point masses. An example is

$$\begin{aligned} \dot{\mathbf{x}}_i &= \mathbf{v}_i\,, \\ \dot{\mathbf{v}}_i &= \gamma\big[(v_i^0)^2 - (\mathbf{v}_i)^2\big]\mathbf{v}_i + \sum_{j\neq i}\mathbf{f}(\mathbf{x}_j,\mathbf{x}_i) + \sum_{j\neq i}\mathbf{g}(\mathbf{x}_j,\mathbf{x}_i|\mathbf{v}_j,\mathbf{v}_i)\,, \end{aligned} \tag{4.50}$$

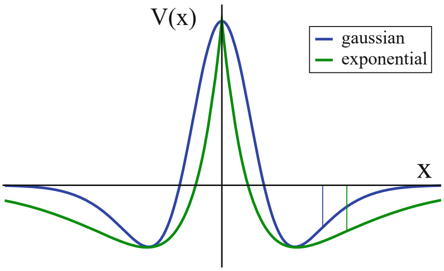


Fig. 4.10 Examples of two typical Mexican–hat potentials (4.51), as generated by superposing two Gaussian or two exponentials. Also indicated are the respective positions of maximal positive slope (thin vertical lines)

with the first term in $\dot{\mathbf{v}}_i$ modelling the preference for moving with a preferred velocity v_i^0. The collective behavior is generated by the pairwise interaction terms **f** and **g**. Reacting to observations needs time and time delays are inherently present in **f** and **g**.

Distance Regulation Animals and humans alike have preferred distances. The regulation of the distance to neighbors, to the side, front, back, above and below, is job of the inter-agent force $f(\mathbf{x}_j, \mathbf{x}_i)$, which may be taken as the derivative of a "Mexican hat potential" $V(z)$,

$$\begin{aligned} f(\mathbf{x}_j, \mathbf{x}_i) = f(\mathbf{x}_j - \mathbf{x}_i) &= -\nabla V(|\mathbf{x}_j - \mathbf{x}_i|) \\ V(z) &= A_1\kappa(z/R_1) - A_2\kappa(z/R_2) \end{aligned} , \tag{4.51}$$

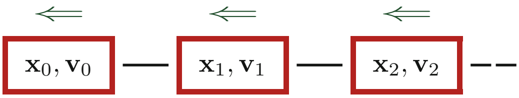
where $\kappa(z)$ is a monotonically decreasing function of its argument, such as an exponential $\sim \exp(-z)$, Gaussian $\sim \exp(-z^2)$, powerlaw $\sim 1/z^\alpha$. For a suitable selection of the amplitudes A_i and of the characteristic distances R_i one can achieve that the potential $V(z)$ is repelling at short distances, having at the same time a stable minimum, as shown in Fig. 4.10.

Alignment of Velocities The function $\mathbf{g}(\mathbf{x}_j, \mathbf{x}_i|\mathbf{v}_j, \mathbf{v}_i)$ in (4.50) expresses the tendency to align one's own velocity with the speed and the direction of the movement of other nearby members of the flock. A suitable functional form would be

$$\mathbf{g}(\mathbf{x}_j, \mathbf{x}_i|\mathbf{v}_j, \mathbf{v}_i) = \big(\mathbf{v}_j - \mathbf{v}_i\big) A_v \kappa(|\mathbf{x}_j - \mathbf{x}_i|/R_v) , \tag{4.52}$$

with the tendency to align vanishing both for identical velocities $\mathbf{v}_j$ and $\mathbf{v}_i$ and for large inter-agent distances $|\mathbf{x}_j - \mathbf{x}_i| \gg R_v$.

Hiking Humans Numerical simulation of equations like (4.50) describe nicely the observed flocking behavior of birds and schools of fishes. For a specific application we ask when a group of humans hiking along a one-dimensional trail will break apart due to differences in the individual preferred hiking speeds v_i^0.



The distance alignment force **f** is asymmetric when walkers pay attention only to the person in front, but not the one next in line. In the simplest setting we may assume furthermore that **g** vanishes, namely that the hikers are primarily concerned to walk close to their own preferred velocities v_i. The group stays together in this case if everybody walks with the speed $v \equiv v_0$ of the leader, we consider here the case that $v_i < v_0$ for $i = 1, 2, \ldots$

The restraining potential illustrated in Fig. 4.10 has a maximal positive slope $f_{\max}$ when the walker lags behind, which implies that a maximal speed difference $v_0 - v_i$ exists, as determined through

$$\gamma\left[(v_i^0)^2 - (v)^2\right]v = -f_{\max}, \qquad (v_i^0)^2 = (v)^2 - f_{\max}/(\gamma v)\,.$$

The group is more likely to break apart when the desire γ to walk with one's own preferred velocity v_i^0 is large. An analogous argument holds for $v_i > v_0$, for which the maximal negative slope of $V(z)$ would enter above expression.

4.3.4 Opinion Dynamics

Opinion dynamics research deals with agents having real-valued opinions x_i, which may change through various processes, like conviction or consensus building.

Bounded Confidence We consider a basic process for consensus building. When meeting, two agents (x_i, x_j) agree on the consensus opinion $\bar{x}_{ij} = (x_i + x_j)/2$, however only when their initial opinions are not too different,

$$(x_i, x_j) \quad \to \quad \begin{cases} (\bar{x}_{ij}, \bar{x}_{ij}) & \text{if } |x_i - x_j| < d \\ (x_i, x_j) & \text{if } |x_i - x_j| > d \end{cases}\,. \tag{4.53}$$

The two agents do not agree to a common opinion if they initially differ beyond the confidence bound d. As a consequence, distinct sets of attracting opinions tend to form, as illustrated in Fig. 4.11.

Master Equation For large populations of agents we may define with $\rho(x, t)$ the density of agents having opinion x, with the time evolution given by the master

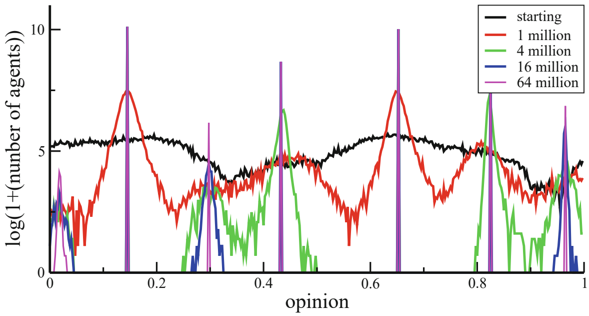


Fig. 4.11 For a confidence interval $d = 0.1$, a simulation of $60 \cdot 10^3$ agents with 1//4/16/64 million pairwise updatings of type (4.53). Opinion attractors with a large number of supporters tend to stabilize faster than attracting states containing smaller number of agents

equation

$$\tau \dot{\rho}(x) = 2 \int_{-d/2}^{d/2} \rho(x+y)\rho(x-y)dy - \int_{-d}^{d} \rho(x)\rho(x+y)dy\,, \tag{4.54}$$

with τ setting the time scale of the consensus dynamics and with the time dependency of $\rho(x,t)$ being suppressed. The first term results from two agents agreeing on x, the second term describes the flow of agents leaving opinion x by agreeing with other agents within the confidence interval d.

Infinitesimal Confidence It is possible to turn (4.54) into an ordinary differential equation. For this purpose we take the limit $d \to 0$ via a Taylor expansion

$$\rho(x+y) \approx \rho(x) + \rho'(x)y + \rho''(x)\frac{y^2}{2} + \dots\,. \tag{4.55}$$

Substituting (4.55) into (4.54) one notices that the terms proportional to y^0 cancel, as the overall number of agents is conserved. The terms $\sim y^1$ also vanish by symmetry and we obtain

$$\begin{aligned}
\tau\dot{\rho} &= 2\int_{-d/2}^{d/2} \left[\rho\rho'' - (\rho')^2\right] y^2 dy - \rho\rho'' \int_{-d}^{d} \frac{y^2}{2} dy \\
&= 4\left[\rho\rho'' - (\rho')^2\right]\frac{1}{3}\frac{d^3}{8} - \rho\rho''\frac{d^3}{3} = -\frac{d^3}{6}\left[\rho\rho'' + (\rho')^2\right].
\end{aligned}$$

With $\partial^2\rho^2/\partial^2 x = 2[\rho'\rho' + \rho\rho'']$ we find

$$\frac{\partial \rho}{\partial t} = -\frac{d^3}{12\tau}\frac{\partial^2 \rho^2}{\partial^2 x}, \qquad \dot{\rho} = -D_o \Delta \rho^2\,, \tag{4.56}$$

with Δ denoting the Laplace operator in one dimension. ρ^2 enters the evolution equation as two agents need to interact for the dynamics to proceed.

One can keep $D_o = d^3/(12\tau)$ in (4.56) constant in the limit $d \to 0$ by appropriately rescaling the time constant τ.[10]

Opinion Current Recasting (4.56) in terms of the continuity equation

$$\dot\rho + \nabla\cdot\mathbf{j}_o = 0, \qquad\qquad \mathbf{j}_o = D_o\nabla\rho^2\,, \tag{4.57}$$

defines the opinion current $\mathbf{j}_o$. With D_0 and ρ being positive this implies that the current strictly flows uphill, an example of a "rich gets richer dynamics" which is evident also in the simulation shown in Fig. 4.11.

Voter Models Large varieties of models describing opinion dynamics have been examined, such as networks of agents having a discrete number of opinions, like 0/1 for conservative/progressive, which can be described by "voter models".[11] Voters change opinion when being 'infected' by the opinion of one or more neighbors. The properties of the underlying social network, like the clustering coefficient, will determine the overall outcome.

4.4 Car Following Models

The flow of cars on a motorway can be modelled by "car following models", akin to the bird flocking model discussed in Sect. 4.3.3. Of central importance is here the interplay between velocity dependent forces and human reaction times.

Chain of Cars We denote with $x_j(t)$ the position of cars on a one-dimensional motorway, $j = 0,\ 1,\ \ldots$ The acceleration $\ddot{x}_j$ of the jth car is given by

$$\ddot{x}_{j+1}(t+T) = \sum_{j\neq i} g(x_j, x_i|\dot{x}_j, \dot{x}_i) = \lambda\big[\dot{x}_j(t) - \dot{x}_{j+1}(t)\big]\,, \tag{4.58}$$

where λ is a reaction constant and T the reaction time.[12] Actions are delayed by T relative to the perception of sensory input. Drivers tend to break when closing in to the car in the front and to accelerate when the distance grows. In car following models, using the notation of (4.50), one considers mostly velocity-dependent forces.

[10] An analogous rescaling is done when deriving the diffusion equation $\dot{p} = D\Delta p$,as discussed in Sect. 3.3.1 of Chap. 3.

[11] The classical voter model is treated in exercise (4.8).

[12] The general theory of dynamical systems with time delays is developed in Sect. 2.5 of Chap. 2.

4.4.1 Linear Flow and Carrying Capacity

One of the most important questions for traffic planning regards the carrying capacity of a road, namely the maximal possible number of cars per hour, q. When all cars advance with identical velocities u, the optimal situation, one would like to evaluate $q(u)$.

Carrying Capacity for the Linear Model Integrating the linear model (4.58) and assuming a steady state, identical velocities $\dot{x}_j \equiv u$ and inter-vehicle distances $x_j - x_{j+1} \equiv s$, one obtains

$$\dot{x}_{j+1} = \lambda[x_j - x_{j+1}] + c_0, \qquad u = \lambda(s - s_0)\,, \tag{4.59}$$

where we did write the constant of integration as $c_0 = -\lambda s_0$, with s_0 being the minimal allowed distance between the cars. The carrying capacity q, the number of cars per time transiting, is given by the product of the mean velocity u and the density $1/s$ of cars,

$$q = \frac{u}{s} = \lambda\left(1 - \frac{s_0}{s}\right) = \frac{u}{s_0 + u/\lambda}, \qquad s = s_0 + \frac{u}{\lambda}\,, \tag{4.60}$$

where we expressed q either as a function of the inter-vehicle distance s, or as a function of the mean velocity u. Above expression suggests that the carrying capacity should be maximal for empty streets, viz when $s \to \infty$, dropping monotonically with increasing traffic until the maximal density $1/s_0$ of cars is reached.

Maximal Velocity The linear model (4.58) cannot be correct, as the velocity $u = \lambda(s - s_0)$, as given by (4.59), would diverge for empty highways, namely when the inter-vehicle distances diverge, $s \to \infty$. The circumstance that real-world cars have an upper velocity $u_{\max}$ implies that the carrying capacity must vanish instead as $u_{\max}/s$ for large inter-vehicle spacings s.

A natural way to overcome this deficiency of the basic model (4.58) would be to consider terms, like for the bird flocking model (4.50), expressing the preference to drive with a certain velocity. An alternative venue, pursued normally when modelling traffic flow, is to consider non-trivial distance dependencies for the velocity dependent force $g(x_j, x_i|\dot{x}_j, \dot{x}_i)$.

Non-Linear Model Driving a car one reacts stronger when the car in front is closer, an observation which can be modeled as

$$\ddot{x}_{j+1}(t+T) = \lambda\, \frac{\dot{x}_j(t) - \dot{x}_{j+1}(t)}{[x_j(t) - x_{j+1}(t)]^{1+\alpha}}, \qquad \alpha > 0\,, \tag{4.61}$$

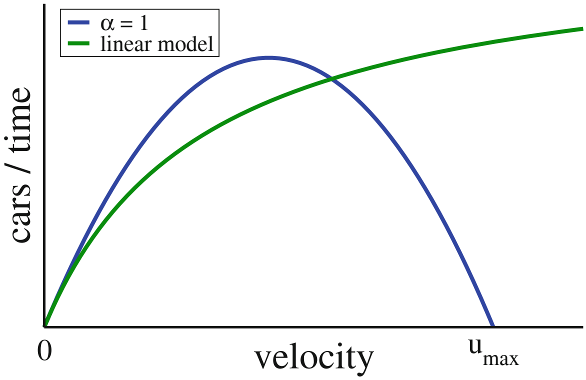


Fig. 4.12 For $\lambda = 3$, the number of cars per hour passing on a highway for the linear (4.60), and for the non-linear car following model (4.63). The carrying capacity vanishes when the road is congested, and the mean velocity vanishes, $u \to 0$. Arbitrary large cruising velocities are possible for the linear model in the limit of empty streets

where we postulated a scale-invariant dependency of the reaction strength with respect to the inter-particle distance $x_j - x_{j+1}$. Integrating (4.61) one obtains, in analogy to (4.59),

$$\dot{x}_{j+1} = \frac{\lambda}{-\alpha}[x_j - x_{j+1}]^{-\alpha} + d_0, \qquad u = \frac{\lambda}{\alpha}\left[\frac{1}{s_0^\alpha} - \frac{1}{s^\alpha}\right], \tag{4.62}$$

with s_0 denoting again the minimal inter vehicle distance and $d_0 = \lambda/(\alpha s_0^\alpha)$. For $\alpha > 0$ the mean velocity u takes a finite value

$$u_{\max} = \frac{\lambda}{\alpha s_0^\alpha}, \qquad u = u_{\max} - \frac{\lambda}{\alpha s^\alpha}, \qquad \frac{1}{s} = (u_{\max} - u)^{1/\alpha}\left(\frac{\alpha}{\lambda}\right)^{1/\alpha}$$

for near to empty streets with $s \to \infty$. The carrying capacity $q = u/s$ is then given by the parabola

$$q(u, \alpha = 1) = u(u_{\max} - u)/\lambda \tag{4.63}$$

for $\alpha = 1$, as illustrated in Fig. 4.12. The traffic volume is maximal for an intermediate mean velocity u, in accordance with daily observations.

4.4.2 Self-Organized Traffic Congestions

A steady flow of cars with $\dot{x}_j(t) \equiv u$ may become unstable when fluctuations propagate along the line of vehicles, a phenomena that can induce traffic congestion even in the absence of external influences, such as an accident.

Moving Frame of Reference The linear car following model (4.58) is the minimal model for analyzing the dynamics for intermediate to high densities of vehicles. We are interested in the evolution of the relative deviations $z_j = x_i/u - 1$ from the steady-state velocity u,

$$\dot{z}_{j+1}(t+T) = \lambda\big[z_j - z_{j+1}\big], \qquad \dot{x}_i(t) = u\big[1 - z_i(t)\big], \tag{4.64}$$

which corresponds to a moving frame of reference.

Slow Perturbations The steady-state configuration $\dot{x}_i(t) \equiv u$ is perturbed when the leading car changes its cruising speeds, e.g. via

$$z_0(t) \;\rightarrow\; \mathrm{e}^{\gamma t}, \qquad \gamma = 1/\tau + i\omega \qquad \tau > 0\,. \tag{4.65}$$

For evaluating the exact time evolution of the following cars one would need to integrate (4.64) piecewise, step by step for the intervals $t \in [nT, (n+1)T]$.

The situation simplifies when assuming that the delay interval T is substantially smaller than the timescale of the perturbation, τ. In this case the system will follow the behavior of the lead car smoothly.

Recursion We make the ansatz that the column follows the leading car with the same time-dependency, as given by (4.65), albeit with individual amplitudes a_j,

$$z_j(t) = a_j\,\mathrm{e}^{\gamma t}, \qquad \gamma\mathrm{e}^{\gamma T} a_{j+1} = \lambda(a_j - a_{j+1})\,.$$

Solving for a_{j+1} one obtains a linear recursion,

$$a_{j+1} = \frac{\lambda}{\lambda + \gamma\mathrm{e}^{T\gamma}}\,a_j, \qquad a_n = \left(\frac{\lambda}{\lambda + \gamma\mathrm{e}^{T\gamma}}\right)^n a_0\,. \tag{4.66}$$

The recursion is stable for real exponents $\gamma = 1/\tau$.

Delay Induced Instabilities We consider harmonic oscillations of the lead car velocity, which corresponds to imaginary exponents $\gamma = i\omega$ in (4.65). Evaluating the norm

$$\left|\lambda + \gamma\mathrm{e}^{T\gamma}\right|^2 = \left|\lambda + i\omega\mathrm{e}^{iT\omega}\right|^2 = \big[\lambda - \omega\sin(T\omega)\big]^2 + \omega^2\cos^2(T\omega)$$

we obtain

$$\left|\frac{\lambda}{\lambda + \gamma\mathrm{e}^{T\gamma}}\right| = \left(\frac{\lambda^2}{\lambda^2 + \omega^2 - 2\lambda\omega\sin(T\omega)}\right)^{1/2} \tag{4.67}$$

for the norm of the prefactor in the recursion (4.66). The recursion is unstable if

$$\lambda^2 + \omega^2 - 2\lambda\omega\sin(T\omega) \;<\; \lambda^2, \qquad \omega \;<\; 2\lambda\sin(T\omega)\,. \tag{4.68}$$

Suitable large time delays, with $T\omega \approx \pi/2$, induce an instability for any values of λ and ω.[13] Non-constant driving may induce traffic jams.

Self-Organized Traffic Jams A traffic instability occurs also in the limit of infinitesimal small frequencies $\omega \to 0$, namely when

$$1/(2\lambda) < T, \qquad \sin(T\omega) \approx T\omega\,. \tag{4.69}$$

When reactions are strong and slow (λ and T large), self-organized traffic jams are likely to emerge.

Propagating Perturbations Lastely, we examine how perturbations grow when $\gamma = 1/\tau$ is real. The deviations $z_n(t)$ from the steady state are

$$z_n(t) = a_0\, C^n\, \mathrm{e}^{t/\tau} = a_0\, \mathrm{e}^{n\log(C)}\, \mathrm{e}^{t/\tau}, \qquad C = \frac{\lambda\tau}{\lambda\tau + \mathrm{e}^{T/\tau}}\,.$$

We are interested in the speed v characterizing the propagation of the perturbation along the line of cars,

$$z_n(t) = a_0\, \mathrm{e}^{(n-vt)\log(C)}, \qquad v = \frac{-1}{\tau\log(C)}\,. \tag{4.70}$$

The speed is positive, $v > 0$, since $0 < C < 1$ and $\log(C) < 0$. Large reaction times T limit propagation speed, since

$$\log(C) \;\to\; -T/\tau, \qquad v \;\to\; 1/T$$

in the limit $T \gg \tau$.

Exercises

(4.1) <u>INSTABILITY OF EULER'S METHOD</u>
Euler's method is in general not suitable for numerically integrating partial differential equations. Discretizing in space and time the diffusion Eq. (4.1) in one dimension reads

$$\frac{\rho(x,t+\Delta t) - \rho(x,t)}{\Delta t} = D\,\frac{\rho(x+\Delta x,t) + \rho(x-\Delta x,t) - 2\rho(x,t)}{(\Delta x)^2}\,.$$

[13] This result is in accordance with our discussion in Sect. 2.5 of Chap. 2, regarding the influence of time delays in ordinary differential equations.

Prove that the resulting explicit time evolution map becomes unstable for $D\Delta t/(\Delta x)^2 > 1/2$ by considering $\rho(x,0) = \cos(\pi x/\Delta x)$ as a particular initial state.

(4.2) EXACT PROPAGATING WAVEFRONT SOLUTION
Find the reaction-diffusion system (4.2) for which the Fermi function

$$\rho^*(x,t) = \frac{1}{1+\mathbf{e}^{\beta(x-ct)}}$$

is an exact particular solution for a solitary propagating wavefront. Determine the reaction term $R(\rho)$ by evaluating the derivatives of ρ^*. Consider appropriate special values for β and for the propagation velocity c.

(4.3) LINEARIZED FISHER EQUATION
Consider the modified reaction-diffusion system

$$\dot{\rho} = \rho(1-\rho) + \rho'' + \frac{2}{1-\rho}(\rho')^2 \tag{4.71}$$

and show that it is equivalent to the linearized Fisher equation (4.18) using the transformation $\rho = 1 - 1/u$.

(4.4) TURING INSTABILITY WITH TWO STABLE NODES
Is it possible that the matrix A_1 entering the Turing instability and defined in (4.26), with positive $\epsilon_a, \epsilon_b > 0$, describes a stable node with both $\lambda_\pm < 0$? If yes, show that the superposition of two stable nodes may generate an unstable direction.

(4.5) EIGENVALUES OF $\mathbf{2} \times \mathbf{2}$ MATRICES
Use the standard expression (4.25) for the eigenvalues of a 2×2 matrix and show that local maxima of the potential $V(x)$ of a one-dimensional mechanical system

$$\dot{x} = y, \qquad \dot{y} = -\lambda(x)y - V'(x)$$

with a space-dependent damping factor $\lambda(x)$ are always saddles.

(4.6) LARGE DENSITIES OF NERVOUS RATS
Evaluate, using the stationarity condition (4.45), the number of nervous rats σ_n per comfort zone for large values of σ. When do all rats become nervous?

(4.7) AGENTS IN THE BOUNDED CONFIDENCE MODEL
Show that the total number of agents $\int \rho(x)dx$ is conserved for the bounded confidence model (4.54) for opinion dynamics.

(4.8) CLASSICAL VOTER MODEL
The basic asynchronous update step,

> "A random voter adapts the opinion of a random neighbor."

is repeated until consensus is reached. For a state with $N_\pm$ voters having voting preferences $\sigma_i = \pm 1$, one defines the magnetization $m = (N_+ -$

$N_-)/(N_+ + N_-)$. Show that m is conserved on the average and that the $\sigma_i \equiv +1$ consensus is reached with probability $(1+m)/2$.

Further Reading

The interested reader is encouraged to take a look at Ellner et al. (2011) for an in-depth treatise on mathematical modelling in biology and ecology, at Walgraef (2012) and Landge et al. (1920) for the generic mathematics of pattern formation in reaction-diffusion systems, and to Petrovskii et al. (2010) for a discussion of exactly solvable models in the field.

We further suggest Hassanien et al. (2018) for a comprehensive textbook on swarm intelligence, Kerner (2004) and Jusup et al. (2022) for traffic modelling and for reviews on opinion dynamics and flocking behaviors. A review of voter models is given in Redner (2019).

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Information Theory of Complex Systems

5

What do we mean, when we say that a given system shows "complex behavior", can we provide precise measures for the degree of complexity? This chapter offers an account of several common measures of complexity together with the relation of complexity to predictability and emergence.

Following a self-contained introduction to information theory and statistics, we will learn about probability distribution functions, Bayesian inference, the law of large numbers, and the central limit theorem. Next, Shannon entropy and mutual information will be discussed, two concepts that play central roles both in the context of time series analysis, and as starting points for the formulation of quantitative measures of complexity. The chapter concludes with a short overview regarding generative approaches to complexity.

5.1 Probability Distribution Functions

Statistics is ubiquitous in everyday life. We are used to chat, e.g. about the probability that our child will have blue or brown eyes, the chances to win a lottery, or those of a candidate to win the presidential elections. Statistics is ubiquitous also throughout the realms of science. Indeed, basic statistical concepts are used abandonedly all over these lecture notes.

Variables and Symbols Probability distribution functions may be defined for continuous or discrete variables, as well as for sets of symbols,

$$x \in [0, \infty], \qquad x_i \in \{1, 2, 3, 4, 5, 6\}, \qquad \alpha \in \{\text{blue, brown, green}\}\,.$$

For example, we may define with $p(x)$ the probability distribution of human life expectancy x, with $p(x_i)$ the chances to obtain x_i when throwing a dice, or with $p(\alpha)$ the probability to meet somebody having eyes of color α. Probabilities are

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https://doi.org/10.1007/978-3-031-55076-8_5

positive definite and the respective distribution functions normalized,

$$p(x),\; p(x_i),\; p(\alpha) \ge 0, \qquad\qquad 1 = \int_0^\infty p(x)\,dx = \sum_i p(x_i) = \sum_\alpha p(\alpha)\,.$$

The notation used for a given variable will indicate in the following its nature, i.e. whether it is a continuous or discrete variable, or denoting a symbol. For continuous variables the distribution $\rho(x)$ represents a probability density function (PDF).

Continuous vs. Discrete Stochastic Variables When discretizing a stochastic variable, e.g. when approximating an integral by a Riemann sum,

$$\int_0^\infty p(x)\,dx \approx \sum_{i=0}^{\infty} p(x_i)\,\Delta x, \qquad\qquad x_i = \Delta x\,(0.5 + i)\,, \tag{5.1}$$

the resulting discrete distribution function $p(x_i)$ is not any more normalized; the properly normalized discrete distribution function is $p(x_i)\Delta x$. The two notations, p_i and $p(x_i)$, are both used for discrete distributions.[1]

Mean, Median and Standard Deviation Common symbols for the average $\langle x\rangle$ are μ and $\bar{x}$. Average and standard deviation σ are given by

$$\langle x\rangle = \int x\, p(x)\,dx, \qquad\qquad \sigma^2 = \int (x-\bar{x})^2\, p(x)\,dx\,. \tag{5.2}$$

Mean and expectation value are synonyms for $\bar{x}$, with σ^2 being the variance.[2] For everyday life situations the median $\tilde{x}$, defined by

$$\int_{x<\tilde{x}} p(x)\,dx = \frac{1}{2} = \int_{x>\tilde{x}} p(x)\,dx\,, \tag{5.3}$$

is somewhat more intuitive than the mean. We have a 50 % chance to meet somebody being smaller/taller than the median height.

[1] The expression $p(x_i)$ is context specific and can denote both a properly normalized discrete distribution function as well as the value of a continuous probability distribution function.

[2] In formal texts on statistics and information theory, the notation $\mu = E(X)$ is used, where X stands for an abstract random variable, with x denoting a particular value and $p_X(x)$ the probability density.

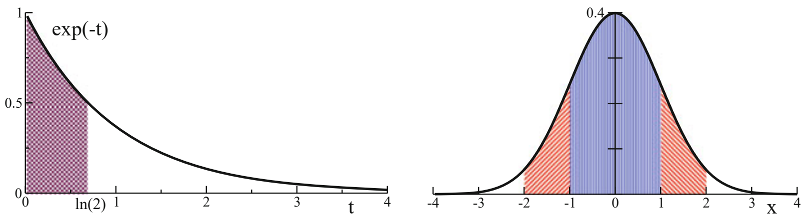

Fig. 5.1 **Left:** For an average waiting time $T = 1$, the exponential distribution $\exp(-t/T)/T$. With 50% probability waiting times are below the the median ln(2) (shaded area). **Right:** For a standard deviation $\sigma = 1$, the normal distribution $\exp(-x^2/2)/\sqrt{2\pi}$. The probability to draw a result within one/two standard deviations of the mean, $x \in [-1, 1]$ and $x \in [-2, 2]$ respectively (shaded regions), is 68 and 95%

Exponential Distribution A first example is the exponential distribution, which describes, e.g. the distribution of waiting times for radioactive decay,

$$p(t) = \frac{1}{T}\,\mathrm{e}^{-t/T}, \qquad\qquad \int_0^\infty p(t)\,dt = 1\,. \tag{5.4}$$

The mean waiting time is

$$\langle t\rangle = \frac{1}{T}\int_0^\infty t\,\mathrm{e}^{-t/T}\,dt = -t\,\mathrm{e}^{-t/T}\Big|_0^\infty + \int_0^\infty \mathrm{e}^{-t/T}\,dt = T\,.$$

Median $\tilde{t}$ and standard deviation σ are evaluated readily as

$$\tilde{t} = T\ln(2), \qquad\qquad \sigma = T\,.$$

In 50 % of times one has to wait less than $\tilde{t} \approx 0.69\,T$, which is smaller than the average waiting time T, compare Fig. 5.1.

Standard Deviation and Bell Curve The standard deviation σ measures the size of the fluctuations around the mean. The standard deviation is especially intuitive for the Gaussian distribution

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\,\mathrm{e}^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad\qquad \langle x\rangle = \mu, \qquad\qquad \langle (x-\bar{x})^2\rangle = \sigma^2\,. \tag{5.5}$$

"Gaussian", "Bell curve", and "normal distribution" all denote (5.5). Bell curves are ubiquitous in daily life, characterizing cumulative processes, as detailed out in Sect. 5.1.1.

Gaussians falls off rapidly with distance from the mean μ, compare Fig. 5.1. The probability to draw a value within n standard deviation of the mean, viz the probability that $x \in [\mu - n\sigma, \mu + n\sigma]$, is 68, 95, 99.7% for $n = 1, 2, 3$. These numbers are valid only for Gaussians, not for general probability distributions.

Probability Generating Functions We recall the basic properties of generating functions,[3]

$$G_0(x) = \sum_k p_k\, x^k\,, \tag{5.6}$$

which are defined for discrete distributions p_k, where $k = 0, 1, 2, \ldots$ For the normalization and the mean $\bar{k} = \langle k \rangle$ one evaluates

$$G_0(1) = \sum_k p_k = 1, \qquad G_0'(1) = \sum_k k\, p_k = \langle k \rangle\,. \tag{5.7}$$

The second moment $\langle k^2 \rangle$,

$$\langle k^2 \rangle = \sum_k k^2\, p_k\, x^k \Big|_{x=1} = x\frac{d}{dx}\big(x\, G_0'(x)\big)\Big|_{x=1}\,, \tag{5.8}$$

allows to express the variance $\sigma^2 = \langle (k - \bar{k})^2 \rangle$ as

$$\begin{aligned} \sigma^2 = \langle k^2 \rangle - \bar{k}^2 &= \frac{d}{dx}\big(x\, G_0'(x)\big)\Big|_{x=1} - \big(G_0'(1)\big)^2 \\ &= G_0''(1) + G_0'(1) - \big(G_0'(1)\big)^2\,. \end{aligned} \tag{5.9}$$

The importance of probability generating functions lies in the fact that the distribution for the sum $k = \sum_\alpha k^\alpha$ of independent stochastic variables k^α is generated by the product of the generating functions $G_0^\alpha(x)$ of the respective individual processes p_k^α, viz

$$G_0(x) = \sum_k p_k x^k = \prod_\alpha G_0^\alpha(x), \qquad G_0^\alpha(x) = \sum_k p_k^\alpha x^k\,.$$

This relation is easily verified, f.i. for the case of two random variables by multiplying out $G_0^1(x) G_0^2(x)$.

[3] Generating functions are discussed further in Sect. 1.3.2, of Chap. 1.

5.1.1 Law of Large Numbers

Throwing a dice many times and adding up the results obtained, the resulting average will be close to $3.5\,N$, where N is the number of throws. This is the typical outcome for cumulative stochastic processes.[4]

> LAW OF LARGE NUMBERS Repeating N times a stochastic process with mean $\bar{x}$ and standard deviation σ, the mean and the standard deviation of the cumulative result will approach $\bar{x}\,N$ and $\sigma\sqrt{N}$ respectively in the thermodynamic limit $N \to \infty$.

The law of large numbers implies, that one obtains $\bar{x}$ as an averaged result, with a standard deviation $\sigma/\sqrt{N}$ for the averaged process. One needs to increase the number of trials by a factor of four in order to improve accuracy by a factor of two.

Proof We prove the law of large numbers for a a discrete process p_k described by the generating functional $G_0(x)$. This is not really a restriction, since probability densities of continuous variables can be discretized with arbitrary accuracy. The generating function of the cumulative process is $G_0^N(x)$, which allows to express the mean as

$$\bar{k}^{(N)} = \frac{d}{dx} G_0^N(x)\Big|_{x=1} = N\, G_0^{N-1}(x)\, G_0'(x)\Big|_{x=1} = N\,\bar{k}\,,$$

with the help of (5.7). For the standard deviation $\sigma^{(N)}$ of the cumulative process we obtain

$$\begin{aligned}
\left(\sigma^{(N)}\right)^2 &= \frac{d}{dx}\left(x\,\frac{d}{dx} G_0^N(x)\right)\Big|_{x=1} - \left(N\bar{k}\right)^2 \\
&= \frac{d}{dx}\left(x\,N\,G_0^{N-1}(x)\,G_0'(x)\right)\Big|_{x=1} - N^2\left[G_0'(1)\right]^2 \\
&= NG_0'(1) + N(N-1)\left[G_0'(1)\right]^2 + NG_0''(1) - N^2\left[G_0'(1)\right]^2 \\
&= N\left(G_0''(1) + G_0'(1) - \left[G_0'(1)\right]^2\right) \equiv N\,\sigma^2\,,
\end{aligned} \tag{5.10}$$

viz the law of large numbers, where (5.9) was used twice.

[4] Please take note of the difference between a cumulative stochastic process, when adding the results of individual trials, and the "cumulative PDF", $F(x)$, defined by $F(x) = \int_{-\infty}^{x} p(x')dx'$.

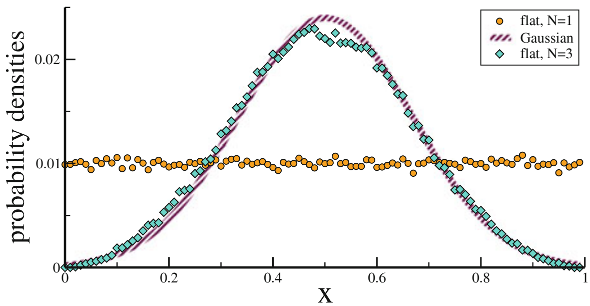

Fig. 5.2 The flat distribution, with variance $\sigma^2 = 1/12$, and the probability density of the sum of $N = 3$ flat distributions. The latter approximates remarkably well the limit Gaussian with $\sigma = 1/\sqrt{3 \cdot 12} = 1/6$, compare (5.11), in accordance with the central limit theorem. 10^5 random samples, with $N_{\text{bin}} = 100$ bins

Central Limit Theorem The law of large numbers states, that the variance σ^2 is additive for cumulative processes, not the standard deviation σ. The "central limit theorem" tells us in addition, that the limit distribution is a Gaussian, as illustrated in Fig. 5.2.

CENTRAL LIMIT THEOREM Given are $i = 1, \ldots, N$ independent random variables x_i, distributed with means μ_i and standard deviations σ_i. For $N \to \infty$, the cumulative distribution of $x = \sum_i x_i$ is described by a Gaussian with mean $\mu = \sum_i \mu_i$ and variance $\sigma^2 = \sum_i \sigma_i^2$.

In most cases one is not interested in the cumulative result, but in averaged quantities, which are obtained by rescaling variables,

$$y = x/N, \qquad \bar{\mu} = \mu/N, \qquad \bar{\sigma} = \sigma/N, \qquad p(y) = \frac{1}{\bar{\sigma}\sqrt{2\pi}}\,\mathrm{e}^{-\frac{(y-\bar{\mu})^2}{2\bar{\sigma}^2}}\,.$$

The rescaled standard deviation scales with $1/\sqrt{N}$. To see this, just consider identical processes with $\sigma_i \equiv \sigma_0$,

$$\bar{\sigma} = \frac{1}{N}\sqrt{\sum_i \sigma_i^2} = \frac{\sigma_0}{\sqrt{N}}\,, \tag{5.11}$$

in accordance with the law of large numbers.

Is Everything Boring Then? One might be tempted to draw the conclusion that systems containing large numbers of variables are boring, since everything seems to average out. This is actually not the case, the law of large numbers holds only for statistically independent processes. Subsystems of distributed complex systems are however dynamically dependent, and it is often the case that dynamical correlations lead to highly non-trivial properties in the thermodynamic limit.

5.1.2 Bayesian Statistics

The notions of statistics considered so far can be easily generalized to the case of more than one random variable. Whenever a certain subset of random variables is considered to be the causing event for the complementary subset of variables one speaks of inference, a domain of the Bayesian approach.

Conditional Probability Events and processes may have dependencies upon each other. A physician will typically have to know, to give an example, the probability that a patient has a certain illness, given that the patient shows a specific symptom.

> CONDITIONAL PROBABILITY The probability that an event x occurs, given that an event y has happened, is the "conditional probability" $p(x|y)$.

Throwing a dice twice, the probability that the first throw resulted in a 1, given that the total result was $4 = 1 + 3 = 2 + 2 = 3 + 1$, is 1/3. One defines with

$$p(x) = \int p(x|y)p(y)dy \tag{5.12}$$

the 'marginal' distribution $p(x)$. The probability of finding x is given by the probability of finding x given y, $p(x|y)$, integrated over the probability of finding y in the first place.

Bayes Theorem The probability distribution of throwing x in the first throw and y in the second throw is determined by the joint distribution $p(x, y)$, which obeys

$$1 = \int p(x, y)dxdy, \qquad p(x) = \int p(x, y)dy\,. \tag{5.13}$$

Together, the two expressions (5.12) and (5.13) for the marginal are equivalent to

$$p(x, y) = p(x|y)p(y) = p(y|x)p(x)\,,$$

or

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} = \frac{p(x|y)p(y)}{\int p(x|y)p(y)dy}\,, \tag{5.14}$$

where (5.12) was used in the second step. This relation is denoted "Bayes theorem". The conditional probability $p(x|y)$ of x happing given that y had occurred, is the "likelihood".

Bayesian Statistics As an exemplary application of Bayes theorem (5.14) consider a medical test.

– The probability of being ill/healthy is given by $p(y)$, $y = \text{ill}/\text{healthy}$.
– The likelihood of passing the test is $p(x|y)$, with $x = \text{positive}/\text{negative}$.

Let's consider an epidemic outbreak with 1 % of the population being infected on the average. Furthermore we assume that the test has an accuracy of 99 %,

$$p(\text{positive}|\text{ill}) = 0.99, \qquad p(\text{positive}|\text{healty}) = 0.02\,,$$

with the latter being the rate of false positives. The probability of a positively tested person of being infected is then actually just 33%,

$$\begin{aligned} p(\text{ill}|\text{pos}) &= \frac{p(\text{pos}|\text{ill})\,p(\text{ill})}{p(\text{pos}|\text{ill})\,p(\text{ill}) + p(\text{pos}|\text{healthy})\,p(\text{healthy})} \\ &= \frac{0.99 \cdot 0.01}{0.99 * 0.01 + 0.02 * 0.99} = \frac{1}{3}\,, \end{aligned}$$

where Bayes theorem (5.14) was used. A second follow-up test is hence necessary.

Statistical Inference We consider again a medical test, but in a slightly different situation. A series of test is performed in a city where an outbreak has occurred, but now with the purpose to estimate the percentage of people being infected.

We can then use expressing (5.12) for the marginal probability $p(\text{positive})$ for obtaining positive test results,

$$p(\text{positive}) = 0.99\,p(\text{ill}) + 0.02\,(1 - p(\text{ill})) \tag{5.15}$$

and solve for our estimate $p(\text{ill})$ of infections. In addition one needs to estimate the confidence of the obtained result, viz the expected fluctuations due to the limited number of tests actually carried out.

Bayesian Inference We start by noting that both sides of Bayes theorem (5.14) are properly normalized,

$$\int p(y|x)\,dy = 1 = \frac{\int p(x|y)p(y)\,dy}{p(x)}\,.$$

For a given x, the probability that any y happens is unity, and vice versa. For a given x we may hence interprete the left-hand side as the probability that y is true. We change the notation slightly,

$$p_1(y) \equiv \frac{p(x|y)\,p_0(y)}{\int p(x|y)\,p_0(y)dy}\,. \tag{5.16}$$

One denotes

- $p_1(y) = p(y|x)$ the "posterior" distribution,
- $p(x|y)$ the likelihood and with
- $p_0(y)$ the "prior".

Equation (5.16) constitutes the basis of Bayesian inference. In this setting one is not interested in finding a self-consistent solution $p_0(y) = p_1(y) = p(y)$. Instead it is premised that one disposes of prior information, viz knowledge and expectations, about the status of the world, $p_0(y)$. Performing an experiment a new result x is obtained which is then used to improve the expectations of the world status through $p_1(y)$, using (5.16).

Bayesian Learning The most common application of Bayesian inference is a situation where inference from a given set of experimental data needs to be drawn, using (5.16) a single time.

Alternatively one can consider (5.16) as the basis of cognitive learning processes, updating the knowledge about the world iteratively with any further observation $x_1, x_2, \ldots, x_n$,

$$p_i(y) \propto p(x_i|y)\, p_{i-1}(y), \qquad \forall y\,.$$

This update procedure of the knowledge $p_i(y)$ about the world is independent of the grouping of observations x_i, viz

$$p_0 \to p_1 \to \cdots \to p_n \qquad \text{and} \qquad p_0 \to p_n$$

yield the same result, due to the multiplicative nature of the likelihood $p(x|y)$, viz when considering in the last relation all consecutive observations $\{x_1, \ldots, x_n\}$ as a single event.

5.1.3 Statistical Binning

Beyond the elementary parameters, like mean and variance, one is interested in many cases in estimating the very probability distribution, in particular for data generated by some known or unknown process, like the temperature measurements of a weather station. When doing so, it is important to keep a few caveats in mind.

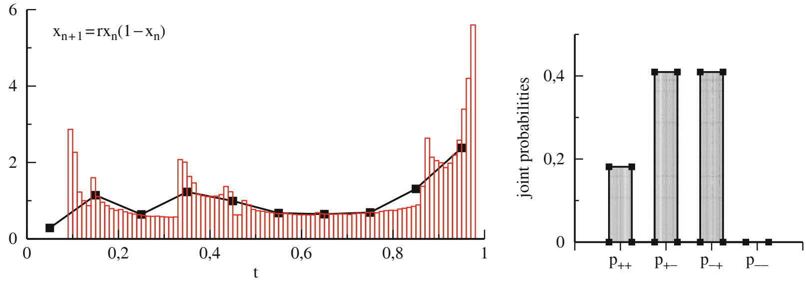

Fig. 5.3 For the logistic map with $r = 3.9$ and $x_0 = 0.6$, two statistical analyses of the identical time series $\{x_n|n = 0, \ldots, N\}$, with $N = 10^6$. **Left:** The distribution $p(x)$ of the x_n. Plotted at the midpoints of the respective bins is $p(x)(N_{\text{bin}}/N)$, for $N_{\text{bin}} = 10$ (square symbols) and $N_{\text{bin}} = 100$ (vertical bars). **Right:** The joint probabilities $p_{\pm\pm}$, as defined by (5.20), of consecutive increases/decreases of x_n. The probability p_{--} that the data decreases consecutively twice vanishes

Binning of Variables Data in the form of a time series of observations is generated typically by a dynamical system. As an example we examine the statistical properties of the logistic map,[5]

$$x_{n+1} = r\, x_n\, (1 - x_n), \qquad x_n \in [0, 1], \qquad r \in [0, 4]\,. \tag{5.17}$$

For systems with continuous readings, as for the logistic map, one needs to bin observations in order to estimate the respective probability distribution. In Fig. 5.3 the statistics of a time series in the chaotic regime is given, here for $r = 3.9$.

Apart from the overall number of bins, N_{bin}, a choice has to be made regarding the positions and the widths of the individual bins. When the data is not uniformly distributed, one may place more bins in a region of interest, generalizing the relation (5.1) through $\Delta x \to \Delta x_i$, with the Δx_i being the width of the individual bins.

For the example shown in Fig. 5.3, we selected $N_{\text{bin}} = 10/100$ equidistant bins. Note that the average number of observations scales with $1/N_{\text{bin}}$. Rescaling the count per bin with N_{bin} allows therefore to compare distributions obtained for different N_{bin}, as done in Fig. 5.3.

The selection of the binning procedure is in general an intricate choice. Fine structure will be lost when N_{bin} is too low, but statistical noise will dominate for large numbers of bins.

[5] A detailed account of period doubling and chaos in the logistic map can be found in Sect. 2.4, of Chap. 2.

Adaptive Binning Classically, number and positions of the set $\{B_i\}$ of bins are predetermined,

$$B_i = \big\{x|x \in [b_i^-, b_i^+]\big\}, \tag{5.18}$$

where $b_i^{\pm}$ is the upper/lower border of the ith bin. Observations $x_n \in B_j$ are added to the count of the jth bin. Results are assigned either to the bare midpoint $(b_i^+ + b_i^-)/2$, or to a weighted average.

For adaptive binning, the desired number N_{obs} of observations per bin is predetermined, not the B_i themselves. For the following we assume that the observations $x_n > 0$ are ordered, with $x_{n+1} \geq x_n$. Starting with $b_1^- = 0$, the following steps are repeated.

- For the ith bin, add data points until N_{obs} is reached. Say, that the last addition is the mth observation.
- Set the border of the current bin halfway to the next data point, $b_i^+ = (x_{m+1} + x_m)/2$.
- Repeat for the next bin, with matching bin borders, $b_{i+1}^- = b_i^+$.

Per construction, the statistical accuracy of all bins are identical when adaptive binning is used. Adaptive binning improves the quality of estimates substantially when the data is distributed unequally over large ranges. This is in particular the case for data showing powerlaw behavior, as for scale-free graphs.[6]

5.1.4 Time Series Characterization

Till now, we implicitly assumed now that the statistical evaluation of a given set of observations is done directly, without further preprocessing. This is however not always the optimal approach.

Symbolization One denotes with “symbolization” the construction of a finite number of symbols suitable for the statistical characterization of a of data of interest.

For a given time series $\{x_t\}$, a standard preprocessing procedure is to extract stepwise changes, such as $x_t - x_{t-1}$. Of interest is also if the data increases or decreases,

$$\delta_t = \text{sign}(x_t - x_{t-1}) = \begin{cases} 1 & x_t > x_{t-1} \\ -1 & x_t < x_{t-1} \end{cases} . \tag{5.19}$$

[6] The data for the in-degree of Internet domains presented in Fig. 1.6 of Chap. 1, has been processed using adaptive binning with $N_{\text{obs}} = 100$.

The consecutive development of the δ_t may also be encoded, using higher-level symbolic stochastic variables. For example, one might be interested in the joint probabilities

$$\begin{aligned} p_{++} &= \big\langle p(\delta_t = 1, \delta_{t-1} = 1)\big\rangle_t, \quad & p_{+-} &= \big\langle p(\delta_t = 1, \delta_{t-1} = -1)\big\rangle_t, \\ p_{-+} &= \big\langle p(\delta_t = -1, \delta_{t-1} = 1)\big\rangle_t, \quad & p_{--} &= \big\langle p(\delta_t = -1, \delta_{t-1} = -1)\big\rangle_t, \end{aligned} \tag{5.20}$$

where p_{++} gives the probability that the data increases at least twice consecutively, etc., and where $\langle \ldots \rangle_t$ denotes the time average. In Fig. 5.3 the values for the joint probabilities $p_{\pm\pm}$ are given for a selected time series of the logistic map in the chaotic regime. The data never decreases twice consecutively, $p_{--} = 0$, a somewhat unexpected result. It tells us, that certain properties of an otherwise chaotic system may be predictable,[7] at times even at a 100% level.

The symbolization procedure selected to analyze a given time series determines the type of information one may hope to extract, as evident from the results presented in Fig. 5.3. The selection of symbolization procedures is given further attention in Sect. 5.2.2.

Self Averaging Per definition, a time series produced by a dynamical system depends on the initial condition. The resulting statistical properties may hence also vary, e.g. when several distinct attracting states are present. We start with a basic example, the XOR series,[8]

$$\sigma_{t+1} = \mathrm{XOR}(\sigma_t, \sigma_{t-1}), \qquad \sigma_t = 0,\ 1\,. \tag{5.21}$$

The four initial conditions 00, 01, 10 and 11, give rise to the following time series',

$$\begin{array}{ll} \ldots 0000000\underline{00} & \qquad \ldots 1011011\underline{01} \\ \ldots 1101101\underline{10} & \qquad \ldots 0110110\underline{11} \end{array} \tag{5.22}$$

where time runs from right to left. In (5.22) the initial conditions σ_1 and σ_0 have been underlined. The typical time series, occurring for 75 % of the initial conditions, is $\ldots 011011011011\ldots$, with $p(0) = 1/3$ and $p(1) = 2/3$ for the probability to find respectively 0/1. When averaging over all four initial conditions, we have on the other hand $(2/3)(3/4) = 1/2$ for the probability to find a 1. Then

$$p(1) = \begin{cases} 2/3 & \text{typical} \\ 1/2 & \text{average} \end{cases}.$$

[7] General aspects of predictability in chaotic systems are developed in Sect. 3.1.3 of Chap. 3.

[8] Remember, that $\mathrm{XOR}(0,0) = 0 = \mathrm{XOR}(1,1)$ and $\mathrm{XOR}(0,1) = 1 = \mathrm{XOR}(1,0)$.

When observing a single time series we are likely to obtain the typical probability, analyzing many time series will result on the other hand in the average probability.

> SELF AVERAGING When the statistical properties of a time series generated by a dynamical process are independent of the respective initial conditions, one says the time series is "self averaging".

The XOR series is not self averaging and one can generally not assume self averaging to occur. An inconvenient situation whenever only a single time series is available, as it is the case for most historical data, e.g. of past climatic conditions.

XOR Series with Noise Most real-world processes involve a certain degree of noise. It is therefore tempting to presume, that noise could effectively restart the dynamics, leading to an implicit averaging over initial conditions. This assumption is not generally valid, it holds however for the XOR process with noise,

$$\sigma_{t+1} = \begin{cases} \mathrm{XOR}(\sigma_t, \sigma_{t-1}) & \text{with probability } 1-\xi \\ \neg\,\mathrm{XOR}(\sigma_t, \sigma_{t-1}) & \text{with probability } \xi \end{cases} \qquad 0 \le \xi \ll 1\,.$$

For low level of noise, $\xi \to 0$, the time series

$$\ldots 000000\underline{0}011011011\underline{0}10110110110\underline{1}1101101101\underline{1}00000000\ldots$$

has stretches of regular behavior interseeded by four types of noise induced dynamics (underlined, time running from right to left). Denoting with p_{000} and p_{011} the probability of finding regular dynamics of type "...000000000..." and "...011011011..." respectively, we obtain the master equation

$$\dot{p}_{011} = \xi p_{000} - \xi p_{011}/3 = -\dot{p}_{000} \tag{5.23}$$

for the noise-induced transition probabilities. In the stationary case $p_{000} = p_{011}/3$ for the XOR process with noise, the same ratio one would obtain for the deterministic XOR series averaged over the initial conditions.

The introduction of noise generally introduces complex dynamics akin to (5.23), which will lead in most cases to self-averaging time series. This is also the case for the OR time series,[9] for which the small noise limit does however not coincide with the time series obtained in the absence of noise.

Time Series Analysis and Cognition Time series analysis is a tricky business whenever the fundamentals of the generative process are unknown, e.g. whether noise is important or not. This is however the setting in which cognitive systems are operative. Our sensory organs, eyes and ears, provide us with a continuous time

[9] For the noisy OR series see exercise (5.4).

series encoding environmental information. Performing an informative and fast time series analysis is paramount for surviving.

ONLINE VS. OFFLINE ANALYSIS If one performs an analysis of a previously recorded time series one speaks of "offline" analysis. An analysis performed on-the-fly, while observing, corresponds to "online" processing.

Animals need to perform online analysis of their sensory data input streams, otherwise they would not survive long enough to react. Training of most machine learning algorithms is however offline.

Trailing Averages Online characterization of a time series in terms of its basic statistical properties, like mean and standard deviation, is quite straightforward.

For a continuous-time input stream $x(t)$ we define with

$$\mu_t = \frac{1}{T}\int_0^\infty d\tau\, x(t-\tau)\, \mathrm{e}^{-\tau/T} \tag{5.24}$$

$$\sigma_t^2 = \frac{1}{T}\int_0^\infty d\tau\, \big(x(t-\tau)-\mu_t\big)^2\, \mathrm{e}^{-\tau/T} \tag{5.25}$$

respectively the "trailing average" μ_t and the trailing variance σ_t^2. Trailing expectation values exponentially discount older data, with the respective moments of the input stream $x(t)$ being recovered in the limit $T \to \infty$. The factor $1/T$ in (5.24) and (5.25) normalizes the respective trailing averages. For the case of a constant, time independent input $x(t) \equiv \bar{x}$, we obtain correctly

$$\mu_t \to \frac{1}{T}\int_0^\infty d\tau\, \bar{x}\, \mathrm{e}^{-\tau/T} = \bar{x}\,.$$

The trailing average can be evaluated by a simple online update rule, there is no need to store past data $x(t-\tau)$. To see this, we evaluate the time dependence

$$\dot{\mu}_t = \frac{1}{T}\int_0^\infty d\tau\, \mathrm{e}^{-\tau/T}\frac{d}{dt}x(t-\tau) = \frac{-1}{T}\int_0^\infty d\tau\, \mathrm{e}^{-\tau/T}\frac{d}{d\tau}x(t-\tau)\,.$$

The last expression can be evaluated by direct partial integration.[10] One obtains

$$\dot{\mu}_t = \frac{x(t)-\mu_t}{T}\,, \tag{5.26}$$

[10] The identical procedure can be used in the context of dynamical systems with distributed time delays, as shown in Sect. 2.5.1 of Chap. 2.

together with an analogous update rule for the variance σ_t^2, by substituting $x \rightarrow (x-\mu)^2$. Expression (5.26) is an archetypical example of an online updating rule for a time averaged quantity, here the trailing average μ_t.

5.2 Entropy and Information

Entropy is a venerated concept from physics encoding the amount of disorder present in a thermodynamic system at a given temperature. The "Second Law of Thermodynamics" states, that entropy can only increase in an isolated, viz closed system. The second law has far reaching consequences, e.g. determining the maximal efficiency of engines and power plants, together with philosophical implications for our understanding of the fundamentals underpinning the nature of life as such.

Entropy and Life Living organisms have a body, which means that they are capable of creating ordered structures from basic chemical constituents. As a consequence, living entities decrease entropy locally, with their bodies, seemingly in violation of the second law. In reality, the local entropy depressions are created at the expense of corresponding entropy increases in the environment, in agreement with the second law of thermodynamics. All living beings need to be capable of manipulating entropy.

Information Entropy and Predictability Entropy is a central concept in information theory, where it is commonly denoted "Shannon entropy" or "information entropy". In this context, one is interested in the amount of information encoded by a sequence of symbols

$$\ldots \sigma_{t+2}, \sigma_{t+1}, \sigma_t, \sigma_{t-1}, \sigma_{t-2}, \ldots ,$$

e.g. when transmitting a message. Typically, in everyday computers, the σ_t are words of bits. Let us consider two time series of bits, e.g.

$$\ldots 101010101010 \ldots, \qquad\qquad \ldots 1100010101100 \ldots . \tag{5.27}$$

The first example is predictable, from the perspective of a time-series, and ordered, from the perspective of an one-dimensional alignment of bits. The second example is unpredictable and disordered respectively.

Information can be transmitted through a time series of symbols only when this time series is not predictable. Talking to a friend, to illustrate this statement, we will not learn anything new when capable of predicting his next joke. We have therefore the following two perspectives,

$$\text{high entropy} \,\hat{=}\, \begin{cases} \text{large disorder} & \text{physics} \\ \text{high information content} & \text{information theory} \end{cases}$$

and vice versa. Only seemingly disordered sequences of symbols are unpredictable and thus potential carriers of information. Note, that the predictability of a given time series, or its degree of disorder, may not necessarily be as self evident as in above example, Eq. (5.27), depending generally on the analysis procedure used, see Sect. 5.2.2.

Extensive Information In complex systems theory, as well as in physics, we are often interested in properties of systems composed of many subsystems.

> EXTENSIVE AND INTENSIVE PROPERTIES For systems composed of N subsystems a property is denoted "extensive" if it scales as $O(N^1)$ and "intensive" when it scales with $O(N^0)$.

A typical extensive property is the mass, a typical intensive property the density. When lumping together two chunks of clay, their mass adds, but the density does not change.

One demands, both in physics and in information theory, that the entropy should be an extensive quantity. The information content of two independent transmission channels should be just the sum of the information carried by the two individual channels.

Shannon Entropy The Shannon entropy $H[p]$ is defined as

$$H[p] = -\sum_{x_i} p(x_i)\,\log_b(p(x_i)) = -\langle\,\log_b(p)\,\rangle, \qquad H[p] \ge 0\,, \tag{5.28}$$

where $p(x_i)$ is a normalized discrete probability distribution function and where the brackets in $H[p]$ denote the functional dependence.[11] Note, that $-p\log(p) \ge 0$ for $0 \le p \le 1$, see Fig. 5.4, the entropy is therefore strictly positive.

b is the base of the logarithm used in (5.28). Common values of b are 2, Euler's number e and 10. The corresponding units of entropy are then termed "bit" for $b = 2$, "nat" for $b = e$ and "digit" for $b = 10$. In physics the natural logarithm is always used and there is an additional constant (the Boltzmann constant k_B) in front of the definition of the entropy. Here we will use $b = 2$ and drop in the following the index b.

Extensiveness of the Shannon Entropy The log-dependence in the definition of the information entropy in (5.28) is necessary for obtaining an extensive quantity. To see this, let us consider a system composed of two independent subsystems. The

[11] A function $f(x)$ is a function of a variable x; a functional $F[f]$ is, on the other hand, functionally dependent on a function $f(x)$. In formal texts on information theory the notation $H(X)$ is often used for the Shannon entropy and a random variable X with probability distribution $p_X(x)$.

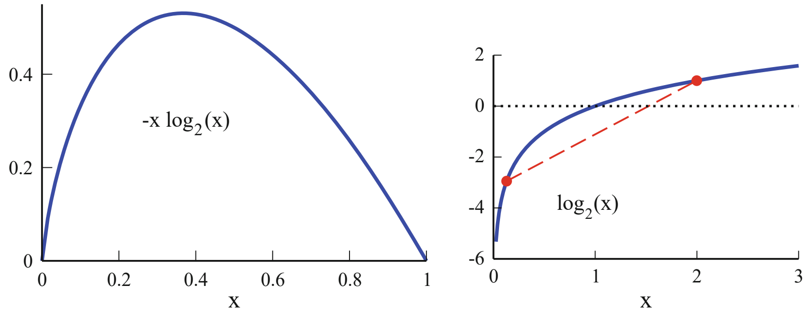

Fig. 5.4 **Left:** Plot of $-x\ \log_2(x)$. **Right:** The logarithm $\log_2(x)$ (full line) is concave, every cord (dashed line) lies below the graph

joint probability distribution is multiplicative,

$$p(x_i, y_j) = p_X(x_i)p_Y(y_j), \qquad \log(p(x_i, y_j)) = \log(p_X(x_i)) + \log(p_Y(y_j))\,.$$

The logarithm is the only function which maps a multiplicative input onto an additive output. Consequently,

$$\begin{aligned}
H[p] &= -\sum_{x_i,y_j} p(x_i, y_j)\log(p(x_i, y_j)) \\
&= -\sum_{x_i,y_j} p_X(x_i)p_Y(y_j)\Big[\log(p_X(x_i)) + \log(p_Y(y_j))\Big] \\
&= -\sum_{x_i} p_X(x_i)\sum_{y_j} p_Y(y_j)\log(p_Y(y_j)) \\
&\quad -\sum_{y_j} p_Y(y_j)\sum_{x_i} p_X(x_i)\log(p_X(x_i)) \\
&= H[p_Y] + H[p_X]\,,
\end{aligned}$$

as necessary for the extensiveness of $H[p]$. Hence the log-dependence in (5.28).

Degrees of Freedom We specialise to a discrete system with $x_i \in [1, \ldots, n]$, having n "degrees of freedom" in physics' slang. If the probability of finding any value is equally likely, as it is the case for a thermodynamic system at infinite temperatures, the entropy is

$$H = -\sum_{x_i} p(x_i)\log(p(x_i)) = -n\frac{1}{n}\log(1/n) = \log(n)\,, \tag{5.29}$$

a celebrated result. The entropy grows logarithmically with the number of degrees of freedom.

Shannon's Source Coding Theorem So far we did show, that (5.28) is the only possible definition, modulo renormalizing factors, for an extensive quantity depending exclusively on the probability distribution. The operative significance of the entropy $H[p]$ in terms of informational content is given by Shannon's theorem.

> SOURCE CODING THEOREM Given is a random variable x with a PDF $p(x)$ and entropy $H[p]$. The cumulative entropy $NH[p]$ is then, for $N \to \infty$, a lower bound for the number of bits necessary when compressing N independent processes drawn from $p(x)$.

If we compress more, we will lose information, the entropy $H[p]$ is therefore a measure of information content.

Entropy and Compression Let's make an example. Consider the four letter alphabet $\{A, B, C, D\}$. Suppose, that these four letters do not occur with the same probability, the relative frequencies being instead

$$p(A) = \frac{1}{2}, \qquad p(B) = \frac{1}{4}, \qquad p(C) = \frac{1}{8} = p(D)\,.$$

When transmitting a long series of words using this alphabet we will have the entropy

$$\begin{aligned} H[p] &= \frac{-1}{2}\log(1/2) - \frac{1}{4}\log(1/4) - \frac{1}{8}\log(1/8) - \frac{1}{8}\log(1/8) \\ &= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75\,, \end{aligned} \tag{5.30}$$

since we are using the logarithm with base $b = 2$. The most naive bit encoding,

$$A \to 00, \qquad B \to 01, \qquad C \to 10, \qquad D \to 11\,,$$

would use exactly 2 bit, which is larger than the Shannon entropy. An optimal encoding would be, on the other hand,

$$A \to 1, \qquad B \to 01, \qquad C \to 001, \qquad D \to 000\,, \tag{5.31}$$

leading to an average length of words transmitted of

$$p(A) + 2p(B) + 3p(C) + 3p(D) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75\,, \tag{5.32}$$

which is the same as the information entropy $H[p]$. The encoding given in (5.31) is actually "prefix-free". When we read the words from left to right, we know where a new word starts and stops,

$$110000010101 \qquad \longleftrightarrow \qquad \text{AADCBB}\,,$$

without ambiguity. Fast algorithms for optimal, or close to optimal encoding are clearly of importance in the computer sciences and for the compression of audio and video data.

Discrete vs. Continuous Variables When defining the entropy, we considered hitherto discrete variables. The information entropy can also be defined for continuous variables. We should be careful though, being aware that the transition from continuous to discrete stochastic variables, and vice versa, is slightly non-trivial, compare (5.1),

$$\begin{aligned} H[p]\Big|_{\text{con}} &= -\int p(x)\log(p(x))\,dx \approx -\sum_i p(x_i)\log(p(x_i))\,\Delta x \\ &= -\sum_i p_i \log(p_i/\Delta x) = -\sum_i p_i \log(p_i) + \sum_i p_i \log(\Delta x) \\ &= H[p]\Big|_{\text{dis}} + \log(\Delta x)\,, \end{aligned} \tag{5.33}$$

where $p_i = p(x_i)\Delta x$ is the properly normalized discretized PDF, see (5.1). The difference $\log(\Delta x)$ between the continuous-variable entropy $H[p]\big|_{\text{con}}$ and the discretized version $H[p]\big|_{\text{dis}}$ diverges as $\Delta x \to 0$, the transition is hence discontinuous.

Entropy of a Continuous PDF If follows from (5.33), that the Shannon entropy $H[p]\big|_{\text{con}}$ can be negative for a continuous probability distribution function. As an example consider a flat distribution in a small interval, $x \in [0, \epsilon]$,

$$p(x) = \begin{cases} 1/\epsilon & \text{for } x \in [0,\epsilon] \\ 0 & \text{otherwise} \end{cases}\,,$$

which leads to the entropy

$$H[p]\Big|_{\text{con}} = -\int_0^\epsilon \frac{1}{\epsilon}\log(1/\epsilon)\,dx = \log(\epsilon) < 0, \qquad \text{for} \quad \epsilon < 1\,.$$

The absolute value of the entropy is hence not meaningful for continuous probability density, only entropy differences. Hence one refers to $H[p]\big|_{\text{con}}$ as the "differential entropy".

5.2.1 Maximal Entropy Distributions

Which kind of distributions maximize entropy, viz information content? Remembering that

$$\lim_{p\to 0,1} p\log(p) = 0, \qquad \log(1) = 0\,,$$

see Fig. 5.4, it is intuitive that a flat distribution might be optimal. This is indeed correct in the absence of any constraint other than the normalization condition $\int p(x)dx = 1$.

Variational Calculus We turn to the task to maximize the functional

$$H[p] = \int f(p(x))\,dx, \qquad f(p) = -p\log(p)\,, \tag{5.34}$$

where the notation used will be of use later on. Maximizing a functional like $H[p]$ is a typical task of variational calculus, which examines the variation $\delta p(x)$ around an optimal function $p_{\mathrm{opt}}(x)$,

$$p(x) = p_{\mathrm{opt}}(x) + \delta p(x), \qquad \delta p(x)\ \text{ arbitrary}\,.$$

At optimality, the dependence of $H[p]$ on the variation δp should be stationary,

$$0 \equiv \delta H[p] = \int f'(p)\,\delta p\,dx, \qquad 0 = f'(p)\,, \tag{5.35}$$

where $f'(p) = 0$ follows from the fact that δp is an arbitrary function.

For the entropy functional $f(p) = -p\log(p)$ we find then with

$$f'(p) = -\log(p) - 1 = 0, \qquad p(x) = \text{const.} \tag{5.36}$$

the expected flat distribution.

Maximal Entropy Distributions with Constraints Under the constraint of a fixed average μ, the maximal entropy is determined by

$$f(p) = -p\log(p) - \lambda x p, \qquad \mu = \int x\,p(x)\,dx\,, \tag{5.37}$$

where λ is a "Lagrange parameter". It is used to enforce a given condition, here that μ takes a predefined value. The stationary condition $f'(p) = 0$ leads to

$$f'(p) = -\log(p) - 1 - \lambda x = 0, \qquad p(x) \propto 2^{-\lambda x} \sim \mathrm{e}^{-x/\mu}\,. \tag{5.38}$$

For a given mean μ, the the exponential distribution (5.38) maximises entropy. The Lagrange parameter λ is determined such that the condition (5.37) is satisfied. For a support $x \in [0, \infty]$, as assumed above, we have $\lambda \log_e(2) = 1/\mu$.

One can generalize this procedure and consider distribution maximizing the entropy under the constraint of a given mean μ and variance σ^2,

$$\mu = \int x\, p(x)\, dx, \qquad \sigma^2 = \int (x-\mu)^2\, p(x)\, dx\,. \tag{5.39}$$

Generalizing the derivation leading to (5.38), one sees that the maximal entropy distribution constrained by (5.39) is a Gaussian,[12] as given by (5.5).

Pairwise Constraints We consider a joint distribution function $p(x_1, \dots, x_n)$ for n variables x_i with pairwise correlations

$$\langle x_i x_j \rangle = \int dx^n x_i x_j\, p(x_1, \dots, x_n)\,. \tag{5.40}$$

Pair correlations can be measured in many instances experimentally, it is hence natural to considered them as constraints for modelling. One can adjust in the maximal entropy distribution

$$p(x_1, \dots, x_n) = \frac{\mathrm{e}^{-H}}{N}, \qquad H = \sum_{ij} J_{ij} x_i x_j + \sum_i \lambda_i x_i \tag{5.41}$$

the $n(n-1)/2$ variational parameters J_{ij} in order to reproduce given $n(n-1)/2$ pairwise correlations $\langle x_i x_j \rangle$, and the Lagrange multiplier λ_i for regulating the respective individual averages $\langle x_i \rangle$.

The maximal entropy distribution (5.41) has the form of a Boltzman factor of statistical mechanics with H representing an Hamiltonian, viz the energy function. It contains coupling constants J_{ij} encoding the strength of pairwise interactions.

5.2.2 Minimal Entropy Principle

The Shannon entropy is a very powerful concept in information theory. The encoding rules are typically known in practical applications, where there is no ambiguity regarding the symbolization procedure to employ when receiving a message via a given technical communication channel. This is however not the case when we are interested in determining the information content of real-world

[12] The derivation of the maximal entropy distribution constrained by μ and σ is treated in exercise (5.6).

processes, such as the time series of certain financial data or the data stream produced by our sensory organs.

Symbolization and Information Content The result obtained for the information content of a real-world time series $\{\sigma_t\}$ depends in general on the symbolization procedure used. Let us consider, as an example, the first time series of (5.27),

$$\ldots 101010101010\ldots . \tag{5.42}$$

When using a 1-bit symbolization procedure, we have

$$p(0) = \frac{1}{2} = p(1), \qquad H[p] = -2\,\frac{1}{2}\log(1/2) = 1\,,$$

as expected. If, on the other hand, we use a 2-bit symbolization, we find

$$p(00) = p(11) = p(01) = 0, \qquad p(10) = 1, \qquad H[p] = -\log(1) = 0\,.$$

When 2-bit encoding is presumed, the time series is predictable and carries no information. This seems intuitively the correct result and the question is: Can we formulate a general guiding principle which tells us which symbolization procedure would yield the more accurate result for the information content of a time series at hand?

Minimal Entropy Principle The Shannon entropy constitutes a lower bound for the number of bits per symbol necessary when compressing the data without information loss. Trying various symbolization procedures, the symbolization procedure yielding the lowest information entropy allows us consequently to represent a given time series lossless with the least number of bits.

> MINIMAL ENTROPY PRINCIPLE The information content of a time series with unknown encoding is given by the minimum (actually the infimum) of the Shannon entropy over all possible symbolization procedures.

The minimal entropy principle then gives us a definite answer with respect to the information content of the time series in (5.42). We have seen that at least one symbolization procedure yields a vanishing entropy, the lowest possible value since $H[p] \geq 0$. This is the expected result, since $\ldots 01010101 \ldots$ is predictable.

Information Content of a Predictable Time Series Note, that a vanishing information content $H[p] = 0$ only implies that the time series is strictly predictable, not that it is constant. One therefore needs only a finite amount of information to encode the full time series, viz for arbitrary lengths $N \to \infty$. When the time series is predictable, the information necessary to encode the series is intensive and not extensive.

Symbolization and Time Horizons The minimal entropy principle is rather abstract. In practice, one may not be able to try out more than a handful of different symbolization procedures. It is therefore important to gain an understanding of the time series at hand.

A core defining aspect of many time series is the intrinsic time horizon τ. Most dynamical processes have characteristic time scales, with the consequence that memories of past states are effectively lost for times exceeding these intrinsic time scales. The symbolization procedure used should therefore match the time horizon τ.

This is what happened when analyzing the time series of (5.42), for which $\tau = 2$. A 1-bit symbolization procedure implicitly presumes that σ_t and σ_{t+1} are statistically independent, missing the intrinsic time scale $\tau = 2$, in contrast to a 2-bit symbolization procedure.

5.2.3 Mutual Information

So far, we have been concerned mostly with individual stochastic processes as well as the properties of cumulative processes generated by the sum of stochastically independent random variables. In order to understand complex systems, we need to develop tools for the description of a large number of interdependent processes. As a first step towards this direction, we study in the following the case of two stochastic processes, which may now be statistically correlated.

Two Channel Markov Process We start with an illustrative example of two correlated channels σ_t and τ_t, with

$$\sigma_{t+1} = \mathrm{XOR}(\sigma_t, \tau_t) \tag{5.43}$$

$$\tau_{t+1} = \begin{cases} \mathrm{XOR}(\sigma_t, \tau_t) & \text{with probability } 1-\xi \\ \neg\mathrm{XOR}(\sigma_t, \tau_t) & \text{with probability } \xi \end{cases} .$$

This dynamics is markovian, as the value for the state $\{\sigma_{t+1}, \tau_{t+1}\}$ depends only on the state at the previous time step,[13] viz on $\{\sigma_t, \tau_t\}$.

> MARKOV PROCESS A discrete-time memory-less dynamical process is denoted a Markov process. The likelihood of future states depends only on the present state, and not on any past states.

When state space is finite, as in our example, one has a Markov chain.

[13] Markov chains are the subject of Sect. 3.3.2 of Chap. 3.

Joint Probabilities A typical instance of the Markov chain specified in (5.43) is

$$\begin{array}{lcl} \ldots\sigma_{t+1}\sigma_t\ldots & : & 0\,0\,0\,1\,0\,0\,0\,0\,0\,0\,0\,1\,0\,1\,0\ldots \\ \ldots\tau_{t+1}\tau_t\ldots & : & 0\,0\,0\,1\,\underline{1}\,0\,0\,0\,0\,0\,0\,1\,\underline{1}\,1\,1\ldots \end{array}$$

where we did underline the loci of noise-induced transitions. For $\xi = 0$ the stationary state is $\{\sigma_t, \tau_t\} = \{0, 0\}$, which is fully correlated. We now calculate the joint probabilities $p(\sigma, \tau)$ for general values of noise ξ, using the transition probabilities

$$\begin{array}{ll} p_{t+1}(0,0) = (1-\xi)\,[p_t(1,1) + p_t(0,0)] & p_{t+1}(1,0) = \xi\,[p_t(0,1) + p_t(1,0)] \\ p_{t+1}(1,1) = (1-\xi)\,[p_t(1,0) + p_t(0,1)] & p_{t+1}(0,1) = \xi\,[p_t(0,0) + p_t(1,1)] \end{array}$$

for the ensemble averaged joint probability distributions $p_t(\sigma, \tau) = \langle p(\sigma_t, \tau_t)\rangle_{\text{ens}}$, where the average $\langle ..\rangle_{\text{ens}}$ denotes the average over an ensemble of time series. For the solution in the stationary case, $p_{t+1}(\sigma, \tau) = p_t(\sigma, \tau) \equiv p(\sigma, \tau)$, we use the normalization

$$p(1,1) + p(0,0) + p(1,0) + p(0,1) = 1\,,$$

finding

$$p(1,1) + p(0,0) = 1 - \xi, \qquad p(1,0) + p(0,1) = \xi\,,$$

by adding the terms $\propto (1-\xi)$ and $\propto \xi$ respectively. It then follows immediately that

$$\begin{array}{ll} p(0,0) = (1-\xi)^2 & p(1,0) = \xi^2 \\ p(1,1) = (1-\xi)\xi \end{array}, \quad \begin{array}{l} \\ p(0,1) = \xi(1-\xi) \end{array}. \tag{5.44}$$

For $\xi = 1/2$ the two channels become 100 % uncorrelated, as the τ-channel is then fully random. The dynamics of the Markov process given in (5.43) is self averaging, which allows to verify (5.44) by a straightforward numerical simulation.

Marginal Distributions Using the notation

$$p_\sigma(\sigma') = \sum_{\tau'} p(\sigma', \tau'), \qquad p_\tau(\tau') = \sum_{\sigma'} p(\sigma', \tau')$$

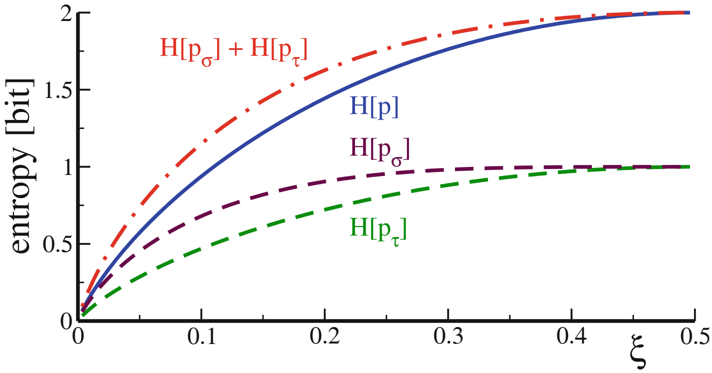

Fig. 5.5 For the two-channel XOR-Markov chain $\{\sigma_t, \tau_t\}$ with noise ξ, see (5.43), the entropy $H[p]$ of the combined process (full line), see (5.47), of the individual channels (dashed lines), see(5.46), $H[p_\sigma]$ and $H[p_\tau]$, together with the sum of the joint entropies (dot-dashed line). Note the positiveness of the mutual information, $I(\sigma,\tau) > 0$, with $I(\sigma,\tau) = H[p_\sigma] + H[p_\tau] - H[p]$

for the marginal distributions p_σ and p_τ, we find from (5.44)

$$\begin{array}{ll} p_\sigma(0) = 1-\xi \\ p_\sigma(1) = \xi \end{array}, \qquad \begin{array}{l} p_\tau(0) = 1-2\xi(1-\xi) \\ p_\tau(1) = 2\xi(1-\xi) \end{array} \tag{5.45}$$

for the distributions of the two individual channels.

Joint and Marginal Entropy We evaluate now two entropies, that of the individual channels, $H[p_\sigma]$ and $H[p_\tau]$, the "marginal entropies", viz

$$H[p_\sigma] = -\langle \log(p_\sigma)\rangle, \qquad H[p_\tau] = -\langle \log(p_\tau)\rangle\,, \tag{5.46}$$

as well as the entropy of the combined process, termed "joint entropy",

$$H[p] = -\sum_{\sigma',\tau'} p(\sigma',\tau')\log(p(\sigma',\tau'))\,. \tag{5.47}$$

In Fig. 5.5 the respective entropies are plotted as a function of noise strength ξ. Some observations.

- In the absence of noise, $\xi = 0$, both the individual channels, as well as the combined process, are predictable and all three entropies, $H[p]$, $H[p_\sigma]$ and $H[p_\tau]$, vanish.
- For maximal noise $\xi = 0.5$, the information content of both individual chains is 1 bit and of the combined process 2 bits, implying statistical independence.

– For general noise strengths $0 < \xi < 0.5$, the two channels are statistically correlated. The information content of the combined process $H[p]$ is consequently smaller than the sum of the information contents of the individual channels, $H[p_\sigma] + H[p_\tau]$.

Mutual Information The degree of statistical dependency of two channels can be measured by comparing the joint entropy with the respective marginal entropies.

MUTUAL INFORMATION For two stochastic processes σ_t and τ_t, the difference

$$I(\sigma, \tau) = H[p_\sigma] + H[p_\tau] - H[p] \tag{5.48}$$

between the sum of the marginal entropies $H[p_\sigma] + H[p_\tau]$ and the joint entropy $H[p]$ is the mutual information $I(\sigma, \tau)$.

When two dynamical processes become correlated, information is lost and this information loss is given by the mutual information. Note, that $I(\sigma, \tau) = I[p]$ is a functional of the joint probability distribution p only, the marginal distribution functions p_σ and p_τ being themselves functionals of p.

Positiveness In the following we refer to the general case of two stochastic processes described by the joint distribution $p(x, y)$ and the respective marginal densities $p_X(x) = \int p(x, y)dy$, and $p_Y(y) = \int p(x, y)dx$. The mutual information

$$\begin{aligned} I(X, Y) &= \langle \log(p) \rangle - \langle \log(p_X) \rangle - \langle \log(p_Y) \rangle \qquad (5.49) \\ &= \int p(x, y) \Big[\log(p(x, y)) - \log(p_X(x)) - \log(p_Y(y)) \Big] dxdy \end{aligned}$$

is strictly positive, $I(X, Y) \geq 0$, as we will show now. Rewriting the mutual information as

$$\begin{aligned} I(X, Y) &= \int p(x, y) \log\left(\frac{p(x, y)}{p_X(x) p_Y(y)} \right) dxdy \\ &= - \int p \log\left(\frac{p_X p_Y}{p} \right) dxdy\,, \end{aligned} \tag{5.50}$$

the positiveness of $I(X, Y)$ follows from the concaveness of the logarithm,

$$\log(p_1 x_1 + p_2 x_2) \geq p_1 \log(x_1) + p_2 \log(x_2), \qquad \forall x_1, x_2 \in [0, \infty]\,, \tag{5.51}$$

as illustrated in Fig. 5.4. The inequality (5.51) holds for $p_1, p_2 \in [0, 1]$, with $p_1 + p_2 = 1$, which expresses that any cord of a concave function lies below the graph. We can regard p_1 and p_2 as the coefficients of a distribution function and generalize,

$$p_1 \delta(x - x_1) + p_2 \delta(x - x_2) \quad \longrightarrow \quad p(x)\,,$$

where $p(x)$ is now a generic, properly normalized probability density. The concaveness condition (5.51) then becomes

$$\log\left(\int p(x)\,x\,dx\right) \ge \int p(x)\log(x)dx, \qquad \Phi\left(\langle x\rangle\right) \ge \langle\, \Phi(x)\,\rangle\,. \tag{5.52}$$

This is the Jensen inequality, which holds for any concave function $\Phi(x)$. It remains valid when substituting $x \to p_X p_Y/p$ for the argument of the logarithm.[14] For the mutual information (5.50) we then obtain

$$\begin{aligned} I(X,Y) &= -\int p\log\left(\frac{p_X p_Y}{p}\right)dxdy \;\ge\; -\log\left(\int p\,\frac{p_X p_Y}{p}dxdy\right) \\ &= -\log\left(\int p_X(x)dx\int p_Y(y)dy\right) = -\log(1) = 0\,, \end{aligned}$$

viz that $I(X,Y)$ is non-negative. Information can only be lost, and not gained, when correlating two previously independent processes.

Conditional Entropy There are various ways to rewrite the mutual information, using Bayes theorem $p(x,y) = p(x|y)p_Y(y)$ between the joint density $p(x,y)$, the conditional probability distribution $p(x|y)$ and the marginal $p_Y(y)$, e.g.

$$\begin{aligned} I(X,Y) &= \left\langle \log\left(\frac{p}{p_X p_Y}\right)\right\rangle = \int p(x,y)\,\log\left(\frac{p(x|y)}{p_X(x)}\right)dxdy \\ &\equiv H(X) - H(X|Y)\,, \end{aligned} \tag{5.53}$$

where we used the notation $H(X) = H[p_X]$ for the marginal entropy, together with the "conditional entropy"

$$H(X|Y) = -\int p(x,y)\log(p(x|y))dxdy\,. \tag{5.54}$$

The conditional entropy is positive for discrete processes, since

$$-\,p(x_i,y_j)\log(p(x_i|y_j)) = -p(x_i|y_j)p_Y(y_j)\log(p(x_i|y_j))$$

is positive, given that $-p\log(p) \ge 0$ holds in the interval $p \in [0,1]$. Compare (5.33) for changing from continuous to discrete variables. Several variants for the conditional entropy can be used to define statistical complexity measures, as discussed in Sect. 5.3.1.

[14] For a proof consider the generic substitution $x \to q(x)$ and a transformation of variables $x \to q$ via $dx = dq/q'$, with $q' = dq(x)/dx$, for the integration in (5.52).

Causal Dependencies For independent processes one has $p(x, y) = p(x)p(y) = p(x|y)p(y)$ and hence

$$p(x|y) = p(x), \qquad H(X|Y) \to H(X)\,.$$

The opposite extreme is realized when the first channel is just a function of the second channel, viz when

$$x_i = f(y_i), \qquad p(x_i|y_i) = \delta_{x_i,f(y_i)}, \qquad p(x_i, y_i) = \delta_{x_i,f(y_i)}\, p(y_i)\,.$$

The conditional entropy (5.54) then vanishes,

$$H(X|Y) = -\sum_{x_i,y_j} \delta_{x_i,f(y_j)}\, p_Y(y_j) \log\left(\delta_{x_i,f(y_j)}\right) = 0\,,$$

since $\delta_{x_i,f(y_j)}$ is either unity, in which case $\log(\delta) = \log(1) = 0$, or zero, in which case $0\log(0)$ vanishes as a limiting process. The conditional entropy $H(X|Y)$ measures hence the amount of information present in the stochastic process X which is not causally related to the process Y. The mutual entropy reduces to the marginal entropy, as a corollary,

$$I(X, Y) \to H(X)\,,$$

for the case that X is fully determined by Y, compare (5.53).

5.2.4 Kullback-Leibler Divergence

One is often interested in comparing two distribution functions $p(x)$ and $q(x)$ with respect to their similarity. When trying to construct a measure for the degree of similarity one is facing the dilemma that probability distributions are positive definite and one can hence not define a scalar product as for vectors; two probability densities cannot be orthogonal. It is nevertheless possible to define a positive definite measure.

KULLBACK-LEIBLER DIVERGENCE Given two probability distribution functions $p(x)$ and $q(x)$, the functional

$$K[p;q] = \int p(x)\log\left(\frac{p(x)}{q(x)}\right)dx \ge 0 \tag{5.55}$$

is a non-symmetric measure for the difference between $p(x)$ and $q(x)$.

The Kullback-Leibler divergence $K[p;q]$ is also denoted "relative entropy." The proof for $K[p;q] \ge 0$ is analogous to the one for the mutual information given in Sect. 5.2.3. The Kullback-Leibler divergence vanishes for identical probability distributions, viz when $p(x) \equiv q(x)$.

Relation to the χ^2 Test We consider the case that the two distribution functions p and q are nearly identical,

$$q(x) = p(x) + \delta p(x), \qquad \delta p(x) \ll 1\,,$$

and expand $K[p;q]$ in powers of $\delta p(x)$, using

$$\log(q) = \log(p + \delta p) \approx \log(p) + \frac{\delta p}{p} - \left(\frac{\delta p}{p}\right)^2 + \ldots\,,$$

obtaining

$$\begin{aligned} K[p;q] &\approx \int dx\, p\Big[\log(p) - \log(p) - \frac{\delta p}{p} + \left(\frac{\delta p}{p}\right)^2\Big] \\ &= \int dx\, \frac{(\delta p)^2}{p} = \int dx\, \frac{(p-q)^2}{p}\,, \end{aligned} \tag{5.56}$$

since $\int \delta p dx = 0$, as a consequence of the normalization conditions $\int p dx = 1 = \int q dx$. This measure for the similarity of two distribution functions is termed χ^2 test. It is actually symmetric under exchanging $q \leftrightarrow p$, up to order $(\delta p)^2$.

Example As a simple example we take two distributions $p(\sigma)$ and $q(\sigma)$ for a binary variable $\sigma = 0/1$,

$$p(0) = 1/2 = p(1), \qquad q(0) = \alpha, \qquad q(1) = 1 - \alpha\,, \tag{5.57}$$

with $p(\sigma)$ being flat and $\alpha \in [0, 1]$. The Kullback-Leibler divergence,

$$\begin{aligned} K[p;q] &= \sum_{\sigma=0,1} p(\sigma)\log\left(\frac{p(\sigma)}{q(\sigma)}\right) = \frac{-1}{2}\log(2\alpha) - \frac{1}{2}\log(2(1-\alpha)) \\ &= \frac{-1}{2}\log(4(1-\alpha)\alpha) \;\geq\; 0\,, \end{aligned}$$

is unbounded, since $\lim_{\alpha\to 0,1} K[p;q] \to \infty$. Interchanging $p \leftrightarrow q$ yields

$$\begin{aligned} K[q;p] &= \alpha\log(2\alpha) + (1-\alpha)\log(2(1-\alpha)) \\ &= \log(2) + \alpha\log(\alpha) + (1-\alpha)\log(1-\alpha) \;\geq\; 0\,, \end{aligned}$$

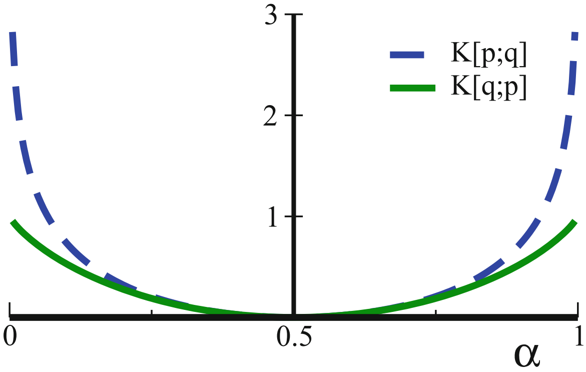

Fig. 5.6 For two distributions p and q parametrized by α, see (5.57), the respective Kullback-Leibler divergences $K[p;q]$ (dashed line) and $K[q;p]$ (full line). Note the maximal asymmetry for $\alpha \to 0, 1$, where $\lim_{\alpha\to 0,1} K[p;q] = \infty$

which is now finite in the limit $\lim_{\alpha\to 0/1}$. The Kullback-Leibler divergence is highly asymmetric, compare Fig. 5.6.

Kullback-Leibler Divergence vs. Mutual Information The mutual information, as defined by (5.50), is a special case of the Kullback-Leibler divergence. We first write (5.2.4) for the case that p and q depend on two variables x and y,

$$K[p;q] = \int p(x,y)\log\left(\frac{p(x,y)}{q(x,y)}\right) dxdy\,. \tag{5.58}$$

This expression is identical to the mutual information (5.50) for the case that $q(x,y)$ is the product of the two marginal distributions of $p(x,y)$,

$$q(x,y) = p(x)p(y), \qquad p(x) = \int p(x,y)dy, \qquad p(y) = \int p(x,y)dx\,.$$

Two independent processes are described by the product of their probability distributions. The mutual information hence measures the distance between a joint distribution $p(x,y)$ and the product of its marginals, viz the distance between correlated and independent processes.

Fisher Information The Fisher information $F(\theta)$ measures the sensitivity of a distribution function $p(y,\theta)$ with respect to a given parametric dependence θ,

$$F(\theta) = \int \left(\frac{\partial}{\partial\theta}\ln(p(y,\theta))\right)^2 p(y,\theta)dy\,. \tag{5.59}$$

In typical applications the parameter θ is a hidden observable one may be interested to estimate.

Kullback-Leibler Divergence vs. Fisher Information The infinitesimal Kullback-Leibler divergence between $p(y, \theta)$ and $p(y, \theta + \delta\theta)$ is

$$K = \int dy\, p(y,\theta) \log\left(\frac{p(y,\theta)}{p(y,\theta+\delta\theta)}\right) \approx -\int dy\, p \log\left(\frac{p + p'\delta\theta}{p}\right)$$
$$= -\int dy\, \frac{\partial p(y,\theta)}{\partial\theta}\delta\theta + \frac{(\delta\theta)^2}{2}\int dy\, \frac{1}{p(y,\theta)}\left(\frac{\partial p(y,\theta)}{\partial\theta}\right)^2 \tag{5.60}$$

with $p = p(y, \theta)$ and $p' = \partial p(y, \theta)/\partial\theta$. The first term in (5.60),

$$(-\delta\theta)\frac{\partial}{\partial\theta}\int dy\, p(y,\theta) = (-\delta\theta)\frac{\partial}{\partial\theta}1 \equiv 0\,,$$

vanishes. The second term in (5.60) contains the Fisher information (5.59). Hence

$$K\big[p(y,\theta);\, p(y,\theta+\delta\theta)\big] = F(\theta)\frac{(\delta\theta)^2}{2}\,, \tag{5.61}$$

which establishes the role of the Fisher information as a metric.

5.3 Complexity Measures

Can we provide a single measure, or a small number of measures, suitable for characterizing the 'degree of complexity' of any dynamical system at hand? This rather philosophical question has fascinated researchers for decades, yet no definitive answer is known.

The quest of complexity measures touches a range of interesting topics in dynamical systems theory, which has led to a number of powerful tools suitable for studying dynamical systems, the original goal of developing a one-size-fits-all measure for complexity seems however not longer a scientifically valid target. Complex dynamical systems can show an extended variety of qualitatively different behaviors, one of the reasons why complex system theory is fascinating, and it is not appropriate to shove all complex systems into a single basket for the purpose of measuring their degree of complexity with a single yardstick.

Intuitive Complexity The task of developing a mathematically well defined measure for complexity is handicapped by the lack of a precisely defined goal. In the following we discuss selected prerequisites and constraints one may postulate for valid complexity measures. In the end it is, however, up to our intuition to decipher whether these requirements are appropriate or not.

An example of a process one may intuitively attribute a high degree of complexity are the intricate spatio-temporal patterns generated by the forest fire model, with

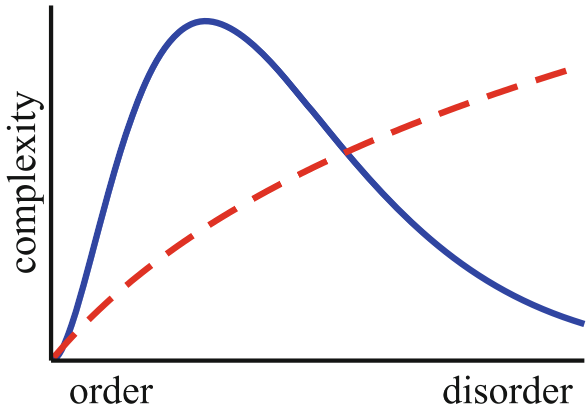

Fig. 5.7 The degree of complexity (full line) should be minimal both in the fully ordered and the fully disordered regime. For some applications it may however be meaningful to consider complexity measures maximal for random states (dashed line)

perpetually changing fronts of fires burning through a continuously regrowing forest.[15]

Complexity vs. Randomness A popular proposal for a complexity measure is the information entropy $H[p]$, as defined by (5.28). It vanishes when the system is regular, which agrees with our intuitive presumption that complexity is low when nothing happens. The entropy is however maximal for random dynamics.

It is a question of viewpoints to which extent one should consider random systems as complex, compare Fig. 5.7. For some considerations, e.g. when dealing with 'algorithmic complexity', which will be treated in Sect. 5.3.2), it makes sense to attribute maximal complexity degrees to fully random sets of objects. In general, however, complexity measures should be concave, attaining minimal values for regular behavior as well as for purely random sequences.

Complexity of Multi-component Systems Complexity should be a positive quantity, like entropy. But what about being extensive or intensive? This is a non-trivial question.

Intuitively one may demand complexity to be intensive, as one would not expect to gain complexity when lotting together N independent dynamical systems. On the other side we cannot rule out that a set of strongly interacting dynamical systems could show more and more complex behavior with an increasing number of subsystems, along the lines of the saying *'quantity has its own quality'*. This purposely a feature of massive machine learning architecture, or of human brains.

There is no simple way out of this quandary when searching for a single one-size-fits-all complexity measure. Both intensive and extensive complexity measures have their areas of validity.

[15] States of the forest fire model are presented in Fig. 6.7, see Chap. 6.

Complexity and Behavior The search for complexity measures is not just an abstract academic quest. As an example consider how bored we are when our environment is repetitive, having low complexity, and how stressed when complexity overwhelms our sensory organs. There are indeed indications that a valid behavioral strategy for highly developed cognitive systems may consist in optimizing the degree of complexity. Well defined complexity measures are necessary in order to quantify this intuitive statement mathematically.

5.3.1 Complexity and Predictability

Interesting complexity measures can be constructed using statistical tools, generalizing concepts like information entropy and mutual information. We consider here time series generated from a finite set of symbols. One may, however, interchange time with space whenever one is concerned with studying the complexity of spatial structures.

Stationary Dynamical Processes As a prerequisite for the analysis of complexity we need stationary dynamical processes, viz dynamical processes which do not change their behavior and their statistical properties qualitatively over time. In practice, this implies that the time series considered has a finite time horizon τ. The system might have several time scales $\tau_i \le \tau$, but for large times $t \gg \tau$ all correlation functions need to fall off exponentially. This is the case for 'normal' systems, but not for critical dynamical systems characterized by dynamical and statistical correlations that do not decay exponentially, but as power laws.[16]

Measuring Joint Probabilities For times $t_0, t_1, \ldots$, a set of symbols X, and a time series containing n elements,

$$x_n,\ x_{n-1},\ \ldots,\ x_2,\ x_1, \qquad x_i = x(t_i), \qquad x_i \in X,, \tag{5.62}$$

we may define the joint probability distribution

$$p_n\ : \qquad p(x_n, \ldots, x_1)\,. \tag{5.63}$$

The joint probability $p(x_n, \ldots, x_1)$ is not given a priori, it needs to be measured from an ensemble of time series. This is a demanding task as $p(x_n, \ldots, x_1)$ has $(N_s)^n$ components, when N_s is the number of symbols in X.

It makes no sense to evaluate joint probabilities p_n for time differences $t_n \gg \tau$, as all joint distributions factorize when time lags become large. For finite values of n large numbers of subsets of length n can be cut out of a complete time series,

[16] An analogous discussion for the autocorrelation function of critical vs. non-critical system is presented in Sect. 6.2. of Chap. 6.

providing the basis for reliable statistical estimates. This is an admissible procedure for stationary dynamical processes.

Entropy Density We recall the definition of the Shannon entropy

$$H[p_n] = -\sum_{x_n,\ldots,x_1\in X} p(x_n,\ldots,x_1)\,\log(p(x_n,\ldots,x_1))$$
$$= -\langle\,\log(p_n)\,\rangle_{p_n}\,, \tag{5.64}$$

which needs to be measured for an ensemble of time series of length n or greater. Of interest is the entropy density in the limit of large times,

$$h_\infty = \lim_{n\to\infty}\frac{1}{n}\,H[p_n]\,, \tag{5.65}$$

which exists for stationary dynamical processes with finite time horizons. The entropy density is the mean number of bits per time step needed for encoding the time series statistically.

Excess Entropy We define the "excess entropy" E as

$$E = \lim_{n\to\infty}\big(H[p_n]-n\,h_\infty\big)\,\ge\,0\,. \tag{5.66}$$

The excess entropy is equivalent to the non-extensive part of the entropy, being the coefficient of the term $\propto n^0$ when expanding the entropy in powers of $1/n$,

$$H[p_n] = n\,h_\infty + E + O(1/n), \qquad n\,\to\,\infty\,, \tag{5.67}$$

compare Fig. 5.8. The excess entropy E is positive as long as $H[p_n]$ is concave as a function of n, which is the case for stationary dynamical processes.[17] For practical purposes, the excess entropy can be approximated using finite differences,

$$h_\infty = \lim_{n\to\infty} h_n, \qquad h_n = H[p_{n+1}] - H[p_n]\,, \tag{5.68}$$

since h_∞ corresponds to the asymptotic slope of $H[p_n]$, compare Fig. 5.8.

– One may use (5.68) together with (5.54) for expressing entropy density h_n in terms of an appropriately generalized conditional entropy.

[17] To prove that the excess entropy is positive is the task of exercise (5.10).

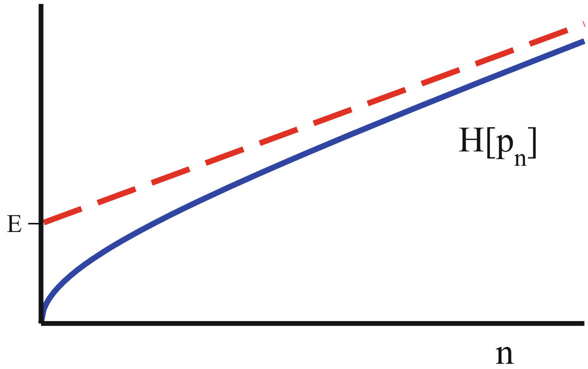

Fig. 5.8 For a time series of length n, the entropy $H[p_n]$ (full line) increases monotonically, with a limiting slope h_∞ (dashed line). For $n \to \infty$ the entropy scales as $H[p_n] \approx E + h_\infty n$, with the excess entropy E given by the intercept of asymptote with the y-axis

– Equation (5.67) allows to rewrite the excess entropy as

$$\sum_n \left[\frac{H[p_n]}{n} - h_\infty \right] .$$

In this form the excess entropy is known as the "effective measure complexity" (EMC) or "Grassberger entropy".

Excess Entropy and Predictability The excess entropy vanishes both for a random and for an ordered system. For a random system

$$H[p_n] = n\, H[p_X] \equiv n\, h_\infty ,$$

where p_X is the marginal probability. The excess entropy (5.66) vanishes consequently. For an example of ordered dynamics we can take a system generating only two types of sequences, say

$$\ldots 000000000000000 \ldots , \qquad\qquad \ldots 111111111111111 \ldots ,$$

respectively with probabilities α and $1 - \alpha$. This kind of dynamics is the natural output of logical AND or OR rules. The joint probability distribution has only two non-zero components,

$$p(0, \ldots, 0) = \alpha, \qquad\qquad p(1, \ldots, 1) = 1 - \alpha, \qquad\qquad \forall n ,$$

all other $p(x_n, \ldots, x_1)$ vanish, which leads to

$$H[p_n] \equiv -\alpha \log(\alpha) - (1 - \alpha) \log(1 - \alpha), \qquad\qquad \forall n .$$

The entropy density h_∞ vanishes for $\alpha \to 0, 1$, viz in the deterministic limit, with the excess entropy E becoming $H[p_n]$.

The excess entropy therefore fulfills the concaveness criteria illustrated in Fig. 5.7, vanishing both in the absence of predictability (random states), and for the case of strong predictability (i.e. for deterministic systems). The excess entropy does however not vanish in above example for $0 < \alpha < 1$, when two predictable states are superimposed statistically in an ensemble of time series. Whether this behavior is compatible with our intuitive notion of complexity is, to a certain extent, a matter of taste.

5.3.2 Algorithmic and Generative Complexity

So far we discussed descriptive approaches using statistical methods for the construction of complexity measures. One may, on the other hand, be interested in modelling the generative process. The question is then, which is the simplest model able to explain the observed data?

Individual Objects For the statistical analysis of a time series we have been concerned with ensembles of time series, as generated by the identical underlying dynamical system, together with the limit of infinitely long times. In this section we will be dealing with individual objects composed of a finite number of n symbols, like

$$0000000000000000000000, \qquad 0010000011101001011001\,.$$

The question is then: which dynamical model can generate the given string of symbols? One is interested, in particular, in strings of bits and in computer codes capable of reproducing them.

Turing Machine In theoretical informatics, the reference computer code is the set of instructions needed for a "Turing machine" to carry out a given computation. The exact definition of a Turing machine is not of relevance here, it is essentially a finite-state machine working on a set of instructions called code. The Turing machine plays a central role in the theory of computability, e.g. when one is interested in examining how hard it is to find the solution to a given set of problems.

Algorithmic Complexity The notion of algorithmic complexity tries to find an answer to the question of how hard it is to reproduce a given time series, in the absence of prior knowledge.

> ALGORITHMIC COMPLEXITY The algorithmic complexity of a string of bits is the length of the shortest program that prints the given string of bits and then halts.

Algorithmic complexity is equivalent to "Kolmogorov complexity". Note, that the involved computer or Turing machine is supposed to start with a blank memory, viz with no prior knowledge.

Algorithmic Complexity and Randomness Algorithmic complexity is a powerful concept for theoretical considerations in the context of optimal computability. It comes however with two drawbacks, being not computable and attributing maximal complexity to random sequences.

Random number generators can only be approximated by finite state machines like the Turing machine, which would need an infinite code length to produce perfectly decorrelated symbols. This is the reason why real-world codes for random number generators generate 'pseudo random numbers', with the degree of randomness to be tested statistically. Algorithmic complexity conflicts therefore with the common postulate for complexity measures to vanish for random states, compare Fig. 5.7.

Deterministic Complexity There is a vast line of research trying to understand the generative mechanism of complex behavior not algorithmically, but from the perspective of dynamical systems theory, in particular for deterministic systems. The question is then: in the absence of noise, which are the features needed to produce intricated trajectories?

Of interest is in this context the sensitivity to initial conditions for systems having a transition between chaotic and regular states in phase space,[18] as well as the effects of bifurcations and non-trivial attractors like strange attractors.[19] Also of relevance are the consequences of feedback and tendencies towards synchronization.[20] This line of research is embedded in the general quest to understand both the properties and the generative causes of complex and adaptive dynamical systems.

Complexity and Emergence Intuitively, we attribute a high degree of complexity to ever changing structure emerging from possibly simple underlying rules, an example being the forest fires burning their way through the forest along self-organized fire fronts. This link between complexity and 'emergence' is, however, not easy to mathematize, as no precise measure for emergence has been proposed to date.

Weak and Strong Emergence On a final note one needs to mention that a vigorous distinction is being made in philosophy between the concept of 'weak emergence', which we treated here, and the scientifically irrelevant notion of 'strong emergence'. Properties of a complex system generated via weak emergence result from the underlying microscopic laws, whereas strong emergence leads to top-level properties which are strictly novel in the sense that they cannot, like magic, linked causally to the underlying microscopic laws of nature.

[18] Transitions between extended chaotic and regular phases occur in boolean networks, see Chap. 7.

[19] For strange attractors and the like consult Chap. 3.

[20] The Kuramoto model is the the standard reference for globally synchronized states, as detailed out in Chap. 9.

Exercises

(5.1) THE LAW OF LARGE NUMBERS
Generalize the derivation for the law of large numbers given in Sect. 5.1.1 for the case of $i = 1, \ldots, N$ independent discrete stochastic processes $p_k^{(i)}$, described by their respective generating functionals $G_i(x) = \sum_k p_k^{(i)} x^k$.

(5.2) CUMMULATIVE DISTRIBUTION FUNCTIONS
Evaluate the distribution function of the sum of $N = 2$ flat distributions and compare the result with the Gaussian (5.11) from the central limit theorem, in analogy to Fig. 5.2.

(5.3) SYMBOLIZATION OF FINANCIAL DATA
Generalize the symbolization procedure defined for the joint probabilities $p_{\pm\pm}$ defined by (5.20) to joint probabilities $p_{\pm\pm\pm}$. For example, p_{+++} would measure the probability of three consecutive increases. Download from the Internet the historical data for your favorite financial asset, like the Dow Jones or the Nasdaq stock indices, and analyze it with this symbolization procedure. Discuss, whether it would be possible, as a matter of principle, to develop in this way a money-making scheme.

(5.4) THE OR TIME SERIES WITH NOISE
Consider the time series generated by a logical OR, akin to (5.21). Evaluate the probability $p(1)$ for finding a 1, with and without averaging over initial conditions, both without noise and in presence of noise. Discuss the result.

(5.5) TRAILING AVERAGES
For a time series $y(t)$, with continuous time $t \in \mathbb{R}$, the trailing average is given by the exponentially discounted mean of its past values,

$$\bar{y}(t) = \frac{1}{T} \int_0^\infty y(t-\tau)\, \mathrm{e}^{-\tau/T} d\tau \,.$$

Show that $y(t)$ can be evaluated by the local updating rule $\frac{d}{dt}\bar{y} = (y - \bar{y})/T$. Is there also a local updating rule for the trailing cos-transform

$$y_c(t,\omega) = \frac{1+(\omega T)^2}{T} \int_0^\infty y(t-\tau)\, \mathrm{e}^{-\tau/T} \cos(\omega\tau)\, d\tau \,,$$

and, analogously, for the trailing sin-transform $y_s(t,\omega)$?

(5.6) MAXIMAL ENTROPY DISTRIBUTION FUNCTION
Determine the probability distribution function $p(x)$, having a given mean μ and a given variance σ^2, compare (5.39), which maximizes the Shannon entropy.

(5.7) <u>TWO-CHANNEL MARKOV PROCESS</u>
Consider, in analogy to (5.43) the two-channel Markov process $\{\sigma_t, \tau_t\}$,

$$\sigma_{t+1} = \mathrm{AND}(\sigma_t, \tau_t), \qquad \tau_{t+1} = \begin{cases} OR(\sigma_t, \tau_t) & \text{probability } 1-\alpha \\ \neg OR(\sigma_t, \tau_t) & \text{probability } \alpha \end{cases} .$$

Evaluate the joint and marginal distribution functions, the respective entropies and the resulting mutual information. Discuss the result as a function of noise strength α.

(5.8) <u>KULLBACK-LEIBLER DIVERGENCE</u>
Try to approximate an exponential distribution function by a scale-invariant density, considering the Kullback-Leibler divergence $K[p;q]$, see (5.2.4), between the two normalized distributions

$$p(x) = \mathrm{e}^{-(x-1)}, \qquad q(x) = \frac{\gamma-1}{x^\gamma}, \qquad x, \gamma > 1 .$$

Which exponent γ minimizes $K[p;q]$? How many times do the graphs for $p(x)$ and $q(x)$ cross?

(5.9) <u>CHI-SQUARED TEST</u>
The quantity

$$\chi^2[p;q] = \sum_{i=1}^{N} \frac{(p_i - q_i)^2}{p_i} \tag{5.69}$$

measures the similarity of two normalized probability distribution functions p_i and q_i. Show, that the Kullback-Leibler divergence $K[p;q]$, Eq. (5.2.4), reduces to $\chi^2[p;q]/2$ if the two distributions are quite similar.

(5.10) <u>EXCESS ENTROPY</u>
Use the representation

$$E = \lim_{n\to\infty} E_n, \qquad E_n \approx H[p_n] - n\big(H[p_{n+1}] - H[p_n]\big) ,$$

to prove that $E \geq 0$, compare (5.66) and (5.68), as long as $H[p_n]$ is concave as a function of n.

(5.11) <u>TSALLIS ENTROPY</u>
The "Tsallis Entropy"

$$H_q[p] = \frac{1}{1-q} \sum_k \left[(p_k)^q - p_k\right], \qquad 0 < q \leq 1 \tag{5.70}$$

of a probability distribution function p is a non-extensive generalization of the Shannon entropy $H[p]$. Prove that

$$\lim_{q\to 1} H_q[p] = H[p], \qquad H_q[p] \ge 0,$$

and the non-extensiveness

$$H_q[p] = H_q[p_X] + H_q[p_Y] + (1-q)\, H_q[p_X]\, H_q[p_Y], \qquad p = p_X p_Y$$

for two statistically independent systems X and Y. For which distribution function p is $H_q[p]$ maximal?

Further Reading

For further readings we recommend introductions to information theory, Pierce (2012), to Bayesian statistics, Bolstad and Curran (2016), and to algorithmic and algebraic complexity, Zenil (2020) and Bürgisser (2013). For the case that the reader may be inclined to delve into the philosophical intricacies of strong and weak emergence, we recommend Wilson (2016).

Of further interest are several review articles, on complexity and predictability, Boffetta et al. (2002), as well as regarding holographic complexity, Alishahiha (2015). A critical assessment of various complexity measures can be found in Olbrich et al. (2008).

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Self-Organized Criticality

6

Classically, a phase transition occurs when the properties of a system change upon tuning an external parameter, like the temperature. Is it possible, that a complex system regulates an internal parameter on its own, self-organized, such that it approaches a critical point all by itself? This is the central question discussed in this chapter.

Starting with an introduction to the Landau theory of phase transitions, particular attention will be devoted to cellular automata, an important and popular class of standardized dynamical systems. Cellular automata allow for an intuitive construction of models, such as the forest fire mode, the game of life, and the sandpile model, which exhibits "self-organized criticality". Mathematically, a further understanding will be attained with the help of random branching theory. The chapter concludes with a discussion of whether self-organized criticality occurs in the most adaptive dynamical system of all, namely in the context of long-term evolution.

6.1 Landau Theory of Phase Transitions

One may describe the physics of thermodynamic phases either microscopically with the tools of statistical physics, or by considering the general properties close to a phase transition. The Landau theory of phase transitions does the latter, providing a general framework valid irrespectively of the microscopic details of the material.

Second-Order Phase Transitions Phase transitions occur regularly in physical systems when the number of components diverges, viz in "macroscopic" systems. Phases have distinct properties, with the key property being the "order parameter", it distinguishes one phase from another. Mathematically one can classify the type of ordering according to the symmetry of the ordering breaks.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_6

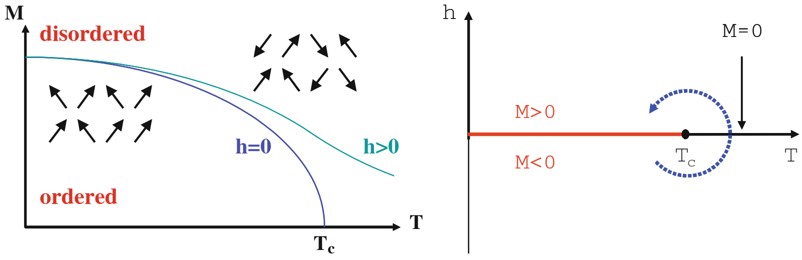

Fig. 6.1 Phase diagram of a magnet in an external magnetic field h. **Left:** The order parameter M, here the magnetization, as a function of temperature across the phase transition. Illustrated are typical arrangements of the local moments (arrows). In the ordered phase there is a net magnetic moment, viz magnetization. For $h = 0/h > 0$ the transition disorder-order is a sharp transition/crossover. **Right:** The $T - h$ phase diagram. A sharp transition occurs only for vanishing external fields h

ORDER PARAMETER In a continuous or "second-order" phase transition, the high-temperature phase has a higher symmetry than the low-temperature phase. The degree of symmetry breaking is characterized by the order parameter ϕ.

Note that all matter is disordered at high enough temperatures, in physical systems ordered phases occur at low to moderate temperatures.

Ferromagnetism in Iron The classical example for a phase transition is that of a magnet like iron. Above the Curie temperature of $T_c = 1,043\,\mathrm{K}$, the elementary magnets are disordered, see Fig. 6.1 for an illustration. They fluctuate strongly and point in random directions. The net magnetic moment vanishes. Below the Curie temperature the moments point on the average to a certain direction creating such a macroscopic magnetic field. Since magnetic fields are generated by circulating currents, with current directions depending on time, magnetic state of ferromagnets like iron break "time-reversal symmetry". Further examples of order parameters characterizing phase transitions in physical systems are listed in Table 6.1.

Table 6.1 Examples of common types of phase transitions in physical systems. When the transition is continuous/discontinuous one speaks of a second-/first-order phase transition. Note, that order parameters can be non-intuitive. The superconducting state, notable for its ability to carry electrical current without dispersion, breaks what one calls the $U(1)$-gauge invariance of the normal (non-superconducting) metallic state

Transition	Type	Order parameter ϕ
Superconductivity	Second-order	U(1)-gauge
Magnetism	Mostly second-order	Magnetization
Ferroelectricum	Mostly second-order	Polarization
Bose–Einstein	Second-order	Amplitude of $k = 0$ state
Liquid-gas	First-order	Density

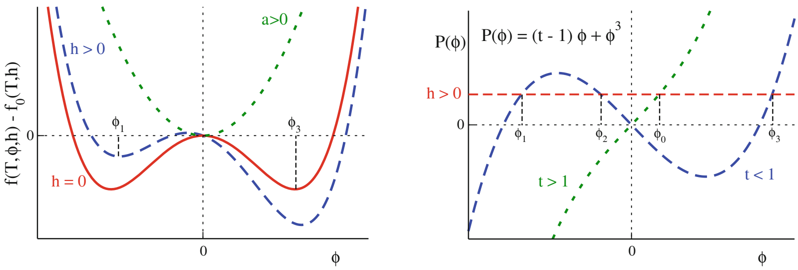

Fig. 6.2 Landau-Ginzburg theory for $t > 1$ (dotted green line) and $t < 1$ (dashed blue line), corresponding respectively to the disordered and ordered phase, where $a = (t-1)/2$. **Left:** The functional dependence of $f(T,\phi,h) - f_0(T,h) = -h\,\phi + a\,\phi^2 + b\,\phi^4$, including the case $h = 0$ (full red line). **Right:** For a non-vanishing field $h \neq 0$, the graphical solution of (6.5), where ϕ_0 is the order parameter in the disordered phase, ϕ_1 and ϕ_3 the stable solutions in the ordered state, and ϕ_2 the unstable solution for $t < 1$

Free Energy A statistical mechanical system takes the configuration with the lowest energy at zero temperature. At finite temperatures $T > 0$, it is however not the energy E which is minimized, but the free energy F, which differs from the energy by a term proportional to the entropy and to the temperature. For the following this difference is however not directly of relevance.

Close to the transition temperature T_c, the order parameter ϕ is small and one assumes within the Landau-Ginzburg model that the free energy density,

$$f = f(T,\phi,h), \qquad f = \frac{F}{V},$$

can be expanded for a small order parameter ϕ and a small external field h,

$$f(T,\phi,h) = f_0(T,h) - h\,\phi + a\,\phi^2 + b\,\phi^4 + \ldots \tag{6.1}$$

where the parameters $a = a(T)$ and $b = b(T)$ are functions of the temperature T. The external field h, e.g. a magnetic field for the case of magnetic systems, is presumed to couple linearly to the order parameter ϕ. For (6.1) to be stable for large ϕ one needs $b > 0$, compare Fig. 6.2.

Spontaneous Symmetry Breaking Odd terms $\sim \phi^{2n+1}$ vanish in the expansion (6.1) for $f(T,\phi,h)$, which is valid for temperatures close to T_c. Odd contributions vanish when the disordered high-temperature state is invariant under inversion symmetry,

$$f(T,\phi,h) = f(T,-\phi,-h), \qquad \phi \leftrightarrow -\phi, \qquad h \leftrightarrow -h\,,$$

which is generically the case. Inversion symmetry is broken when the temperature is lowered below T_c, viz when the order parameter ϕ acquires a finite expectation value. This is the phenomenon of "spontaneous symmetry breaking", which is inherent to second-order phase transitions.[1]

Variational Approach The Landau-Ginzburg functional (6.1) expresses the free energy for all potentially possible values of ϕ. The true physical state, namely "thermodynamical stable state", is obtained by finding the value of ϕ minimizing $f(T, \phi, h)$,

$$0 = \delta f = \left(-h + 2a\phi + 4b\,\phi^3\right)\delta\phi,$$
$$0 = -h + 2a\phi + 4b\phi^3\,, \tag{6.2}$$

where δf and $\delta\phi$ denote small variations respectively of the free energy and of the order parameter. The stationary condition (6.2) corresponds to a minimum of the free energy if

$$\delta^2 f > 0, \qquad \delta^2 f = \left(2a + 12b\phi^2\right)(\delta\phi)^2\,. \tag{6.3}$$

In this case the solution is "locally stable", since any change in ϕ from its optimal value would increase the free energy.

Solutions for $h = 0$ When there is no external field, for $h = 0$, the solution of (6.2) is

$$\phi = \begin{cases} 0 & \text{for } a > 0 \\ \pm\sqrt{-a/(2b)} & \text{for } a < 0 \end{cases}\,. \tag{6.4}$$

The trivial solution $\phi = 0$ is stable,

$$\left(\delta^2 f\right)_{\phi=0} = 2a(\delta\phi)^2\,,$$

if $a > 0$. The nontrivial solutions are given by $\phi^2 = -a/(2\,b)$, they are stable for $a < 0$,

$$\left(\delta^2 f\right)_{\phi\neq 0} = -4a(\delta\phi)^2\,.$$

[1] Spontaneous symmetry breaking is not present for first-order transitions, at which properties change discontinuous, see Table 6.1.

Graphically this is immediately evident,[2] see Fig. 6.2. For $a > 0$ there is a single global minimum at $\phi = 0$, whereas one has two symmetric minima. when $a < 0$.

Continuous Phase Transition Given (6.4), one has that the Ginzburg-Landau functional (6.1) describes a continuous phase transition when $a = a(T)$ changes sign at the critical temperature T_c. We may expand $a(T)$ for small $T - T_c$,

$$a(T) \sim T - T_c, \qquad a = a_0\,(t-1), \qquad t = T/T_c, \qquad a_0 > 0\,,$$

where we used $a(T_c) = 0$. For $T < T_c$, in the ordered phase, the solution takes the form

$$\phi = \pm\sqrt{\frac{a_0}{2b}\,(1-t)}, \qquad t < 1, \qquad T < T_c\,.$$

The sqrt-increase of the order parameter is a telltale characteristics of a second-order phase transition on a mean-field level.

Simplification by Rescaling One can always rescale the order parameter ϕ, the external field h and the free energy density f, such that $a_0 = 1/2$ and $b = 1/4$. This leads to

$$a = \frac{t-1}{2}, \qquad f(T,\phi,h) - f_0(T,h) = -h\,\phi + \frac{t-1}{2}\,\phi^2 + \frac{1}{4}\,\phi^4$$

and

$$\phi = \pm\sqrt{1-t}, \qquad t = T/T_c$$

for the non-trivial solution.

Solutions for h ≠ 0 The solutions of (6.2) are determined in rescaled form by

$$h = (t-1)\,\phi + \phi^3 \equiv P(\phi)\,, \tag{6.5}$$

as illustrated in Fig. 6.2. In general one finds three solutions $\phi_1 < \phi_2 < \phi_3$ and one can show that the intermediate solution is always locally unstable,[3] with ϕ_3 (ϕ_1) being globally stable for $h > 0$ ($h < 0$).

[2] From a dynamical systems point of view, the transition shown in Fig. 6.2 is equivalent to a pitchfork bifurcation, as detailed out Sect. 2.2.2 of Chap. 2.

[3] The stability of the three solutions are treated in exercise (6.1).

First-Order Phase Transition We note, see Fig. 6.2, that the solution ϕ_3 for $h > 0$ remains locally stable when we vary the external field slowly, viz adiabatically,

$$(h > 0) \quad \to \quad (h = 0) \quad \to \quad (h < 0)\,,$$

in the ordered state $T < T_c$. At a certain critical field, see Fig. 6.3, the order parameter changes sign abruptly, jumping from the branch corresponding to $\phi_3 > 0$ to the branch $\phi_1 < 0$. One speaks of hysteresis, a phenomenon typical for first-order phase transitions.

Susceptibility When the system approaches the phase transition from above in the disordered state, it has an increased sensitivity towards ordering under the influence of an external field h.

SUSCEPTIBILITY The susceptibility χ of a system denotes its response to an external field,

$$\chi = \left(\frac{\partial\phi}{\partial h}\right)_T\,,$$

where the subscript T indicates that the temperature is kept constant.

The susceptibility is a prime candidate for experimental investigations, as it measures the increase of ordering, $\phi = \phi(h)$, induced by an external field h.

Diverging Response Taking the derivative with respect to the external field h in (6.5), $h = (t-1)\,\phi + \phi^3$, we find

$$1 = \left[(t-1) + 3\,\phi^2\right]\frac{\partial\phi}{\partial h}, \qquad \chi(T)\Big|_{h\to 0} = \frac{1}{t-1} = \frac{T_c}{T-T_c} \tag{6.6}$$

for the disordered phase, since $\phi(h=0)=0$ for $T > T_c$. The susceptibility diverges at the phase transition for $h = 0$, see Fig. 6.3. This divergence is a typical precursor of ordering for a second-order phase transition. Exactly at T_c, viz at criticality, the response of the system is formally infinite.

A non-vanishing external field $h \neq 0$ induces a finite amount of ordering $\phi \neq 0$ at all temperatures, which masks the phase transition, compare Fig. 6.1. In this case, the susceptibility is a smooth function of temperature, see (6.6) and Fig. 6.3.

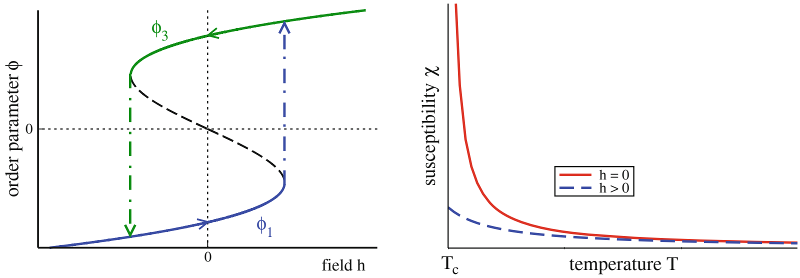

Fig. 6.3 **Left:** Discontinuous phase transition and hysteresis in the Landau model. Plotted is the solution $\phi = \phi(h)$ of $h = (t-1)\phi + \phi^3$ in the ordered phase, viz for $t < 1$, when changing the field h. **Right:** The susceptibility $\chi = \partial\phi/\partial h$ for $h = 0$ (solid line) and $h > 0$ (dotted line). The susceptibility divergence in the absence of an external field, compare (6.6)

6.2 Criticality in Dynamical Systems

Length Scales Pphysical or complex system normally dispose of well defined time and space scales. As an example we take a look at the Schrödinger equation for the hydrogen atom,

$$i\hbar\frac{\partial\Psi(t,\boldsymbol{r})}{\partial t} = H\,\Psi(t,\boldsymbol{r}), \qquad H = -\frac{\hbar^2\Delta}{2m} - \frac{Ze^2}{|\boldsymbol{r}|},$$

where

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

is the Laplace operator. One does not need to know the physical significance of the parameters to realize that one can rewrite the differential operator H, namely the "Hamilton operator", as

$$H = -E_R\left(a_0^2\Delta + \frac{2a_0}{|\boldsymbol{r}|}\right), \qquad E_R = \frac{mZ^2e^4}{2\hbar^2}, \qquad a_0 = \frac{\hbar^2}{mZe^2}.$$

The length scale $a_0 = 0.53\,\mathring{A}/Z$ is called "Bohr radius", with the energy scale $E_R = 13.6\,\text{eV}$ being the "Rydberg energy". The energy scale E_R determines both the ground state energy and excitation. The Bohr radius a_0 sets the scale for the mean radius of the ground state wavefunction and all other radius-dependent properties.

Similar length scales can be defined for essentially all dynamical systems defined by a set of differential equations. The damped harmonic oscillator and the diffusion equations, e.g. are given respectively by

$$\ddot{x}(t) - \gamma\dot{x}(t) + \omega^2 x(t) = 0, \qquad \frac{\partial\rho(t,\boldsymbol{r})}{\partial t} = D\Delta\rho(t,\boldsymbol{r})\,. \tag{6.7}$$

The parameters $1/\gamma$ and $1/\omega$ determine respectively the time scales for relaxation and oscillation, with D being the diffusion constant.

Correlation Function A suitable quantity to measure and discuss the properties of dynamical systems like the ones defined by (6.7), is the equal-time correlation function $S(r)$, which is the expectation value

$$S(r) = \langle\rho(t_0,\boldsymbol{x})\rho(t_0,\boldsymbol{y})\rangle, \qquad r = |\boldsymbol{x}-\boldsymbol{y}|\,. \tag{6.8}$$

Here $\rho(t_0,\boldsymbol{x})$ denotes the particle density, for the case of the diffusion equation or when considering a statistical mechanical system of interacting particles. The exact expression for $\rho(t_0,\boldsymbol{x})$ in general depends on the type of dynamical system considered, for the Schrödinger equation $\rho(t,\boldsymbol{x}) = \Psi^*(t,\boldsymbol{x})\Psi(t,\boldsymbol{x})$, i.e. the probability to find the particle at time t at the point $\boldsymbol{x}$.

The equal-time correlation function measures the probability to find a particle at position $\boldsymbol{x}$ when there is one at $\boldsymbol{y}$. Often one is interested in the deviation of the correlation from the average behaviour. In this case one considers $\langle\rho(\boldsymbol{x})\rho(\boldsymbol{y})\rangle - \langle\rho(\boldsymbol{x})\rangle\langle\rho(\boldsymbol{y})\rangle$ instead of (6.8).

Correlation Length Of interest is the behavior of the equal-time correlation function $S(r)$ for large distances $r \to \infty$. In general we have two possibilities,

$$S(r)\Big|_{r\to\infty} \quad \sim \quad \begin{cases} \mathrm{e}^{-r/\xi} & \text{non-critical} \\ 1/r^{d-2+\eta} & \text{critical} \end{cases}\,. \tag{6.9}$$

In normal, viz non-critical systems, correlations over arbitrary large distances cannot be built up, which implies that the correlation function decays exponentially, over a scale determined by the correlation length ξ. The notation $d-2+\eta > 0$ for the decay exponent of the critical system is a convention from statistical physics, where $d = 1, 2, 3, \ldots$ is the dimensionality of the embedding system.

Scale-Invariance and Self-Similarity If a control parameter of a physical system, often the temperature, is tuned such that it sits exactly at the point of a phase transition, the system is said to be critical. At this point there are no characteristic length scales.

SCALE INVARIANCE If a measurable quantity, like the correlation function, decays like a power of the distance $\sim (1/r)^\delta$, with a critical exponent δ, the system is said to be critical or scale-invariant.

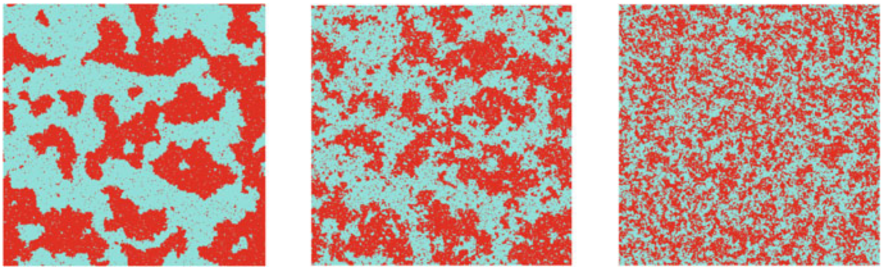

Fig. 6.4 Simulation of the 2D-Ising model on a square lattice, $H = \sum_{\langle i,j\rangle} \sigma_i \sigma_j$, with nearest-neighbor interactions $\langle i, j\rangle$. There are two magnetization orientations $\sigma_i = \pm 1$ (dark/light dots). Shown is the ordered state $T < T_c$ (left), the critical state $T \approx T_c$ (middle), and the disordered state $T > T_c$ (right). Note the occurrence of fluctuations at all length scales at criticality, reflfecting self similarity

Power laws have no scale, they are self-similar,[4] because

$$S(r) = c_0 \left(\frac{r_0}{r}\right)^{\delta} \equiv c_1 \left(\frac{r_1}{r}\right)^{\delta}, \qquad c_0\, r_0^{\delta} = c_1\, r_1^{\delta},$$

holds for arbitrary distance scales r_0 and r_1.

Universality at the Critical Point The equal-time correlation function $S(r)$ is scale invariant at criticality, compare (6.9). This is a surprising statement, since we have seen before that the differential equations determining dynamical systems have well defined time and length scales. How then does the solution of a dynamical system become effectively independent of the parameters entering its governing equations?

Scale invariance implies that fluctuations occur over all length scales, albeit with varying probabilities. This can be seen by observing snapshots of statistical mechanical simulations of simple models. The case of the two-dimensional Ising model is given in Fig. 6.4.

The scale invariance of the correlation function at criticality is a central result of the theory of phase transitions within statistical physics. Properties of systems close to a phase transition are not determined by the exact values of their parameters, but by the structure of the governing equations and their symmetries. This circumstance is denoted “universality”, it constitutes one of the reasons for classifying phase transitions according to the symmetry of their order parameters, see Table 6.1.

[4] Power laws in terms of scale-free degree distributions are a cornerstone of network theory, please consult Sect. 1.6 of Chap. 1.

Autocorrelation Function The equal-time correlation function $S(r)$ measures real-space correlations. The corresponding quantity in time domain is the autocorrelation function

$$\Gamma(t) = \frac{\langle A(t+t_0)A(t_0)\rangle - \langle A\rangle^2}{\langle A^2\rangle - \langle A\rangle^2}\,, \tag{6.10}$$

which can be defined for any time-dependent measurable quantity A, such as $A(t) = \rho(t,\vec{r})$. Note that observations are defined relative to $\langle A\rangle^2$, viz relative to the mean fluctuations. The denominator in (6.10) is a normalization convention, it leads to $\Gamma(0) \equiv 1$.

In the non-critical state, that is in the diffusive regime, long-term memory is absent. As a consequence, information about the initial state is lost exponentially,

$$\Gamma(t) \sim \mathrm{e}^{-t/\tau}, \qquad t \to \infty\,, \tag{6.11}$$

where τ is the relaxation or autocorrelation time. It corresponds to the time scale of the diffusion process.

Dynamical Critical Exponent The relaxation time entering (6.11) diverges at criticality, as does the real-space correlation length ξ entering (6.9). One can then define an appropriate exponent z, dubbed "dynamical critical exponent" z, in order to relate the two power laws for τ and ξ, via

$$\tau \sim \xi^z, \qquad \xi = \frac{1}{|T-T_c|^\nu}, \qquad \xi\big|_{T\to T_c} \to \infty\,.$$

The autocorrelation time is divergent in the critical state $T \to T_c$.

Self-Organized Criticality We have seen that phase transitions can be characterized by a set of exponents describing the respective power laws of various quantities, like the correlation function or the autocorrelation function. The phase transition occurs generally at a single point, viz $T = T_c$ for a thermodynamical system. At the phase transition the system becomes effectively independent of the details of its governing equations, being determined by symmetries.

It may then come as a surprise that there should exist complex dynamical systems that attain a critical state for a finite range of parameters. This possibility, denoted "self-organized criticality", is somewhat counter-intuitive. We can regard the parameters entering the evolution equation as given externally. Self-organized criticality then implies that the system adapts to changes in the external parameters, e.g. to changes in the given time and length scales, in such a way that the stationary state becomes effectively independent of those changes.

6.2.1 1/f Noise

So far we have discussed the occurrence of critical states in classical thermodynamics and statistical physics. We now ask ourselves for experimental evidence that criticality might play a central role in certain time-dependent phenomena.

$1/f$ Noise The power spectrum of the noise generated by many real-world dynamical processes falls off inversely with frequency f. This $1/f$ noise has been observed for various biological activities, like the heart beat rhythms, for functioning electrical devices or for meteorological data series. The origin of "pink noise", another word for a $1/f$ power spectrum, could be self-organized phenomena. Within this view, one describes pink noise as being generated by a continuum of weakly coupled damped oscillators representing the environment.

Power Spectrum of a Single Damped Oscillator A system with a single relaxation time τ, see (6.7), and eigenfrequency ω_0, has a Lorentzian power spectrum

$$\begin{aligned} S(\omega,\tau) &= \mathrm{Re}\int_0^\infty dt\, \mathrm{e}^{i\omega t}\mathrm{e}^{-i\omega_0 t - t/\tau} = \mathrm{Re}\frac{-1}{i(\omega-\omega_0)-1/\tau} \\ &= \frac{\tau}{1+\tau^2(\omega-\omega_0)^2}\,. \end{aligned}$$

For large frequencies $\omega \gg 1/\tau$, the power spectrum $S(\omega,\tau)$ falls off like $1/\omega^2$. Being interested in the large-f behavior, we neglect ω_0 in the following.

Distribution of Oscillators The combined power or frequency spectrum of a continuum of oscillators is determined by the distribution $D(\tau)$ of relaxation times τ. For a critical system relaxation occurs over all time scales, as discussed in Sect. 6.2. We may hence assume a scale-invariant distribution

$$D(\tau) \approx \frac{1}{\tau^\alpha} \tag{6.12}$$

for the relaxation times τ. This distribution of relaxation times leads to a frequency spectrum

$$\begin{aligned} S(\omega) &= \int d\tau\, D(\tau)\frac{\tau}{1+(\tau\omega)^2} \sim \int d\tau \frac{\tau^{1-\alpha}}{1+(\tau\omega)^2} \\ &= \frac{1}{\omega\,\omega^{1-\alpha}}\int d(\omega\tau)\frac{(\omega\tau)^{1-\alpha}}{1+(\tau\omega)^2} \sim \frac{1}{\omega^{2-\alpha}}\,. \end{aligned} \tag{6.13}$$

For $\alpha = 1$ we obtain $1/\omega$, the typical behavior of $1/f$ noise.

The question is then how assumption (6.12) can be justified. The wide-spread appearance of $1/f$ noise can only happen when scale-invariant distribution of relaxation times are ubiquitous, viz if they were self-organized. The $1/f$ noise therefore constitutes an interesting motivation for the search of possible mechanisms leading to self-organized criticality.

6.3 Cellular Automata

Cellular automata are finite state lattice systems with discrete local update rules.

$$z_i \to f_i(z_i, z_{i+\delta}, \ldots), \qquad z_i \in [0, 1, \ldots, n], \tag{6.14}$$

where $i + \delta$ denote neighboring sites of site i. Each site or "cell" of the lattice follows a prescribed rule evolving in discrete time steps. At each step the new value for a cell depends only on the current state of itself and on the state of its neighbors. Cellular automata differ from general dynamical networks in two aspects.[5]

- The update functions are all identical: $f_i() \equiv f()$, viz they are translational invariant.
- The number n of states per cell is usually larger than two, the boolean case.

Cellular automata can give rise to exceedingly complex behavior despite their deceptively simple dynamical structure. We note that cellular automata are always updated synchronously and never sequentially or randomly. The state of all cells is renewed simultaneously.

Number of Update Rules The number of possible update rules is exponentially large. For a two-dimensional model, the square lattice, we consider cells taking one of two possible states,

$$z_i = 0 \quad \text{(dead)}, \qquad z_i = 1 \quad \text{(alive)}\,.$$

For a basic type of rules the evolution of a given cell to the next time step depends only on the current state of the cell and on the values of each of its eight nearest neighbors. In this case there are

$$2^9 = 512 \text{ configurations}, \qquad 2^{512} = 1.3 \times 10^{154} \text{ possible rules}\,,$$

since any one of the 512 configurations can be mapped independently to 'live' or 'dead'. For comparison note that the age universe is of the order of 3×10^{17} seconds.

[5] A particularly important class of dynamical networks are studied in Chap. 7.

Totalistic Update Rules It clearly does not make sense to explore systematically the consequences of arbitrary updating rules. One simplification is to consider a mean field approximation that results in a subset of rules termed "totalistic". For mean field rules, the new state of a cell depends only on the total number of living neighbors, besides its own state. An eight-cell neighborhood has

$$\text{9 possible total occupancy states of neighboring sites,}$$

$$2 \cdot 9 = 18 \text{ configurations,} \qquad 2^{18} = 262{,}144 \text{ totalistic rules}\,.$$

This is a large number, but it is exponentially smaller than the number of all possible update rules for the same neighborhood.

6.3.1 Conway's Game of Life

The "game of life" takes its name because it attempts to simulate the reproductive cycle of a species. It is formulated on a square lattice with an update rule involving the eight-cell neighborhood. A new offspring needs exactly three parents in its neighborhood. A living cell dies of loneliness if it has less than two live neighbors, and of overcrowding if it has more than three live neighbors. A living cell feels comfortable with two or three live neighbors; in this case it survives. The complete set of updating rules is listed in Table 6.2.

Living Isolated Sets The time evolution of an initial set of a cluster of living cells can show varied types of behavior. Fixpoints of the updating rules, such as a square

$$\Big\{(0,0),\,(1,0),\,(0,1),\,(1,1)\Big\}$$

of four neighboring live cells, survive unaltered. There are many configurations of living cells which oscillate, such as three live cells in a row or column,

$$\Big\{(-1,0),\,(0,0),\,(1,0)\Big\}, \qquad \Big\{(0,-1),\,(0,0),\,(0,1)\Big\}\,.$$

Table 6.2 Updating rules for the game of life, with $z_i = 0, 1$ corresponding to empty and living cells. An "x" as an entry denotes what is going to happen for the respective number of living neighbors

		Number of living neighbors				
$z_i(t)$	$z_i(t+1)$	0	1	2	3	4…8
0	1				x	
	0	x	x	x		x
1	1			x	x	
	0	x	x			x

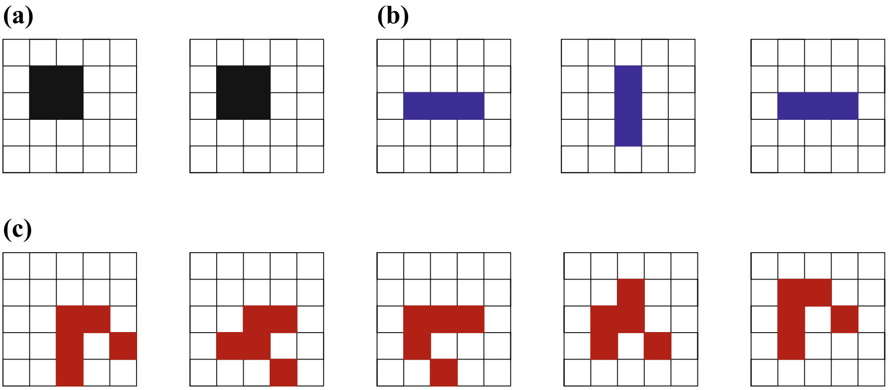

Fig. 6.5 Time evolution of some living configurations for the game of life, see Table 6.2. **(a)** The "block"; it quietly survives. **(b)** The "blinker"; it oscillates with period 2. **(c)** The "glider"; it shifts by $(-1, 1)$ after four time steps

It constitutes a fixpoint of $f(f(.))$, alternating between a vertical and a horizontal bar. The configuration

$$\Big\{(0,0), (0,1), (0,2), (1,2), (2,1)\Big\}$$

is dubbed "glider", since it returns to its initial shape after four time steps but is displaced by $(-1, 1)$, see Fig. 6.5. It constitutes a fixpoint of $f(f(f(f(.))))$ times the translation by $(-1, 1)$. The glider continues to propagate until it encounters a cluster of other living cells.

Game of Life as a Universal Computer It is interesting to investigate, from an engineering point of view, all possible interactions between initially distinct sets of living cells in the game of life. In this context one finds that it is possible to employ gliders for the propagation of information over arbitrary distances. One can prove that arbitrary calculations can be performed by the game of life, when identifying the gliders with bits. Suitable and complicated initial configurations are necessary for this purpose, in addition to dedicated living subconfigurations performing logical computations, in analogy to electronic gates, when hit by one or more gliders.

6.3.2 Forest Fire Model

The forest fires automaton is a simplified model of real-world forest fires. It is formulated on a square lattice with three possible states per cell,

$$z_i = 0 \quad \text{(empty)}, \qquad z_i = 1 \quad \text{(tree)}, \qquad z_i = 2 \quad \text{(fire)}\,.$$

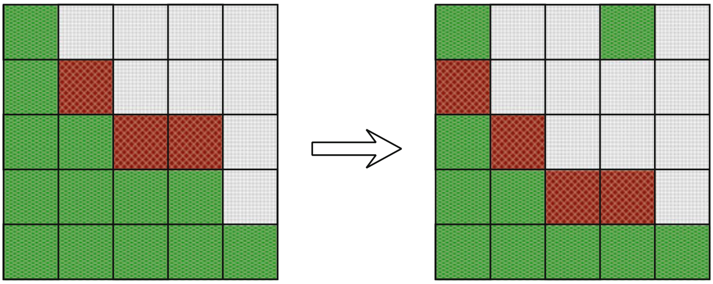

Fig. 6.6 For the forest fire model, the time evolution (from left to right) of a configuration of living trees (green), burning trees (red) and of places burnt down (grey). Places burnt down can regrow spontaneous with a small rate, the fire always spreads to nearest neighboring trees

A tree sapling can grow on every empty cell, with probability $p < 1$. There is no need for nearby parent trees, as sperms are carried by wind over wide distances. Trees do not die in this model, but they catch fire from any burning nearest neighbor tree. The rules are illustrated in Fig. 6.6.

The forest fire automaton differs from typical rules, such as Conway's game of life, because it has a stochastic component. In order to have an interesting dynamics one needs to adjust the growth rate p as a function of system size, so as to keep the fire burning continuously. The fires burn down the whole forest when trees grow too fast. When the growth rate is too low, on the other hand, the fires, being surrounded by ashes, may die out completely.

$z_i(t)$	$z_i(t+1)$	Condition
Empty	Tree	With probability $p < 1$
Tree	Tree	No fire close by
Tree	Fire	At least one fire close by
Fire	Empty	Always

When adjusting the growth rate properly, one reaches a steady state, the system having fire fronts continually sweeping through the forest, as it is observed for real-world forest fires, see Fig. 6.7 for an illustration. In large systems stable spiral structures form and set up a steady rotation.

Criticality and Lightning As defined above, the forest fire model is not critical, since the characteristic time scale $1/p$ for the regrowth of trees governs the dynamics. This time scale translates into a characteristic length scale $1/p$, which can be observed in Fig. 6.7, via the propagation rule for the fire.

Self-organized criticality can, however, be induced in the forest fire model when introducing an additional rule, namely that a tree might ignite spontaneously with a small probability f, when struck by lightning, causing also small patches of forest

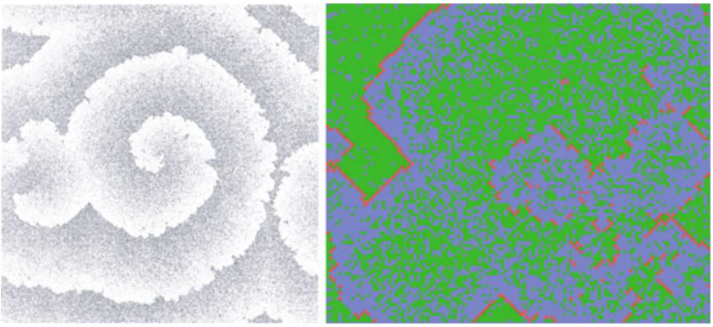

Fig. 6.7 Simulations of the forest fire model. **Left:** Fires burn in characteristic spirals for a growth probability $p = 0.005$ and no lightning, $f = 0$. Reprinted from Clar et al. (1996) with permissions, © 1996 IOP Publishing Ltd. **Right:** A snapshot of the forest fire model with a growth probability $p = 0.06$ and a lightning probability $f = 0.0001$. Note the characteristic fire fronts with trees in front and ashes behind

to burn. We will not discuss this mechanism in detail here, treating instead in the next section the occurrence of self-organized criticality in the sandpile model on a firm mathematical basis.

6.4 Sandpile Model and Self-Organized Criticality

Self-Organized Criticality The notion of 'life at the edge of chaos' states,[6] that certain dynamical and organizational aspects of living organisms may be critical. Normal physical and dynamical systems show criticality however only for selected parameters, e.g. $T = T_c$, see Sect. 6.1. For criticality to be biologically relevant, the system must evolve into a critical state starting from a wide range of initial states, the defining trait of "self-organized criticality".

Sandpile Model Per Bak and coworkers introduced a simple cellular automaton, the BTW model, that mimics the properties of sandpiles. Every cell is characterized by a force

$$z_i = z(i, j) = 0, 1, 2, \ldots, \qquad i, j = 1, \ldots, L$$

on a finite $L \times L$ lattice. There is no one-to-one correspondence of the sandpile model to real-world sandpiles. Loosely speaking one may identify the force z_i with the slope of real-world sandpiles. But this analogy is not rigorous, as the slope

[6] The concept of life at the edge of chaos was first develop in the context of information-processing networks, as discussed Chap. 7.

of a real-world sandpile is a continuous variable. The slopes belonging to two neighboring cells should therefore be similar, whereas the values of z_i and z_j on two neighboring cells can differ by an arbitrary amount within the sandpile model.

The sand begins to topple when slopes gets too big,

$$z_j \to z_j - \Delta_{ij}, \qquad \text{if} \qquad z_i > K\,,$$

where K is the threshold slope and Δ_{ij} The toppling matrix,

$$\Delta_{i,j} = \begin{cases} \;\;4 \; i = j & \\ -1 \; i,j & \text{nearest neighbors} \\ \;\;0 & \text{otherwise} \end{cases} . \tag{6.15}$$

This update rule is valid for the four-cell neighborhood $\{(0, \pm 1), (\pm 1, 0)\}$. The threshold K is arbitrary, a shift in K simply shifts z_i. It is customary to consider $K = 3$. Any initial random configuration will eventually relax into a steady-state with

$$z_i = 0, 1, 2, 3, \qquad\qquad \text{(stable state)}\,.$$

Phases with $z_i \le K$ are absorbing, the subject of Sect. 6.4.1.

Open Boundary Conditions Without boundaries, the update rule (6.15) is conserving, with the number of grains remaining constant.

> CONSERVING QUANTITY If there is a quantity that is not changed by the update rule, it is said to be conserving.

The sandpile model is locally conserving and hence said to be “abelian”. The total height $\sum_j z_j$ is constant due to $\sum_j \Delta_{i,j} = 0$. Globally, however, it is not conserving, as one uses open boundaries, at which excess sand is lost, toppling down from the table. Sand is lost when a site at the boundary topples.

However, there is only a tenuous relation of automatata models to real-world sandpiles. The conserving nature of the sandpile model mimics the fact that sand grains cannot be lost in real-world sandpiles, but this interpretation contrasts with the previously assumed correspondence of z_i with the slope of real-world sandpiles.

Avalanches When starting from a random initial state with $z_i \le K$, the system settles in a steady-state dynamic state when adding “grains of sand” for a while. When a grain is dropped onto a site with $z_i = K$

$$z_i \to z_i + 1, \qquad\qquad z_i = K\,,$$

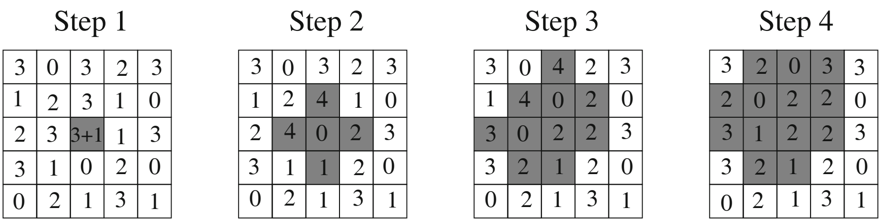

Fig. 6.8 The progress of an avalanche, with duration $t = 3$ and size $s = 13$, for a sandpile configuration on a 5×5 lattice with $K = 3$. The height of the sand in each cell is indicated by the numbers. Also shown is avalanche progression (shaded region). The avalanche stops after step 3

a toppling event is induced, which may in turn lead to a whole series of topplings. The resulting avalanche is characterized by its duration t and the size s of affected sites. It continues until a new stable configuration is reached. In Fig. 6.8 a small avalanche is shown.

Distribution of Avalanches We define with $D(s)$ and $D(t)$ the distributions of the size and of the duration of avalanches. One finds that they are scale free,

$$D(s) \sim s^{-\alpha_s}, \qquad D(t) \sim t^{-\alpha_t}, \tag{6.16}$$

as we will discuss in the next section. The scaling relations (6.16) express the essence of self-organized criticality, which we expect to be valid for a wide range of cellular automata with conserving dynamics, independent of the special values of the parameters entering the respective update functions. Numerical simulations and analytic approximations for $d = 2$ dimensions yield

$$\alpha_s \approx \frac{5}{4}, \qquad \alpha_t \approx \frac{3}{4}.$$

Conserving Dynamics and Self-Organized Criticality Given that toppling events are locally conserving, avalanches of arbitrary large sizes must occur, as sand can be lost only at the boundary of the system. Vice versa, only avalanches reaching the boundary contribute to the powerlaw scaling expressed by (6.16). Self-organized criticality breaks down as soon as there is a small but non-vanishing probability to lose sand somewhere inside the system.

Features of the Critical State The empty board, when all cells are initially empty, $z_i \equiv 0$, is not critical. The system remains in the frozen phase when adding sand, as long as most $z_i < K$. Adding one sand corn after the other, the critical state is slowly approached. There is no way to avoid the critical state.

Once the critical state is achieved the system remains critical. The critical state is paradoxically also the point at which the system is dynamically most unstable. It has an unlimited susceptibility to the external driving, viz to the addition of grains, using the terminology of Sect. 6.1, as a single added grain of sand can trip avalanches of arbitrary size.

6.4.1 Absorbing Phase Transitions

For a variation of the original sandpile model, we disregard both dissipation and external driving. Instead of losing sand at the boundaries, periodic boundary conditions are implemented, with the consequence that the number of grains is conserved both locally as well as globally. Starting with a random configuration $\{z_i\}$, there will be a rush of initial avalanches, following the toppling rules (6.15), until the system settles into either an active or an absorbing state.[7]

- All topplings will stop eventually whenever the average number of grains is too small. The resulting inactive configuration is called "absorbing state".
- For a large average number of grains, the redistribution of grains will never terminate, resulting in a continously active state.

Adding externally a single grain to an absorbing state will lead at most to a single avalanche with the transient acitivity terminating in another absorbing state. In this picture, the grain of sand added has been absorbed.

Transition from Absorbing to Active State The average number of particles $\rho = \langle z_i \rangle$ controls the transition from absorbing to active phase. The active state is characterized by the mean number of sites with heights z_i greater then the threshold K, the active sites. The avalanche shown in Fig. 6.8, has 1/2/2/0 active sites respectively, to give an example, at time steps 1/2/3/4. The transition from the absorbing to the active state is of second order, as illustrated in Fig. 6.9, with the number of active sites acting as an order parameter.[8]

Self-Organization Towards the Critical Density Based on a separation of time scales, there is a deep relation between absorbing state transitions and the concept of self-organized criticality.

The external drive, the addition of grains of sand, one by one, is infinitesimal slow in the sandpile model. The reason is that the external drive is stopped once an avalanche starts, and resumed only once the avalanche has terminated. Relative to the time scale of the external drive, the avalanche is hence instantaneous. Slowly adding one particle after another continuously increases the mean particle

[7] Generically, absorbing states are final configurations of Markov chains, as discussed in Sect. 3.3.2 of Chap. 3.

[8] A fully self consistent mean-field theory of absorbing phase transitions in the presence of uncorrelated external driving is presented in Sect. 10.2.2 of Chap. 10.

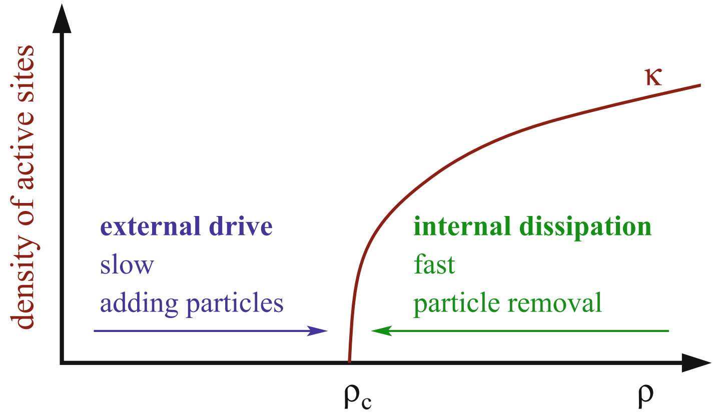

Fig. 6.9 Changing the mean particle density ρ, an absorbing phase transition may occur, whith the density of active sites acting as an order parameter. The system self-organizes towards the critical particle density ρ_c through the balancing of a slow external drive, realized by adding grains in the sandpile model, and a fast internal dissipative process, when loosing grains of sand at the boundaries

number, driving the system towards criticality from below, as shown in Fig. 6.9. Particles surpassing the critical density are instantaneously dissipated through large avalanches reaching the boundaries of the systems, the mean particle density is hence pinned at criticality.

6.5 Random Branching Theory

Branching theory deals with the growth of networks via branching. Networks generated by branching processes are loopless; they typically arise in theories of evolutionary processes.

6.5.1 Branching Theory of Self-Organized Criticality

Avalanches have an intrinsic relation to branching processes; at every time step the avalanche can either continue or stop. Random branching theory is hence a suitable method for studying self-organized criticality.

Branching in Sandpiles A typical update during an avalanche is of the form

$$\begin{array}{lll} \text{time 0:} & z_i \to z_i - 4 & z_j \to z_j + 1 \\ \text{time 1:} & z_i \to z_i + 1 & z_j \to z_j - 4 \end{array}$$

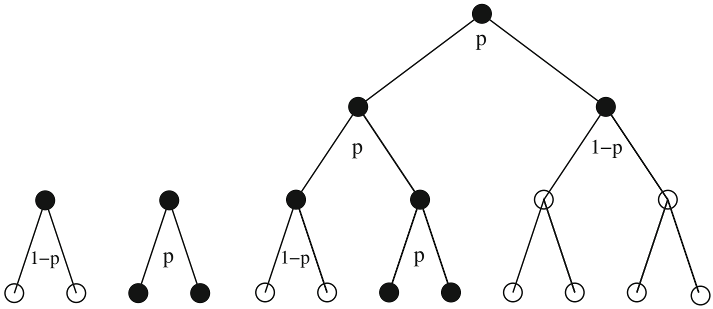

Fig. 6.10 Branching processes. **Left:** The two possible processes of order $n = 1$. **Right:** A generic process of order $n = 3$ with an avalanche of size $s = 7$

when two neighboring cells i and j initially have $z_i = K + 1$ and $z_j = K$. This implies that an avalanche typically intersects with itself. However, for a d-dimensional hypercubic lattice with $K = 2d - 1$, self-interaction of avalanches becomes unimportant when $1/d \to 0$. In large dimensions, avalanches can be mapped rigorously to a random branching process.[9]

Binary Random Branching The notion of neighbors loses meaning in large dimensions. All that is important when $d \to \infty$, is the question whether an avalanche continues, increasing its size continuously, or whether it stops. This can be expressed by binary branching, the case that toppling events create two new active sites.

> BINARY BRANCHING An active site of an avalanche topples with the probability p creating two new active sites.

For $p < 1/2$ the number of new active sites decreases on the average, with avalanches dying out; $p_c = 1/2$ is the critical state with on the average conserving dynamics. See Fig. 6.10 for some examples of branching processes.

Distribution of Avalanche Sizes The properties of avalanches are determined by the probability distribution,

$$P_n(s, p), \qquad \sum_{s=1}^{\infty} P_n(s, p) = 1 \, ,$$

[9] An analogous situation occurs in the context of high-dimensional and/or random graphs, as discussed in Sect. 1.3.3 of Chap. 1, which are also loopless in the thermodynamic limit.

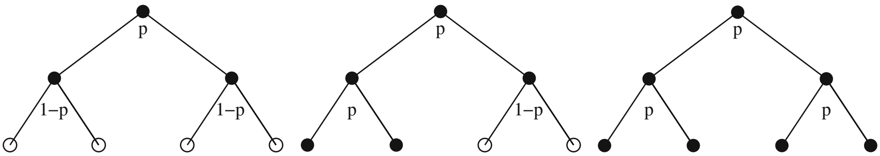

Fig. 6.11 Branching processes of order $n = 2$ with avalanches of sizes $s = 3, 5, 7$ (left/middle/right) and boundaries $\sigma = 0, 2, 4$

describing the probability to find an avalanche of size s for a branching process of order n. Here s is the (odd) number of sites inside the avalanche, see Figs. 6.10 and 6.11 for some examples.

Small Avalanches For small s and large n one can evaluate the probability for small avalanches to occur by hand,

$$P_n(1,p) = 1-p, \qquad P_n(3,p) = p(1-p)^2, \qquad P_n(5,p) = 2p^2(1-p)^3\,,$$

compare Figs. 6.10 and 6.11. Note that $P_n(1,p)$ is the probability to find an avalanche of just one site.

Generating Function Formalism The generating function formalism for probability distributions is very useful when one has to deal with independent stochastic processes, as the joint probability of two independent stochastic processes is equivalent to the simple multiplication of the corresponding generating functions.[10] We define with

$$f_n(x,p) = \sum_s P_n(s,p)\, x^s, \qquad f_n(1,p) = \sum_s P_n(s,p) = 1 \tag{6.17}$$

the generating functional $f_n(x,p)$ for the probability distribution $P_n(s,p)$. We note that

$$P_n(s,p) = \frac{1}{s!}\,\frac{\partial^s f_n(x,p)}{\partial x^s}\bigg|_{x=0}, \qquad n,\ p \text{ fixed}\,. \tag{6.18}$$

Recursion Relation For generic n the recursion relation

$$f_{n+1}(x,p) = x\,(1-p) \;+\; x\,p\,f_n^2(x,p) \tag{6.19}$$

[10] For an introduction to generating functions for probability distributions we refer to Sect. 1.3.2 of Chap. 1.

is valid. To see why, one builds the branching network backwards, adding a site at the top.

– With the probability $(1-p)$
  one adds a single-site avalanche described by the generating functional x.
– With the probability p
  one adds a site, described by the generating functional x, which generated two active sites, described each by the generating functional $f_n(x, p)$.

In the terminology of branching theory, one speaks of the decomposition of the branching process after its first generation, a standard procedure.

Self-Consistency Condition For large n and finite x the generating functionals $f_n(x, p)$ and $f_{n+1}(x, p)$ become identical, leading to the self-consistency condition

$$f_n(x,p) = f_{n+1}(x,p) = x\,(1-p) + x\,p\,f_n^2(x,p)\,,$$

with the solution

$$f(x,p) \equiv f_n(x,p) = \frac{1-\sqrt{1-4x^2p(1-p)}}{2xp} \tag{6.20}$$

for the steady-state generating functional $f(x, p)$. The normalization condition

$$f(1,p) = \frac{1-\sqrt{1-4^2p(1-p)}}{2p} = \frac{1-\sqrt{(1-2p)^2}}{2p} = 1$$

is fulfilled for $p \in [0, 1/2]$. For $p > 1/2$ the last step in above equation would not be correct.

Subcritical Solution Expanding (6.20) in powers of x^2 we find terms like

$$\frac{1}{p}\left[4p(1-p)\right]^k \frac{\left(x^2\right)^k}{x} = \frac{1}{p}\left[4p(1-p)\right]^k x^{2k-1}\,.$$

Comparing this with the definition of the generating functional (6.17) we note that $s = 2k-1$, which is equivalent to $k = (s+1)/2$. This implies

$$P(s,p) \sim \frac{1}{p}\sqrt{4p(1-p)}\left[4p(1-p)\right]^{s/2} \sim \mathrm{e}^{-s/s_c(p)} \tag{6.21}$$

for the probability to find an avalance of size s, where we have used the relation

$$a^{s/2} = \mathrm{e}^{\ln(a^{s/2})} = \mathrm{e}^{-s(\ln a)/(-2)}, \qquad a = 4p(1-p)$$

together with the definition

$$s_c(p) = \frac{-2}{\ln[4p(1-p)]}, \qquad \lim_{p\to 1/2} s_c(p) \to \infty$$

of the avalanche correlation length $s_c(p)$. For $p < 1/2$, the correlation length $s_c(p)$ is finite and the avalanche is consequently not scale-free, compare Sect. 6.2. The characteristic size of an avalanche $s_c(p)$ diverges for $p \to p_c = 1/2$. Note that $s_c(p) > 0$ for $p \in]0,1[$.

Critical Solution We now consider the critical case with

$$p = 1/2, \qquad 4p(1-p) = 1, \qquad f(x,p) = \frac{1-\sqrt{1-x^2}}{x}\,.$$

The expansion of $\sqrt{1-x^2}$ with respect to x is

$$\sqrt{1-x^2} = \sum_{k=0}^{\infty} \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\cdots\left(\frac{1}{2}-k+1\right)}{k!}\left(-x^2\right)^k$$

in (6.20) and therefore

$$P_c(k) \;\equiv\; P(s{=}2k{-}1,\, p{=}1/2) = \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\cdots\left(\frac{1}{2}-k+1\right)}{-k!}\,(-1)^k\,.$$

This expression is still unhandy. However we are only interested in the asymptotic behavior for large avalanche sizes s. For this purpose we consider the recursive relation

$$P_c(k+1) = \frac{1/2-k}{k+1}(-1)P_c(k) = \frac{1-1/(2k)}{1+1/k}\,P_c(k)$$

in the limit of large $k = (s+1)/2$, where $1/(1+1/k) \approx 1-1/k$,

$$P_c(k+1) \approx \Big[1-1/(2k)\Big]\Big[1-1/k\Big]\,P_c(k) \approx \Big[1-3/(2k)\Big]\,P_c(k)\,.$$

This asymptotic relation leads to

$$\frac{P_c(k+1)-P_c(k)}{1} = \frac{-3}{2k}\,P_c(k), \qquad \frac{\partial P_c(k)}{\partial k} = \frac{-3}{2k}\,P_c(k)\,,$$

with the solution

$$P_c(k) \sim k^{-3/2}, \qquad D(s) = P_c(s) \sim s^{-3/2}, \qquad \alpha_s = \frac{3}{2}, \tag{6.22}$$

for large k, s, since $s = 2k - 1$.

Distribution of Avalanche Duration The distribution of the duration n of avalanches can be evaluated in a similar fashion. For this purpose one defines the probability distribution function

$$Q_n(\sigma, p)$$

for an avalanche of duration n to have σ cells at the boundary, see Fig. 6.11. One can derive a recursion relation for the corresponding generating functional, in analogy to (6.19), and solve it self-consistently.[11]

The distribution of avalanche durations is determined from the generating function via $Q_n = Q_n(\sigma = 0, p = 1/2)$, i.e. by the probability that the avalanche stops after n steps. One finds

$$Q_n \sim n^{-2}, \qquad D(t) \sim t^{-2}, \qquad \alpha_t = 2\,. \tag{6.23}$$

Tuned or Self-Organized Criticality? The random branching model discussed in this section had only one free parameter, the probability p. This model is critical only for $p \to p_c = 1/2$, giving rise to the impression that one has to fine tune the parameters in order to obtain criticality, just like for ordinary phase transitions.

This is actually not the case. As an example we could generalize the sandpile model to continuous forces $z_i \in [0, \infty]$, together with the update rule

$$z_j \to z_j - \Delta_{ij}, \qquad \text{if} \qquad z_i > K\,,$$

and

$$\Delta_{i,j} = \begin{cases} K \; i = j & \\ -c\,K/4 \; i, j & \text{nearest neighbors} \\ -(1-c)\,K/8 \; i, j & \text{next-nearest neighbors} \\ 0 & \text{otherwise} \end{cases} \tag{6.24}$$

[11] The evaluation of avalanche durations is the topic of exercise (6.8).

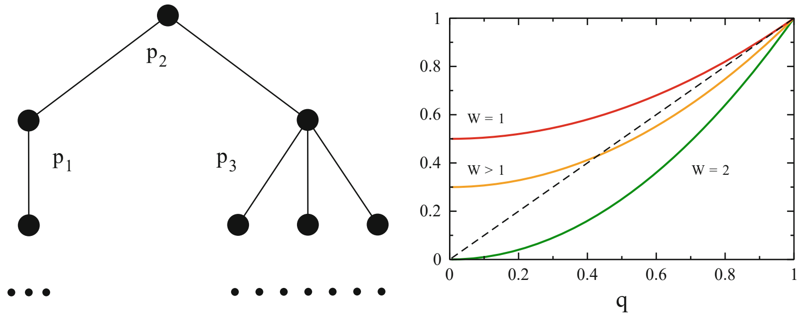

Fig. 6.12 Galton–Watson processes. **Left:** Example of a reproduction tree, with p_m being the probabilities of having $m = 0, 1, \ldots$ offsprings. **Right:** Graphical solution for the fixpoint equation (6.27), for various average numbers of offsprings W

for a square-lattice with four nearest neighbors and eight next-nearest neighbors. The update rules are conserving,

$$\sum_j \Delta_{ij} = 0, \qquad \forall c \in [0, 1] .$$

For $c = 1$ this model corresponds to the continuous field generalization of the original sandpile model. The model defined by (6.24) can be expected to map in the limit $d \to \infty$ to an appropriate random branching model with $p = p_c = 1/2$; it will be critical for all values of the parameters K and c, due to its conserving dynamics.

6.5.2 Galton–Watson Processes

Galton–Watson processes are generalizations of the binary branching processes considered so far, with interesting applications in evolution theory and some everyday experiences.

History of Family Names Family names are handed down traditionally from father to son. Family names regularly die out, leading over the course of time to a substantial reduction of the pool of family names. This effect is especially pronounced in countries looking back on millenia of cultural continuity, like China, where 22 % of the population are sharing only three family names.

The evolution of family names is described by a Galton–Watson process, with a key quantity of interest being the extinction probability, viz the probability that the last person bearing a given family name dies without descendants.

Galton–Watson Process The basic reproduction statistics determines the evolution of family names, see Fig. 6.12.

We denote with p_m the probability that an individual has m offsprings and with with $p_D^{(n)}$ the probability of finding a total of D descendants in the n-th generation. The respective generating functions,

$$G_0(x) = \sum_m p_m x^m, \qquad G^{(n)}(x) = \sum_D p_D^{(n)} x^D \,,$$

obey the recursion relation

$$G^{(n+1)}(x) = \sum_D p_D^{(n)} \left[G_0(x)\right]^D = G^{(n)}(G_0(x)) \,.$$

We rewrite the recursion relation as

$$G^{(n)}(x) = G_0(G_0(\ldots G_0(x)\ldots)) = G_0\left(G^{(n-1)}(x)\right) \,, \tag{6.25}$$

which can be solved together with the starting point $G^{(0)}(x) = x$. This representation is the basis for all further considerations; we consider here the extinction probability q.

Extinction Probability The reproduction process dies out when there is a generation with zero members. The probability of having zero persons bearing a specific family name in the n-th generation is

$$q = p_0^{(n)} = G^{(n)}(0) = G_0\left(G^{(n-1)}(0)\right) = G_0(q) \,, \tag{6.26}$$

where we have used the recursion relation (6.25) and the stationarity condition $G^{(n)}(0) \approx G^{(n-1)}(0)$. The extinction probability q is hence given by the fixpoint $q = G_0(q)$ of the generating functional $G_0(x)$ of the reproduction probability.

Binary Branching as a Galton–Watson Process As an example we take the case that

$$G_0(x) = 1 - \frac{W}{2} + \frac{W}{2}x^2, \qquad G_0'(1) = W \,,$$

viz that people may have either zero or two sons, respectively with probabilities $1 - W/2$ and $W/2 < 1$. The expected number of offsprings W corresponds to the fitness of an individuum in evolution theory.[12] This setting corresponds to the case of binary branching, see Fig. 6.10, with $W/2$ being the branching probability, e.g. as it would hold for the reproductive dynamics of unicellular organisms.

[12] See Chap. 8.

The self-consistency condition (6.26) for the extinction probability $q = q(W)$ then reads

$$q = 1 - \frac{W}{2} + \frac{W}{2}q^2, \qquad q(W) = \frac{1}{W} \pm \sqrt{\frac{1}{W^2} - \frac{(2-W)^2}{W^2}}, \tag{6.27}$$

with the smaller root being here of relevance. The extinction probability vanishes for a reproduction rate of two,

$$q(W) = \begin{cases} 0 & W = 2 \\ q \in]0,1[& 1 < W < 2 \\ 1 & W \le 1 \end{cases}$$

and is unity for a fitness below one, compare Fig. 6.12.

6.6 Application to Long-Term Evolution

An application of the techniques developed in this chapter can be used to study a model for the evolution of species proposed by Bak and Sneppen.

Fitness Landscapes Evolution deals with the adaption of species and their fitness relative to the ecosystem they live in.

> FITNESS LANDSCAPES The function that determines the chances of survival of a species, its fitness, is the fitness landscape.

In Fig. 6.13 a simple fitness landscape is illustrated, in which there is only one dimension in the genotype and/or phenotype space.[13]

Due to the presence of fitness barriers between adjacent local peaks, the population will spend most of its time in a local fitness maximum whenever the mutation rate is low with respect to the selection rate, see Fig. 6.13. Mutations are random processes that induce evolutionary transitions from one local fitness maximum to the next through stochastic escape.[14]

Coevolution It is important to keep in mind for the following discussion that an ecosystem, and with it the respective fitness landscapes, is not static on long time scales. The ecosystem is the result of the combined action of geophysical factors,

[13] The term "genotype" denotes the ensemble of genes. The actual form of an organism, the "phenotype", is determined by the genotype plus environmental factors, like food supply during growth.

[14] Stochastic escape is discussed in Sect. 3.5.2 of Chap. 3.

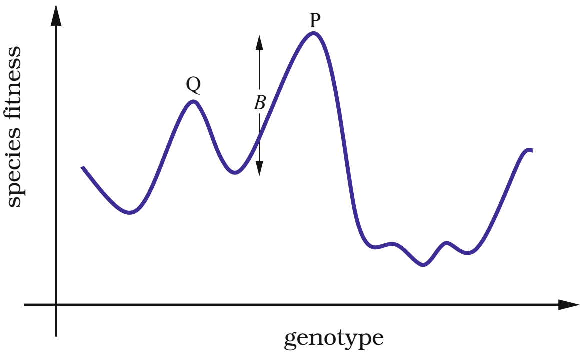

Fig. 6.13 Illustration of a one-dimensional fitness landscape. A species evolving from an adaptive peak P to a new adaptive peak Q needs to overcome the fitness barrier B

such as average rainfall and temperature, and biological influences, namely the properties and actions of the constituting species. The evolutionary progress of one species will therefore be likely to trigger adaption processes in other species appertaining to the same ecosystem, a process denoted "coevolution".

Evolutionary Time Scales In the model of Bak and Sneppen there are no explicit fitness landscapes like the one illustrated in Fig. 6.13. Instead, the model proposes that a single number, the "fitness barrier", can be used as a proxy for the influence of all other species making up the ecosystem. The time needed for a stochastic escape from one local fitness optimum increases exponentially with barrier height. We may therefore assume that the average time t it takes to mutate across a fitness barrier of height B scales as

$$t = t_0\, \mathrm{e}^{B/T}\,, \tag{6.28}$$

where t_0 and T are appropriate constants. The value of t_0 is arbitrary, as it merely sets the time scale. The parameter T depends on the mutation rate, and the assumption that mutation is low implies that T is small compared to the typical barrier heights B in the landscape. In this case the time scales t for crossing slightly different barriers are distributed over many orders of magnitude and only the lowest barrier is relevant.

Bak–Sneppen Model The Bak–Sneppen model is a phenomenological model for the evolution of barrier heights for a fixed number N of species. Each species has a barrier,

$$B_i = B_i(t) \in [0, 1], \qquad t = 0, 1, 2, \dots\,,$$

for its further evolution. The initial $B_i(0)$ are drawn randomly from $[0, 1]$. The evolutionary dynamics generated by the model consists of the repetition of two steps.

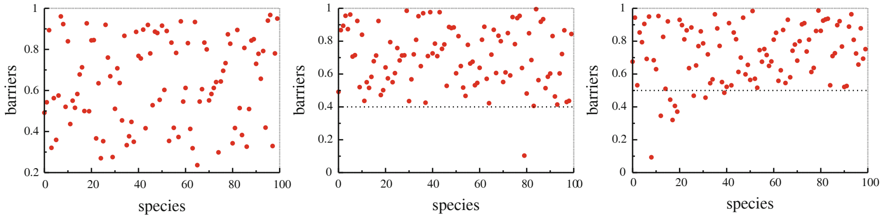

Fig. 6.14 The barrier values (dots) for a 100 species one-dimensional Bak–Sneppen model with $K = 2$ after 50, 200 and 1600 steps of a simulation. Included in each frame is the approximate position of the upper edge of the "gap" (horizontal line). A few species have barriers below this level, indicating that they were involved in an avalanche at the moment when the snapshot of the system was taken

– The times for a stochastic escape are exponentially distributed, see (6.28). It is therefore reasonable to assume that the species with the lowest barrier B_i mutates and escapes first. After escaping, it will adapt quickly to a new local fitness maximum and acquire a new barrier for mutation, which is assumed to be uniformly distributed in [0, 1].
– The fitness function for a species i is determined by the ecological environment it lives in, which is made up of the other species. When a given species mutates it therefore influences the fitness landscape for a certain number of other species. Within the Bak–Sneppen model this translates into assigning new random barriers B_j for $K - 1$ neighbors of the mutating species i.

With these two rules, the Bak–Sneppen model tries to capture two essential ingredients of long-term evolution: The exponential distribution of successful mutations and the interaction of species when one constituting species evolves.

Random Neighbor Model The topology of the interaction between species in the Bak–Sneppen model is unclear. It might be chosen to be two-dimensional, if the species are thought to live geographically separated, or one-dimensional in a toy model. In reality, the topology is complex and can be assumed to be random, in first approximation, resulting in the soluble random neighbor model.

Evolution of Barrier Distribution We start with a qualitative discussion of the redistribution of barrier heights under the sequential repetition of the two steps above, as shown Fig. 6.14. The initial barrier heights are uniformly distributed over the interval [0, 1] and the lowest barrier, removed in the first step is small. The new heights reassigned subsequently will therefore lead, on average, to an increase in the average barrier height with passing time.

With increasing average barrier height the characteristic lowest barrier is also raised, and eventually a steady state will be reached, just as in the sandpile model discussed previously. It turns out that the characteristic value for the lowest barrier is about $1/K$ at equilibrium when a the mean-field approximation is used and that the steady state is critical.

Molecular Field Theory The barrier distribution,

$$p(x,t)\,,$$

is the probability to find a barrier of hight $x \in [0,1]$ at time step $t = 1, 2, \ldots$. In addition, we define with $Q(x)$ the probability to find a barrier above x:

$$Q(x) = \int_x^1 dx'\, p(x'), \qquad Q(0) = 1, \qquad Q(1) = 0\,. \tag{6.29}$$

The dynamics is governed by the size of the smallest barrier. The distribution function $p_1(x)$ for the lowest barrier is

$$p_1(x) = N\, p(x)\, Q^{N-1}(x)\,, \tag{6.30}$$

it is given by the probability $p(x)$ for one barrier (out of the N barriers) to have the barrier height x, while all the other $N-1$ barriers are larger. $p_1(x)$ is normalized,

$$\int_0^1 dx\, p_1(x) = (-N)\int_0^1 dx\, Q^{N-1}(x)\frac{\partial Q(x)}{\partial x} = -Q^N(x)\Big|_{x=0}^{x=1} = 1\,,$$

where we used $p(x) = -Q'(x)$, $Q(0) = 1$ and $Q(1) = 0$.

Time Evolution of Barrier Distribution The time evolution for the barrier distribution consists in taking away one (out of N) barrier, the lowest, via

$$p(x,t) - \frac{1}{N}\, p_1(x,t)\,,$$

and by removing randomly $K-1$ barriers from the remaining $N-1$ barriers, and adding K random barriers,

$$\begin{aligned} p(x,t+1) = p(x,t) - \frac{1}{N}\, p_1(x,t) \\ -\frac{K-1}{N-1}\left(p(x,t) - \frac{1}{N}\, p_1(x,t)\right) + \frac{K}{N}\,. \end{aligned} \tag{6.31}$$

We note that $p(x, t + 1)$ is normalized whenever $p(x, t)$ and $p_1(x, t)$ were normalized correctly,

$$\int_0^1 dxp(x, t+1) = 1 - \frac{1}{N} - \frac{K-1}{N-1}\left(1 - \frac{1}{N}\right) + \frac{K}{N}$$
$$= \left(1 - \frac{K-1}{N-1}\right)\frac{N-1}{N} + \frac{K}{N} = \frac{N-K}{N} + \frac{K}{N} \equiv 1\,.$$

Stationary Distribution Iterating (6.31), the barrier distribution will approach a stationary solution $p(x, t + 1) = p(x, t) \equiv p(x)$, as can be observed from the numerical simulation shown in Fig. 6.14. The stationary distribution corresponds to the fixpoint condition

$$0 = p_1(x)\frac{1}{N}\left(\frac{K-1}{N-1} - 1\right) - p(x)\,\frac{K-1}{N-1} + \frac{K}{N}$$

of (6.31). With $p_1 = Np\,Q^{N-1}$ we obtain

$$0 = Np(x)\,Q^{N-1}(x)(K-N) - p(x)\,(K-1)N + K(N-1)\,,$$

which can be further simplified using $p(x) = -\partial Q(x)/\partial x$,

$$0 = N(N-K)\frac{\partial Q(x)}{\partial x}Q^{N-1} + (K-1)N\frac{\partial Q(x)}{\partial x} + K(N-1)\,,$$

or, equivalently,

$$0 = N(N-K)\,Q^{N-1}\,dQ + (K-1)N\,dQ + K(N-1)\,dx\,.$$

One can integrate the last expression with respect to x,

$$0 = (N-K)\,Q^N(x) + (K-1)N\,Q(x) + K(N-1)\,(x-1)\,, \tag{6.32}$$

where we took care of the boundary condition $Q(1) = 0$, $Q(0) = 1$.

Solution in the Thermodynamic Limit The polynomial (6.32) simplifies in the thermodynamic limit, with $N \to \infty$ and $K/N \to 0$, to

$$0 = Q^N(x) + (K-1)\,Q(x) - K\,(1-x)\,. \tag{6.33}$$

We note that $Q(x) \in [0, 1]$ and that $Q(0) = 1$, $Q(1) = 0$. There must therefore be some $x \in]0, 1[$ for which $0 < Q(x) < 1$. Then

$$Q^N(x) \to 0, \qquad Q(x) \approx \frac{K}{K-1}\,(1-x)\,. \tag{6.34}$$

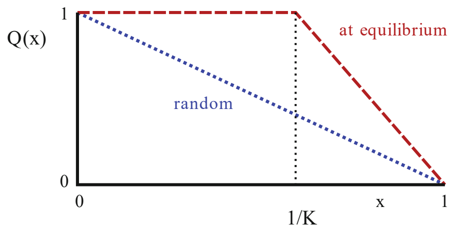

Fig. 6.15 The distribution $Q(x)$ to find a fitness barrier larger than $x \in [0, 1]$ for the Bak–Sneppen model, for the case of random barrier distribution (dotted blue line) and the stationary distribution (dashed red line), compare (6.35)

This relation remains valid as long as $Q < 1$, or $x > x_c$:

$$1 = \frac{K}{K-1}(1 - x_c), \qquad x_c = \frac{1}{K}.$$

We then have in the limit $N \to \infty$

$$\lim_{N\to\infty} Q(x) = \begin{cases} 1 & \text{for } x < 1/K \\ (1-x)K/(K-1) & \text{for } x > 1/K \end{cases}, \tag{6.35}$$

compare Fig. 6.15, which translates to

$$\lim_{N\to\infty} p(x) = \begin{cases} 0 & \text{for } x < 1/K \\ K/(K-1) & \text{for } x > 1/K \end{cases}, \tag{6.36}$$

when using $p(x) = -\partial Q(x)/\partial x$. This result compares qualitatively well with the numerical results presented in Fig. 6.14. Note, however, that the mean-field solution (6.36) does not predict the exact critical barrier height for non-random neighbors, which is somewhat larger for a one-dimensional arrangement of neighbors, as in Fig. 6.14.

Distribution of the Lowest Barrier Equation (6.36) cannot be rigorously true for $N < \infty$, since there is a finite probability for barriers with $B_i < 1/K$ to reappear at every step. If the barrier distribution is zero below the self-organized threshold $x_c = 1/K$ and constant above, then the lowest barrier must be below x_c with equal probability:

$$p_1(x) \;\to\; \begin{cases} K & \text{for } x < 1/K \\ 0 & \text{for } x > 1/K \end{cases}, \qquad \int_0^1 dx\, p_1(x) = 1. \tag{6.37}$$

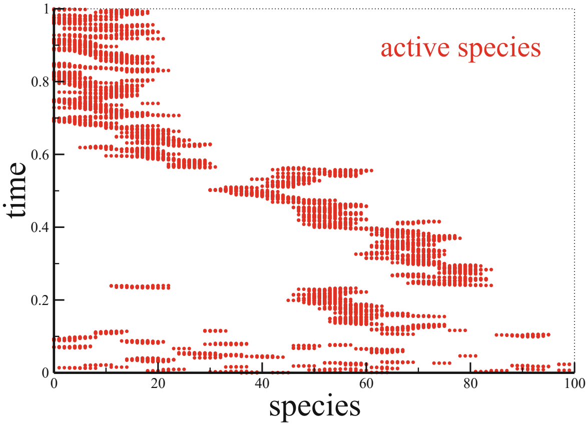

Fig. 6.16 A time series of evolutionary activity in a simulation of the one-dimensional Bak–Sneppen model with $K = 2$ showing coevolutionary avalanches interrupting the punctuated equilibrium. Shown are the new barrier values (dots), viz the evolving species

Coevolution and Avalanches When the species with the lowest barrier mutates, we assign new random barrier heights to it and to its $K - 1$ neighbors. This causes an avalanche of evolutionary adaptations whenever one of the new barriers becomes the new lowest fitness barrier. One calls this phenomenon "coevolution" since the evolution of one species drives the adaption of other species belonging to the same ecosystem. In Fig. 6.16 this process is illustrated for the one-dimensional model. The avalanches in the system are clearly visible and well separated in time. In between the individual avalanches the barrier distribution does not change appreciably; one speaks of a "punctuated equilibrium".

Critical Coevolutionary Avalanches In Sect. 6.5 we discussed the connection between avalanches and random branching. The branching process is critical when it continues with a probability of 1/2. To see whether the coevolutionary avalanches within the Bak–Sneppen model are critical we calculate the probability p_{bran} that at least one of the K new, randomly selected, fitness barriers will be the new lowest barrier.

With probability x one of the new random barriers is in $[0, x]$ and below the actual lowest barrier, which is distributed with $p_1(x)$, see (6.37). We then have

$$p_{\text{bran}} = K \int_0^1 p_1(x)\, x\, dx = K \int_0^{1/K} K\, x\, dx = \frac{K^2}{2}\, x^2 \Big|_0^{1/K} \equiv \frac{1}{2},$$

viz the avalanches are critical. The distribution of the size s of the coevolutionary avalanches is then

$$D(s) \sim \left(\frac{1}{s}\right)^{3/2},$$

as evaluated within the random branching approximation, see (6.22), and independent of K. The size of a coevolutionary avalanche can be arbitrarily large and involve, in extremis, a finite fraction of the ecosystem, compare Fig. 6.16.

Features of the Critical State The sandpile model evolves into a critical state under the influence of an external driving by adding one grain of sand after another. The critical state is characterized by a distribution of slopes (or heights) z_i that has a discontinuity; there is a finite fraction of slopes with $z_i = Z - 1$, but no slope with $z_i = Z$, apart from some of the sites transiently participating in an avalanche.

In the Bak–Sneppen model the same process occurs, but without external drivings. At criticality, the barrier distribution $p(x) = \partial Q(x)/\partial x$ has a discontinuity at $x_c = 1/K$, see Fig. 6.15. One could say, cum grano salis, that the system has developed an "internal phase transition", namely a transition in the barrier distribution $p(x)$, which is an internal variable. The emergent state for $p(x)$ is a many-body or collective effect, since it results from the reciprocal interactions of the species participating in the formation of the ecosystem.

Exercises

(6.1) SOLUTIONS OF THE LANDAU–GINZBURG FUNCTIONAL
Determine the order parameter for $h \neq 0$ and $t < 1$ via (6.5) and Fig. 6.2. Discuss the local stability condition (6.3) for the three possible solutions. You may select special values for h and t, such that $P(\phi) - h$ factorizes.

(6.2) THERMODYNAMICS OF THE LANDAU-GINZBURG MODEL
Determine the entropy $S(T) = \frac{\partial F}{\partial T}$ and the specific heat $c_V = T\frac{\partial S}{\partial T}$ within the Landau–Ginzburg theory (6.1) for phase transitions.

(6.3) THE GAME OF LIFE
Consider the evolution of the following states, see Fig. 6.5, under the rules for Conway's game of life:

$$\begin{array}{rl} \big\{(0,0),(1,0),(0,1),(1,1)\big\} & \big\{(0,-1),(0,0),(0,1)\big\} \\ \big\{(0,0),(0,1),(1,0),(-1,0),(0,-1)\big\} & \big\{(0,0),(0,1),(0,2),(1,2),(2,1)\big\} \end{array} \tag{6.38}$$

The predictions can be checked with online simulation applications.

(6.4) GAME OF LIFE ON A SMALL-WORLD NETWORK
Write a program to simulate the game of life on a 2D lattice. Consider this lattice as a network with every site having edges to its eight neighbors. Rewire the network such that (a) the local connectivities $z_i \equiv 8$ are retained for every site and (b) a small-world network is obtained. This can be achieved by cutting

two arbitrary links with probability p and rewiring the four resulting stubs randomly.[15]
Define an appropriate dynamical order parameter and characterize the changes as a function of the rewiring probability.

(6.5) FOREST FIRE MODEL
Develop a mean-field theory for the forest fire model by introducing appropriate probabilities to find cells with trees, fires and ashes. Find the critical number of nearest neighbors Z for fires to continue burning.

(6.6) REALISTIC SANDPILE MODEL
Propose a cellular automata model that simulates the physics of real-world sandpiles somewhat more realistically than the original model. The cell value $z(x, y)$ should correspond to the local height of sand. Write a program to simulate the model.

(6.7) RECURSION RELATION FOR AVALANCHE SIZES
Use the definition (6.17) for the generating functional $f_n(x, p)$ of avalanche sizes in (6.19) and derive a recursion relation for the probability $P_s(n, p)$ of finding an avalanche of size s in the nth generation, given a branching probability p. How does this recursion relation change when the branching is not binary but, as illustrated in Fig. 6.12, determined by the probability p_m of generating m offsprings?

(6.8) RANDOM BRANCHING MODEL
Derive the distribution of avalanche durations (6.23) in analogy to the steps explained in Sect. 6.5, by considering a recursion relation for the integrated duration probability $\tilde{Q}_n = \sum_{n'=0}^{n} Q_n(0, p)$, viz for the probability that an avalanche last maximally n time steps.

(6.9) GALTON–WATSON PROCESS
Use the fixpoint condition (6.26) and show that the extinction probability is unity if the average reproduction rate is smaller than one.

Further Reading

Regarding cellular automata in general, and the game of life in particular, we recommend Wolfram (1986), Creutz (2004), and Rennard (2002), with the later text specifying how to implement logical functions in the game of life. It is pointed out in Drossel (2000) that scaling results from avalanches reaching the boundary, with the relation to random branching theory being treated in Zapperi et al. (1995). In Calcaterra (2022), Lenia is disucssed, a generalization of the game of life to continuous systems.

For a review of the forest fire and several related models, see Clar et al. (1996), for a review on absorbing phase transitions Hinrichsen (2000). The forest fire model

[15] This procedure corresponds to a 2D Watts and Strogatz model, which is discussed in Sect. 1.5 of Chap. 1.

with lightning was introduced by Drossel and Schwabl (1992). A comprehensive review on power laws in theory and nature is presented in Marković et al. (2014), for a textbook on statistical physics and phase transitions see Sinai (2014). The formulation of the Bak and Sneppen (1993) model for long-term coevolutionary processes and its mean-field solution are discussed by Flyvbjerg et al. (1993).

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Random Boolean Networks

7

Complex systems theory deals with dynamical systems containing a very large number of variables. The resulting dynamical behavior can be arbitrary complex and sophisticated. It is therefore important to have well controlled benchmarks, dynamical systems which can be investigated and understood in a controlled way for large numbers of variables.

Networks of interacting binary variables, i.e. boolean networks, constitute such canonical complex dynamical systems. They allow the formulation and investigation of important concepts, among others regarding information retention and loss and the occurrence of phase transition in the resulting dynamical states. Boolean networks are recognized to be the starting points for the modeling of gene expression and protein regulation networks; the fundamental networks at the basis of life.

7.1 Introduction

Boolean Networks In this chapter, we describe the dynamics of sets of N "binary variables", namely of variables having only two possible values, typically 0 and 1. The actual values are irrelevant; ± 1 is an alternative popular choice. The elements interact with each other according to given interaction rules, the coupling functions.

> BOOLEAN COUPLING FUNCTION A boolean function $\{0, 1\}^K \to \{0, 1\}$ maps K boolean variables onto a single one.

The dynamics of the system is considered to be discrete, $t = 0, 1, 2, \ldots$. The value of the variables at the next time step are determined by the choice of boolean coupling functions.

> BOOLEAN NETWORK The set of boolean coupling functions interconnecting the N boolean variables can be represented graphically by a directed network, the boolean network.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_7

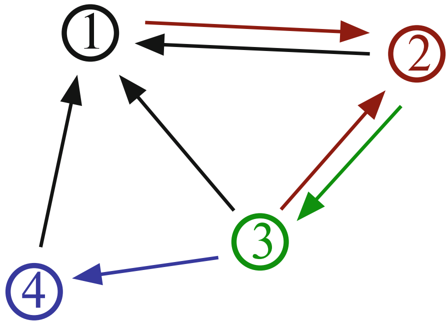


Fig. 7.1 A boolean network with $N = 4$ sites. $\sigma_1(t+1)$ is determined by $\sigma_2(t)$, $\sigma_3(t)$ and $\sigma_4(t)$, $\sigma_2(t+1)$ by $\sigma_1(t)$ and $\sigma_3(t)$, etc. The number K_i of controlling elements of σ_i is 3/2/1/1 (color coded)

In Fig. 7.1 a small boolean network is illustrated. Boolean networks at first sight seem to be quite esoteric, devoid of the practical significance for real-world phenomena. Why are they then studied so intensively?

Cell Differentiation in Terms of Stable Attractors The field of boolean networks was given a first big boost by a seminal study by Kauffman in the late 1960s. Kauffman casted the problem of gene expression in terms of a gene regulation network, introducing the N–K model in this context. The question regards the how of cell differentiation, i.e. the fact that a skin cell differs from a muscle cell, albeit having the identical genetic markup. This can be due only to differences in gene activity patterns in the respective cells. Kauffman proposed that different stable attractors, viz cycles in gene expression networks, would correspond to distinct cells in the bodies of animals.

The notion would hence be that cell types correspond to dynamical states of a complex system, i.e. of the gene expression network. This proposal by Kauffman has received on one side strong support from experimental studies. In Sect. 7.5.2 the case of the yeast cell division cycle will be discussed, supporting the notion that gene regulation networks constitute the underpinnings of life. Regarding the mechanisms of cell differentiation in multicellular organisms, the situation is, on the other side, less clear. Cell types are mostly determined by DNA methylation, which affects the respective gene expression on long time scales.

Boolean Networks Are Everywhere Kauffman's original work on gene expression networks has been generalized to a wide spectrum of applications, such as, to give a few examples, the modeling of neural networks by random boolean networks and of the "punctuated equilibrium" in long-term evolution.[1]

Classical dynamical systems theory deals with dynamical systems containing a comparatively compact number of variables. Dynamical systems with large numbers of variables need in part new toolsets for analysis and control. In this

[1] Punctuated equilibrium will feature in Chap. 8.

field, random boolean networks are of prototypical importance, as they provide a well defined class of dynamical systems for which the thermodynamical limit $N \to \infty$ can be taken. They show chaotic as well as regular behavior, despite their apparent simplicity, along with other typical phenomena of dynamical systems. In the thermodynamic limit a phase transition between chaotic and regular regimes appears, with distinct consequences for information flow. These are the issues studied in this chapter.

N–K Networks There are several types of random boolean networks. The most simple realization is the N–K model. It is made up of N boolean variables, each variable interacting exactly with K other randomly chosen variables. The respective coupling functions are also chosen randomly from the set of all possible boolean functions mapping K boolean inputs onto a single boolean output.

An ideal realization of the N–K model in nature is not known. All real physical or biological problems have specific couplings determined by the structure and the physical and biological interactions of the system in question. In many instances, the topology of the couplings is complex and mostly unknown, it is then often a good starting point to model a real-world system using a generic framework, like the N–K model.

Binary Variables Modeling real-world systems by a collection of interacting binary variables is often a simplification, as real-world variables are mostly continuous. For the case of the gene expression network, one just keeps two possible states for every single gene, active or inactive.

Thresholds, viz parameter regimes at which the dynamical behavior changes qualitatively, are widespread in biological systems. Examples are neurons, which fire or do not fire depending on the total strength of presynaptic activity. Similar thresholds occur in metabolic networks in the form of activation potentials for the chemical reactions involved. Modeling real-world systems based on threshold dynamics with binary variables is a viable first step towards an understanding.

7.2 Random Variables and Networks

Boolean networks have a rich variety of possible concrete model realizations. The most important types are discussed in the following.

7.2.1 Boolean Variables and Graph Topologies

State Space The N binary variables

$$\sigma_i \in \{0, 1\}, \qquad i = 1, 2, \ldots, N$$

constitute the state of the system Σ_t at time t,

$$\Sigma_t = \{\sigma_1(t), \sigma_2(t), \ldots, \sigma_N(t)\} ,$$

which can be thought of as a vector pointing to one of the $\Omega = 2^N$ edges of a N-dimensional hypercube, where Ω is the number of possible configurations. For numerical implementations and simulations, it can be useful to encode Σ_t as the binary representation of an integer number $0 \leq \Sigma_t < 2^N$.

Controlling Elements Time is assumed to be discrete,

$$\sigma_i = \sigma_i(t), \qquad t = 1, 2, \ldots .$$

The value of a given boolean element σ_i at the next time step is determined by the values of K controlling variables.

CONTROLLING ELEMENTS The controlling elements $\sigma_{j_1(i)}$, $\sigma_{j_2(i)}$, …, $\sigma_{j_{K_i}(i)}$ of a boolean variable σ_i determine its time evolution.

The next state of an element is hence a boolean function of its controlling elements,

$$\sigma_i(t+1) = f_i(\sigma_{j_1(i)}(t), \sigma_{j_2(i)}(t), \ldots, \sigma_{j_{K_i}(i)}(t)) . \tag{7.1}$$

Here f_i is a boolean function associated with σ_i. The set of controlling elements might include σ_i itself. Some exemplary boolean functions are listed in Table 7.1.

Table 7.1 Examples of boolean functions of three arguments. Given is a particular random function and a canalizing function of the first argument. For the latter, the function value is 1 when $\sigma_1 = 0$. If $\sigma_1 = 1$, the output can be either 0 or 1. For the additive function shown, the output is 1 (active) if at least two inputs are active. The generalized XOR is true when the number of true-bits is odd

			$f(\sigma_1, \sigma_2, \sigma_3)$			
σ_1	σ_2	σ_3	Random	Canalizing	Additive	Gen. XOR
0	0	0	0	1	0	0
0	0	1	1	1	0	1
0	1	0	1	1	0	1
0	1	1	0	1	1	0
1	0	0	1	0	0	1
1	0	1	0	1	1	0
1	1	0	1	0	1	0
1	1	1	1	0	1	1

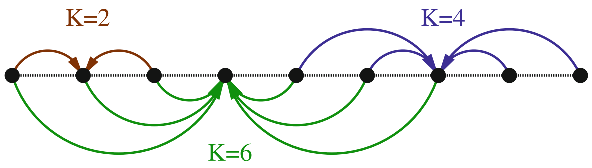


Fig. 7.2 Translational invariant linkages for a completely ordered one-dimensional lattice with connectivities $K = 2, 4, 6$

Model Definition For a complete definition of the system several parameters need to be specified.

– CONNECTIVITY
The first step is to select the connectivity K_i of each element, i.e. the number of its controlling elements. With

$$\langle K \rangle = \frac{1}{N} \sum_{i=1}^{N} K_i$$

the average connectivity is defined. Mostly we will consider the case in which the connectivity is identical for all nodes, $K_i = K$.
– LINKAGES
The second step is to select the specific set of controlling elements

$$\left\{\sigma_{j_1(i)}, \sigma_{j_2(i)}, \ldots, \sigma_{j_{K_i}(i)}\right\}$$

on which the element σ_i depends. See Fig. 7.1 for an illustration.
– EVOLUTION RULE
The third step is to choose the boolean function f_i determining the value of $\sigma_i(t+1)$ from the values of the linkages $\left\{\sigma_{j_1(i)}(t), \sigma_{j_2(i)}(t), \ldots, \sigma_{j_{K_i}(i)}(t)\right\}$.

Geometry of the Network The way the linkages are assigned determines the topology of the network and networks can have highly diverse topologies. It is custom to consider two special cases.

LATTICE ASSIGNMENT The boolean variables σ_i are assigned to the nodes of a regular lattice. The K controlling elements $\left\{\sigma_{j_1(i)}, \sigma_{j_2(i)}, \ldots, \sigma_{j_K(i)}\right\}$ are then chosen in a regular, translational invariant manner, as in Fig. 7.2.

Complectly regular contrasts with fully random, viz uniform.

UNIFORM ASSIGNMENT For a uniform assignment, the set of controlling elements are randomly drawn from all N sites of the network. This is the case for the N–K model, the original Kauffman net. In terms of graph theory one also speaks of an Erdös–Rényi random graph.

Intermediate cases are possible. Small-world networks, to give an example, with regular short-distance links and random long-distance links are popular models in network theory, such are scale-free topologies.[2]

7.2.2 Coupling Functions

Number of Coupling Functions The coupling function

$$f_i : \qquad \left\{\sigma_{j_1(i)}, \ldots, \sigma_{j_K(i)}\right\} \to \sigma_i$$

has 2^K different arguments. To each argument value one can assign either 0 or 1. Thus, there are a total of

$$N_f = 2^{(2^K)} = 2^{2^K} = \begin{cases} 2 & K = 0 \\ 4 & K = 1 \\ 16 & K = 2 \\ 256 & K = 3 \end{cases}$$

possible coupling functions. In Table 7.1 several examples are presented for the case $K = 3$, out of the $2^{2^3} = 256$ distinct $K = 3$ boolean functions.

Classification of Coupling Functions For small numbers of connectivity K one can completely classify all possible coupling functions.

- $K = 0$
  There are only two constant functions, $f = 1$ and $f = 0$.
- $K = 1$
  Apart from the two constant functions, forming the class $\mathcal{A}$, there is the identity and the negation, lumped together into class $\mathcal{B}$.

σ	Class $\mathcal{A}$		Class $\mathcal{B}$	
0	0	1	0	1
1	0	1	1	0

[2] Erdös–Rényi graphs are discussed in conjunction with small-world and scale-free networks in Chap. 1.

– $K = 2$
  There are four classes of functions $f(\sigma_1, \sigma_2)$, as listed in Table 7.2, with each class being invariant under the interchange $0 \leftrightarrow 1$ in either the arguments or the value of f.

For $K = 2$ the classes are

$$\begin{aligned} \mathcal{A} &: \text{Constant functions.} \\ \mathcal{B}_1 &: \text{Fully canalizing functions.} \\ \mathcal{B}_2 &: \text{Normal canalizing functions, see also Table 7.1.} \\ \mathcal{C} &: \text{Non-canalizing functions, also denoted “reversible functions”.} \end{aligned}$$

For fully canalizing functions one of the arguments determines the output deterministically.

Types of Coupling Ensembles A range of distinct choices for the probability distribution of the coupling functions are possible. Here the most important examples.

– Uniform Distribution
  As introduced originally by Kauffman, the uniform distribution specifies all possible coupling functions to occur with the same probability $1/N_f$.
– Magnetization Bias[3]
  The probability of a coupling function to occur is proportional to p if the outcome is 0 and proportional to $1 - p$ if the outcome is 1.
– Forcing Functions
  For forcing or canalizing functions, the function value is determined when one of its arguments, say $m \in \{1, \ldots, K\}$, has a specific value, say $\sigma_m = 0$. In contrast, the function value is not specified if the forcing argument has a different value, here when $\sigma_m = 1$. Compare Table 7.1.

Table 7.2 The 16 boolean functions for $K = 2$. For the definition of the various classes see Sect. 7.2.2 and Aldana-Gonzalez et al. (2003)

σ_1	σ_2	Class $\mathcal{A}$		Class $\mathcal{B}_1$					Class $\mathcal{B}_2$									Class $\mathcal{C}$	
0	0	1	0	0	1	0	1		1	0	0	0	0	1	1	1		1	0
0	1	1	0	0	1	1	0		0	1	0	0	1	0	1	1		0	1
1	0	1	0	1	0	0	1		0	0	1	0	1	1	0	1		0	1
1	1	1	0	1	0	1	0		0	0	0	1	1	1	1	0		1	0

[3] Magnetic moments may have only two possible directions, up or down in the language of spin-1/2 particles. A compound is hence magnetic when more moments point into one of the two possible directions, viz if the two directions are populated unequally.

– Additive Functions
  In order to simulate the additive properties of inter-neural synaptic activities one can choose

$$\sigma_i(t+1) = \Theta\big(\tilde{f}_i(t)\big), \qquad \tilde{f}_i(t) = -h + \sum_{j=1}^{N} w_{ij}\,\sigma_j(t)$$

  where $\Theta(x)$ is the Heaviside step function, h the threshold for activating the neuron and w_{ij} the synaptic weight connecting the pre- and post-synaptic neurons j and i. The value of $\sigma_i(t+1)$ depends only on a weighted sum of its controlling elements at time t.

7.2.3 Dynamics

Model Realizations A given set of linkages and boolean functions $\{f_i\}$ defines what one calls a "realization" of the model, with the dynamics following from (7.1). For the updating of the elements during a time step one has several choices.

– Synchronous Update
  All variables $\sigma_i(t)$ are updated simultaneously.
– Asynchronous updating
  A single variable is updated at every step. This variable may be picked at random or by some predefined ordering scheme. Hence also the name "serial update".

The choice of updating does not affect thermodynamic properties, like the phase diagram discussed in Sect. 7.3.2. The occurrence and the properties of cycles and attractors, as discussed in Sect. 7.4, depends however crucially on the form of update.

Selection of the Model Realization There are several alternatives for choosing the model realization during numerical simulations.

– Quenched Model[4]
  One specific realization of coupling functions is selected at the beginning and kept throughout all time.

[4] An alloy made up of two or more substances is said to be "quenched" when it is cooled so quickly that it remains stuck in a specific atomic configuration, which does not change anymore subsequently.

– Annealed Model[5]
  A new realization is randomly selected at each time step. Then either the linkages or the coupling functions or both change with every update, depending on the choice of the algorithm.
– Genetic Algorithm
  If the network is thought to approach a predefined goal, one may employ a genetic algorithm in which the system slowly modifies its realization with passing time.

Real-world systems are normally modeled by quenched systems with synchronous updating. Interactions are fixed for all times.

Cycles and Attractors Boolean dynamics correspond to a trajectory within a finite state space of size $\Omega = 2^N$. Any trajectory generated by a dynamical system with unmutable dynamical update rules, as for the quenched model, will eventually lead to a cyclical behavior. No trajectory can generate more than Ω distinct states in a row. Once a state is revisited,

$$\Sigma_t = \Sigma_{t-T}, \qquad T < \Omega ,$$

part of the original trajectory is retraced and cyclic behavior follows. The resulting cycle acts as an attractor for a set of initial conditions.

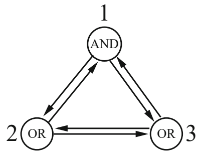
Cycles of length one are fixpoint attractors. The fixpoint condition $\sigma_i(t+1) = \sigma_i(t)$ $(i = 1, \ldots, N)$ is independent of the updating rules, viz synchronous vs. asynchronous. The order of updating the individual σ_i is irrelevant when none of them changes.

An Example In Fig. 7.3 a network with $N = 3$ and $K = 2$ is fully defined. The time evolution of the $2^3 = 8$ states Σ_t is given for synchronous updating. One can observe one cycle of length 2 and two cycles of length 1 (fixpoints).

7.3 Dynamics of Boolean Networks

We will now examine how we can characterize the dynamical state of boolean networks in general and of N–K nets in particular. Two concepts will turn out to be of central importance, the relation of robustness to the flow of information and the characterization of the overall dynamical state, which we will find to be either frozen, critical or chaotic.

[5] A compound is said to be "annealed" when it has been kept long enough at elevated temperatures such that the thermodynamic stable configuration has been achieved.



OR

1 2	3
0 0	0
0 1	1
1 0	1
1 1	1

OR

1 3	2
0 0	0
0 1	1
1 0	1
1 1	1

AND

2 3	1
0 0	0
0 1	0
1 0	0
1 1	1

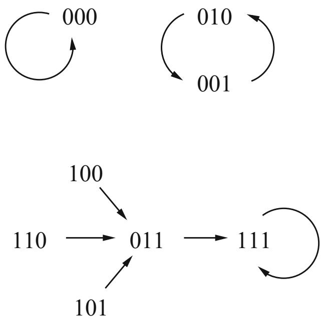


Fig. 7.3 A boolean network with $N = 3$ sites and connectivities $K_i \equiv 2$. **Left:** Definition of the network linkage and coupling functions. **Right:** The complete network dynamics. Reprinted from Luque and Sole (2000) with permissions, © 2000 Elsevier Science B.V

7.3.1 Flow of Information Through a Network

Response to Changes Biological systems need to be robust. A gene regulation network, to give an example, for which even small damage routinely results in the death of the cell, will be at an evolutionary disadvantage with respect to a more robust set-up. Here we will examine the sensitivity of the dynamics with regard to initial conditions. A system is robust if two similar initial conditions lead to similar long-time behavior. This approach is well defined for quenched models.

Hamming Distance Between Orbits For two distinct initial states,

$$\Sigma_0 = \{\sigma_1(0), \sigma_2(0), \ldots, \sigma_N(0)\}, \qquad \tilde{\Sigma}_0 = \{\tilde{\sigma}_1(0), \tilde{\sigma}_2(0), \ldots, \tilde{\sigma}_N(0)\},$$

we are typically interested in the case when Σ_0 and $\tilde{\Sigma}_0$ are close, viz when they differ in the values of only a few elements. A suitable measure for the distance is the "Hamming distance" $D(t) \in [0, N]$,

$$D(t) = \sum_{i=1}^{N} \Big(\sigma_i(t) - \tilde{\sigma}_i(t)\Big)^2, \tag{7.2}$$

which is just the sum of the elements that differ in Σ_0 and $\tilde{\Sigma}_0$. As an example take

$$\Sigma_1 = \{1, 0, 0, 1\}, \qquad \Sigma_2 = \{0, 1, 1, 0\}, \qquad \Sigma_3 = \{1, 0, 1, 1\}.$$

We have 4 for the Hamming distance Σ_1–Σ_2 and 1 for the Hamming distance Σ_1–Σ_3. If the system is robust, two close-by initial conditions will never move far apart with time passing, in terms of the Hamming distance.

Normalized Overlap The normalized overlap $a(t) \in [0, 1]$ between two configurations is defined as

$$a(t) = 1 - \frac{D(t)}{N} = 1 - \frac{1}{N}\sum_{i=1}^{N}\left(\sigma_i^2(t) - 2\sigma_i(t)\tilde{\sigma}_i(t) + \tilde{\sigma}_i^2(t)\right)$$

$$\approx \frac{2}{N}\sum_{i=1}^{N}\sigma_i(t)\tilde{\sigma}_i(t)\,, \tag{7.3}$$

where we did assume the absence of any magnetization bias, namely that

$$\frac{1}{N}\sum_i \sigma_i^2 \;\approx\; \frac{1}{2} \;\approx\; \frac{1}{N}\sum_i \tilde{\sigma}_i^2\,,$$

in the last step. The normalized overlap (7.3) corresponds to a rescaled scalar product between Σ and $\tilde{\Sigma}$. On the average, two arbitrary states have a Hamming distance of $N/2$, which translates to a normalized overlap $a = 1 - D/N$ of 1/2.

Information Retention for Long Times The difference between two initial states Σ and $\tilde{\Sigma}$ can also be interpreted as an information for the system. One then has two possible behaviors.

- Loss of Information
  $\lim_{t\to\infty} a(t) \to 1$, which implies that two states are identical in the thermodynamic limit, or that they differ only by a finite number of elements. This happens when two states are attracted by the same cycle. All information about the starting states is lost.
- Information Retention
  $\lim_{t\to\infty} a(t) = a^* < 1$, which means that the system 'remembers' that the two configurations were initially different, with the difference measured by the respective Hamming distance.

The system is robust when information is routinely lost, which holds for $a^* = 1$. Robustness depends on the value of a^* when information is kept. If $a^* > 0$ then two trajectories retain a certain similarity for all time scales.

Percolation of Information for Short Times Above we discussed how information present in initial states evolves over extended time scales. Alternatively one may ask a typical question of dynamical systems theory, namely how information is processed for short times. We write

$$D(t) \approx D(0)\,\mathrm{e}^{\lambda t}\,, \tag{7.4}$$

where $0 < D(0) \ll N$ is the initial Hamming distance, with λ being the Lyapunov exponent.[6]

The question is then whether two initially close trajectories converge or diverge initially. One may generally distinguish between three different types of behaviors or phases.

- CHAOTIC PHASE
  $\lambda > 0$: The Hamming distance grows exponentially, i.e. information is transferred to an exponentially large number of elements. Two initially close orbits soon become different. This behavior, found for large connectivities K, is not suitable for real-world biological systems.
- FROZEN PHASE
  $\lambda < 0$: Two close trajectories typically converge, as they are attracted by the same attractor. This behavior arises for small connectivities K. The system is locally robust.
- CRITICAL PHASE
  $\lambda = 0$: When present, exponential time dependencies dominate all other contributions. There is no exponential time dependence when the Lyapunov exponent λ vanishes. In this case, the Hamming distance will depend algebraically on time, $D(t) \propto t^{\gamma}$.

All three phases can be found in the N–K model when $N \to \infty$. This is our next step.

7.3.2 Mean-Field Phase Diagram

With in a mean-field approach, aka "molecular-field theory", a microscopic model is treated by averaging the influence of distinct components, lumping them together into a single molecular field. Mean-field theories are ubiquitous.[7]

Mean-Field Theory We recall that we are working with two initial states,

$$\Sigma_0, \qquad \tilde{\Sigma}_0, \qquad D(0) = \sum_{i=1}^{N} \left(\sigma_i - \tilde{\sigma}_i\right)^2,$$

and that the Hamming distance $D(t)$ measures the number of elements differing in Σ_t and $\tilde{\Sigma}_t$. An illustration of the following arguments is presented in Fig. 7.4.

[6] Stability in terms of Lyapunov exponents is treated in Sect. 2.2 of Chap. 2.

[7] A reference molecular-field framework is the Landau theory of phase transitions, which is discussed in detail in Sect. 6.1 of Chap. 6.

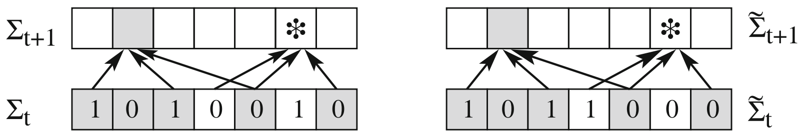


Fig. 7.4 The time evolution of the overlap between two states Σ_t and $\tilde{\Sigma}_t$ (left/right panels). The vertices (squares) have values 0 or 1. Vertices with identical values in both states, Σ_t and $\tilde{\Sigma}_t$, are highlighted (gray background). The values of vertices at the next time step, $t+1$, can only differ if the corresponding arguments are not identical. It is indicated whether vertices at time $t+1$ must have the same value in both states (grey), or whether they can be different (star)

- For the N–K model, every boolean coupling function f_i is as likely to occur.
- On average, variables are controlling elements for K other variables.
- The variables differing in Σ_t and $\tilde{\Sigma}_t$ affect on the average $KD(t)$ coupling functions.
- In the absence of a magnetization bias, coupling functions change their value with probability $1/2$.

Taken together, we conclude that the number of elements different in Σ_{t+1} and $\tilde{\Sigma}_{t+1}$, viz the Hamming distance $D(t+1)$, is given by

$$D(t+1) = \frac{K}{2}\, D(t), \qquad D(t) = \left(\frac{K}{2}\right)^t D(0) = D(0)\, \mathrm{e}^{t\ln(K/2)}\,, \qquad (7.5)$$

which identifies $\lambda = \ln(K/2)$ as the local Lyapunov exponent.

Classification of Phases The connectivity K then determines the phase of the N–K network.

- CHAOTIC
  $K > 2$: Two initially close orbits diverge, the number of different elements, i.e. the relative Hamming distance grows exponentially with time t.
- FROZEN
  $K < 2$: The two orbits approach each other exponentially. All initial information contained $D(0)$ is lost.
- CRITICAL
  $K = K_c = 2$: The evolution of Σ_t relative to $\tilde{\Sigma}_t$ is driven by fluctuations. The power laws typical for critical regimes cannot be deduced within mean-field theory, which discards fluctuations.

The mean-field theory takes only average quantities into account. The evolution law $D(t+1) = (K/2)D(t)$ holds only on the average. Fluctuations, viz the deviation of the evolution from the mean-field prediction, are however of importance only close to a phase transition, i.e. close to the critical point $K = 2$.

The mean-field approximation generally works well for lattice physical systems in high spatial dimensions and fails in low dimensions. The Kauffman network has no dimension per se, but the connectivity K plays an analogous role.

Phase Transitions in Dynamical Systems and the Brain The notion of a 'phase transition' originally comes from physics, where it denotes the transition between two or more different physical phases, like ice, water and gas, which are well characterized by their respective order parameters.

Classically, the term phase transition denotes therefore a transition between two stationary states. The phase transition discussed here involves the characterization of the overall behavior of a dynamical system. They are well defined phase transitions in the sense that $1 - a^*$ plays the role of an order parameter; its value uniquely characterizes the frozen phase and the chaotic phase in the thermodynamic limit.

An interesting, completely open and unresolved question is in this context whether dynamical phase transitions play a role in the most complex information processing system known, the mammalian brain. It is tempting to speculate, e.g., that the phenomenon of consciousness may result from a dynamical state characterized by a yet unknown order parameter. In case, consciousness would be an 'emergent' state.

7.3.3 Bifurcation Phase Diagram

In deriving (7.5), we assumed that the coupling functions f_i of the system acquire the values 0 and 1 with the identical probabilities $p = 1/2$. We generalize this approach to the case of a magnetic bias, as defined by

$$f_i = \begin{cases} 0 & \text{with probability} \quad p \\ 1 & \text{with probability} \quad 1-p \end{cases} .$$

For a given value of the bias p and connectivity K, there are critical values

$$K_c(p), \qquad\qquad p_c(K)\,,$$

such that for $K < K_c$ ($K > K_c$) the system is in the frozen phase (chaotic phase). Vice versa, keeping the connectivity fixed and vary p, we have that $p_c(K)$ separates the system into a chaotic and frozen phase.

Evolution of the Overlap We note that the overlap $a(t) = 1 - D(t)/N$ between two states Σ_t and $\tilde{\Sigma}_t$ at time t is the probability that two vertices have the same value both in Σ_t and in $\tilde{\Sigma}_t$.

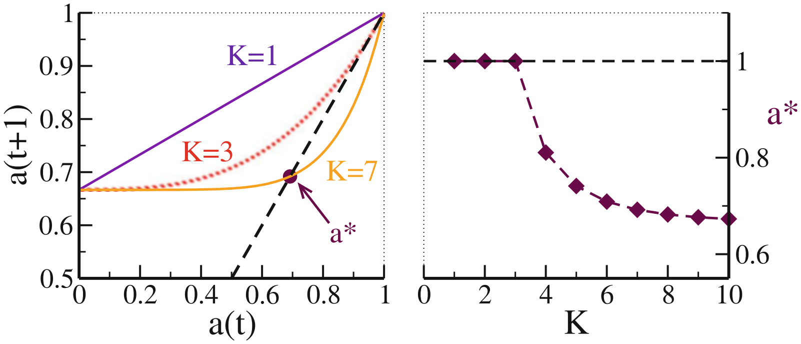


Fig. 7.5 Solution of the self-consistency condition $a^* = 1 - \left[1 - (a^*)^K\right]/K_c$, see (7.9). **Left:** Graphical solution equating both sides. **Right:** Numerical result for a^* for $K_c = 3$. The fixpoint $a^* = 1$ becomes unstable for $K > K_c = 3$

– The probability that all arguments of the function f_i will be the same is both configurations is

$$\rho_K = \left[\, a(t) \,\right]^K . \tag{7.6}$$

– As illustrated by Fig. 7.4, the values at the next time step differ with a probability $2p(1-p)$, but only if the arguments of the coupling functions are non-different.
– The probability that at least one controlling element has different values in Σ_t and $\tilde{\Sigma}_t$, $1 - \rho_K$, this gives the probability, $(1-\rho_K)2p(1-p)$, of values being different in the next time step.

We then have

$$a(t+1) = 1 - (1-\rho_K)\,2p(1-p) = 1 - \frac{1 - [a(t)]^K}{K_c} , \tag{7.7}$$

where K_c is given in terms of p as

$$K_c = \frac{1}{2p(1-p)}, \qquad\qquad p_c^{1,2} = \frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{1}{2K}} . \tag{7.8}$$

The fixpoint a^* of (7.7) obeys

$$a^* = 1 - \frac{1 - [a^*]^K}{K_c} . \tag{7.9}$$

This self-consistency condition for the normalized overlap can be solved graphically or numerically by simple iterations, compare Fig. 7.5.

Stability Analysis The trivial fixpoint

$$a^* = 1$$

always constitutes a solution of (7.9). We examine its stability under the time evolution, Eq. (7.7), by considering a small deviation $0 < \Delta a_t \ll 1$ from the fixpoint solution, $a_t = a^* - \Delta a_t$:

$$1 - \Delta a_{t+1} = 1 - \frac{1 - [1 - \Delta a_t]^K}{K_c}, \qquad \Delta a_{t+1} \approx \frac{K \Delta a_t}{K_c}\,. \tag{7.10}$$

The trivial fixpoint $a^* = 1$ therefore becomes unstable for $K/K_c > 1$, viz when $K > K_c = \big(2p(1-p)\big)^{-1}$.

Bifurcation Equation (7.9) has two solutions for $K > K_c$, a stable fixpoint $a^* < 1$ and the unstable solution $a^* = 1$. One speaks of a bifurcation, which is shown in Fig. 7.5. We note that

$$K_c\Big|_{p=1/2} = 2\,,$$

in agreement with our previous mean-field result, Eq. (7.5). For large connectivities one finds

$$\lim_{K\to\infty} a^* = \lim_{K\to\infty} \left(1 - \frac{1 - [a^*]^K}{K_c}\right) = 1 - \frac{1}{K_c} = 1 - 2p(1-p)\,,$$

since $a^* < 1$ for $K > K_c$, compare Fig. 7.5. Note that $a^* = 1/2$ for $p = 1/2$ corresponds to the average normalized overlap for two completely unrelated states in the absence of a magnetization bias, $p = 1/2$. Two initial similar states eventually become completely uncorrelated for $t \to \infty$ when also the connectivity K diverges.

Rigidity of the Kauffman Net We can connect the results for the phase diagram of the N–K network illustrated in Fig. 7.6 with our discussion on robustness, see Sect. 7.3.1.

– CHAOTIC PHASE
  $K > K_c$: The infinite time normalized overlap a^* is less than unity even when two trajectories Σ_t and $\tilde{\Sigma}_t$ start out very close to each other. However, a^* always remains above the value expected for two completely unrelated states. The reason is that the Hamming distance remains constant, modulo small-scale fluctuations that do not contribute in the thermodynamic limit, $N \to \infty$, after the two orbits did enter two distinct attractors.

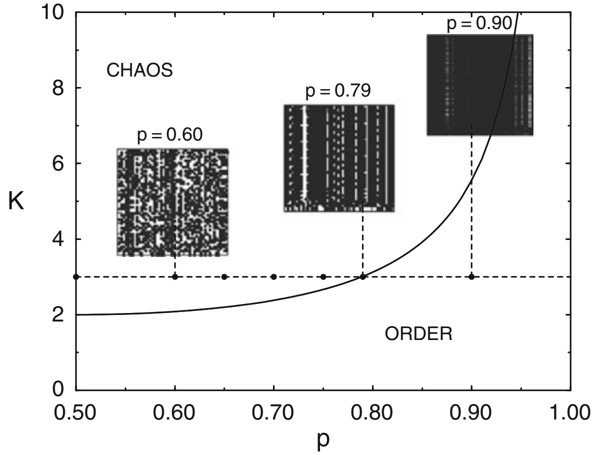


Fig. 7.6 Phase diagram for the N–K model. Shown is the separation $K_c = [2p(1-p)]^{-1}$ between the chaotic and order/frozen phase (solid line). The insets are simulations for $N = 50$ networks with $K = 3$ and $p = 0.60$ (chaotic phase), $p = 0.79$ (on the critical line) and $p = 0.90$ (frozen phase). The site index runs horizontally, the time vertically. Note the fluctuations for $p = 0.79$. Reprinted from Luque and Sole (2000) with permissions, © 2000 Elsevier Science B.V

– FROZEN PHASE
  $K < K_c$: The infinite time overlap a^* is exactly one. All trajectories approach essentially the same configuration independently of the starting point, apart from fluctuations that vanish in the thermodynamic limit. The system is said to "order".

In the frozen phase, close-by orbits are attracted by the same cyclic attractor, in the chaotic state by different attracting states.

Lattice Versus Random Networks The complete loss of information in the ordered phase observed for the Kauffman net does not occur for lattice networks, for which $a^* < 1$ for any $K > 0$. The finite range of the linkages in lattice systems allows to store information about the initial data in spatially finite subsections of the system, specific to the initial state. For Kauffman graphs, every region of the network is equally close to any other and local storage of information is impossible.

Percolation Transition in Lattice Networks For lattice boolean networks, the frozen and chaotic phases cannot be distinguished examining the value of the long-term normalized overlap a^*, as it is always smaller than unity. The lattice topology, allows however to connect with percolation theory. One considers a finite system, e.g. a 100×100 square lattice, and two states Σ_0 and $\tilde{\Sigma}_0$ that differ only along one edge.

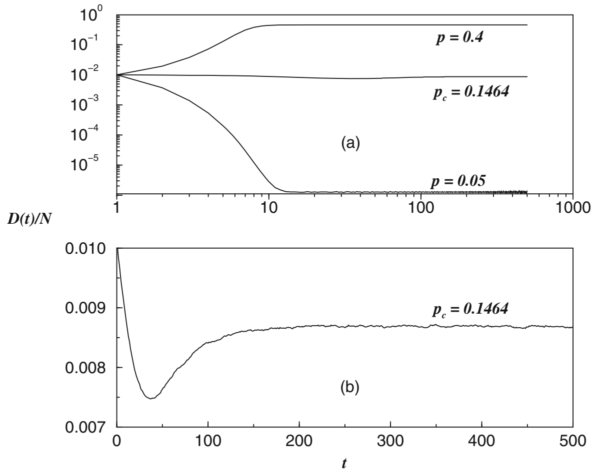


Fig. 7.7 Normalized Hamming distance $D(t)/N$ for a Kauffman net with $N = 10,000$ variables, connectivity $K = 4$ and $D(0) = 100$, viz $D(0)/N = 0.01$. **Top:** Frozen phase ($p = 0.05$), critical ($p_c \simeq 0.1464$) and chaotic ($p = 0.4$) phases, plotted with a logarithmic scale. **Bottom:** Hamming distance for the critical phase ($p = p_c$) but on a non-logarithmic scale. Reprinted from Aldana-Gonzalez et al. (2003) with permissions, © 2003 Springer-Verlag New York, Inc

– If the damage, viz the difference in between Σ_t and $\tilde{\Sigma}_t$, spreads for long times to the opposite edge, then the system is said to be percolating and in the chaotic phase.
– If the damage never reaches the opposite edge, then the system is in the frozen phase.

Numerical simulations indicate, a critical $p_c \simeq 0.298$ for the two-dimensional square lattice with connectivity $K = 4$.

Numerical Simulations The results of the mean-field solution for the Kauffman net are confirmed by numerical solutions of finite-size networks. In Fig. 7.7 the normalized Hamming distance, $D(t)/N$, is plotted for a Kauffman graph with connectivity $K = 4$ containing $N = 10{,}000$ elements. The, results are shown for parameters corresponding to the frozen phase and to the chaotic phase, in addition to a parameter close to the critical line. Note that $1 - a^* = D(t)/N \to 0$ in the frozen phase.

7.3.4 Scale-Free Boolean Networks

The Kauffman model is a reference model which can be generalized in various ways, e.g. by considering small-world or scale-free networks.

Scale-Free Connectivity Distributions Scale-free connectivity distributions,[8]

$$P(K) = \frac{1}{\zeta(\gamma)} K^{-\gamma}, \qquad \zeta(\gamma) = \sum_{K=1}^{\infty} K^{-\gamma}, \qquad \gamma > 1\,, \tag{7.11}$$

abound in real-world networks. Here $P(K)$ denotes the probability to draw a coupling function $f_i(\cdot)$ having Z arguments. The distribution (7.11) is normalizable for $\gamma > 1$. The average connectivity $\langle K \rangle$ is

$$\langle K \rangle = \sum_{K=1}^{\infty} K\; P(K) = \begin{cases} \infty & \text{if} \;\; 1 < \gamma \le 2 \\ \frac{\zeta(\gamma-1)}{\zeta(\gamma)} < \infty & \text{if} \;\; \gamma > 2 \end{cases}, \tag{7.12}$$

where $\zeta(\gamma)$ is the Riemann zeta function.

Annealed Approximation As before, we examine two states Σ_t and $\tilde{\Sigma}_t$ together with the respective normalized overlap,

$$a(t) = 1 - D(t)/N\,,$$

which is identical to the probability that two vertices in Σ and $\tilde{\Sigma}$ have the same value. For a magnetization bias p, we derived in Sect. 7.3.3 that

$$a(t+1) = 1 - (1-\rho_K)\, 2p(1-p) \tag{7.13}$$

holds for the time-evolution of $a(t)$, where

$$\rho_K = [a(t)]^K \quad \to \quad \sum_{K=1}^{\infty} [a(t)]^K\; P(K) \tag{7.14}$$

is the average probability that the $K = 1, 2, \ldots$ controlling elements of the coupling function $f_i()$ are all identical. In (7.14) we generalized (7.6) to a non-constant connectivity distribution $P(K)$, which leads to

$$a(t+1) = 1 - 2p(1-p) \left\{ 1 - \sum_{K=1}^{\infty} a^K(t)\; P(K) \right\} \equiv F(a(t))\,, \tag{7.15}$$

compare (7.7). The statistical averaging used in (7.14) corresponds effectively to an annealed model.

[8] The theory of scale free graphs is developed in Sect. 1.6 of Chap. 1.

Fixpoints Within the Annealed Approximation In the limit $t \to \infty$, Eq. (7.15) reduces to the self-consistency equation

$$a^* = F(a^*)\,,$$

for the fixpoint a^*, where $F(a)$ is defined as the right-hand side of (7.15). Again, $a^* = 1$ is always a fixpoint of (7.15), since $\sum_K P(K) = 1$, per definition.

Stability of the Trivial Fixpoint We repeat the stability analysis of the trivial fixpoint $a^* = 1$, as in Sect. 7.3.3, and assume a small deviation $\Delta a > 0$ from a^*,

$$a^* - \Delta a = F(a^* - \Delta a) = F(a^*) - F'(a^*)\Delta a, \qquad \Delta a = F'(a^*)\Delta a\,.$$

The fixpoint a^* becomes unstable if $F'(a^*) > 1$. The critical point is determined by

$$\begin{aligned} 1 = \lim_{a\to 1^-} \frac{dF(a)}{da} &= 2p(1-p)\sum_{k=1}^{\infty} K\, P(K) \\ &= 2p(1-p)\,\langle K\rangle\,. \end{aligned} \tag{7.16}$$

For the classical N–K model all elements have the same connectivity, $K_i = \langle K\rangle = K$, with (7.16) reducing to (7.10).

Frozen and Chaotic Phases for the Scale-Free Model For $1 < \gamma \le 2$ the average connectivity is infinite, see (7.12). $F'(1) = 2p(1-p)\,\langle K\rangle$ is then always larger than unity and $a^* = 1$ unstable, as illustrated in Fig. 7.8. Equation (7.15) then has a stable fixpoint $a^* \neq 1$; the system is in the chaotic phase for all $p \in]0, 1[$.

For $\gamma > 2$ the first moment of the connectivity distribution $P(K)$ is finite and the phase diagram is identical to that of the N–K model shown in Fig. 7.6, with K replaced by $\zeta(\gamma_c - 1)/\zeta(\gamma_c)$. The phase diagram in γ–p space is presented in Fig. 7.8. One finds that $\gamma_c \in [2, 2.5]$ for any value of p. There is no chaotic scale-free network for $\gamma > 2.5$. It is interesting to note that $\gamma \in [2, 3]$ for many real-world scale-free networks.

7.4 Cycles and Attractors

We emphasized so far the general properties of boolean networks, such as the phase diagram. We now turn to a more detailed inspection of the dynamics, in particular regarding the structure of the attractors.

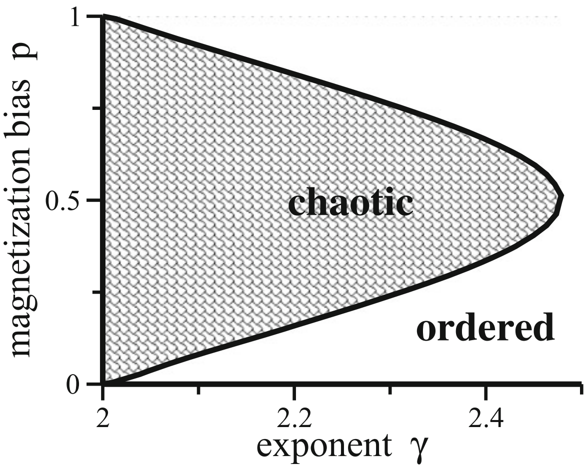


Fig. 7.8 Phase diagram for a scale-free boolean network with connectivity distribution $\propto K^{-\gamma}$, as given by (7.16). The average connectivity diverges for $\gamma < 2$ and the network is chaotic for all p

7.4.1 Quenched Boolean Dynamics

Self-Retracting Orbits From now on we consider quenched systems for which the coupling functions $f_i(\sigma_{i_1}, \ldots, \sigma_{i_K})$ are fixed for all times. Orbits will eventually retrace themself, at least partly, since state space $\Omega = 2^N$ is finite. Long-term trajectories are therefore cyclic.

ATTRACTOR An attractor A_0 of a discrete dynamical system is a region $\{\Sigma_t\} \subset \Omega$ in phase space that maps completely onto itself under the time evolution, $A_{t+1} = A_t \equiv A_0$.

In practice, this means that

$$\Sigma^{(1)} \quad \to \quad \Sigma^{(2)} \quad \to \quad \ldots \quad \to \quad \Sigma^{(1)},$$

see Figs. 7.3 and 7.9 for some examples. Fixed points are cycles of length 1.

ATTRACTION BASIN The attraction basin B of an attractor A_0 is the set $\{\Sigma_t\} \subset \Omega$ for which there is a time $T < \infty$ such that $\Sigma_T \in A_0$.

For randomly drawn initial conditions, the probability of ending up in a given cycle is directly proportional to the size of its basin of attraction. The three-site network illustrated in Fig. 7.3 is dominated by the fixpoint $\{1, 1, 1\}$, which is reached with probability 5/8 for random initial starting states.

Attractors are Everywhere Attractors and fixpoints are generic features of dynamical systems and core to their characterization, as they dominate the time evolution in state space within their respective basins of attraction. Random boolean networks allow for detailed studies of the structure of attractors and of

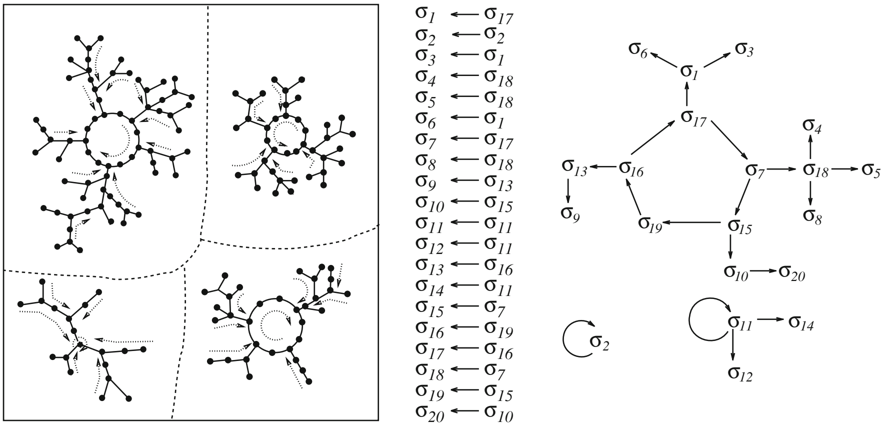


Fig. 7.9 Cycles and linkages. **Left:** sketch of the state space where every bold point stands for a state $\Sigma_t = \{\sigma_1, \ldots, \sigma_N\}$. The state space decomposes into distinct attractor basins for each cyclic- or fixpoint attractor. **Right:** linkage loops for an $N = 20$ model with $K = 1$. The controlling elements are listed in the center column. Each arrow points from the controlling element toward the direct descendant. There are three modules of uncoupled variables. Reprinted from Aldana-Gonzalez et al. (2003) with permissions, © 2003 Springer-Verlag New York, Inc

the connection to network topology. Of special interest is how various properties of the attractors, like the cycle length and the size of the attractor basins, relate to the thermodynamic differences between the frozen and the chaotic phase. These are the issues that we shall discuss now.

Linkage Loops, Ancestors and Descendants Any variable σ_i may appear as an argument in the coupling functions for other elements; it is said to act as a controlling element. Graphically, the collections of all linkages corresponds to a directed graph, as illustrated in Figs. 7.1, 7.3 and 7.9, with the vertices representing the individual binary variables. A given element σ_i can influence potential a large number of different states during the continued time evolution.

> ANCESTORS AND DESCENDANT. The elements a vertex affects consecutively via the coupling functions are its descendants. Going backwards in time one finds the ancestors of an element.

In the 20-site network illustrated in Fig. 7.9 the descendants of σ_{11} are σ_{11}, σ_{12} and σ_{14}.

When an element is its own descendant (and ancestor) it is said to be part of a "linkage loop". Different linkage loops can overlap, as is the case for the linkage loops

$$\sigma_1 \to \sigma_2 \to \sigma_3 \to \sigma_4 \to \sigma_1, \qquad \sigma_1 \to \sigma_2 \to \sigma_3 \to \sigma_1$$

shown in Fig. 7.1. Linkage loops are disjoint for $K = 1$, compare Fig. 7.9.

Modules and Time Evolution The set of ancestors and descendants determines the overall dynamical dependencies.

> MODULE The collection of all ancestors and descendants of a given element σ_i is the module (or component) to which σ_i belongs.

If we go through all variables σ_i, $i = 1, \ldots, N$ we find all modules, with every element belonging to one and only one specific module. Otherwise stated, disjoint modules correspond to disjoint subgraphs, the set of all modules making up the full linkage graph. The time evolution is block-diagonal in terms of modules. For all times t an element $\sigma_i(t)$ is independent of all variables not belonging to its own module.

In lattice networks the clustering coefficient, the density of three-site loops, is large and closed linkage loops occur frequently. For big lattice systems with a small mean linkage K we expect far away spatial regions to evolve independently, due the lack of long-range connections.

Relevant Nodes and Dynamic Core Taking a look at dynamics of the 20-site model illustrated in Fig. 7.9, we notice that the elements σ_{12} and σ_{14} just follow the dynamics of σ_{11}, they are "enslaved" by σ_{11}. These two elements do not control any other element and one can delete them from the system without qualitative changes to the overall dynamics.

> RELEVANT NODE A node is termed relevant if its state is not constant and if it controls at least one other relevant element (eventually itself).

An element is constant if it evolves, independently of the initial conditions, always to the same state and not constant otherwise. The set of relevant nodes, the dynamic core, controls the overall dynamics. The dynamics of all other nodes can be disregarded without changing the attractor structure. The node σ_{13} of the 20-site network illustrated in Fig. 7.9 is relevant if the boolean function connecting it to itself is either the identity or the negation.

The concept of a dynamic core is relevant for practical applications. Gene expression networks may be composed of thousands of nodes, with a relatively small dynamic core controlling overall network dynamics. This is the case for the gene regulation network controlling the yeast cell cycle discussed in Sect. 7.5.2.

Lattice Nets vs. Kauffman Nets Linkages are short-ranged for lattice systems and whenever a given element σ_j acts as a controlling element for another element σ_i, there is a high probability that also the reverse holds, viz that σ_i is an argument of f_j.

The linkages are generally non-reciprocal for the Kauffman net; the probability for reciprocity is K/N, which vanishes in the thermodynamic limit for finite K. The number of disjoint modules in a random network therefore grows more slowly than the system size. For lattice systems, on the other hand, the number of modules is proportional to the size of the system. The differences between lattice and Kauffman networks translate to different cycle structures, as every periodic orbit for the full system is constructed out of the individual attractors of all modules present in the network.

7.4.2 K = 1 Kauffman Network

We start our discussion of the cycle structure of Kauffman nets with the case $K = 1$, which can be solved exactly. The maximal length $l_{\max}$ for a linkage loop is on the average of the order of

$$l_{\max} \sim N^{1/2} . \tag{7.17}$$

At every of the l steps of a sequence of length l, the next node σ_i could be one of the l sites visited previously. Overall, the typical cycle length is reached when the probability to close, $\sim l\,(l/N)$, becomes of order one, which leads to (7.17). This derivation can be made rigorous following the line of arguments we will develop in Sect. 7.4.4 for large-K nets.

Three-Site Linkage Loop with Identities Linkage loops determine the cycle structure together with the choice of the coupling ensemble. As an example we take the case $N = 3$.

For $K = 1$ there are only two non-constant coupling functions, i.e. the identity I and the negation $\neg$. When all three coupling functions are the identity, we have

$$ABC \rightarrow CAB \rightarrow BCA \rightarrow ABC \rightarrow \ldots ,$$

where we denoted with A, B, C the values of the binary variables σ_i, $i = 1, 2, 3$. There are two cycles of length one in which all elements are identical. When the three elements are not identical, the cycle length is three. The complete dynamics is then

$$\begin{array}{ll} 000 \rightarrow 000 \qquad\qquad & 100 \rightarrow 010 \rightarrow 001 \rightarrow 100 \\ 111 \rightarrow 111 \qquad\qquad & 011 \rightarrow 101 \rightarrow 110 \rightarrow 011 \end{array} ,$$

as illustrated in Fig. 7.10.

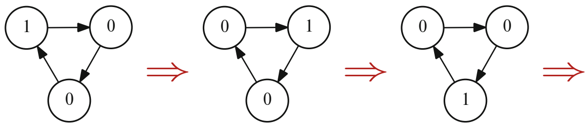


Fig. 7.10 A $K = 1$ linkage loop with $N = 3$ sites and identities as Boolean coupling functions

Three-Site Linkage Loops with Negations Let us consider now the case that all three coupling functions are negations:

$$ABC \to \bar{C}\bar{A}\bar{B} \to BCA \to \bar{A}\bar{B}\bar{C} \to \ldots \qquad \bar{A} = \neg A, \quad \text{etc.}.$$

The cycle length is two if all elements are identical

$$000 \to 111 \to 000$$

and of length 6 if they are not,

$$100 \to 101 \to 001 \to 011 \to 010 \to 110 \to 100\,.$$

The complete state space $\Omega = 2^3 = 8$ decomposes into two cycles, one of length 6 and one of length 2.

Three-Site Linkage Loops with a Constant Function Let us see what happens if any of the coupling functions is a constant function. For illustration purposes we consider the case of two constant functions 0 and 1 and the identity,

$$ABC \to 0A1 \to 001 \to 001\,. \tag{7.18}$$

Generally it holds that the cycle length is one if any of the coupling functions is the identity, in this case only a single fixpoint attractor exists. Equation (7.18) holds for all $A, B, C \in \{0, 1\}$; the basin of attraction for 001 is hence the whole state space, with 001 being a global attractor.

The Kauffman net contains very large linkage loops for $K = 1$, see (7.17). The probability that a given linkage loop contains at least one constant function is consequently very high, which implies that the average cycle length remains short.

Loops and Attractors Attractors are made up of the set of linkage loops, which illustrate by means of a five-site network with two linkage loops,

$$A \xrightarrow{I} B \xrightarrow{I} C \xrightarrow{I} A, \qquad D \xrightarrow{I} E \xrightarrow{I} D\,,$$

with all coupling functions being the identity I. The states

$$00000, \qquad 00011, \qquad 11100, \qquad 11111$$

are fixpoints in phase space $\Sigma = ABCDE$. Examples of cyclic attractors of length 3 and 6 are

$$10000 \to 01000 \to 00100 \to 10000$$

and

$$10010 \to 01001 \to 00110 \to 10001 \to 01010 \to 00101 \to 10010\,.$$

In general, the length of an attractor is given by the least common multiple of the periods of the constituent loops. This relation holds for $K = 1$ boolean networks, for general K the attractors are composed of the cycles of the constituent set of modules.

Critical $K = 1$ Boolean Networks When the coupling ensemble is selected uniformly, as defined in Sect. 7.2.2, the $K = 1$ network is in the frozen state. If we do however restrict our coupling ensemble, allowing only identity and negation, the value of one node is just copied or inverted to exactly one other node. Information is retained, and not lost, when the two constant $K = 1$ coupling functions are not present. The information is not multiplied either, being transmitted to exactly one and not more nodes. The network is hence critical in terms of the framework developed in Sect. 7.3.1.

7.4.3 K = 2 Kauffman Network

The $K = 2$ Kauffman net is critical, as discussed in Sects. 7.3.1 and 7.3.2. When physical systems undergo a second-order phase transition, right at the point of transition response functions will be scale free, following power laws. It is therefore natural to expect the same for critical dynamical systems, such as a random boolean network.

Initially, this expectation was born out by a series of mostly numerical investigations, which indicated that both the typical cycle lengths, as well as the mean number of different attractors, would grow algebraically with N, namely like $\sqrt{N}$. It was therefore tempting to relate power laws seen in natural organisms to the behavior of critical random boolean networks.

Undersampling of State Space However, the problem to determine the number and the length of cycles turned out to be numerically exceedingly demanding. In order to extract power laws one has to simulate systems with large N, which implies that one needs to deal with an exponentially growing state space $\Omega = 2^N$. An exhaustive enumeration of all cycles is hence impossible and one has to resort to a weighted sampling of state space for a given network realization, extrapolating subsequently from a small fraction of states sampled to the full state space. This method yielded the $\sqrt{N}$ dependence referred to above.

As a matter of principle, weighted sampling may however undersample state space. The number of cycles found in the average state space may not be representative for the overall number of cycles, as there might be small fractions of state space with very high number of attractors dominating the total number of attractors.

This is indeed the case. One can prove rigorously that the number of attractors grows faster than any power for the $K = 2$ Kauffman net. However, one may still argue that results for the "average state space" are relevant for biological applications, as biological systems are not too big anyway. Hormone and gene regulation networks of mammals contain respectively of the order of 100 and 20,000 elements.

Observational Scale Invariance Experimental observations of a dynamical system are typically equivalent to a random sampling of its phase space. Experimental results will hence reflect the properties of the attractors with large basins of attractions, which dominate phase space. For the case of the $K = 2$ Kauffman net an external observer would hence find scale invariance. Given that the response of the system to a random perturbation will also involve the dominating attractors, one may considered the $K = 2$ net as "effectively scale free".

7.4.4 K = N Kauffman Network

Within mean-field theory, which holds for the fully connected $K = N$ network, one can evaluate the average number and the length of cycles using probability arguments.

Random Walk Through Configuration Space We examine orbits starting from an arbitrary configuration Σ_0 at time $t = 0$. The time evolution generates a series of states

$$\Sigma_0,\ \Sigma_1,\ \Sigma_2,\ \dots$$

through configuration space, with size $\Omega = 2^N$. For a large connectivity, as in the limit $K = N$, one can assume the Σ_t to be uncorrelated, forming a random walk.

Closing the Random Walk The walk through configuration space continues until it hits a previously visited point, as illustrated in Fig. 7.11. We define with

- q_t : the probability that the trajectory remains unclosed after t steps;
- P_t : the probability of terminating the excursion exactly at time t.

Including Σ_0 and Σ_t, $t + 1$ different sites have been visited if the trajectory is still open at time t. Therefore, there are $t + 1$ ways of terminating the walk at the next time step and the relative probability of termination is $\rho_t = (t + 1)/\Omega$. The overall

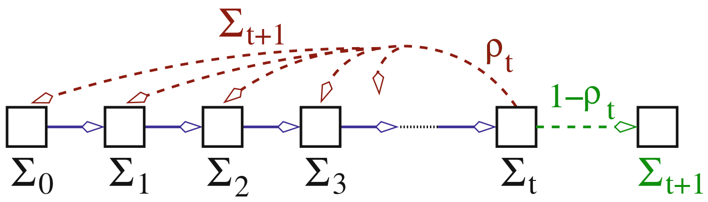


Fig. 7.11 A random walk in configuration space. The relative probability $\rho_t = (t+1)/\Omega$ of closing the loop at time t is the probability that $\Sigma_{t+1} \equiv \Sigma_{t'}$, with a certain $0 \le t' \le t$

probability P_{t+1} to terminate the random walk at time $t+1$ is

$$P_{t+1} = \rho_t \, q_t = \frac{t+1}{\Omega} \, q_t \, .$$

In analogy, the probability for the trajectory to remain open after $t+1$ steps is given by

$$q_{t+1} = q_t(1-\rho_t) = q_t \left(1 - \frac{t+1}{\Omega}\right) = q_0 \prod_{i=1}^{t+1} \left(1 - \frac{i}{\Omega}\right), \qquad q_0 = 1 \, .$$

Diverging Phase Space In the thermodynamic limit, $N \to \infty$, phase space diverges exponentially, $\Omega = 2^N \to \infty$, which implies that the approximation

$$q_t = \prod_{i=1}^{t} \left(1 - \frac{i}{\Omega}\right) \approx \prod_{i=1}^{t} \mathrm{e}^{-i/\Omega} = \mathrm{e}^{-\sum_i i/\Omega} = \mathrm{e}^{-t(t+1)/(2\Omega)} \tag{7.19}$$

becomes exact. For large times t we may use $t(t+1)/(2\Omega) \approx t^2/(2\Omega)$. The overall probability

$$\sum_{t=1}^{\Omega} P_t \simeq \int_0^\infty dt \, \frac{t}{\Omega} \, \mathrm{e}^{-t^2/(2\Omega)} = 1$$

for the random walk to close at all is unity.

Cycle Length Distribution The average number $\langle N_c(L) \rangle$ of cycles of length L is

$$\big\langle N_c(L) \big\rangle = \frac{q_{t=L-1}}{\Omega} \, \frac{\Omega}{L} = \frac{\exp[-L^2/(2\Omega)]}{L} \, , \tag{7.20}$$

where we used (7.19), with $\langle \cdots \rangle$ denoting an ensemble average over realizations. In deriving (7.20) we used the following considerations.

– The probability that Σ_{t+1} is identical to Σ_0 is $1/\Omega$.
– There are Ω possible starting points (factor Ω).
– Factor $1/L$ corrects for the overcounting of cycles when considering the L possible starting sites of the L-cycle.

Average Number of Cycles From (7.20) the mean number $\bar{N}_c$ of cycles can be extracted,

$$\bar{N}_c = \sum_{L=1}^{N} \langle N_c(L) \rangle \simeq \int_1^\infty dL \, \langle N_c(L) \rangle \,. \tag{7.21}$$

When going from the sum $\sum_L$ to the integral $\int dL$ in (7.21) we neglected terms of order unity. Rescaling variables by $u = L/\sqrt{2\Omega}$, one obtains

$$\bar{N}_c = \int_1^\infty dL \, \frac{\exp[-L^2/(2\Omega)]}{L} = \underbrace{\int_{1/\sqrt{2\Omega}}^{1} du \, \frac{e^{-u^2}}{u}}_{\equiv I_1} + \underbrace{\int_1^\infty du \, \frac{e^{-u^2}}{u}}_{\equiv I_2} \,.$$

A separation $\int_{1/\sqrt{2\Omega}}^\infty = \int_{1/\sqrt{2\Omega}}^c + \int_c^\infty$ of the integral was performed, for simplicity with $c = 1$, any other finite value for c would also do the job. The second integral, I_2, does not diverge as $\Omega \to \infty$. For I_1 we have

$$\begin{aligned} I_1 &= \int_{1/\sqrt{2\Omega}}^{1} du \, \frac{e^{-u^2}}{u} = \int_{1/\sqrt{2\Omega}}^{1} du \, \frac{1}{u}\left(1 - u^2 + \frac{1}{2}u^4 + \ldots\right) \\ &\approx \ln(\sqrt{2\Omega}) \,, \end{aligned}$$

since all further terms $\propto \int_{1/\sqrt{2\Omega}}^1 du \, u^{n-1} < \infty$ for $n = 2, 4, \ldots$ and $\Omega \to \infty$. The average number of cycles is then

$$\bar{N}_c = \ln(\sqrt{2^N}) \; + \; O(1) \simeq \frac{N \ln 2}{2} \,, \tag{7.22}$$

which holds for $N = K$ Kauffman nets approaching the thermodynamic limit $N \to \infty$.

Mean Cycle Length On the average, the length $\bar{L}$ of a random cycle is

$$\begin{aligned} \bar{L} &= \frac{1}{\bar{N}_c} \sum_{L=1}^{\infty} L \, \langle N_c(L) \rangle \approx \frac{1}{\bar{N}_c} \int_1^\infty dL \, L \, \frac{\exp[-L^2/(2\Omega)]}{L} \\ &= \frac{1}{\bar{N}_c} \int_1^\infty dL \, e^{-L^2/(2\Omega)} = \frac{\sqrt{2\Omega}}{\bar{N}_c} \int_{1/\sqrt{2\Omega}}^\infty du \, e^{-u^2} \end{aligned} \tag{7.23}$$

after rescaling with $u = L/\sqrt{2\Omega}$ and using (7.20). The last integral on the right-hand-side of (7.23) converges for $\Omega \to \infty$, with the consequence that the mean cycle length $\bar{L}$ scales as

$$\bar{L} \sim \Omega^{1/2}/N = 2^{N/2}/N\,, \tag{7.24}$$

where we used (7.22), namely that $\bar{N}_c \sim N$.

7.5 Applications

7.5.1 Living at the Edge of Chaos

Gene Expression Networks and Cell Differentiation Kauffman introduced the N–K model in the late 1960s for the purpose of modeling dynamics and time evolution of networks of interacting genes, i.e. the gene expression network. In this model an active gene might influence the expression of any other gene, e.g. when the protein transcripted from the first gene influences the expression of the second gene.

Gene expression networks of real-world cells are not random. However, the web of linkages and connectivities among the genes in a living organism is intricate and one may considered it to be a good 0-th order approximation to model gene-gene interactions as random. The purpose would be to gain generic insights into the properties of gene expression networks, namely results that are independent of the particular set of linkages and connectivities realized in a particular living cell.

Dynamical Cell Differentiation Whether random or not, in order to keep the cell functioning, the dynamics resulting from a gene expression network needs to be stable. Humans have only a few hundred different cell types in their bodies. Considering the fact that every single cell contains the identical genetic material, in full, Kauffman proposed that cell types would correspond to a distinct dynamical state of the underlying gene expression network. It is natural to assume that these states correspond to attractors, viz in general to cycles. In the chaotic phase the average length $\bar{L}$ of a cycle in a N–K Kauffman net is

$$\bar{L} \sim 2^{\alpha N}\,,$$

e.g. for $N = K$ where $\alpha = 1/2$, see (7.24). Considering that $N \approx 20{,}000$ for the human genome, an exponentially large mean cycle length $\bar{L}$ is somewhat unsettling, a single cell would take the universe's lifetime to complete just a single cycle. It then follows that operational gene expression networks of living organisms cannot be in the chaotic phase.

Living at the Edge of Chaos There are but two possibilities left if the gene expression network cannot operate in the chaotic phase, the frozen phase together with the critical point. In the frozen phase, the average cycle length is short and the dynamics stable, see Sect. 7.4.2. The system is consequently resistant to damage of linkages.

But what about Darwinian evolution? Is too much stability good for the adaptability of cells in a changing environment? Kauffman suggested that gene expression networks would operate "at the edge of chaos", an expression that became legendary. By this Kauffmann meant that networks close to criticality benefit from the stability properties of the close-by frozen phase, being at the same time sensible to changes in the network structure, such that Darwinian adaption remains possible.

But how can a system reach criticality by itself? For the N–K network there is no extended critical phase, only a single critical point, $K = 2$. One speaks of "self-organized criticality" when internal mechanisms allow adaptive systems to evolve autonomously in such a way that they approach the critical point.[9] This would be the case if Darwinian evolution trims the gene expression networks towards criticality. Cells close to the critical point would have the highest fitness, as cells in the chaotic phase die because they are operationally unstable, with cells deep in the frozen phase being selected out in the course of time, given that they are unable to adapt to environmental changes.

7.5.2 Yeast Cell Cycle

Cell Division Process Cells have two tasks: to survive and to multiply. When a living cell grows too big, a cell division process starts. The cell cycle has been studied intensively for budding yeast. In the course of the division process, the cell goes through a distinct set of states

$$G_1 \to S \to G_2 \to M \to G_1\,,$$

with G_1 being the "ground state" in physics slang, viz the normal cell state; the chromosome division takes place during the M phase. These states are characterized by distinct gene activities, i.e. by the kinds of proteins active in the cell. All eukaryote cells have similar cell division cycles.

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[9] See Chap. 6.

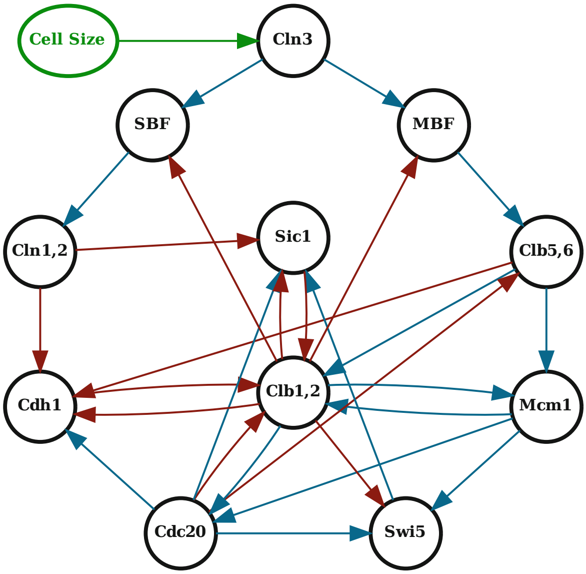

Fig. 7.12 The $N = 11$ core network responsible for the yeast cell cycle. Acronyms denote protein names, arrows excitatory/inhibitory connections (blue/red). Cln3 is inactive in the resting state G_1, becoming active when the cell reaches a certain size (top), which initiates the cell division process. Data from Li et al. (2004)

Yeast Gene Expression Network From the $\sim$800 genes involved only 11–13 core genes are actually responsible for regulating the part of the gene expression network responsible for the division process; all other genes are more or less just descendants of the core genes. The cell dynamics contains certain checkpoints, where the cell division process can be stopped if something was to go wrong. When eliminating the checkpoints a core network with only 11 elements remains, as shown in Fig. 7.12.

Boolean Dynamics The full dynamical dependencies are not known in detail for the yeast gene expression network. The simplest model is to assume

$$\sigma_i(t+1) = \begin{cases} 1 & \text{if } a_i(t) > 0 \\ 0 & \text{if } a_i(t) \le 0 \end{cases}, \qquad a_i(t) = \sum_j w_{ij}\sigma_j(t)\,, \tag{7.25}$$

i.e. a boolean dynamics[10] for the binary variables $\sigma_i(t) = 0/1$ representing the activation/deactivation of protein i, with couplings $w_{ij} = \pm 1$ for an excitatory/inhibitory functional relation.

[10] Genes are boolean variables in the sense that they are either expressed or not. The quantitative amount of proteins produced by a given active gene is regulated via a separate mechanism involving microRNA, small RNA snippets.

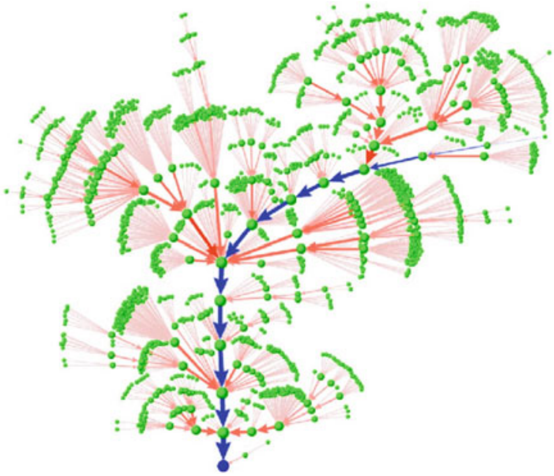

Fig. 7.13 The yeast cell cycle as an attractor trajectory of the gene expression network. Out of the $2^{11} = 2,048$ states in phase space, shown are the 1764 states (green dots), that make up the basin of attraction of the biologically stable G_1-state (bottom). After starting with the excited G_1 normal state (the first state in the biological pathway), compare Fig. 7.12, the boolean dynamics runs through the known intermediate states (blue arrows), until the G_1 states attractor is again reached, this time representing the two daughter cells. Reprinted from Li et al. (2004) with permissions, © 2004 by The National Academy of Sciences, U.S.A

Fixpoints The 11-site network has 7 attractors, all cycles of length one, viz fixpoints. The dominating fixpoint has an attractor basin of 1764 states, representing about 72% of the state space $\Omega = 2^{11} = 2{,}048$. Remarkably, the protein activation pattern of the dominant fixpoint corresponds exactly to that of the experimentally determined G_1 ground state of the living yeast cell.

Cell Division Cycle In the G_1 ground state the protein Cln3 is inactive. When the cell reaches a certain size it becomes expressed, i.e. it becomes active. For the network model one then just starts the dynamics by setting

$$\sigma_{\mathrm{Cln3}} \to 1, \qquad \text{at} \quad t = 0$$

in the G_1 state. The ensuing simple boolean dynamics, induced by (7.25), is depicted in Fig. 7.13.

The remarkable result is that the system follows an attractor pathway that runs through all experimentally known intermediate cell states, reaching the ground state G_1 in 12 steps.

Comparison with Random Networks The properties of the boolean network depicted in Fig. 7.12 can be compared with those of a random boolean network. A random network of the same size and average connectivity would have more attractors with correspondingly smaller basins of attraction. Living cells clearly need a robust protein network to survive in harsh environments.

Nevertheless, the yeast protein network shows more or less the same susceptibility to damage as a random network. The core yeast protein network has an average connectivity of $\langle K \rangle = 27/11 \simeq 2.46$. The core network has only $N = 11$ sites, a number far too small to allow comparison with the properties of N–K networks in the thermodynamic limit $N \to \infty$. Nevertheless, an average connectivity of 2.46 is remarkably close to $K = 2$, i.e. the critical connectivity for N–K networks.

Life as an Adaptive Network Living beings are complex and adaptive dynamical systems.[11] The here discussed insights in the yeast gene expression network indicate that this statement is not just an abstract notion. Adaptive regulative networks constitute the core of all living.

7.5.3 Application to Neural Networks

Time Encoding by Random Neural Networks There is a certain debate in neuroscience whether, and to which extent, time encoding is used in neural processing.

– ENSEMBLE ENCODING
  Ensemble encoding entails that the activity of a sensory input is transmitted via the firing of certain ensembles of neurons. Distinct sensory inputs, like the different smells sensed by the nose, is encoded by dedicated neural ensembles.
– TIME ENCODING
  Time encoding is present if the same neurons transmit more than one piece of sensory information by changing their respective firing patterns.

Cyclic attractors are an obvious tool to generate time encoded information. One would envision that appropriate initial conditions corresponding to certain activity patterns of the primary sensory organs start the dynamics of neural and/or random boolean networks, which will settle eventually into a cycle, as discussed in Sect. 7.4. The random network may then be used to encode initial firing patterns by the time

[11] In this context see Chap. 3, together with Chap. 8.

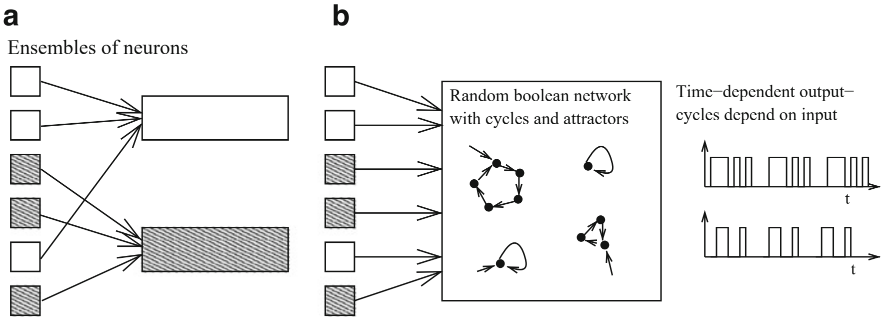

Fig. 7.14 Illustration of ensemble and time encoding (left/right). **Left:** Neuronal receptors corresponding to the same class of input signals are combined, as occuring in the nose for different odors. **Right:** The primary input signals are mixed together by a random neural network close to criticality, the relative weights of odor components are subsequently time encoded by the output signal

sequence of neural activities resulting from the firing patterns of the corresponding limiting cycle, see Fig. 7.14.

Critical Sensory Processing The processing of incoming information is qualitatively different in the various phases of the N–K model, as discussed in Sect. 7.3.1.

The chaotic phase is unsuitable for information processing, any input results in an unbounded response and saturation. The response in the frozen phase is strictly proportional to the input and is therefore well behaved, but also relatively uninteresting. The critical state, on the other hand, has the possibility of nonlinear signal amplification.

Sensory organs in animals can routinely process physical stimuli, such as light, sound, pressure, or odorant concentrations, which vary by many orders of magnitude in intensity. The primary sensory cells, e.g. the light receptors in the retina, have, however a linear sensibility to the intensity of the incident light, with a relatively small dynamical range. It is therefore conceivable that the huge dynamical range of sensory information processing of animals is a collective effect, as it occurs in a random neural network close to criticality. This mechanism, which is plausible from the view of possible genetic encoding mechanisms, is illustrated in Fig. 7.15.

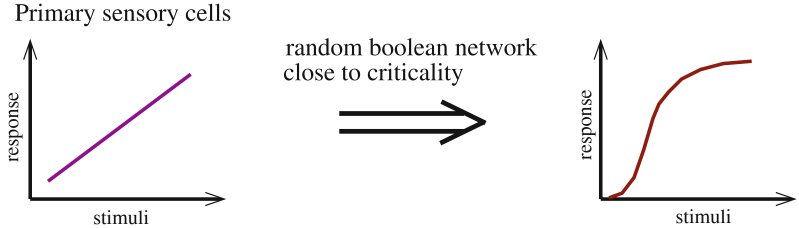

Fig. 7.15 The primary response of sensory receptors can be enhanced by many orders of magnitude using the non-linear amplification properties of a random neural network close to criticality

Exercises

(7.1) $K = 1$ KAUFFMAN NET
Analyze selected $N = 3$ Kauffman nets connected via a $K = 1$ cyclic linkage tree, $\sigma_1 = f_1(\sigma_2)$, $\sigma_2 = f_2(\sigma_3)$, $\sigma_3 = f_3(\sigma_1)$. Consider

- $f_1 = f_2 = f_3 =$ identity,
- $f_1 = f_2 = f_3 =$ negation, and
- $f_1 = f_2 =$ negation, $f3 =$ identity.

Construct all cycles and their attraction basins.

(7.2) $N = 4$ KAUFFMAN NET
Consider the $N = 4$ graph illustrated in Fig. 7.1. Assume all coupling functions to be generalized XOR-functions (1/0 if the number of input-1's is odd/even). Find all cycles.

(7.3) SYNCHRONOUS VS. ASYNCHRONOUS UPDATING
Consider the dynamics of the three-site network illustrated in Fig. 7.3 under sequential asynchronous updating. At every time step first update σ_1 then σ_2 and then σ_3. Determine the full network dynamics, find all cycles and fixpoints and compare with the results for synchronous updating shown in Fig. 7.3.

(7.4) LOOPS AND ATTRACTORS
Consider, as in Sect. 7.4.2, a $K = 1$ network with two linkage loops,

$$A \xrightarrow{I} B \xrightarrow{\neg} C \xrightarrow{I} A, \qquad D \xrightarrow{\neg} E \xrightarrow{\neg} D,$$

with I denoting the identity coupling and $\neg$ the negation, compare Sect. 7.2.2. Find all attractors by considering first the dynamics of the individual linkage loops. Is there any state in phase space which is not part of any cycle?

(7.5) Relevant Nodes and Dynamic Core
How many constant nodes does the network shown in Fig. 7.3 have? Replace then the AND function with XOR and calculated the complete dynamics. How many relevant nodes are there now?

(7.6) The Huepe and Aldana Network
Solve the boolean neural network with uniform coupling functions and noise,

$$\sigma_i(t+1) = \begin{cases} \text{sign}\left(\sum_{j=1}^{K} \sigma_{i_j}(t)\right) & \text{with probability} \quad 1-\eta \\ -\text{sign}\left(\sum_{j=1}^{K} \sigma_{i_j}(t)\right) & \text{with probability} \quad \eta \end{cases}$$

via mean-field theory, where $\sigma_i = \pm 1$, by considering the order parameter

$$\Psi = \lim_{T\to\infty} \frac{1}{T} \int_0^T |s(t)|\, dt, \qquad s(t) = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \sigma_i(t)\,.$$

See Huepe and Aldana-González (2002) and additional hints in the solutions section.

(7.7) Lower Bound for Bond Percolation
When covering the individual bonds of a d-dimensional hypercubic lattice with a probability p there is giant connected component above a certain critical $p_c(d)$ and none below. Define with

$$\lambda(d) = \lim_{n\to\infty} \left\{ \left[\sigma_n(d)\right]^{1/n} \right\} \tag{7.26}$$

the connectivity constant $\lambda(d)$ of the original lattice, where $\sigma(n)$ is the number of paths of length n starting from a given site. Find an upper bound for $\sigma_n(d)$ and hence for the connectivity constant. For percolation to happen the bond probability p must be at least as large as $1/\lambda(d)$. Why? Use this argument to find a lower bound for $p_c(d)$.

Further Reading

The interested reader may want to take a look at Kauffman's seminal work on random boolean networks, Kauffman (1969), or to study his well-known book, Kauffman (1993). For reviews on boolean networks please consult Aldana-Gonzalez et al. (2003) and Schwab et al. (2020). See also Hopfensitz et al. (2013), for a tutorial on attractors in boolean nets, and Wang et al. (2012), for boolean network modeling of biological systems.

Original studies of potential interest include a numerical investigation of Kauffman nets, Bastolla and Parisi (1998), together with an investigation of the scaling of the number of attractors with size, Samuelsson and Troein (2003). The modeling of the yeast reproduction cycle by boolean networks is given in Li et al. (2004). For the concept of observational scale invariance, and of nonlinear signal amplification close to criticality, see respectively Marković and Gros (2013) and Kinouchi and Copelli (2006).

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8 Darwinian Evolution, Hypercycles and Game Theory

Adaptation and evolution are quasi synonymous in popular language, it is hence little surprising that the theory of Darwinian evolution is infused with complex systems concepts. We will discuss when a species will manage to successfully mutate while escaping its local fitness optimum, or when it may spiral towards extinction in the wake of an “error catastrophe”. Venturing into the mysteries surrounding the origin of life, we will investigate the possible advent of a quasispecies in terms of mutually supporting hypercycles. The basic theory of evolution is furthermore closely related to game theory, the mathematical theory of socially interacting agents, viz of rationally acting economic persons. The tragedy of the commons occurring in this context describes the over-exploitation of resources.

We will learn in this chapter that complex systems come with respective individual characteristics. For Darwinian evolution the core notions are fitness, selection and mutation. On this basis, generic concepts from dynamical systems theory unfold, like the phenomenon of stochastic escape. Furthermore we will show that evolutionary processes affect not only the welfare of an individual species, but the pattern of species abundances of entire ecosystems, which will be discussed in this context the neutral theory of macroecology.

8.1 Introduction

Population Genetics The ecosystem of the earth is a complex and adaptive system. It formed via Darwinian evolution through species differentiation and adaptation to a changing environment. Reproduction success is based on a set of inheritable traits, the genome, that is passed from parent to offspring, with the interplay between random mutations and natural selection playing a central role. This process

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_8

is denoted “microevolution”.[1] Gene frequencies shifts may be induced also by stochastic fluctuations, like “genetic drift”.

POPULATION GENETICS Population genetics involves the evolution of the frequency of distinct gene variants, as induced primarily by microevolution and genetic drift.

Our focus will be on microevolution, with some excursions into noise-induced processes.

Basic Terminology Let us introduce some basic variables and concepts.

– POPULATION
  Mostly we will assume that the number of individuals M does not change with time, which models the competition for a limited supply of resources.
– GENOME
  A genome of size N encodes the inheritable traits by a set of N binary variables,

$$\mathbf{s} = (s_1, s_2, \ldots, s_N), \qquad s_i = \pm 1\,.$$

  Here, N is considered fixed.
– GENERATIONS
  For simplicity, time sequences of non-overlapping generations will be assumed, like in a wheat field. The population present at time t is replaced by their offspring at generation $t + 1$.
– ASEXUAL REPRODUCTION
  Asexual reproduction is present when individuals have just one parent, our basic assumption.

We will treat predominantly models for asexual reproduction, though most concepts can be easily generalized to the case of sexual reproduction. In Table 8.1 some

Table 8.1 Genome size N and the spontaneous mutation rates μ, compare (8.2), per base for two RNA-based bacteria and DNA-based eukaryotes. From Jain and Krug (2006) and Drake et al. (1998).

Organism	Genome size	Rate per base	Rate per genome
Bacteriophage $Q\beta$	4.5×10^3	1.4×10^{-3}	6.5
Bacteriophage λ	4.9×10^4	7.7×10^{-8}	0.0038
E. Coli	4.6×10^6	5.4×10^{-10}	0.0025
C. Elegans	8.0×10^7	2.3×10^{-10}	0.018
Mouse	2.7×10^9	1.8×10^{-10}	0.49
Human	3.2×10^9	5.0×10^{-11}	0.16

[1] Evolutionary processes transcending a single species, like the radiation of lineages, would be the domain of “macroevolution”.

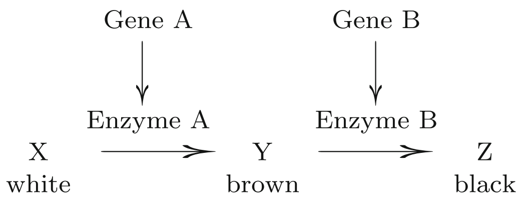

Fig. 8.1 A simple form of epistatic interaction occurs when the influence of one gene builds on the outcome of another. In this fictitious example black hair can only be realized when the gene for brown hair is also present

typical values for the size N of genomes are listed. Note the three orders of magnitude between simple eukaryotic life forms and the human genome.

State of the Population The state of the population at time t can be described by specifying the genomes of all the individuals,

$$\{\mathbf{s}^{\alpha}(t)\}, \qquad \alpha = 1 \ldots M, \qquad \mathbf{s} = (s_1, \ldots, s_N)\,,$$

which leads with

$$X_{\mathbf{s}}(t), \qquad \sum_{\mathbf{s}} X_{\mathbf{s}}(t) = M\,, \tag{8.1}$$

to the definition of the number $X_{\mathbf{s}}$ of individuals with genome $\mathbf{s}$ for each of the 2^N points $\mathbf{s}$ in genome space. Typically, most of these occupation numbers vanish, biological populations are extremely sparse in genome space.

Combinatorial Genetics of Alleles Classical genetics focuses on the presence/absence of a few characteristic traits. These traits are determined by specific sites in the genome, the "loci". The genetic realizations of these specific loci are "alleles". Popular examples are alleles for blue, brown, and green eyes.

Combinatorial genetics in the form of "Mendelian inheritance" deals with the evolution of the frequency of appearance of a given allele, as resulting from environmental changes during the adjustment process. Most visible evolutionary changes are due to a remixing of alleles, as mutation induced changes in the genome are rare. For a comparison, we list typical mutation rates in Table 8.1.

Beanbag Genetics Without Epistatic Interactions "Epistasis" is present when a specific allele affects the expression of other alleles, or genes in general, as illustrated in Fig. 8.1. Classical genetics neglects epistatic interactions. The resulting picture is often called "beanbag genetics", as if the genome was nothing but a bag with a selection of different alleles.

Genotype and Phenotype The physical appearance of an organism is not exclusively determined by gene expression. One distinguishes between genotype and phenotype.

- GENOTYPE
  The genotype of an organism is the class to which that organism belongs as determined by the DNA that was passed to the organism by its parents at conception.
- PHENOTYPE
  The phenotype of an organism is the class to which that organism belongs as determined by the physical and behavioral characteristics of the organism, for example its size and shape, its metabolic activities and its pattern of movement.

Strictly speaking, selection acts upon phenotypes, but only the genotype is bequeathed. Variations in phenotypes act therefore as a source of noise for the selection process. Pheno- and genotypes are set to be identical in the following.

Speciation "Speciation" is the process leading to the differentiation of an initial species into ore or more distinct species. Speciation occurs most often due to adaptation to different ecological niches, in particular in distinct geographical environments. We will not treat the various theories proposed for speciation here.

8.2 Mutations and Fitness in a Static Environment

Constant Environment On short time scales, the environment can be though to be static, our standard setting, which is assumed to include predicable variations, like the day-night cycle. This presumption breaks down for long time scales, since the evolutionary change of one species might lead to repercussions all over the ecosystem to which it appertains.

Independent Individuals An important issue in the theory of evolution is the emergence of specific kinds of social behavior, which arises when individuals of the same population interact. Several phenomena arising in a social context will be discussed within a game theoretical framework in Sect. 8.7. Until then, we assume non-interacting individuals, which implies that the fitness of a given genetic trait is independent of the frequency of this and other alleles, apart from the overall competition for resources.

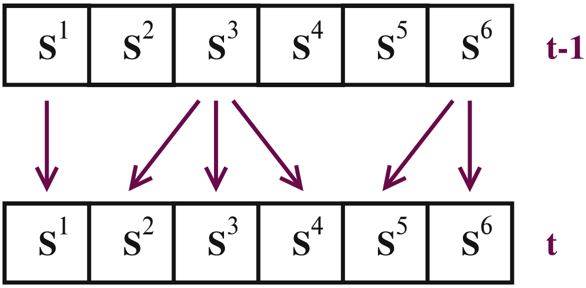

Fig. 8.2 Illustration of a basic reproduction process proceeding from generation $t-1$ to t, with individuals $1,\ 3,\ 6$ having $1,\ 3$ and 2 descendants respectively

Constant Mutation Rates We furthermore assume that the mutation rates are

- Constant over time,
- Independent of the locus in the genome, and
- Not subject to genetic control.

Alternative frameworks would require refined modeling, which is beyond our scope.

Stochastic Evolution The evolutionary process can be modeled as a three-stage stochastic process.

- Reproduction
  The individual α at generation t is the offspring of an individual α' living at generation $t-1$. Reproduction is thus represented as a stochastic map

$$\alpha \quad \longrightarrow \quad \alpha' = G_t(\alpha) \,,$$

  where $G_t(\alpha)$ is the parent of the individual α, which is chosen at random among the M individuals living at generation $t-1$. For the reproduction process illustrated in Fig. 8.2, one has $G_t(1) = 1$, $G_t(2) = 3$, and so on.
- Mutation
  The genomes of the offspring differ from the respective genomes of their parents through random changes.
- Selection
  The number of surviving offspring of an individual depends on its genome via the "fitness" associated with the genome, being directly proportional to the fitness.

Most mutations are neutral in real life, which leads to a "neutral regime" when selection pressure is low, as further discussed in Sect. 8.4.

Point Mutations and Mutation Rate The basic theory is based on independent point mutations, namely that every element of the genome is modified independently of the other elements,

$$s_i^{\alpha}(t) = -s_i^{G_t(\alpha)}(t-1) \qquad \text{with probability } \mu\,, \tag{8.2}$$

where the parameter $\mu \in [0,1/2]$ is the microscopic "mutation rate".[2] In real organisms, more complex phenomena take place, like global rearrangements of the genome, copies of some part of the genome, displacements of blocks of elements from one location to another, and so on. The values for the real-world mutation rates μ for various species listed in Table 8.1 are therefore to be considered as effective mutation rates.

Fitness and Fitness Landscape The fitness $W(\mathbf{s})$, also called "Wrightian fitness", of a genotype trait $\mathbf{s}$ is proportional to the average number of offspring an individual possessing the trait $\mathbf{s}$ has. It is strictly positive and can therefore be written as

$$W(\mathbf{s}) = \mathrm{e}^{kF(\mathbf{s})} \quad \propto \quad \text{average number of offspring of } \mathbf{s}\,. \tag{8.3}$$

Selection acts in first place upon phenotypes, but we neglect here the difference, considering the variations in phenotypes as a source of noise, as discussed above. Several key parameters and functional dependencies enter (8.3).

$W(\mathbf{s})$	:	Wrightian fitness
$F(\mathbf{s})$	:	Fitness landscape
k	:	Inverse selection temperature
$w(\mathbf{s})$	:	Malthusian fitness

The inverse selection temperature takes it name from physics slang.[3] For the Malthusian fitness, one rewrites (8.3) as $W(\mathbf{s}) = \mathrm{e}^{w(\mathbf{s})\Delta t}$, where Δt is the generation time. We will work here with discrete time, viz with non-overlapping generations, making use only of the Wrightian fitness $W(\mathbf{s})$.

Distribution of Fitness Effects In general, the "distribution of fitness effects" describes if new mutations are advantageous, neutral or deleterious, and to which proportions. For modeling frameworks based on concrete mutation processes, our approach here, there is no need to treat the distribution of fitness effects separately.

[2] Point mutations can be thought to correspond to "single-nucleotide mutations" (SNM).

[3] The probability to find a state with energy E in a thermodynamic system with temperature T is proportional to the Boltzmann factor $\exp(-\beta E)$. The inverse temperature is $\beta = 1/(k_B T)$, with k_B being the Boltzmann constant.

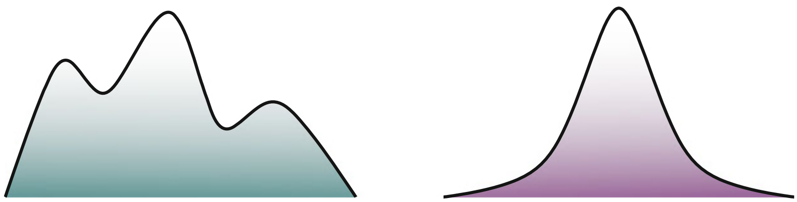

Fig. 8.3 Illustration of idealized (smooth) one-dimensional model fitness landscapes $F(\mathbf{s})$. In contrast, real-world fitness landscapes are likely to contain discontinuities. **Left:** A fitness landscape with peaks and valleys, metaphorically denoted "rugged landscape". **Right:** A fitness landscape containing a single smooth peak, as described by (8.19)

We remark that fitness is a concept defined at the level of individuals in a homogeneous population. The resulting fitness of a species, as an averaged quantity, needs to be explicitly evaluated and is model dependent.

Fitness Ratios The assumption of a constant population size makes the reproductive success a relative notion. Only the ratios

$$\frac{W(\mathbf{s}_1)}{W(\mathbf{s}_2)} = \frac{\mathrm{e}^{kF(\mathbf{s}_1)}}{\mathrm{e}^{kF(\mathbf{s}_2)}} = \mathrm{e}^{k[F(\mathbf{s}_1)-F(\mathbf{s}_2)]} \tag{8.4}$$

are important. It follows that the quantity $W(\mathbf{s})$ is defined up to a proportionality constant and, accordingly, the fitness landscape $F(\mathbf{s})$ only up to an additive constant, much like the energy in physics.

Fitness Landscape A graphical representation of the fitness function $F(\mathbf{s})$ is not truely possible for real-world fitness functions, due to the high dimensionality of the genome space, which contains 2^N states. It is nevertheless customary to draw fitness landscapes, like the ones shown in Fig. 8.3. However, one must bear in mind that these illustrations are not to be taken at face value, apart for model considerations.

Fundamental Theorem of Natural Selection The fundamental theorem of natural selection, first stated by Fisher in 1930, deals with adaptation in the absence of mutations when the population size is large, close to the thermodynamic limit $M \to \infty$, which allows to neglect fluctuations.

The theorem states that the average fitness of the population cannot decrease in time under these circumstances, and that the average fitness becomes stationary only when all individuals in the population have the maximal reproductive fitness.

The proof is straightforward. We define with

$$\langle W \rangle_t \;\equiv\; \frac{1}{M}\sum_{\alpha=1}^{M} W(\mathbf{s}^{\alpha}(t)) = \frac{1}{M}\sum_{\mathbf{s}} W(\mathbf{s})X_{\mathbf{s}}(t)\,, \tag{8.5}$$

the average fitness of the population, with $X_{\mathbf{s}}$ being the number of individual having the genome $\mathbf{s}$. Note that the $\sum_{\mathbf{s}}$ in (8.5) contains 2^N terms. The evolution equations are given in the absence of mutations by

$$X_{\mathbf{s}}(t+1) = \frac{W(\mathbf{s})}{\langle W\rangle_t} X_{\mathbf{s}}(t)\,, \tag{8.6}$$

where $W(\mathbf{s})/\langle W\rangle_t$ is the relative reproductive success. The overall population size remains constant,

$$\sum_{\mathbf{s}} X_{\mathbf{s}}(t+1) = \frac{1}{\langle W\rangle_t} \sum_{\mathbf{s}} X_{\mathbf{s}}(t) W(\mathbf{s}) = M\,,$$

where we have used (8.5) for $\langle W\rangle_t$. Then,

$$\begin{aligned}
\langle W\rangle_{t+1} &= \frac{1}{M}\sum_{\mathbf{s}} W(\mathbf{s}) X_{\mathbf{s}}(t+1) = \frac{\frac{1}{M}\sum_{\mathbf{s}} W^2(\mathbf{s}) X_{\mathbf{s}}(t)}{\frac{1}{M}\sum_{\mathbf{s}'} W(\mathbf{s}') X_{\mathbf{s}'}(t)} \\
&= \frac{\langle W^2\rangle_t}{\langle W\rangle_t} \;\ge\; \langle W\rangle_t\,,
\end{aligned}$$

since $\langle W^2\rangle_t - \langle W\rangle_t^2 = \langle \Delta W^2\rangle_t \ge 0$. The steady state,

$$\langle W\rangle_{t+1} = \langle W\rangle_t, \qquad \langle W^2\rangle_t = \langle W\rangle_t^2\,,$$

is only realizable when all individuals $1\ldots M$ in the population have the same fitness. Modulo degeneracies, this implies identical genotypes.

Baldwin Effect Variations in the phenotype may be induced not only via stochastic influences of the environment, but also through adaption of the phenotype itself to the environment, viz through cognitive learning. Learning can speed up evolution whenever the underlying fitness landscape is very rugged, by smoothing it out and providing a stable gradient towards the genotype with the maximal fitness. One speaks of the “Baldwin effect”.

8.3 Deterministic Evolution

Mutations are random events, which implies that evolution is inherently a stochastic process. This does not hold in the limit of infinite population size $M \to \infty$, for which stochastic fluctuations average out, becoming irrelevant. In this limit the equations governing evolution are deterministic and governed only by average transition rates. This allows to study in detail the condition necessary for adaptation to occur for various mutation rates.

8.3.1 Evolution Equations

Mutation Matrix The mutation matrix

$$Q_\mu(\mathbf{s}' \to \mathbf{s}), \qquad \sum_{\mathbf{s}} Q_\mu(\mathbf{s}' \to \mathbf{s}) = 1 \tag{8.7}$$

denotes the probabilities of obtaining a genotype $\mathbf{s}$ when attempting to reproduce an individual with genotype $\mathbf{s}'$. The mutation rates $Q_\mu(\mathbf{s}' \to \mathbf{s})$ may depend on a parameter μ determining the overall mutation rate. The mutation matrix includes the absence of any mutation, viz the transition $Q_\mu(\mathbf{s}' \to \mathbf{s}')$. It is normalized.

Deterministic Evolution with Mutations We generalize (8.6), which is valid in the absence of mutations, by including the effect of mutations via the mutation matrix $Q_\mu(\mathbf{s}' \to \mathbf{s})$,

$$\frac{X_{\mathbf{s}}(t+1)}{M} = \left(\sum_{\mathbf{s}'} X_{\mathbf{s}'}(t) W(\mathbf{s}') Q_\mu(\mathbf{s}' \to \mathbf{s})\right) \Big/ \left(\sum_{\mathbf{s}'} W_{\mathbf{s}'} X_{\mathbf{s}'}(t)\right),$$

or

$$x_{\mathbf{s}}(t+1) = \frac{\sum_{\mathbf{s}'} x_{\mathbf{s}'}(t) W(\mathbf{s}') Q_\mu(\mathbf{s}' \to \mathbf{s})}{\langle W \rangle_t}, \qquad \langle W \rangle_t = \sum_{\mathbf{s}'} W_{\mathbf{s}'} x_{\mathbf{s}'}(t), \tag{8.8}$$

where we have introduced the normalized population variables

$$x_{\mathbf{s}}(t) = \frac{X_{\mathbf{s}}(t)}{M}, \qquad \sum_{\mathbf{s}} x_{\mathbf{s}}(t) = 1.$$

The evolution dynamics (8.8) retains the overall size $\sum_{\mathbf{s}} X_{\mathbf{s}}(t)$ of the population, due to the normalization of the mutation matrix $Q_\mu(\mathbf{s}' \to \mathbf{s})$, via (8.7).

Hamming Distance The Hamming distance

$$d_{\mathrm{H}}(\mathbf{s}, \mathbf{s}') = \sum_{i=1}^{N} \frac{(s_i - s_i')^2}{4} = \frac{N}{2} - \frac{1}{2} \sum_{i=1}^{N} s_i s_i' \tag{8.9}$$

measures the number of units that are different in two genome configurations $\mathbf{s}$ and $\mathbf{s}'$, e.g. before and after the effect of a mutation event.

Mutation Matrix for Point Mutations We examine the effects of the simplest mutation process, namely the case that a genome of fixed length N is affected by random transcription errors afflicting only individual loci. For this case, namely point mutations, the overall mutation probability

$$Q_\mu(\mathbf{s}' \to \mathbf{s}) = \mu^{d_\mathrm{H}}(1-\mu)^{N-d_\mathrm{H}} \tag{8.10}$$

is the product of the independent mutation probabilities for all loci $i = 1, \ldots, N$, with d_H denoting the Hamming distance $d_\mathrm{H}(\mathbf{s}, \mathbf{s}')$ given by (8.9) and μ the mutation rate μ defined in (8.2). One has

$$\sum_{\mathbf{s}} Q_\mu(\mathbf{s}' \to \mathbf{s}) = \sum_{d_H} \binom{N}{d_H} (1-\mu)^{N-d_H} \mu^{d_H} = (1-\mu+\mu)^N \equiv 1\,,$$

the mutation matrix defined by (8.10) is consequently normalized. Discarding a normalization factor, we rewrite the mutation matrix as

$$\begin{aligned} Q_\mu(\mathbf{s}' \to \mathbf{s}) &\propto \exp\Big(\big[\log(\mu) - \log(1-\mu)\big] d_H\Big) \\ &\propto \exp\left(\beta \sum_i s_i s_i'\right)\,, \end{aligned} \tag{8.11}$$

where we denoted by β an effective inverse temperature, defined by

$$\beta = \frac{1}{2}\log\left(\frac{1-\mu}{\mu}\right).$$

The relation of the evolution equation (8.11) to the partition function of a thermodynamical system, hinted at by the terminology "inverse temperature" will become evident below.

Evolution Equations for Point Mutations With the help of the exponential representations for both the fitness, $W(\mathbf{s}) = \exp[kF(\mathbf{s})]$, see (8.3,) and for the mutation matrix $Q_\mu(\mathbf{s}' \to \mathbf{s})$, we may express the evolution equation (8.8) as

$$x_\mathbf{s}(t+1) = \frac{1}{N_t} \sum_{\mathbf{s}'} x_{\mathbf{s}'}(t) \exp\left(\beta \sum_i s_i s_i' + kF(\mathbf{s}')\right) \tag{8.12}$$

in a form that is suggestive of a statistical mechanics analogy. The normalization N_t in (8.12) takes care of both the average fitness $\langle W\rangle_t$, as appearing in (8.8), and of the normalization of the mutation matrix, compare (8.11).

Evolution Equations in Linear Form The evolution equation (8.12) is non-linear in the dynamical variables $x_\mathbf{s}(t)$, which determine the average fitness $\langle W\rangle_t$ entering the normalization N_t. A suitable change of variables allows however to cast the evolution equation into a linear form. For this purpose we introduce non-normalized variables $y_{\mathbf{s}}(t)$ implicitly via

$$x_{\mathbf{s}}(t) = \frac{y_{\mathbf{s}}(t)}{\sum_{\mathbf{s}'} y_{\mathbf{s}'}(t)} . \tag{8.13}$$

The idea is that the normalization $\sum_{\mathbf{s}'} y_{\mathbf{s}'}(t)$ can be selected freely for every generation $t = 1, 2, 3, \ldots$. The evolution (8.12) becomes

$$y_{\mathbf{s}}(t+1) = Z_t \sum_{\mathbf{s}'} y_{\mathbf{s}'}(t) \exp\left(\beta \sum_i s_i s_i' + kF(\mathbf{s}')\right) , \tag{8.14}$$

where

$$Z_t = \frac{1}{N_t} \frac{\sum_{\mathbf{s}'} y_{\mathbf{s}'}(t+1)}{\sum_{\mathbf{s}'} y_{\mathbf{s}'}(t)} .$$

Choosing a different normalization for $y_{\mathbf{s}}(t)$ and for $y_{\mathbf{s}}(t+1)$ we may achieve $Z_t \equiv 1$. Equation (8.14) is then linear in $y_{\mathbf{s}}(t)$.

Evolution Hamiltonian In the following we will make use of analogies to notations commonly used in statistical mechanics. Readers unfamiliar with the mathematics of the one-dimensional Ising model may skip the mathematical details and concentrate on the interpretation of the results.

The form

$$y_{\mathbf{s}}(t+1) = \sum_{\mathbf{s}'} \mathrm{e}^{\beta H[\mathbf{s},\mathbf{s}']} \, y_{\mathbf{s}'}(t)$$

linear evolution equation (8.14) involves an effective "Hamiltonian"[4]

$$\beta H[\mathbf{s}, \mathbf{s}'] = \beta \sum_i s_i s_i' + kF(\mathbf{s}') , \tag{8.15}$$

which is a function of the binary variables $\mathbf{s}$ and $\mathbf{s}'$. Given that $\mathbf{s}$ and $\mathbf{s}'$ run over the genome space of subsequent generations, we may rename variables, $\mathbf{s} \to \mathbf{s}(t+1)$ and $\mathbf{s}' \to \mathbf{s}(t)$, which leads to the notation

$$y_{\mathbf{s}(t+1)} = \sum_{\mathbf{s}(t)} \mathrm{e}^{\beta H[\mathbf{s}(t+1),\mathbf{s}(t)]} \, y_{\mathbf{s}(t)} . \tag{8.16}$$

[4] In classical thermodynamics, the Hamiltonian $H(\alpha)$, or energy function, determines the probability that a given state α is populated. This probability is proportional to the Boltzmann factor $\exp(-\beta H(\alpha))$, where $\beta = 1/(k_B t)$ is the inverse temperature.

Statistical Mechanics of the Ising Model Evolution progresses generation per generation, which is equivalent to an iterative solution of (8.16),

$$y_{\mathbf{s}(t+1)} = \sum_{\mathbf{s}(t),\ldots,\mathbf{s}(0)} e^{\beta H[\mathbf{s}(t+1),\mathbf{s}(t)]} \,\ldots\, e^{\beta H[\mathbf{s}(1),\mathbf{s}(0)]}\, y_{\mathbf{s}(0)}\,, \tag{8.17}$$

which defines in turn a two-dimensional Ising-type Hamiltonian[5]

$$\beta H = \beta \sum_{i,t} s_i(t+1)s_i(t) + k \sum_t F(\mathbf{s}(t))\,. \tag{8.18}$$

The two dimensions are genome space, $i = 1, \ldots, N$, and time t, viz the sequence of subsequent generations. This expression will be used in the next sections.

8.3.2 Beanbag Genetics: Evolution Without Epistasis

Fujiyama Landscape The fitness function

$$F(\mathbf{s}) = \sum_{i=1}^{N} h_i s_i, \qquad W(\mathbf{s}) = \prod_{i=1}^{N} \mathrm{e}^{k\, h_i s_i}\,, \tag{8.19}$$

is denoted the "Fujiyama landscape", since it corresponds to a single smooth peak as illustrated in Fig. 8.3. To see why, we consider the case $h_i > 0$ and rewrite (8.19) as

$$F(\mathbf{s}) = \mathbf{s}_0 \cdot \mathbf{s}, \qquad \mathbf{s}_0 = (h_1, h_2, \ldots, h_N)\,.$$

The fitness of a given genome $\mathbf{s}$ is directly proportional to the scalar product with the master sequence $\mathbf{s}_0$, with a well defined gradient pointing towards the optical genom.

The Fujiyama Hamiltonian Epistatic interactions are absent in the smooth peak landscape (8.19). In terms of the corresponding Hamiltonian, see (8.18), this fact expresses itself as

$$\beta H = \beta \sum_{i=1}^{N} H_i, \qquad H_i = \sum_t s_i(t+1)s_i(t) + \frac{kh_i}{\beta} \sum_t s_i(t)\,. \tag{8.20}$$

[5] Any system of binary variables is equivalent to a system of interacting Ising spins, which retains only the classical contribution to the energy of interacting quantum mechanical spins (the magnetic moments).

Every locus i is independent, corresponding to the one-dimensional Ising model βH_i in an effective uniform magnetic field kh_i/β. Loci are hence propagated independently in generation time, $t = 1, 2, \ldots$.

Transfer Matrix The Hamiltonian (8.20) does not contain interactions between different loci of the genome. It is enough, as a consequence, to examine the evolution of a single gene locus i, together with the associated Hamiltonian H_i. We define with

$$\mathbf{y} = \big(y_+(t), y_-(t)\big), \qquad \mathbf{y}^T = \begin{pmatrix} y_+(t) \\ y_-(t) \end{pmatrix}$$

the non-normalized densities of individuals having gene states $s_i = \pm 1$, for a specific locus i. The iterative solution (8.17) takes the form

$$\mathbf{y}(t+1) = \left(\prod_{t'=0}^{t} T\right) \mathbf{y}(0) = T^{t+1}\, \mathbf{y}^T(0)\,, \tag{8.21}$$

where we defined the 2×2 "transfer matrix"

$$T = \mathrm{e}^{\beta H_i[s_i(t+1), s_i(t)]}, \qquad T = \begin{pmatrix} \mathrm{e}^{\beta + kh_i} & e^{-\beta} \\ \mathrm{e}^{-\beta} & e^{\beta - kh_i} \end{pmatrix}. \tag{8.22}$$

For the last expression $s, s' = \pm 1$ was used, together with the symmetrized form

$$\beta H_i = \beta \sum_t s_i(t+1) s_i(t) + \frac{kh_i}{2} \sum_t \Big[s_i(t+1) + s_i(t) \Big]$$

of the one-dimensional Ising model. The matrix T transfers the state of the system from one generation to the next, from time t to $t + 1$, hence the naming.

Eigenvalues of the Transfer Matrix For simplicity we take

$$h_i \equiv 1, \qquad \mathbf{s}_0 = (1, 1, \ldots, 1)\,,$$

and evaluate the eigenvalues ω of the transfer matrix,

$$\omega^2 - 2\omega\, \mathrm{e}^{\beta} \cosh(k) + \mathrm{e}^{2\beta} - \mathrm{e}^{-2\beta} = 0\,.$$

The solutions are

$$\omega_{1,2} = \mathrm{e}^{\beta} \cosh(k) \pm \sqrt{\mathrm{e}^{2\beta} \cosh^2(k) - \mathrm{e}^{2\beta} + \mathrm{e}^{-2\beta}}\,.$$

In the absence of a fitness landscape, namely when $k = 0$, the eigenvalues are $e^{\beta} \pm e^{-\beta}$. Adaption does not occur, with the eigenvectors $\mathbf{y} = (1, \pm 1)$ of the transfer matrix attributing equal weight to all states in genome space. In the following we consider with $k > 0$ non-trivial fitness landscapes.

Dominance of the Largest Eigenvalue The genome distribution $\mathbf{y}$ is determined by large powers of the transfer matrix, $\lim_{t\to\infty} T^t$, as entering (8.21). The difference between the two eigenvalues of the transfer matrix is

$$\Delta\omega = \omega_1 - \omega_2 = 2\sqrt{e^{2\beta}\sinh^2(k) + e^{-2\beta}}\,, \tag{8.23}$$

which vanishes only if $k = 0$ and $\beta \to \infty$. For $\Delta\omega > 0$ the larger eigenvalue ω_1 dominates in the limit $t \to \infty$ and $\mathbf{y} = (y_+, y_-)$ is given the eigenstate corresponding to ω_1, if $y_+ > y_-$.

Adaption in the Absence of Epistatic Interactions The sequence $\mathbf{s}_0 = (1, \ldots, 1)$ is the state with maximal fitness. The probability to find individuals with genome $\mathbf{s}$ at a Manhatten distance d_H from $\mathbf{s}_0$ is proportional to

$$(y_+)^{N-d_H}\,(y_-)^{d_H} \;\propto\; \left(\frac{y_-}{y_+}\right)^{d_H} .$$

The entire population is hence within a finite distance of the optimal genome sequence $\mathbf{s}_0$ whenever $y_-/y_+ < 1$, viz for $\Delta\omega > 0$. We recall that

$$\beta = \frac{1}{2}\log\left(\frac{1-\mu}{\mu}\right), \qquad W(\mathbf{s}) = e^{kF(\mathbf{s})}\,,$$

where μ is the mutation rate for point mutations. Thus we see that there is some degree of adaptation whenever the fitness landscape does not vanish ($k > 0$). This is the case even for a maximal mutation rate $\mu \to 1/2$, which corresponds to $\beta \to 0$.

8.3.3 Epistatic Interactions and the Error Catastrophe

The result of the previous Sect. 8.3.2, i.e. the occurrence of adaptation in a smooth fitness landscape for any non-trivial model parameter, is due to the absence of epistatic interactions in the smooth fitness landscape. Epistatic interactions introduce a phase transition to a non-adapting regime once the mutation rate becomes too high.

Sharp Peak Landscape One possibility to study this phenomenon is the limiting case of strong epistatic interactions. In this limit, individual elements of the genotype do not provide information on the value of the fitness, as per se, which is defined by

$$W(\mathbf{s}) = \begin{cases} 1 & \text{if } \mathbf{s} = \mathbf{s}_0 \\ 1-\sigma & \text{otherwise} \end{cases}, \tag{8.24}$$

with $\sigma > 0$. In this case, all genome sequences but one have the identical fitness, which is lower than the fitness of the master sequence $\mathbf{s}_0$. The corresponding landscape $F(\mathbf{s})$, defined by $W(\mathbf{s}) = \mathrm{e}^{kF(\mathbf{s})}$, is equally discontinuous, with (8.24) corresponding to a fitness landscape with a "tower". This landscape is devoid of gradients pointing towards the master sequence of maximal fitness.

At first sight, the presence of epistatic interactions in (8.24) may not be evident. For this we recall that the fitness landscape $F(\mathbf{s})$ of beanbag genetics is strictly additive. Epistatic interactions are hence present whenever the fitness $F(\mathbf{s})$ is not strictly multiplicative, which is the case for the sharp-peak model (8.24).

Distance from Optimality We define by x_k the fraction of the population whose genotype has a Hamming distance k from the preferred genotype,

$$x_k(t) = \frac{1}{M} \sum_{\mathbf{s}} \delta_{d_{\mathrm{H}}(\mathbf{s},\mathbf{s}_0),k}\, X_{\mathbf{s}}(t)\,.$$

The evolution equations can be formulated entirely in terms of the x_k, which correspond to the fraction of the population being k point mutations away from the master sequence.

Infinite Genome Limit For the limit $N \to \infty$ we rescale the rate for point mutations, μ, as

$$\mu = u/N\,, \tag{8.25}$$

compare (8.2). In this limit, the mean number of mutations

$$u = N\mu$$

occurring per genome remains finite. Experimentally, this approximation holds well for microbial taxa, as "Drake's rule" states, with $u \approx 0.003$ for base substitutions.

Absence of Back Mutations Starting from the optimal genome $\mathbf{s}_0$ we examine the effect of successive mutations. Successful mutations increase the distance k from the optimal genome $\mathbf{s}_0$. Assuming $u \ll 1$ in (8.25) leads to several consequences.

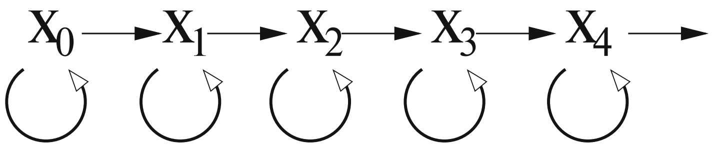

Fig. 8.4 The linear chain model for the tower landscape, as defined by (8.26), (8.27), and (8.28), with k denoting the number of point mutations necessary to reach the optimal genome. The population fraction $x_k(t+1)$ is influenced only by $x_{k-1}(t)$ and $x_k(t)$, its own value at time t

- Per Generation, multiple mutations do not appear.
- One can neglect back mutations, viz mutations reducing the value of k.

The relative probability of back mutations,

$$\frac{k}{N-k} \ll 1,$$

vanishes in the infinite-genome limit $N \to \infty$ for genomes within a finite distance k from the master sequence. As a consequence, the tower model has the structure of a linear chain, with $k = 0$ being the starting point of the chain.

Linear Chain Model We have two parameters; u, which measures the mutation rate, and σ, setting the strength of the selection. Remembering that the fitness $W(\mathbf{s})$ is proportional to the number of offspring, see (8.24), we find

$$x_0(t+1) = \frac{1}{\langle W \rangle}\Big[x_0(t)\,(1-u)\Big], \tag{8.26}$$

$$x_1(t+1) = \frac{1}{\langle W \rangle}\Big[u x_0(t) + (1-u)\,(1-\sigma)\,x_1(t)\Big], \tag{8.27}$$

$$x_k(t+1) = \frac{1}{\langle W \rangle}\Big[u x_{k-1}(t) + (1-u) x_k(t)\Big](1-\sigma), \qquad k > 1, \tag{8.28}$$

where $\langle W \rangle$ is the average fitness. These equations describe a linear chain model, as illustrated in Fig. 8.4. The population of individuals with the optimal genome x_0 is constantly losing members, due to mutations. But it also has a higher number of offspring than all other populations, due to its larger fitness.

Stationary Solution The average fitness of the population is given by

$$\langle W \rangle = x_0 + (1-\sigma)(1-x_0) = 1 - \sigma(1-x_0). \tag{8.29}$$

When looking for the stationary distribution $\{x_k^*\}$, one notes that the equation for x_0^* does not involve x_k^* with $k > 0$,

$$x_0^* = \frac{x_0^*(1-u)}{1-\sigma(1-x_0^*)}, \qquad\qquad 1-\sigma(1-x_0^*) = 1-u\,,$$

which leads to

$$x_0^* = \begin{cases} 1-u/\sigma & \text{if } u < \sigma \\ 0 & \text{if } u \geq \sigma \end{cases}, \tag{8.30}$$

due to the normalization condition $0 \leq x_0^* \leq 1$. For $u > \sigma$ the model becomes ill defined. The stationary solutions for the x_k^* follow, for $k = 1$ one has

$$x_1^* = \frac{u}{1-\sigma(1-x_0^*)-(1-u)(1-\sigma)}\, x_0^*\,,$$

compare (8.27) and (8.29). The stationary probability x_k^* to find genomes at a generic distance from optimality, $k > 1$, is

$$x_k^* = \frac{(1-\sigma)u}{1-\sigma(1-x_0^*)-(1-u)(1-\sigma)}\, x_{k-1}^*\,, \tag{8.31}$$

which follows from (8.28) and (8.29).

Distinct Phases Two regimes can be distinguished, as determined by the magnitude of the mutation rate $\mu = u/N$ relative to the fitness parameter σ, with

$$u = \sigma$$

being the transition point. The density x_0^* of optimal genomes can be taken as an order parameter,[6] being zero for $u \geq \sigma$, and finite otherwise. In physics language, epistasis corresponds to many-body interactions, which can be thought to induce a phase transition in the sharp peak model. In this view, the absence of many-body interactions in the smooth fitness landscape model is the underlying reason for the likewise absence of distinct phases, compare Sect. 8.3.2.

Quasispecies Formation in the Adaptive Regime In the regime of small mutation rates, $u < \sigma$, one has $x_0^* > 0$, in fact the whole population lies within a finite distance from the preferred genotype. To see why, we note that

$$\sigma(1-x_0^*) = \sigma(1-1+u/\sigma) = u$$

[6] Order parameters characterize the different between ordered and disordered phases, as explained in Sect. 6.1 of Chap. 6.

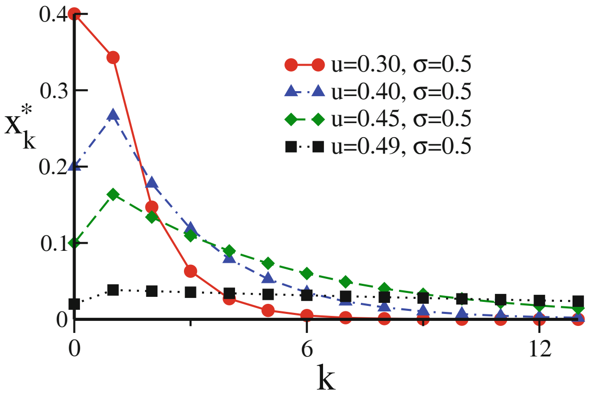

Fig. 8.5 Quasispecies formation within the sharp peak fitness landscape, as defined by (8.24). The stationary population density x_k^*, see (8.31), is peaked around the genome with maximal fitness, located at $k = 0$. The population tends to spread out in genome space when the overall mutation rate u approaches the critical point $u \to \sigma$

and take a look at (8.31),

$$\frac{(1-\sigma)u}{1-u-(1-u)(1-\sigma)} = \left(\frac{1-\sigma}{1-u}\right)\left(\frac{u}{\sigma}\right) \le 1, \qquad \text{for } u < \sigma .$$

The x_k^* therefore form a geometric series,

$$x_k^* \sim \left(\frac{1-\sigma}{1-u}\frac{u}{\sigma}\right)^k ,$$

which is summable whenever $u < \sigma$. In this adaptive regime the population forms what Manfred Eigen denoted a "quasispecies", see Fig. 8.5.

QUASISPECIES A quasispecies is a population of genetically close, but not identical individuals.

The quasispecies concept can be thought to contrast the classical view of genetics, as dealing with a limited number of competing alleles. It shapes current views on how primordial life may have evolved, a topic treated in Sect. 8.5.1. Equally, present-day pathogentic viral popluations are often spread out thinly in genome space, a key factor for overcoming resistences of potential hosts.

Wandering Regime and Error Threshold In the regime of a large mutation rate, $u > \sigma$, we have $x_k^* = 0, \forall k$, which is equivalent to a situation where the population is distributed in an essentially uniform way over genotype space. Fitness will drop to very low levels, given that the whole population lies far away from the preferred genotype. The size of the population will drop equally, until the preconditions of above analysis are no longer valid and a "wandering regime" is reached. In this state, the effects of finite population size are prominent.

ERROR CATASTROPHE The transition from the adaptive (quasispecies) regime to the wandering regime is denoted the "error threshold" or "error catastrophe".

Quasispecies theory is intrinsically linked to the the notion of an error catastrophe, independent of the exact nature of the fitness landscape containing epistatic interactions. A quasispecies will not be able to adapt when mutation rates become too large. Most of the time, an error catastrophe implies extinction in the real world.

Drift Barrier Hypothesis So far, our line of arguments has been based on short-term adaptive pressures, namely on a fitness landscape that is defined in relation to a static environment. In this situation, the more individuals possess the optimal genome, the better. However, environments change on longer time scales, which is bad news for a well adapted species. Too few variations in the extant gene pool severely restricts adaptability to new environmental forcings.

In order to be able to adapt to changing environments a species needs, importantly, only an intensive number of individuals with suitable genomes, not a finite fraction of the population. The overall degree of genetic variability is hence proportional to

$$uM = \frac{M}{N}\,\mu$$

where μ is the mutation rate per base pair, in analogy to (8.2). If genetic variability can be assumed to be similar between species, it follows that optimal mutation rates scale as $1/M$, viz inversely with population size. This is indeed seen in multicellular eukaryotic species. There is however a catch, $u \sim 1/M$ will cross the "drift barrier", as the error threshold is called in this context, when the population size M becomes too small. This is the "drift barrier hypothesis", which can be seen as an instance of Kauffman's notion of "life at the edge of chaos".[7] Both concepts deal with the tradeoff between short- and long-term survivability.

8.4 Finite Populations and Stochastic Escape

Punctuated Equilibrium Evolution is not a steady process, there are regimes of rapid increase of the fitness and phases of relative stasis. This interplay of dynamical behaviors is denoted "punctuated equilibrium". In this context, adaptation can result either from local optimization of the fitness of a single species or via coevolutionary

[7] The notion of life at the edge of chaos features prominently in Chap. 7.

avalanches,[8] which occur in an ecosystems when the evolution of one species induces successive adaptive changes in a series of other species.

NEUTRAL REGIME The stage where evolution is driven essentially by random mutations is called the neutral, wandering, or drifting regime.

The quasispecies model is inconsistent in the neutral regime, in which the population spreads out in genome space. In this situation, fluctuations of the reproductive process in a finite population have to be taken into account.

Deterministic Versus Stochastic Evolution Evolution is driven by stochastic processes, since mutations are random events. Nevertheless, randomness averages out and the evolution process becomes deterministic in the thermodynamic limit, as discussed in Sect. 8.3, when the number M of individuals diverges, $M \to \infty$.

Evolutionary processes in populations with a finite number of individuals differ from deterministic evolution quantitatively and sometimes also qualitatively, the latter being here our focus of interest.

STOCHASTIC ESCAPE Random mutations in a finite population might lead to a decrease in the fitness with the fixation of potentially detrimental traits.

We will discuss in some detail under which circumstances stochastic escape is important for evolutionary processes in small populations.[9]

8.4.1 Adaptive Climbing Under Strong Selective Pressure

Our first inroad into the population dynamics of finite populations is based on three basic assumptions.

- Finite populations.
- Strong selective pressure.
- Small mutation rates.

Strong selective pressure implies that one can represent the population by a single point in genome space, namely that the genomes of all individuals are taken to be equal.

Adaptive Walks The evolutionary dynamics is taken to consist of three basic steps.

[8] Coevolutionary avalanches are likely to lead to a self-organized critical state, as shown in Sect. 6.6 of Chap. 6.

[9] A general account of the theory of stochastic escape is given in Sect. 3.5.2 of Chap. 3.

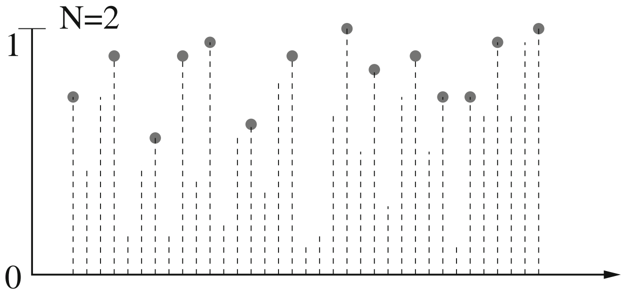

Fig. 8.6 Local fitness optima in a one-dimensional random fitness distribution. The illustration shows that random distributions may exhibit enormously large numbers of local optima (filled circles). For these, all sequences with a hamming distance of one point mutations have lower fitness values

- At each time step, only a single locus of the genome of a certain individual mutates.
- If the mutation leads to a genotype with higher fitness, the new genotype spreads rapidly throughout the entire population, which moves altogether to the new position in genome space.
- If the fitness of the new genotype is lower, the mutation is rejected and the population remains at the old position.

Physicists would call this type of dynamics a Monte Carlo process at zero temperature. Further below we will relax the condition that unfavorable mutations are relaxed.

Random Energy Model As formulated so far, our rules do not allow to transverse a valley with low fitness, viz to pass from one local optimum to the next. As a first step it hence important to investigate the statistical properties of local optima, which depend on the properties of the fitness landscape. A suitable approach is to assume that fitness values are randomly distributed.

> RANDOM ENERGY MODEL The fitness landscape $F(\mathbf{s})$ is uniformly distributed between zero and one.

The random energy model is illustrated in Fig. 8.6. It captures an ingredient expected for real-world fitness landscapes, namely a large number of local fitness optima close to the global fitness maximum.

Local Optima in the Random Energy Model Let us denote by N the number of genome elements. The probability that a sequence with fitness $F(\mathbf{s})$ has a higher fitness than all its N neighbors is given by

$$F^N = F^N(\mathbf{s}),$$

since we have to impose that every one of the N nearest neighbors,

$$(s_1, \ldots, -s_i, \ldots, s_N), \qquad (i = 1, \ldots, N), \qquad \mathbf{s} = (s_1, \ldots, s_N)\,,$$

has a fitness less than F. The probability that a point in genome space is a local optimum is consequently given by

$$P_{\text{local optimum}} = \int_0^1 F^N dF = \frac{1}{N+1}\,,$$

since the fitness F is equally distributed in [0, 1]. There are therefore many local optima, namely $2^N/(N+1)$. A schematic picture of the large number of local optima in a random distribution is given in Fig. 8.6.

Average Fitness of a Local Optimum The typical fitness of a local optimum is

$$\begin{aligned} F_{\text{typ}} &= \frac{1}{1/(N+1)} \int_0^1 F\,F^N dF = \frac{N+1}{N+2} = \frac{1+1/N}{1+2/N} \\ &\approx 1 - \frac{1}{N}\,, \end{aligned} \tag{8.32}$$

viz very close to the global optimum of 1, when the genome length N is large. At every successful step, a mutation followed by selection, on average the distance from the top is divided by a factor of two.

Successful Mutations The typical fitness attained after ℓ successful steps is therefore of the order of

$$1 - \frac{1}{2^{\ell+1}}\,,$$

when starting with $\ell = 0$ from an average initial fitness of 1/2. It follows that the typical number of successful mutations after which an optimum is approximately attained is

$$F_{\text{typ}} = 1 - \frac{1}{N} \approx 1 - \frac{1}{2^{\ell_{\text{typ}}+1}}, \qquad \ell_{\text{typ}} + 1 = \frac{\log N}{\log 2}\,, \tag{8.33}$$

which is logarithmically small. The climbing process is illustrated in Fig. 8.7.

Time Needed for One Successful Mutation Even though the number of successful mutations (8.33) needed to arrive at the local optimum is small, the time needed to climb to the local peak can be very long. In first place, a given mutation must be successful before it can contribute to the climbing process. For the random energy

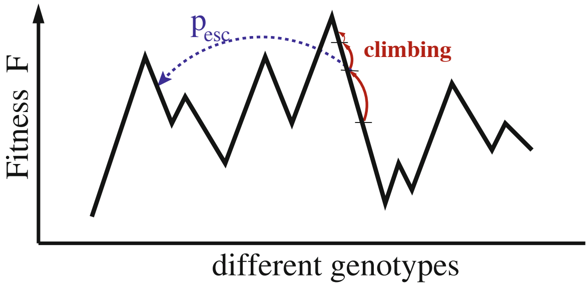

Fig. 8.7 Adaptive climbing vs. stochastic escape. The higher the fitness, the more difficult it becomes to climb further. With the escape probability p_{esc} the population jumps somewhere else, escaping from the local optimum

model this is only rarely the case close to the top, because mutations lead to a randomly distributed fitness of the offspring.

We define with

$$t_F = \sum_n n\, P_n, \qquad n: \text{ number of generations},$$

the average number of generations necessary for a population with fitness F to attain a single successful mutation. We denoted with

$$P_n = (1 - F)F^{n-1}$$

the probability that it takes exactly n generations, which is the case if $n-1$ mutation attempts with lower fitness are followed by one with fitness in $[F, 1]$. The climbing time is evaluated as

$$\begin{aligned} t_F &= \frac{1-F}{F} \sum_{n=0}^{\infty} n\, F^n = \frac{1-F}{F} \left(F \frac{\partial}{\partial F} \sum_{n=0}^{\infty} F^n \right) \\ &= (1-F) \frac{\partial}{\partial F} \frac{1}{1-F} = \frac{1}{1-F}\,. \end{aligned} \tag{8.34}$$

The average number of generations necessary to increase fitness further by a successful mutation diverges close to the global optimum $F \to 1$.

Total Climbing Time Every successful mutation decreases the distance $1 - F$ to the top by 1/2. Vice versa the factor $1/(1 - F)$ increases on the average by two. We define with

$$T_{\text{opt}} = t_F \sum_{\ell=0}^{\ell_{\text{typ}}} 2^{\ell}$$

the expected total number of generations to arrive at a local optimum, where ℓ_{typ} is the typical number of successful mutations needed to arrive at a local optimum, see (8.33). Using (8.34) for the average number of generations necessary for a single successful mutation, t_F, we obtain

$$\begin{aligned} T_{\text{opt}} &= t_F \, \frac{1-2^{\ell_{\text{typ}}+1}}{1-2} \approx t_F \, 2^{\ell_{\text{typ}}+1} = t_F \, e^{(\ell_{\text{typ}}+1)\log 2} \\ &\approx t_F \, e^{\log N} = \frac{N}{1-F} \approx 2N \,, \end{aligned} \tag{8.35}$$

where we used $\ell_{\text{typ}} + 1 = \log N/\log 2$, setting $F \approx 1/2$ in the last step, our choice for a typical starting fitness. When starting randomly, the number of generations needed to climb to a local maximum in the random fitness landscape is hence proportional to the length of the genome.

8.4.2 Adaptive Climbing vs. Stochastic Escape

The focus of Sect. 8.4.1 has been on the average properties of adaptive climbing. We now take fluctuations in the reproductive process into account. Our aim to compare two typical time scales, the number of generations needed for a stochastic escape and for adaptive climbing.

Escape Probability Under the condition of strong selection limit, as assumed in our model, favorable mutations spread instantaneously into the whole population. In first order, it is therefore sufficient to restrict the analysis to populations of size M in which all individuals have an identical fitness.

Mutations occur with probability u per genome, leading with probability F for descendants having a lower fitness. The probability for this to happen, viz the probability p_{esc} of stochastic escape, is consequently

$$p_{\text{esc}} \;\propto\; (Fu)^M \;\approx\; u^M, \qquad\qquad F \approx 1 \,.$$

The escape can only happen when an adverse mutation occurs for every member of the population within the same generation. A single individual not mutating retains a higher fitness, the one of the present local optimum, with the consequence that all other mutations are discarded when the selective pressure is strong, as assumed here. The same holds whenever a single positive mutation in $[F, 1]$ shows up. Compare also Fig. 8.7).

Stochastic Escape and Stasis As in the previous section, we consider a population that is close to a local optimum, but still climbing. The probability that the fitness of a given individual increases is $(1 - F)u$. It needs to mutate with a probability u

and to achieve a higher fitness, when mutating, which happens $1 - F$ of times. We denote with

$$a = 1 - (1 - F)u$$

the probability that the fitness of an individual does not increase with respect to the current fitness F of the population. The probability q_{pos} that at least one better genotype is found is then given by

$$q_{\mathrm{pos}} = 1 - a^M .$$

Real-world populations are typically close to a local optimum. In this situation, we can then distinguish between two evolutionary regimes.

– Adaptive Walk
  The escape probability p_{esc} is much smaller than the probability to increase the fitness, $q_{\mathrm{pos}} \gg p_{\mathrm{esc}}$. The population continuously increases its fitness via small mutations.
– Wandering Regime
  The adaptive dynamics slows down close to a local optimum. The probability of stochastic escape p_{esc} may then become comparable to that of an adaptive process, $p_{\mathrm{esc}} \approx q_{\mathrm{pos}}$.

In the "drifting regime", as the second case is also called, the population wanders around in genome space, starting a new adaptive walk after every successful escape.

Typical Escape Fitness The fitness F increases steadily during the adaptive walk regime, until it reaches a certain typical fitness, F_{esc}, for which the probability of stochastic escape becomes substantial, i.e. when $p_{\mathrm{esc}} \approx q_{\mathrm{pos}}$,

$$\begin{aligned} p_{\mathrm{esc}} &= (F_{\mathrm{esc}}u)^M \\ &= 1 - [1 - (1 - F_{\mathrm{esc}})u]^M = q_{\mathrm{pos}} . \end{aligned}$$

As $1 - F_{\mathrm{esc}}$ is small, we can set $(F_{\mathrm{esc}}u)^M \approx u^M$ and expand the right-hand-side of above expression with respect to $1 - F_{\mathrm{esc}}$,

$$u^M \approx 1 - [1 - M(1 - F_{\mathrm{esc}})u] = M(1 - F_{\mathrm{esc}})u ,$$

obtaining

$$1 - F_{\mathrm{esc}} = u^{M-1}/M . \tag{8.36}$$

The fitness F_{esc} necessary for the stochastic escape to become relevant is exponentially close to the global optimum $F = 1$ for large populations M.

Relevance of Stochastic Escape In general, stochastic escape occurs when the population is at a local optimum, or close to one. We may estimate the importance of the escape process relative to that of the adaptive walk by comparing the typical fitness F_{typ} of a local optimum achieved by a typical climbing process with the typical fitness F_{esc} needed for the escape process to become important,

$$F_{\text{typ}} = 1 - \frac{1}{N} \equiv F_{\text{esc}} = 1 - \frac{u^{M-1}}{M}, \qquad \frac{1}{N} = \frac{u^{M-1}}{M},$$

where we have used (8.32) for F_{typ}. The last expression is independent of the details of the fitness landscape, containing only the measurable parameters N, M and u. This condition can be fulfilled only when the number of individuals M is much smaller than the genome length N, as $u < 1$. The phenomenon of stochastic escape occurs only for very small populations.

8.5 Prebiotic Evolution

Prebiotic evolution deals with the questions surrounding the origin of life. Is it possible to define chemical autocatalytic networks in the primordial soup having properties akin to those of the metabolic reaction networks powering the workings of every living cell?

In this scenario, a key point regards the cell membrane, which acts as a defining boundary, separating body and environment on a physical level. Precursors of life would not have membranes, but one may hypothesize that chemical regulation networks emerging within a primordial soup of macromolecules did evolve into the protein regulation networks of living cells once enclosed by a membrane.

8.5.1 Quasispecies Theory

In Sect. 8.3.3 we discussed the concept of a quasispecies as a collection of closely related genomes in which no single genotype is dominant. This situation is presumably also typical for prebiotic evolutionary processes. In this context, Eigen formulated the quasispecies theory for a system of information carrying macromolecules through a set of equations for chemical kinetics,

$$\dot{x}_i = W_{ii}x_i + \sum_{j \neq i} W_{ij}x_j - x_i\phi(t), \tag{8.37}$$

where the x_i denote the concentrations of $i = 1 \dots N$ molecules. W_{ii} is the (autocatalytic) self-replication rate, with the off-diagonal terms W_{ij} $(i \neq j)$ corresponding to effective mutation rates.

In a first step, we will generalize the concept of a quasi-species to a collection of macromolecules using (8.37). Next, in Sect. 8.5.2, we will turn to a somewhat more realistic model of interacting chemical reactions, which will be, hopefully, functionally closer to the precursors of the protein regulation network of living cells.

Mass Conservation We can choose the flux $-x\phi(t)$ in Eigen's equation (8.37) for prebiotic evolution such that the total concentration

$$C = \sum_i x_i, \qquad \dot{C} = \sum_{ij} W_{ij} x_j - C\,\phi,$$

is conserved for long times. Demanding $\dot{C} \to 0$ and $C \to 1$ for large times suggests

$$\phi(t) = \sum_{ij} W_{ij} x_j(t) \tag{8.38}$$

for a suitable choice for the field $\phi(t)$, which leads in turn to

$$\dot{C} = \phi\,(1 - C), \qquad \frac{d}{dt}(C - 1) = -\phi\,(C - 1)\,. \tag{8.39}$$

The total concentration $C(t)$ will therefore approach 1 for $t \to \infty$ for $\phi > 0$, which we assume to be the case in the first case, implying total mass conservation. In this case the autocatalytic rates W_{ii} dominate with respect to the transmolecular mutation rates W_{ij} $(i \neq j)$.

Eigenstates We can write the evolution equation (8.37) in matrix form,

$$\frac{d}{dt}\mathbf{x}(t) = \big(W - \mathbf{1}\phi\big)\mathbf{x}(t), \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}, \tag{8.40}$$

where W is the matrix W_{ij}. Assuming for simplicity a symmetric mutation matrix $W_{ij} = W_{ji}$, the solution of the linear differential equation (8.40) is given in terms of the respective eigenvectors $\mathbf{e}_\lambda$,

$$W\mathbf{e}_\lambda = \lambda\,\mathbf{e}_\lambda, \qquad \mathbf{x}(t) = \sum_\lambda a_\lambda(t)\,\mathbf{e}_\lambda, \qquad \dot{a}_\lambda = [\lambda - \phi(t)]\,a_\lambda\,.$$

The eigenvector $\mathbf{e}_{\lambda_{\max}}$ with the largest eigenvalue $\lambda_{\max}$ will dominate for $t \to \infty$, due to the overall mass conservation, as enforced by (8.39). The flux will likewise adapt to the largest eigenvalue,

$$\lim_{t\to\infty} \Big(\lambda_{\max} - \phi(t)\Big) \;\to\; 0\,,$$

leading to the stationary condition $\dot{x}_i = 0$ for the evolution equation (8.40) in the long time limit.

Quasispecies The eigenvectors $\mathbf{e}_\lambda$ will contain only a single non-zero entry when W is diagonal in (8.40), viz when mutations are absent, such as $(0, \ldots, 1, \ldots, 0)$. In this case, a single macromolecule remains in the primordial soup for $t \to \infty$.

EXTENSIVE/INTENSIVE VECTORS A normalized vector of length N is extensive, respectively intensive, when its entries scale as $1/\sqrt{N}$, respectively $\sim N^0$.

Intensive vectors are dominated by a finite number of entries, or by entries that are summable.[10] A quasispecies of distinct but closely related macromolecules emerges when $\mathbf{e}_{\lambda_{\max}}$ is intensive, which is the case if the mutation rates W_{ij} $(i \neq j)$ are either absent, or finite but small.

Error Catastrophe Eigen's model is consistent when a quasispecies of macromolecules stabilizes. $\mathbf{e}_{\lambda_{\max}}$ is intensive in this case and the flux ϕ finite, which means that the mass conservation condition (8.39) can be fulfilled. One finds, in close analogy to the error catastrophe discussed in Sect. 8.3.3, that these two properties break down at a certain point when the off-diagonal matrix elements W_{ij} $(i \neq j)$ are consecutively increased.[11] At the threshold, $\mathbf{e}_{\lambda_{\max}}$ becomes extensive and the population disperses, which is the telltale sign of an error catastrophe. The flux ϕ diverges at the same time, with the consequence that total mass conservation is no longer possible, see (8.39). The underlying mechanism occurs also in models for the emergence of hypercycles, which we will examine next within an analytically tractable framework.

8.5.2 Hypercycles and Autocatalytic Networks

RNA World The macromolecular evolution equations (8.37) do not contain terms describing the catalysis of molecule i by molecule j. This process is, however, important both for the prebiotic evolution, as stressed by Manfred Eigen, as well as for the protein reaction network in living cells.

HYPERCYCLES Two or more molecules may form a stable catalytic (hyper) cycle when the respective intermolecular catalytic rates are large enough to mutually support their respective synthesis.

An illustration of selected hypercycles is given in Figs. 8.8 and 8.9. A likely chemical candidate for the constituent molecules is RNA, functioning both enzy-

[10] A sequence of elements is summable if the sum does not diverge when the length is successively increased. For vectors, the sum of the squared entries is of relevance.

[11] This detailed out in exercise (8.2).

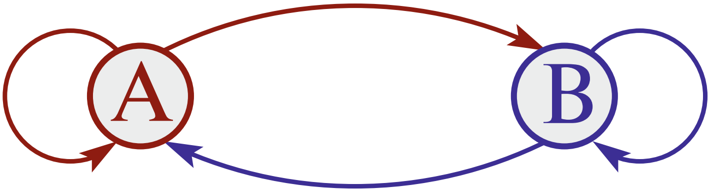

Fig. 8.8 The simplest hypercycle. Here, A/B are self-replicating molecules, with A acting as a catalyst for B, and vice versa. The replication rate of one species increases hence with the concentration of the other

matically and as a precursor of the genetic material. One speaks also of an "RNA world".

Reaction Networks Concentrating on the properties of catalytic reaction equations, we disregard mutations. Our basic reaction network is

$$\dot{x}_i = x_i \left(\lambda_i + \sum_j \kappa_{ij} x_j - \phi \right) , \tag{8.41}$$

where x_i is the concentration of reactant i. The autocatalytic growth rate is λ_i, with κ_{ij} encoding the transmolecular catalytic rates. Total concentration $C = \sum_i x_i$ remains constant if the flux ϕ is set to

$$\phi = \sum_k x_k \left(\lambda_k + \sum_j \kappa_{kj} x_j \right) . \tag{8.42}$$

In analogy to (8.39) one has

$$\begin{aligned} \dot{C} = \sum_i \dot{x}_i = \sum_i x_i \left(\lambda_i + \sum_j \kappa_{ij} x_j \right) - C\,\phi \\ = (1 - C)\,\phi \;\to\; 0 , \end{aligned}$$

which implies that $C \to 1$ in equilibrium.

Homogeneous Network An analytically solvable realization of (8.41) is given when the interactions $\kappa_{i \neq j}$ are homogeneous,

$$\kappa_{i \neq j} = \kappa, \qquad \kappa_{ii} = 0, \qquad \lambda_i = \alpha\, i , \tag{8.43}$$

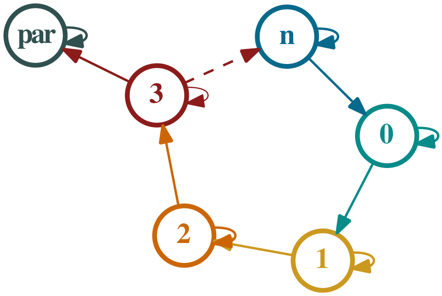

Fig. 8.9 A hypercycle of order n consisting of n cyclically coupled self-replicating molecules, with each molecule providing catalytic support for the subsequent molecule. Parasitic self-replicating molecules, indicated by 'par', receive catalytic support from the hypercycle without contributing to it

where we also presumed uniformly distributed autocatalytic growth rates λ_i. The reaction equations take the form

$$\dot{x}_i = x_i \left(\lambda_i + \kappa \sum_{j \neq i} x_j - \phi \right) = x_i \Big(\lambda_i + \kappa - \kappa x_i - \phi \Big) , \tag{8.44}$$

where we have used $\sum_i x_i = 1$. The stationary concentrations $\{x_i^*\}$ of (8.44) are

$$x_i^* = \begin{cases} (\lambda_i + \kappa - \phi)/\kappa \\ 0 \end{cases} \qquad \lambda_i = \alpha,\ 2\alpha,\ \ldots,\ N\alpha\,, \tag{8.45}$$

as illustrated in Fig. 8.10, where the non-zero solution is valid when the respective x_i^* is positive, namely for $\lambda_i - \kappa - \phi > 0$. The flux ϕ is determined self-consistently via (8.42).

Self-Consistent Quasispecies Formation Dynamically, the $x_i(t)$ with the largest growth rates λ_i will dominate and obtain a non-zero steady-state concentration x_i^*, as given by (8.45) However, total mass $1 = C = \sum_i x_i^*$ needs to be conserved. We may therefore assume that there exists an $N^* \in [1, N]$ such that

$$x_i^* = \begin{cases} (\lambda_i + \kappa - \phi)/\kappa & \text{for} \quad N^* \le i \le N \\ 0 & \text{for} \quad 1 \le i < N^* \end{cases} , \tag{8.46}$$

as shown in Fig. 8.10, where N^* and ϕ are determined by the normalization condition $1 = \sum_i x_i^*$. Remarkably, the self organization of the reaction network leads to a clear distinction. Part of the reactants die out, with the remainder forming a stable, mutually supporting catalytic network.

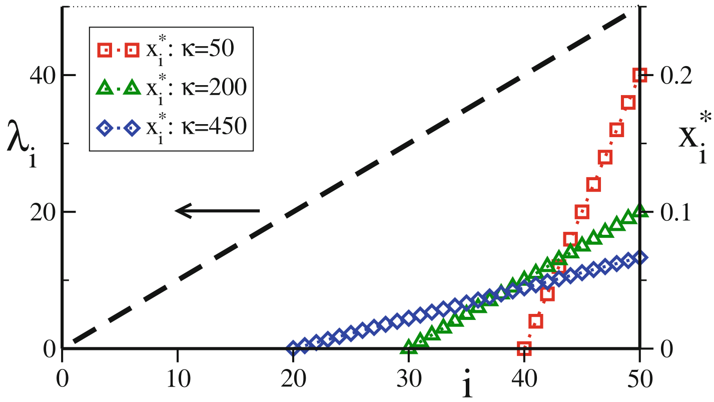

Fig. 8.10 The autocatalytic growth rates λ_i (left axis), as in (8.43) with $\alpha = 1$. The stationary concentrations x_i^* (right axis) constitutes a prebiotic quasispecies, compare (8.46). Shown are results for various mean transcatalytic rates κ. The individual molecules $i = 1, \ldots, 50$ are indexed along the horizontal axis

Termination Condition Mass conservation dictates

$$1 = \sum_i x_i^* = \sum_{i=N^*}^{N} \frac{\lambda_i + \kappa - \phi}{\kappa} = \frac{\alpha}{\kappa} \sum_{i=N^*}^{N} i + \left[\frac{\kappa - \phi}{\kappa}\right]\left(N + 1 - N^*\right)$$

$$= \frac{\alpha}{2\kappa}\left[N(N+1) - N^*(N^*-1)\right] + \left[\frac{\kappa - \phi}{\kappa}\right]\left(N + 1 - N^*\right),$$

where $\lambda_i = \alpha i$ has been used. The termination condition, $x_i^* = 0$ for $i = N^* - 1$,

$$0 = \frac{\lambda_{N^*-1} + \kappa - \phi}{\kappa} = \frac{\alpha(N^*-1)}{\kappa} + \frac{\kappa - \phi}{\kappa}, \tag{8.47}$$

can be used to eliminate $(\kappa - \phi)/\kappa$. We take the limit of large numbers of reactants N and N^*,

$$\frac{2\kappa}{\alpha} \simeq N^2 - \left(N^*\right)^2 - 2N^*\left(N - N^*\right)$$

$$= N^2 - 2N^*N + \left(N^*\right)^2 = (N - N^*)^2 .$$

The number of surviving reactants $N - N^*$ is therefore

$$N - N^* \simeq \sqrt{\frac{2\kappa}{\alpha}}, \tag{8.48}$$

which is non-zero for a finite and positive inter-molecular catalytic rate κ. A hypercycle of mutually supporting species (or molecules) has formed.

Origin of Life Aeons have passed since life emerged on earth. The scientific discussions concerning its origin continue to be controversial and it remains speculative whether hypercycles played a central role. The basic theory of hypercycles treated here, describing closed systems of chemical reactions, would be needed to be extended to non-equilibrium situations, with constant in- and outflows of both molecules and energy. In fact, a defining feature of biological activities is the buildup of local structures, which is possible when entropy is reduced locally at the expense of an overall increase of environmental entropy. Life, as we understand it today, is based on open systems driven by a constant flux of energy.

It is nevertheless interesting to point out that (8.48) implies a clear division between molecules $i = N^*, \ldots, N$ that can be considered to form a primordial "life form", and the remainder molecules $i = 1, \ldots, N^* - 1$. These reactants are 'outside', viz belonging to the environment, as their concentrations drop to zero 'inside'. This clear separation between participating and non-participating substances is a result of the non-linearity of the reaction equation (8.41). The linear evolution equation (8.37) would, on the other hand, result in a continuous density distribution, as illustrated in Fig. 8.5 for the case of the sharp peak fitness landscape. One could then conclude that life is possible only via cooperation, resulting from non-linear evolution equations.

8.6 Macroecology and Species Competition

In Macroecology one disregards both the genetic basis of evolutionary processes as well as specific species interdependencies. One is interested in formulating general principles and models describing the overall properties of extended communities of species.

Neutral Theory A central topic in ecology is to explain the distribution of species abundances, as illustrated in Fig. 8.11 for the count of trees in a tropical rainforest plot. In this example, for most species around 32 individual trees did grow in the patch examined, there are fewer species with more/less individuals. Similar species abundance distributions are found in virtually all ecosystems studied.

The neutral theory, as formulated originally by Hubbel, proposes that a handful of universal principles are enough to explain the species abundance distributions observed in nature. The framework is "neutral" as it abstracts from attributes that are specific to the members of the taxa examined. The two central principles of the neutral theory can be implemented mathematically in various fashions.

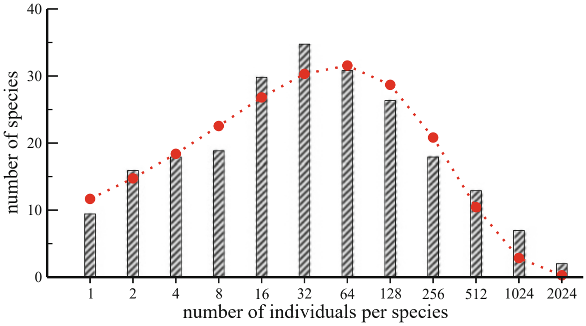

Fig. 8.11 The abundance of trees in a 50 ha patch of tropical rainforest in Panama (bars), in comparison with the prediction of neutral theory (filled circles), see (8.52), namely a Gamma distribution. Data from Volkov et al. (2003)

– DETERMINISTIC COMPETITION
  In between species, resource competition is deterministic.
– STOCHASTIC EVENTS
  Births, deaths, migration, and speciation, are treated on a stochastic level.

Stochasticity is thought to lead to random fluctuations in population sizes.

Stochastic Walk Through Population Space We consider a species performing a random walk in population space. The master equation for the probability $p_x(t)$ to observe x individuals of the species at time t is given by birth and death events, proportional to b_x and d_x respectively,

$$\dot{p}_x(t) = b_{x-1}\, p_{x-1}(t) + d_{x+1}\, p_{x+1}(t) - \big[b_x + d_x\big] p_x(t)\,. \tag{8.49}$$

The birth and death processes contain both intensive and extensive terms, respectively $\propto (x)^0$ and $\propto (x)^1$,

$$b_x = \tilde{b}_0 + \tilde{b}_1 x, \qquad d_x = \tilde{d}_0 + \tilde{d}_1 x\,, \tag{8.50}$$

where the intensive contributions $\tilde{b}_0$ ($\tilde{d}_0$) model the cumulative effects of immigration (emigration) and speciation (extinction), which occur on the level of individuals. The extensive terms, $\tilde{b}_1$ ($\tilde{d}_1$), are generated by reproduction processes actual births and deaths, which are proportional to the population size.

Fokker–Planck Equation of Macroecology For large populations we may treat x as a continuous variable and approximate the difference $d_{x+1}p_{x+1} - d_x p_x$ occurring on the right hand side of (8.49) through a Taylor expansion,

$$d_{x+1}\, p_{x+1} - d_x\, p_x \;\simeq\; \frac{\partial}{\partial x}\big(d_x\, p_x\big) + \frac{1}{2}\frac{\partial^2}{\partial x^2}\big(d_x\, p_x\big) + \ldots\,,$$

where $x+1 = x+\Delta x$ was used implicitly with $\Delta x = 1$. An analogous relation holds for the birth processes. Using the notation $p(x,t) = p_x(t)$ we obtain with

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial}{\partial x}\big(d_x - b_x\big)\, p(x,t) + \frac{1}{2}\frac{\partial^2}{\partial x^2}\big(d_x + b_x\big)\, p(x,t)$$

the "Fokker–Planck equation of macroecology",[12] which we rewrite as

$$\frac{\partial p(x,t)}{\partial t} = \frac{\partial}{\partial x}\left(\frac{x}{\tau} - b\right)\, p(x,t) + D\frac{\partial^2}{\partial x^2}\big(x\, p(x,t)\big)\,, \tag{8.51}$$

where we defined

$$\tau = \left(\tilde{d}_1 - \tilde{b}_1\right)^{-1} > 0, \qquad b = \tilde{b}_0 - \tilde{d}_0 = 2\tilde{b}_0 > 0, \qquad D = \frac{\tilde{b}_1 + \tilde{d}_1}{2}\,,$$

when restricting to the case $b_0 = -d_0 > 0$.

Competition vs. Diffusion The parameters introduced in (8.51) have the following interpretations:

- D induces fluctuations of the order $\sqrt{x}$ in a population of size x.
- b corresponds to the net influx, caused either by immigration or by speciation.
- τ is a measure of the strength of interaction effects in the ecosystem, expressed as the time scale the system needs to react to perturbations.

In order to understand the effect of τ in more detail, we start with the case $b = 0 = D$, for which the Fokker–Planck equation (8.51) reduces to

$$\tau\dot{p} = p + xp', \qquad p \sim e^{-t/T}x^{\beta}, \qquad -\tau/T = 1 + \beta\,.$$

The distribution is normalizable for $\beta < -1$, which implies $T > 0$. The ecosystem would slowly die out, on a time scale T, as a consequence of the competition between the species, when not counterbalanced by the diffusion D and the external source b. Note, that $\tau > 0$ implies that $\tilde{d}_1 > \tilde{b}_1$, namely that there are more deaths than births, on average, for all species and independent of population sizes.

Solution of the Fokker–Planck Equation The steady-state solution $\dot{p}(x,t) = 0$ of (8.51) is the Gamma distribution

$$p_0(x) = A\, x^{b/D-1}\, e^{-x/(D\tau)}\,, \tag{8.52}$$

[12] The theory behind the Fokker–Planck equation is developed in Sect. 3.5.2 of Chap. 3.

where A is a normalization constant. For a verification note that the population current J, defined via the continuity equation $\dot{p}(x,t) = -\nabla J$, vanishes in the steady state, $J = 0$, and hence

$$0 = \left(\frac{x}{\tau} - b\right) p_0(x) + D\frac{\partial}{\partial x}\big(x p_0(x)\big) = \left(\frac{x}{\tau} + D - b\right) p_0(x) + D x p_0'(x)\,,$$

which is satisfied by the solution (8.52). The steady state solution (8.52) fits the real-world data quite well, see Fig. 8.11.

Microscopic Models The master equation (8.49) of macroevolution can be derived from microscopic models. A specific microscopic update rule will lead to particular expressions regarding the dependence of the birth and death rates b_x and d_x on population size, and other microscopic parameters, which may be similar, but not identical, to the relations postulated in (8.50).

An example of a microscopic model would be an ecosystem of S species containing an overall number of N individuals, with a pairwise relation between the species. For every pair one of the species dominates the other species with probability ρ, with probability $1-\rho$ their relation is neutral.

At every time step a pair of individuals belonging to two different species S and S' is considered.

- Stochastic process. With probability μ the number of individuals in species S (S') is increased by one (decreased by one).
- Competition. With probability $1-\mu$ the number of individuals in the dominating (inferior) species is increased by one (reduced by one). No update is performed if their relation is neutral.

These update rules are conserving with respect to the total number of individuals. The steady-state distribution obtained by this model is similar to the one obtained for the neutral model defined by the birth and death rates (8.50), shown in Fig. 8.11, with the functional form of their respective species abundance distributions differing in details.

8.7 Coevolution and Game Theory

The average number of offspring, viz the fitness, is the single relevant reward function within Darwinian evolution. There is hence a direct connection between evolutionary processes and ecology with game theory, which deals with interacting agents trying to maximize a single reward function, denoted utility. Several types of games may be considered in this context, namely games of interacting species giving rise to coevolutionary phenomena or games of interacting members of the same species, pursuing distinct behavioral strategies.

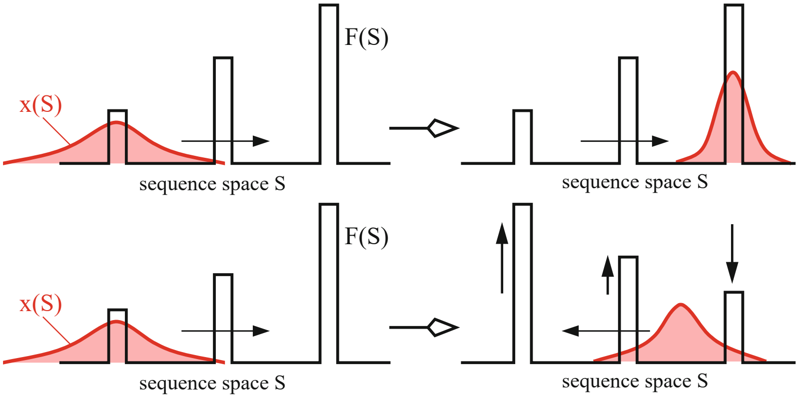

Fig. 8.12 **Top:** An evolutionary process of a single species in a static fitness landscape here with tower-like structures, as defined in (8.24). The density of individuals in sequence space, $x(\mathbf{S})$, will reorganize accordingly. **Bottom:** Coevolutionary processes are present when the adaption of one species changes the fitness landscapes $F(\mathbf{S})$ of other species

Coevolution The larger part of this chapter has been devoted to the discussion of the evolution of a single species. In Sect. 8.5.2 we ventured into the stabilization of 'ecosystems' composed of a hypercycle of mutually supporting species, before turning in Sect. 8.6 to general macroecological principles. We now go back to the level of a few interdependent species.

> COEVOLUTION When two or more species form an interdependent ecosystem the evolutionary progress of part of the ecosystem will produce new evolutionary forcings for other species.

One can view the coevolutionary process also as a change in the respective fitness landscapes, as illustrated in Fig. 8.12. A prominent outcome of reciprocal coevolution forcings is the "red queen" phenomenon.

> RED QUEEN PHENOMENON When two or more species are interdependent then "It takes all the running, to stay in place" (from Lewis Carroll's children's book "Through the Looking Glass").

A well-known example of the red queen phenomenon is the "arms race" between predator and prey commonly observed in real-world ecosystems. Snakes becoming ever more poisonous, with frogs developing successively higher resistance levels.

Green World Hypothesis Plants abound in real-world ecosystems, which are, geology and climate permitting, rich and green. Naively one may expect that herbivores should proliferate when food is plenty, keeping vegetation constantly down. This does not seem to happen in the world, as a result of coevolutionary interdependencies. The likely culprit are "trophic cascades", where predators keep herbivores substantially below the support level of the bioproductivity of the plants.[13] This "green world hypothesis" arises naturally in evolutionary models featuring coevolutionary interdependencies.

Avalanches and Punctuated Equilibrium We discussed previously,[14] that coevolutionary avalanches may be observed within a state of punctuated equilibrium.

PUNCTUATED EQUILIBRIUM Most of the time ecosystems are in a steady-state neutral phase. Outside influences, or rare stochastic processes may induce periods of rapid coevolutionary processes.

The term punctuated equilibrium was coined to describe a characteristic feature of the evolution of simple traits observed in fossil records. In contrast to the gradualistic view of evolutionary changes, these traits typically show long periods of stasis interrupted by sequences of rapid changes.

Strategies and Game Theory In contrast to the mostly stochastic considerations discussed so far, one is often interested in the evolutionary processes giving rise to specific survival strategies. These are questions that can be addressed within game theory, which deals with strategically interacting agents in economics and beyond. When an animal meets another animal it has to decide, to give an example, whether confrontation, cooperation, or defection is the best strategy. Some basic elements.

- UTILITY
  Every participant, the agent, plays for himself, solely maximizing its own utility.
- STRATEGY
  Every participant follows a set of rules of what to do when encountering an opponent; the strategy.
- ADAPTIVE GAMES
  In adaptive games, the participants change their strategy in order to maximize future return. This change can be either deterministic or stochastic.
- ZERO-SUM GAMES
  When the sum of utilities is constant, you can only win what the others lose.
- NASH EQUILIBRIUM
  Any strategy change by participants leads to a reduction of its utility.

[13] The basic model for a trophic cascade, the Lotka-Volterra Model for rabbits and foxes, is treated in Sect. 3.2.1 of Chap. 3.

[14] See Sect. 6.6 of Chap. 6, where the Bak and Sneppen model of coevolutionary cascades is analyzed.

Hawks and Doves This simple evolutionary game tries to model competition in terms of expected utilities between aggressive behavior (by the "hawk") and peaceful demeanor (by the "dove"). The rules are given in Table 8.2.

The expected returns, the utilities, are collected together in the "payoff matrix",

$$A = \begin{pmatrix} A_{\mathrm{HH}} & A_{\mathrm{HD}} \\ A_{\mathrm{DH}} & A_{\mathrm{DD}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(V-C) & V \\ 0 & \frac{V}{2} \end{pmatrix} . \tag{8.53}$$

which specifies what happens if agents meet. Does it pay to be peaceful, or aggressive, and under which conditions? This is the question.

Adaptation by Evolution The introduction of reproductive capabilities for the participants turns the Hawks and Doves game into an evolutionary game. In this context one presumes that behavioral strategies are generated by distinct alleles.

The average number of offspring of a player is proportional to its fitness, which is given in turn by the expected utility,

$$\begin{aligned} \dot{x}_{\mathrm{H}} &= \big(A_{\mathrm{HH}}x_{\mathrm{H}} + A_{\mathrm{HD}}x_{\mathrm{D}} - \phi(t)\big)\, x_H \\ \dot{x}_{\mathrm{D}} &= \big(A_{\mathrm{DH}}x_{\mathrm{H}} + A_{\mathrm{DD}}x_{\mathrm{D}} - \phi(t)\big)\, x_D \end{aligned} , \tag{8.54}$$

where x_{D} and x_{H} are the densities of doves and hawks. The flux ϕ is given by

$$\phi(t) = x_{\mathrm{H}}A_{\mathrm{HH}}x_{\mathrm{H}} + x_{\mathrm{H}}A_{\mathrm{HD}}x_{\mathrm{D}} + x_{\mathrm{D}}A_{\mathrm{DH}}x_{\mathrm{H}} + x_{\mathrm{D}}A_{\mathrm{DD}}x_{\mathrm{D}}$$

it ensures an overall constant population, $x_{\mathrm{H}} + x_{\mathrm{D}} = 1$.

Steady State Solution We are interested in the steady-state solution of (8.54), with $\dot{x}_{\mathrm{D}} = 0 = \dot{x}_{\mathrm{H}}$. Setting

$$x_{\mathrm{H}} = x, \qquad x_{\mathrm{D}} = 1 - x ,$$

we find

$$\phi(t) = \frac{x^2}{2}(V-C) + Vx(1-x) + \frac{V}{2}(1-x)^2 = \frac{V}{2} - \frac{C}{2}x^2$$

Table 8.2 Rules of the Hawk & Dove game, together with the entries A_{DD}, etc., of the payoff matrix defined in (8.53)

Dove meets Dove	$A_{\mathrm{DD}} = V/2$	They divide the territory
Hawk meets Dove	$A_{\mathrm{HD}} = V$, $A_{\mathrm{DH}} = 0$	The Hawk gets all the territory, the Dove retreats and gets nothing
Hawk meets Hawk	$A_{\mathrm{HH}} = (V-C)/2$	They fight, get injured, and win half the territory

for the flux. The update rule for the density of hawks is

$$\begin{aligned}\dot{x} &= \left(\frac{V-C}{2}x + V(1-x) - \phi(t)\right)x \;=\; \left(\frac{V}{2} - \frac{V}{2}x + \frac{C}{2}\left(x^2 - x\right)\right)x \\ &= \frac{C}{2}\,x\,\left(x^2 - \frac{C+V}{C}x + \frac{V}{C}\right) \;=\; \frac{C}{2}\,x\,(x-1)\left(x - \frac{V}{C}\right) \\ &\equiv -\frac{d}{dx}V(x)\,,\end{aligned}$$

where we defined the potential

$$V(x) = -\frac{x^2}{4}V + \frac{x^3}{6}(V+C) - \frac{x^4}{8}C\,.$$

The steady-state solution is given by

$$V'(x) = 0, \qquad\qquad x = V/C\,,$$

apart from the trivial states $x = 0/1$ (no/only hawks).

– For $V > C$ doves are eliminated from the population, it does not pay to be peaceful.
– For $V < C$ hawks and doves coexists, with respective densities $x = V/C$ and $1 - V/C$.

A population consisting exclusively of cooperating doves, $x = 0$, is unstable against the intrusion of hawks.

Prisoner's Dilemma The payoff matrix of the prisoner's dilemma is

$$A = \begin{pmatrix} R & S \\ T & P \end{pmatrix} \qquad \begin{array}{c} T > R > P > S \\ 2R > S + T \end{array} \qquad \begin{array}{r} \text{cooperator} \mathrel{\hat{=}} \text{dove} \\ \text{defector} \mathrel{\hat{=}} \text{hawk} \end{array}\,. \tag{8.55}$$

Here "cooperation" between the two prisoners is implied, and not cooperation between a suspect and the police. The prisoners are best off if both keep silent. The standard values are

$$T = 5, \qquad R = 3, \qquad P = 1, \qquad S = 0\,.$$

The maximal global utility NR is obtained when everybody cooperates. However, in a situation where agents interact independently, the only stable Nash equilibrium is defection, with a global utility NP.

$$\text{reward for cooperators} = R_c = \Big[RN_c + S(N - N_c)\Big]/N\,,$$

$$\text{reward for defectors} = R_d = \Big[TN_c + P(N - N_c)\Big]/N\,,$$

where N_c is the number of cooperators and N the total number of agents. The difference is

$$R_c - R_d \;\sim\; (R - T)N_c + (S - P)(N - N_c) \;<\; 0\,,$$

as $R - T < 0$ and $S - P < 0$. The reward for cooperation is always smaller than that for defecting.

Evolutionary Games on a Lattice The adaptive dynamics of evolutionary games can change completely when the individual agents are placed on a regular lattice. A strong impact has also the possibility to adapt strategies based on past observations. We give a possible set rules.

– At each generation (time step) agents evaluate their payoff together with the payoff of the neighbors.
– Next, agents compare their payoff one by one with the payoffs obtained by the neighbors.
– Agents switche strategy (cooperate/defect) to the strategy of the neighbor with the highest payoff.

Despite their simplicity, this set of rules leads to surprisingly complex real-space patterns of defectors intruding patches of cooperators, as illustrated in Fig. 8.13. Further details depend on the value chosen for the payoff matrix.

Nash Equilibria vs. Coevolutionary Avalanches Coevolutionary games on a lattice eventually lead to an equilibrium state, which by definition has to be a Nash equilibrium. If such a state is perturbed from the outside, a self-critical coevolutionary avalanche may follow, in close relation to the sandpile model.[15]

Game Theory and Memory Standard game theory deals with an anonymous society of agents, with agents having no memory of previous encounters. Generalizing this standard setup it is possible to empower the agents with a memory of their past strategies and achieved utilities. Considering additionally individualized societies,

[15] The occurrence of self-organized criticality in the sandpile model is discussed in Chap. 6.

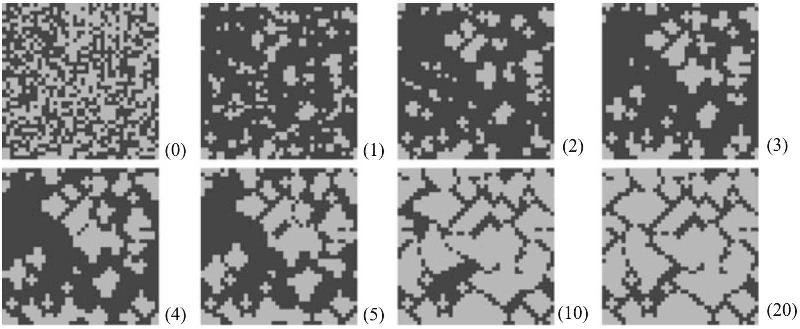

Fig. 8.13 Time series of the spatial distribution of cooperators (gray) and defectors (black) on a lattice of size $N = 40 \times 40$. Time is given as the numbers of generations (in brackets). Initial condition: equal number of defectors and cooperators, randomly distributed. Parameters for the payoff matrix, $\{T; R; P; S\} = \{3.5; 3.0; 0.5; 0.0\}$. Reprinted from Schweitzer et al. (2002) with permissions, © 2002 World Scientific Publishing Co Pte Ltd

this memory may include the names of the opponents encountered previously. This type of games provides the basis for studying the emergence of sophisticated survival strategies, like altruism, via evolutionary processes.

Opinion Dynamics Agents in classical game theory aim to maximize their respective utilities. However, this does not necessarily reflect daily social interactions. When encountering somebody else, explicit maximization of rewards or utilities is not always the priority.

An example of reward-free games is given by opinion dynamics models. In a basic model $i = 1, \ldots, N$ agents have continuous opinions $x_i = x_i(t)$. When two agents interact, they change their respective opinions according to

$$x_i(t+1) = \begin{cases} [x_i(t) + x_j(t)]/2 & |x_i(t) - x_j(t)| < \theta \\ x_i(t) & |x_i(t) - x_j(t)| \geq \theta \end{cases}, \tag{8.56}$$

where θ is the confidence interval. This is the "bounded confidence model" we discussed before.[16] Consensus can be reached step by step only when the initial opinions are not too contrarian. Relative to the initial scatter of opinions, global consensus will be reached for large confidence intervals θ. Clusters of distinct opinions emerge on the other side for a small confidence interval.

[16] Bounded confidence opinion dynamics is treated in detail in Sect. 4.3.4 of Chap. 4.

8.7.1 Tragedy of the Commons

Ecological settings correspond in many instances to a game theoretical problem. Competition between agents can be either direct, as for the Hawks and Doves game, or indirect. The latter occurs when agents compete for a common resource, a situation described by the celebrated "tragedy of the commons".

Common Pool Resources A village may have a pond without access restrictions. Everybody with a fishing rod may go to the pond and fish at any time. Overfishing is the likely outcome, with the consequence that the common pool resource is depleted. When this happens, the "tragedy of the commons" is in the making. However, free access does not imply that fishing does not incur costs. One has to invest money, for the equipment, and time, for the trip to the pond and for the actual activity. People will stop fishing once returns become too small.

Diverse situations are described by the tragedy of the commons, such as the overfishing of an ocean by fleets of trawlers, or the common exploitation of an underground aquifer. Importantly, pollution of a common resource, like CO_2 emissions into earth's atmosphere, fall into the same category.

Reference Model Agents invest into the commons as long as their payoffs E_i remain positive, where $i = 1, .., N$, with N being the number of agents. The quantity invested, $x_i \geq 0$, can viewed as time or money. As usual, payoffs are given by the difference between nominal return and investment costs,

$$E_i = \left(\mathrm{e}^{-x_\mathrm{tot}} - c_i\right) x_i, \qquad x_\mathrm{tot} = \sum_j x_j\,, \tag{8.57}$$

where $c_i > 0$ are per-unit investing costs. The factor $\exp(-x_\mathrm{tot})$ specifies how the productivity of the commons degrades when total investment x_tot increases.

Profitable Agents Selfish agents increase their investments x_i as long as their gradients dE_i/dx_i remain positive. The equilibrium condition is

$$(1 - x_i)\mathrm{e}^{-x_\mathrm{tot}} = c_i, \qquad x_i = 1 - c_i \mathrm{e}^{x_\mathrm{tot}}\,. \tag{8.58}$$

which can be rewritten as as

$$x_i = 1 - \frac{c_i}{c_\mathrm{max}}, \qquad c_\mathrm{max} = \mathrm{e}^{-x_\mathrm{tot}}\,, \tag{8.59}$$

where c_max is the profitability barrier. Agents with $c_i > c_\mathrm{max}$ would have negative x_i and returns, which is not profitable. It is hence sufficient to restrict the analysis to cost-efficient agents, with $c_i \leq c_\mathrm{max}$..

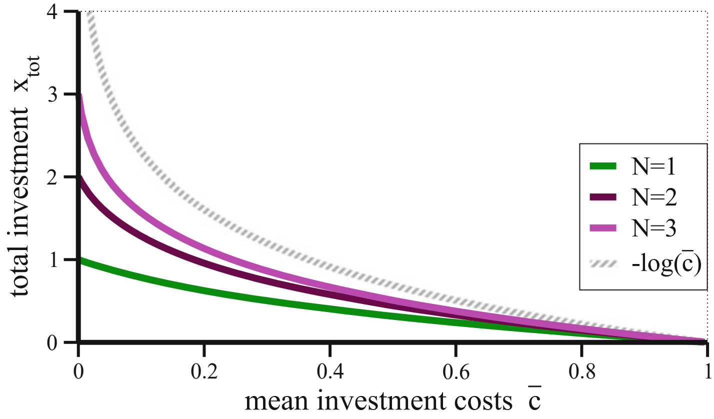

Fig. 8.14 For the tragedy of the commons, the optimal total investment x_{tot}. For N agents, total investment as a function of mean investment costs $\bar{c}$ is given, see (8.61). Limiting behaviors are $x_{tot} = N$ at $\bar{c} = 0$ and $x_{tot} = 0$ for $\bar{c} = 1$. Also shown is the case $N \to \infty$, for which $x_{tot} = \log(1/\bar{c})$ diverges logarithmically for vanishing $\bar{c} \to 0$

Dispersion Relation For the stationary state, combining (8.59) and (8.57) leads to

$$E_i = \left(c_{\max} - c_i\right)\left(1 - \frac{c_i}{c_{\max}}\right) = c_{\max}\left(1 - \frac{c_i}{c_{\max}}\right)^2, \tag{8.60}$$

which constitutes a dispersion relation $E_i = E(c_i)$. The profitability barrier $c_{\max} = \exp(-x_{tot})$ is determined in turn by the self-consistency condition

$$\left(1 - \frac{x_{tot}}{N}\right) e^{-x_{tot}} = \bar{c}, \qquad \bar{c} = \frac{1}{N}\sum_i c_i\,, \tag{8.61}$$

which is obtained by summing (8.58) over all agents. Here, $\bar{c}$ are the average investment costs of profitable agents. The resulting x_{tot} is plotted in Fig. 8.14. As expected, the productivity $\exp(-x_{tot})$ of the common resource degrades progressively with increasing N and decreasing $\bar{c}$.

Catastrophic Poverty We expand the self-consistency condition (8.61) for large numbers of agents N,

$$\frac{\bar{c}}{c_{\max}} = \frac{\bar{c}}{e^{-x_{tot}}} = 1 - \frac{x_{tot}}{N} \approx 1 - \frac{\log(1/\bar{c})}{N}\,, \tag{8.62}$$

where we used that $\lim_{N\to\infty} x_{\text{tot}} = \log(1/\bar{c})$. For an average agent, with $c_i = \bar{c}$, the large-N expansion of the payoff $E(\bar{c})$ is consequently

$$E(\bar{c}) \approx c_{\max}\left(\frac{\log(1/\bar{c})}{N}\right)^2 \sim \frac{1}{N^2}\,, \tag{8.63}$$

where the dispersion relation (8.60) has been used. Intuitively one could have expected that payoffs would scale instead as $1/N$. The fact that the average payoff scales worse, denoted "castastrophic poverty", can be traced to a progressively deteriorating state of the commons.

Oligarchs At face value, Eq. (8.63) holds only for the average agent. One can however prove that the payoffs of the majority of all agents scale as $(1/N)^2$. A notable exception are "oligarchs", namely agents with investement costs substantially below $\bar{c}$. Oligarchs have payoffs that scale as $(1/N)^0$. Only a finite number of oligarchs may howver be present.

Exercises

(8.1) THE ONE-DIMENSIONAL ISING MODEL
Solve the one-dimensional Ising model

$$H = J\sum_i s_i s_{i+1} + B\sum_i s_i$$

by the transfer matrix method presented in Sect. 8.3.2. Calculate the free energy $F(T,B)$, the magnetization $M(T,B)$ and the susceptibility $\chi(T) = \lim_{B\to 0}\frac{\partial M(T,B)}{\partial B}$.

(8.2) ERROR CATASTROPHE
For the prebiotic quasispecies model (8.40), consider tower-like autocatalytic reproduction rates W_{jj} and mutation rates W_{ij} $(i \neq j)$ of the form

$$W_{ii} = \begin{cases} 1 & i = 1 \\ 1-\sigma & i > 1 \end{cases}, \qquad W_{ij} = \begin{cases} u_+ & i = j+1 \\ u_- & i = j-1 \\ 0 & i \neq j \quad \text{otherwise} \end{cases},$$

with $\sigma, u_\pm \in [0,1]$. Determine the error catastrophe for the two cases $u_+ = u_- \equiv u$ and $u_+ = u$, $u_- = 0$. Compare with the results for the tower landscape discussed in Sect. 8.3.3.
Hint: For the stationary eigenvalue (8.40), with $\dot{x}_i = 0$ $(i = 1, \ldots)$, write x_{j+1} as a function of x_j and x_{j-1}. This two-step recursion relation leads to a 2×2 matrix. Consider the eigenvalues/vectors of this matrix, the initial condition

for x_1, and the normalization condition $\sum_i x_i < \infty$ valid in the adapting regime.

(8.3) COMPETITION FOR RESOURCES

The competition for scarce resources has been modeled in the quasispecies theory, see (8.37), by an overall constraint on population density. With

$$\dot{x}_i = W_{ii}x_i \qquad W_{ii} = fr_i - d, \qquad \dot{f} = a - f\sum_i r_i x_i \tag{8.64}$$

one models the competition for the resource f explicitly. Here a and fr_i are the regeneration rates of the resource f, respectively for species i, with d being the mortality rate. Equation (8.64) does not contain mutation terms $\sim W_{ij}$, describing a stationary ecosystem.

Which is the steady-state value of the total population density $C = \sum_i x_i$ and of the resource level f? Is the ecosystem stable?

(8.4) COMPETITIVE POPULATION DYNAMICS

Consider the symmetric Lotka-Volterra system

$$\dot{x} = x\big[(1-ky) - x\big], \qquad \dot{y} = y\big[(1-kx) - y\big] \tag{8.65}$$

modeling populations of sheeps and rabbits with populations densities x and y competing for the same food resource. Find the fixpoints and consider the case of weak competition $k < 1$ and of strong competition $k > 1$. When do sheeps and rabbits live happily together?

Equation (8.65) is invariant under an exchange $x \leftrightarrow y$. Can this symmetry be spontaneously broken?

(8.5) HYPERCYCLES

Examine a reaction system for $N = 2$ molecules and a homogeneous network, as defined by (8.41) and (8.42). Find the fixpoints and discuss their stability.

(8.6) PRISONER'S DILEMMA ON A LATTICE

Consider the stability of intruders in the prisoner's dilemma (8.55) on a square lattice, as the one illustrated in Fig. 8.13. Presume either a single intruder or two adjacent intruders, both for defectors/cooperators in a background of cooperators/defectors. Who survives?

(8.7) NASH EQUILIBRIUM

Examine the Nash equilibrium and its optimality for the following two-player game:

Each player acts either cautiously or riskily. A player acting cautiously always receives a low payoff. A player playing riskily gets a high payoff if the other player also takes a risk. Otherwise, the risk-taker obtains no reward.

(8.8) WAR OF ATTRITION

When the primary investment is time, like in the competition of animals for nesting grounds, the agent staying longest may win. A strategy S is defined in terms of the probability $p(x)$ to invest $x \in [0, \infty]$. Agents bidding m against

S have the expected payoff

$$E_m = v \int_0^m p(x)dx - c(m) \int_m^\infty p(x)dx - \int_0^m c(x)p(x)dx \,, \qquad (8.66)$$

where v is the bare reward and $c(x) = x^\gamma$ a generic cost function. When losing, the invested time is not recovered. Find the optimal strategy for the war of attrition, as defined by (8.66).

(8.9) TRAGEDY OF THE COMMONS
For the analysis of the tragedy of the commons a productivity function $P(x_{\text{tot}}) = \exp(-x_{\text{tot}})$ had beed used in (8.57). Show that catastrophic poverty is present for a generic $P(x_{\text{tot}})$.

Further Reading

There are excellent review articles on statistical approaches to Darwinian evolution and ecology, see Azaele et al. (2016), Blythe and McKane (2007), and Drossel (2001). The interested reader may want to consult Bull et al. (2005) regarding the notion of quasispecies, Higgs and Lehman (2015) with respect to the RNA world, Heasley et al. (2021), for the punctuated equilibrium, and Wilkinson et al. (2016) for the green world hypothesis. General textbooks on evolution, game theory and hypercycles are Sigmund (2017) and Eigen and Schuster (2012). The tragedy of the commons is treated together with the notion of catastrophic poverty in Gros (2023).

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Synchronization Phenomena

9

Complex systems are based on interacting local computational units may show non-trivial emerging behaviors. Examples are the time evolution of an infectious disease in a certain city that is mutually influenced by an ongoing outbreak of the same disease in another city, or the case of a neuron firing spontaneously while processing the effects of afferent axon potentials.

A fundamental question is whether the time evolutions of interacting local units remain dynamically independent of each other, or whether they will change their states simultaneously, following identical rhythms. This is the notion of synchronization, which we will study throughout this chapter. Starting with the paradigmatic Kuramoto model we will learn that synchronization processes may be driven either by averaging dynamical variables, or through causal mutual influences. On the way, we will visit piecewise linear dynamical systems and the reference model for infectious diseases, the SIRS model.

9.1 Frequency Locking

In this chapter, we will be dealing mostly with autonomous dynamical systems that may synchronize spontaneously. A dynamical system can also be driven by outside influences, being forced to follow the external signal synchronously.

Driven Harmonic Oscillator For a start, we recapitulate the dynamical states of the driven harmonic oscillator,

$$\ddot{x} + \gamma\,\dot{x} + \omega_0^2\,x = F\left(\mathrm{e}^{i\omega t} + c.c.\right), \qquad \gamma > 0\,, \tag{9.1}$$

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_9

where 'c.c.' stands for conjugate complex. Of relevance will be the relation between the eigenfrequency ω_0, damping γ, and the driving frequency ω. In the absence of an external forcing, $F \equiv 0$, the solution is

$$x(t) \sim \mathrm{e}^{\lambda t}, \qquad \lambda_{\pm} = -\frac{\gamma}{2} \pm \frac{1}{2}\sqrt{\gamma^2 - 4\omega_0^2}, \tag{9.2}$$

which is damped/critical/overdamped for $\gamma < 2\omega_0$, $\gamma = 2\omega_0$ and $\gamma > 2\omega_0$.

Frequency Locking In the limit of long times, $t \to \infty$, the dynamics of the system follows the external driving. Due to the damping $\gamma > 0$, this holds for all $F \neq 0$. For the ansatz

$$x(t) = a\mathrm{e}^{i\omega t} + c.c., \tag{9.3}$$

the amplitude a may contain, as a matter of principle,[1] an additional time-independent phase. Applying to (9.1) yields

$$\begin{aligned} F &= a\left(-\omega^2 + i\omega\gamma + \omega_0^2\right) = -a\left(\omega^2 - i\omega\gamma - \omega_0^2\right) \\ &= -a\,(\omega + i\lambda_+)\,(\omega + i\lambda_-)\ , \end{aligned}$$

where the eigenfrequencies $\lambda_{\pm}$ are given by (9.2). The solution for the amplitude a can then be written in terms of $\lambda_{\pm}$ or alternatively as

$$a = \frac{-F}{\left(\omega^2 - \omega_0^2\right) - i\omega\gamma}. \tag{9.4}$$

The response becomes divergent, viz $a \to \infty$, at resonance $\omega = \omega_0$ and small dampings $\gamma \to 0$.

General Solution The driven, damped harmonic oscillator (9.1) is an inhomogeneous linear differential equation. The general solution is hence given by the superposition of the special solution (9.4) with the general solution of the homogeneous system (9.2). The latter dies out for $t \to \infty$ and the system synchronizes with the external driving frequency ω.

[1] "Secular perturbation theory" deals with time-dependent amplitudes, $a = a(t)$. See Sect. 3.2 of Chap. 3.

9.2 Coupled Oscillators and the Kuramoto Model

Whenever their dynamical behaviors are similar and the mutual couplings substantial, sets of local dynamical systems may synchronize. It is not necessary that the individual units show periodic dynamics on their own, coupled chaotic attractors can also synchronize. We start by discussing the simplest non-trivial set-up, the case of harmonically coupled uniform oscillators.

Limit Cycles In suitable coordinates, free rotations such as

$$\mathbf{x}(t) = r\big(\cos(\theta(t)), \sin(\theta(t))\big), \qquad \theta(t) = \omega t + \theta_0, \qquad \dot{\theta} = \omega\,,$$

describe periodic attractors, viz limit cycles. In the following we take the radius r as given, using the phase variable $\theta = \theta(t)$ for an effective description.

Coupled Dynamical Systems We study the behavior of a collection of individual dynamical systems $i = 1, \ldots, N$, each characterized by a limit cycle with natural frequency ω_i. The coupled system obeys

$$\dot{\theta}_i = \omega_i + \sum_{j=1}^{N} \Gamma_{ij}(\theta_i, \theta_j), \qquad i = 1, \ldots, N\,, \tag{9.5}$$

where Γ_{ij} are suitable coupling constants.

Kuramoto Model A particularly tractable choice for the coupling constants Γ_{ij} has been proposed by Kuramoto:

$$\Gamma_{ij}(\theta_i, \theta_j) = \frac{K}{N}\,\sin(\theta_j - \theta_i)\,, \tag{9.6}$$

where $K \geq 0$ is the coupling strength. The factor $1/N$ ensures that the model is well behaved in the limit $N \to \infty$.

Two Coupled Oscillators We consider first the case $N = 2$,

$$\dot{\theta}_1 = \omega_1 + \frac{K}{2}\,\sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = \omega_2 + \frac{K}{2}\,\sin(\theta_1 - \theta_2)\,, \tag{9.7}$$

or

$$\Delta\dot{\theta} = \Delta\omega - K\,\sin(\Delta\theta), \qquad \Delta\theta = \theta_2 - \theta_1, \qquad \Delta\omega = \omega_2 - \omega_1\,. \tag{9.8}$$

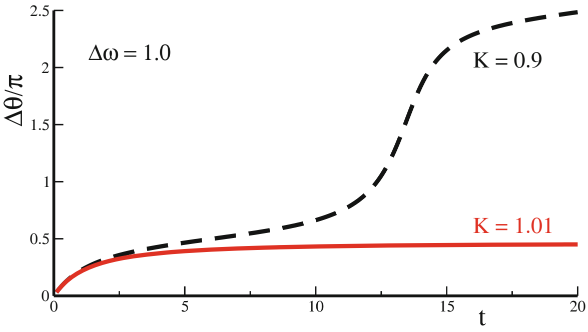

Fig. 9.1 The relative phase $\Delta\theta(t)$ of two coupled oscillators, obeying (9.8), with $\Delta\omega = 1$ and a critical coupling strength $K_c = 1$. For an undercritical coupling, $K = 0.9$, the relative phase increases steadily, for an overcritical coupling $K = 1.01$ it locks

The system has a fixpoint $\Delta\theta^*$ with regard to the relative phase, determined by

$$\frac{d}{dt}\Delta\theta^* = 0, \qquad \sin(\Delta\theta^*) = \frac{\Delta\omega}{K}, \tag{9.9}$$

which leads to

$$\Delta\theta^* \in [-\pi/2, \pi/2], \qquad K > |\Delta\omega| . \tag{9.10}$$

This condition is valid for attractive coupling constants $K > 0$. For repulsive $K < 0$ anti-phase states are stabilized. We analyze the stability of the fixpoint using $\Delta\theta = \Delta\theta^* + \delta$ and (9.8), obtaining

$$\frac{d}{dt}\delta = -\left(K\cos\Delta\theta^*\right)\delta, \qquad \delta(t) = \delta_0\, \mathrm{e}^{-K\cos\Delta\theta^* t} .$$

The fixpoint is stable since $K > 0$ and $\cos\Delta\theta^* > 0$, due to (9.10). Consequently, a bifurcation is present.

- For $K < |\Delta\omega|$ there is no phase coherence between the two oscillators, which are drifting with respect to each other.
- For $K > |\Delta\omega|$ phase locking is observed, which means that the two oscillators rotate together with a constant phase difference.

The situation is illustrated in Fig. 9.1.

Natural Frequency Distribution We now examine what happens if we have a larger number of coupled oscillators, $N \to \infty$. We select the probability distribution $g(\omega)$ of the natural frequencies ω_i, to be symmetric,

$$g(\omega) = g(-\omega), \qquad \int_{-\infty}^{\infty} g(\omega)\, d\omega = 1 , \tag{9.11}$$

The choice of a zero average frequency,

$$\int_{-\infty}^{\infty} \omega\, g(\omega)\, d\omega = 0\,,$$

implicit in (9.11) is generally possible, as the dynamical equations (9.5) and (9.6) are invariant under a global translation

$$\omega \to \omega + \Omega, \qquad\qquad \theta_i \to \theta_i + \Omega t\,,$$

with Ω being the initial non-zero mean frequency.

Order Parameter The complex order parameter

$$r\, \mathrm{e}^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} \mathrm{e}^{i\theta_j} \tag{9.12}$$

is a macroscopic quantity that can be interpreted as the collective rhythm produced by the assembly of the interacting oscillating systems. The radius $r(t)$ measures the degree of phase coherence, with $\psi(t)$ corresponding to the average phase.

Molecular Field Representation We rewrite the order parameter defined in (9.12) as

$$r\, \mathrm{e}^{i(\psi-\theta_i)} = \frac{1}{N}\sum_{j=1}^{N} \mathrm{e}^{i(\theta_j-\theta_i)}, \qquad\qquad r\sin(\psi-\theta_i) = \frac{1}{N}\sum_{j=1}^{N}\sin(\theta_j-\theta_i)\,,$$

retaining in the second step the imaginary component. Inserting the second expression into the governing equation (9.5), we find

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_j \sin(\theta_j-\theta_i) = \omega_i + Kr\sin(\psi-\theta_i)\,. \tag{9.13}$$

The motion of individual oscillators $i = 1, \ldots, N$ is coupled to the other oscillators exclusively through the mean-field phase ψ, with a coupling strength that is proportional to the mean-field amplitude r.

The individual phases θ_i are drawn towards the self-consistently determined mean phase ψ, as can be seen in the numerical simulations presented in Fig. 9.2. Mean-field theory is exact for the Kuramoto model. It is nevertheless non-trivial to solve, as the self-consistency condition (9.12) needs to be fulfilled.

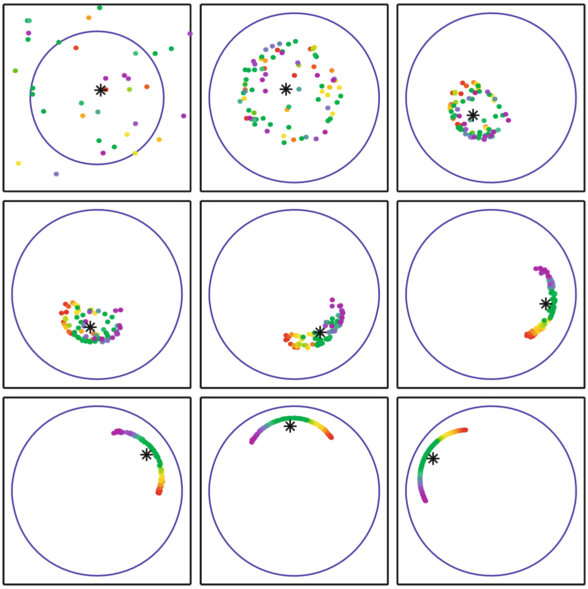

Fig. 9.2 Spontaneous synchronization in a network of limit cycle oscillators with distributed individual frequencies. Color coding with slowest–fastest (red–violet) natural frequencies. With respect to (9.5), an additional distribution of individual radii $r_i(t)$ has been assumed. These are slowly relaxing. Also shown is the mean field $re^{i\psi} = \sum_i r_i e^{i\theta_i}/N$ (asterix), compare (9.12). Reprinted from Strogatz (2001) with permissions, © 2001 Macmillan Magazines Ltd

Rotating Frame of Reference The order parameter $re^{i\psi}$ performs a free rotation for long times $t \to \infty$ in the thermodynamic limit. With

$$r(t) \to r, \qquad \psi(t) \to \Omega t, \qquad N \to \infty,$$

and one can transform via

$$\theta_i \to \theta_i + \psi = \theta_i + \Omega t, \qquad \dot{\theta}_i \to \theta_i + \Omega, \qquad \omega_i \to \omega_i + \Omega$$

to the rotating frame of reference. The governing equation (9.13) then becomes

$$\dot{\theta}_i = \omega_i - Kr\sin(\theta_i). \tag{9.14}$$

This expression is identical to the one for the case of two coupled oscillators, see (9.8), when substituting Kr by K. It then follows directly that $\omega_i = Kr$ constitutes a special point.

Drifting and Locked Components Eq. (9.14) has a fixpoint θ_i^* for which $\dot{\theta}_i = 0$, as determined by

$$Kr\sin(\theta_i^*) = \omega_i, \qquad |\omega_i| < Kr, \qquad \theta_i^* \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]. \tag{9.15}$$

The interval for θ_i^* is selected such that $\sin(\theta_i^*) \in [-1, 1]$.

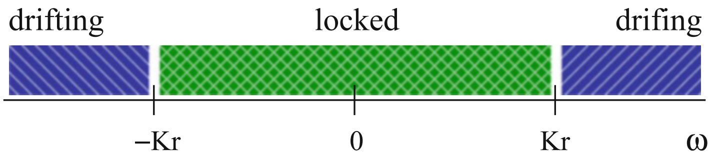

Fig. 9.3 The region of locked and drifting natural frequencies ω within the Kuramoto model

– LOCKED UNITS
  As we are working in the rotating frame of reference, $\dot{\theta}_i = 0$ means that the participating limit cycles oscillate for $|\omega_i| < Kr$ with the average frequency ψ, they are "locked" to ψ.
– DRIFTING UNITS
  For $|\omega_i| > Kr$, the participating limit cycle drifts, i.e. $\dot{\theta}_i$ never vanishes. They do, however, slow down when they approach the locked oscillators, compare (9.14).

The distinct dynamics of locked and drifting units is illustrated in Figs. 9.1, 9.2, and 9.3.

Stationary Frequency Distribution We denote with

$$\rho(\theta, \omega)\, d\theta$$

the fraction of oscillators with natural frequency ω that lie between θ and $\theta + d\theta$. Of relevance for the following will be the continuity equation for $\rho = \rho(\theta, \omega)$,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta}\left(\rho\, \dot{\theta}\right) = 0\,,$$

where $\rho\dot{\theta}$ is a current density, it generalizes the generic expression "density times velocity". The frequency distribution $\rho(\theta, \omega)$ will be inversely proportional to the speed

$$\dot{\theta} = \omega - Kr\sin(\theta)$$

in the stationary case, when $\dot{\rho} = 0$. The individual oscillators pile up at slow places, thinning out at fast places on the circle. Hence

$$\rho(\theta, \omega) = \frac{C}{|\omega - Kr\sin(\theta)|}, \qquad \int_{-\pi}^{\pi} \rho(\theta, \omega)\, d\theta = 1\,, \tag{9.16}$$

for $\omega > 0$, where C is an appropriate normalization constant.

Formulation of the Self-Consistency Condition We rewrite the self-consistency condition (9.12) for the molecular field as

$$\langle e^{i\theta}\rangle = \langle e^{i\theta}\rangle_{\text{locked}} + \langle e^{i\theta}\rangle_{\text{drifting}} = r\, e^{i\psi} \equiv r\,, \tag{9.17}$$

where the brackets $\langle\cdot\rangle$ denote population averages. In the last step we used the fact that one can set the average phase ψ to zero.

Locked Contribution The locked contribution to (9.17) is

$$\langle e^{i\theta}\rangle_{\text{locked}} = \int_{-Kr}^{Kr} e^{i\theta^*(\omega)} g(\omega)\, d\omega = \int_{-Kr}^{Kr} \cos\big((\theta^*(\omega)\big)\, g(\omega)\, d\omega\,,$$

where we have assumed $g(\omega) = g(-\omega)$ for the distribution $g(\omega)$ of the natural frequencies within the rotating frame of reference. Using (9.15),

$$\omega = Kr\sin(\theta_i), \qquad d\omega = Kr\cos(\theta^*)d\theta^*\,,$$

we obtain

$$\begin{aligned} \langle e^{i\theta}\rangle_{\text{locked}} &= \int_{-\pi/2}^{\pi/2} \cos(\theta^*)\, g(Kr\sin\theta^*)\, Kr\, \cos(\theta^*)\, d\theta^* \\ &= Kr \int_{-\pi/2}^{\pi/2} \cos^2(\theta^*) g(Kr\sin\theta^*)\, d\theta^*\,. \end{aligned} \tag{9.18}$$

We will make use of this expression further on.

Drifting Contribution The drifting contribution to the order parameter

$$\langle e^{i\theta}\rangle_{\text{drifting}} = \int_{-\pi}^{\pi} d\theta \int_{|\omega|>Kr} d\omega\, e^{i\theta} \rho(\theta,\omega) g(\omega) = 0$$

vanishes. Physically this is clear: oscillators that are not locked to the mean field cannot contribute to the order parameter. Mathematically, it follows from $g(\omega) = g(-\omega)$, together with $\rho(\theta+\pi,-\omega) = \rho(\theta,\omega)$ and $e^{i(\theta+\pi)} = -e^{i\theta}$.

Critical Coupling With the drifting component vanishing, the population average $\langle e^{i\theta}\rangle$ of the order parameter,

$$r = \langle e^{i\theta}\rangle \equiv \langle e^{i\theta}\rangle_{\text{locked}} = Kr \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta)\, g(Kr\sin\theta)\, d\theta\,. \tag{9.19}$$

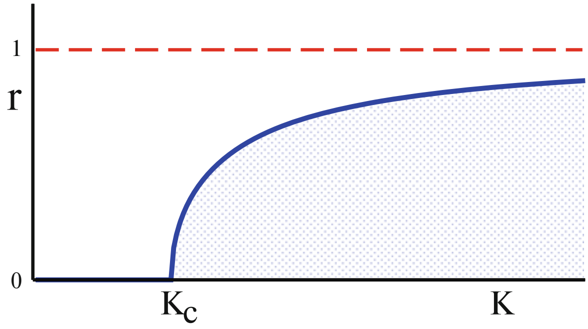

Fig. 9.4 The solution (9.22) for the order parameter in the Kuramoto model, $r = R_0\sqrt{1 - K_c/K}$, here with $R_0 = 1$

is given by the locked contribution (9.18). We simplified notation, using $\theta^* \to \theta$. When the coupling strength K is small, below a certain critical value K_c, only the trivial solution $r = 0$ exists. Above, for $K > K_c$, a finite order parameter $r > 0$ is stabilized, as illustrated in Fig. 9.4.

The critical coupling strength K_c is obtained considering the limes $r \to 0+$ in (9.19),

$$1 = K_c\, g(0) \int_{|\theta|<\frac{\pi}{2}} \cos^2(\theta)\, d\theta = K_c\, g(0) \frac{\pi}{2}, \qquad K_c = \frac{2}{\pi g(0)}, \tag{9.20}$$

where $\lim_{r\to 0} g(Kr\sin\theta) = g(0)$ has been used.[2]

Expansion around Criticality For the functional dependence of the order r around $K = K_c$ we expand (9.19) with respect to $r \ll 1$,

$$\begin{aligned} 1 &\simeq K \int_{|\theta|<\frac{\pi}{2}} d\theta\, \cos^2(\theta) \left[g(0) + \frac{g''(0)}{2}\big(Kr\sin\theta\big)^2 \right] \\ &= K\, g(0)\frac{\pi}{2} + K^3 r^2\, g''(0)\frac{\pi}{2\cdot 8} \end{aligned} \tag{9.21}$$

which holds in light of our previous assumption, namely that integrals antisymmetric in θ vanish.[3] Multiplying with K_c/K, we rewrite (9.21) as

$$1 - \frac{K_c}{K} = \frac{r^2}{R_0^2}, \qquad \frac{1}{R_0^2} = \frac{K^2 K_c\, |g''(0)|}{16/\pi},$$

[2] Note, that $\int dx \cos^2(x) = [\cos(x)\sin(x) + x]/2$, modulo a constant.

[3] One has $\int dx \cos^2(x)\sin^2(x) = x/8 - \sin(4x)/32$, plus an integration constant.

where we assumed $g''(0) < 0$, namely that the frequency distribution is locally maximal. Expanding $R_0 = R_0(K)$ into powers of $K - K_c$ would lead to higher-order corrections, one can hence set $R_0 = R_0(K_c)$. Together, we find

$$r = R_0\sqrt{1 - \frac{K_c}{K}}, \qquad K_c = \frac{2}{\pi g(0)}, \tag{9.22}$$

as illustrated in Fig. 9.4.

- One observes sqrt-scaling, namely that $r \sim \sqrt{K - K_c}$, which is the classical mean-field result.[4]
- For large K, the order parameter $r = \langle e^{i\theta} \rangle$ will approach unity, $\lim_{K\to\infty} r = 1$. For (9.22), the limit $K \to \infty$ would be recovered for $R_0 = 1$, generally this is however not the case.

Physics of Rhythmic Applause A well known application of the Kuramoto model concerns synchronization of the clapping of an audience after a performance, which happens when everybody claps at a slow frequency and in tact. In this case the distribution of "natural clapping frequencies" is quite narrow, and $K > K_c \propto 1/g(0)$.

When an individual wants to express particular satisfaction with the performance he/she increases the clapping frequency by about a factor of two, as measured experimentally. This increases the noise level, which depends primarily on the clapping frequency. Measurements have shown, see Fig. 9.5, that the distribution of natural clapping frequencies is broader when the clapping is fast. The resulting drop in $g(0)$ induces $K < K_c \propto 1/g(0)$, which implies in turn that synchronized clapping is dynamically unstable when the applause is intense.

Chimera States The original chimera, the one from Greek mythology, was a hybrid animal, part lion, goat and snake. In the world of synchronization phenoma, a "chimera" is a hybrid state, supporting both synchronous and asynchronous behaviors in networks of identical coupled oscillators. A prototypical system is

$$\dot{\theta}_i = \omega + \sum_{j\neq i} w_{ij} \sin(\theta_j - \theta_i - \alpha), \tag{9.23}$$

where the w_{ij} are coupling matrix elements of varying strength. Translation invariance, namely that $w_{ij} = w_{i-j}$, is allowed, e.g. for units that are distributed regularly in real space. Of relevance is a non-zero phase lag α, which induces reciprocal frustration. For large numbers N of participating units, strictly speaking when $N \to \infty$, partially synchronized states are stabilized for possibly exponentially

[4] More about critical scaling in Sect. 6.1 of Chap. 6.

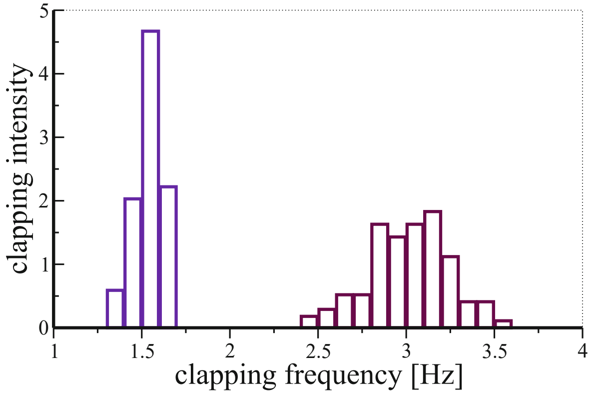

Fig. 9.5 Normalized distribution for the frequencies of the clappings of 100 individuals. Data from Néda et al. (2000a)

large transients. This is somewhat surprising, given that the natural frequencies are all identical.

9.3 Synchronization in the Presence of Time Delays

Synchronization phenomena need the exchange of signals from one subsystem to another. Real-world information exchange typically needs a certain time, which introduces time delays that become important when they are comparable to the intrinsic time scales of the individual subsystems.[5] Here we discuss the effect of time delays on the synchronization process.

Kuramoto Model with Time Delays Two limit-cycle oscillators, coupled via a time delay T are described by

$$\dot{\theta}_1(t) = \omega_1 + \frac{K}{2}\,\sin\big[\theta_2(t-T) - \theta_1(t)\big],$$

$$\dot{\theta}_2(t) = \omega_2 + \frac{K}{2}\,\sin\big[\theta_1(t-T) - \theta_2(t)\big].$$

In the steady state,

$$\theta_1(t) = \omega t, \qquad \theta_2(t) = \omega t + \Delta\theta^*, \tag{9.24}$$

[5] An introduction into the intricacies of time-delayed dynamical systems is given in Sect. 2.5 of Chap. 2.

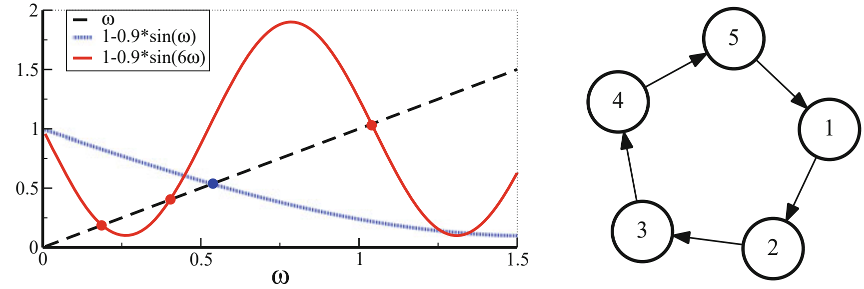

Fig. 9.6 **Left:** Graphical solution of the self-consistency condition (9.27), for time delays $T = 1/6$ (shaded/full line), having respectively one/three intersections (filled circles) with the diagonal (dashed line). The coupling constant is $K = 1.8$. **Right:** An example of a directed ring, containing five sites

there is a synchronous oscillation with a yet to be determined locking frequency ω and phase slip $\Delta\theta^*$. Using $\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$, we find

$$\omega = \omega_1 + \frac{K}{2}\big[-\sin(\omega T)\cos(\Delta\theta^*) + \cos(\omega T)\sin(\Delta\theta^*)\big], \tag{9.25}$$

$$\omega = \omega_2 + \frac{K}{2}\big[-\sin(\omega T)\cos(\Delta\theta^*) - \cos(\omega T)\sin(\Delta\theta^*)\big].$$

Taking the difference leads to

$$\Delta\omega = \omega_2 - \omega_1 = K\sin(\Delta\theta^*)\cos(\omega T), \tag{9.26}$$

which generalizes (9.9) to the case of a finite time delay T. Together, (9.25) and (9.26) determine the locking frequency ω and the phase slip $\Delta\theta^*$.

Multiple Synchronization Frequencies For finite time delays T, there are generally more than one solution for the synchronization frequency ω. For concreteness, we consider

$$\omega_1 = \omega_2 \equiv 1, \qquad \Delta\theta^* \equiv 0, \qquad \omega = 1 - \frac{K}{2}\sin(\omega T), \tag{9.27}$$

compare (9.26) and (9.25). This equation can be solved graphically, as shown in Fig. 9.6.

For $T \to 0$ the two oscillators are phase locked, oscillating with the original natural frequency $\omega = 1$. A finite time delay leads to a change of the synchronization frequency and, eventually, for large enough time delay T and couplings K, to multiple solutions for the locking frequency, with every second intersection shown in Fig. 9.6 being stable/unstable. The introduction of time delays induces consequently qualitative changes regarding the structuring of phase space.

Rings of Delayed-Coupled Oscillators An example illustrating the complexity potential arising from delayed couplings is given by rings of N oscillators, as illustrated in Fig. 9.6. Unidirectional couplings correspond to

$$\dot{\theta}_j = \omega_j + K \sin\left[\theta_{j-1}(t-T) - \theta_j(t)\right], \qquad j = 1, \ldots, N, \tag{9.28}$$

where periodic boundary conditions are implied, namely that $\theta_{N+1} \equiv \theta_1$. Specializing to the uniform case $\omega_j \equiv 1$, the network becomes invariant under discrete rotations, which allows for plane-wave solutions with frequency ω and momentum k,[6]

$$\theta_j = \omega t - k\, j, \qquad k = n_k \frac{2\pi}{N}, \qquad n_k = 0, \ldots, N-1, \tag{9.29}$$

where $j = 1, \ldots, N$. With this ansatz, the locking frequency ω is determined by the self-consistency condition

$$\omega = 1 + K \sin(k - \omega T). \tag{9.30}$$

In analogy to (9.27), a set of solutions with distinct frequencies ω can be found for a given momentum k. The resulting dynamics is characterized by complex spatio-temporal symmetries, in term of the constituent units $\theta_j(t)$, which oscillate fully in phase only for vanishing momentum $k \to 0$.

The plane-wave ansatz (9.29) describes phase-locked oscillations. Additional solutions with drifting units cannot be excluded, they may show up in numerical simulations. It is important to remember in this context that initial conditions in the entire interval $t \in [-T, 0]$ need to be provided.[7]

9.4 Synchronization Mechanisms

The synchronization of limit cycle oscillators discussed in Sect. 9.2 is mediated by a molecular field, viz by an averaged quantity. Averaging plays a central role in many synchronization processes and may act both on a local basis and on a global level.

[6] In the complex plane, $\psi_j(t) = \mathrm{e}^{i\theta_j(t)} = \mathrm{e}^{i(\omega t - kj)}$ corresponds to a plane wave on a periodic ring. Eq. (9.28) is then equivalent to the phase evolution of the wavefunction $\psi_j(t)$. The system is invariant under translations $j \to j+1$, which implies that the discrete momentum k is a good quantum number, in the jargon of quantum mechanics. The periodic boundary condition $\psi_{j+N} = \psi_j$ is satisfied for $k = 2\pi n_k/N$.

[7] Initial functions for time delay systems are discussed in Sect. 2.5 of Chap. 2.

Alternatively, synchronization can be driven by the casual influence of temporally well defined events, a route to synchronization we will discuss in Sect. 9.4.2.

9.4.1 Aggregate Averaging

The coupling term of the Kuramoto model, see (9.6), contains differences $\theta_i - \theta_j$ in the respective dynamical variables θ_i and θ_j. With an appropriate sign of the coupling constant, this coupling corresponds to a driving force towards the mean,

$$\theta_1 \to \frac{\theta_1 + \theta_2}{2}, \qquad \theta_2 \to \frac{\theta_1 + \theta_2}{2} ,$$

which competes with the time development of the individual oscillators whenever the respective natural frequencies ω_i and ω_j are distinct. On their own, the individual units would rotate with different angular velocities. As carried out in Sect. 9.2, a detailed analysis is necessary when studying this competition between the synchronizing effect of the coupling and the de-synchronizing influence of a non-trivial natural frequency distribution.

Aggregate Variables Generalizing above considerations, we consider a generic set of dynamical variables x_i. When isolated the evolution rule $\dot{x}_i = f_i(x_i)$ holds. The geometry of the couplings will be determined by the normalized weighted adjacency matrix,[8]

$$A_{ij}, \qquad \sum_j A_{ij} = 1 .$$

The matrix elements are $A_{ij} > 0$ if the units i and j are coupled, and zero otherwise, with A_{ij} representing the relative weight of the link. We define now aggregate variables $\bar{x}_i = \bar{x}_i(t)$ by

$$\bar{x}_i = (1 - \kappa_i)x_i + \kappa_i \sum_j A_{ij} x_j , \tag{9.31}$$

where $\kappa_i \in [0, 1]$ is a local coupling strength. The aggregate variables $\bar{x}_i$ correspond to a superposition of x_i with the weighted mean activity $\sum_j A_{ij} x_j$ of all its neighbors.

[8] The connection of the adjacency matrix to the graph spectrum is discussed in Sect. 1.2 of Chap. 1.

Coupling via Aggregate Averaging A general class of dynamical networks can be formulated in terms of aggregate variables, through

$$\dot{x}_i = f_i(\bar{x}_i), \qquad i = 1, \dots, N, \tag{9.32}$$

with the $\bar{x}_i$ given by (9.31). The f_i describe local dynamical systems which could be, e.g., harmonic oscillators, relaxation oscillators, or chaotic systems.

Expansion Around the Synchronized State In order to expand (9.32) around the globally synchronized state, we first rewrite the aggregate variables as

$$\begin{aligned} \bar{x}_i &= (1-\kappa_i)x_i + \kappa_i \sum_j A_{ij}(x_j - x_i + x_i) \\ &= x_i\Big(1 - \kappa_i + \kappa_i \sum_j A_{ij}\Big) + \kappa_i \sum_j A_{ij}(x_j - x_i) \\ &= x_i + \kappa_i \sum_j A_{ij}(x_j - x_i), \end{aligned} \tag{9.33}$$

where we have used the normalization $\sum_j A_{ij} = 1$. The differences in activities $x_j - x_i$ are small close to the synchronized state,

$$f_i(\bar{x}_i) \approx f_i(x_i) + f_i'(x_i)\,\kappa_i \sum_j A_{ij}(x_j - x_i). \tag{9.34}$$

Differential couplings $\sim (x_j - x_i)$ between the nodes of the network are hence equivalent, close to synchronization, to the aggregate averaging of the local dynamics via the respective $\bar{x}_i$.

General Coupling Functions One may go one step further and define with

$$\dot{x}_i = f_i(x_i) + h_i(x_i) \sum_j g_{ij}(x_j - x_i) \tag{9.35}$$

a general system of $i = 1, \dots, N$ dynamical units interacting via coupling functions $g_{ij}(x_j - x_i)$, which are respectively modulated through the $h_i(x_i)$. Expanding (9.35) close to the synchronized orbit yields

$$\dot{x}_i \approx f_i(x_i) + h_i(x_i) \sum_j g_{ij}'(0)(x_j - x_i), \qquad h_i(x_i)g_{ij}'(0) \mathrel{\hat{=}} f_i'(x_i)\kappa_i A_{ij}.$$

The equivalence of $h_i(x_i)g_{ij}'(0)$ and $f_i'(x_i)\kappa_i A_{ij}$, compare (9.34), is only local in time, which is however sufficient for a local stability analysis. The synchronized state of the system with differential couplings, see (9.35), is hence locally stable whenever the corresponding system with aggregate couplings is equally stable against perturbations, compare (9.32).

Synchronization via Aggregated Averaging The equivalence of (9.32) and (9.35) tells us that the driving forces leading to synchronization are aggregated averaging processes of neighboring dynamical variables.

Till now we considered exclusively globally synchronized states. Synchronization processes may lead however also to more complicated states, we will mention alternative possibilities. For all cases, above discussion concerning the role of aggregate averaging retains its validity.

– We saw, when discussing the Kuramoto model in Sect. 9.2, that generically not all nodes of a network participate in a synchronization process. For the Kuramoto model the oscillators with natural frequencies far away from the average do not lock to the time development of the order parameter, see Fig. 9.3, retaining drifting trajectories.
– Generically, synchronization takes the form of coherent time evolution with phase lags, we have seen an example when discussing two coupled oscillators in Sect. 9.2. The synchronized orbit is then

$$x_i(t) = x(t) + \Delta x_i\,, \qquad \Delta x_i \text{ const.}\,,$$

which implies that all elements $i = 1, \ldots, N$ are locked.

Partially synchronized states involving an aggregated averaging over particularly large numbers of distinct units are "chimera states".[9]

Stability Analysis of the Synchronized State The stability of a globally synchronized state, $x_i(t) = x(t)$ for $i = 1, \ldots, N$, can be determined by considering small perturbations, viz

$$x_i(t) = x(t) + \delta_i\, c^t\,, \qquad |c|^t = e^{\lambda t}\,, \tag{9.36}$$

where λ is a Lyapunov exponent. The eigenvector $(\delta_1, \ldots, \delta_N)$ of the perturbation is determined by the equations of motion linearized at a given point around the synchronized trajectory. There is one Lyapunov exponent for every eigenvector, N in all,

$$\lambda^\alpha, \qquad (\delta_1^\alpha, \ldots, \delta_N^\alpha), \qquad \alpha = 1, \ldots, N\,.$$

[9] Chimera states are shortly discussed on page 336.

One of the exponents characterizes the flow along the synchronized direction. The synchronized state is stable if all the remaining λ^j $(j = 2, \ldots, N)$ Lyapunov exponents are negative when averaged over the orbit.

Coupled Logistic Maps As an example for aggregate averaging we consider two coupled logistic maps,[10]

$$x_i(t+1) = r\,\bar{x}_i(t)\big(1 - \bar{x}_i(t)\big), \qquad i = 1,\ 2, \qquad r \in [0, 4]\,, \tag{9.37}$$

with

$$\bar{x}_1 = (1-\kappa)x_1 + \kappa x_2, \qquad \bar{x}_2 = (1-\kappa)x_2 + \kappa x_1\,,$$

where $\kappa \in [0, 1]$ is the coupling strength. Using (9.36) as an ansatz we obtain

$$c\begin{pmatrix}\delta_1\\ \delta_2\end{pmatrix} = r\big(1 - 2x(t)\big)\begin{pmatrix}(1-\kappa) & \kappa\\ \kappa & (1-\kappa)\end{pmatrix}\begin{pmatrix}\delta_1\\ \delta_2\end{pmatrix}$$

in linear order in the δ_i around the synchronized orbit $x = \bar{x}_i = x_i$, for $i = 1, 2$. The expansion factor c corresponds to an eigenvalues of the Jacobian, the usual situation for discrete maps. As a result, we find two local pairs of eigenvalues and eigenvectors, namely

$$c_1 = r(1-2x) \qquad\qquad (\delta_1, \delta_2) = \frac{1}{\sqrt{2}}(1, 1)$$

$$c_2 = r(1-2x)(1-2\kappa) \qquad\qquad (\delta_1, \delta_2) = \frac{1}{\sqrt{2}}(1, -1)$$

corresponding to the respective local Lyapunov exponents, $\lambda = \log|c|$,

$$\lambda_1 = \log|r(1-2x)|, \qquad \lambda_2 = \log|r(1-2x)(1-2\kappa)|\,. \tag{9.38}$$

As expected, $\lambda_1 > \lambda_2$, since λ_1 corresponds to a perturbation along the synchronized orbit. The overall stability of the synchronized trajectory can be examined by averaging above local Lyapunov exponents over the full time development, which defines the "maximal Lyapunov exponent".[11]

Synchronization of Coupled Chaotic Maps The Lyapunov exponents defined by (9.38) need to be evaluated numerically. However, one can obtain an lower bound

[10] For an illustration of the logistic map see Fig. 2.18 in Chap. 2.

[11] Maximal Lyapunov exponents are discussed together with the theory of discrete maps in Chap. 2.

for the coupling strength κ necessary for stable synchronization by observing that $|1-2x| \le 1$ and hence

$$|c_2| \le r|1-2\kappa| .$$

The synchronized orbit is stable for $|c_2| < 1$. For $\kappa \in [0, 1/2]$ we obtain

$$|c_2| \le r(1-2\kappa), \qquad \kappa > \frac{r-1}{2r}$$

for the lower bound for κ. For the maximal reproduction rate, $r = 4$, synchronization is possible for $3/8 < \kappa \le 1/2$. Given that the logistic map is chaotic for $r > r_\infty \approx 3.57$, this results proves that chaotic coupled systems may synchronize. Interestingly, synchronization through aggregate averaging is achieved in a single step for $\kappa = 1/2$. This holds as $\bar{x}_1 = \bar{x}_2$ for $\kappa = 1/2$.

9.4.2 Causal Signaling

The synchronization of limit cycle oscillators discussed in Sect. 9.2 is in general slow, see Fig. 9.2, as the information between distinct oscillators is exchanged only indirectly, namely via a molecular field, which is in turn an averaged quantity. Synchronization may be sustantially faster when local dynamical units influence each other with precisely timed signals, the route to synchronization discussed here.

Relaxation oscillators, as studied in the next section, have non-uniform cycles, which implies that the timing of the stimulation of one element by another is important. This is a characteristic property of real-world neurons, which emit short pluses, the spikes, along the efferent axon. The corresponding artificial neuron models are called “integrate-and-fire” neurons. Relaxation oscillators are hence well suited to study the phenomena of synchronization via causal signaling.

Terman–Wang Oscillators There are many variants of relaxation oscillators relevant for describing integrate-and-fire neurons, starting from the classical Hodgkin-Huxley equations. We discuss here a particularly transparent dynamical system, as introduced originally by Terman and Wang, namely

$$\begin{array}{ll} \dot{x} = f(x) - y + I & \qquad f(x) = 3x - x^3 + 2 \\ \dot{y} = \epsilon\big(g(x) - y\big) & \qquad g(x) = \alpha\big(1 + \tanh(x/\beta)\big) \end{array} . \tag{9.39}$$

In this setting, x corresponds in neural terms to the membrane potential, with I representing an external stimulation of the neural oscillator. The amount of

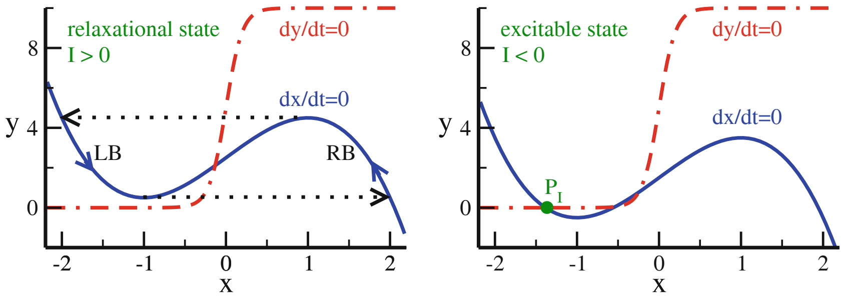

Fig. 9.7 The $\dot{y} = 0$ (red dashed-dotted lines), and $\dot{x} = 0$ (blue full lines) isocline of the Terman–Wang oscillator (9.39), here for $\alpha = 5$, $\beta = 0.2$, $\epsilon = 0.1$. **Left:** $I = 0.5$, shown is the relaxation cycle for $\epsilon \ll 1$ (dotted line, arrows). **Right:** $I = -0.5$, which has a stable fixpoint, P_I (filled green dot)

dissipation is given by[12]

$$\frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} = 3 - 3x^2 - \epsilon = 3(1 - x^2) - \epsilon\,.$$

For small $\epsilon \ll 1$ the system takes up energy for membrane potentials $|x| < 1$, dissipating energy for $|x| > 1$.

Fixpoints The fixpoints of (9.39) are determined via

$$\begin{aligned} \dot{x} &= 0 \qquad & y &= f(x) + I \\ \dot{y} &= 0 \qquad & y &= g(x) \end{aligned}$$

by the intersection of the two functions $f(x) + I$ and $g(x)$, as illustrated in Fig. 9.7. Hence, there are two parameter regimes.

– For $I \geq 0$ a single unstable fixpoint (x^*, y^*) exists, with $x^* \simeq 0$.
– For $I < 0$ and $|I|$ large enough, two additional fixpoints show up, as given by the crossing of the sigmoid $\alpha\big(1 + \tanh(x/\beta)\big)$ with the left branch (LB) of the cubic $f(x) = 3x - x^3 + 2$. One of the new fixpoints is stable.

The stable fixpoint P_I is indicated in Fig. 9.7.

[12] The divergence of the flow is equivalent to the relative contraction/expansion of phase space, $\Delta V/V$, as discussed in Sect. 3.1.1 of Chap. 3.

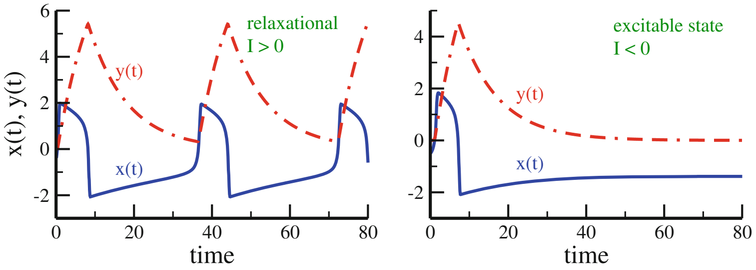

Fig. 9.8 Sample trajectories of $x(t)$ (blue line) and $y(t)$ (dashed-dotted line). Results for the Terman–Wang oscillator (9.39), together with $\alpha = 5$, $\beta = 0.2$, $\epsilon = 0.1$. **Left:** Spiking behavior for $I = 0.5$, characterized by silent/active phases for negative/positive x. **Right:** Relaxation to a stable fixpoint for $I = -0.5$

– RELAXATION REGIME
  For the $I > 0$ the Terman–Wang oscillator relaxes to a periodic solution, see Fig. 9.7, which is akin to the relaxation oscillation observed for the Van der Pol oscillator.[13]
– EXCITABLE STATE
  For $I < 0$ the system settles into a stable fixpoint, becoming however active again when an external input shifts I into positive region. The system is said to be "excitable".

The presence of an excitable state is a precondition for neuronal models.

Silent and Active Phases In the relaxation regime, the periodic solution jumps rapidly, for $\epsilon \ll 1$, between trajectories that approach closely the right branch (RB), and the left branch (LB) of the $\dot{x} = 0$ isocline. The times spent respectively on the two branches may however differ substantially, as indicated in Figs. 9.7 and 9.8. Indeed, the limit cycle is characterized by two distinct types of flows.

– SILENT PHASE
  The LB ($x < 0$) of the $\dot{x} = 0$ isocline is comparatively close to the $\dot{y} = 0$ isocline, the system spends therefore considerable time on the LB. This leads to a "silent phase", which is equivalent to a refractory period.
– ACTIVE PHASE
  The RB ($x > 0$) of the $\dot{x} = 0$ isocline is far from the $\dot{y} = 0$ isocline, when $\alpha \gg 1$, see Fig. 9.7. The time development $\dot{y}$ is hence fast on the RB, which defines the "active phase".

[13] The van der Pol oscillator, treated in Sect. 3.2 of Chap. 3, has to regimes, corresponding respectively to small/large adaptive term.

The relative rate of the time development through $\dot{y}$ between silent and active phase is determined by the parameter α, as defined in (9.39).

Spikes as Activity Bursts In its relaxation phase, the Terman–Wang oscillator can be considered as a spontaneously spiking neuron, see Fig. 9.8. When $\alpha \gg 1$, the active phase undergoes a compressed burst, viz a "spike".

The differential equation defined in (9.39) are examples of a standard technique within dynamical systems theory, the coupling of a slow variable to a fast variable, which results in a separation of time scales. When the slow variable $y = y(t)$ relaxes below a certain threshold, see Fig. 9.8, the fast variable $x = x(t)$ responds rapidly, resetting on the way the slow variable.

Synchronization via Fast Threshold Modulation Limit cycle oscillators can synchronize, albeit slowly, via a common molecular field, as discussed in Sect. 9.2. Substantially faster synchronization can be achieved for networks of interacting relaxation oscillators via "fast threshold synchronization".

The idea is simple. Relaxation oscillators are characterized by distinct phases during their cycle, which we denoted respectively silent and active states. These two states take functionally distinct roles when assuming that a neural oscillator is able to influence the threshold I of other units during its active phase, namely by emitting a short burst of activity, viz by spiking. For our basic equations (9.39), this concept translates into

$$I \to I + \Delta I, \qquad \Delta I > 0\,,$$

such that the second neural oscillator changes from an excitable state to the oscillating state. This process is illustrated graphically in Fig. 9.9, it corresponds to a signal sent from the first to the second dynamical unit. In neural terms, when the first neuron fires, the second neuron follows suit.

Activity Propagation A basic model illustrating the propagation of activity is given by a line of $i = 1, \dots, N$ coupled oscillators $(x_i(t), y_i(t))$,

$$\boxed{1} \Rightarrow \boxed{2} \Rightarrow \boxed{3} \Rightarrow \dots$$

which are assumed to be initially all in the excitable state, with $I_i \equiv -0.5$. Inter-unit coupling via fast threshold modulation corresponds to

$$\Delta I_i(t) = \Theta(x_{i-1}(t))\,, \tag{9.40}$$

where $\Theta(x)$ is the Heaviside step function. That is, we define an oscillator i to be in its active phase whenever $x_i > 0$. The resulting dynamics is shown in Fig. 9.10. The

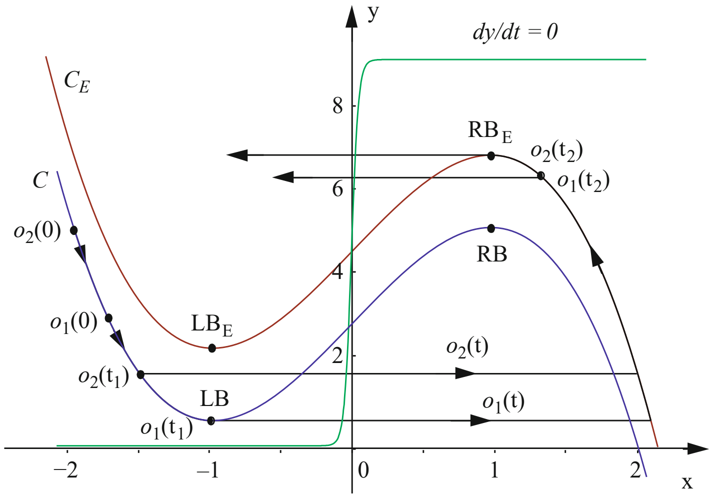

Fig. 9.9 Fast threshold modulation for two excitatory coupled relaxation oscillators, see (9.39), denoted symbolically as $o_1 = o_1(t)$ and $o_2 = o_2(t)$. When o_1 jumps at $t = t_1$ from the LB to the RB, it becomes active. The cubic $\dot{x} = 0$ isocline for o_2 is consequently raised from C to C_E. This induces o_2 to jump as well from left to right. Note that the jumping from the right branch (RB and RB_E) back to the left branches occurs in the reverse order, o_2 jumps first

chain is driven by setting the first oscillator of the chain into the spiking state for a certain period of time. All other oscillators spike consecutively in rapid sequence.

9.5 Piecewise Linear Dynamical Systems

For an understanding of a dynamical system of interest, it is often desirable to study a simplified version. A standard technique involves to approximate non-linearities in the flow by piecewise linear functions. This technique allows at times to obtain analytic expressions for the orbits, which is a highly non-trivial outcome. For the case of limit-cycle oscillators, this approach can be carried out for our reference system, as defined by (9.39).

Piecewise Linear Terman–Wang Oscillators We approximate the nullclines of (9.39) as

$$f(x) \to \begin{cases} 1 - x & \text{for } x > 0 \\ -1 - x & \text{for } x < 0 \end{cases}, \qquad g(x) \to \begin{cases} g_0 & \text{for } x > 0 \\ -g_0 & \text{for } x < 0 \end{cases}, \tag{9.41}$$

as illustrated in Fig. 9.11. No fixpoint exists when $g_0 > 1$. In comparison to the original functionality shown in Fig. 9.7, the $\dot{x} = 0$ and $\dot{y} = 0$ isoclines are now discontinuous, but otherwise piecewise linear or constant functions.

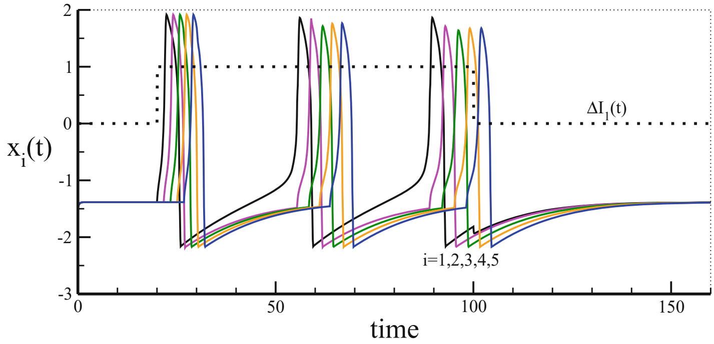

Fig. 9.10 As an example of synchronization via causal signaling, shown are sample trajectories $x_i(t)$ (lines) for a line of coupled Terman–Wang oscillators, as defined by (9.39). The relaxation oscillators are in excitable states, with $\alpha = 10$, $\beta = 0.2$, $\epsilon = 0.1$ and $I = -0.5$. For $t \in [20, 100]$ a driving current $\Delta I_1 = 1$ is added to the first oscillator (dotted line). In consequence, x_1 starts to spike, driving the other oscillators one by one via fast threshold modulation

Piecewise Linear Dynamics With $I = 0$, the system becomes inversion symmetric. It is then sufficient to consider with

$$\dot{x} = 1 - x - y, \qquad \dot{y} = \epsilon(g_0 - y) \tag{9.42}$$

positive $x > 0$. As a further simplification we set $\epsilon = 1$. The solution of (9.42) takes the form

$$x = \big(\tilde{x} - \tilde{y}t\big)\mathrm{e}^{-t} + 1 - g_0, \qquad y = \tilde{y}\mathrm{e}^{-t} + g_0\,, \tag{9.43}$$

which can be verified by direct differentiation. The trajectory is determined by three free parameters, $\tilde{x}/\tilde{y}$ and the yet unkown period T of the limit cycle.

Matching Conditions The individual segments of a piecewise linear system need to be joined once explicit expressions for the solutions have been found. The resulting "matching conditions" determine the parameters appearing in the respective analytic solution.

Starting at $x(0) = 0$, the orbit crosses the $x = 0$ line the next time after half-a-period, which is evident from Fig. 9.11, with the y-component changing in between. The matching conditions are therefore

$$x(0) = 0 = x(T/2), \qquad y(T/2) = -y(0)\,. \tag{9.44}$$

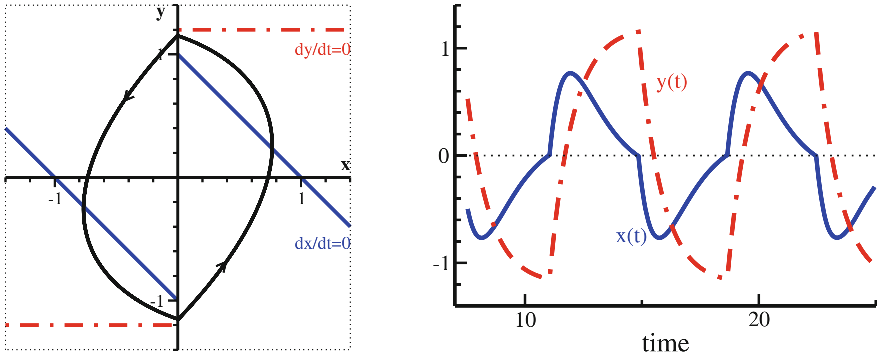

Fig. 9.11 The piecewise linear Terman–Wang system, as described by (9.41) and (9.42). The parameters are $g_0 = 1.2$ and $\epsilon = 1$. **Left:** The phase space (x, y). Shown are the piecewise linear $\dot{x} = 0$ isocline (blue), the piecewise constant $\dot{y} = 0$ isolcine (red), and the resulting limit cycle (black). **Right:** As a function of time, kinks show up when the orbit crosses $x = 0$. Compare Figs. 9.7, and 9.8

The starting condition $x(0) = 0$ yields $\tilde{x} = g_0 - 1$, see (9.43), which leads to

$$\begin{aligned} g_0 - 1 &= \left(g_0 - 1 - \tilde{y}\frac{T}{2}\right) \mathrm{e}^{-T/2}, \\ 2g_0 &= -\tilde{y}\left[\mathrm{e}^{-T/2} + 1\right]. \end{aligned} \tag{9.45}$$

As a result one has two condition for two parameters, $\tilde{y}$ and T.

Limit Cycle Period Given g_0, the only free model parameter, one can eliminate either $\tilde{y}$ or T from (9.45). Doing so, one obtains a self-consistency equation for the remaining parameter that can be solved numerically.

Alternatively we may ask which g_0 would lead to a given period T. In this case T is fixed und one eliminates $\tilde{y}$. From

$$\tilde{y} = \frac{2}{T}\left[1 - \mathrm{e}^{T/2}\right](g_0 - 1), \qquad \tilde{y}\left[1 + \mathrm{e}^{-T/2}\right] = -2g_0$$

one finds

$$-2g_0 = \frac{2}{T}(g_0 - 1)\left[1 - \mathrm{e}^{T/2}\right]\left[1 + \mathrm{e}^{-T/2}\right]$$

and hence

$$g_0 = \frac{2\sinh(T/2)}{2\sinh(T/2) - T}, \tag{9.46}$$

a remarkably simple expression. E.g., the orbit shown in Fig. 9.11 has a period of $T \approx 7.6$, which is consistent with $g_0 = 1.2$,.

Slow/Fast Oscillations The advantage of an analytic result is that it allows to study limiting behaviors. On the positive y-axis, the orbit has to cross $x = 0$ in the interval $[1, g_0]$, as evident from Fig. 9.11, given that $y = 1$ and $y = g_0$ mark the upper/lower border of respectively the $\dot{x} = 0$ and $\dot{y} = 0$ isoclines. From (9.46) one finds

$$\lim_{T\to\infty} g_0 = 1\,, \tag{9.47}$$

an expected relation. The oscillatory behavior slows down progressively when $g_0 \to 1$, viz when the orbit needs to squeeze through the closing gap between the two isoclines. In the opposite limit, when $T \to 0$, the Taylor expansion $\sinh(z) \approx z + z^3/6$ leads to

$$g_0 \sim \frac{3}{T^2}, \qquad T \sim \sqrt{\frac{3}{g_0}}, \qquad T \ll 1\,. \tag{9.48}$$

The flow accelerates as $1/\sqrt{g_0}$ when g_0 becomes large.

9.6 Synchronization Phenomena in Epidemics

There are illnesses, like measles, that come and go recurrently. Looking at the statistics of local measle outbreaks presented in Fig. 9.12, one can observe that outbreaks may occur in quite regular time intervals within a given city. Interestingly though, these outbreaks can be either in phase (synchronized) or out of phase between different cities.

The oscillations in the number of infected persons are definitely not harmonic, sharing instead many characteristics with relaxation oscillations, which typically have silent and active phases, compare Sect. 9.4.2.

SIRS Model The reference model for infectious diseases is the SIRS model. It contains three compartments, in the sense that individuals belong at any point in time to one of three possible classes.

S : susceptible,
I : infected,
R : recovered.

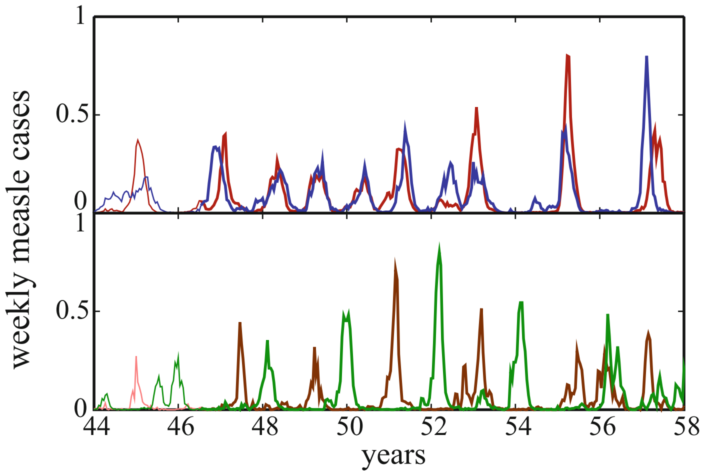

Fig. 9.12 Observation of the number of infected persons in a study on illnesses. **Top:** Weekly cases of measle cases in Birmingham and Newcastle (red/blue lines). **Bottom:** Weekly cases of measle cases in Cambridge and Norwich (green/brown lines). Data from He and Stone (2003)

The dynamics is governed by the following rules:

– INFECTION PROCESS
  With a certain probability, susceptibles pass to the infected state after coming into contact with one infected individual.
– RECOVERING
  Infected individuals recover from the infection after a given period of time, τ_I, passing to the recovered state.
– IMMUNITY
  For a certain period, τ_R, recovered individuals are immune, returning to the susceptible state once when immunity is lost.

These three steps complete the infection circle S $\rightarrow$ I $\rightarrow$ R $\rightarrow$ S. When $\tau_R \rightarrow \infty$ (lifelong immunity) the model reduces to the SIR-model.

Sum Rule The SIRS model can be implemented both for discrete and for continuous time, we start with the former. The infected phase is normally short, which allows to use τ_I as the unit of time, setting $\tau_I = 1$. The recovery time τ_R is then a multiple of $\tau_I = 1$. We define with

x_t the fraction of infected individuals at time $t = 1, 2, 3, \ldots$, and with
s_t the percentage of susceptible individuals at time t.

S	S	S	I	R	R	R	S	S	state
1	2	3	4	5	6	7	8	9	time

Fig. 9.13 Example of the course of an individual infection within the discrete-time SIRS model with an infection period $\tau_I = 1$ and a recovery duration $\tau_R = 3$. The number of individuals recovering at time t is the sum of infected individuals at times $t-1$, $t-2$ and $t-3$, compare (9.49)

Normalizing the total number of individuals to one, the sum rule

$$s_t = 1 - x_t - \sum_{k=1}^{\tau_R} x_{t-k} = 1 - \sum_{k=0}^{\tau_R} x_{t-k} \tag{9.49}$$

holds, as the fraction of susceptible individuals is just one minus the number of infected individuals minus the number of individuals in the recovery state, compare Fig. 9.13.

Discrete Time SIRS Model We denote with a the rate of transmitting an infection when there is a contact between an infected individual and a susceptible individual. Using the recursion relation (9.49) one obtains with

$$x_{t+1} = a x_t s_t = a x_t \left(1 - \sum_{k=0}^{\tau_R} x_{t-k}\right) \tag{9.50}$$

the discrete-time SIRS model.

Relation to the Logistic Map For $\tau_R = 0$, the discrete time SIRS model (9.50) reduces to the logistic map,

$$x_{t+1} = a x_t \left(1 - x_t\right) .$$

The trivial fixpoint $x_t \equiv 0$ is globally attracting when $a < 1$, which implies that the illness dies out. The non-trivial steady state is

$$x^{(1)} = 1 - \frac{1}{a}, \qquad \text{for} \quad 1 < a < 3 .$$

At $a = 3$ a Hopf bifurcation is observed, beyond an oscillation of period two. Increasing the growth rate a further, a transition to deterministic chaos takes place.

The SIRS model (9.50) shows somewhat similar behaviors. Due to the memory terms x_{t-k}, the resulting oscillations may however depend on the initial condition.

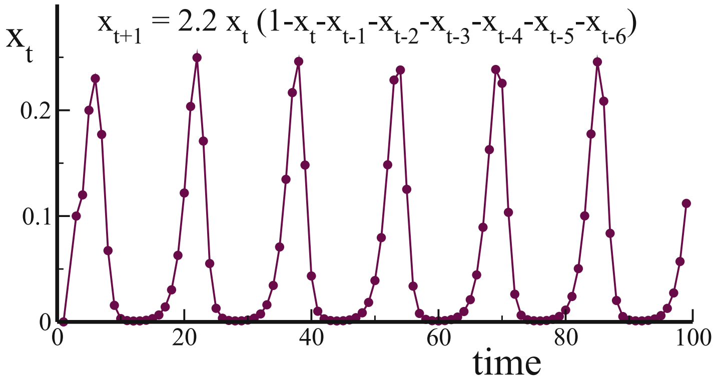

Fig. 9.14 Example of a solution to the SIRS model, see (9.50), for $\tau_R = 6$. The number of infected individuals might drop to very low values during the silent phase in between two outbreaks, as most of the population is first infected and then immunized during an outbreak

For extended immunities, $\tau_R \gg \tau_I$, features characteristic of relaxation oscillators are found, compare Fig. 9.14.

Two Coupled Epidemic Centers What happens when two epidemic centers are weakly coupled? We use

$$s_t^{(1,2)}, \qquad x_t^{(1,2)}$$

to denote the fraction of susceptible/infected individuals in the respective cities. Different dynamical couplings are conceivable, via exchange or visits of susceptible or infected individuals. With

$$\begin{aligned} x_{t+1}^{(1)} &= a\left(x_t^{(1)} + e\, x_t^{(2)}\right) s_t^{(1)}, \\ x_{t+1}^{(2)} &= a\left(x_t^{(2)} + e\, x_t^{(1)}\right) s_t^{(2)}, \end{aligned} \tag{9.51}$$

one describes a situation where a small fraction e of infected individuals visits the respective other center. In addition, one needs to apply the sum rule (9.49) to both centers.

For $e = 1$ there is no distinction between the two centers, which means that the dynamics described by (9.51) can be merged via $x_t = x_t^{(1)} + x_t^{(2)}$ and $s_t = s_t^{(1)} + s_t^{(2)}$. A single combined epidemic population remains. The situation is similar to the case of to coupled logistic maps, as given by (9.37). This is not surprising, since the coupling term in (9.51) is based on aggregate averaging.

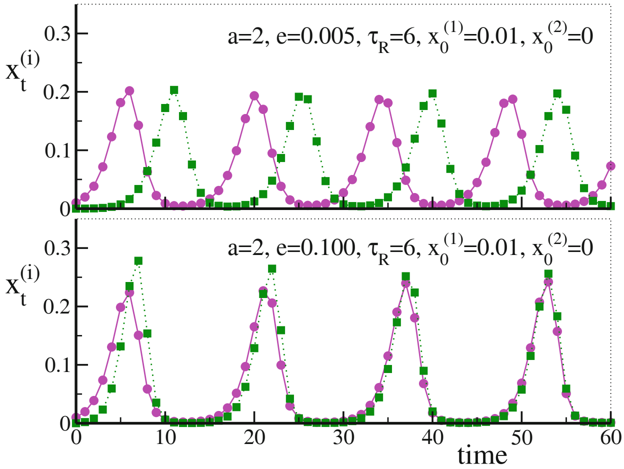

Fig. 9.15 Time evolution of the fraction of infected individuals $x^{(1)}(t)$ and $x^{(2)}(t)$ within the SIRS model, see (9.51), for two epidemic centers $i = 1, 2$ with recovery times $\tau_R = 6$ and infection rates $a = 2$, see (9.50). For a very weak coupling $e = 0.005$ (top) the outbreaks occur out of phase, for a moderate coupling $e = 0.1$ (bottom) in phase

In Phase Versus Out of Phase Synchronization We have seen in Sect. 9.2, that relaxation oscillators coupled strongly during their active phases synchronize rapidly. Here the active phase corresponds to an outbreak of the illness, which is however not pre-given, but a self-organized event. In this sense one implements with (9.51) a coupling equivalent to a fast threshold modulation, as discussed in Sect. 9.4.2, albeit based on a self-organized process.

In Fig. 9.15 we present the results from a numerical simulation of the coupled model, illustrating the typical behavior. We see that the epidemic outbreaks occur in the SIRS model indeed in phase for moderate to large coupling constants e. In contrast, for very small couplings e between the two centers of epidemics, the synchronization phase flips to antiphase. This phenomenon is observed in reality, compare Fig. 9.12.

Time Scale Separation The reason for the occurrence of out of phase synchronization is the emergence of two separate time scales in the limit $t_R \gg 1$ and $e \ll 1$. A small seed $\sim eax^{(1)}s^{(2)}$ of infections in the second city needs substantial time to induce a full-scale outbreak, even via exponential growth, when e is too small. But in order to remain in phase with the current outbreak in the first city the outbreak occurring in the second city may not lag too far behind. When the dynamics is symmetric under exchange $1 \leftrightarrow 2$ the system then settles in antiphase cycles.

9.6.1 Continuous Time SIRS Model

Changing notation slightly, we use S, I, and $R = 1 - I - S$ for the densities of susceptible, infected, and recovered.

Continuous Time Limit Substituting the discrete-time approximation for the derivative,

$$\dot{I} \approx \frac{I_{t+\Delta t} - I_t}{\Delta t}, \qquad \Delta t = 1\,,$$

in (9.50) yields

$$\dot{I} = aI(1 - I - R) - I = aIS - I\,, \tag{9.52}$$

which has a simple interpretation. The number of infected increases with a rate that is proportional to the infection probability a and to the densities I and S of infected and susceptibles. The term $-I$ describes per-time recovering.

Continuous Time SIRS Model For the discrete case, we took the duration of the illness as the unit of time, as illustrated in Fig. 9.13. Using (9.52) and rescaling time by $T = 1/\lambda$, one obtains with

$$\dot{S} = -gIS + \delta R, \qquad \dot{I} = gIS - \lambda I, \qquad \dot{R} = +\lambda I - \delta R \tag{9.53}$$

the standard continuous time SIRS model, where $g = \lambda a$ and $\delta = \lambda/\tau_R$. Recall that τ_R denotes the average time an individual remains immune, as measured in units of illness duration T.

Endemic Fixpoint The endemic state is defined by the fixpoint condition

$$S = \frac{\lambda}{g}, \qquad I = \frac{g - \lambda}{\delta + \lambda}, \qquad R = \frac{\lambda}{\delta}\,\frac{g - \lambda}{\delta + \lambda}\,, \tag{9.54}$$

which follows directly from (9.53). It exists when the infection rate g is larger than the recovery rate λ, namely when $g > \lambda$. Otherwise, the illness dies out. One can show that the endemic state is always stable. The discrete SIRS model supports in contrast solutions corresponding to periodic outbreaks, as shown in Fig. 9.13.

Analytic Solution of the SIR Model Life-long immunity, which corresponds to setting $\delta = 0$ in (9.53), reduces the SIRS model to the SIR model. Dividing the equation for $\dot{I}$ by the one for $\dot{S}$ yields

$$\frac{\dot{I}}{\dot{S}} = \frac{dI}{dS} = -1 + \frac{\lambda}{g}\,\frac{1}{S}\,, \tag{9.55}$$

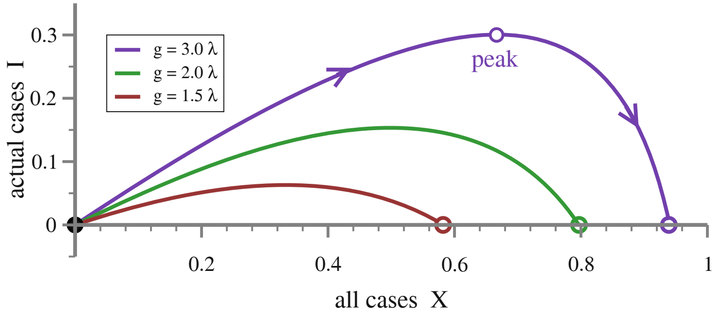

Fig. 9.16 For the SIR model, the current number of infected I as a function of the cumulative number of cases $X = 1 - S$. The curves are described by the analytic solution (9.56), using $R_0 = 0 = I_0$ and $S_0 = 1$. The outbreak starts at $(X, I) = (0, 0)$, ending when I vanishes again (open circles). A sizeable fraction of the population is never infected

which has the solution

$$I(t) = 1 - R_0 - S(t) + \frac{\lambda}{g}\,\log\left(\frac{S(t)}{S_0}\right), \tag{9.56}$$

where $S_0 = S(0)$, $I_0 = I(0)$ and $R_0 = 1 - S_0 - I_0$ denote the starting configuration. The functional dependence of $I = I(X)$ is shown in Fig. 9.16, where $X = I + R = 1 - S$ is the cumulative number of cases. The illness dies out before the entire population has been infected.

Exercises

(9.1) THE DRIVEN HARMONIC OSCILLATOR
Solve the driven harmonic oscillator, see (9.1), for all times t and compare it with the long time solution $t \to \infty$, Eqs. (9.3) and (9.4).

(9.2) SELF-SYNCHRONIZATION
Consider an oscillator with time delay feedback,

$$\dot{\theta}(t) = \omega_0 + K\,\sin[\theta(t - T) - \theta(t)]\,.$$

Discuss the self-synchronization in analogy to the stability of the steady-state solutions of time delay systems.[14] Which is the auto-locking frequency in the limit of strong self-coupling, $K \to \infty$?

[14] The methods of time delay systems are laid down in Sect. 2.5 of Chap. 2.

(9.3) KURAMOTO MODEL WITH THREE OSCILLATORS
Consider the Kuramoto model, as defined by (9.5) and (9.6), for three identical oscillators. Start by showing that the natural frequencies can be disregarded when $\omega_i \equiv \omega$. Use relative variables $\alpha = \theta_2 - \theta_1$ and $\beta = \theta_3 - \theta_2$, for the determination of the fixpoints. Which ones are stable/unstable for $K > 0$ and/or $K < 0$?

(9.4) LYAPUNOV EXPONENTS ALONG SYNCHRONIZED ORBITS
Evaluate, for two coupled oscillators, see (9.7), the longitudinal and the orthogonal Lyapunov exponents, with respect to the synchronized orbit. Does the average of the longitudinal Lyapunov exponent over one period vanish?

(9.5) SYNCHRONIZATION OF CHAOTIC MAPS
The Bernoulli shift map $f(x) = ax \bmod 1$ with $x \in [0, 1]$ is chaotic for $a > 1$. Consider with

$$\begin{aligned} x_1(t+1) &= f\big([1-\kappa]x_1(t) + \kappa x_2(t-T)\big) \\ x_2(t+1) &= f\big([1-\kappa]x_2(t) + \kappa x_1(t-T)\big) \end{aligned} \qquad (9.57)$$

two coupled chaotic maps, with $\kappa \in [0, 1]$ being the coupling strength and T the time delay, compare (9.32). Discuss the stability of the synchronized states $x_1(t) = x_2(t) \equiv \bar{x}(t)$ for general time delays T. What drives the synchronization process?

(9.6) TERMAN–WANG OSCILLATOR
Discuss the stability of the fixpoints of the Terman–Wang oscillator, see (9.39). Linearize the differential equations around the fixpoint solution and consider the limit $\beta \to 0$.

(9.7) PULSE COUPLED LEAKY INTEGRATOR NEURONS
The membrane potential $x(t)$ of a leaky-integrator neuron can be thought to increase over time like

$$\dot{x} = \gamma(S_0 - x), \qquad x(t) = S_0\left(1 - \mathrm{e}^{-\gamma t}\right) . \qquad (9.58)$$

When the membrane potential reaches a certain threshold, say $x_\theta = 1$, it spikes and resets its membrane potential to $x \to 0$ immediately after spiking. For two pulse coupled leaky integrators $x_A(t)$ and $x_B(t)$ the spike of one neuron induces an increase in the membrane potential of the other neuron by a finite amount ϵ. Discuss and find the limiting behavior for long times.

(9.8) SIRS MODEL
Find the fixpoints $x_t \equiv x^*$ of the discrete-time SIRS model, see (9.50), for all τ_R. As a function of a, study the stability for $\tau_R = 0, 1$.

(9.9) EPIDEMIOLOGY OF ZOMBIES
If humans and zombies meet, there are only two possible outcomes. Either the zombie bites the human, or the human kills the zombie, Defining with $\alpha \in [0, 1]$ and $1 - \alpha$ the respective probabilities, the evolution equations for

the two population densities H and Z are

$$\dot{H} = -\alpha HZ, \qquad \dot{Z} = (2\alpha - 1)HZ\,. \tag{9.59}$$

Find and discuss the analytic solution when starting from initial densities H_0 and $Z_0 = 1 - H_0$. Which is the difference to the SIRS model?

Further Reading

The reader may consult a textbook containing examples for synchronization processes by Pikovsky et al. (2003), and an informative review of the Kuramoto model by Pikovsky and Rosenblum (2015), which contains in addition historical annotations. Some of the material discussed in this chapter requires a certain background in theoretical neuroscience, see e.g. Sterrat et. al (2023). An introductory review to chimera states is given in Panaggio and Abrams (2015).

We recommend that the interested reader takes a look at some of the original research literature, such as the exact solution relaxation oscillators, Terman and Wang (1995), the concept of fast threshold synchronization, Somers and Kopell (1993), the physics of synchronized clapping, Néda et al. (2000a,b), and synchronization phenomena within the SIRS model of epidemics, He and Stone (2003). For synchronization with delays see D'Huys et al. (2008).

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10 Complexity of Machine Learning

Without doubt, the brain is the most complex adaptive system known to humanity, arguably also a complex system about which we know little. In both respects, the brain faces increasing competition from machine learning architectures.

We present an introduction to basic neural network and machine learning concepts, with a special focus on the connection to dynamical systems theory. Starting with point neurons and the XOR problem, the relation between the dynamics of recurrent networks and random matrix theory will be developed. The somewhat counter-intuitive notion of continuous numbers of network layers is shown next to lead to neural differential equations, respectively for information processing and error backpropagation. Approaches aimed at understanding learning processes in deep architectures make often use of the infinite-layer limit. As a result, machine learning can be described by Gaussian processes together with neural tangent kernels. Finally, the distinction between information processing and information routing will be discussed, with the latter being the task of the attention mechanism, the core component of transformer architectures.

10.1 Computation Units

Modern day computation devices and architectures are constructed using large numbers of local computation units. Examples are transistors and quantum gates for classical and quantum computers, or artificial neurons for deep learning algorithms.

Dimensionality Reduction Information processing by local computation units intrinsically involves dimensionality reduction, as illustrated in Fig. 10.1. Per output, only a single feature of the input data stream can be analyzed. Local units are hence feature extractors.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_10

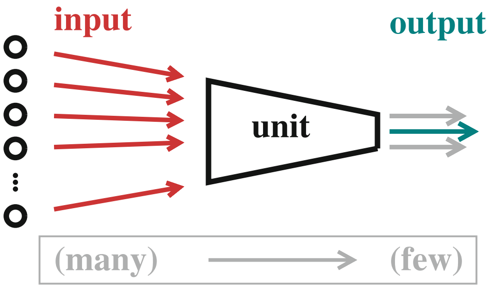

Fig. 10.1 Artificial neurons and other local computation units receive many inputs, with the subsequent processing reducing dimensionality. Only a few output streams are produced, in most cases only a single one. Computation units are hence equivalent to local feature extractors

Commonly used computation units generate just a single output stream. An exception are quantum gates, which need to be unitary. Quantum gates have hence identical numbers of inputs and outputs, normally just two.

Perceptron Unit We start with the classical artificial neuron, denoted here "perceptron unit". Time is discrete, with individual in- and outputs corresponding to real numbers, often with restricted domains. Dimensionality reduction is achieved by taking the scalar product of the input vector **x** with a weight vector **w**,

$$y = \sigma(a(x - b)), \qquad x = \mathbf{w} \cdot \mathbf{x}\,. \tag{10.1}$$

The output y is generated via a transfer function $\sigma(z)$. In machine learning, the threshold b is defined most of the time with an opposite sign. When the aim is to study a given set of weights **w**, one keeps the gain a as a free parameter. If not, one can set it to unity, $a \to 1$, as done in machine learning. Geometrically, the isocline $x = b$,

$$\mathbf{w} \cdot \mathbf{x} = b\,, \tag{10.2}$$

corresponds to a plane in input space. Percetron units act hence as linear classifiers.

Planes divide space into two half-spaces, say S_A and S_B. As feature extractors, perceptron units output information about whether the input vector is part of either S_A or S_B.

Geometric Features Spaces When images or visual data are processed, input units may be associated with specific geometric locations $\mathbf{R}_i$, usually corresponding to pixels in a two-dimensional plane. After adaption, the weights w_i of a specific classifying neuron,

$$w_i(\mathbf{R}_i) \;\to\; w(\Delta\mathbf{R}_i), \qquad \Delta\mathbf{R}_i = \mathbf{R}_i - \mathbf{R}_y\,, \tag{10.3}$$

will be a function of the geometric location $\mathbf{R}_i$ of the afferent neuron, normally relative to the location $\mathbf{R}_y$ of the classifying neuron.

Geometric feature extraction is the prime task of the visual cortex. Examples are linear or center-surround contrasts and gratings, both in the black-white and in the color domain. Features can be highly non-linear geometric objects, classification is nevertheless linear in the space of input activities.

Transfer Functions The transfer function $\sigma(z)$ entering (10.1) comes in a range of varieties. Two classical variants are

$$\sigma_{\text{sig}}(z) = \frac{1}{1+\text{e}^{-z}}, \qquad \sigma_{\tanh}(z) = \tanh(z)\,, \tag{10.4}$$

where the first, the "sigmoidal", works in the range $\sigma_{\text{sig}} \in [0, 1]$. For the second we have $\sigma_{\tanh} \in [-1, 1]$. Both transfer functions are monotonically increasing and bounded, which makes the output a measure for the degree to which the input has been classified. The infinite-gain limit

$$\lim_{a\to\infty} \sigma_{\text{sig}}(a(x-b)) \;=\; \begin{cases} 0.0 & x < b \\ 0.5 & x = b \\ 1.0 & x > b \end{cases} \tag{10.5}$$

leads to binary classification. The same holds for $\sigma_{\tanh}$.

ReLU Units The transfer function of "rectified linear units",

$$\sigma_{\text{ReLU}}(z) = \begin{cases} 0 & z < 0 \\ z & z > 0 \end{cases}, \tag{10.6}$$

is piecewise linear.[1] Negative arguments are shunted. On the positive side, gradients will not vanish for large arguments, as they do for σ_{sig} and $\sigma_{\tanh}$, remaining finite for all $z > 0$. This helps to avoid the vanishing gradient problem when propagating errors backwards, a subject treated further below in Sect. 10.3.

Boltzmann Units So far, deterministic updating in discrete time has been used. Stochastic dynamics,[2]

$$\sigma_{\text{Boltz}}(z) = \begin{cases} 1 & \text{with probability } \; p \\ 0 & \text{with probability } \; 1-p \end{cases}, \tag{10.7}$$

is however also an option. In the context of "Boltzmann machines" one relates the transition probability p to the difference between the two suitably defined energies,

[1] See Sect. 9.5 of Chap. 9, for the general theory of piecewise linear dynamical systems.

[2] For the general theory of stochastic dynamical systems see Chap. 3.

respectively of starting/final states. For

$$p = p(z) = \sigma_{\mathrm{sig}}(z) \tag{10.8}$$

one recovers on average the sigmoidal transfer function.

10.1.1 Structured Units

The computation units discussed so far correspond to "point neurons", in the sense that the output depends exclusively on a single aggregate quantity, namely $x = \mathbf{w}\cdot\mathbf{x}$, respectively $z = a(x - b)$. A possible extension are detailed biological models that are characterized by a potentially large number of biophysical compartments. "Compartmental neurons" are of relevance in the neurosciences. Remaining on an abstract level, we discuss here several options for internally structured computation units.

Internal Memories In general there is a tradeoff between increasing either the number of units or the complexity of the constituent elements. The latter is a viable option in particular when qualitative new features are added. A prime example is the introduction of an internal variable acting as a local memory, viz a memory cell. An influential version in this respect is LSTM, a unit with "long short-term memory".

When an internal memory is present, the unit needs a mechanism deciding to which extend past activities influence the new output, together with the updated value of the memory cell. For a LSTM unit, the corresponding mechanisms are encoded as gates, respectively for input, output and forgetting. Gating parameters are adaptable, viz learned during training.

Gated Units Gating occurs when a certain data stream controls the processing of a second data stream. A basic example is the "gated linear unit", GLU,

$$\sigma_{\mathrm{GLU}}(\mathbf{x}) = (x_w - b_w)\,\sigma_{\mathrm{sig}}(x_v - b_v), \qquad x_w = \mathbf{w}\cdot\mathbf{x}, \qquad x_v = \mathbf{v}\cdot\mathbf{x}\,, \tag{10.9}$$

which contains two adaptable weight matrices and thresholds, $\mathbf{w}$, $\mathbf{v}$ and b_w, b_v. The output is a linear function of the input vector $\mathbf{x}$, which is however modulated by a sigmoidal, $\sigma_{\mathrm{sig}}(x_v - b_v) \in [0, 1]$. After training, $\mathbf{w}$ and $\mathbf{v}$ may encode distinct features. If this happens, the processing of a given feature may proceed only when the second feature is also present. This type of gating is hence equivalent to a coincidence detector for self-selected features. It holds as a corollary that a single GLU unit can express all logical gates, including the XOR gate.[3]

[3] See Exercise (10.1).

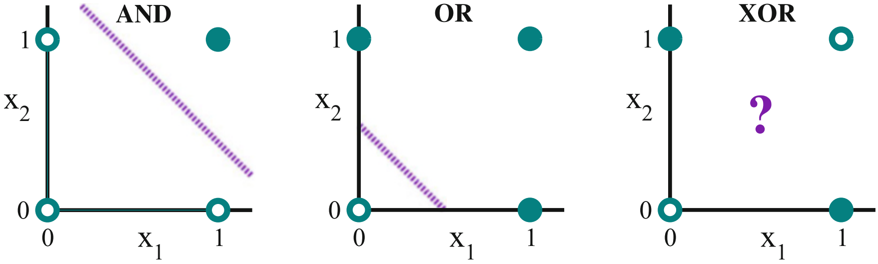

Fig. 10.2 The AND (left) and OR (middle) gates are examples of linearly separable logical gates (dashed diagonal line). The XOR gate is not linearly separable (right). Filled/open bullets denote logical true/false

10.1.2 The XOR Problem

The term "artificial intelligence" was coined at a 1956 Dartmouth workshop. Hopes of rapidly achieving human-level AI were however crushed when it was realized 13 years later that linear classifiers,[4] like point neurons, may not perform non-linear classification tasks, such as representing an XOR gate; see Fig. 10.2. This was considered unfortunate, given that universal computation hinges on being able to express all logical gates, inclusively the XOR gate. The "XOR problem", as it was dubbed, is thought to have initiated an extended period of disillusionment with neuronal networks, the "AI winter".

Hidden Layers In machine learning jargon, whatever is not part of input or output is "hidden". A classical solution to the XOR problem is to add a hidden layer.

As an example consider that the weights and thresholds of the two hidden units are such that they define respectively the planes

$$x_1 + x_2 = 0.5, \qquad x_1 + x_2 = 1.5\,, \tag{10.10}$$

as illustrated in Fig. 10.3. For binary neurons with $\sigma(z) = \theta(z)$, where $\theta(z)$ is the Heaviside step function, the mapping $(x_1, x_2) \to (y_1, y_2)$ reads

$$\begin{aligned} (0,0) &\to (0,0) \qquad & (1,1) &\to (1,1), \\ (0,1) &\to (1,0) \qquad & (1,0) &\to (1,0)\,, \end{aligned} \tag{10.11}$$

which implies that $(0, 1)$ and $(1, 0)$ are mapped to the same point in the activity space of the two hidden-layer units, (y_1, y_2). At this step, the XOR configuration becomes linearly separable.

[4] M. Minsky, S. Papert, "Perceptrons: An Introduction to Computational Geometry" (1969).

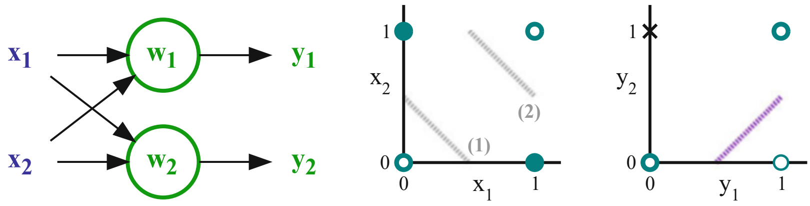

Fig. 10.3 **Left:** Receiving inputs x_1 and x_2, two hidden layer neurons produce outputs y_1 and y_2, the hidden-layer activity. Specific to the hidden layer are the weight vectors $\mathbf{w}_1$ and $\mathbf{w}_1$, together with the respective thresholds (not shown). **Middle:** In input space (x_1, x_2), the classifying planes (gray dashed lines) of the two hidden neurons; see (10.10). **Right:** The activity space of the hidden neurons, (y_1, y_2). Shown is the mapping (10.11) for binary neurons. The XOR configuration is now linearly separable (dashed diagonal line)

Dimensional Embedding Classification becomes simpler when a given problem is embedded in an enlarged state space. We keep a single point neuron, extending however the input space via $(x_1, x_2) \to (x_1, x_2, x_3)$. As an example we set $x_3 = \pm\Delta$, together with

$$
\begin{aligned}
(0,0) &\to (0,0,+\Delta) \qquad & (1,1) &\to (1,1,+\Delta),\\
(0,1) &\to (1,0,-\Delta) \qquad & (1,0) &\to (1,0,-\Delta)\,.
\end{aligned}
\tag{10.12}
$$

Consequently, the $x_3 = 0$ plane can be used for the classification of the XOR gate. Above example is constructed by hand. When using random embeddings, a common approach, larger embedding dimensions may be necessary. The brain makes extensive use of dimensional embedding,

Universal Function Approximators Solutions to the XOR problem abound. An option mentioned further above are gated units, see (10.9), which can do the job all alone. Neural networks are "Turing complete" when logical gates can be represented, acting as "universal function approximators". Given enough units, arbitrary complex functional dependencies can be modeled with increasing accuracy.

10.2 Recurrent Neural Networks

Point neurons connected via synaptic weights form a network of interconnected units.[5] Networks corresponding to "feedforward nets" are free of loops, "recurrent

[5] Generic network theory is developed in Chap. 1.

nets" not. In general, the "membrane potential" x_i of a point neuron is given by

$$x_i = e_i + \sum_j w_{ij} y_j, \qquad y_i = \tanh(a_i(x_i - b_i)), \tag{10.13}$$

where e_i is an external driving and w_{ij} the entries of the synaptic weight matrix $\hat{w}$. A tanh-transfer function has been selected.

Lyapunov Exponents Recurrent nets feed back on themselves, which gives raise to a trailing memory. In the autonomous case, when $e_i \equiv 0$, the largest eigenvalue of $\hat{w}$, the largest "Lyapunov exponent" of the map defined by (10.13), determines if activities tend to increase or decrease in linear order.[6] This issue will be treated in detail in Sect. 10.2.2.

10.2.1 Random Matrix Theory

The statistics of the entries of a matrix can be used for estimating the corresponding spectrum of eigenvalues. When the entries are only weakly correlated, as defined below, random matrix theory applies, providing precise results. Often, matrix elements can be assumed to be independently distributed, in particular when the rank of the matrix is large, the typical case for both neuroscience and machine-learning applications.

Ensemble Average Random matrix theory deals with matrices with elements that are drawn from a given probability distribution, normally a Gaussian. Drawing all elements once generates a specific realization. "Ensemble average" denotes the average over realizations. In practice one is however interested in the spectrum of a specific matrix. For large matrices the ensemble average can be substituted by an average of all entries, the venue taken here.

Off-Diagonal Correlations In its basic form, random matrix theory can be applied only when the entries w_{ij} are statistically fully independent. It holds however also when the off-diagonal cross-correlation Γ,

$$\Gamma = \frac{\sum_{i,j}\big(w_{ij}-\mu_w\big)\big(w_{ji}-\mu_w\big)}{\sum_{i,j}\big(w_{ij}-\mu_w\big)^2}, \qquad \Gamma \in [-1, 1], \tag{10.14}$$

[6] See Chap. 2.

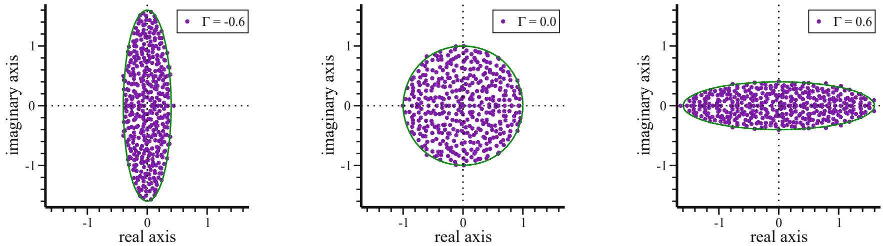

Fig. 10.4 The distribution of eigenvalues of an elliptic 400×400 matrix. The eigenvalues (filled dots) in the complex plane are shown for $\Gamma = -0.6/0.0/0.6$ (left/middle/right), as defined by (10.14). Included are the corresponding analytic results (10.15), which are valid in the thermodynamic limit (lines)

is non-zero. For a $N \times N$ matrix, we defined the mean $\mu_w = \sum_{ij} w_{ij}/N^2$. The inequality $|\Gamma| \le 1$ follows from $(a \pm b)^2 \ge 0$, or $a^2 + b^2 \ge \mp 2ab$. Three special cases are of interest.

- $\Gamma = 0$: Fully uncorrelated, complex eigenvalues.
- $\Gamma = 1$: Symmetric matrix with $w_{ij} = w_{ji}$; real spectrum.
- $\Gamma = -1$: Antisymmetric matrix with $w_{ij} = -w_{ji}$; imaginary eigenvalues.

One speaks of "elliptic matrices" when $\Gamma \neq 0$.

Elliptic Random Matrices We set $\mu_w = 0$, without loss of generality, and assume $\sigma_w = 1/\sqrt{N}$ for the standard deviation σ_w of the matrix elements. Random matrix theory states that the spectrum of the eigenvalues λ is uniformly distributed within an ellipse defined by

$$E_\Gamma = \left\{ \lambda = x + iy, \quad \frac{x^2}{(1+\Gamma)^2} + \frac{y^2}{(1-\Gamma)^2} \le 1 \right\} . \tag{10.15}$$

The half widths are $1 \pm \Gamma$ respectively along the real and imaginary axis. All eigenvalues are real/imaginary for $\Gamma = \pm 1$. In Fig. 10.4 a comparison between numerical spectra and (10.15) is presented. The agreement would be perfect in the limit of infinite-large matrices.

Finite Mean For $\Gamma = 0$ the ellipse defined by (10.15) reduces to a circle and one recovers the "circular law of random matrix theory", namely that the eigenvalues are uniformly distributed within a circle. When the mean $\mu_w = \sum_{ij} w_{ij}/N^2$ is finite, an additional isolated eigenvalue λ_μ is generated. For $\Gamma = 0$ one can derive that

$$\lambda_\mu = \tilde{\mu}_w + \frac{\tilde{\sigma}_w^2}{\tilde{\mu}_w}, \qquad \mu_w = \frac{\tilde{\mu}_w}{N}, \qquad \sigma_w^2 = \frac{\tilde{\sigma}_w^2}{N} . \tag{10.16}$$

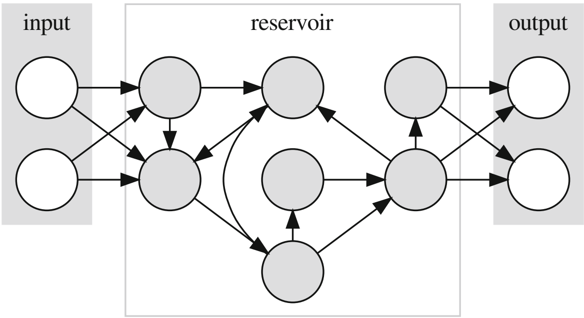

Fig. 10.5 Recurrent nets are the centerpiece of reservoir computing, mapping data input streams into a high-dimensional reservoir of non-linear features. Only the weights of linear output units are trained

The additional contribution, $\tilde{\sigma}_w^2/\tilde{\mu}_w$, can be interpreted as an interaction between mean and continuum.

Spectral Radius When the eigenvalue spectrum is bounded, there exists a $R_w > 0$ such that

$$|\lambda_i| \leq R_w, \qquad \forall i \tag{10.17}$$

The "spectral radius" R_w determines the long-term evolution when applying the mapping in question repeatedly. For elliptic random matrices with zero mean $R_w = \tilde{\sigma}_w(1 + |\Gamma|)$.

Variance Control An interesting point is that the spectral radius R_w scales with $\tilde{\sigma}_w = \sigma_w\sqrt{N}$, which is directly proportional to $\sqrt{N}$ and to the standard deviation σ_w of the matrix elements. Standard learning algorithms, like Hebbian plasticity in biology or backpropagation in machine learning, treat each synaptic weight however as an independent adaptable parameter. The variance σ_w^2 resulting from the learning process is hence neither optimized nor controlled.

Reservoir Computing Spectral radius regulation is central for "reservoir computing", as illustrated in Fig. 10.5. A recurrent net with otherwise fixed synaptic weights is used to transform the input into a multitude of random non-linear features. Only the weights of a subsequent linear output layer are adapted during training, typically for prediction tasks.

10.2.2 Criticality in Recurrent Networks

The brain is autonomously active, which implies that recurrent connections are functionally important. Does the level of internally produced activities impact the capability of the brain to process information? We have seen in Chap. 4, that critical dynamical systems exhibit non-trivial properties. Indeed, it can be shown that the long-range correlations characteristic of systems poised on a bifurcation point improve the processing of incoming data streams. This is captured by the "critical brain hypothesis". The quantitative impact of being close to criticality can be examined via a self-consistent theory for recurrent neural nets. We start with some preliminaries.

Self-Consistent Variance Theory We have seen that the variance of a random matrix determines its spectrum (10.15). It is hence clear that self-consistent approaches to recurrent nets will be on the level of variances.

Starting, we recall elementary relations for two statistically independent random variables z_1/z_2 with means μ_1/μ_2 and standard deviations σ_1/σ_2. The variances of the sum $z_1 + z_2$ and respectively of the product $z_1 z_2$ are given by

$$\sigma_{1+2}^2 = \sigma_1^2 + \sigma_2^2, \qquad \sigma_{1\cdot 2}^2 = \big(\sigma_1^2 + \mu_1^2\big)\big(\sigma_2^2 + \mu_2^2\big) - \mu_1^2\mu_2^2\,. \tag{10.18}$$

The first relation is the standard sum rule for the variance,[7] the second relation is treated in exercise (10.2).

Statistically Independent Units For a network with N units we define with

$$\sigma_w = \frac{\tilde{\sigma}_w}{\sqrt{N}}, \qquad \sigma_x, \qquad \sigma_y, \qquad \sigma_{\text{ext}}\,, \tag{10.19}$$

the standard deviations of respectively the synaptic weights, of the membrane potential x, of the neural activity y, and of the external input e, see (10.13). In reality, the respective distributions would be specific to the individual unit. The central approximation used here is that all units are statistically independent, and otherwise equivalent. It is then sufficient to work with (10.19).

Balanced Activities As a further simplification we consider networks with balanced activity and weight distributions, which implies vanishing means μ_w, μ_x, μ_y and μ_{ext}. The definition (10.13) of the membrane potential, $x_i = e_i + \sum_j w_{ij} y_j$, leads to

$$\sigma_x^2 = \sigma_{\text{ext}}^2 + N\sigma_w^2\sigma_y^2 = \sigma_{\text{ext}}^2 + \tilde{\sigma}_w^2\sigma_y^2\,, \tag{10.20}$$

where we used (10.18) and (10.19) and that units receive N recurrent inputs. The variance of the neural activity is determined by

$$\sigma_y^2 + \mu_y^2 = \int dx\, \tanh^2(x-b)\, P(x)\,, \tag{10.21}$$

[7] More about random variables in Chap. 5.

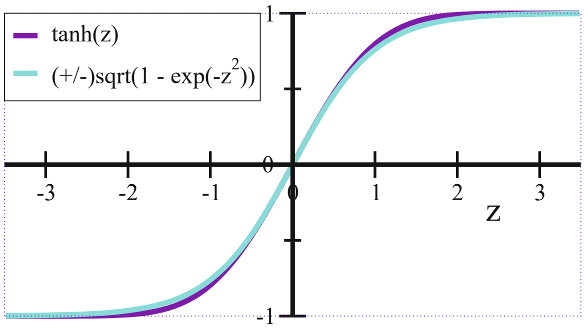

Fig. 10.6 Comparison between $\tanh(z)$ and the Gaussian transfer functions defined in (10.23)

where $P(x)$ is the distribution of the membrane potential. We did set $a = 1$ and made use of $\langle(y-\mu_y)^2\rangle = \langle y^2\rangle - \mu_y^2$. For balanced activities the threshold vanishes, $b = 0$, together with $\mu_y = 0 = \mu_x$.

Central Limit Theorem The central limit theorem states that sums of large numbers of random variables converges to a normal distribution. We can hence assume that the probability distribution $P(x)$ of the membrane potential entering (10.21) is normal distributed with mean μ_x and variance σ_x^2,

$$P(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}}\,\mathrm{e}^{-(x-\mu_x)^2/(2\sigma_x^2)}, \qquad 1 = \int dx\,P(x)\,. \tag{10.22}$$

A last step is needed before the integral in (10.21) can be evaluated analytically.

Gaussian Transfer Function We define with

$$\sigma_{\mathrm{Gauss}} = \pm\sqrt{\sigma^2_{\mathrm{Gauss}}}, \qquad \sigma^2_{\mathrm{Gauss}}(z) = 1 - \mathrm{e}^{-z^2} \tag{10.23}$$

a "Gaussian transfer function". The comparison between the tanh-transfer function and σ_{Gauss} presented in Fig. 10.6 shows that both transfer functions are functionally similar, with identical slopes at $z = 0$. One can therefore use one as an approximation for the other.

Variance of the Neural Activity We use the Gaussian transfer function for a balanced system, with b, μ_x, μ_y and μ_w vanishing, together with (10.22) in (10.21). The resulting expression,

$$\sigma_y^2 = \frac{1}{\sqrt{2\pi\sigma_x^2}}\int dx\left(1 - \mathrm{e}^{-x^2}\right)\mathrm{e}^{-x^2/(2\sigma_x^2)}\,, \tag{10.24}$$

leads via

$$1-\sigma_y^2 = \frac{1}{\sqrt{2\pi\sigma_x^2}}\,\frac{\sigma_{\text{eff}}}{\sigma_{\text{eff}}}\int dx\, \mathrm{e}^{-x^2/(2\sigma_{\text{eff}}^2)}, \qquad \frac{1}{\sigma_{\text{eff}}^2} = \frac{1+2\sigma_x^2}{\sigma_x^2}$$

to

$$1-\sigma_y^2 = \frac{\sigma_{\text{eff}}}{\sigma_x} = \frac{1}{\sqrt{1+2\sigma_x^2}}, \qquad 2\sigma_x^2 = \frac{1}{(1-\sigma_y^2)^2} - 1\,. \tag{10.25}$$

Note that $y^2 < 1$ and hence $\sigma_y^2 < 1$.

Variance Self-Consistency For $\Gamma = 0$, the case considered here, one can substitute $\tilde{\sigma}_w$ by the spectral radius $R_w = \tilde{\sigma}_w$. Together, (10.20) and (10.25) lead to the self-consistency condition

$$2\left(\sigma_{\text{ext}}^2 + R_w^2\sigma_y^2\right) = 2\sigma_x^2 = \frac{1}{(1-\sigma_y^2)^2} - 1\,,$$

or

$$2R_w^2\sigma_y^2\left(1-\sigma_y^2\right)^2 = 1 - \left(1+2\sigma_{\text{ext}}^2\right)\left(1-\sigma_y^2\right)^2, \tag{10.26}$$

which is quite remarkable. It states that the neural activity, in terms of the variance σ_y^2, is determined exclusively by the spectral radius R_w of the weight matrix, together with an eventually present external input, σ_{ext}^2.

Absorbing Phase Transition The system is autonomous when $\sigma_{\text{ext}} = 0$. In this case one obtains

$$2R_w^2\sigma_y^2 = 2\sigma_y^2 + \text{ higher powers}$$

when expanding both sides of (10.26) in powers of σ_y^2. Consequently, the system undergoes an "absorbing phase transition"[8] at $R_w = 1$.

- The activity dies out for $R_w < 1$.
- The activity steadily increases for $R_w > 1$, until limited by the non-linearity of the transfer function.

The state with vanishing neural activity is the final state of all orbits when $R_w < 1$, whatever the starting configuration. It is hence termed "absorbing". The active state

[8] Absorbing phase transitions are treated in depth in Sect. 6.4.1 of Chap. 6.

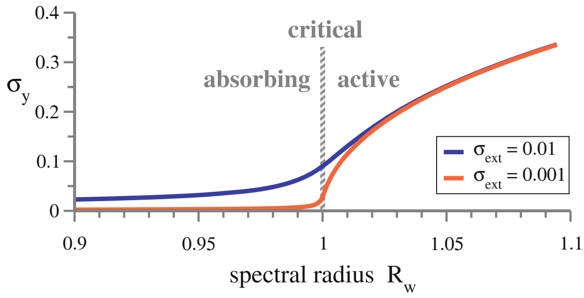

Fig. 10.7 The solution of the variance self-consistency condition (10.26). An absorbing phase transition occurs at $R_w = 1$ in the absence of external stimuli, viz when $\sigma_{\text{ext}} = 0$

is chaotic when performing simulations with a specific weight matrix. This is a consequence of $R_w > 1$, as discussed in Chap. 2.

Criticality vs. External Driving In Fig. 10.7 numerical solutions of (10.26) are presented. The phenomenology of a second-order phase transition is reproduced,[9] with the input variance σ_{ext}^2 acting as an external field.

Interestingly, Fig. 10.7 shows that the neural activity is substantial at criticality $R_w = 1$ even when external stimuli are comparatively weak. This result raises some doubts with regard to the critical brain hypothesis. Information processing will still benefit quantitatively from being close to criticality even when the variance σ_{ext}^2 of the data input stream is not negligible small. On a qualitative level, the working regime of the network does however not change as a function of the spectral radius in the presence of an external driving.

Feed-Forward Networks Eq. (10.25) can be interpreted as a mapping of σ_x to σ_y across a non-linearity. Adding (10.20) one obtains a mapping from $\sigma_{y,n-1}$ to $\sigma_{y,n}$, when adding layer indices, which holds also across a feed-forward layer,

$$\sigma_{y,n}^2 = 1 - \frac{1}{\sqrt{1 + 2\sigma_{x,n}^2}} = 1 - \frac{1}{\sqrt{1 + 2\big(\sigma_{\text{ext},n}^2 + \tilde{\sigma}_w^2 \sigma_{y,n-1}^2\big)}}, \tag{10.27}$$

where $\sigma_{\text{ext},n}^2$ is an additional input to layer n, if existing. Above expression is valid when all averages are zero, which would not be the case in a practical application. The generalization to non-zero μ_y, μ_x, μ_w is straightforward, but a bit cumbersome, see exercise (10.5).

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[9] See Sect. 6.1 of Chap. 6 for an introduction to the Landau theory of phase transitions. The connection is made in exercise (10.3).

10.3 Neural Differential Equations

Deep networks correspond to a mapping

$$\mathbf{f} = \mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta})$$

of an input $\mathbf{x} = (x_1, x_2, ..)$ to an output $\mathbf{f} = (f_1, f_2, ..)$. The parameters $\boldsymbol{\vartheta} = (\vartheta_1, \vartheta_2, ..)$ are adaptable, viz determined during training.

Training Data Training data consists of a set of N_α pairs of inputs $\mathbf{x}^\alpha$ and target outputs $\mathbf{F}^\alpha$. The overall goal is to minimize a loss function L, e.g.

$$L = \sum_\alpha \big(\mathbf{F}^\alpha - \mathbf{f}^\alpha\big)^2, \qquad \mathbf{f}^\alpha = \mathbf{f}(\mathbf{x}^\alpha, \boldsymbol{\vartheta}), \tag{10.28}$$

which can be done by following the gradient, $\dot{\boldsymbol{\vartheta}} \sim -\nabla_{\boldsymbol{\vartheta}} L$. For layered systems, as the one illustrated in Fig. 10.3, the gradient is evaluated recursively.

Error Backpropagation As a basic deep learning architecture we consider a system made up of N_L identical layers,

$$\mathbf{y}_l = \mathbf{f}(\mathbf{x}_l, \boldsymbol{\vartheta}_l), \qquad \mathbf{x}_l = \mathbf{y}_{l-1}, \tag{10.29}$$

where $\boldsymbol{\vartheta}_l$ denotes the adaptable parameters of the lth layer, like thresholds and synaptic weights. In order to simplify the notation, we set $N_L = 3$. For a given training pair $(\mathbf{x}^\alpha, \mathbf{F}^\alpha)$, the gradient of the loss function with respect to the parameters $\boldsymbol{\vartheta}_3$ of the last layer is

$$\nabla_{\boldsymbol{\vartheta}_3} L = \nabla_{\boldsymbol{\vartheta}_3}\mathbf{f}(\mathbf{x}_3, \boldsymbol{\vartheta}_3)\cdot\mathbf{E}_3, \qquad \mathbf{E}_3 = (-2)\big(\mathbf{F}^\alpha - \mathbf{f}^\alpha\big).$$

We used here the compressed notation $\nabla\mathbf{A}\cdot\mathbf{B} \equiv \sum_i (\nabla A_i)B_i$. With the chain rule, we have

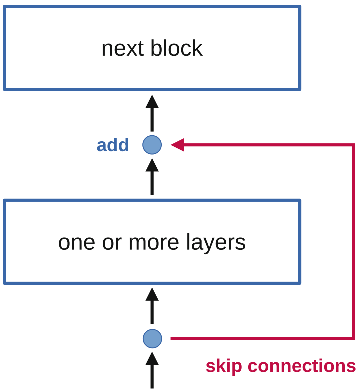
$$\nabla_{\boldsymbol{\vartheta}_2} L = \nabla_{\boldsymbol{\vartheta}_2}\mathbf{f}(\mathbf{x}_2, \boldsymbol{\vartheta}_2)\cdot\mathbf{E}_2, \qquad \mathbf{E}_2 = \nabla_{\mathbf{x}_3}\mathbf{f}(\mathbf{x}_3, \boldsymbol{\vartheta}_3)\mathbf{E}_3 \tag{10.30}$$

for the gradient of the loss function with respect to $\boldsymbol{\vartheta}_2$. The error $\mathbf{E}_2$ is a linear function of $\mathbf{E}_3$, the error signal of the next layer. Errors are hence propagated from top to down, which can be continued recursively. This is denoted "backpropagation".

10.3.1 Residual Nets

Layers process their inputs typically via (10.1), which involves a non-linear transfer function $\sigma(\cdot)$. The gradient of $\sigma(\cdot)$ vanishes for large inputs when the transfer

Fig. 10.8 Skip connections short-circuit the information processing by one or more layers. The input $\mathbf{x}_l$ is added to the output $\mathbf{y}_l$ of a given block, $\mathbf{x}_{l+1} = \mathbf{y}_l + \mathbf{x}_l$



function is limited, as is it the case, e.g., for the sigmoidal. The magnitude of the error backpropagated via (10.30) decreases hence in magnitude layer by layer, a phenomenon denoted "vanishing gradient problem".

Skip Connections A popular workaround for the vanishing gradient problem are ReLU units, see (10.6), for which the gradient is constant for positive arguments. An alternative are "skip connections",

$$\mathbf{y}_l = \mathbf{x}_l + \mathbf{f}(\mathbf{x}_l, \boldsymbol{\vartheta}_l), \qquad \mathbf{x}_l = \mathbf{y}_{l-1}\,, \tag{10.31}$$

which corresponds to add the identity to the forward pass, see Fig. 10.8. Learning by adapting the parameters $\boldsymbol{\vartheta}_l$ has now the task to generate the correct 'residual' $\mathbf{y}_l - \mathbf{x}_l$, hence the name ResNet, "residual net".

Neural Differential Equation It is tempting, with $\mathbf{x}_{l+1} = \mathbf{y}_l$, to rewrite (10.31) as

$$\mathbf{x}_{l+1} - \mathbf{x}_l = \mathbf{f}(\mathbf{x}_l, \boldsymbol{\vartheta}_l), \qquad \frac{d\mathbf{x}}{dt} \approx \mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta})\,, \tag{10.32}$$

where the layer index l was substituted by a pro forma time t, using $\mathbf{x}_l \to \mathbf{x}(t)$ and $\boldsymbol{\vartheta}_l \to \boldsymbol{\vartheta}(t)$. Layer time $t \in [0, T_L]$ is continuous, whereas the layer index $l = 1, 2 \,\ldots$ was discrete. The "neural differential equation" (10.32) corresponds hence to a continuous layer representation of a residual network.

Continuous Backpropagation As such, the neural differential equation (10.32) contains an unknown function, $\boldsymbol{\vartheta}(t)$, which specifies the value of the parameters for any layer time t. With skip connections, the recursive relation (10.30) for the error becomes

$$\mathbf{E}_{l-1} = \big(\nabla_{\mathbf{x}_l}\mathbf{f}(\mathbf{x}_l, \boldsymbol{\vartheta}_l) + \mathbf{1}\big)\mathbf{E}_l, \qquad \frac{d\mathbf{E}}{dt} \approx -\nabla_{\mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta}) \cdot \mathbf{E}\,. \tag{10.33}$$

Backpropagated errors tend to increase, given that $\nabla_{\mathbf{x}}\mathbf{f}$ is normally positive. In the continuous formulation, the error function $\mathbf{E} = \mathbf{E}(t)$ is obtained by integrating back in layer time. When adapting parameters via gradient minimization, $\Delta\boldsymbol{\vartheta} = -\eta\nabla_{\boldsymbol{\vartheta}} L$, one can use (10.30) to obtain

$$\Delta\boldsymbol{\vartheta} = -\eta\nabla_{\boldsymbol{\vartheta}}\mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta}) \cdot \mathbf{E}\,, \tag{10.34}$$

which leads in turn to two interdependent differential equations with respect to layer time,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta}), \qquad \dot{\mathbf{E}} = -\nabla_{\mathbf{x}}\mathbf{f}(\mathbf{x}, \boldsymbol{\vartheta}) \cdot \mathbf{E}\,. \tag{10.35}$$

In the adiabatic approximation, when the learning rate η is small, one takes $\boldsymbol{\vartheta}$ as constant when integrating $\mathbf{x}$ up from $\mathbf{x}(0) = \mathbf{x}^{\alpha}$. Then the error function $\mathbf{E}$ is evaluated downwards from $\mathbf{E}(T_L) = (-2)(\mathbf{F}^{\alpha} - \mathbf{x}(T_L))$. In the end, the parameters are updated.

It may seem a bit unusual, at first sight, to work with continuously stacked layers, which is however a viable option. Which one to use, continuous or discrete layers, is in the end a question of performance.

10.4 Gaussian Processes

In Sect. 10.2.2 we made extensive use of normal distributed random variables. Gaussian processes also involve normal distributed objects, however in the form of functions, and not variables. Starting with the basics, we recapitulate "multivariate" normal distributions, viz the case that more than one degree of freedom is relevant.

10.4.1 Multivariate Gaussians

The probability density for a set of N independent normal distributed random variables,

$$P_0(\mathbf{x}) = \prod_{i=1}^{N} \frac{\mathrm{e}^{-(x_i-\mu_i)^2/(2\sigma_i^2)}}{\sqrt{2\pi\sigma_i^2}} \tag{10.36}$$

is a special case of

$$P(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^N\sqrt{\det(\Sigma)}}\,\mathrm{e}^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})} \tag{10.37}$$

where Σ is a $N \times N$ symmetric matrix denoted "covariance matrix". The vector of means is $\boldsymbol{\mu}$, with Σ^{-1} denoting the inverse, $\Sigma\Sigma^{-1} = \Sigma^{-1}\Sigma = \mathbf{1}$.

Diagonalization We assume that the covariance matrix is positive definite, with eigenvalues σ_i^2. There exists hence an orthogonal transformation U such that

$$U^T \Sigma U = \begin{pmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_N^2 \end{pmatrix} \equiv \Sigma_0, \qquad U^{-1} = U^T\,, \tag{10.38}$$

with $\det(\Sigma_0) = \prod_i \sigma_i^2 = \det(\Sigma)$. The columns of U are the eigenvectors of Σ. We recall that the eigenvectors $\mathbf{e}_i$ of a non-singular matrix are also eigenvectors of the inverse,

$$\Sigma\,\mathbf{e}_i = \sigma_i^2\,\mathbf{e}_i \qquad \Leftrightarrow \qquad \frac{1}{\sigma_i^2}\,\mathbf{e}_i = \Sigma^{-1}\mathbf{e}_i\,,$$

with the precondition that none of the eigenvalues vanish, viz that Σ is non-singular. Equivalently, the relation

$$U^T \Sigma^{-1} U = \begin{pmatrix} \frac{1}{\sigma_1^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\sigma_N^2} \end{pmatrix} \tag{10.39}$$

holds. For a quick check one multiplies both sides of (10.38) and (10.39).

Normalization The properties of a multivariate Gaussian distribution can be evaluated using U for transforming the variables,

$$\mathbf{x} = U\mathbf{y}, \qquad \det(U) = 1\,. \tag{10.40}$$

The normalization of $P(x)$ follows directly,

$$\int d\mathbf{x}\, P(\mathbf{x}) = \int d\mathbf{y} \prod_i \frac{1}{\sqrt{2\pi\sigma_i^2}}\, \mathrm{e}^{-y_i/(2\sigma_i^2)} = 1\,,$$

where we eliminated $\boldsymbol{\mu}$ via an appropriate shift of the origin. It also follows that (10.36) and (10.37) are identical, modulo an orthogonal variable transformation.

Covariance Matrix We have to show that Σ honors its name, namely that the general definition of the covariance matrix,

$$\big\langle (x_i-\mu_i)(x_j-\mu_j)\big\rangle = \int d\mathbf{x}\,(x_i-\mu_i)(x_j-\mu_j)\,P(\mathbf{x})\,, \tag{10.41}$$

coincides with Σ_{ij}. For statistically independent variables, distributed according to P_0, see (10.36), the covariance matrix is diagonal,

$$\big\langle (x_i-\mu_i)(x_j-\mu_j)\big\rangle_0 = \Sigma_0\,,$$

with Σ_0 defined in (10.38). Applying the orthogonal transformation U^T to both sides maps $P_0(\mathbf{x})$ to $P(\mathbf{x})$, and Σ_0 to Σ, the preposition we wanted to prove.

Regression A central machine learning task is regression, namely to extrapolate or interpolate training data. The goal is to make an educated guess about something not yet seen. This can be quantified in particular when assuming that values are drawn from multivariate Gaussians. We enlarge the set of variables,

$$\mathbf{x} = \begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}, \qquad \boldsymbol{\mu} = \begin{bmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{bmatrix}, \qquad \Sigma = \begin{bmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}, \tag{10.42}$$

where the Σ_{ij} are matrices, not matrix elements. The probability of observing $\mathbf{x}_1$, the prediction we want to make after having observed $\mathbf{x}_2$, is denoted as

$$P(\mathbf{x}_1|\mathbf{x}_2)\,,$$

reading, "the probability of observing $\mathbf{x}_1$ given $\mathbf{x}_2$".

Bayes Theorem Per definition, the relation

$$P(\mathbf{x}_1|\mathbf{x}_2)P(\mathbf{x}_2) = P(\mathbf{x}_1,\mathbf{x}_2), \qquad P(\mathbf{x}_1|\mathbf{x}_2) = \frac{P(\mathbf{x}_1,\mathbf{x}_2)}{P(\mathbf{x}_2)}\,, \tag{10.43}$$

holds. It is also denoted "Bayes theorem", in particular together with the recursive substitution $P(\mathbf{x}_1,\mathbf{x}_2) = P(\mathbf{x}_2|\mathbf{x}_1)P(\mathbf{x}_1)$. Given the relations,

$$1 = \int d\mathbf{x}_1 P(\mathbf{x}_1|\mathbf{x}_2), \qquad P(\mathbf{x}_1) = \int d\mathbf{x}_2 P(\mathbf{x}_1,\mathbf{x}_2)\,,$$

one has that $P(\mathbf{x}_1|\mathbf{x}_2) \ge 0$ is a properly normalized probability distribution, for any $\mathbf{x}_2$, denoted "posterior". Idem for the "marginal", $P(\mathbf{x}_1)$. Posterior distributions are the object of desire when the information gained from an observation is to be expressed quantitatively, a process denoted "inference".[10]

Variational Inference For practical calculations, the distributions involved in Bayesian inference need to be parameterized. One speaks of "variational inferences" when a certain functional form is stipulated together with appropriate variational parameters. The approach is "variational" when the parameters are determined by minimizing an information-theoretical objective function.

Multivariate Gaussians are prime candidates for variational inference. One can show that the posterior $P(\mathbf{x}_1|\mathbf{x}_2)$ is also a multivariate Gaussian, with mean vector and covariance matrix, $\boldsymbol{\mu}_{1|2}$ and $\Sigma_{1|2}$, given by

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12}\ \Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \qquad \Sigma_{1|2} = \Sigma_{11}\ - \Sigma_{12}\ \Sigma_{22}^{-1}\Sigma_{21}\ . \tag{10.44}$$

The derivation involves skilled manipulations of Gaussian distributions. A motivation using neural tangent kernels is given in Sect. 10.4.3 One can generalize (10.44) to the case that observations are noisy and not accurate.

10.4.2 Correlated Stochastic Functions

In machine learning, a key objective is to generate increasingly accurate function approximations. Say we have a first guess, denoted "prior", which we want to refine during training. Being a guess, the prior cannot be accurately defined. Therefore, an algorithm for modeling stochastically broadened functional dependencies is needed. That's what stochastic processes are about.

Smooth Stochastic Functions We denote with f the stochastic function to be modeled. For every argument x from the support, the function is stochastically distributed around a mean value, $m(x)$, e.g. $m(x) = x^2$. The notation

$$f \sim G_p(m, k), \qquad k = k(x, x')\,, \tag{10.45}$$

defines a Gaussian process for f. The covariance matrix $k(x, x')$ has two tasks.

- $k(x, x)$ is the variance of the probability distribution $f(x)$, for a given argument x.
- $k(x, x')$ makes sure that strong discontinuities are rare, when drawing function values for $x \neq x'$.

[10] More about Bayesian statistics in Sect. 5.1 of Chap. 5.

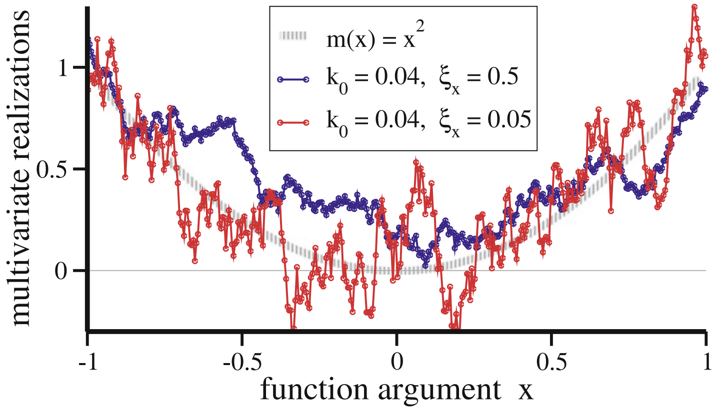


Fig. 10.9 The Gaussian process defined by (10.48), with a diagonal variance $k_0 = 0.04$. Shown is the mean function $m(x) = x^2$ (shaded line) and two realizations of the corresponding multivariate Gaussian distribution, respectively for a correlation length $\xi_x = 0.5/0.05$ (blue/red). The function estimate is smoother for longer-ranged correlations between the $N = 400$ arguments

Estimates are independent for each argument x when the covariance matrix is diagonal.

Discrete Supports In practice, function estimates are evaluated on a discrete set of arguments, x_i, as in the example presented in Fig. 10.9. On a given set of N arguments, $\mathbf{x} = (x_1, \ldots, x_N)$, a multivariate Gaussian distribution,

$$\mu_i = m(x_i), \qquad \Sigma_{i,j} = k(x_i, x_j)\,, \tag{10.46}$$

is recovered, where $i, j \in [1, N]$. For every argument x_i, the function values f_i are distributed according to (10.37),

$$P(\mathbf{f}) = \frac{1}{\sqrt{2\pi}^N \sqrt{\det(\Sigma)}}\, \mathrm{e}^{-\frac{1}{2}(\mathbf{f}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{f}-\boldsymbol{\mu})} \tag{10.47}$$

where $\mathbf{f} = (f_1, \ldots, f_N)$ and with $\boldsymbol{\mu}$ and Σ given by (10.46). In Fig. 10.9 we illustrate a quadratic mean function,

$$m(x) = x^2, \qquad k(x, x') = k_0\, \mathrm{e}^{-|x-x'|/\xi_x}\,, \tag{10.48}$$

together with an exponentially decaying covariance matrix. The stochastic function approximation becomes smoother with increasing correlation length ξ_x.

Inference The expressions (10.44) for the mean and the covariance matrix of the posterior hold in analogy for Gaussian processes. We enlarge the number of arguments and function values as

$$\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2), \qquad \mathbf{f} = (\mathbf{f}_1, \mathbf{f}_2)\,,$$

where $\mathbf{x}_1$ is the N-dimensional vector of arguments covering the entire support. The data vector $\mathbf{x}_2$, of length $N_d < N$, includes all arguments for which training data is available, with corresponding function values $\mathbf{f}_2$. Inference consists of predicting $\mathbf{f}_1$.

In analogy to (10.44), the parameters of the posterior $P(\mathbf{f}_1|\mathbf{f}_2)$ are

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12}\ \Sigma_{22}^{-1}(\mathbf{f}_2 - \boldsymbol{\mu}_2), \qquad \Sigma_{1|2} = \Sigma_{11}\ - \Sigma_{12}\ \Sigma_{22}^{-1}\Sigma_{21}\ . \tag{10.49}$$

On a formal level, there is a difference regarding the term $\mathbf{f}_2 - \boldsymbol{\mu}_2$, as we are describing the distribution of function values $f_i = m(x_i)$, and not of the arguments.

Radial Basis Functions Correlations decay exponentially for non-critical processes, the reason we used an exponentially decaying covariance matrix in (10.48). Quadratic exponents,

$$m(x) \to 0, \qquad k(x, x') = k_0\, \mathrm{e}^{-(x-x')^2/(2\sigma_k^2)}\,, \tag{10.50}$$

are however more covenient for frameworks based on multivariate Gaussians. Covariance matrices of this type are denoted "radial basis functions". Also included in (10.50) is a standard selection, namely a vanishing mean function $m(x)$. An alternative would be the average of the training data. When inference is performed, the mean of the posterior will be updated via (10.49), viz based on the training data. A non-trivial prior may bias this process.

In Fig. 10.10 an example of inference with Gaussian processes is presented. The variance $[\sigma_{1|2}(x)]^2$ of the prediction is given by $\Sigma_{1|2}(x, x)$, which has been included in terms of a confidence interval of two standard deviations. It is evident that inference with Gaussian processes is much better at interpolating data than extrapolating. The latter because predictions revert to the prior away from the data, here to $m(x) = 0$. The estimated confidence intervals depend sensibly on the positioning of the training data.

Intra-Domain Inference As a matter of principle, inference can be performed between different domains. Out-of domain inference requires however knowledge about semantic cross-correlations, which cannot be modeled with radial basis functions, or another generic ansatz for the covariance matrix.

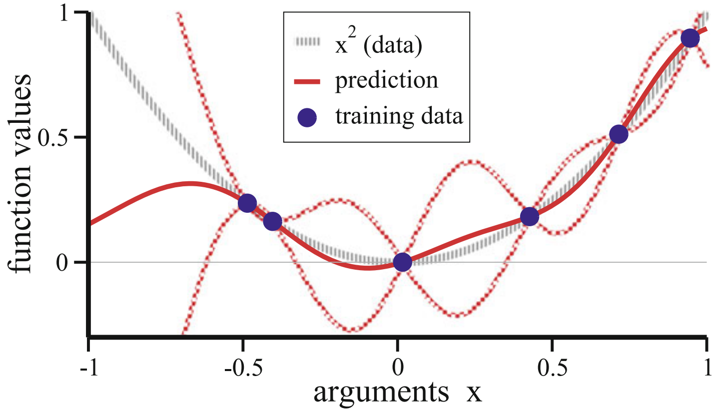


Fig. 10.10 Predictions (red line) inferred via (10.44) from a randomly drawn set of data points (blue dots). The covariance matrix is $k(x, x') = \exp(-(x - x')^2/0.2)$. The function values for the training data (blue bullets) follow $f_2(x) = x^2$. The mean function $m(x) = 0$ vanishes, in contrast to Fig. 10.9. Also included are 2σ confidence estimators (checkerborad red lines)

Here we work exclusively with intra-domain inference, such as predicting a function value when observing another, as in Fig. 10.10. The corollary is that all components of the covariance matrix have identical dependencies on their arguments.

Intra-domain inference includes the case of predicting the value for an observed position, viz that $\mathbf{x}_1 \to \mathbf{x}_2$. When this happens, one would expect that the prediction concurs with the observation, as it happens for the example shown in Fig. 10.10. This is easily confirmed analytically when both $\mathbf{x}_1$ and $\mathbf{x}_2$ are scalars, which implies $\Sigma_{11} = \Sigma_{12} = \Sigma_{21} = \Sigma_{22} \equiv k(0)$ when $x_1 = x_2$. The predicted mean is then

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\, \Sigma_{22}^{-1}(x_2 - \mu_2) \;\to\; x_2\,,$$

given that $\mu_1 = m(x_1)$ coincides with $\mu_2 = m(x_2)$ when $x_1 \to x_2$. The inferred mean function value, $\mu_{1|2}$, coincides hence with the training value x_2. This will not be the case when the data is noisy, which is described by adding $\sigma_D^2 \delta_{ij}$ to the covariance matrix Σ_{ij} in (10.46).

10.4.3 Machine Learning with Neural Tangent Kernels

Deep learning architectures typically involve large numbers of computational units. It is hence natural to assume that Gaussian distributions abound. This is indeed the case.

Infinite Layer Limit A basic setup is that of a single layer neural network,

$$f = \sum_i v_i y_i, \qquad y_i = \sigma(x_i - b_i), \qquad x_i = \sum_j w_{ij} x_j^\alpha, \tag{10.51}$$

where f is a classifying unit, the function to be computed, and $\mathbf{y} = (y_1, y_2, ..)$ the activity of the hidden layer. The training samples $\mathbf{x}^\alpha = (x_1^\alpha, x_2^\alpha, ..)$ are specific to the problem at hand. All internal parameters are taken to be random but frozen, viz corresponding to a specific realization.

The components of the training vectors, like x_i^α and x_j^α, are not necessarily uncorrelated, which implies that the membrane potentials x_i are independent random variables, but possibly not normal distributed. The same holds for the y_i. As a sum of independent distributions, f becomes however Gaussian for large hidden layers. This line of arguments, denoted the "infinite layer limit", can be extended to deep architectures.

Learning Correlations Learning is all about extracting information from data samples. Inputs can be differentiated, viz classified, only when differences can be quantified in terms of correlations. In leading order this regards two-point correlations. Given that the output is Gaussian distributed in the infinite-layer limit, the output f of a deep learning architecture can be modeled in leading order by a Gaussian process over input space,

$$f \sim G_p(m, k), \qquad m = m(\mathbf{x}), \qquad k = k(\mathbf{x}, \mathbf{x}'), \tag{10.52}$$

compare (10.45). Mean function and covariance matrix are averages over random sets $\boldsymbol{\vartheta}$ of parameters,

$$\begin{aligned} m(\mathbf{x}) &= \big\langle f(\mathbf{x}; \boldsymbol{\vartheta}) \big\rangle_{\boldsymbol{\vartheta}} \\ k(\mathbf{x}, \mathbf{x}') &= \big\langle (f(\mathbf{x}; \boldsymbol{\vartheta}) - m(\mathbf{x})(f(\mathbf{x}'; \boldsymbol{\vartheta}) - m(\mathbf{x}') \big\rangle_{\boldsymbol{\vartheta}} . \end{aligned} \tag{10.53}$$

The output of the network is Gaussian because the parameters $\boldsymbol{\vartheta}$ act as random variables, hence the average over $\boldsymbol{\vartheta}$. The functional dependencies of the covariance matrix were central for the application of Gaussian processes discussed before. This is however not a relevant point for the "neural network gaussian process" defined by (10.53), which serves mainly as a conceptual background.

Supervised Learning During training, the output of the network,

$$f^\alpha = f(\mathbf{x}^\alpha; \boldsymbol{\vartheta})$$

is compared to the target F^α associated with the specific training sample $\mathbf{x}^\alpha$. The set of network parameters $\boldsymbol{\vartheta}$ are adapted by minimizing the loss function

$L = \sum_\alpha (F^\alpha - f^\alpha)^2$ via gradient descent,

$$\frac{d}{dt}\boldsymbol{\vartheta} = \eta \sum_\alpha \left(\nabla_{\boldsymbol{\vartheta}} f^\alpha\right) \left(F^\alpha - f^\alpha\right), \tag{10.54}$$

with η being the learning rate. The right-hand-side includes the contribution of all N_α training samples. Training is "supervised" because the target F^α is explicitly provided. For a general input $\mathbf{x}$, the output $f = f(\mathbf{x}; \boldsymbol{\vartheta})$ evolves as

$$\begin{aligned} \frac{d}{dt} f &= \big(\nabla_{\boldsymbol{\vartheta}} f\big) \cdot \Big(\frac{d}{dt}\boldsymbol{\vartheta}\Big) \equiv \eta \sum_\alpha \Theta(\mathbf{x}, \mathbf{x}^\alpha)\big(F^\alpha - f^\alpha\big) \\ \Theta(\mathbf{x}, \mathbf{x}^\alpha) &= \nabla_{\boldsymbol{\vartheta}} f(\mathbf{x}; \boldsymbol{\vartheta}) \cdot \nabla_{\boldsymbol{\vartheta}} f(\mathbf{x}^\alpha; \boldsymbol{\vartheta}) \end{aligned} \tag{10.55}$$

when using the gradient update rule (10.54). So far we did not make use of the infinite layer limit. Equations (10.54) and (10.55) are valid for any least-square minimization via gradient descent.

Tangent Kernels As a function of its arguments, $\mathbf{x}^\alpha$ and $\mathbf{x}^\beta$, the "neural tangent kernel" $\Theta(\mathbf{x}^\alpha, \mathbf{x}^\beta)$ is a symmetric matrix. With the matrix elements being given by a scalar product, the neural tangent kernel is also positive definite.[11] For large networks the tangent kernel is

- constant during training and
- independent of the initial distribution of network parameters.

One says that $\Theta(\mathbf{x}^\alpha, \mathbf{x}^\beta)$ is "deterministic". We do not delve into the proof of these two prepositions, which involve inductive argumentation tailored to the architecture at hand. Systems become increasingly over-parametrized when network sizes diverge, which leads to the emergence of an exponentially large number of optimal parameter configurations. The probability for the starting parameter configuration to be close to an optimum will therefore increase steadily with layer width. Individual parameters need to change just a little when their number diverges.

Training Space Given that the tangent kernels are constant during training, one can solve the time evolution of $f = f(\mathbf{x}; \boldsymbol{\vartheta})$ explicitly via (10.55). First one needs to determine the time-dependency of the $f^\alpha = f(\mathbf{x}^\alpha; \boldsymbol{\vartheta})$, which is done in the space of training data,

$$\mathbf{F} = (F^1, \ldots, F^{N_\alpha}), \qquad \mathbf{f}_2 = (f^1, \ldots, f^{N_\alpha}),$$

[11] See exercise (10.7).

where we used the subscript '2' to indicate training inputs, in equivalence to (10.49). Specifying (10.55) to the training space yields

$$\frac{d}{dt} f^\beta = \eta \sum_\alpha \Theta(\mathbf{x}^\beta, \mathbf{x}^\alpha)\big(F^\alpha - f^\alpha\big), \qquad \frac{d}{dt}\mathbf{f}_2 = \eta\, \Theta_{22}\big(\mathbf{F} - \mathbf{f}_2\big)\,, \tag{10.56}$$

which constitutes a closed set of equations for N_α dynamical variables, $\mathbf{f}_2$. The projection of the tangent kernel to the training space, Θ_{22}, is a symmetric $N_\alpha \times N_\alpha$ matrix. The solution,

$$\mathbf{f}_2(t) = \mathbf{e}^{-\eta\, \Theta_{22} t}\, \mathbf{f}_2(0) + \big(\mathbf{1} - \mathbf{e}^{-\eta\, \Theta_{22} t}\big)\, \mathbf{F}\,, \tag{10.57}$$

holds when Θ_{22} is constant. One verifies (10.57) by direct differentiation. The constant of integration has been selected such that $\mathbf{f}_2(t = 0) = \mathbf{f}_2(0)$. Perfect learning, $\mathbf{f}_2(t) \to \mathbf{F}$, is recovered in the limit $t \to \infty$.

Tangent Kernel Inference We define with $\mathbf{f}_1$ the vector of network outputs to be predicted and use (10.57) to rewrite (10.55) as

$$\frac{d}{dt}\mathbf{f}_1 = \eta\, \Theta_{12}\big(\mathbf{F} - \mathbf{f}_2\big) = \eta\, \Theta_{12}\, \mathbf{e}^{-\eta \Theta_{22} t}\big(\mathbf{F} - \mathbf{f}_2(0)\big)\,,$$

where the first and second arguments of Θ_{12} are respectively generic inputs and training data. Starting from $\mathbf{f}_1(0)$ and $\mathbf{f}_2(0)$, direct integration yields

$$\mathbf{f}_1(t) = \mathbf{f}_1(0) + \Theta_{12}\, \Theta_{22}^{-1}\left(\mathbf{1} - \mathbf{e}^{-\eta \Theta_{22} t}\right)\big(\mathbf{F} - \mathbf{f}_2(0)\big)\,, \tag{10.58}$$

which is a quite remarkable result.

- The output of the network is fully specified by (10.58), including the entire training procedure.
- When training is complete, viz for $t \to \infty$, the predictions obtained using the infinite layer limit coincide formally with the corresponding expression for Bayesian inference, Eq. (10.49).

As for Bayesian inference, one has that $\mathbf{f}_1 \to \mathbf{F}$ when predictions are made for training data. This is seen easily for the case that $\mathbf{f}_1$ spans the entire training space, which leads to $\Theta_{12} \to \Theta_{22}$ and $\mathbf{f}_1(0) \to \mathbf{f}_2(0)$.

Note, that the formal equivalence of (10.58) and (10.49) does not imply that the tangent kernel $\Theta(\mathbf{x}, \mathbf{x}')$ and the covariance matrix $k(\mathbf{x}, \mathbf{x}')$ are identical. They are not. The covariance matrix is defined via (10.41) independently in terms of two-point correlations of the output $f = f(\mathbf{x}, \boldsymbol{\vartheta})$, in contrast to the definition of the neural tangent kernel, as given by (10.55). It is however possible, though rather cumbersome, to express $k(\mathbf{x}, \mathbf{x}')$ in terms of $\Theta(\mathbf{x}, \mathbf{x}')$.

Confidence Intervals The tangent kernel approach generates precisely defined predictions for inputs $\mathbf{x}$ not seen during training. Given that tangent kernels are deterministic in the infinite layer limit, this result may seem at odds with the statement that the output is Gaussian distributed, which holds for the same limit. Indeed, there is no contradiction.

The tangent kernel predictions (10.58) depend on $\mathbf{f}_1(0)$ and $\mathbf{f}_2(0)$, namely on the outputs generated before training did start, which are determined in turn by the starting parameter configuration. Given that $\boldsymbol{\vartheta}$ is drawn randomly, confidence intervals are obtained by sampling (10.58) with respect to $\boldsymbol{\vartheta}$, viz by sampling $\mathbf{f}_1(0)$ and $\mathbf{f}_2(0)$ with respect to the initial set of parameters.

The Linear Paradox Training is "lazy" when parameters hardly change, as in the infinite layer limit. The first order Taylor expansion,

$$f(\mathbf{x}, \boldsymbol{\vartheta}) \approx f(\mathbf{x}, \boldsymbol{\vartheta}_0) + \big(\nabla_{\boldsymbol{\vartheta}} f(\mathbf{x}, \boldsymbol{\vartheta}_0)\big)\Delta\boldsymbol{\vartheta}, \qquad \Delta\boldsymbol{\vartheta} = \boldsymbol{\vartheta} - \boldsymbol{\vartheta}_0\,, \tag{10.59}$$

becomes then exact. The "linear model" (10.59) is a bit paradoxical. It is still non-linear in $\mathbf{x}$, but linear in parameter space. This is possible as one works in a regime of a diverging number of parameters, one is hence dealing with overfitting.

10.5 Attention Induced Information Routing

Classical deep learning is about information processing. When data sets become complex, as for language processing, it may become important to focus processing resources to subsets of the input stream, the very task of "attention". As such, attention has long been studied both in the neurosciences and in psychology. The first self-contained computational model was however developed from a machine learning perspective.

Top-Down Attention Signals Classically, attention is mediated in the neurosciences by a top-down signal. An example would be the task to concentrate on red objects in the visual stream. This can be done f.i. by gain control, viz by modulating the gain a of the neural transfer mapping $\sigma(a(x - b))$, see (10.1). The gain of neurons processing red/non-red objects would be increased/decreased.

Another example for neural filtering is biased competition, which involves mutually inhibiting clusters of neurons. Clusters encoding red objects will dominate the competition when receiving an additional boost, e.g. in the form of additional top-down inputs.

The Homunculus Problem From a computational perspective, neuroscience models like gain control or biased competition have two core shortcomings.

Generally, attention signals are taken to be generated by some unknown process in higher brain areas, which corresponds on a functional level to presume the existence of an identity, an “homunculus”, deciding whether or not to pay attention to a certain type of objects or data. This is a problem, since theories not including the generation of attention signals are incomplete. Related is the question of how higher brain areas know about lower-area specializations, f.i. with regard to the color of objects within the input stream.

Training Attention The homunculus problem is related conceptional to an assumption often made implicitly in the neurosciences, as well as in psychology. The view is that advanced cognitive systems first develop an understanding of the world, via data acquisition and world modeling, with attention being an add-on. In this view, a cognitive system can pay attention to red objects only once the system understands concepts like ‘red’ and ‘object’.

The approach taken by machine learning is radically different. Here, all parts of the system are jointly trained. This includes attention, when incorporated, as in transformer architectures. The view is that attention can develop in a meaningful manner only together with representation extractions and world modeling.

10.5.1 Transformer

Functionally, attention corresponds to information routing. Only part of the incoming data is processed further. We have seen that gated units, as described by (10.9), are capable to regulate the processing on a local level. Attention works in contrast on higher level, deciding whether somewhat substantial chunks of data are of interest or not.

Tokenization We have seen previously that it may not be desirable to work directly with raw data.[12] For time series’, this amounts to extract meaningful symbols, a procedure denoted “tokenization” in the context of machine learning. For text processing, the most widely used tokens are word equivalents, with token vocabularies having typically about $5 \cdot 10^4$ entries. Tokens are mapped to high-dimensional activity states $\mathbf{x}_i$ in $\boldsymbol{R}^D$, e.g. with $D = 2^9$. Further processing will benefit if embedding schemes respect semantic relations between tokens.

Pairwise Attention Attention involves pairs of tokens where one token needs to decide if another token carries relevant information. This decision cannot be carried out directly on the level of the respective token activities, say $\mathbf{x}_i$ and $\mathbf{x}_j$, with the reason being that activity encodings are constrained by the nature of the overall task. For a working attention mechanism, tokens therefore need to generate associated

[12] See the discussion on time series characterization, Sect. 5.1.4 of Chap. 5.

objects. Given that attention is inherently asymmetric, at least two additional objects are necessary, "query" and "key". In the end, a given token will receive relevant information not just from a single other token, but from a larger number. A third object is hence needed for the alignment of the respective representations, "value".

Query, Key, Value The constituent units of attention-based machine learning models, such as transformers, are tokens, with high-dimensional activity vectors $\mathbf{x}_i$. Tokens come with three matrices,

$$\hat{Q}_i, \qquad \hat{K}_i, \qquad \hat{V}_i ,$$

denoted query, key and value. The entries of these matrices are adapted via backpropagation during training. The three associated query, key and value vectors,

$$\mathbf{Q}_i = \hat{Q}_i \cdot \mathbf{x}_i, \qquad \mathbf{K}_i = \hat{K}_i \cdot \mathbf{x}_i, \qquad \mathbf{V}_i = \hat{V}_i \cdot \mathbf{x}_i , \tag{10.60}$$

are activity dependent. In order to decide whether token j carries information that is relevant to i, the respective queries and keys are compared, viz $\mathbf{Q}_i$ with $\mathbf{K}_j$.

Dot Product Attention As a matter of principle, many methods would be suitable for comparing query and key vectors, say $\mathbf{Q}_i$ with $\mathbf{K}_j$. Given that the involved query and key matrices are anyhow adapted during training, the respective scalar product,

$$e_{ij} = \mathbf{Q}_i \cdot \mathbf{K}_j ,$$

is sufficient. For a measure of how relevant j is to i, we need to map e_{ij} to a value $\alpha_{ij} \in [0, 1]$, e.g. via

$$\alpha_{ij} = \frac{1}{Z_i}\,\mathrm{e}^{-\beta e_{ij}}, \qquad Z_i = \sum_{j\le i} \mathrm{e}^{-\beta e_{ij}}, \qquad \sum_{j\le i} \alpha_{ij} = 1 , \tag{10.61}$$

where β is an effective inverse temperature. The Boltzmann distribution (10.61) is denoted "softmax" in machine learning jargon.

Causal Attention In (10.61) attention is paid only to past events $j \le i$, the causality condition. Attention is not an all-or-nothing affair, but a graded process, with weights that are given by the entries α_{ij} of the attention matrix. The output activity $\mathbf{y}_i$ of token i is determined via

$$\mathbf{y}_i = \sum_{j\le i} \alpha_{ij} \mathbf{V}_j \tag{10.62}$$

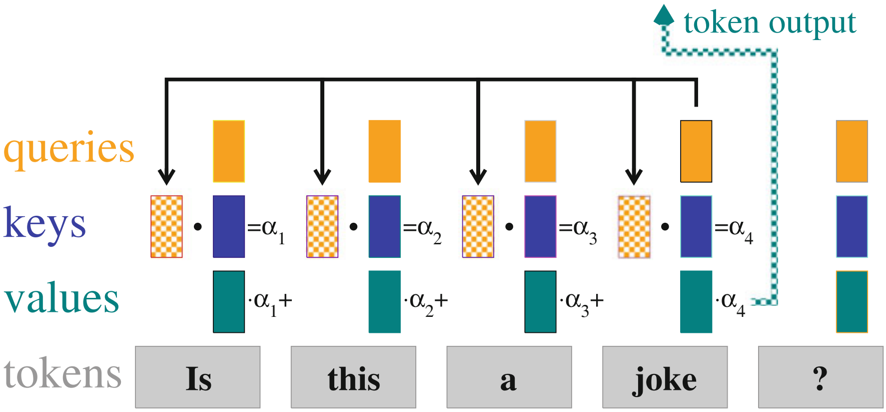


Fig. 10.11 Attention in machine learning. Tokens with activities $\mathbf{x}_i$ generate query/key/value vectors as $\mathbf{Q}_i = \hat{Q}_i \cdot \mathbf{x}_i$, $\mathbf{K}_i = \hat{K}_i \cdot \mathbf{x}_i$ and $\mathbf{V}_i = \hat{V}_i \cdot \mathbf{x}_i$. Causality implies to attend only to previous words in a sentence. For a given token, here 'joke', the scalar product of its query vector with all previous key vectors generates weights α_i. The sum of the weighted values is the output of the attention layer. See (10.61) and (10.62)

by the amount of attention paid to what happened before. An illustration is given in Fig. 10.11. The past becomes irrelevant in the limiting case $\alpha_{ij} = \delta_{ij}$, which is included in (10.62).

Transformer The primary task of attention is information routing. Once selected, information needs also to be processed. For transformer models this is done on a token-per-token level, by feeding $\mathbf{y}_i$ into a perceptron with a single hidden layer. Transformer layers consist hence of two sublayers, attention and a classical token-wise neural net.

Linearized Attention Attention is a non-linear process scaling quadratically with the number of tokens per layer, N_T, the context length. There are $N_\mathrm{T}^2/2$ scalar products $\mathbf{Q}_i \cdot \mathbf{K}_j$ to be computed, which are individually quadratic in the respective token activities, $\mathbf{x}_i$ and $\mathbf{x}_j$. Linearized versions of attention forfeit the mapping (10.61) to a normalized distribution, using instead

$$\mathbf{y}_i\big|_{\mathrm{lin}} = \sum_{j\le i} (\mathbf{Q}_i \cdot \mathbf{K}_j)\mathbf{V}_j = \mathbf{Q}_i \cdot \sum_{j\le i} \mathbf{K}_j \otimes \mathbf{V}_j \tag{10.63}$$

for the new token activity, modulo a normalization factor, where $\otimes$ denotes the outer product.

Recurrent Linear Attention We rewrite (10.63) as

$$\mathbf{y}_i\big|_{\text{lin}} = \mathbf{Q}_i \cdot \hat{S}_i, \qquad\qquad \hat{S}_i = \sum_{j\le i} \gamma^{i-j}\, \mathbf{K}_j \otimes \mathbf{V}_j$$

where we introduced with $\gamma \in [0, 1]$ an additional decay factor. It plays a role equivalent to the spectral radius of recurrent networks, see Sect. 10.2.1. Past events are explicitly discounted when $\gamma < 1$. The update relation

$$\hat{S}_{i+1} = \mathbf{K}_{i+1} \otimes \mathbf{V}_{i+1} + \gamma\,\hat{S}_i \tag{10.64}$$

shows that the inference effort scales linearly with context length N_{T}. This formulation is also denoted "retentive network".

One can interpret (10.64) as an abstract recurrent network unfolding in 'context time', with the entries of the key-value matrix $\hat{S}_i$ corresponding to neuronal activities. The new network activity $\hat{S}_{i+1}$ is given by the sum of two terms, the input $\mathbf{K}_{i+1} \otimes \mathbf{V}_{i+1}$ and the old activity $\hat{S}_i$, with the latter being discounted by the spectral radius γ of the network.

Exercises

(10.1) GATED LINEAR UNITS
A symmetrized version of (10.9) is

$$\sigma_{\text{GLU}}(\mathbf{x}) = \tanh(x_w - b_w)\,\tanh(x_v - b_v)\,, \tag{10.65}$$

where $\sigma_{\tanh}$ has been used for the transfer function. Show that a single unit can encode the AND and the XOR gate.

(10.2) VARIANCE OF A PRODUCT
Derive (10.18) for the product of two statistically independent random variables. Use $x = \Delta x + \mu_x$, where $\Delta x = x - \mu_x$, equivalently for both variables.

(10.3) WIGNER SEMI-CIRCULAR LAW
Show that (10.15) reduces to the Wigner semi-cricular law (1.15) for the density of state of the adjacency matrix of an Erdös–Rényi graph with coordination number z.

(10.4) LANDAU THEORY FOR ABSORBING PHASE TRANSITIONS
The variance σ_y^2 of the neuronal activity is small close to the critical point $R_w = 1$. An effective theory can be obtained by expanding the self-consistency condition (10.26) with respect to powers of σ_y^2. Show that the corresponding expression (6.2) for the Landau theory of phase transitions is recovered when σ_{ext}^2 is measured in units of σ_y.

(10.5) <u>Layer Variance Mapping</u>
Generalize the mapping (10.27) of the variance of unit activity from one layer to the next to non-zero means μ_x, μ_y and μ_w.

(10.6) <u>Neural Differential Equations</u>
Analytically solve the neural differential equations (10.35) for the case of a single linear unit per layer, with $f(x, \vartheta) = \vartheta x$. A single training pair (x^α, F^α) is given. Is it consistent with the update rule (10.34) that ϑ is independent of layer time?

(10.7) <u>Dot Products are Kernel Functions</u>
Kernels are (semi-) positive definite symmetric functions of two inputs. Prove that scalar products are kernels.

(10.8) <u>Linear Attention</u>
Assume that the elements of the $D \times D$ query, key and value matrices are randomly initialized with zero mean and standard deviation σ_0. Use (10.63) to evaluate the variance of the activity $\mathbf{y}_i$ when inputs are random, having zero mean and standard deviation σ_x. How should σ_0 scale with the embedding dimension D?

Further Reading

We recommend reviews and textbooks on the notion of brain criticality, Gros (2021), classical deep learning concepts, Biehl (2023), and the defining features of attention in psychology and the neurosciences, Lindsay (2020). Of interest to the reader may be furthermore introductions to Gaussian processes, Williams and Rasmussen (2006), to random matrix theory, Akjouj et al. (2022), and recurrent neural architectures, (Yu et al., 2019).

Original literature relevant to the subjects treated concerns gated linear units, Dauphin et al. (2017), elliptic random matrices, Sommers et al. (2017), and the notion of flow control, Schubert & Gros. (2019). Neural tangent kernels are treated in Jacot et al. (2018), neural ordinary differential equations in Chen et al. (2018). Transformer models were introduced in the seminal ‘Attention is All You Need’ paper Vaswani et al. (2017), retentive networks in Sun et al. (2023).

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11 Solutions

The nature of the exercises given at the end of the respective chapters varies. Some exercises are concerned with the extension of formulas and material, others deal with the investigation of models through computer simulations. Follow the index to reach the solution to any exercise.

C. Gros, *Complex and Adaptive Dynamical Systems*,
https://doi.org/10.1007/978-3-031-55076-8_11

11.1 Solutions to the Exercises of Chap. 1

(1.1) Network of cliques
The respective networks are shown in Fig. 11.1. The network diameter D is 2 (3) for the network of companies (managers), the average degree $18/6 = 3$ ($36/9 = 4$) and the clustering coefficient $5/12 \approx 0.42$ ($163/270 \approx 0.6$). For comparison, the clustering would have been $C_{\text{rand}} = z/(N-1)$, see (1.3), for random graphs, viz $3/5 = 0.6$ ($4/8 = 0.5$). The network of managers contains the respective board compositions as cliques, which explains the large value of C.

(1.2) Ensemble Fluctuations
The probability that a given vertex has degree k is provided by a Binomial distribution, see (1.4). Therefore, the probability that R vertices have degree k, viz $X_k = R$, is given by

$$P(X_k = R) = \binom{N}{R} p_k^R (1-p_k)^{N-R} . \tag{11.1}$$

Considering the large-N limit with $N \gg R$, and $\binom{N}{R} \sim (N/R)^R$, we find

$$P(X_k = R) = e^{-\lambda_k} \frac{(\lambda_k)^R}{R!}, \qquad \lambda_k = N p_k = \langle X_k \rangle , \tag{11.2}$$

viz the binomial distribution. Equation (11.1) reduces for $p_k \ll 1$ to the Poisson distribution, namely to (11.2), in the thermodynamic limit $N \to \infty$.

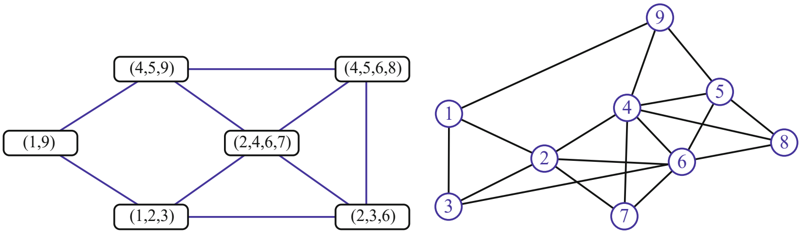


Fig. 11.1 A network of cliques (left) and the associated network of nodes (right). Nodes correspond to managers sitting on the board of several companies, the cliques

(1.3) CLUSTERING COEFFICIENT AND DEGREE SEQUENCE

The expression (1.29) for the clustering coefficient has the form

$$C = \frac{1}{Nz}\left[\sum_k k q_k\right]^2 = \frac{1}{Nz^3}\left[\sum_k k(k+1)p_{k+1}\right]^2$$

where we have used $q_k = (k+1)p_{k+1}/z$ for the excess degree distribution q_k. We hence find

$$C = \frac{1}{Nz^3}\left[\sum_k (k-1)kp_k\right]^2 = \frac{1}{Nz^3}\left[\frac{1}{N}\sum_i (k_i-1)k_i\right]^2, \qquad (11.3)$$

when expressing C as a function of the degree sequence $\{k_i\}$. For a homogeneous network, with $k_i = z$, for all sites, the formula (11.3) yields $N^2(z-1)^2z^2/(N^3z^3) \approx z/N$, in agreement with (1.3).

A simple star-shaped network with a single central site has the degree sequence

$$k_1 = N-1, \qquad k_i = 1, \qquad i = 2, \ldots, N,$$

with an intensive coordination number $z = 2(N-1)/N \approx 2$, for large number of sites N. The statistical formula (11.3) for the clustering coefficient would, on the other hand, diverge like

$$\approx \frac{1}{N2^3}\left[\frac{(N-2)(N-1)}{N}\right]^2 \sim \frac{N}{2^3}$$

in the thermodynamic limit $N \to \infty$. When deriving the expression (11.3) one did disregard all dependencies of the respective degrees of nearest-neighbor sites. A large experimental value indicates therefore the presence of strong inter-site degree correlations.

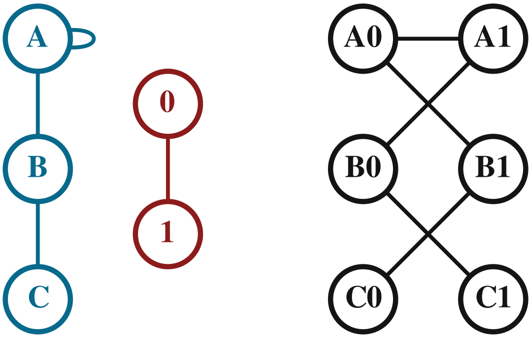


Fig. 11.2 The Kronecker graph $K = G \otimes H$ (right) is the outer product of G (left) and H (middle). A pair of sites in K are linked only if the constituent sites of G and H are both connected

(1.4) <u>TSALLIS–PARETO DISTRIBUTION</u>

The mean μ of the Tsallis–Pareto distribution (1.78) was already evaluated in Sect. 1.1.3 as

$$\mu = (\alpha - 1)\int_0^X dx\, \frac{x+1-1}{(1+x)^\alpha} = \frac{\alpha-1}{2-\alpha}(1+x)^{2-\alpha}\Big|_0^X - 1$$
$$= \begin{cases} \infty, & \alpha < 2, \quad X \to \infty \\ 1/(\alpha-2), & \alpha > 2, \quad X \to \infty \end{cases} \tag{11.4}$$

where the normalization $1 = \int p(x)dx$ for $\alpha > 1$ was used. With $x^2 = (x+1)^2 - 2x - 1$ we find

$$\langle x^2 \rangle + 2\langle x \rangle + 1 = (\alpha - 1)\int_0^X dx\, \frac{(x+1)^2}{(1+x)^\alpha} = \frac{\alpha-1}{3-\alpha}(1+x)^{3-\alpha}\Big|_0^X$$

for the second moment, which simplifies for $\alpha > 3$ to

$$\langle x^2 \rangle = \frac{\alpha-1}{\alpha-3} - \frac{2}{\alpha-2} - 1 = \frac{2}{\alpha-3} - \frac{2}{\alpha-2} = \frac{2}{(\alpha-3)(\alpha-2)}\,. \tag{11.5}$$

The standard deviation σ^2 is then

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{\alpha-1}{(\alpha-3)(\alpha-2)^2}\,. \tag{11.6}$$

(1.5) <u>KRONECKER GRAPHS</u>

An illustrative example of the Kronecker graph $K = G \otimes H$ of a 3-site graph G and a two-site graph H is given in Fig. 11.2. The resulting network K has $3 \cdot 2 = 6$ sites.

Denoting by $p^G(k)$ and $p^H(k)$ the respective degree distributions one can evaluate the degree distribution of the K as

$$p^K(k) = \sum_{l,l'} \delta_{k,l\cdot l'}\, p^G(l)\, p^H(l') \;=\; \int_0^\infty dl\, dl'\, \delta(k - l \cdot l')\, p^G(l)\, p^H(l')$$
$$= \int_0^\infty \frac{dl}{l}\, p^G(l)\, p^H(k/l)\,,$$

since a given site x_{ij} in K is linked to $k_i \cdot k_j$ other sites, where k_i and k_j are the respective degrees of the constituent sites in G and H. In above derivation we have used $\delta(ax) = \delta(x)/|a|$. The average degree and hence the coordination number is multiplicative,

$$\langle k \rangle_K = \sum_k k\, p^K(k) = \sum_{l,l'} l \cdot l'\, p^G(l)\, p^H(l') = \langle k \rangle_G\, \langle k \rangle_H\,.$$

(1.6) <u>SELF-RETRACING PATH APPROXIMATION</u>
This exercise needs a background in Green's functions. One needs to find the one-particle Green's function $G(\omega)$ of single particle hopping with amplitude $t = 1$ (the entries of the adjacency matrix) on a random lattice. We denote by

$$G_0(\omega) = \frac{1}{\omega}, \qquad G(\omega) = \frac{1}{\omega - \Sigma(\omega)},$$

the single-site Green's function $G_0(\omega)$, viz the Green's function on an isolated vertex and with $\Sigma(\omega)$ the respective one-particle self-energy.

We may now expand the self-energy in terms of hopping processes, with the lowest-order process being to hop to the next site and back. Once the next site has been reached the process can be iterated. We then have

$$\Sigma(\omega) = zG(\omega), \qquad G(\omega) = \frac{1}{\omega - zG(\omega)},$$

which is just the starting point for the semi-circle law, Eq. (1.14).

(1.7) <u>PROBABILITY GENERATING FUNCTIONS</u>
We start by evaluating the variance σ_K^2 of the distribution p_k generated by $G_0(x) = \sum_k p_k\, x^k$, considering

$$G_0'(x) = \sum_k k\, p_k\, x^{k-1}, \qquad G_0''(x) = \sum_k k(k-1)\, p_k\, x^{k-2}$$

together with

$$G_0'(1) = \langle k\rangle, \qquad G_0''(1) = \langle k^2\rangle - \langle k\rangle .$$

Hence

$$\sigma_K^2 = \langle k^2\rangle - \langle k\rangle^2 = G_0''(1) + \langle k\rangle - \langle k\rangle^2 . \tag{11.7}$$

For a cummulative process generated by $G_C(x) = G_N(G_0(x))$ we have

$$\frac{d}{dx}G_C(x) = G_N'(G_0(x))\, G_0'(x), \qquad \mu_C = \langle n\rangle\langle k\rangle ,$$

where we have denoted with μ_C the mean of the cummulative process. For the variance σ_C^2 we find w

$$\frac{d^2}{dx^2}G_C(x) = G_N''(G_0(x))\left(G_0'(x)\right)^2 \,+\, G_N'(G_0(x))\, G_0''(x) ,$$

which leads with (11.7) to

$$\begin{aligned}\sigma_C^2 &= G_C''(1) + \mu_C - \mu_C^2\\ &= \left(\sigma_N^2 - \langle n\rangle + \langle n\rangle^2\right)\langle k\rangle^2 + \langle n\rangle\left(\sigma_K^2 - \langle k\rangle + \langle k\rangle^2\right) + \langle n\rangle\langle k\rangle - \langle n\rangle^2\langle k\rangle^2\\ &= \langle k\rangle^2\sigma_N^2 \,+\, \langle n\rangle\,\sigma_K^2 .\end{aligned}$$

If p_n is deterministic, e.g. if we throw the dice always exactly N times, then $\sigma_N^2 = 0$ and $\sigma_C^2 = N\sigma_K^2$, in agreement with the law of large numbers, Eq. (5.10).

(1.8) CLUSTERING COEFFICIENT OF REGULAR LATTICES

One Dimension We start proving (1.60),

$$C = \frac{3(z-2)}{4(z-1)} ,$$

which holds for the clustering coefficient of a one-dimensional lattice with coordination number z. In this case, a site is connected to all sites within a distance of $z/2$ (included), both to the left and to the right.

The clustering coefficient C is defined by the average fraction of pairs of neighbors of a vertex, which are also neighbors of each other. Therefore, we remind ourselves that

$$\sum_{k=1}^{z-1} k = \frac{1}{2}z(z-1)$$

is the number of pairs of neighbors of a given vertex. Next, we evaluate the connected pairs of neighbors for a given node. Ordering the neighbors of a given node linearly, compare Fig. 1.13, in a chain of lengths $z + 1$, one has $3z/4$ starting nodes, each connected to $z/2 - 1$ sites (in one direction, left or right), which results in a total of

$$\frac{3z}{4}\left(\frac{z}{2} - 1\right)$$

connected pairs of neighbors. Thus the result for C is

$$C\frac{1}{2}z(z-1) = 3\frac{z}{4}\left(\frac{z}{2} - 1\right), \qquad C = \frac{3(z-2)}{4(z-1)}\,.$$

General Dimensions The arguments above can be generalized for lattices in arbitrary dimension d using relatively simple arguments. Consider that we are now dealing with d coordinate or lattice lines traversing a certain node. Thus, in order to calculate the cluster coefficient, we confine ourselves to a one-dimensional subspace substitute z by z/d, the connectivity on the line, yielding

$$C_d = \frac{3(z/d-2)}{4(z/d-1)} = \frac{3(z-2d)}{4(z-d)}\,.$$

(1.9) <u>Robustness of Graphs Against Focused Attacks</u>
Following Sect. 1.4, we define with

$$F_0(x) = \sum_{k=0}^{\infty} p_k b_k x^k = \sum_{k=0}^{k_c} p_k x^k, \qquad F_1(x) = \frac{F_0'(x)}{z}$$

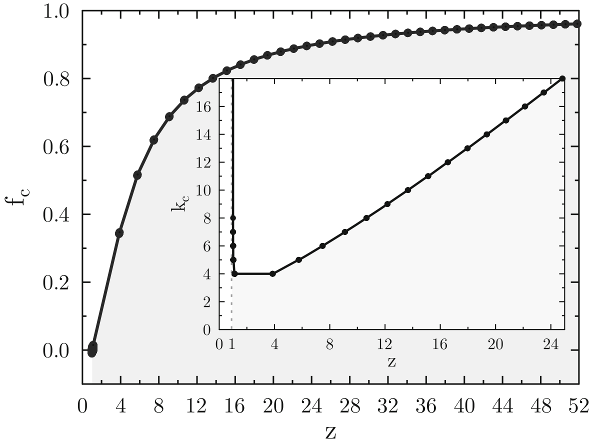
the non-normalized generating function for the degree distribution of the active nodes. The percolation threshold occurs for

$$F_1'(1) = 1, \qquad F_1'(1) = \frac{1}{z}\sum_{k=2}^{k_c} p_k k(k-1)\,.$$

Using the degree distribution for Erdös–Rényi graphs we obtain, see Fig. 11.3,

$$p_k = \mathrm{e}^{-z}\frac{z^k}{k!}, \qquad \mathrm{e}^z = z\sum_{k'=0}^{k_c-2}\frac{z^{k'}}{k'!}\,. \qquad (11.8)$$

Fig. 11.3 Critical fraction f_c for Erdös–Rényi graphs to retain functionality, as a function of coordination number z. **Inset:** Critical degree k_c. See (11.9) and (11.8) respectively



The network will fall below the percolation threshold when the fraction of removed vertices exceeds the critical fraction

$$f_c = 1 - \sum_{k=0}^{k_c} p_k = 1 - \frac{1}{z} - e^{-z}\left(\frac{z^{k_c-1}}{(k_c-1)!} + \frac{z^{k_c}}{k_c!}\right), \qquad (11.9)$$

where we have used (11.8). The result is plotted in Fig. 11.3. The critical fraction f_c approaches unity in the limit $z \to \infty$.

(1.10) <u>EPIDEMIC SPREADING IN SCALE-FREE NETWORKS</u>
This task has the form of a literature study. One defines with $\rho_k(t)$ the probability that an individual having k social connections is ill, viz infected (i.e. k is the degree). It changes via

$$\dot{\rho}_k(t) = -\rho_k(t) + \lambda k\,[1-\rho_k]\,\Theta(\lambda)\,. \qquad (11.10)$$

The first term on the right-hand-side describes the recovering process, the second term the infection via the infection rate λ. The infection process is proportional to the number of social contacts k, to the probability $[1-\rho_k]$ of having been healthy previously, and to the probability

$$\Theta(\lambda) = \sum_k \frac{k\rho_k(t)\,p_k}{\sum_j j p_j} \qquad (11.11)$$

that a given social contact is with an infected individual, where p_k is the degree distribution. The arguments leading to above functional form for $\Theta(\lambda)$ are very similar to those for the excess degree distribution q_k discussed in Sect. 1.3.1. Note, that all edges leading to the social contact enter $\Theta(\lambda)$, whereas only the outgoing edges contributed to q_k.

Numerically one can the solve (11.10) and (11.11) self-consistently for the stationary state $\dot{\rho}_k = 0$ and a given scale-invariant degree distribution p_k, as explained in the reference given. The central question then regards the existence of a threshold: Does the infection rate λ need to be above a finite threshold for an outbreak to occur or does an infinitesimal small λ suffice?

11.2 Solutions to the Exercises of Chap. 2

(2.1) Jacobians with degenerate eigenvalues
The Jacobian and the eigenvalues of (2.72) are

$$\begin{pmatrix} -1 & 1 \\ 0 & -r \end{pmatrix}, \qquad \lambda_1 = -1, \qquad \lambda_2 = -r, \qquad \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

with $\mathbf{e}_1$ being the eigenvector for λ_1. In order to determine $\mathbf{e}_2$, we consider the ansatz

$$\mathbf{e}_2 = \begin{pmatrix} a \\ 1 \end{pmatrix}, \qquad -a + 1 = -ra, \qquad a = \frac{1}{1-r}.$$

The result is, that $\mathbf{e}_2$ aligns more and more with $\mathbf{e}_1$ in the limit $r \to 1$ (when normalized), viz when the two eigenvalues become degenerate.

(2.2) Catastrophic Superlinear Growth
No Damping For $\gamma = 0$ in (2.73), the polynomial growth equation

$$\dot{x} = x^{\beta}, \qquad \dot{x}x^{-\beta} = 1, \qquad \frac{x^{1-\beta}}{1-\beta} = t + \frac{x_0^{1-\beta}}{1-\beta}$$

has the solution

$$x(t) = \left[(1-\beta)t + x_0^{1-\beta}\right]^{1/(1-\beta)} = \left[\frac{1}{1/x_0^{\beta-1} - (\beta-1)t}\right]^{\frac{1}{\beta-1}}, \tag{11.12}$$

where $x_0 = x(t=0)$. The long-time behavior is regular for sublinear growth $\beta < 1$. For superlinear growth, $\beta > 1$, one finds instead a singularity t_s with $\lim_{t\to t_s} x(t) = \infty$

$$x(t) = \left[\frac{1/(\beta-1)}{t_s - t}\right]^{\frac{1}{\beta-1}}, \qquad t_s = \frac{1}{\beta-1}\frac{1}{x_0^{\beta-1}}. \tag{11.13}$$

With Damping A finite $\gamma = 1$ in (2.73),

$$\dot{x} = x^\beta - x = x(x^{\beta-1} - 1). \tag{11.14}$$

may change the flow qualitatively.

- For a superlinear $\beta > 1$ one finds $x(t) \to 0$ for all starting $x_0 \in [0, 1]$, since then $x^{\beta=1} < 1$, approaching however again a singularity akin to (11.13) for starting resources $x_0 > 1$.
- For a sublinear $\beta < 1$ on finds however $x(t) \to 0$ for all starting x_0, since the Lyapunov exponent of the fixpoint $x^* = 1$ is $\beta - 1$.
- Note, that superlinear growth processes have been described for some cultural phenomena, like the growth of cities, with exponents $\beta \approx 1.2$ being slightly above unity. See, e.g., Bettencourt et al. (2007).

Sublinear Decay Finally we consider a sublinear decay process,

$$\dot{x} = -x^\beta, \qquad \dot{x}x^{-\beta} = -1, \qquad \frac{x^{1-\beta}}{1-\beta} = -t + \frac{x_0^{1-\beta}}{1-\beta},$$

which has the solution

$$x(t) = (1-\beta)^{\frac{1}{1-\beta}}\left[t_0 - t\right]^{\frac{1}{1-\beta}}, \qquad t_0 = \frac{x_0^{1-\beta}}{1-\beta}. \tag{11.15}$$

For $\beta \in [0, 1]$, the resource is fully depleted when t reaches t_0.

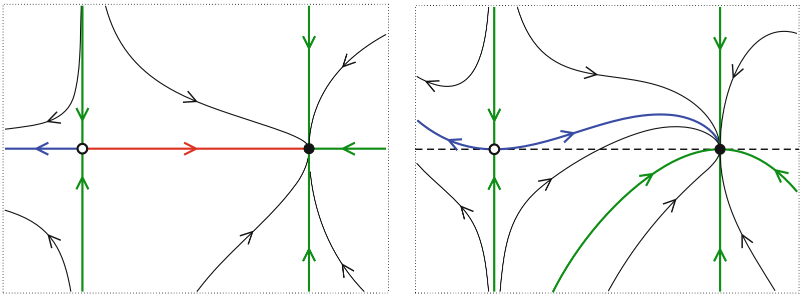


Fig. 11.4 Sample trajectories of the system (11.16) for $\epsilon = 0$ (left) and $\epsilon = 0.2$ (right). Shown are the stable manifolds (thick green lines), the unstable manifolds (thick blue lines) and the heteroclinic orbit (thick red line)

(2.3) <u>HETEROCLINIC ORBITS AND SYMMETRIES</u>
The system

$$\begin{aligned}\dot{x} &= 1 - x^2 \\ \dot{y} &= -y + \epsilon\big(1 - x^2\big)^2\end{aligned} \qquad J(x^*) = \begin{pmatrix} -2x^* & 0 \\ 0 & -1 \end{pmatrix} \tag{11.16}$$

has the two fixpoints $\mathbf{x}^*_\pm = (x^*, 0)$, where $x^* = \pm 1$. The eigenvectors of the Jacobian $J(x^*)$ are aligned with the x and the y axis respectively, just as for the system (2.18), with $(-1, 0)$ being now a saddle and $(1, 0)$ a stable node. The flow diagram is illustrated in Fig. 11.4, it is invariant when inverting $x \leftrightarrow (-x)$. The system contains the $y = 0$ axis as a mirror line for a vanishing $\epsilon = 0$ and there is a heteroclinic orbit connecting the unstable manifold of $\mathbf{x}^*_-$ to one of the stable manifolds of $\mathbf{x}^*_+$.
A finite ϵ removes the mirror line $y = 0$ present at $\epsilon = 0$ and destroys the heteroclinic orbit. The unstable manifold of $\mathbf{x}^*_-$ is still attracted by $\mathbf{x}^*_+$, however now as generic trajectory.

(2.4) <u>THE SUBCRITICAL PITCHFORK TRANSITION</u>
The fastest way to understand the behavior of the subcritical pitchfork transition,

$$\dot{x} = ax + x^3, \qquad U(x) = -\frac{a}{2}x^2 - \frac{x^4}{4}, \tag{11.17}$$

is to look at the respective bifurcation potential $U(x)$, which is illustrated in Fig. 11.5. The fixpoints $\pm\sqrt{-a}$ exist for $a < 0$ and are unstable, the trivial fixpoint is stable (unstable) for $a < 0$ ($a > 0$).

(2.5) <u>ENERGY CONSERVATION IN LIMIT CYCLES</u>
The numerical integration of the energy change $\int_0^t \dot{E}\, dt'$ for the Taken-Bogdanov system (2.33) is presented in Fig. 11.6. Energy is continuously taken up and dissipated even along the limit cycle, but the overall energy

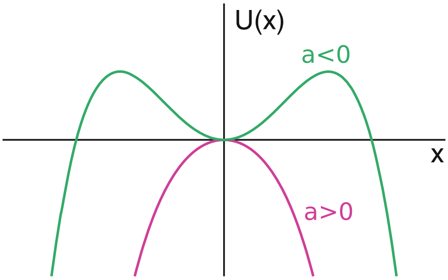


Fig. 11.5 The bifurcation potential $U(x)$ of the subcritical pitchfork transition, as defined by (11.17). The trivial fixpoint $x^* = 0$ becomes unstable for $a > 0$

balance vanishes once one cycle has been completed. Simple integrator methods like Euler's method may not provide adequate numerical precision.

(2.6) MAXIMAL LYAPUNOV EXPONENT
We consider the definition

$$\lambda^{(\max)} = \lim_{n\gg 1} \frac{1}{n} \log \left| \frac{df^{(n)}(x)}{dx} \right|, \qquad f^{(n)}(x) = f\big(f^{(n-1)}(x)\big).$$

of the maximal Lyapunov exponent (2.55), together with

$$\frac{df^{(n)}(x)}{dx} = \frac{df\big(f^{(n-1)}(x)\big)}{dx} = f'\big(f^{(n-1)}(x)\big)\,\frac{df^{(n-1)}(x)}{dx}$$
$$= f'\big(f^{(n-1)}(x)\big)\,\frac{df\big(f^{(n-2)}(x)\big)}{dx}.$$

We hence obtain

$$\lambda^{(\max)} = \lim_{n\gg 1} \frac{1}{n} \sum_{j=0}^{n-1} \log \left|f'(f^{(j)})\right|, \qquad f^{(0)}(x) = x,$$

which is just the definition of the time average of the local Lyapunov exponent (2.54).

(2.7) LYAPUNOVEXPONENTS ALONG PERIODIC ORBITS
A perturbation $\delta\varphi$ of a solution $\varphi^*(t)$ of (2.35) evolves in linear order as

$$\frac{d}{dt}\varphi^* + \frac{d}{dt}\delta\varphi = 1 - K\cos(\varphi^*) + K\sin(\varphi^*)\delta\varphi, \qquad \frac{d}{dt}\delta\varphi = \lambda_\varphi \delta\varphi,$$

with the longitudinal (local) Lyapunov exponent being $\lambda_\varphi = K\sin(\varphi*)$. There are two fixpoints for a supercritical $K > 1$, compare Fig. 2.13, with the upper/lower one being stable/unstable, in agreement with λ_φ being positive/negative.

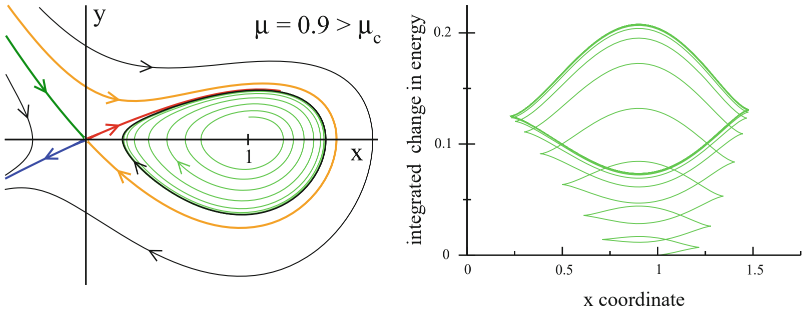


Fig. 11.6 **Left:** The flow for the Taken-Bogdanov system (2.33), for $\mu = 0.9 > \mu_c$. The thick black line is the limit cycle, compare Fig. 2.12. **Right:** The integrated energy balance (2.34) along a trajectory starting at $(x, y) = (1, 0.2)$ (in light green in the left panel), using forth-order Runge-Kutta integration

For $K < K_c$ the trajectory is periodic and $\lambda_\varphi = K\sin(\varphi*)$ varies along the unit circle. The mean exponent,

$$\lambda_T = \int_0^{2\pi} \lambda_\varphi d\varphi = K\int_0^{2\pi} \sin(\varphi)d\varphi = 0\,,$$

vanishes however. λ_T is the "global" or "maximal" Lyapunov exponent. This result is not a coincidence. Two periodic solutions $\varphi^*(t + t_{1,2})$ with starting angles $\varphi^*_{1,2} = \varphi^*(t_{1,2})$ will have a separation $\Delta\varphi = \varphi^*_2 - \varphi^*_1$ which oscillates with the same period T, viz

$$\Delta\varphi(t+T) = \Delta\varphi(t), \qquad \varphi^*(t+T) = \varphi^*(t)\,. \tag{11.18}$$

Linearizing (11.18) results in a vanishing global Lyapunov exponent λ_T.

(2.8) <u>GRADIENT DYNAMICAL SYSTEMS</u>

For the two dimensional system (11.16),

$$\begin{aligned}\dot{x} &= f_x(x, y) = 1 - x^2 + g(x, y) \equiv -\frac{\partial U(x,y)}{\partial x} \\ \dot{y} &= f_y(x, y) = -y + \epsilon\big(1 - x^2\big)^2 \equiv -\frac{\partial U(x,y)}{\partial y}\end{aligned}\,, \tag{11.19}$$

to be a gradient system, as defined by (2.42), we need to select the term $g(x, y)$ such that

$$\frac{\partial f_x}{\partial y} = \frac{\partial f_y}{\partial x} = -4\epsilon x(1 - x^2), \qquad g(x, y) = -4\epsilon x(1 - x^2)y\,.$$

The potential U is then

$$U(x,y) = \frac{y^2}{2} + \frac{x^3}{3} - x - \epsilon\big(1-x^2\big)^2 y\,.$$

The system (11.19) retains its fixpoints $(\pm 1, 0)$ even in the presence of $g(x,y)$. The respective Jacobians also remain unchanged, since

$$\frac{\partial g}{\partial x}\Big|_{\mathbf{x}=(\pm 1,0)} = 0 = \frac{\partial g}{\partial y}\Big|_{\mathbf{x}=(\pm 1,0)}\,.$$

An additional fixpoint is however generated by $g(x,y)$. Which one is it?

(2.9) <u>LIMIT CYCLES IN GRADIENT SYSTEMS</u>
Let's denote with $x^*(t)$ a closed trajectory with period T and $x^*(T) = x^*(0)$. The flow is given by $\dot{\mathbf{x}} = -\nabla V(\mathbf{x})$, which leads to

$$0 = V\big(\mathbf{x}^*(T)\big) - V\big(\mathbf{x}^*(0)\big) = \int_0^T \frac{dV}{dt}dt$$
$$= \int_0^T \big(\nabla V \cdot \dot{\mathbf{x}}\big)dt = -\int_0^T \big(\dot{\mathbf{x}}\big)^2 dt\,.$$

The flow $\dot{\mathbf{x}}$ must hence vanish along any closed trajectory for gradient limit cycles, which is only possible if this trajectory consists of an isolated fixpoint.

(2.10) <u>DELAY DIFFERENTIAL EQUATIONS</u>
Probing for a harmonic orbit, we have

$$\dot{y}(t) = -ay(t) - by(t-T), \qquad -i\omega = -a - be^{i\omega T}, \qquad y(t) = e^{-i\omega t}\,.$$

Taking the real and the imaginary part leads to

$$-\frac{a}{b} = \cos(\omega T), \qquad \frac{\omega}{b} = \sin(\omega T), \qquad a^2 + \omega^2 = b^2\,,$$

where we have used $\cos^2 + \sin^2 = 1$ in the last step. For $a = 0$ we obtain

$$\omega T = \frac{\pi}{2} + \pi n, \qquad n = 0,\ \pm 1,\ \pm 2,\ \ldots$$

and $b = \pm\omega$ for even/odd values of n.

(2.11) <u>PERIOD-3 CYCLES IN THE LOGIST MAP</u>
The 3-fold iterated logistic map is illustrated in Fig. 11.7. At $r = r^*$ it is tangential to the identity, hence the name "tangent transition". For r slightly larger than r^* there are three pairs of stable/unstable fixpoints. At criticality their values merge to

$$x_1 \approx 0.5144, \qquad x_2 \approx 0.9563, \qquad x_3 \approx 0.1510\,,$$

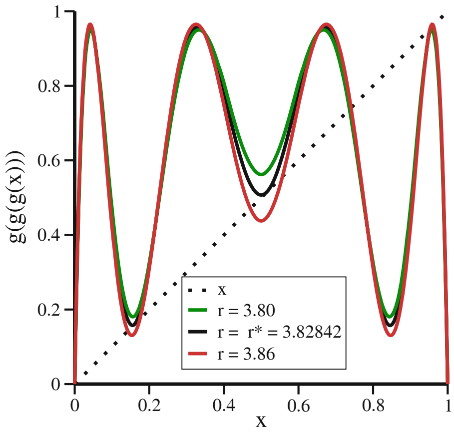


Fig. 11.7 The functional dependence of the 3-fold iterated logistic map. A saddle-node bifurcation occurs at $r = r^* = 1 + 2\sqrt{2}$. At this point the iterated map is tangential to the identity map

which can be found from the roots of $x = g(g(g(x)))$, with $g(x) = rx(1-x)$. The derivative of the flow,

$$\frac{d}{dx}\,g(g(g(x))) = g'(x_3)g'(x_2)g'(x_1) \;\to\; 1$$

becomes neutral, as one can verify numerically. As expected, the Lyapunov exponent vanishes at criticality.

(2.12) Car-Following Model
We consider $\dot{x} = v_0 + \Delta\dot{x}$ in (2.76), which reads $\ddot{x}(t+T) = \alpha(v(t) - \dot{x}(t))$, and obtain

$$\frac{d}{dt}\Delta\dot{x}(t+T) = -\alpha\Delta\dot{x}(t), \qquad -\lambda \mathrm{e}^{-\lambda T} = -\alpha, \qquad \Delta\dot{x}(t) = e^{-\lambda t}\,.$$

The steady-state solution is stable for $\lambda > 0$, which is the case when both α and T are positive. It is fun to simulate the model numerically for non-constant $v(t)$.

11.3 Solutions to the Exercises of Chap. 3

(continued)

(3.1) SHIFT MAP

Transformation With $x = \big[1 - \cos(\pi\theta)\big]/2$ we find

$$4x(1-x) = (1-\cos(\pi\theta))(1+\cos(\pi\theta)) = 1-\cos^2(\pi\theta)) = \sin^2(\pi\theta))$$

for the logistic map $g(x) = rx(1-x)$ at $r = 4$, which takes the form

$$\begin{aligned} x_{n+1} &= \sin^2(\pi\theta_n)) \\ \cos(\pi\theta_{n+1}) &= 1 - 2\sin^2(\pi\theta_n) = \cos^2(\pi\theta_n) - \sin^2(\pi\theta_n) \\ &= \cos(2\pi\theta_n)\,. \end{aligned}$$

The arguments of this expression lead directly to the shift map (3.73), which is shown in Fig. 11.8.

Stationary Distribution The distribution functions $p(x)$ and $p(\theta)$ are related by

$$p(x)dx \;=\; p(\theta)d\theta, \qquad \frac{dx}{d\theta} = \frac{\pi}{2}\,\sin(\pi\theta), \qquad \cos(\pi\theta) = 1-2x\,.$$

With $\sin(\pi\theta) = \sqrt{1-\cos^2(\pi\theta)}$ we find

$$p(x) = \frac{p(\theta)}{dx/d\theta} = \frac{2}{\pi}\frac{p(\theta)}{\sqrt{1-(1-2x)^2}} = \frac{1}{\pi}\frac{p(\theta)}{\sqrt{x(1-x)}}\,, \tag{11.20}$$

which is illustrated in Fig. 11.8. One can convince oneself easily that $p(\theta) = 1$, namely that all angles θ occur with identical probabilities.

(3.2) FIXPOINTS OF THE LORENZ MODEL

Linearizing the differential equations (3.8) around the fixpoint (x^*, y^*, z^*) yields

$$\begin{aligned} \dot{\tilde{x}} &= -\sigma\,\tilde{x} \,+\, \sigma\,\tilde{y}, \\ \dot{\tilde{y}} &= (-z^* + r)\,\tilde{x} \,-\, \tilde{y} - x^*\tilde{z}, \\ \dot{\tilde{z}} &= y^*\,\tilde{x} \,+\, x^*\,\tilde{y} \,-\, b\,\tilde{z}\,, \end{aligned}$$

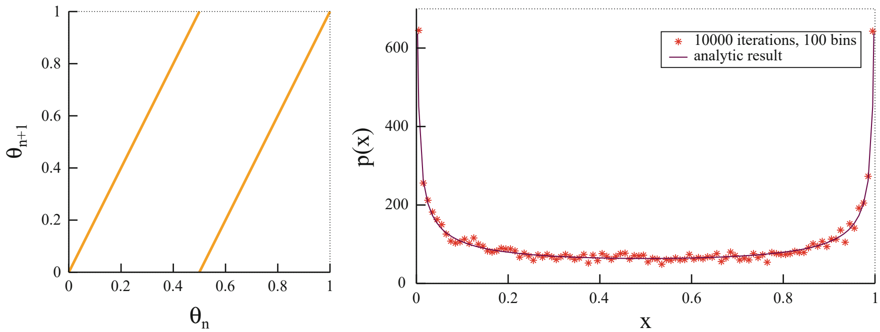


Fig. 11.8 For $r = 4$ the logistic map (2.46) is equivalent to the shift map (left), as defined by (3.73). The probability distribution $p(x)$ can be evaluated exactly, see (11.20), and compared with a simulation (right)

where $\tilde{x} = x - x^*$, $\tilde{y} = y - y^*$, and $\tilde{z} = z - z^*$ are small perturbations around the fixpoint, here for $\sigma = 10$ and $\beta = 8/3$. Using the ansatz $\tilde{x} \sim \tilde{y} \sim \tilde{z} \sim e^{\lambda t}$ we can determine the eigenvalues λ_i in the above equation.
For the fixpoint $(x^*, y^*, z^*) = (0, 0, 0)$ we find the eigenvalues

$$(\lambda_1, \lambda_2, \lambda_3) = \left(-\frac{8}{3}, \frac{-11 - \sqrt{81 + 40\,r}}{2}, \frac{-11 + \sqrt{81 + 40\,r}}{2}\right) .$$

For $r < 1$ all three eigenvalues are negative and the fixpoint is stable. For $r = 1$ the last eigenvalue $\lambda_3 = (-11 + 11)/2 = 0$ is marginal. For $r > 1$ the fixpoint becomes unstable.

The stability of the non-trivial fixpoints, Eq. (3.9), for $1 < r < r_c$ can be proven in a similar way, leading to a cubic equation. You can either find the explicit analytical solutions via Cardano's method or solve them numerically, e.g. via Mathematica, Maple or Mathlab, and determine the critical r_c, for which at least one eigenvalue turns positive.

(3.3) HAUSDORFF DIMENSION OF THE CANTOR SET

Dimension of a Line To cover a line of length l we need one circle of diameter l. If we reduce the diameter of the circle to $l/2$, we require two circles to cover the line. Generally we require a factor of 2 more circles if we reduce the diameter to a half. From the definition (3.12) of the Hausdorff dimension we obtain

$$D_H = -\frac{\log[N(l)/N(l')]}{\log[l/l']} = -\frac{\log[1/2]}{\log[2]} = 1 ,$$

where we used $N(l) = 1$ and $N(l' = l/2) = 2$. Therefore, the line is one-dimensional.

Dimension of the Cantor Set If we reduce the diameter of the circles from l to $l/3$, we require a factor of two more circles to cover the Cantor set. Therefore we obtain the Hausdorff dimension

$$D_H = -\frac{\log[N(l)/N(l')]}{\log[l/l']} = -\frac{\log[1/2]}{\log[3]} \approx 0.6309\,,$$

where we used $N(l) = 1$ and $N(l'{=}l/3) = 2$.

(3.4) Driven Harmonic Oscillator

In the long time limit the system oscillates with the frequency of the driving force. Hence, we can use the ansatz

$$x(t) = x_0 \cos(\omega t + \phi)\,,$$

where we have to determine the amplitude x_0 and the phase shift ϕ. Using this ansatz for the damped harmonic oscillator we find

$$(\omega_0^2 - \omega^2)\, x_0\ \cos(\omega t + \phi) -\ \gamma\, x_0\, \omega\ \sin(\omega t + \phi) = \epsilon\ \cos(\omega t)\,.$$

The amplitude x_0 and the phase shift ϕ can now be found by splitting above equation into $\sin(\omega t)$ terms and $\cos(\omega t)$ terms and comparing the prefactors. For the case $\omega = \omega_0$ we obtain $\phi = -\pi/2$ and $x_0 = \epsilon/(\gamma\omega_0)$. Note that $x_0 \to \infty$ for $\gamma \to 0$.

(3.5) Markov Chain of Umbrellas

Let p and $q = 1 - p$ be the probability of raining and not raining respectively, with $n \in \{0, 1, 2, 3, 4\}$ being the number of umbrellas in the place Lady Ann is currently staing (which could be either the office or her apartment). This process is illustrated in Fig. 11.9.

Neglecting the time indices for the stationary distribution $\rho(n)$ for the number of umbrellas, we have

$$\begin{aligned}
0 &= \dot\rho(0) = q\rho(4) - \rho(0)\\
0 &= \dot\rho(4) = \rho(0) + p\rho(1) - (p+q)p(4) \ \ = p\big(\rho(1) - \rho(4)\big)\\
0 &= \dot\rho(1) = p\rho(4) + q\rho(3) - (p+q)p(1) = q\big(\rho(3) - \rho(1)\big)\\
0 &= \dot\rho(3) = q\rho(1) + p\rho(2) - (p+q)p(3) = p\big(\rho(2) - \rho(3)\big)\\
0 &= \dot\rho(2) = p\big(\rho(3) - p(2)\big)
\end{aligned}$$

and hence $\rho(4) = \rho(1) = \rho(3) = \rho(2)$, which leads to

$$1 = \rho(0) + 4\rho(4) = \rho(0) + 4\rho(0)/q, \qquad \rho(0) = \frac{q}{q+4}\,. \tag{11.21}$$

The probability for Lady Ann to get wet is therefore $p \cdot \rho(0) = pq/(4+q)$.

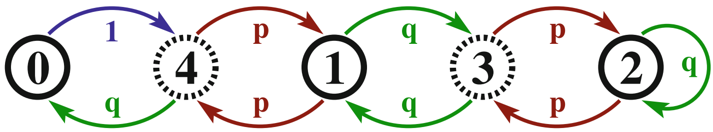


Fig. 11.9 Graphical representation of the Markov chain corresponding to the $N = 4$ umbrella problem. Being at office/home (solid/dashed circles), there can be $n = 0, 1, 2, 3, 4$ umbrellas in the respective place, being taken with probability p (when it rains) to the next location. When it does not rain, with probability $q = 1-p$, no umbrella changes place. Note that the sum of outgoing probabilities is one for all vertices

(3.6) <u>Biased Random Walk</u>

The master equation (3.31) generalizes to

$$p_{t+1}(x) = \big[(1+\alpha)\, p_t(x-1) \;+\; (1-\alpha)\, p_t(x+1)\big]/2\,,$$

with $(1 \pm \alpha)/2$ being the probabilities for a walker to jump to the right or to the left, and hence

$$\begin{aligned}\frac{p_{t+\Delta t}(x) - p_t(x)}{\Delta t} &= \frac{(\Delta x)^2}{2\Delta t}\,\frac{p_t(x+\Delta x) + p_t(x-\Delta x) - 2p_t(x)}{(\Delta x)^2} \\ &\quad - \frac{\alpha\Delta x}{\Delta t}\,\frac{p_t(x+\Delta x) - p_t(x-\Delta x)}{2\Delta x}\,,\end{aligned}$$

which results in

$$\dot p(x,t) = Dp''(x,t) \;-\; vp'(x,t), \qquad D = \frac{(\Delta x)^2}{2\Delta t}, \qquad v = \frac{\alpha\Delta x}{\Delta t}\,. \tag{11.22}$$

Rewriting the expression for the ballistic contribution v as

$$v = \frac{2\alpha}{\Delta x}\,\frac{(\Delta x)^2}{2\Delta t} \;\to\; \frac{2\alpha}{\Delta x}\,D, \qquad \alpha \propto \Delta x \tag{11.23}$$

we see that a finite propagation velocity v is recovered in the limit $\Delta t \to 0$ and $\Delta x \to 0$ whenever α is linear in Δx.

For a vanishing $D \to 0$ the solution of (11.22) is given by a ballistically travelling wave $p(x,t) = u(x - vt)$.

(3.7) <u>Continuous-Time Logistic Equation</u>

The continuous-time logistic equation is

$$\dot y(t) = \alpha y(t)\big[1 - y(t)\big], \qquad \frac{\dot y}{y} + \frac{\dot y}{1-y} = \alpha\,,$$

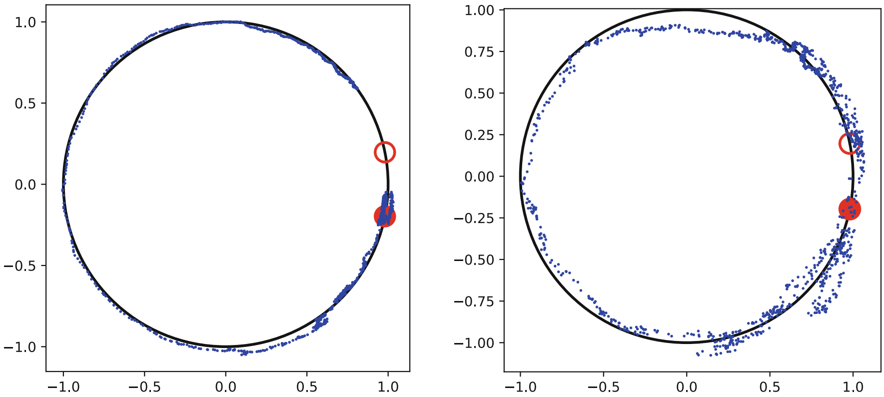


Fig. 11.10 Numerical simulation of (3.74), with $K = 1.02$ and $g = 1.2$, using Euler integration with a time step $\Delta t = 0.01$. Noise, as defined by (3.75), with a standard deviation $\sigma = 0.003/0.009$ (left/right plot) has been employed. The filled/open circle denotes the locus of the stable/unstable fixpoint, $1 = K\cos(\varphi_\pm)$, compare Fig. 2.13

with the solution

$$\log(y) - \log(1-y) = \alpha t + c, \qquad y(t) = \frac{1}{e^{-\alpha t - c} + 1},$$

where c is a constant of integration. Clearly $\lim_{t\to\infty} y(t) = 1$. For a comparison with the logistic map, Eq. (2.46), we discretize time via

$$\frac{y(t+\Delta t) - y(t)}{\Delta t} = \alpha y(t)\big[1 - y(t)\big],$$

$$y(t+\Delta t) = (\alpha\Delta t + 1)y(t)\left[1 - \frac{\alpha\Delta t}{\alpha\Delta t + 1}y(t)\right],$$

$$x(t+\Delta t) = r\,x(t)\big[1 - x(t)\big],$$

when renormalizing $x(t) = (\alpha\Delta t)y(t)/(\alpha\Delta t + 1)$. Then $x \to 0$ for $y \to 1$ in the limit $\Delta t \to 0$, which implies that the continuous time logistic differential equation corresponds to $r = (\alpha\Delta t + 1) \to 1$, compare (2.47).

(3.8) NOISE-INDUCED QUASI-PERIODIC ORBITS
Two illustrative numerical orbits are shown in Fig. 11.10. The orbit will make another revolution once it has been able to jump from the stable to the unstable fixpoint, which corresponds to a stochastic escape process (see Sect. 3.5.2 of Chap. 3).

11.4 Solutions to the Exercises of Chap. 4

(4.1) Instability of Euler's method

The discretized diffusion equation reads

$$\rho(x,t+\Delta t) = (1-2r)\rho(x,t) \;+\; r\big[\rho(x+\Delta x,t)+\rho(x-\Delta x,t)\big]$$

where $r = D\Delta t/(\Delta x)^2$. Considering the periodic initial state

$$\rho(x,0) = \cos(\pi x/\Delta x) \;=\; (-1)^n, \qquad x = n\Delta x$$

we find

$$\rho(n\Delta x, \Delta t) = (1-2r)(-1)^n + 2r(-1)^{n\pm 1} = (1-4r)(-1)^n$$

for the next state with $t = \Delta t$. This map is stable,[1] whenever

$$|1-4r| \;<\; 1, \qquad 0 \le r \le 1/2\,.$$

You may verify this result numerically.

(4.2) Exact propagating wavefront solution

The Fermi function an its first order derivative are

$$\rho^*(x,t) = \frac{1}{1+\mathrm{e}^{\beta(x-ct)}}, \qquad \frac{\partial\rho^*}{\partial t} = \beta c\rho^*(1-\rho^*)\,,$$

and, equivalently

$$\frac{\partial\rho^*}{\partial x} = -\beta\rho^*(1-\rho^*)\,, \qquad \frac{\partial^2\rho^*}{\partial x^2} = \beta^2(1-2\rho^*)\rho^*(1-\rho^*)\,.$$

[1] See Sect. 2.2 of Chap. 2.

Setting $\rho^* \to \rho$ and using the definition (4.2) of a reaction-diffusion system, we find with

$$
\begin{aligned}
R(\rho) = \frac{\partial \rho}{\partial t} - D\frac{\partial^2 \rho}{\partial x^2} &= \rho(1-\rho)\big[\beta c - \beta^2 D(1-2\rho)\big] \\
&= \beta\rho(1-\rho)\big[(c - \beta D + 2\beta D\rho\big] \\
&= r\rho(1-\rho)(\rho-\theta)
\end{aligned}
$$

the "Zeldovich equation", where we have defined $r = 2\beta^2 D$ and the threshold $\theta = (\beta D - c)/(2\beta)$. For $\theta = 0$ one finds

$$
R(\rho) = r\rho^2(1-\rho), \qquad c = \beta D\,. \tag{11.24}
$$

The result, $R(\rho) = \rho^2(1-\rho)$, describes a specific type of bisexual logistic growth, which can be seen by considering a population composed of $\alpha\rho$ females and $(1-\alpha)\rho$ males. The number of offsprings will be $\sim \alpha\rho$ (logistic growth) if mating strategies lead to all females being reproductive, and $\sim \alpha(1-\alpha)\rho^2$ if males and females meet randomly.

(4.3) Linearized Fisher Equation

With $\rho = 1 - 1/u$ the derivatives are

$$
\dot\rho = \frac{\dot u}{u^2}, \qquad \rho' = \frac{u'}{u^2}, \qquad \rho'' = \frac{-2(u')^2}{u^3} - \frac{u''}{u^2}
$$

and (4.71) becomes

$$
\frac{\dot u}{u^2} = \left(1 - \frac{1}{u}\right)\frac{1}{u} - \frac{2(u')^2}{u^3} - \frac{u''}{u^2} + \frac{2}{1/u}\frac{(u')^2}{u^4}
$$

or

$$
\dot u = (u-1) + u'', \qquad \dot v = v + v''\,,
$$

which is equivalent to the linearized Fisher equation (4.18) when shifting via $v = u - 1$ the zero.

(4.4) Turing instability with two stable nodes

For the two eigenvalues to be real and both negative (or both positive) the determinant $\Delta_1 = 1 - \epsilon_a\epsilon_b$ need to be positive, compare (4.25), which implies

$$
(\epsilon_a - \epsilon_b)^2 \;>\; 4\Delta_1 = 4 - 4\epsilon_a\epsilon_b, \qquad (\epsilon_a + \epsilon_b)^2 \;>\; 4\,.
$$

This is easily possible, e.g. using $\epsilon_a = 2$ and $\epsilon_b = 1/4$ with the determinant $\Delta_1 = 1 - 2/4 = 1/2$ remaining positive, and with the trace $\epsilon_b - \epsilon_a$ remaining negative. A Turing instability with two stable nodes is hence possible, e.g. with the matrices

$$A_1 = \begin{pmatrix} -2 & 1 \\ -1 & 1/4 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} -8 & 0 \\ 0 & -1/8 \end{pmatrix}, \qquad A = \begin{pmatrix} -10 & 1 \\ -1 & 1/8 \end{pmatrix}.$$

The eigenvalues $\lambda_\pm(A_1) = (-7 \pm \sqrt{17})/8$ are both attracting and the determinant of $A = A_1 + A_2$ is with $\Delta = -1/4$ negative.

(4.5) <u>EIGENVALUES OF **2** × **2** MATRICES</u>
For a fixpoint $\mathbf{x}^* = (x^*, 0)$ we find for the Jacobian J

$$J = \begin{pmatrix} 0 & 1 \\ -V''(x^*) & -\lambda(x^*) \end{pmatrix}, \qquad \dot{x} = y, \qquad \dot{y} = -\lambda(x)y - V'(x)\,.$$

The determinant

$$\Delta = V''(x^*) \;<\; 0$$

is negative for a local maximum and $\mathbf{x}^*$ hence a saddle, compare (4.25).

(4.6) <u>LARGE DENSITIES OF NERVOUS RATS</u>
The stationarity condition (4.45) reads

$$\sigma_n \;=\; \mathrm{e}^{\sigma}\left(1 - \mathrm{e}^{-\sigma_n}\right)(\sigma - \sigma_n) \approx \mathrm{e}^{\sigma}(\sigma - \sigma_n), \qquad \sigma \gg 1, \qquad \sigma_n \to \sigma$$

and hence

$$\sigma_n\left(1 + \mathrm{e}^{\sigma}\right) \;\approx\; \sigma\,\mathrm{e}^{\sigma}, \qquad \sigma_n \;\approx\; \frac{\sigma}{1 + \mathrm{e}^{-\sigma}}\,. \tag{11.25}$$

The number of calm rats per comfort zone,

$$\sigma_c = \sigma - \sigma_n \;\approx\; \frac{\sigma\,\mathrm{e}^{-\sigma}}{1 + \mathrm{e}^{-\sigma}}$$

becomes exponentially small, but never vanishes.

(4.7) <u>AGENTS IN THE BOUNDED CONFIDENCE MODEL</u>
The total number of agents does not change when $\int \dot{\rho}(x)dx = 0$, viz when

$$2\int_{-d/2}^{d/2} dy \int dx \rho(x+y)\rho(x-y) = -\int_{-d}^{d} dy \int dx \rho(x)\rho(x+y)\,, \tag{11.26}$$

compare (4.54). Using $x = \tilde{x} - y/2$ and $y = 2\tilde{y}$ we rewrite the second term in (11.26) as

$$\int_{-d}^{d} dy \int d\tilde{x}\,\rho(\tilde{x} - y/2)\rho(\tilde{x} + y/2) = 2\int_{-d/2}^{d/2} d\tilde{y} \int d\tilde{x}\,\rho(\tilde{x} - \tilde{y})\rho(\tilde{x} + \tilde{y}),$$

which has exactly the same form a the first term in (11.26).

(4.8) CLASSICAL VOTER MODEL

The probability p_i that a given agent changes opinion is

$$p_i = \frac{1}{2}\left(1 - \frac{\sigma_i}{z}\sum_j \sigma_j\right),$$

where z is the coordination number of the underlying network, viz the average number of neighbors. The sum $\sum_j$ runs over the neighbors of the ith agent. Averaging over all realizations one has

$$\dot{m} = \frac{d}{dt}\langle\sigma_i\rangle = -2\langle\sigma_i p_i\rangle = -\langle\sigma_i\rangle + \frac{1}{z}\sum_j\langle\sigma_j\rangle, \qquad \sigma_i^2 = 1\,.$$

Whenever a voter flips with probability p_i, its preference reverses, changing by $-2\sigma_i$. On average, $\langle\sigma_i\rangle = \langle\sigma_j\rangle$, which implies that the magnetization $m = \langle\sigma_i\rangle$ remains constant when using $z = \sum_j 1$ in above expression for $\dot{m}$. Magnetization is conserved only as an average over realizations. For a specific realization of consecutive opinion flips, consensus will eventually be reached, either with $\sigma_i \equiv 1$ or $\sigma_i \equiv -1$, respectively with probabilities P_+ and $P_- = 1 - P_+$. On the average, magnetization remains constant,

$$m = P_+ - P_- = 2P_+ - 1, \qquad P_+ = \frac{1+m}{2},$$

which we wanted to prove.

11.5 Solutions to the Exercises of Chap. 5

(continued)

(5.1) THE LAW OF LARGE NUMBERS

For the generating function $G(x)$ of a cumulative of stochastically independent variables we have

$$G(x) = \prod_i G_i(x), \quad G_i(x) = \sum_k p_k^{(i)} x^k, \quad \sigma_i^2 = G_i''(1) + G_i'(1) - \big(G_i'(1)\big)^2$$

and $\mu_i = G_i'(1)$, see (5.7) and (5.9). Then

$$\mu = G'(1) = \sum_j \left[\left(\prod_{i \neq j} G_i(1) \right) G_j'(1) \right] = \sum_j \mu_j\,.$$

The mean of the cumulative process is just the sum of the means of the individual processes. For the evaluation of the variance we need

$$\begin{aligned} G''(1) &= \sum_{l,j \neq l} \left[\left(\prod_{i \neq j,l} G_i(1) \right) G_j'(1) G_l'(1) \right] + \sum_j \left[\left(\prod_{i \neq j} G_i(1) \right) G_j''(1) \right] \\ &= \sum_{l,j \neq l} \mu_j \mu_l + \sum_j G_j''(1) = \left(\sum_j \mu_j \right)^2 - \sum_j \mu_j^2 \\ &\qquad + \sum_j \left(\sigma_i^2 - \mu_i + \mu_i^2 \right) \\ &= \left(\sum_j \mu_j \right)^2 + \sum_j \left(\sigma_i^2 - \mu_i \right) = \mu^2 + \sum_j \sigma_i^2 - \mu\,, \end{aligned}$$

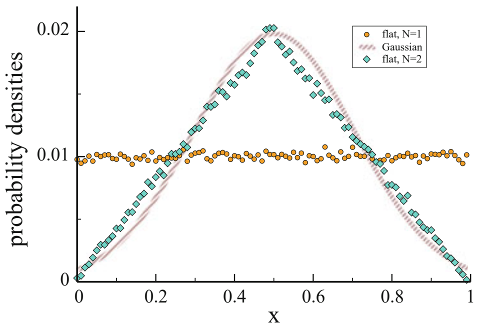


Fig. 11.11 The flat distribution, which has a variance of $\sigma^2 = 1/12$ is shown together with the probability density of the sum of $N = 2$ flat distribution. Also shown is the corresponding limiting Gaussian with $\sigma = 1/\sqrt{12N}$, compare (5.11). 100 bins and a sampling of 10^5 have been used

where we have used $\sigma_i^2 = G_i''(1) + \mu_i - \mu_i^2$. Then

$$\sigma^2 = G''(1) + \mu - \mu^2 = \sum_j \sigma_j^2 ,$$

which is the law of large numbers: the variance is additive.

(5.2) CUMMULATIVE DISTRIBUTION FUNCTIONS

The distribution $p_2(x)$ for the sum of two drawings from a given PDF $p(x)$ is

$$p_2(y) = \int dx_1 dx_2 \delta(y - x_1 - x_2) p(x_1) p(x_2) = \int_0^1 dx_1 dx_2 \delta(y - x_1 - x_2) ,$$

where we have assumed that $p(x)$ is the flat distribution, for $x \in [0, 1]$, which has the variance

$$\sigma^2 = \int_0^1 x^2 dx = \frac{1}{12} .$$

Changing to $x_1 + x_2$ and $x_1 - x_2$ as integration variables one finds, that $p_2(y)$ is a triangle, as illustrated in Fig. 11.11.

(5.3) SYMBOLIZATION OF FINANCIAL DATA

For the Dow Jones index in the twentieth-century, from 1900 to 1999, the joint probabilities are

$$\begin{array}{llll} p_{+++} = 0.15, & p_{+-+} = 0.11, & p_{-++} = 0.13, & p_{--+} = 0.12 \\ p_{++-} = 0.13, & p_{+--} = 0.12, & p_{-+-} = 0.10, & p_{---} = 0.11 \end{array} ,$$

as extracted from historical daily data. The default would be $1/8 = 0.125$. Of interest are the ratios

$$\begin{array}{ll} p_{++-}/p_{+++} = 0.87, & p_{+--}/p_{+-+} = 1.09 \\ p_{-+-}/p_{-++} = 0.77, & p_{---}/p_{--+} = 0.92 \end{array} .$$

The conditional probabilities $p(+|-+)$ and $p(-|-+)$ of a third-day increase/decrease, given that the Dow Jones index had decreased previously on the first day and increased on the second day, are evaluated via

$$p(+|-+) + p(-|-+) = 1, \qquad \frac{p(+|-+)}{p(-|-+)} = \frac{p_{-++}}{p_{-+-}} = \frac{1}{0.77},$$

yielding $p(+|-+) = 0.57$ and $p(-|-+) = 0.43$. There had been consequently, in the twentieth-century, a somehow substantially larger chance of a third day increase in the Dow Jones index, when the index had decreased previously on the first day and increased on the second day. This kind of information could be used, at least as a matter of principle, for a money-making scheme. But there are many caveats: There is no information about the size of the prospective third-day increase in this analysis, the standard deviation of the $p_{\pm\pm\pm}$ may be large, and there may be prolonged periods of untypical statistics. We do recommend not to invest real world money using home made money-making schemes on the basis of this or related statistical analysis.

(5.4) THE OR TIME SERIES WITH NOISE

Recall that $\mathrm{OR}(0,0) = 0$, together with $\mathrm{OR}(0,1) = \mathrm{OR}(1,0) = \mathrm{OR}(1,1) = 1$. For the initial condition $\sigma_1 = 0$, $\sigma_0 = 0$, the time series is ..0000000.., for all other initial conditions the time series is ..1111111..,

$$p(1) = \begin{cases} 1 & \text{typical} \\ 3/4 & \text{average} \end{cases}.$$

The time series always returns to the typical sequence,

$$\ldots 11111\underline{0}111111111\underline{0}1111111111111\underline{1}0000000000000\ldots$$

for finite but low levels of noise, where we underlined noise-induced transitions (time runs from right to left). We have therefore $p(1) \to 1$ when noise is present at low levels. For larger amounts of noise the dynamics becomes, on the other hand, completely random, with $p(1) \to 1/2$.

(5.5) TRAILING AVERAGES

Using partial integration one can derive local (in time) updating rules,

$$\begin{aligned}\frac{d}{dt}\bar{y}(t) &= \frac{1}{T}\int_0^\infty d\tau\, y'(t-\tau)\,\mathrm{e}^{-\tau/T} = -\frac{1}{T}\int_0^\infty d\tau\,\mathrm{e}^{-\tau/T}\frac{d}{d\tau}y(t-\tau) \\ &= -\frac{1}{T}\,\mathrm{e}^{-\tau/T}\,y(t-\tau)\Big|_0^\infty - \frac{1}{T^2}\int_0^\infty d\tau\,\mathrm{e}^{-\tau/T}y(t-\tau) \\ &= \frac{y(t)-\bar{y}(t)}{T}, \end{aligned} \tag{11.27}$$

which proves the usual equivalence between an exponential kernel and (11.27). The trailing averages are normalized, which can be seen when specializing to constant inputs,

$$\bar{y}(t)\Big|_{y(t)\equiv y_0} = y_0, \qquad y_c(t)\Big|_{y(t)\equiv y_0} = y_0, \qquad y_s(t)\Big|_{y(t)\equiv y_0} = y_0\,,$$

which determines the respective prefactors. The normalized trailing sin-transform is given by

$$y_s(t,\omega) = \frac{1+(\omega T)^2}{\omega T^2}\int_0^\infty d\tau\; y(t-\tau)\, \mathrm{e}^{-\tau/T}\sin(\omega\tau)\,.$$

Again, the update rules can be derived using a partial integration. One finds

$$\dot{y}_c(t,\omega) = \frac{y(t)-y_c(t,\omega)}{T} + (\omega T)^2\,\frac{y(t)-y_s(t,\omega)}{T} \tag{11.28}$$

$$\dot{y}_s(t,\omega) = \frac{y_c(t,\omega)-y_s(t,\omega)}{T}\,. \tag{11.29}$$

These update rules become numerically unstable for large frequencies, due to the factor $(\omega T)^2$ in (11.28). They are however useful when dealing with low-pass filters.

(5.6) MAXIMAL ENTROPY DISTRIBUTION FUNCTION

The two conditions (5.39), of a given mean μ and a given varicance σ^2, can be enforced by two Lagrange parameters λ_1 and λ_2 respectively, via

$$f(p) = -p\log(p) - \lambda_1 x p - \lambda_2 (x-\mu)^2 p\,.$$

The stationary condition (5.35), $f'(p)=0$, leads to

$$-\log(p) - 1 - \lambda_1 x - \lambda_2(x-\mu)^2 \;=\; 0, \qquad \log(p) = \log(\text{const.}) - \lambda_2(x-\tilde{\mu})^2\,,$$

where $\tilde{\mu} = \mu - \lambda_1/(2\lambda_2)$. Therefore

$$p(x)\;\propto\; 2^{-\lambda_2(x-\tilde{\mu})^2}\;\sim\;\mathrm{e}^{\frac{(x-\tilde{\mu})^2}{2\sigma^2}}\,,$$

with $\sigma^2 = 1/(2\lambda_2\log_e(2))$, which determines the Lagrange parameter λ_2. The first condition in (5.39), demands the mean to equal μ, viz $\tilde{\mu}\equiv\mu$ and consequently $\lambda_1 = 0$.

(5.7) TWO-CHANNEL MARKOV PROCESS

We remind ourselves that $\mathrm{OR}(0,1) = \mathrm{OR}(1,1) = \mathrm{OR}(1,1) = 1$, together with $\mathrm{OR}(0,0) = 0$, and that $\mathrm{AND}(0,1) = \mathrm{AND}(1,0) = \mathrm{AND}(0,0) = 0$, with $\mathrm{AND}(1,1) = 1$. We consider first $\alpha = 0$ (no noise), and the four initial conditions (underlined),

$$
\begin{array}{rlll}
\ldots \sigma_{t+1}\sigma_t & : \ldots 0\,0\,0\,0\,0\,\underline{0} & \quad \ldots 1\,1\,1\,1\,1\,\underline{1} \\
\ldots \tau_{t+1}\tau_t & : \ldots 0\,0\,0\,0\,0\,\underline{0} & \quad \ldots 1\,1\,1\,1\,1\,\underline{1}
\end{array},
$$

and

$$
\begin{array}{rlll}
\ldots \sigma_{t+1}\sigma_t & : \ldots 0\,0\,0\,0\,0\,\underline{0} & \quad \ldots 0\,0\,0\,0\,0\,\underline{1} \\
\ldots \tau_{t+1}\tau_t & : \ldots 1\,1\,1\,1\,1\,\underline{1} & \quad \ldots 1\,1\,1\,1\,1\,\underline{0}
\end{array}.
$$

In consequence, one finds three types of stationary states. The master equations for the respective joint probabilities are

$$p_{t+1}(0,0) = (1-\alpha)p_t(0,0) + \alpha\,[p_t(1,0) + p_t(0,1)] \tag{11.30}$$

$$p_{t+1}(1,1) = (1-\alpha)p_t(1,1) \tag{11.31}$$

and

$$p_{t+1}(1,0) = \alpha p_t(1,1) \tag{11.32}$$

$$p_{t+1}(0,1) = (1-\alpha)\,[p_t(1,0) + p_t(0,1)] + \alpha p_t(0,0) \tag{11.33}$$

The stationary condition is $p_{t+1} = p_t \equiv p$. From (11.31) and (11.32) it then follows, for $0 < \alpha < 1$,

$$p(1,1) = 0, \qquad p(1,0) = 0\,.$$

Using this result we obtain

$$p(0,0) = p(0,1) \equiv \frac{1}{2}$$

from (11.30) and (11.33), where we normalized the result in the last step. This finding is independent of the noise level α, for any $\alpha \neq 0, 1$. The marginal distribution functions are

$$
\begin{array}{ll}
p_\sigma(0) = 1 & \qquad p_\tau(0) = 1/2 \\
p_\sigma(1) = 0 & \qquad p_\tau(1) = 1/2
\end{array}.
$$

Lastly, we find

$$H[p] = 1, \qquad H[p_\sigma] = 0, \qquad H[p_\tau] = 1$$

for the respective entropies. The mutual information $I(\sigma,\tau) = H[p_\sigma] + H[p_\tau] - H[p]$ vanishes since the σ-channel becomes deterministic for any finite noise level α.

(5.8) KULLBACK-LEIBLER DIVERGENCE
The Kullback-Leibler divergence for the two normalized PDFs

$$p(x) = \mathrm{e}^{-(x-1)}, \qquad q(x) = \frac{\gamma - 1}{x^\gamma}, \qquad x,\ \gamma\ >\ 1$$

is

$$\begin{aligned} K[p;q] &= \int_1^\infty p(x) \log_2\left(\frac{p(x)}{q(x)}\right) dx \\ &= -H[p] - \int_1^\infty \mathrm{e}^{-(x-1)} \log_2\left((\gamma-1)x^{-\gamma}\right) \\ &= -H[p]\ -\ \log_2(\gamma - 1)\ +\ \gamma \int_1^\infty \mathrm{e}^{-(x-1)} \log_2(x)\,, \end{aligned}$$

which is stationary for

$$\frac{\partial K[p;q]}{\partial \gamma} = 0$$

viz

$$\frac{1}{\gamma - 1} = \ln(2) \int_1^\infty \mathrm{e}^{-(x-1)} \log_2(x) = \int_1^\infty \mathrm{e}^{-(x-1)} \ln(x)\,,$$

where we have used

$$\log_2(x) = \ln(x)/\ln(2), \qquad \frac{d}{dx}\log_2(x) = \frac{1}{\ln(2)}\frac{d}{dx}\ln(x) = \frac{1}{\ln(2)}\frac{1}{x}\,.$$

Numerically one finds

$$\int_1^\infty \mathrm{e}^{-(x-1)} \ln(x)\ \approx\ 0.596, \qquad \gamma = \frac{1}{0.596}\ +\ 1\ \approx\ 2.678\,.$$

The graphs intersect twice, $q(x) > p(x)$ both for $x = 1$ and $x \to \infty$.

(5.9) CHI-SQUARED TEST
We rewrite the Kullback Leibler divergence, see (5.2.4), as

$$\begin{aligned} K[p;q] &= \sum_i p_i \log\left(\frac{p_i}{q_i}\right) = -\sum_i p_i \log\left(\frac{q_i}{p_i}\right) \\ &= -\sum_i p_i \log\left(\frac{q_i - p_i + p_i}{p_i}\right) - \sum_i p_i \log\left(1 + \frac{q_i - p_i}{p_i}\right) \end{aligned}$$

$$\approx -\sum_i (q_i - p_i) + \sum_i \frac{(q_i - p_i)^2}{2p_i} = \sum_i \frac{(p_i - q_i)^2}{2p_i}$$

$$\equiv \frac{\chi^2[p;q]}{2},$$

where we used the Taylor expansion $\log(1+x) \approx x - x^2/2$ together with the normalization conditions $\sum_i p_i = 1 = \sum_i q_i$. Note the relation to the Fisher information, see (5.59) and (5.61).

(5.10) <u>EXCESS ENTROPY</u>

We show that E_n,

$$E_n \approx H[p_n] - n\big(H[p_{n+1}] - H[p_n]\big)$$

is monotonically increasing with n as long as $H[p_n]$ is concave,

$$\begin{aligned} E_n - E_{n-1} &= H[p_n] - n\big(H[p_{n+1}] - H[p_n]\big) \\ &\qquad - H[p_{n-1}] + (n-1)\big(H[p_n] - H[p_{n-1}]\big) \\ &= n\Big(2H[p_n] - H[p_{n+1}] - H[p_{n-1}]\Big) \\ &= 2n\Big(H[p_n] - \frac{1}{2}\big(H[p_{n+1}] - H[p_{n-1}]\big)\Big), \end{aligned}$$

where the last term in the brackets is positive as long as $H[p_n]$ is concave as a function of n, compare Fig. 5.4.

(5.11) <u>TSALLIS ENTROPY</u>

For small $1-q$ we may develop the Tsallis entropy $H_q[p]$, as defined by (5.70), as

$$H_q[p] = \frac{1}{1-q}\sum_k p_k\Big[\big(p_k\big)^{q-1} - 1\Big] = \frac{1}{1-q}\sum_k p_k\Big[e^{(q-1)\ln(p_k)} - 1\Big]$$

$$\approx -\sum_k p_k \ln(p_k) \equiv H[p].$$

With $p_k \in [0,1]$, $q \in]0,1]$ and $q - 1 \le 0$ we have

$$\big(p_k\big)^{q-1} \ge 1, \qquad H_q[p] = \frac{1}{1-q}\sum_k p_k\Big[\big(p_k\big)^{q-1} - 1\Big] \ge 0.$$

The joint distribution function for two statistically independent systems p is just the product of the individual distributions, $p = p_X p_Y$, which leads to

$$\sum_{x_i,y_j} \Big[p_X(x_i)p_Y(y_j)\Big]^q = \sum_{x_i} p^q(x_i) \sum_{y_j} p^q(y_j)$$
$$= \Big[(1-q)H_q[p_X]+1\Big]\Big[(1-q)H_q[p_Y]+1\Big],$$

where a substitution equivalent to $\sum_k p_k = 1$ was used for $H_q[p_X]$ and $H_q[p_Y]$. Hence we have

$$H_q[p] = \frac{1}{1-q}\left(\sum_k (p_k)^q - 1\right)$$
$$= H_q[p_X] + H_q[p_Y] + (1-q)H_q[p_X]H_q[p_Y],$$

which shows that the Tsallis entropy is extensive only in the limit $q \to 1$. The Tsallis entropy is maximal for an equiprobability distribution, which follows directly from the general formulas (5.34) and (5.36).

11.6 Solutions to the Exercises of Chap. 6

(6.1) Solutions of the Landau–Ginzburg Functional
We want that the cubic equation (6.5) factorizes,

$$0 = P(\phi) - h = \phi^3 + (t-1)\phi - h \equiv (\phi-\phi_3)(\phi-\phi_2)(\phi-\phi_1)\,.$$

This is the case, e.g., selecting

$$\phi_3 = 1, \qquad \phi_2 = \frac{-1}{4}, \qquad \phi_1 = \frac{-3}{4}, \qquad t = \frac{3}{16}, \qquad h = \frac{3}{16}.$$

Local stability is determined by the derivative $P'(\phi)$, as illustrated in Fig. 6.2. An example is

$$P'(\phi_2) = \frac{d}{d\phi}\big[P(\phi) - h\big]_{\phi_2} = (\phi_2 - \phi_3)(\phi_2 - \phi_1) < 0\,,$$

which implies that ϕ_2 is an unstable fixpoint. In analogy, ϕ_1 and ϕ_3 are locally stable.

(6.2) <u>THERMODYNAMICS OF THE LANDAU-GINZBURG MODEL</u>
In th ordered state, the free energy density is given by

$$f(T,\phi,h) - f_0(T,h) = \frac{t-1}{2}\,\phi^2 + \frac{1}{4}\,\phi^4 \;=\; -\frac{(t-1)^2}{4}\,,$$

where we used $\phi^2 = 1 - t$. It follows that

$$\frac{\partial F}{\partial t} = -V\,\frac{t-1}{2} \qquad \Rightarrow \qquad S = \frac{\partial F}{\partial T} \;=\; \frac{V}{T_c}\frac{1-t}{2}$$

for the entropy S, with $t = T/T_c$. The specific heat C_V is

$$C_V = T_c\frac{\partial S}{\partial T} = -\frac{V}{2T_c}, \qquad\qquad T < T_c\,.$$

For $T > T_c$ the specific heat C_V vanishes, there is a jump at $T = T_c$.

(6.3) <u>THE GAME OF LIFE</u>
The solutions have already been given in Fig. 6.5, apart from the cross $\{(0,0), (0,1), (1,0), (-1,0), (0,-1)\}$. For an illustration of its development see Fig. 11.12.

(6.4) <u>GAME OF LIFE ON A SMALL-WORLD NETWORK</u>
The construction of a small-world net with conserving local connectivities $k_i \equiv 8$ is shown and explained in Fig. 11.13. An appropriate dynamical order parameter would be the density of life cells $\rho(t)$ at time t.[2]

(6.5) <u>FOREST FIRE MODEL</u>
We define by x_t, x_f and x_e the densities of cells with trees, fires and ashes (empty), with $x_t + x_f + x_e = 1$. A site burns if there is at least one fire on one of the Z nearest-neighbor cells. The probability that none of Z cells is

[2] This problem has been surveyed in detail by Huang et al. (2003).

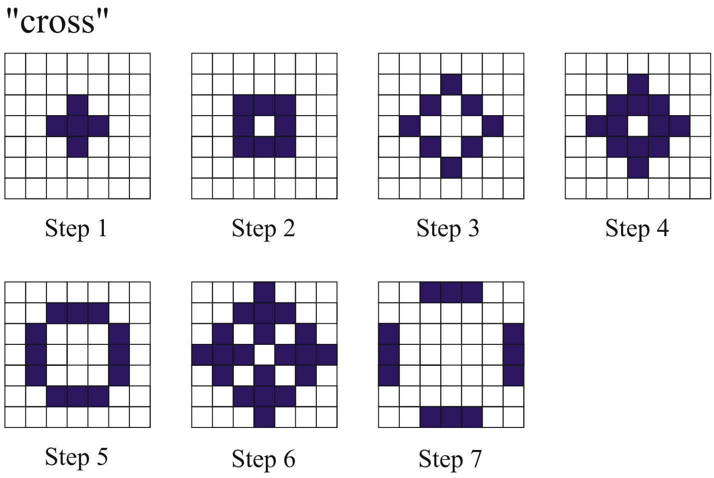


Fig. 11.12 Evolution of the pattern "cross" in the game of life: after seven steps it gets stuck in a fixed state with four blinkers

burning is $(1-x_f)^Z$, the probability that at least one out of Z is burning is $1-(1-x_f)^Z$. We have then the update rules

$$x_f(t+1) = \left[1-(1-x_f(t))^Z\right] x_t(t), \qquad x_e(t+1) = x_f(t) - p x_e(t)\,,$$

where p is the probability for a sampling to grow. Compare Fig. 6.6. The stationary densities $x_e(t+1) = x_e(t) \equiv x_e^*$, etc., are

$$(1+p)x_e^* = x_f^*, \qquad 1 = x_f^* + x_t^* + x_f^*/(1+p), \qquad x_t^* = 1 - \frac{2+p}{1+p} x_f^*\,,$$

which leads to self-consistency condition for the stationary density x_f^* of fires,

$$x_f^* = \left[1-(1-x_f^*)^Z\right] \left(1 - \frac{2+p}{1+p} x_f^*\right)\,. \tag{11.34}$$

In general one needs to solve (11.34) numerically. For small densities of fires we expand

$$\begin{aligned} 1-(1-x_f^*)^Z &\approx 1 - \left(1 - Z x_f^* + \frac{Z(Z-1)}{2}\,(x_f^*)^2\right) \\ &= Z x_f^* - \frac{Z(Z-1)}{2}\,(x_f^*)^2\,, \end{aligned}$$

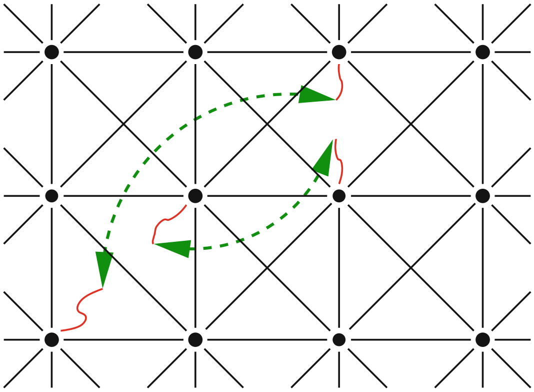


Fig. 11.13 Construction of a small-world network out of the game of life on a 2D-lattice: One starts with a regular arrangement of vertices where each one is connected to its eight nearest neighbors. Two arbitrarily chosen links (wiggled lines) are cut with probability p and the remaining stubs are rewired randomly as indicated (dashed arrows). The result is a structure showing clustering as well as a fair amount of shortcuts between far away sites, as in the Watts and Strogatz model, Fig. 1.14, but with conserved connectivities $k_i \equiv 8$

finding

$$\frac{1}{Z} = \left(1 - \frac{Z-1}{2}x_f^*\right)\left(1 - \frac{2+p}{1+p}x_f^*\right) \approx 1 - \left(\frac{Z-1}{2} + \frac{2+p}{1+p}\right)x_f^*$$

for (11.34). The minimal number of neighbors for fires to burn continuously is $Z > 1$ in mean-field theory.

(6.6) <u>Realistic Sandpile Model</u>

The variable z_i should denote the true local height of a sandpile; the toppling starts when the slope becomes too big after adding grains of sand randomly, i.e. when the difference $z_i - z_j$ between two neighboring cells exceeds a certain threshold K. Site i then topples in the following fashion:

– Look at the neighbor j of site i for which $z_i - z_j$ is biggest (and positive) and transfer one grain of sand from i to j,

$$z_i \to z_i - 1, \qquad z_j \to z_j + 1\,. \tag{11.35}$$

If more than one neighbor satisfies the criteria, select one randomly.

– Repeat the first step as long as there is at least a single neighbor j of i satisfying the condition $z_i \geq z_j + 1$.

The toppling process mimics a local instability, being at the same time conserving. Sand is lost only at the boundaries. This model leads to true sandpiles, in the sense that it is highest in the center and lowest at the

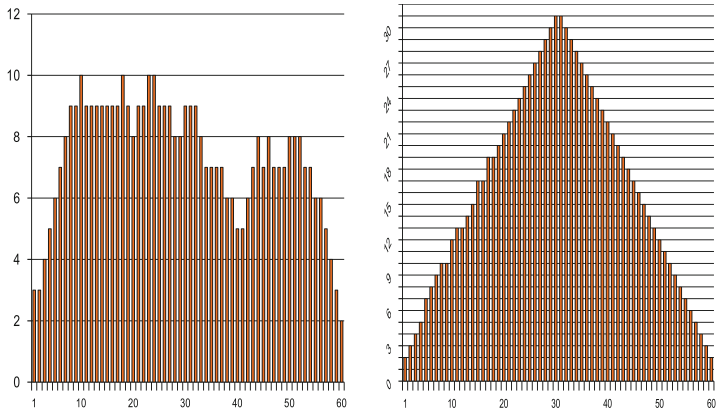


Fig. 11.14 Example of a simulation of a one-dimensional realistic sandpile model, see (11.35), with 60 cells, after 500 (left) and 2000 (right) time step

boundaries, compare Fig. 11.14. Note that there is no upper limit to z_i, only to the slope $|z_i - z_j|$.

(6.7) <u>Recursion Relation for Avalanche Sizes</u>
Substituting $f_n(x) = \sum_s P_n(s)\,x^s$ (where we dropped the functional dependence on the branching probability p), as defined by (6.17), into the recursion relation (6.19) for the generating functional $f_n(x)$, one obtains

$$\sum_s P_{n+1}(s)\,x^s = x\,(1-p) + x\,p\left(\sum_s P_n(s)\,x^s\right)^2 . \tag{11.36}$$

The recursion relation for the probability $P_n(s)$ of finding an avalanche of size s is obtained by Taylor-expanding the right-hand side of (11.36) in powers of x^s, and comparing prefactors,

$$x^0: \qquad P_{n+1}(0) = 0,$$
$$x^1: \qquad P_{n+1}(1) = (1-p) + p\big(P_n(0)\big)^2 = 1-p\,,$$

for $s = 0, 1$. There is at least one site, hence $P_n(0) = 0$. We find

$$P_{n+1}(s) = p\sum_{s'} P_n(s')P_n(s-s'-1), \qquad s > 1 \tag{11.37}$$

for general avalanche sizes. This expression is easy to evaluate numerically by evaluating all $P_n(s)$ with increasing n, the numerical effort increases like $O(s^2)$.

For a general branching process, described by the probability p_m of having m offsprings, the recursion relation (6.19) takes the form

$$f_{n+1}(x) = x \sum_m p_m \big(f_n(x)\big)^m . \tag{11.38}$$

Also in this case a recursion relation, in analogy to (11.37), could be derived, involving however multiple summations, with the computational effort scaling like $O(s^M)$, where M is the maximal number of descendants. One has $p_m = 0$ for $m > M$.

(6.8) <u>RANDOM BRANCHING MODEL</u>
For the probability $\tilde{Q}_n = \sum_{n'=0}^{n} Q_n(0, p)$ for the avalanche to last $1 \ldots n$ time steps, the recursion ansatz

$$\tilde{Q}_{n+1} = (1-p) + p\, \tilde{Q}_n^2 \tag{11.39}$$

is valid, in analogy to the recursion relation (6.19) for the generating function of avalanche sizes. The case here is simpler, as one can work directly with probabilities: The probability $\tilde{Q}_{n+1}$ to find an avalanche of duration $1 \ldots (n+1)$ is the probability $(1-p)$ to find an a avalanche of length 1 plus the probability $p\, \tilde{Q}_n^2$ to branch one time step, generating two avalanches of length $1 \ldots n$.

In the thermodynamic limit we can replace the difference $\tilde{Q}_{n+1} - \tilde{Q}_n$ by the derivative $\frac{\mathrm{d}\tilde{Q}_n}{\mathrm{d}n}$ leading to the differential equation

$$\frac{\mathrm{d}\tilde{Q}_n}{\mathrm{d}n} = \frac{1}{2} + \frac{1}{2}\tilde{Q}_n^2 - \tilde{Q}_n, \qquad \text{for} \qquad p = p_c = \frac{1}{2}, \tag{11.40}$$

which can easily be solved by separation of variables. The derivative of the solution $\tilde{Q}_n$ with respect to n is the probability of an avalanche to have a duration of exactly n steps.

$$\tilde{Q}_n = \frac{n}{n+2}, \tag{11.41}$$

$$D(t=n) = \frac{\mathrm{d}\tilde{Q}_n}{\mathrm{d}n} = \frac{2}{(n+2)^2} \sim n^{-2} .$$

Check that (11.41) really solves (11.40), with the initial condition $\tilde{Q}_0 = 0$.

(6.9) <u>GALTON–WATSON PROCESS</u>
The generating functions $G_0(x)$ are generically qualitatively similar to the ones shown in Fig. 6.12, due to the normalization condition $G(1) = 1$.

The fixpoint condition $G(q) = q$, Eq. (6.26), has a solution for $q < 1$ whenever

$$G'(x)\Big|_{x=1} > 1, \qquad W = G'(1) > 1\,,$$

since $G'(1)$ is the expectation value of the distribution function, which is the mean number of offsprings W.

11.7 Solutions to the Exercises of Chap. 7

(7.1) $K=1$ Kauffman Net
The solutions are illustrated in Fig. 11.15.

(7.2) $N=4$ Kauffman Net
The attractors of the $N = 4$ graph illustrated in Fig. 7.1 are given in Fig. 11.15.

(7.3) Synchronous vs. Asynchronous Updating
Asynchronous updating of the three-site network shown in Fig. 7.3 leads to the attractors illustrated in Fig. 11.16.

(7.4) Loops and Attractors
The attractors are defined by the set of linkage loops. There are two loops given which write out to C_1

$$ABC \to CA\bar{B} \to \bar{B}C\bar{A} \to \bar{A}\bar{B}\bar{C} \to \bar{C}\bar{A}B \to B\bar{C}A \to ABC\,,$$

and C_2

$$DE \to \bar{E}\bar{D} \to DE\,.$$

In its general form the attractor cycle is

$$\begin{aligned} ABCDE &\to CA\bar{B}\bar{E}\bar{D} \to \bar{B}C\bar{A}DE \to \bar{A}\bar{B}\bar{C}\bar{E}\bar{D} \\ &\to \bar{C}\bar{A}BDE \to B\bar{C}A\bar{E}\bar{D} \to ABCDE\,. \end{aligned}$$

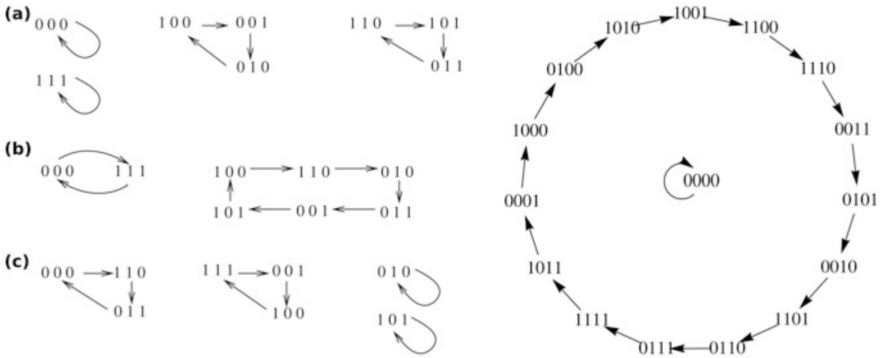


Fig. 11.15 **Left:** Solution of three $K = 1$, $N = 3$ boolean networks with a cyclic linkage tree $\sigma_1 = f_1(\sigma_2)$, $\sigma_2 = f_2(\sigma_3)$, and $\sigma_3 = f_3(\sigma_1)$. **(a)** All coupling functions are the identity. **(b)** All coupling functions are the negation. **(c)** f_1 and f_2 = negation, f_3 identity. **Right:** Solution for the $N = 4$ Kauffman nets shown in Fig. 7.1, $\sigma_1 = f(\sigma_2, \sigma_3, \sigma_4)$, $\sigma_2 = f(\sigma_1, \sigma_3)$, $\sigma_3 = f(\sigma_2)$, $\sigma_4 = f(\sigma_3)$, with all coupling functions $f(\ldots)$ being the generalized XOR functions, which count the parity of the controlling elements

Depending on the initial state, the loops C_1 und C_2 lead to two cycles each. Cycles L_1, L_2 emerge from C_1, L_3 and L_4 from C_2,

$$
\begin{aligned}
L_1 &: \quad 000 \to 001 \to 101 \to 111 \to 110 \to 010 \to 000 \\
L_2 &: \quad 100 \to 011 \to 100 \\
L_3 &: \quad 00 \to 11 \to 00 \\
L_4 &: \quad 01 \to 10 \to 01
\end{aligned}
$$

The possible combinations of (L_1, L_2) and (L_3, L_4) construct all possible attractor cycles. This leads to two attractor cycles of length two and two attractor cycles of length six,

$$
\begin{aligned}
L_1L_3 &: \; 00000 \to 00111 \to 10100 \to 11111 \to 11000 \to 01011 \to 00000 \\
L_1L_4 &: \; 00001 \to 00110 \to 10101 \to 11110 \to 11001 \to 01010 \to 00001 \\
L_2L_3 &: \; 10000 \to 01111 \to 10000 \\
L_2L_4 &: \; 10001 \to 01110 \to 10001
\end{aligned}
$$

(7.5) RELEVANT NODES AND DYNAMIC CORE
The attractors of the network illustrated in Fig. 7.3 are (000), (111) and (010) $\leftrightarrow$ (001). None of the elements σ_i, $i = 1, 2, 3$, has the same value in all three attractors, there are no constant nodes.

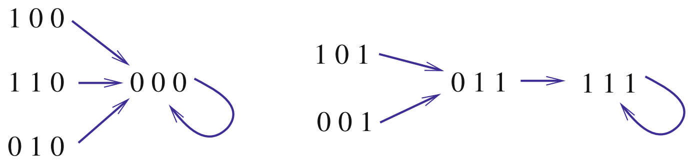


Fig. 11.16 Solution of the $N = 3$, $Z = 2$ network defined in Fig. 7.3, when using sequential asynchronous updating. The cycles completely change in comparison to the case of synchronous updating shown in Fig. 7.3

The dynamics obtained when substituting the AND by an XOR is shown in Fig. 11.17. One obtains two fixpoints, (000) and (011). The first element, σ_1 is hence constant, with the dynamic core being made-up from σ_2 and σ_3.

(7.6) <u>THE HUEPE AND ALDANA NETWORK</u>

As the exact solution can be found in the paper Huepe and Aldana-González (2002), we confine ourselves to some hints. You should start with the fraction of elements $\phi_N(t)$ with $+1$ at time t, which reduces to the probability $\phi(t)$ for $\sigma_i = +1$ in the $N \to \infty$ limit. You will then find that

$$s(t) = 2\phi(t) - 1\,.$$

Afterwards one has to consider the probability I(t) for the output function to be positive, which gives us the recursion equation

$$\phi(t+1) = I(t)(1-\eta) + (1 - I(t))\eta\,.$$

The relation between $I(t)$ and $\phi(t)$ is still unknown but can be calculated via

$$I(t) = \int_0^\infty P_{\xi(t)}(x)\mathrm{d}x$$

with $P_{\xi(t)}$ being the probability density function of the sum $\xi(t) = \sum_{j=1}^{K} \sigma_{i_j}(t)$, which can be represented as the K-fold of $P_{\sigma(t)}$ in Fourier space,

$$\hat{P}_{\xi(t)} = \left[\hat{P}_{\sigma(t)}\right]^K\,.$$

For the probability density of $\sigma(t)$ the proper ansatz is

$$P_{\sigma(t)} = \phi(t)\delta(x-1) + [1-\phi(t)]\,\delta(x+1)\,.$$

After some calculus you should finally obtain the recursion relation for $s(t)$ and find both its fixed points and the critical value η_c.

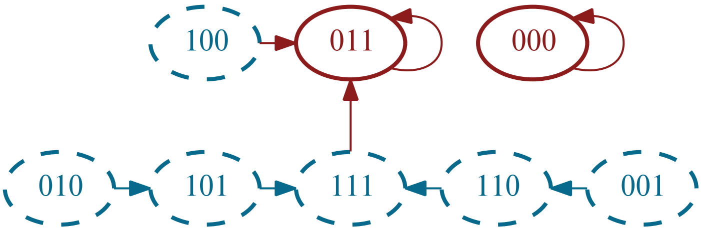


Fig. 11.17 Solution for the 3-site network shown in Fig. 7.3, with the AND function substituted by an XOR

(7.7) LOWER BOUND FOR BOND PERCOLATION
An upper bound of the number $\sigma_n(d)$ of paths of length n on a d-dimensional hypercubic lattice is

$$\sigma_n(d) \;<\; 2d(2d-1)^{n-1}, \qquad \lambda(d) = \lim_{n\to\infty}\left\{\left[\sigma_n(d)\right]^{1/n}\right\} \;<\; 2d-1\,,$$

as there are $2d$ possibilities for the first step and $2d-1$ for each subsequent step. Self-retracing paths do however occur and not all of the such constructed paths are hence distinct.
An exponentially small (as a function of n) fraction of $\sigma_n(d)$ of paths of length n and starting from a given site will be covered with bonds if the bond covering probability p is lower than $1/\lambda(d)$ (because than $p\lambda(d) < 1$) and the percolation transition hence occurs at $p_c(d) = 1/\lambda(d)$. We hence find the lower bound

$$p_c(d) = \frac{1}{\lambda(d)} \;>\; \frac{1}{2d-1}\,. \tag{11.42}$$

11.8 Solutions to the Exercises of Chap. 8

(continued)

(8.1) The One-Dimensional Ising Model

For the one-dimensional Ising system the energy is

$$E = -J \sum_i S_i S_{i+1} - B \sum_i S_i \,, \tag{11.43}$$

with $S_i = \pm 1$. The partition function

$$\mathcal{Z}_N = \mathcal{Z}_N(T, B) = \sum_n e^{-\beta E_n} = \sum_{S_1} \dots \sum_{S_N} T_{S_1,S_2} T_{S_2,S_3} \dots T_{S_N,S_1}$$

can be evaluated with the help of the 2×2 transfer matrix

$$T = \begin{pmatrix} e^{\beta(J+B)} & e^{-\beta J} \\ e^{\beta J} & e^{\beta(J-B)} \end{pmatrix} . \tag{11.44}$$

It has the eigenvalues

$$\lambda_{1,2} = e^{\beta J} \cosh \beta B \pm \sqrt{e^{2\beta J} \cosh^2 \beta B - 2 \sinh 2\beta J} \,,$$

leading to

$$\mathcal{Z}_N(T, B) = \sum_{S1=S2} \left(T^N\right)_{S1,S2} = (\lambda_1)^N + (\lambda_2)^N \simeq (\lambda_1)^N \,,$$

for large N and $\lambda_1 \gg \lambda_2$. The free energy per particle is given by

$$\frac{F(T, B)}{N} = \frac{-kT}{N} \ln \mathcal{Z}_N(T, B) \,,$$

and the magnetization per particle by

$$\frac{M(T, B)}{N} = -\frac{\partial F(T, B)}{\partial B} = \left(1 + \frac{e^{-4\beta J}}{\sinh^2 \beta B}\right)^{-1/2} .$$

(8.2) ERROR CATASTROPHE
Case $u_- = 0$, $u_+ = u$ The fixpoint conditions read

$$
\begin{aligned}
0 &= (1-\sigma)x_i + ux_{i-1} - \phi x_i, \qquad i > 1\,, \\
0 &= x_1 - \phi x_1\,,
\end{aligned}
$$

where the x_i are the respective concentrations. Hence we can immediately write down the $N \times N$ reproduction rate matrix W:

$$
W = \begin{pmatrix}
1 & 0 & 0 & 0 \cdots \\
u & (1-\sigma) & 0 & 0 \cdots \\
0 & u & (1-\sigma) & 0 \cdots \\
\vdots & \vdots & \vdots & \vdots \ddots
\end{pmatrix},
$$

whose diagonal elements obviously represent the eigenvalues. The largest eigenvalue is $1 = W_{11}$, and the corresponding eigenvector

$$
e_1 = \frac{1}{\sqrt{N}}\left(1, \frac{u}{\sigma}, \left(\frac{u}{\sigma}\right)^2, \cdots, \left(\frac{u}{\sigma}\right)^{N-1}\right).
$$

This eigenvector is normalizable only for $u < \sigma$, viz $u = \sigma$ is the error threshold.
Case $u_- = u_+ = u$ The fixpoint conditions are

$$
\begin{aligned}
0 &= (1-\sigma)x_i + ux_{i-1} + ux_{i+1} - \phi x_i, \qquad i > 1, \\
0 &= x_1 + ux_2 - \phi x_1\,.
\end{aligned}
$$

The first equation is equivalent to

$$
x_{i+1} = \frac{\phi + \sigma - 1}{u} x_i - x_{i-1}, \qquad i > 1\,, \tag{11.45}
$$

which can be cast into the 2×2 recursion form

$$
\begin{pmatrix} x_i \\ x_{i+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & \frac{\phi+\sigma-1}{u} \end{pmatrix} \begin{pmatrix} x_{i-1} \\ x_i \end{pmatrix}.
$$

The largest eigenvalues of the above recursion matrix determine the scaling of x_i for large i. For the determination of the error threshold you may solve for the x_i numerically, using (11.45) and the mass-normalization condition $\sum_i x_i = 1$ for the self-consistent determination of the flux ϕ.

(8.3) COMPETITION FOR RESOURCES
Summing in (8.64) over the species i one obtains

$$\dot{C} = f\sum_i r_i x_i - dC, \qquad \dot{f} = a - f\sum_i r_i x_i, \qquad 0 = a - dC\,,$$

for the steady-state case $\dot{C} = 0 = \dot{f}$, viz $C = \sum_i x_i = a/d$; no life without regenerating resources. The growth rates $W_{ii} = fr_i - d$ need to vanish (or to be negative) in the steady state and hence

$$\lim_{t\to\infty} f(t) \;\to\; \min_i \left(\frac{d}{r_i}\right)\,.$$

Only a single species survives the competition for the common resource (the one with the largest growth rate r_i), the ecosystem is unstable.

(8.4) COMPETITIVE POPULATION DYNAMICS
The fixpoints of the symmetric sheeps and rabbits Lotka–Volterra model (8.65) are $\mathbf{x}_0^* = (0,0)$, $\mathbf{x}_x^* = (1,0)$, and $\mathbf{x}_y^* = (0,1)$, together with the non-trivial fixpoint

$$\mathbf{x}_k^* = \frac{1}{1+k}\big(1,1\big), \qquad J = \begin{pmatrix} (1-2x-ky) & -kx \\ -ky & (1-2y-kx) \end{pmatrix}\,.$$

For $\mathbf{x} = \mathbf{x}_k^*$ the Jacobian J and the Lyapunov exponents $\lambda_\pm$ are

$$\frac{1}{1+k}\begin{pmatrix} -1 & -k \\ -k & -1 \end{pmatrix}, \qquad \lambda_\pm = \frac{-1\pm k}{1+k}\,.$$

The non-trivial fixpoint is stable for weak competition $k < 1$. In this case, sheeps and rabbits coexist.

$\mathbf{x}_k^*$ becomes however a saddle for strong competition $k > 1$ and the phenomenon of spontaneous symmetry breaking occurs. Depending on the initial conditions, the system will flow either to $\mathbf{x}_x^*$ or to $\mathbf{x}_y^*$, even though the constituting equations of motion (8.65) are symmetric under an exchange of x and y, with $\mathbf{x}_x^*$ and $\mathbf{x}_y^*$ having the identical two Lyapunov exponents (-1) and $(1-k)$.

The trivial fixpoint $\mathbf{x}_0^*$ is always unstable with a degenerate eigenvalue of $(+1)$.

(8.5) HYPERCYCLES
The fixpoints (x_1^*, x_2^*) of the reaction network (8.41) are determined by

$$\begin{aligned} 0 &= x_1^*\,(\alpha + \kappa\, x_2^* - \phi) \\ 0 &= x_2^*\,(2\alpha + \kappa\, x_1^* - \phi) \\ \phi &= \alpha x_1^* + 2\alpha x_2^* + 2\kappa x_1^* x_2^* \end{aligned}$$

where ϕ is the flux. The last equation, see (8.42), is used to satisfy the constraint $x_1 + x_2 = 1$, together with $x_1^*, x_2^* \geq 0$.
Solving above equations, one finds $x_1^* = \frac{\kappa-\alpha}{2\kappa}$ and $x_2^* = \frac{\kappa+\alpha}{2\kappa}$ for $\kappa > \alpha$. Otherwise, only the trivial solutions $(x_1^*, x_2^*) = (0, 1)$ and $(x_1^*, x_2^*) = (1, 0)$ are fixpoints. Linearizing the equations around the fixpoints leads us to the matrix

$$M = \begin{pmatrix} (\kappa-\alpha)x_2^* - 4\kappa x_1^* x_2^* & (\kappa-\alpha)x_1^* - 2\kappa(x_1^*)^2 \\ \kappa x_2^* - 2\kappa(x_2^*)^2 & \alpha + \kappa x_1^* - 2\alpha x_2^* - 4\kappa x_1^* x_2^* \end{pmatrix}.$$

For $(x_1^*, x_2^*) = (1, 0)$ the biggest eigenvalue of M is $\kappa + \alpha$, which is positive for positive growth rates, so the fixpoint is unstable. For (0, 1) one finds the condition $\kappa < \alpha$, which guarantees that all eigenvalues are negative. For the analysis of $(\frac{\kappa-\alpha}{2\kappa}, \frac{\kappa+\alpha}{2\kappa})$, computer algebra systems like Maple, Mathematica or Mathlab, may be used.

(8.6) PRISONER'S DILEMMA ON A LATTICE
We use first a general payoff matrix and then, specifically, $\{T; R; P; S\} = \{3.5; 3.0; 0.5; 0.0\}$ as in Fig. 8.13. We consider the four cases separately:

- ONE DEFECTOR IN THE BACKGROUND OF COOPERATORS
The payoffs are

$$\begin{aligned} \text{intruding defector:} \quad & 4 \times T && = 4 \times 3.5 && = 14 \\ \text{cooperating neighbors:} \quad & 3 \times R + 1 \times S && = 3 \times 3 + 0 && = 9 \end{aligned}$$

Therefore, the neighboring cooperators will become defectors in the next step.

- TWO ADJACENT DEFECTORS IN THE BACKGROUND OF COOPERATORS
The payoffs are:

$$\begin{aligned} \text{intruding defectors:} \quad & 3 \times T + 1 \times P && = 3 \times 3.5 + 0.5 && = 11 \\ \text{cooperating neighbors:} \quad & 3 \times R + 1 \times S && = 3 \times 3 + 0 && = 9 \end{aligned}$$

Therefore, the neighboring cooperators will become defectors in the next step.

- ONE COOPERATOR IN THE BACKGROUND OF DEFECTORS
The payoffs are:

$$\begin{aligned} \text{intruding cooperator:} \quad & 4 \times S && = 4 \times 0 && = 0 \\ \text{defecting neighbors:} \quad & 3 \times P + 1 \times T && = 3 \times 0.5 + 3.5 && = 5 \end{aligned}$$

The cooperating intruder will die and in the next step only defectors will be present.

– TWO ADJACENT COOPERATORS IN THE BACKGROUND OF DEFECTORS
The payoffs are:

$$\begin{aligned} \text{intruding cooperators:} &\quad 3 \times S + 1 \times R = 4 \times 0 + 3 \qquad = 3 \\ \text{defecting neighbors:} &\quad 3 \times P + 1 \times T = 3 \times 0.5 + 3.5 = 5 \end{aligned}$$

The cooperating intruders will die and in the next step only defectors will be present.

One can go one step further and consider the case of three adjacent intruders. Not all intruders will then survive for the case of defecting intruders and not all intruders will die for the case of cooperating intruders.

(8.7) NASH EQUILIBRIUM
The payoff matrix of this game is given by

$$A = \begin{pmatrix} L & L \\ 0 & H \end{pmatrix}, \qquad L < H\,,$$

for the cautious/risky player, where L signifies the low payoff and H the high payoff. Denoting the number of cautious players by N_c we can compute the reward for participants playing cautiously or riskily, respectively and from this the global reward G:

$$\begin{aligned} R_c &= [LN_c + L(N - N_c)]\,/N = L\,, \\ R_r &= [0 \cdot N_c + H(N - N_c)]\,/N = H(N - N_c)/N\,, \\ G(N_c) &= \frac{(N - N_c)^2}{N} H + N_c L\,. \end{aligned}$$

The function $G(N_c)$ has two local maxima at $N_c = 0$ and $N_c = N$ representing the Nash equilibria. The first case is optimal for all players, with the maximal global utility being NH.

(8.8) WAR OF ATTRITION
For the stationary solution, the payoffs E_m in (8.66) must be independent of the investment m, a condition which holds generically for games with continuous investments. The outcome is a "mixed evolutionary stable strategy", as determined by $\partial E_m/\partial m = 0$,

$$0 = vp(m) - c'(m) \int_m^\infty p(x)dx, \qquad \frac{p(m)}{c'(m)} = \frac{1}{v} \int_m^\infty p(x)dx\,.$$

Differentiation leads to

$$\frac{p'(x)}{c'(x)} - \frac{p(x)}{(c'(x))^2}c''(x) = \frac{-1}{v}p(x)\,, \tag{11.46}$$

when changing variables from m to x. The solution of (11.46) is

$$p(x) = \frac{c'(x)}{v}\mathrm{e}^{-c(x)/v}, \qquad \int_0^\infty p(x)dx = 1\,.$$

A linear cost function $c(x) = x$ corresponds to an exponentially distributed evolutionary stable strategy.

(8.9) <u>TRAGEDY OF THE COMMONS</u>

For a generic productivity of the commons, the payoff function is

$$E_i = x_i\big[P(x_{\text{tot}}) - c_i\big], \qquad P(0) = 1, \qquad P'(x_{\text{tot}}) < 0\,.$$

Payoffs are optimal when

$$P(x_{\text{tot}}) - c_i = -x_i P'(x_{\text{tot}}), \qquad E_i = -P'(x_{\text{tot}})\,x_i^2\,, \tag{11.47}$$

where x_{tot} is determined by averaging the first equation over agents,

$$P(x_{\text{tot}}) - \bar{c} = -\frac{x_{\text{tot}}}{N}P'(x_{\text{tot}})\,. \tag{11.48}$$

Maximal investments costs c_{max} are derived from the limit $x_i \to 0$ in (11.47),

$$c_{\text{max}} = P(x_{\text{tot}}), \qquad x_i = \frac{c_{\text{max}} - c_i}{-P'(x_{\text{tot}})}\,, \tag{11.49}$$

which leads to

$$E_i = \frac{(c_{\text{max}} - c_i)^2}{-P'(x_{\text{tot}})}\,.$$

Again, the dispersion relation is strictly quadratic, compare the original case, Eq. (8.60). That it depends explicitly on x_{tot} is not relevant, as it follows from (11.48) that $c_{\text{max}} - \bar{c} \sim 1/N$, which leads directly to catastrophic poverty. Catastrophic poverty arises hence whenever the productivity function $P(x_{\text{tot}})$ is monotonically decaying.

11.9 Solutions to the Exercises of Chap. 9

(9.1) THE DRIVEN HARMONIC OSCILLATOR

For all times, the solution can be obtained by combining the homogeneous solution (no external force) and one special solution, such as the long-time ansatz from (9.3) and (9.4). Since the homogeneous solution is given by

$$x(t) \sim e^{\lambda t}, \qquad \lambda_\pm = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2},$$

with damping γ, this contribution vanishes in the limit $t \to 0$ and only the special solution survives.

(9.2) SELF-SYNCHRONIZATION

We use the ansatz

$$\dot\theta(t) = \omega_0 + K \sin[\theta(t-T) - \theta(t)], \qquad \theta(t) = \omega t + \epsilon e^{\lambda t}$$

for the steady-state solution $\propto \omega t$, together with a small perturbation $\propto \epsilon e^{\lambda t}$. The steady-state solution is stable whenever the real part of λ is negative.[3] Expanding for $\epsilon \ll 1$ we obtain

$$\omega = \omega_0 - K \sin(\omega T), \qquad \lambda = K \cos(\omega T)\left[e^{-\lambda T} - 1\right], \tag{11.50}$$

for the contributions $O(\epsilon^0)$ and $O(\epsilon^1)$ respectively, which specializes (9.27) to the case of self-synchronization. There is always a solution $\lambda = 0$ to (11.50), indicating marginal stability. A graphical analysis of the second

[3] Compare Sect. 2.5 of Chap. 2.

equation of (11.50) shows that $\lambda = 0$ remains the only solution for $K\cos(\omega T) > 0$, and that a solution with $\lambda < 0$ appears for

$$-1 < TK\cos(\omega T) < 0\,.$$

Further on, for $TK\cos(\omega T) < -1$, the exponent λ becomes positive and the steady-state trajectory $\propto \omega t$ becomes unstable.

Considering for the locking frequency ω graphically the limit $K \to \infty$ in (11.50) on finds that

$$\omega T \;\to\; n\,2\pi, \qquad \omega = n\,\frac{2\pi}{T}, \qquad n = 0,\ 1,\ 2,\ \ldots\,.$$

The natural frequency ω_0 becomes irrelevant and the system self locks with respect to the delay time T.

(9.3) <u>KURAMOTO MODEL WITH THREE OSCILLATORS</u>
For the Kuramoto system,

$$\dot\theta_i = \omega_i + \frac{K}{N}\sum_{j\neq i}\sin(\theta_j - \theta_i)\,, \tag{11.51}$$

with identical natural frequencies $\omega_i \equiv \omega$, one can set $\omega = 0$ without loss of generality. This is achieved by transforming onto the rotating frame of reference, via $\theta_i \to \theta_i + \omega t$. Setting $\tau = 3/K$ one obtains

$$\tau\dot\theta_1 = \sin(\theta_2 - \theta_1) + \sin(\theta_3 - \theta_1)$$
$$\tau\dot\theta_2 = \sin(\theta_1 - \theta_2) + \sin(\theta_3 - \theta_2)$$
$$\tau\dot\theta_3 = \sin(\theta_1 - \theta_3) + \sin(\theta_2 - \theta_3)$$

for the case of $N = 3$ oscillators. Using relative variables $\alpha = \theta_2 - \theta_1$ and $\beta = \theta_3 - \theta_2$, which implies $\theta_3 - \theta_1 = \beta + \alpha$, leads to a closed set of equations for α and β:

$$\tau\dot\alpha = -2\sin(\alpha) + \sin(\beta) - \sin(\alpha+\beta), \tag{11.52}$$

$$\tau\dot\beta = -2\sin(\beta) + \sin(\alpha) - \sin(\alpha+\beta)\,. \tag{11.53}$$

For the fixpoints, determined by $\dot\alpha = 0 = \dot\beta$, we add above equations after multiplying one of them by a factor of two:

$$\sin(\alpha) = -\sin(\alpha+\beta), \qquad \sin(\beta) = -\sin(\alpha+\beta)\,. \tag{11.54}$$

The fixpoint are

$$(0,0), \qquad (\pi,0), \qquad (0,\pi), \qquad (\pi,\pi), \qquad \pm\frac{2\pi}{3}(1,1)\,, \tag{11.55}$$

of which $(0,0)$ (fully synchronized) is stable for $K > 0$ (attractive coupling). The last state, $(1,1)2\pi/3$, corresponds to three oscillators separated respectively by 120°, which is the stable configuration for $K < 0$ (repulsive coupling). The remainder three fixpoints, $(0,\pi)$, $(\pi,0)$ and (π,π) are unstable; they correspond to configurations in which two oscillators are synchronized, with the third one being anti-synchronized.

(9.4) LYAPUNOV EXPONENTS ALONG SYNCHRONIZED ORBITS
For two coupled harmonic oscillators

$$\dot\theta_1 = \omega_1 + \frac{K}{2}\sin(\theta_2-\theta_1), \qquad \dot\theta_2 = \omega_2 + \frac{K}{2}\sin(\theta_1-\theta_2),$$

see (9.7), the Jacobian at the fixpoint (9.9) is

$$\frac{K}{2}\cos(\theta_2-\theta_1)\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}, \qquad \sin(\theta_2-\theta_1) = \frac{\omega_2-\omega_1}{K}\,.$$

The eigenvalues and eigenvectors are then

$$\lambda_1 = 0, \qquad \mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad \lambda_2 = -\kappa, \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix},$$

where $\kappa = K\cos(\theta_2-\theta_1)/2 > 0$, since $\theta_2-\theta_1 \in [-\pi/2,\pi/2]$ (at the attracting fixpoint). The first Lyapunov exponent describes a neutral flow along the synchronized orbits, with $\Delta\theta = \theta_2-\theta_1$ remaining constant. The second Lyapunov exponent is negative, describing the attraction along the orthogonal direction. The periodicity condition, namely that the average of the first Lyapunov exponent over one period vanishes, is fulfilled.

(9.5) SYNCHRONIZATION OF CHAOTIC MAPS
For (9.57), we are looking for solutions of type

$$x_1(t) = \bar x(t) + \epsilon c^t, \qquad x_2(t) = \bar x(t) - \epsilon c^t\,,$$

with $\epsilon \ll 1$. That is we are looking for perturbations perpendicular to the synchronized trajectory $x_1(t) = x_2(t)$. For the Bernoulli shift map, $f(x) = ax \bmod 1$ with $x \in [0,1]$, one obtains

$$\bar x(t+1) = f\Big((1-\kappa)\bar x(t) + \kappa\bar x(t-T)\Big), \qquad c = (1-\kappa)a - \kappa a c^{-T}\,. \tag{11.56}$$

The solution $x_1(t) = x_2(t) = \bar{x}(t)$ is stable for $|c| < 1$, viz for $\kappa > \kappa_c$. Setting $c = 1$ in (11.56) we find

$$1 = a - 2\kappa_c a, \qquad \kappa_c = \frac{a-1}{2a},$$

which is actually independent of the time delay T. It is instructive to plot $x_1(t)$ and $x_2(t)$, using a small script or program. Note however, that the synchronization process may take more and more time with increasing delay T.

The synchronization process is driven by an averaging procedure, which is most evident for $\kappa = 1/2$ and $T = 0$. For this setting of parameters, perfect synchronization is achieved in a single time step.

(9.6) TERMAN–WANG OSCILLATOR

We linearize (9.39) around the fixpoint (x^*, y^*), taking at the same time the limit $\beta \to 0$,

$$\lim_{\beta\to\infty} \tanh(x/\beta) = -1 + 2\Theta(x) = \begin{cases} -1 & (x < 0) \\ \ \ 1 & (x > 0) \end{cases} .$$

We find, since $x^* < 0$ (compare Fig. 9.7),

$$\dot{\tilde{x}} = 3\,(1 - x^{*2})\,\tilde{x} \;-\; \tilde{y}$$
$$\dot{\tilde{y}} = -\,\epsilon\,\tilde{y}$$

where $\tilde{x} = x - x^*$ and $\tilde{y} = y - y^*$ are small perturbations around the fixpoint.

The ansatz $\tilde{x} \sim \tilde{y} \sim e^{\lambda t}$ determines the eigenvalues λ in above set of equations. We obtain $\lambda_1 = 3\,(1 - x^{*2})$ and $\lambda_2 = -\epsilon$. The fixpoint $x^* \simeq 0$ is unstable, since $\lambda_1 \simeq 3 > 0$ for this case. The fixpoint at $|x^*| > 1$ is stable, since $\lambda_1 < 0$, $\lambda_2 < 0$, which implies that $\tilde{x} \sim \tilde{y} \sim e^{\lambda t}$ decays in the long time limit.

(9.7) PULSE COUPLED LEAKY INTEGRATOR NEURONS

To be specific, we select $x_A(t = 1) = 1$ in (9.58), which implies $S_0 = 1/(1 - \exp(-\gamma))$. For the first neuron, the equation of motion reads then

$$\dot{x}_A = \gamma(S_0 - x_A), \qquad x_A(t) = \left(1 - \mathrm{e}^{-\gamma t}\right)/(1 - \mathrm{e}^{-\gamma})\,, \tag{11.57}$$

compare Fig. 11.18. For two coupled neurons $x_A(t)$ and $x_B(t)$ the membrane potential of the second neuron gets a kick ϵ when the first neuron spikes, and vice versa.

When the two neurons are synchronized, say A spikes just before B, the membrane potentials differ by ϵ just after the spike. Once x_A reaches the firing threshold, the difference $x_A(t) - x_B(t)$ is smaller than ϵ, due to the

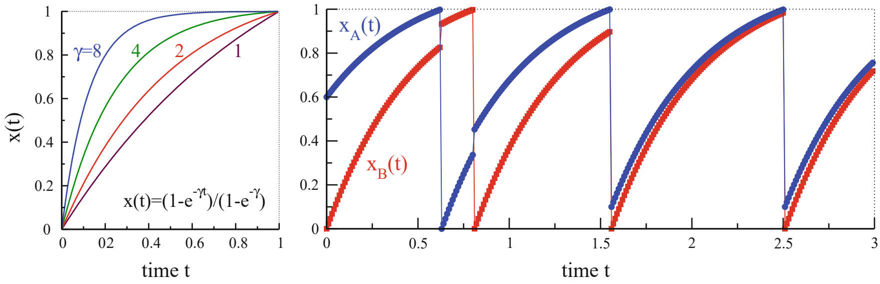


Fig. 11.18 **Left:** The time development $x(t)$ of a single leaky integrator (11.57). It spikes when $x > 1$, resetting the membrane potential x to zero. **Right:** For $\gamma = 2$, two coupled leaky integrators. When one neuron spikes, the membrane potential of the other neuron is increased by $\epsilon = 0.1$

concaveness of the solution (11.57), compare Fig. 11.18. The resulting limit cycle is stable.

If the two neurons are not synchronized the concaveness of $x(t)$ reduces the difference of the spiking times for A and B until the limit cycle is reached.

(9.8) SIRS MODEL

The fixpoint equation reads

$$x^* = ax^* \left[1 - (\tau_R + 1)x^*\right] ,$$

which leads to the solution

$$x^* = 0 \qquad \text{or} \qquad x^* = \frac{a-1}{a(\tau_R + 1)} ,$$

where $\tau_R = 0, 1, 2, \ldots$ is the recovery time. We examine the stability of x^* against a small perturbation $\tilde{x}_n$ by linearization using $x_n = x^* + \tilde{x}_n$,

$$\tilde{x}_{n+1} = -ax^* \sum_{k=0}^{\tau_R} \tilde{x}_{n-k} + a\tilde{x}_n \Big[1 - (\tau_R + 1)x^*\Big]. \tag{11.58}$$

No Immunity For $\tau_R = 0$ the stability condition (11.58) for the trivial fixed point $x^* = 0$ reduces to

$$\tilde{x}_{n+1} = a\tilde{x}_n, \qquad \text{leading to the stability condition} \quad a < 1 .$$

The analysis for the second fixed point with $\tau_R = 0$ runs analogously to the computation concerning the logistic map.[4]

[4] See Sect. 2.4 of Chap. 2.

Immunity of One Time Step For $\tau_R = 1$, the situation becomes somewhat more complicated,

$$\tilde{x}_{n+1} = \frac{1}{2}(3-a)\tilde{x}_n + \frac{1}{2}(1-a)\tilde{x}_{n-1}\,.$$

With the common ansatz $\tilde{x}_n = \lambda^n$ for linear recurrence relations, one finds the conditions

$$\left|-\frac{1}{4}a + \frac{3}{4} + \frac{1}{4}\sqrt{a^2 - 14a + 17}\right| < 1, \qquad \text{and} \qquad a^2 - 14a + 17 > 0$$

for small perturbations to remain small and not to grow exponentially, i.e. $|\lambda| < 1$. In consequence, a has to fulfill

$$1 \;<\; a \;<\; 7 - 4\sqrt{2} \;\approx\; 1.34\,.$$

(9.9) <u>EPIDEMIOLOGY OF ZOMBIES</u>
Dividing the two equations

$$\dot{H} = -\alpha H Z, \qquad \dot{Z} = (2\alpha - 1)HZ \tag{11.59}$$

yields

$$\frac{dH}{dZ} = \frac{\alpha}{1-2\alpha}, \qquad H = \frac{(1-\alpha)H_0 - \alpha}{1-2\alpha} + \frac{\alpha}{1-2\alpha}Z\,, \tag{11.60}$$

where the constant of integration has been chosen such that the initial condition $Z_0 = 1 - H_0$ is respected. Whether zombies or humans are more effective depends on the value of α

- $\alpha > 1/2$
  Zombies win with $\alpha/(1-2\alpha)$ being negative. Humans go extinct when the population of zombies reaches

$$Z^* = \frac{\alpha - (1-\alpha)H_0}{\alpha}\,, \tag{11.61}$$

  which is positive for $\alpha > 1/2$ and all $H_0 \in [0, 1]$.
- $\alpha < 1/2$
  Whether humans manage to clinch on depends on the size of their initial population. For $Z \to 0$ the formally surviving human population is

$$H^* = \frac{(1-\alpha)H_0 - \alpha}{1-2\alpha} = \left\{\begin{matrix} >0 \\ <0 \end{matrix}\right\} \quad \text{for} \quad H_0 \left\{\begin{matrix} > \\ < \end{matrix}\right\} \frac{\alpha}{1-\alpha}\,. \tag{11.62}$$

Humans always survive when the initial zombie population $Z_0 = 1 - H_0$ is small.

Zombies do not recover by themselves, in contrast to the infected population within the SIRS model, they need to be eliminated by humans (the susceptibles). For a further discussion of the epidemiology of zombies see Alemi (2015).

11.10 Solutions to the Exercises of Chap. 10

(10.1) Gated Linear Units

The input vector of (10.65) is taken to be two dimensional, $\mathbf{x} = (x_1, x_2)$, with $x_i = \pm 1$. We set

$$\mathbf{w} = (1, 0), \qquad \mathbf{v} = (0, -1), \qquad b_w = 0 = b_v\,. \tag{11.63}$$

With $\sigma_\pm \equiv \tanh(\pm 1)$ we have

$$\sigma_{\mathrm{GLU}}(1,1) = \sigma_+\sigma_-, \qquad \sigma_{\mathrm{GLU}}(1,-1) = \sigma_+\sigma_+\,,$$

and equivalently for $(-1,-1)$ and $(-1,1)$. This is, apart from an overall normalization, the definition of the XOR gate.

For the AND gate one sets, e.g., $v_1 = 0 = v_2$, $b_v = -1$, which leads to constant second factor, and $w_1 = 1 = w_2$ and $b_w = 1.5$.

11.10 Variance of a Product

Trivially, $x = \Delta x + \mu_x$ holds, where $\Delta x = x - \mu_x$, and the same for y. Consequently one has

$$xy - \mu_x\mu_y = (\Delta x + \mu_x)(\Delta y + \mu_y) - \mu_x\mu_y = \Delta x \Delta y + \Delta x \mu_y + \Delta y \mu_x\,.$$

Per definition $\langle \Delta x \rangle = 0 = \langle \Delta y \rangle$, which leads to

$$\begin{aligned}\left\langle (xy - \mu_x \mu_y)^2 \right\rangle &= \left\langle \left[\Delta x \Delta y + \Delta x \mu_y + \Delta y \mu_x \right]^2 \right\rangle \\ &= \left\langle (\Delta x \Delta y)^2 \right\rangle + \left\langle (\Delta x \mu_y)^2 \right\rangle + \left\langle (\Delta y \mu_x)^2 \right\rangle .\end{aligned}$$

Lastly it suffices to note that x and y are independent. One then obtains

$$\sigma_{xy}^2 = \left(\sigma_x^2 + \mu_x^2\right)\left(\sigma_y^2 + \mu_y^2\right) - \mu_x^2 \mu_y^2 ,$$

the formula to be proven.

(10.3) <u>WIGNER SEMI-CIRCULAR LAW</u>
Adjacency matrices are symmetric, $\Gamma = 1$, with entries that are either zero or unity. With z being the coordination number, we have

$$\mu_w = \frac{z}{N}, \qquad \sigma_w^2 = \frac{1}{N}\left[(N-z)\left(\frac{z}{N}\right)^2 + z\left(1 - \frac{z}{N}\right)^2 \right] \approx \frac{z}{N} .$$

The boundaries,

$$(1+\Gamma)\sqrt{N}\sigma_w = 2\sqrt{z} ,$$

do hence agree. Given that (10.15) describes a uniformly filled ellipse, it follows that the density of state is also an ellipse.

(10.4) <u>LANDAU THEORY FOR ABSORBING PHASE TRANSITIONS</u>
Making use of the Taylor series $1/(1-x)^2 \approx 1 + 2x + 3x^2 + \ldots$ the self-consistency condition (10.26) takes the form

$$2R_w^2 \sigma_y^2 \approx 1 + 2\sigma_y^2 + 3\sigma_y^4 - \left(1 + 2\sigma_{\text{ext}}^2\right), \qquad \sigma_y^2 = x ,$$

or

$$2\sigma_{\text{ext}}^2 = 2(1 - R_w^2)\sigma_y^2 + 3\sigma_y^4 . \tag{11.64}$$

Substituting the relations

$$h \equiv 2\,\frac{\sigma_{\text{ext}}^2}{\sigma_y}, \qquad a = 1 - R_w^2, \qquad b = \frac{4}{3}, \qquad \phi \equiv \sigma_y \tag{11.65}$$

into (11.64) leads to

$$h = 2a\phi + wb\phi^3 ,$$

which is exactly Eq. (6.2), which determines the relation between the order parameter ϕ and the external field h, where $a = 1 - R_w^2 = a \sim T - T_c$.

For $h = 0$ the order parameter is finite when $a < 0$, viz when $T < T_c$ or $R_w^2 > 1$.

Interestingly, the correspondence to the Landau theory of phase transitions is recovered only when σ_{ext}^2 is measured in units of σ_y, as expressed by (11.65), with the relation $h = 2\sigma_{\text{ext}}^2/\sigma_y$ involving a self-consistency loop. In practice, one determines first σ_{ext}^2 from (11.64).

(10.5) LAYER VARIANCE MAPPING

Disregarding layer indices, the mean of the membrane potential is

$$\mu_x = \mu_{\text{ext}} + N\mu_w\mu_y = \mu_{\text{ext}} + \tilde{\mu}_w\mu_y, \qquad \mu_w = \frac{\tilde{\mu}_w}{N}, \tag{11.66}$$

where we made use of (10.16). The $1/N$ scaling for the mean implies that it drops out of the relation (10.18) for the variance,

$$\begin{aligned}\sigma_{\mathbf{w}\cdot\mathbf{y}}^2 &= N\Big[\big(\sigma_w^2 + \mu_w^2\big)\big(\sigma_y^2 + \mu_y^2\big) - \mu_w^2\mu_y^2\Big] \\ &= \tilde{\sigma}_w^2\big(\sigma_y^2 + \mu_y^2\big), \qquad \sigma_w = \frac{\tilde{\sigma}_w}{\sqrt{N}}.\end{aligned} \tag{11.67}$$

The probability distribution $P(x)$ is therefore a Gaussian with mean μ_x and variance $\sigma_x^2 = \sigma_{\text{ext}}^2 + \sigma_{\mathbf{w}\cdot\mathbf{y}}^2$. Equation (10.24) takes hence the from

$$\sigma_y^2 + \mu_y^2 = \frac{1}{\sqrt{2\pi\sigma_x^2}} \int dx \left(1 - \mathrm{e}^{-(x-b)^2}\right) \mathrm{e}^{-(x-\mu_x)^2/(2\sigma_x^2)},$$

or

$$1 - \sigma_y^2 - \mu_y^2 = \sqrt{2\pi\sigma_b^2} \int dx N(x|\mu_b, \sigma_b) N(x|\mu_x, \sigma_x) \tag{11.68}$$

$$= \frac{\sqrt{2\pi\sigma_b^2}}{\sqrt{2\pi(\sigma_b^2 + \sigma_x^2)}}\, \mathrm{e}^{-\frac{(\mu_x-\mu_b)^2}{2(\sigma_b^2+\sigma_x^2)}}, \tag{11.69}$$

when writing the integrand as the product of two Gaussians, where $\mu_b = b$ and $2\sigma_b^2 = 1$. In the last step we used the common formula for the product of two Gaussians.

Adding layer indices we obtain a mapping $(\sigma_{y,n-1}, \mu_{x,n-1}) \to (\sigma_{y,n}, \mu_{x,n})$, with

$$\begin{aligned}\mu_{x,n} &= \mu_{\text{ext},n} + \tilde{\mu}_{w,n}\mu_{y,n-1} \\ \sigma_{x,n}^2 &= \sigma_{\text{ext},n}^2 + \tilde{\sigma}_{w,n}^2\big(\sigma_{y,n-1}^2 + \mu_{y,n-1}^2\big)\end{aligned}$$

and

$$\mu_{y,n} = \int dx \sigma_{\text{Gaus}}(x - b_n) N(x|\mu_{x,n}, \sigma_{x,n}) \approx \sigma_{\text{Gaus}}(\mu_{x,n} - b_n)$$

$$1 - \sigma_{y,n}^2 - \mu_{y,n}^2 = \frac{1}{\sqrt{1 + 2\sigma_{x,n}^2}} \mathrm{e}^{-\frac{(\mu_{x,n} - b_n)^2}{1+2\sigma_{x,n}^2}}$$

where $\sigma_{\text{Gaus}}(z)$ is the Gaussian transfer function (10.23).

(10.6) <u>NEURAL DIFFERENTIAL EQUATIONS</u>

For a single linear unit per layer the neural differential equations (10.35) read

$$\dot{x} = \vartheta x, \qquad \dot{E} = -\vartheta E, , \qquad x(0) = x^\alpha \qquad E(1) = 2[F^\alpha - x(1)] , \tag{11.70}$$

where we normalized layer time, $t \in [0, 1]$. Also included are the boundary conditions for $x(t)$ (starting) and $E(t)$ (ending). Here we used a different sign convention for $E(t)$ than in the main text. It is possible to assume that $\vartheta(t) \equiv \vartheta$ is uniform when training starts, given that the update procedure for ϑ is independent of layer time,

$$\Delta\vartheta = -\eta x E, \qquad \frac{d}{dt}\big(xE\big) = (\vartheta - \vartheta)xE = 0 ,$$

when using (11.70) and (10.34), which implies that $xE = 2f^\alpha(F^\alpha - f^\alpha)$ for all layers. Here $f^\alpha = x(1)$. The layer-dependency of the activity $x = x(t)$ and the error $E = E(t)$ are

$$x(t) = x^\alpha \mathrm{e}^{\vartheta t}, \qquad E(t) = 2(F^\alpha - f^\alpha)\mathrm{e}^{-\vartheta(t-1)}, \qquad f^\alpha = x^\alpha \mathrm{e}^{\vartheta} .$$

Training stops once $\vartheta = \log(F^\alpha/x^\alpha)$, the expected outcome.

(10.7) <u>DOT PRODUCTS ARE KERNEL FUNCTIONS</u>

A feature function $\mathbf{g}(\mathbf{x})$ maps an input vector $\mathbf{x}$ to a real vector. The dot-product of the discretized function is

$$\Theta_{\alpha,\beta} = \sum_i g_i^\alpha g_i^\beta , \qquad \mathbf{g}^\alpha = \mathbf{g}(\mathbf{x}^\alpha), \qquad \alpha = 1, \ldots, N_\alpha .$$

For the tangent kernel, $\sum_i$ involves all adaptable parameters, with N_α being the training data size. The expectation value of any vector $\mathbf{h} = (h^1, \ldots, h^{N_\alpha})$ is

$$\langle\Theta_{\alpha,\beta}\rangle_{\mathbf{h}} = \sum_{\alpha,\beta} h^\alpha \Theta_{\alpha,\beta} h^\beta = \sum_i c_i^2 , \qquad c_i = \sum_\alpha h^\alpha g_i^\alpha ,$$

which is always positive (or zero). This statement holds in particular for the eigenvectors of Θ, which proves that the eigenvalues are positive (or zero).

(10.8) LINEAR ATTENTION
With (10.18), the variance of the entries of the query, key and value vectors is

$$\sigma_{QKV}^2 = D\sigma_0^2\sigma_x^2\,.$$

An individual key-query scalar product has hence a variance $D(\sigma_{QKV}^2)^2 = D^3\sigma_0^4\sigma_x^4$. Multiplied with a value vectors, the geometric sum (10.63) yields

$$\sigma_y^2 = \frac{1}{1-\gamma}\left(D\sigma_{QKV}^2\right)\sigma_{QKV}^2 = \frac{1}{1-\gamma}\,D^4\sigma_0^6\sigma_x^6$$

for the variance of the elements of the output activity vector. It is desirable that σ_y remains constant when D increases.

– When the normalization of the inputs activities does not scale with D, e.g. when $\sigma_x = 1$, one has $\sigma_0 \sim (1/D)^{2/3}$.
– When input vectors are normalized, such that $|\mathbf{x}_i| = 1$, one has $\sigma_x^2 \sim 1/D$ and hence $\sigma_0 \sim (1/D)^{1/6}$.

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